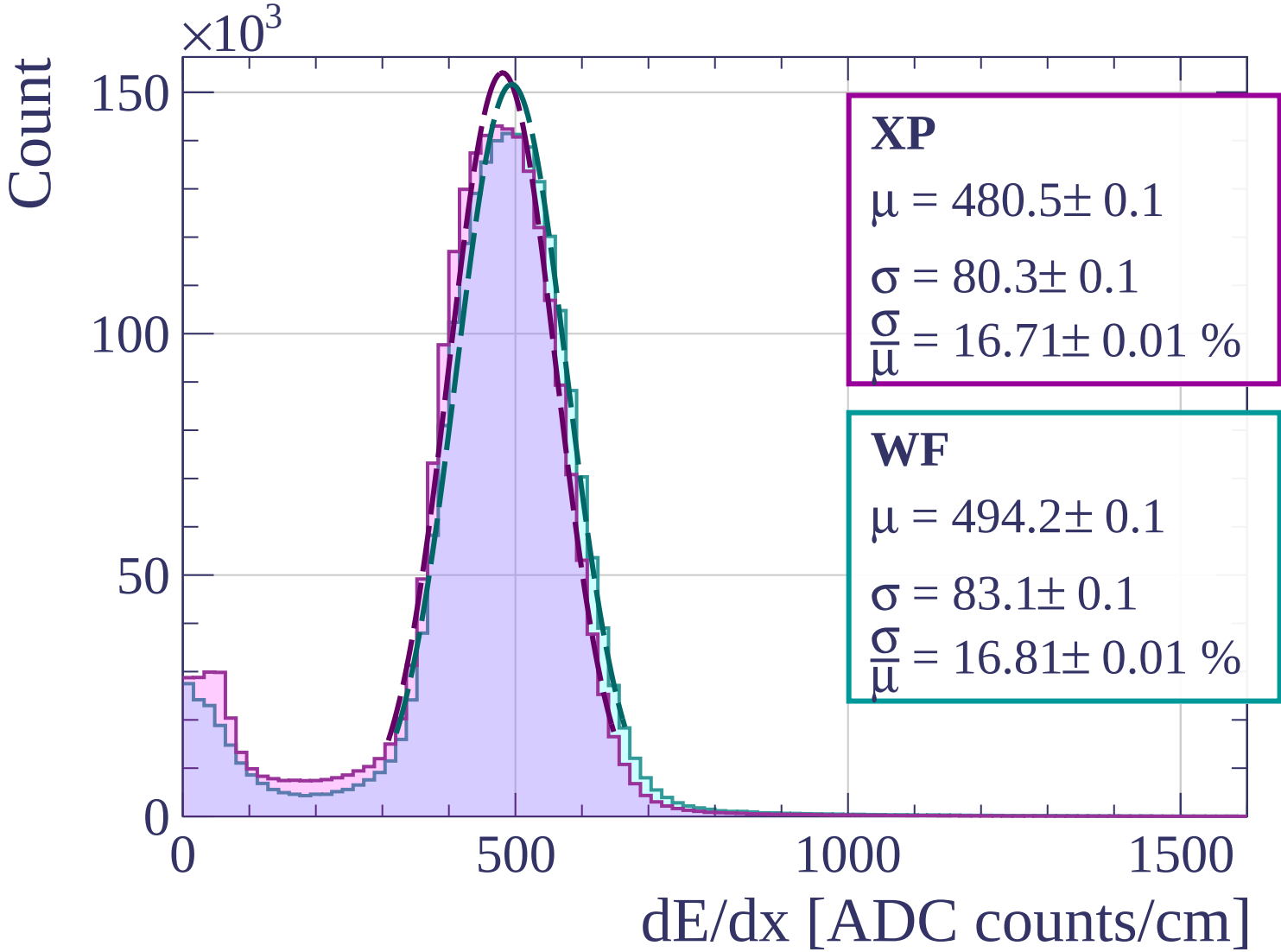
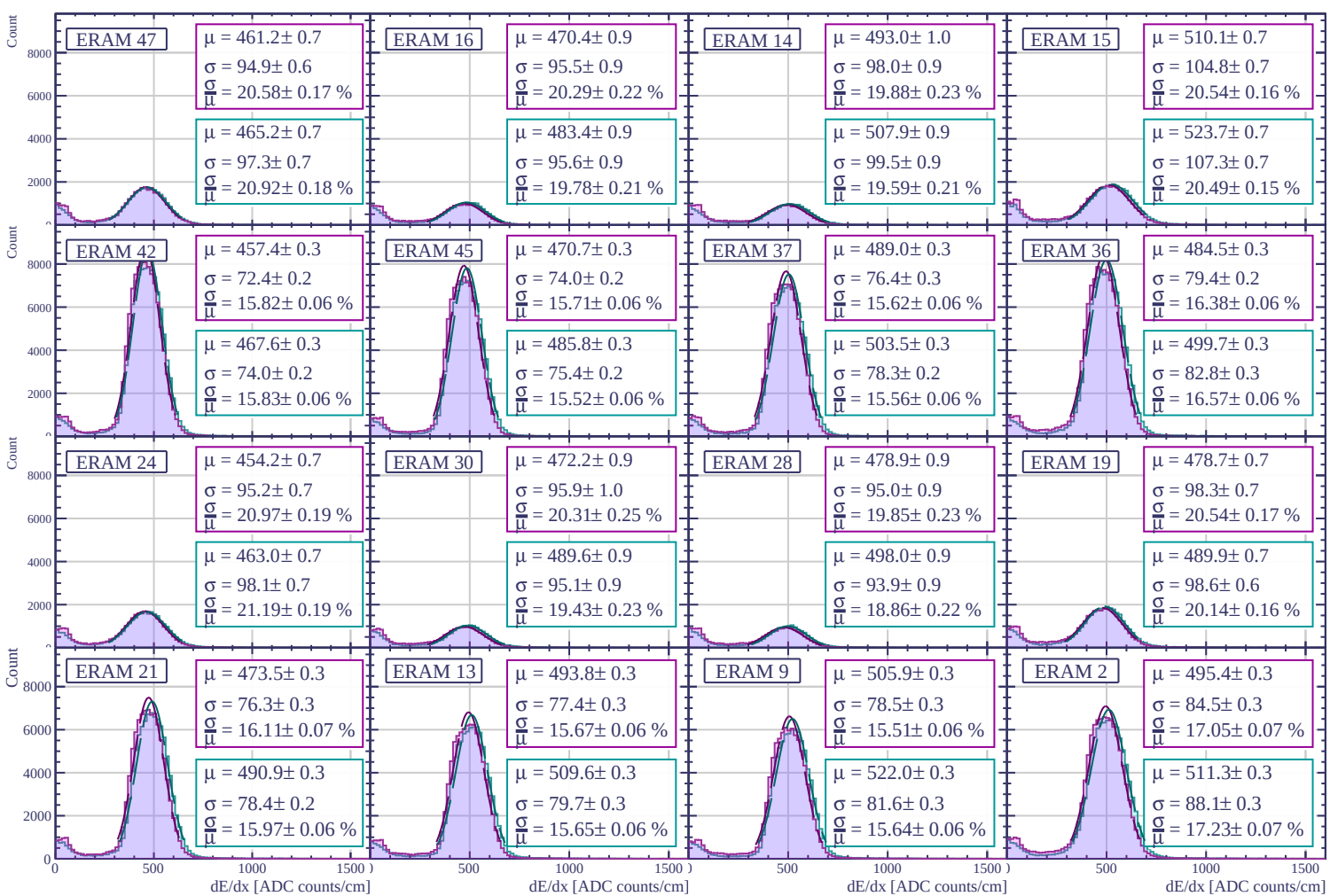
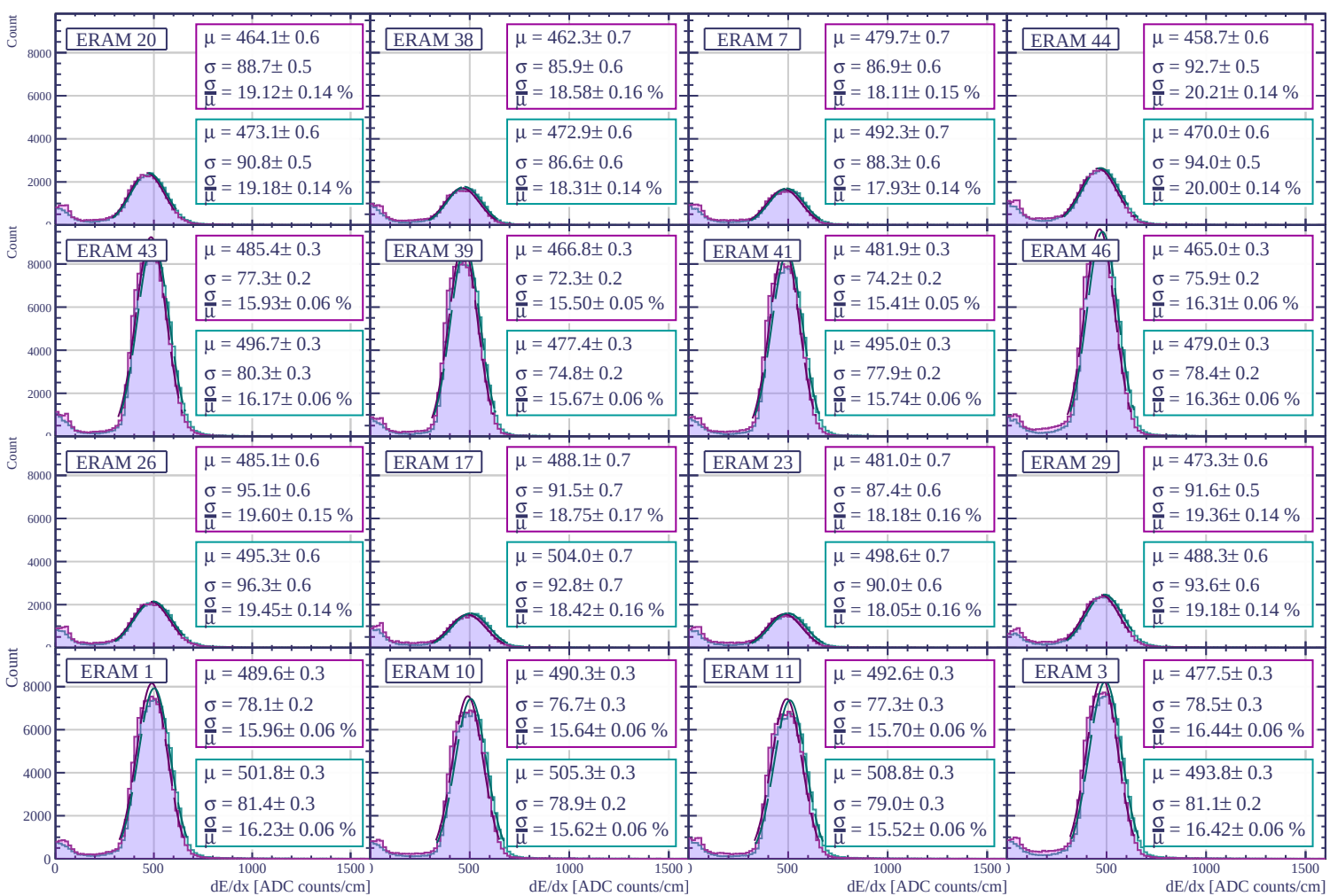
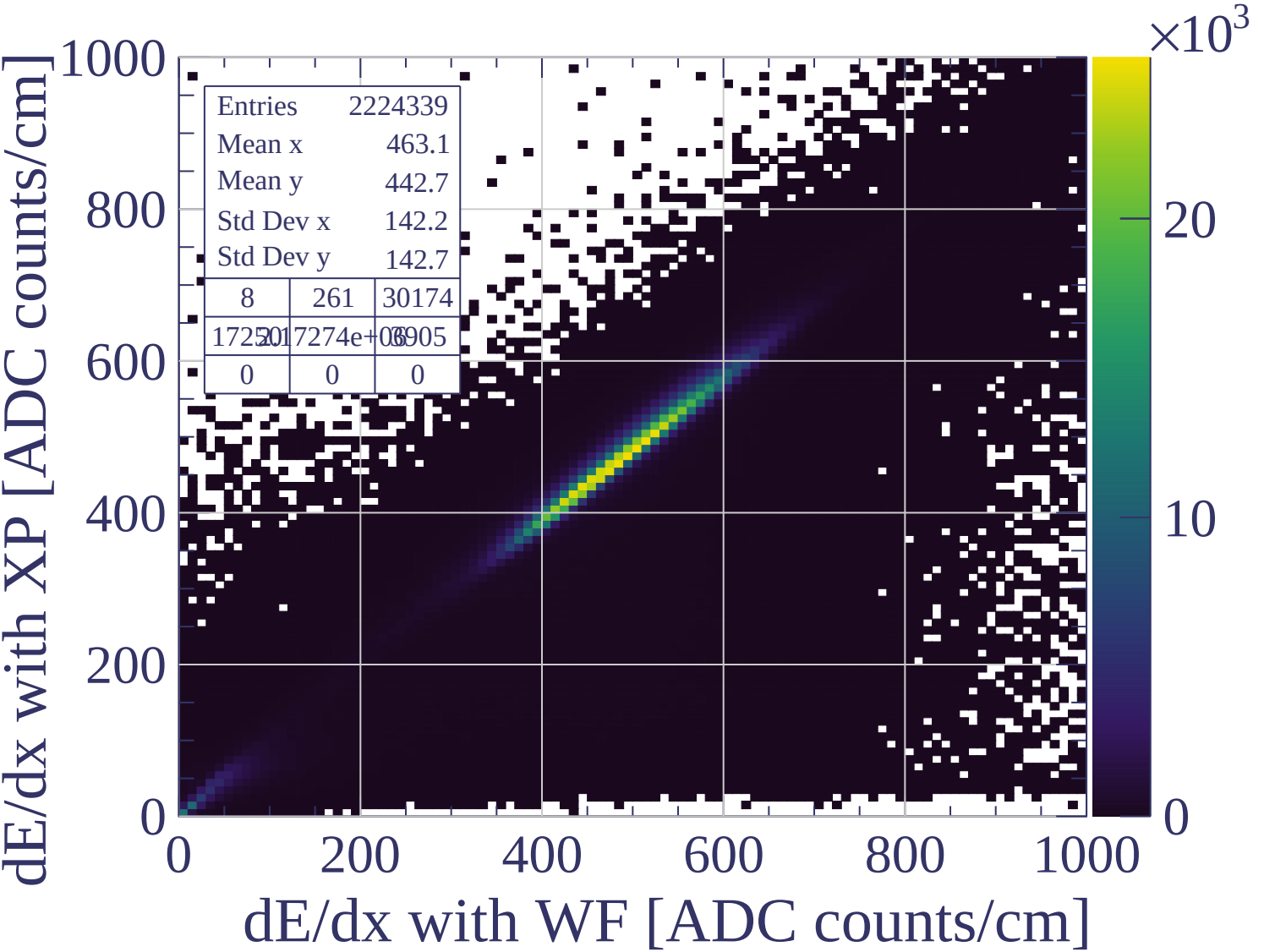
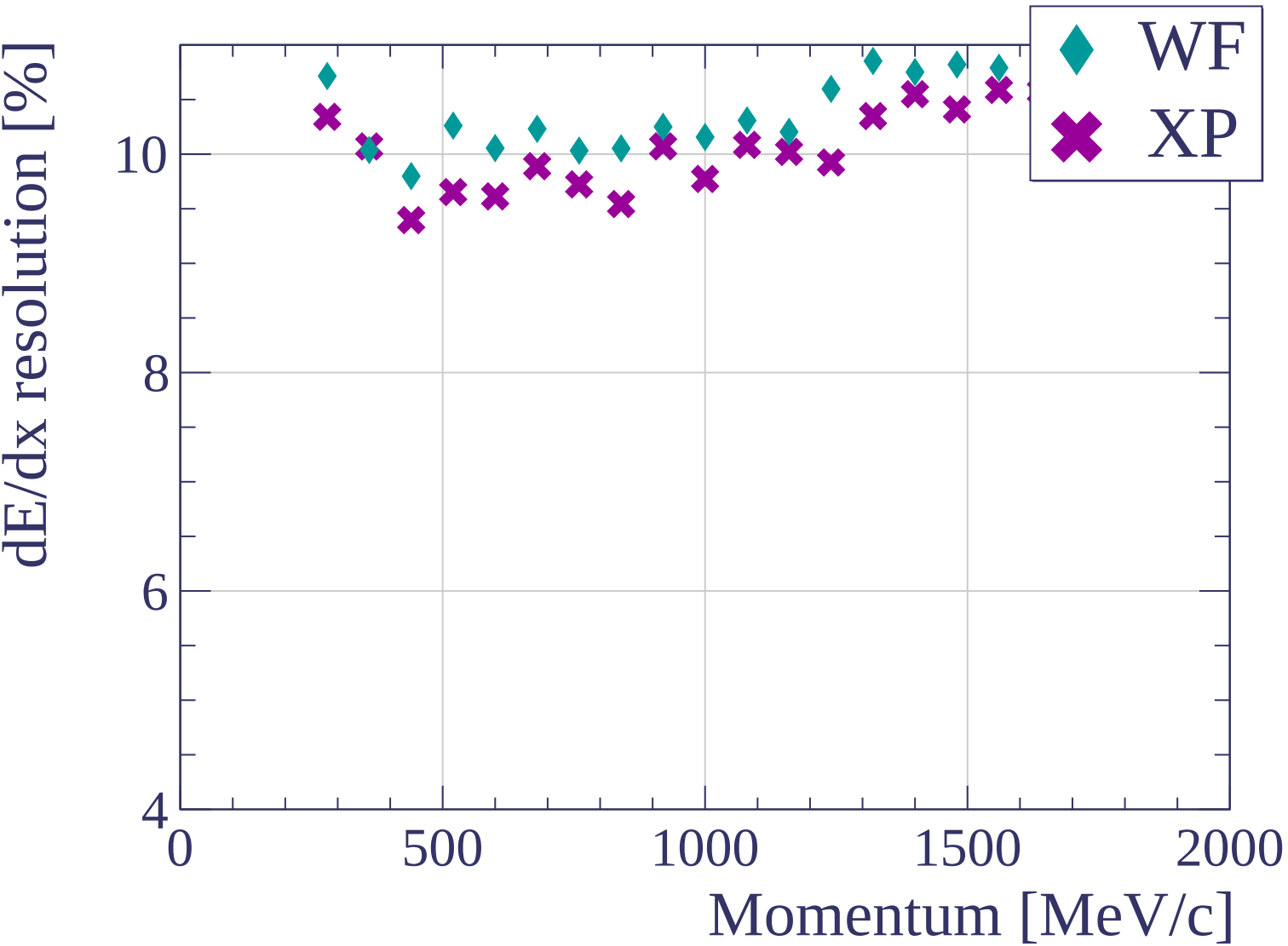


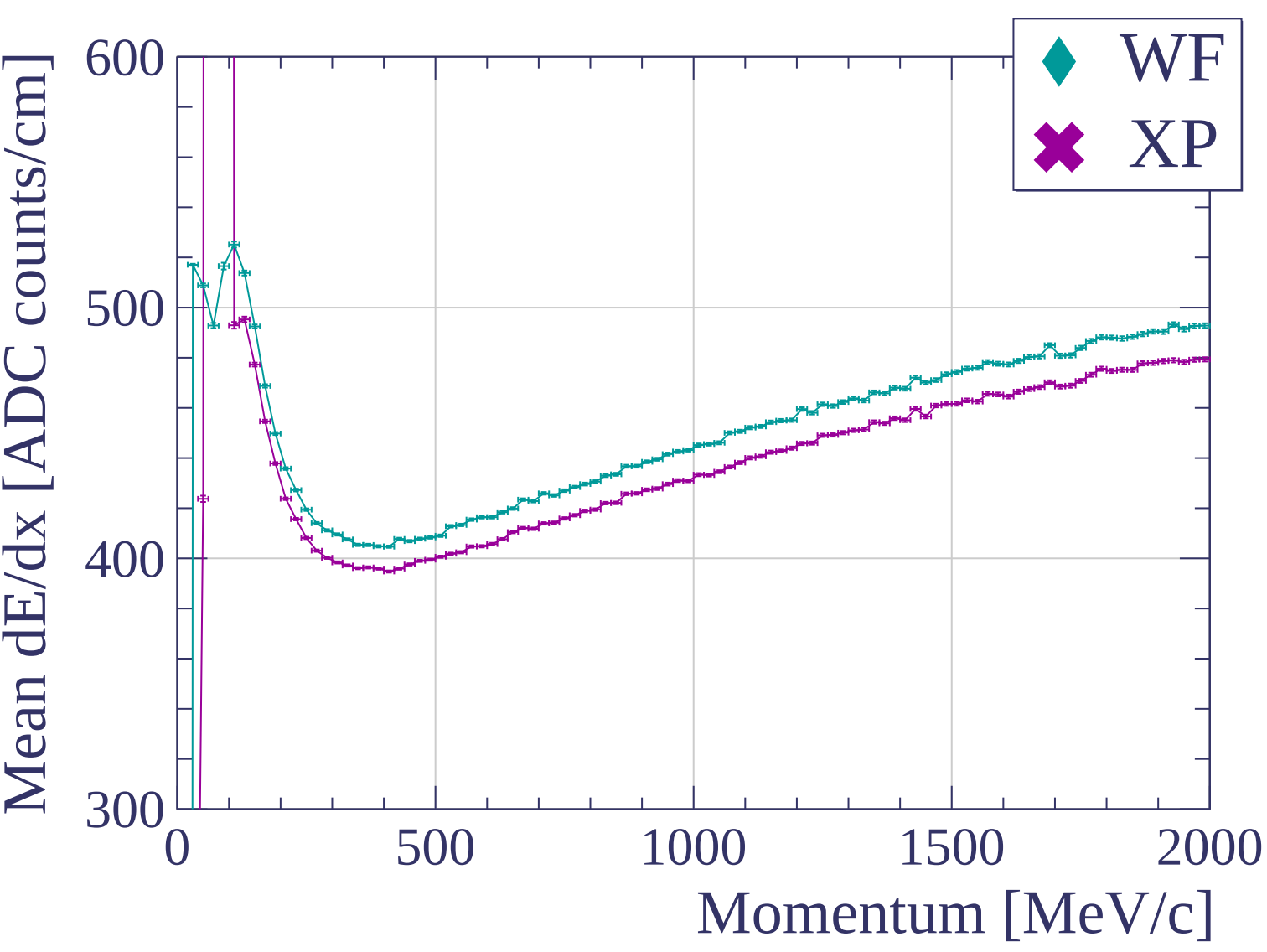


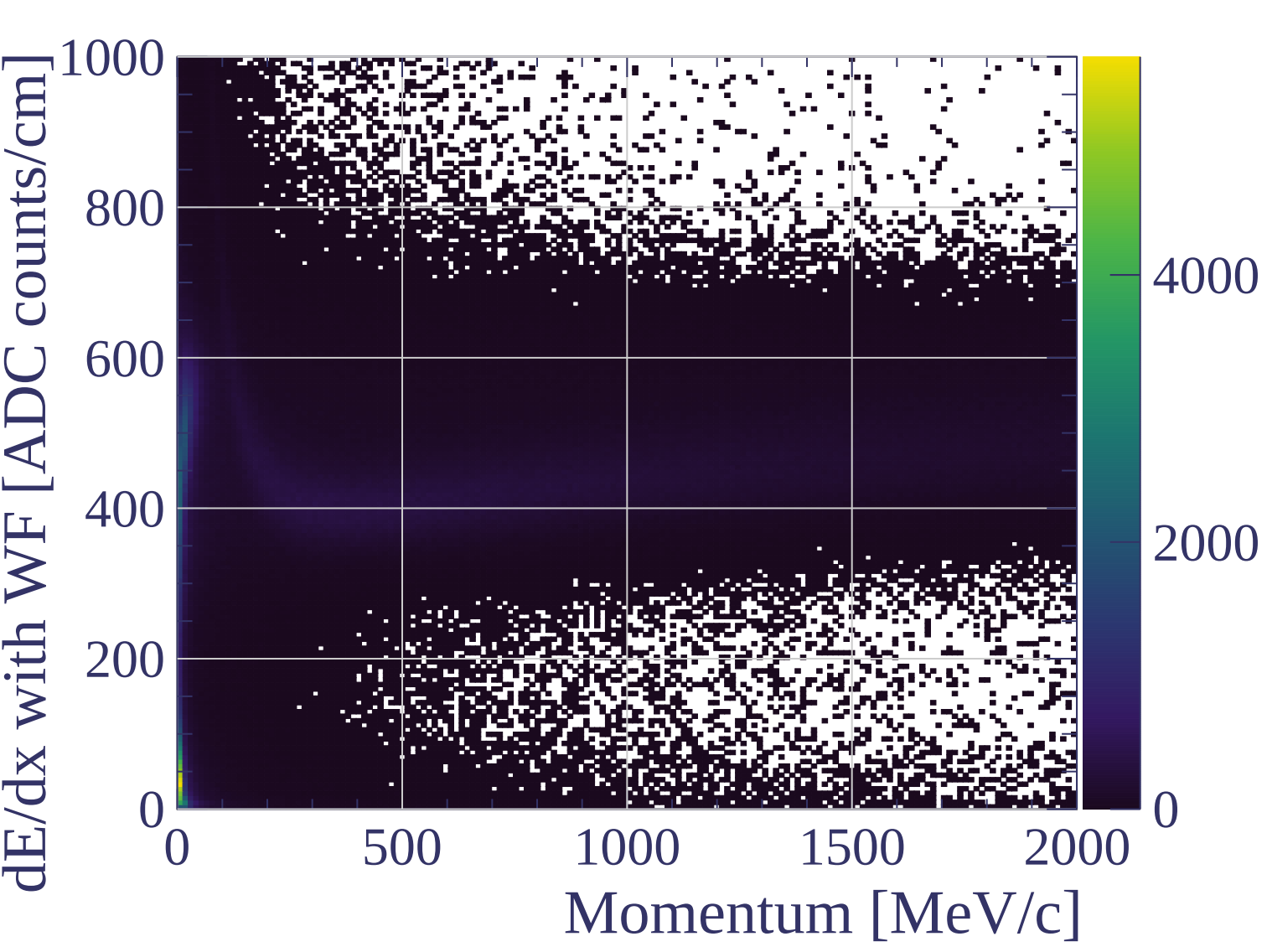


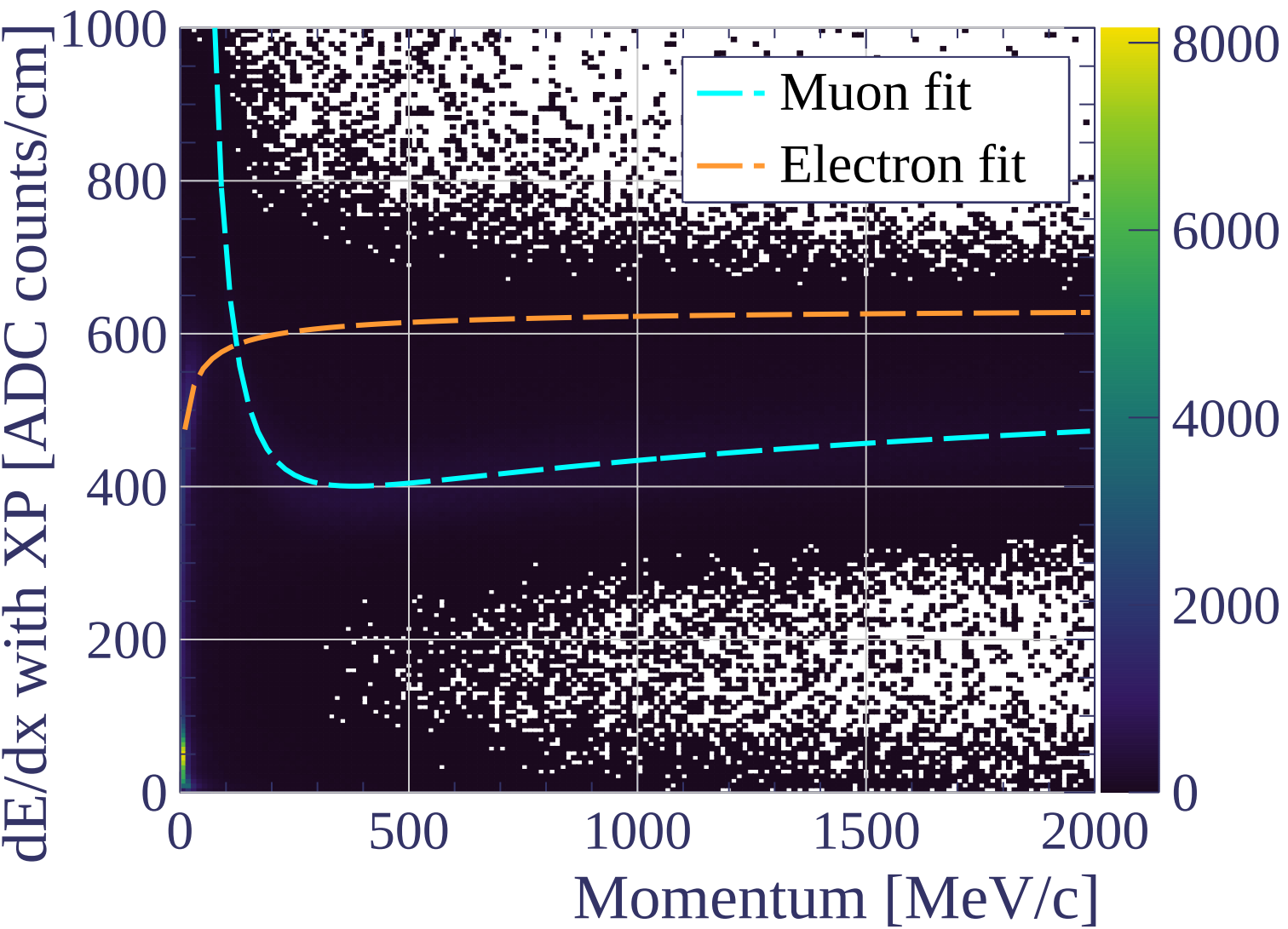




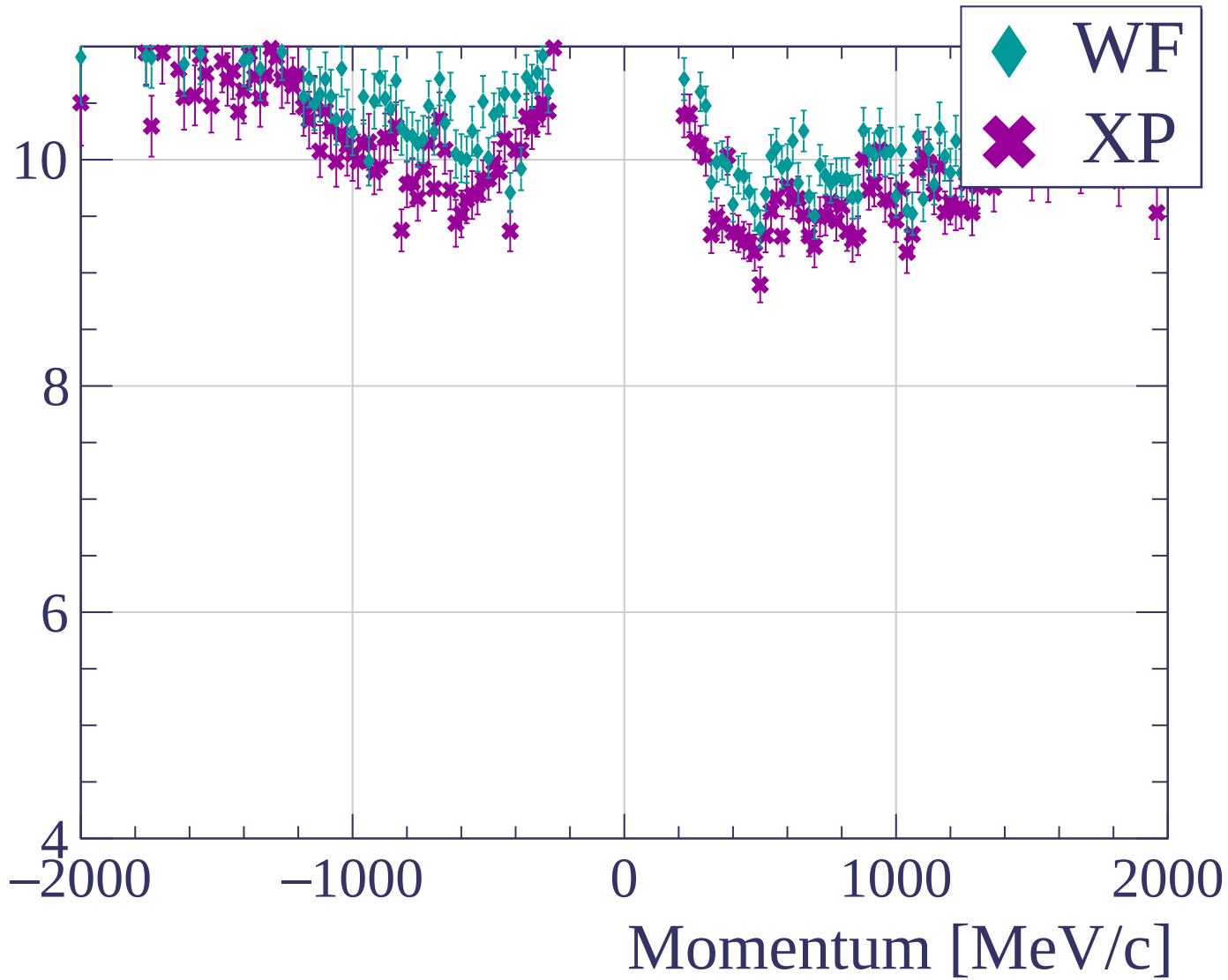


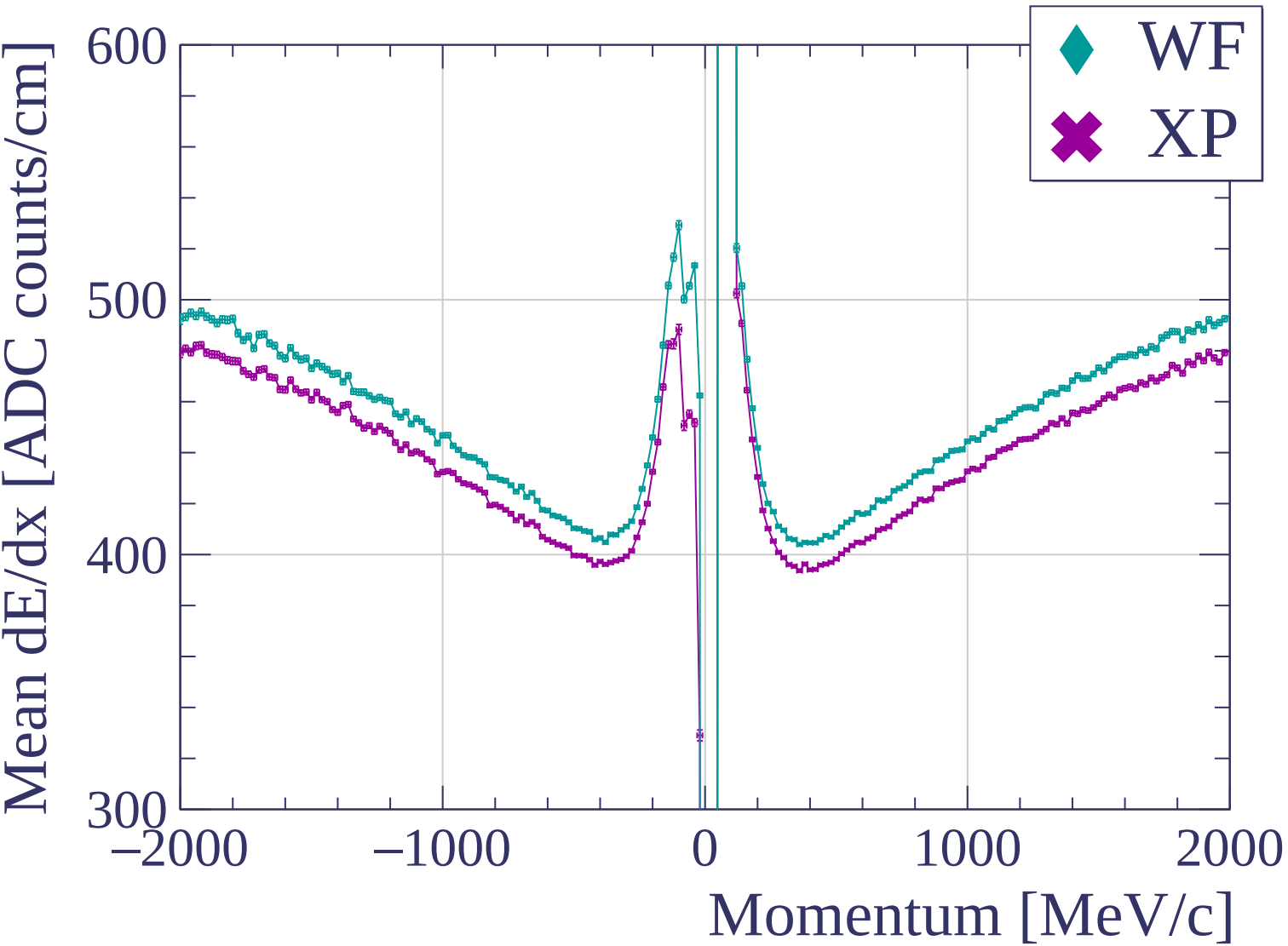


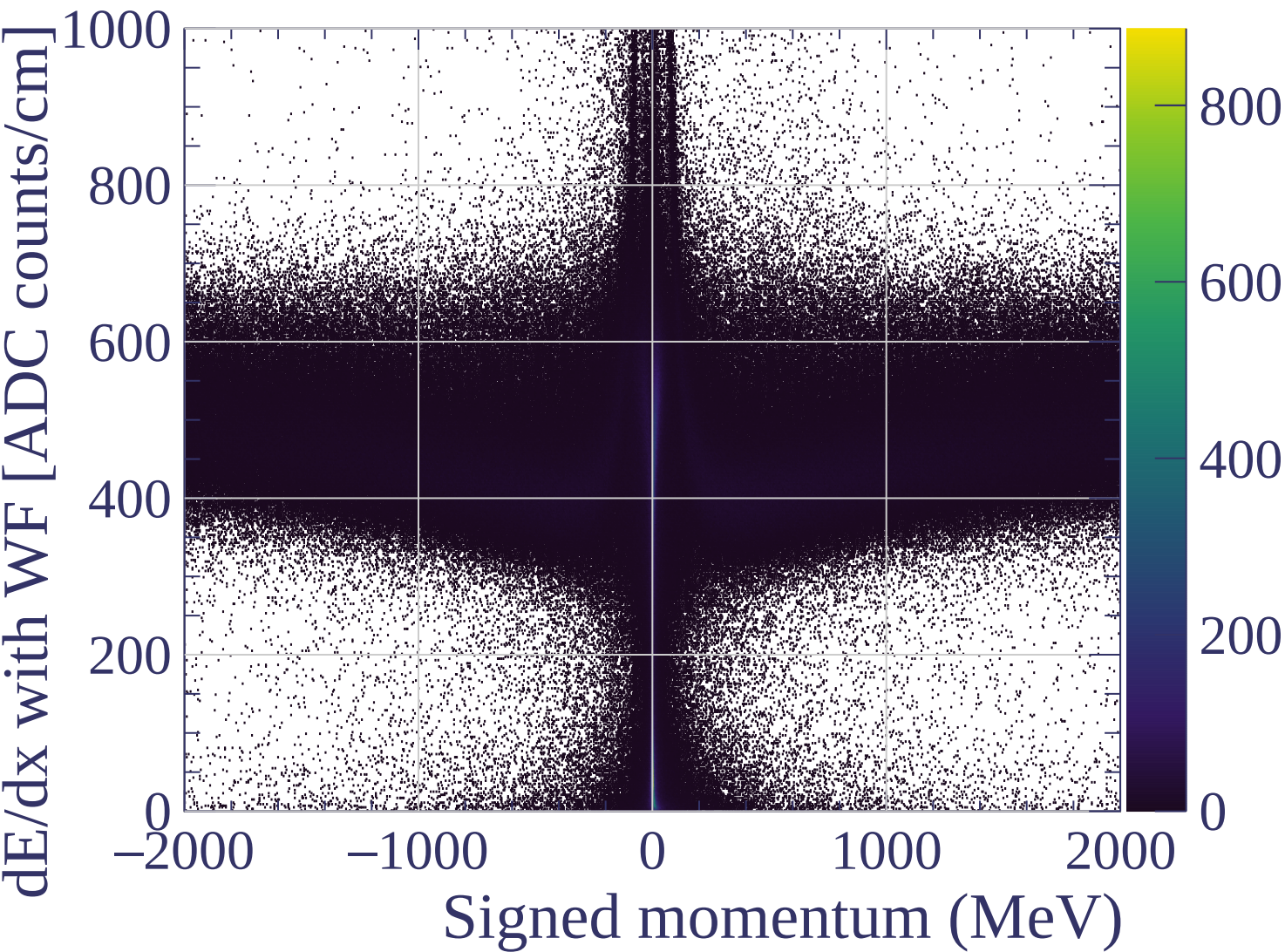


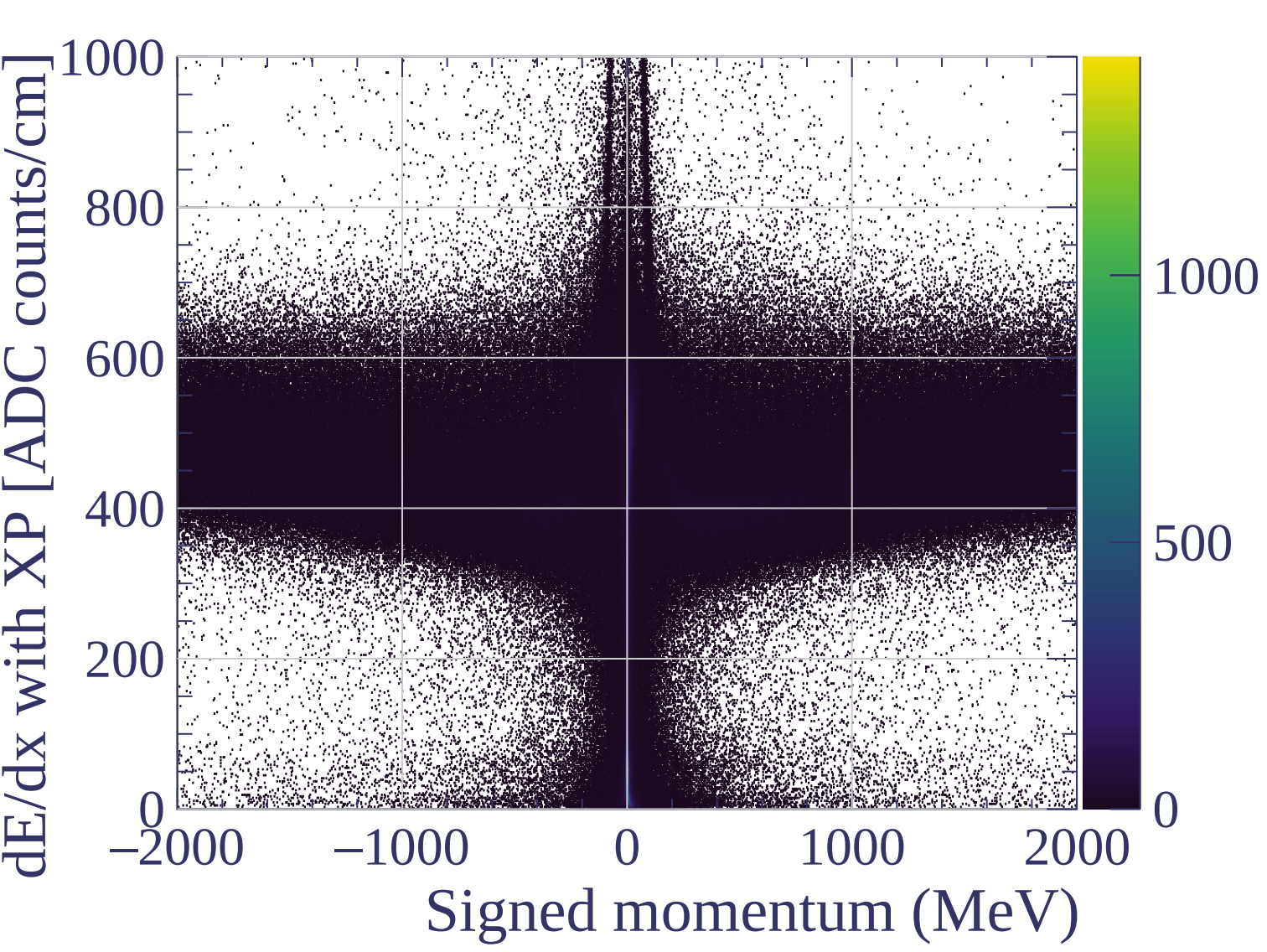


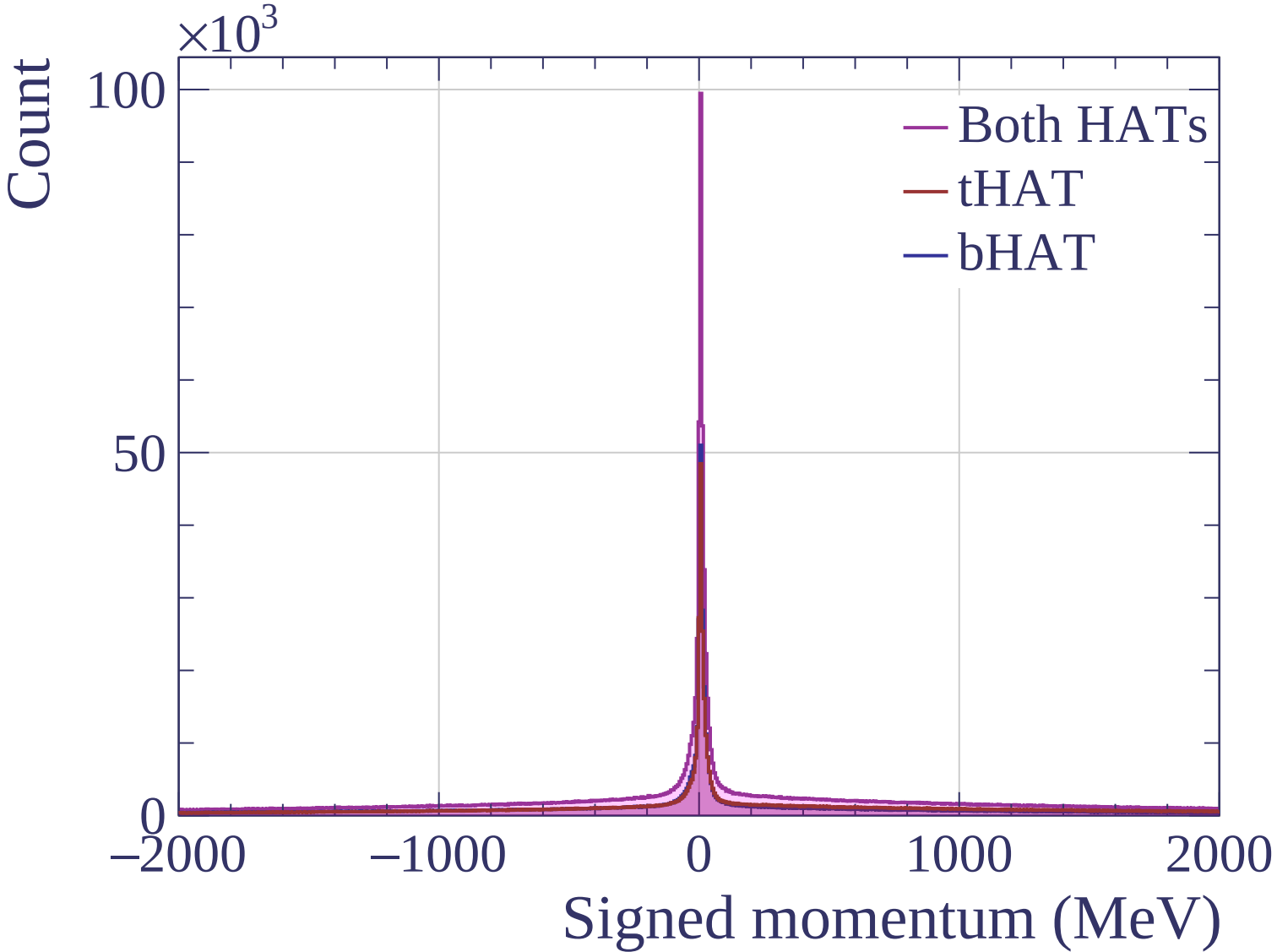
dE/dx resolution [%]

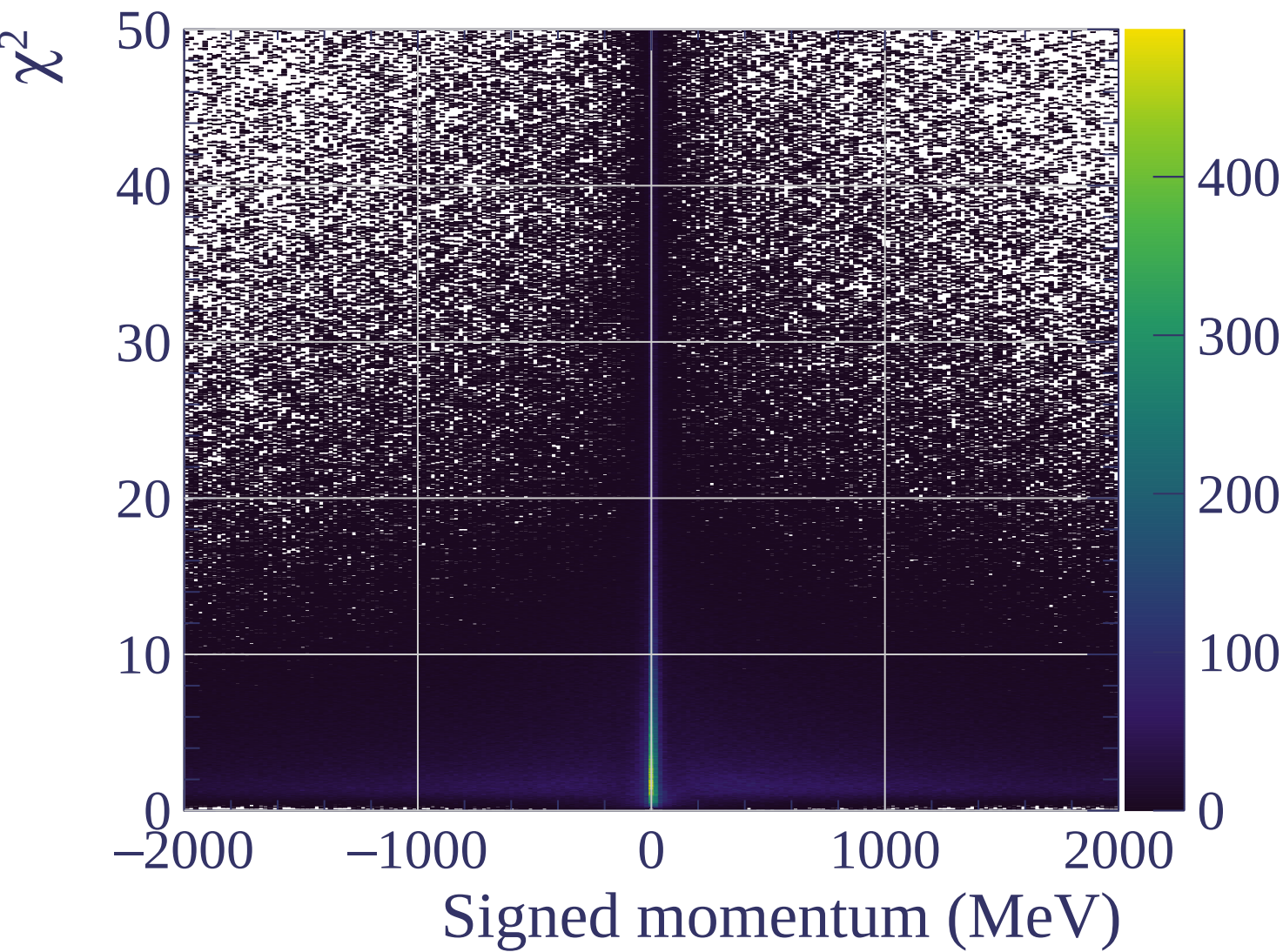


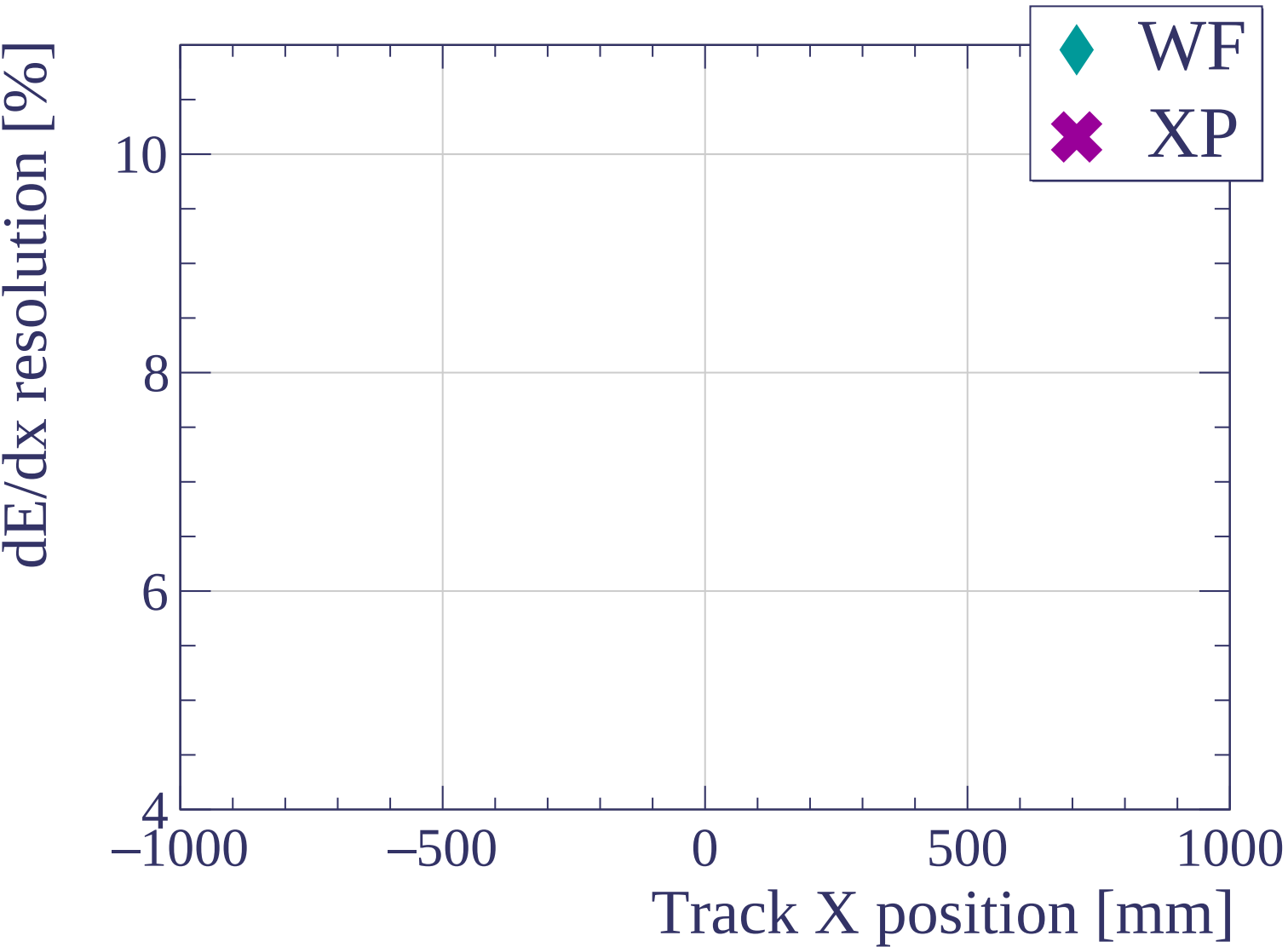


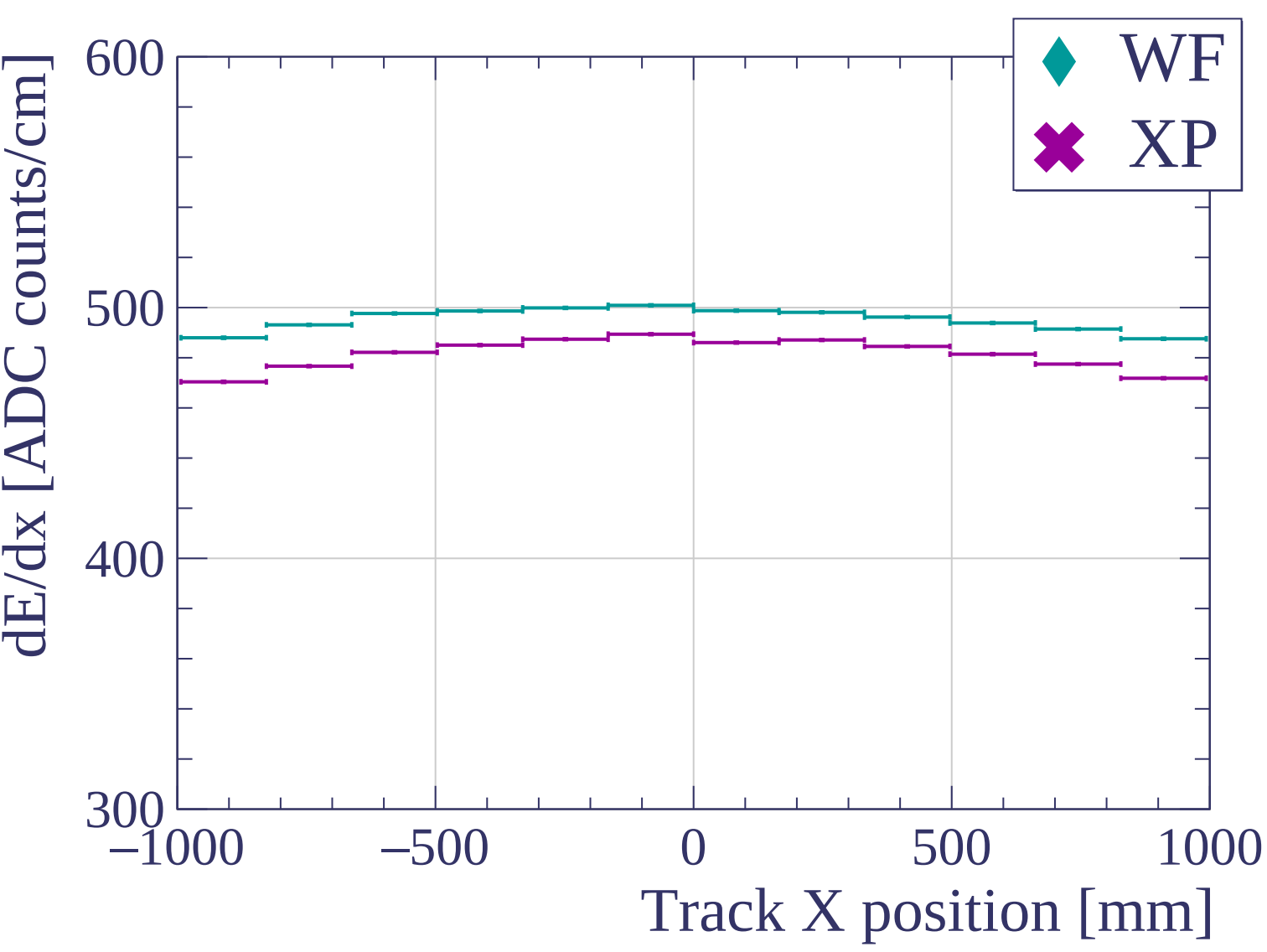


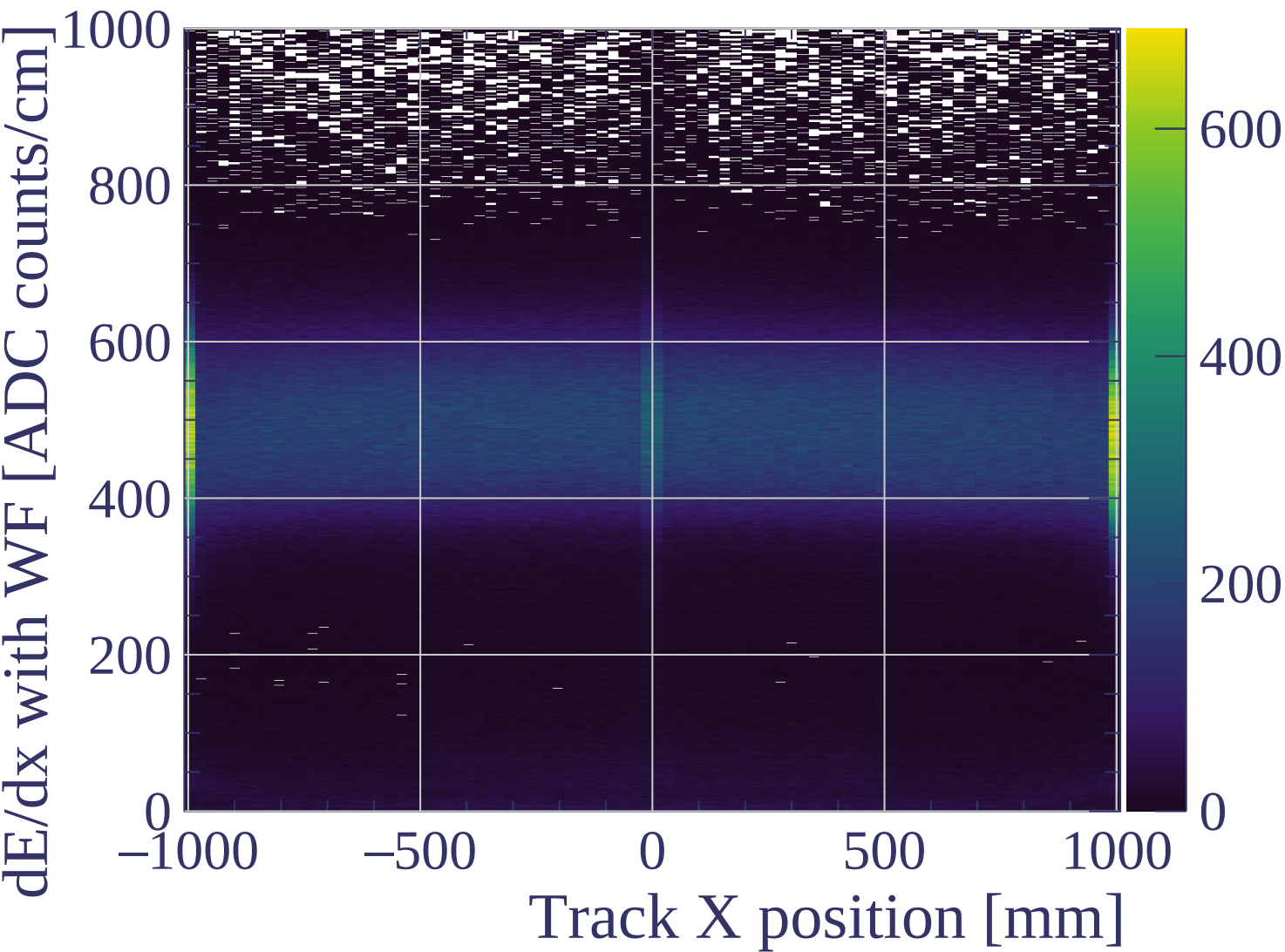


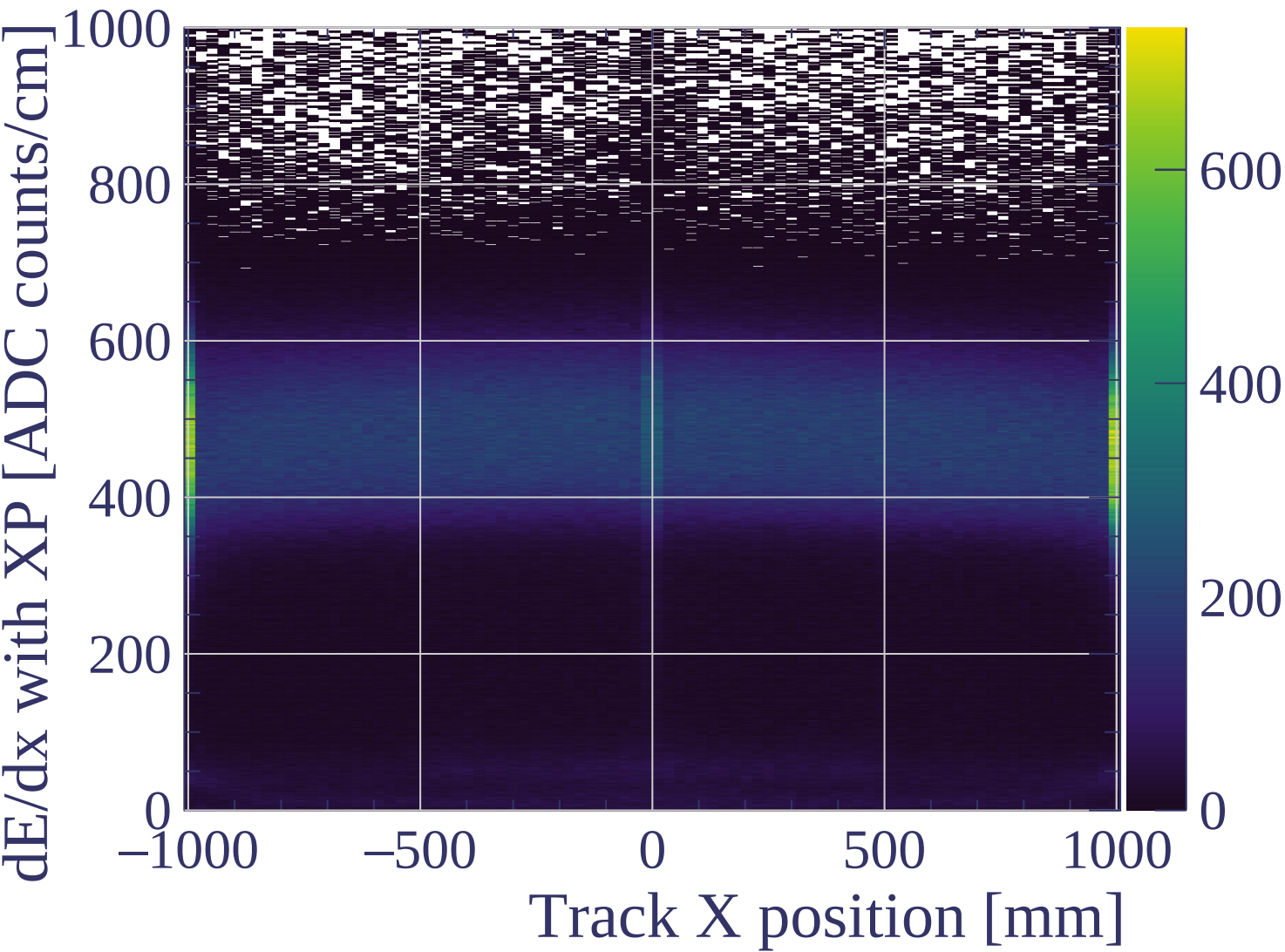


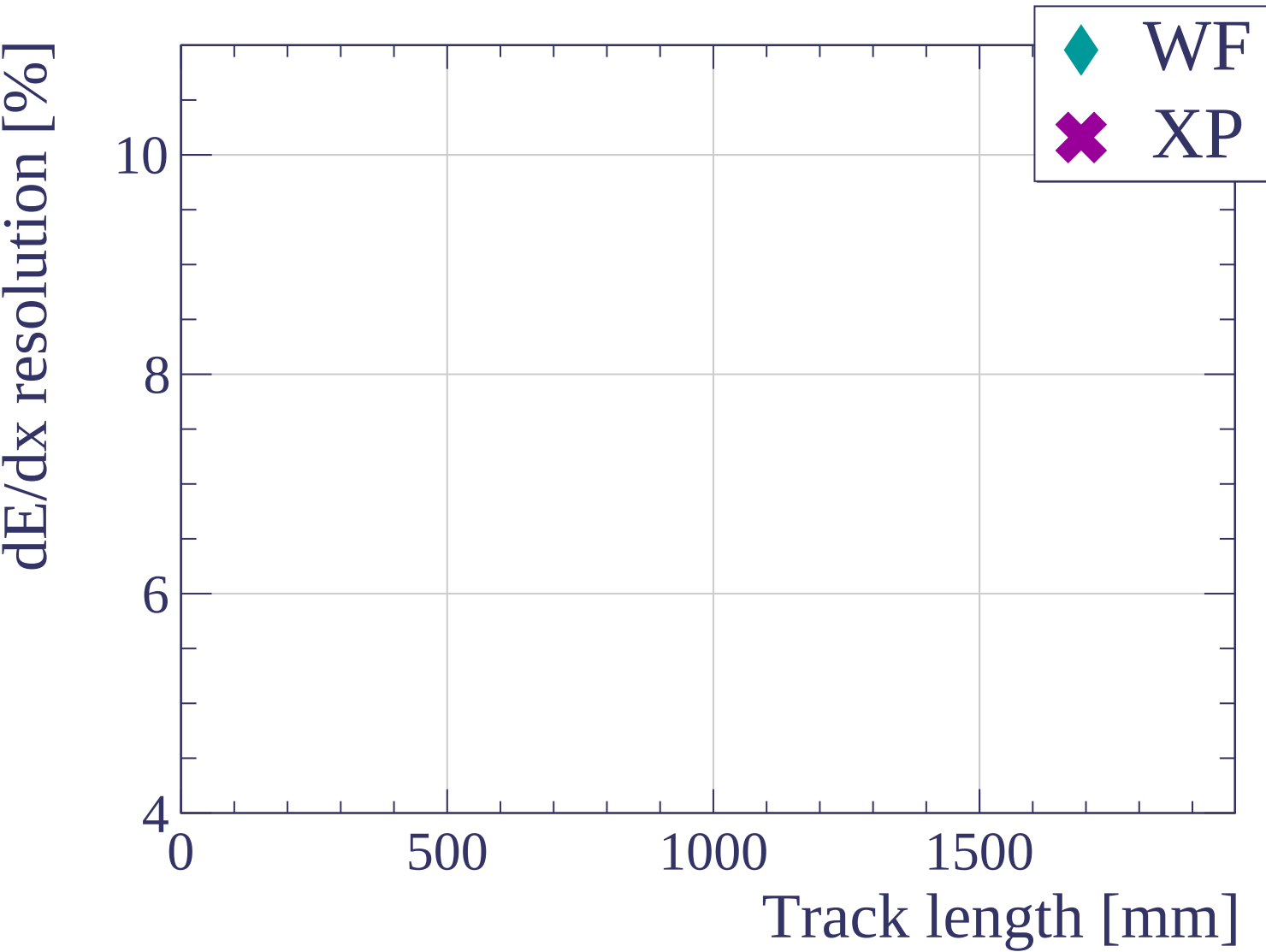


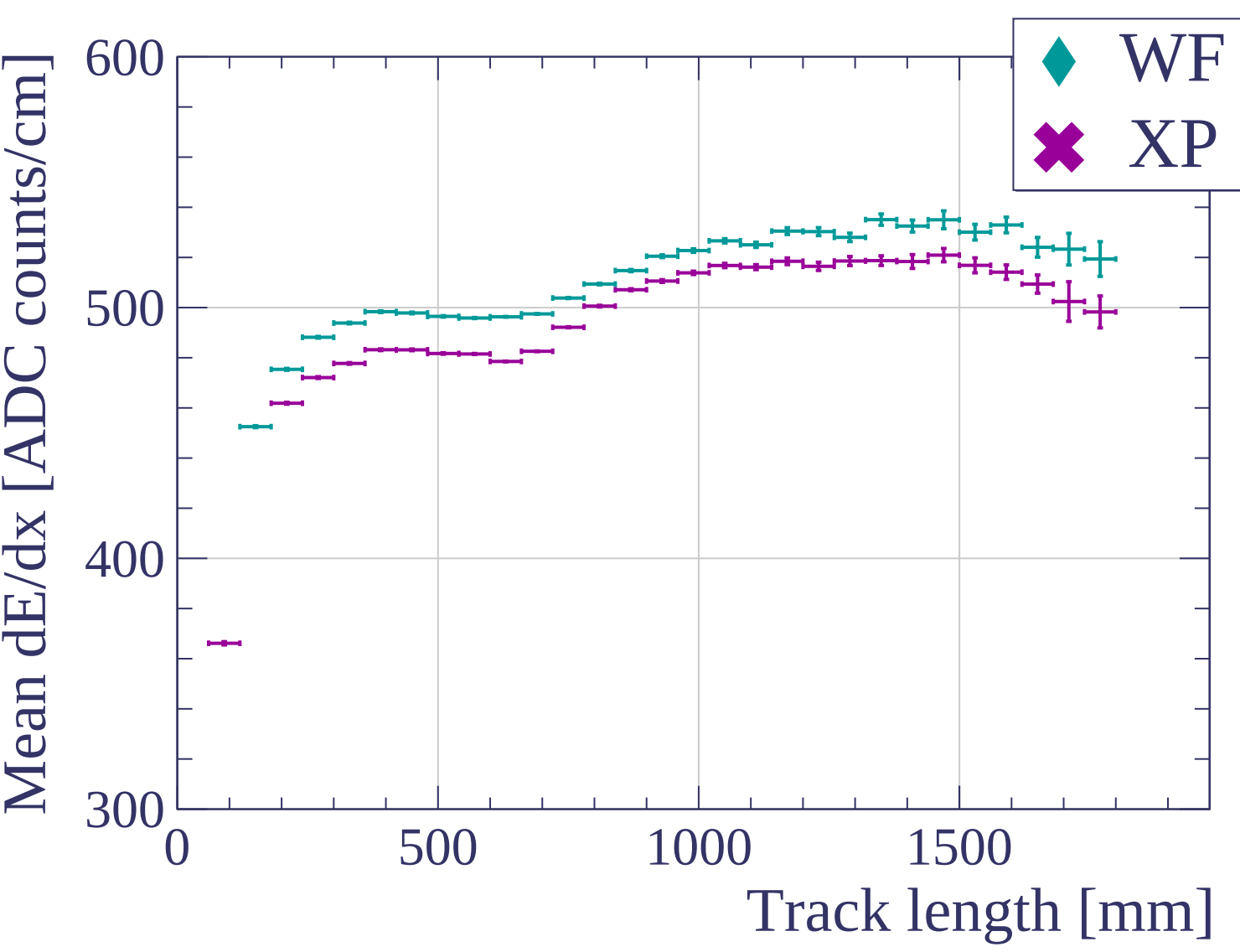


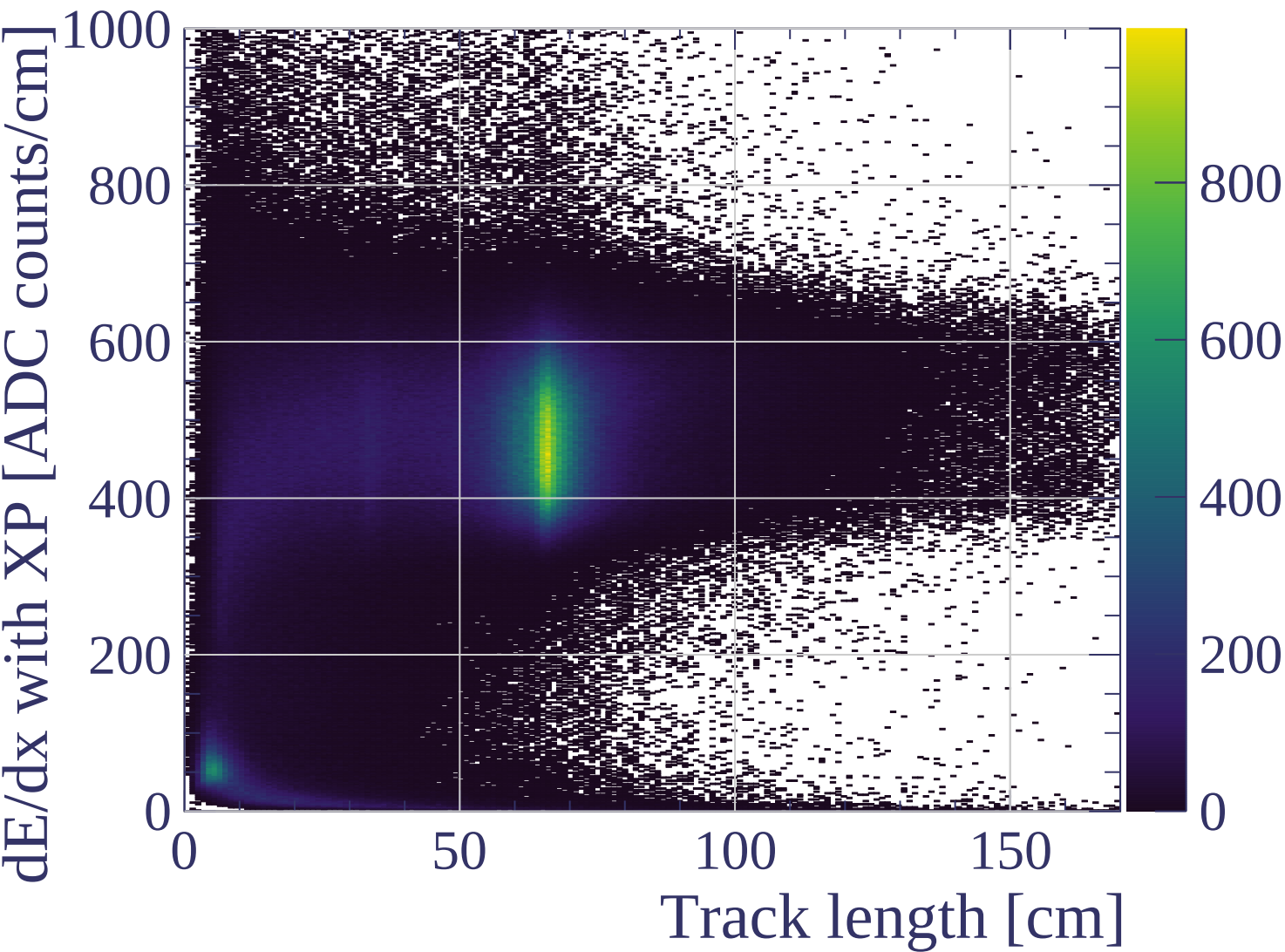


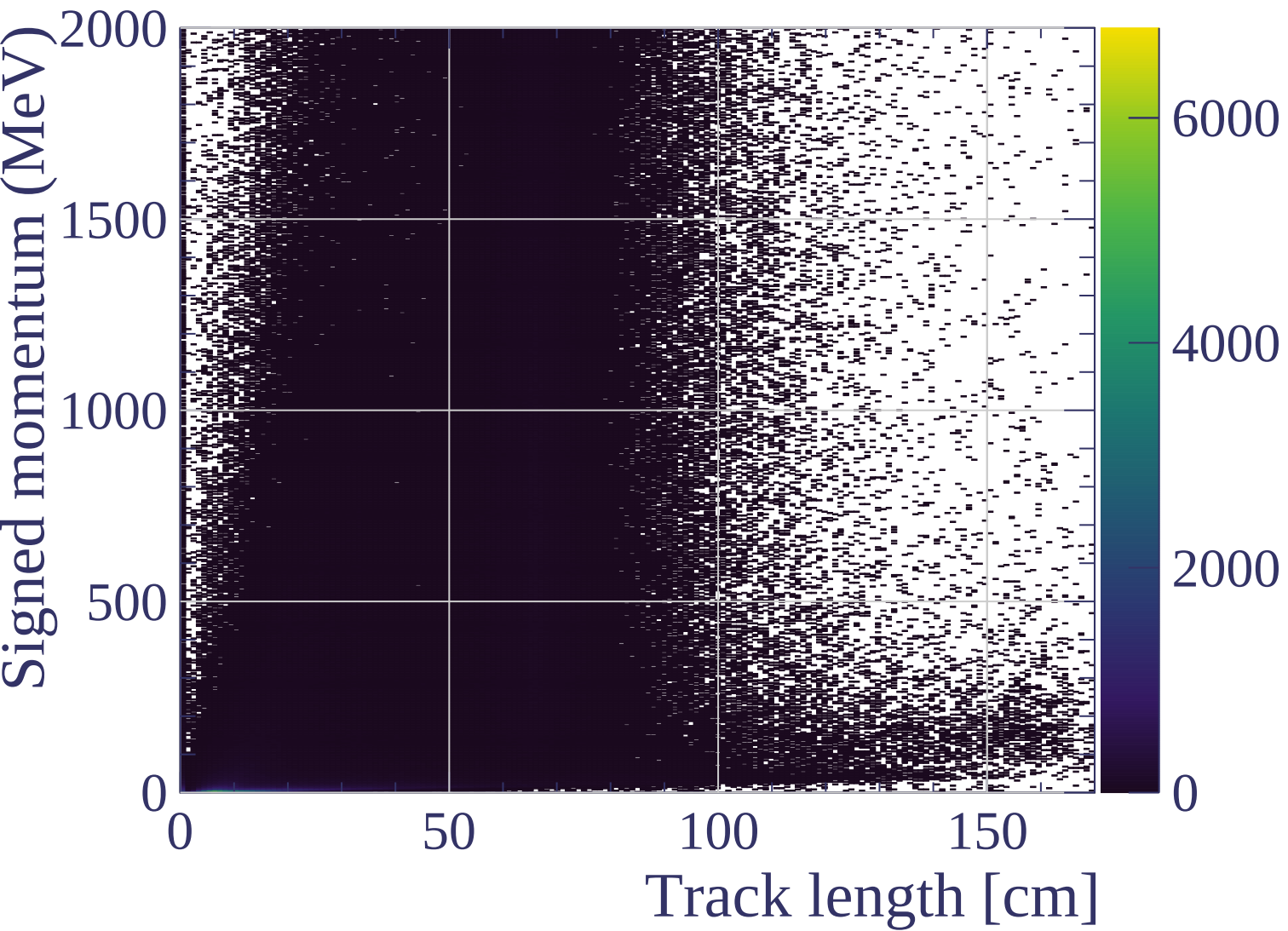


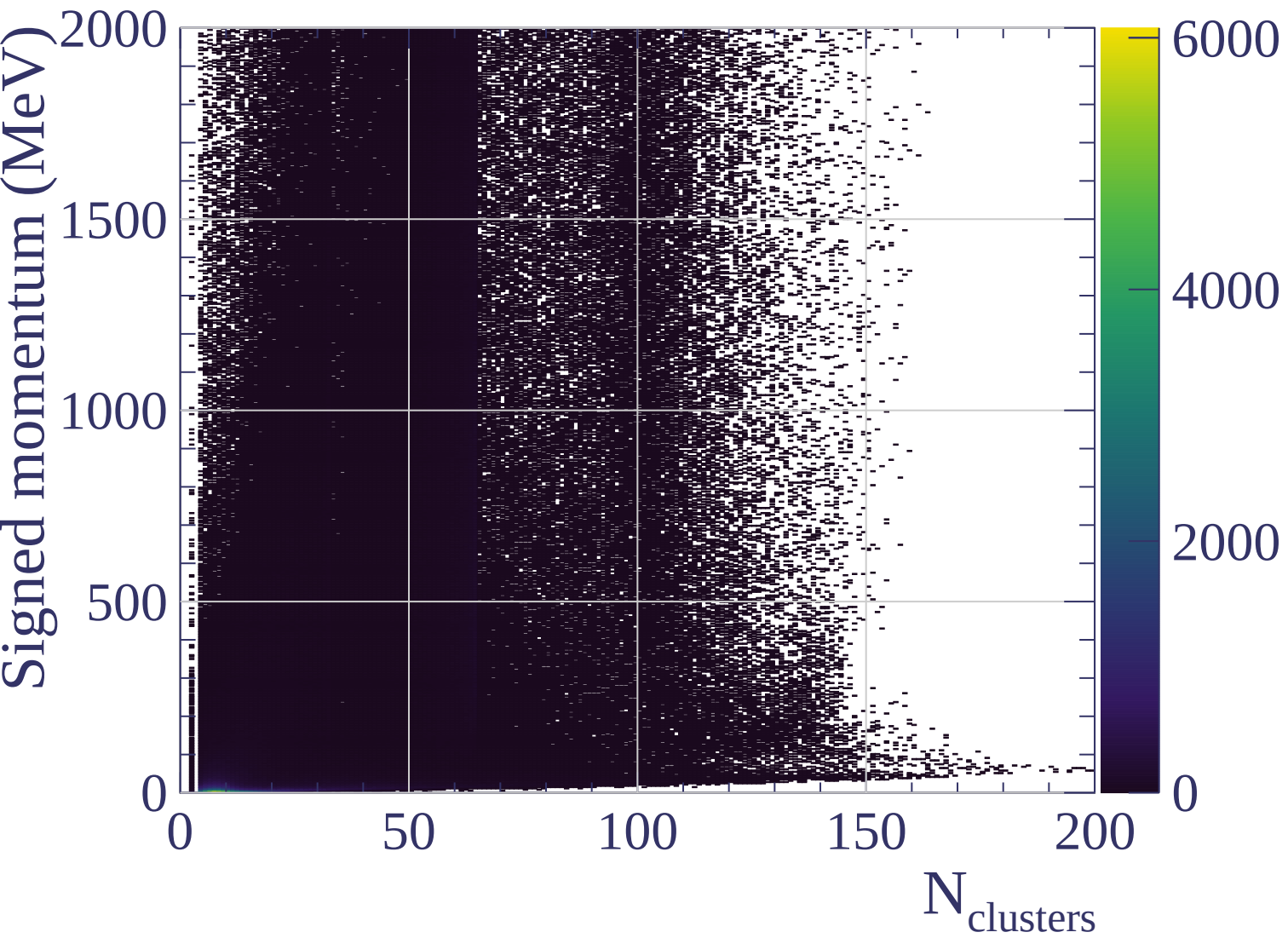


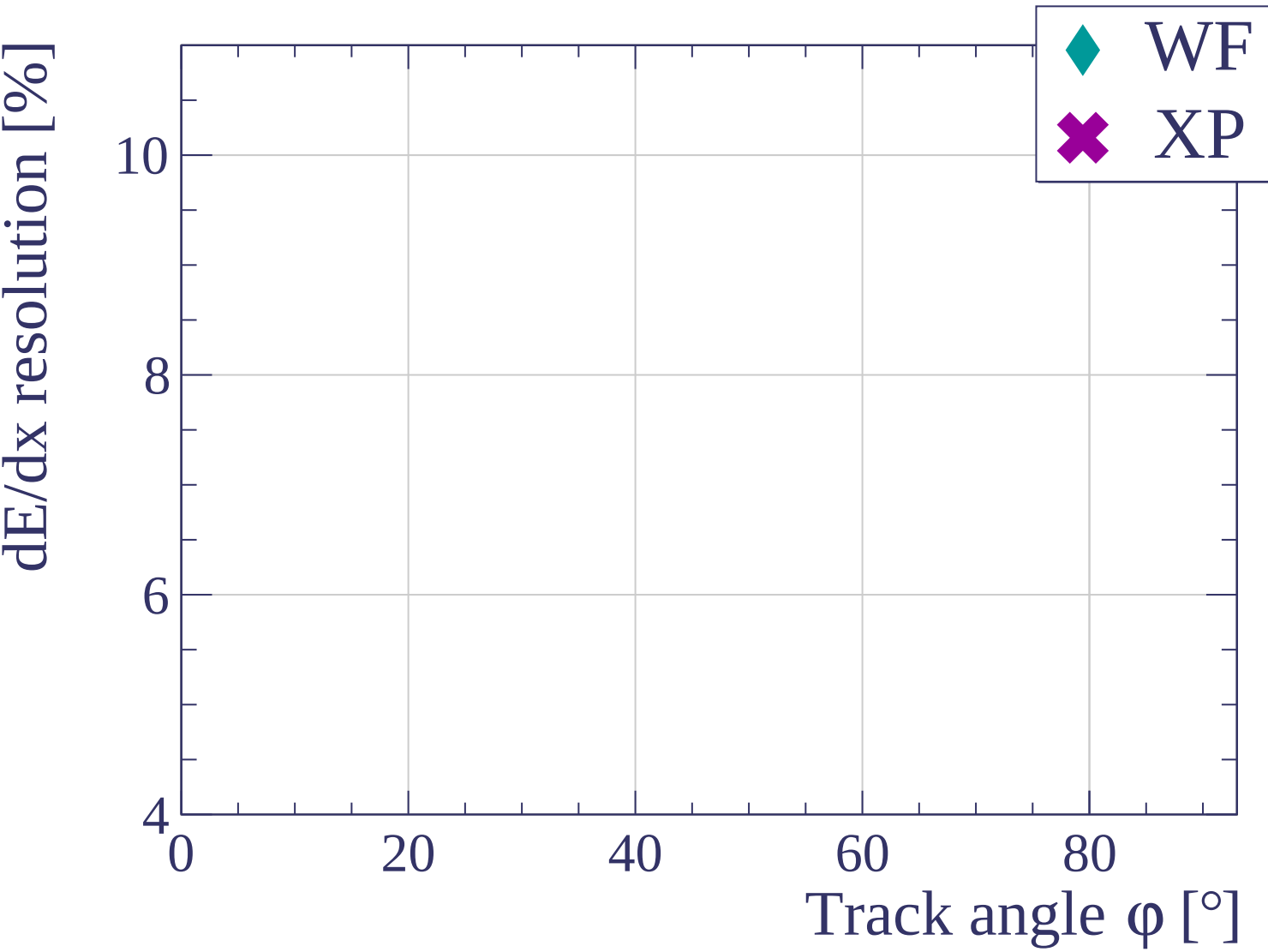


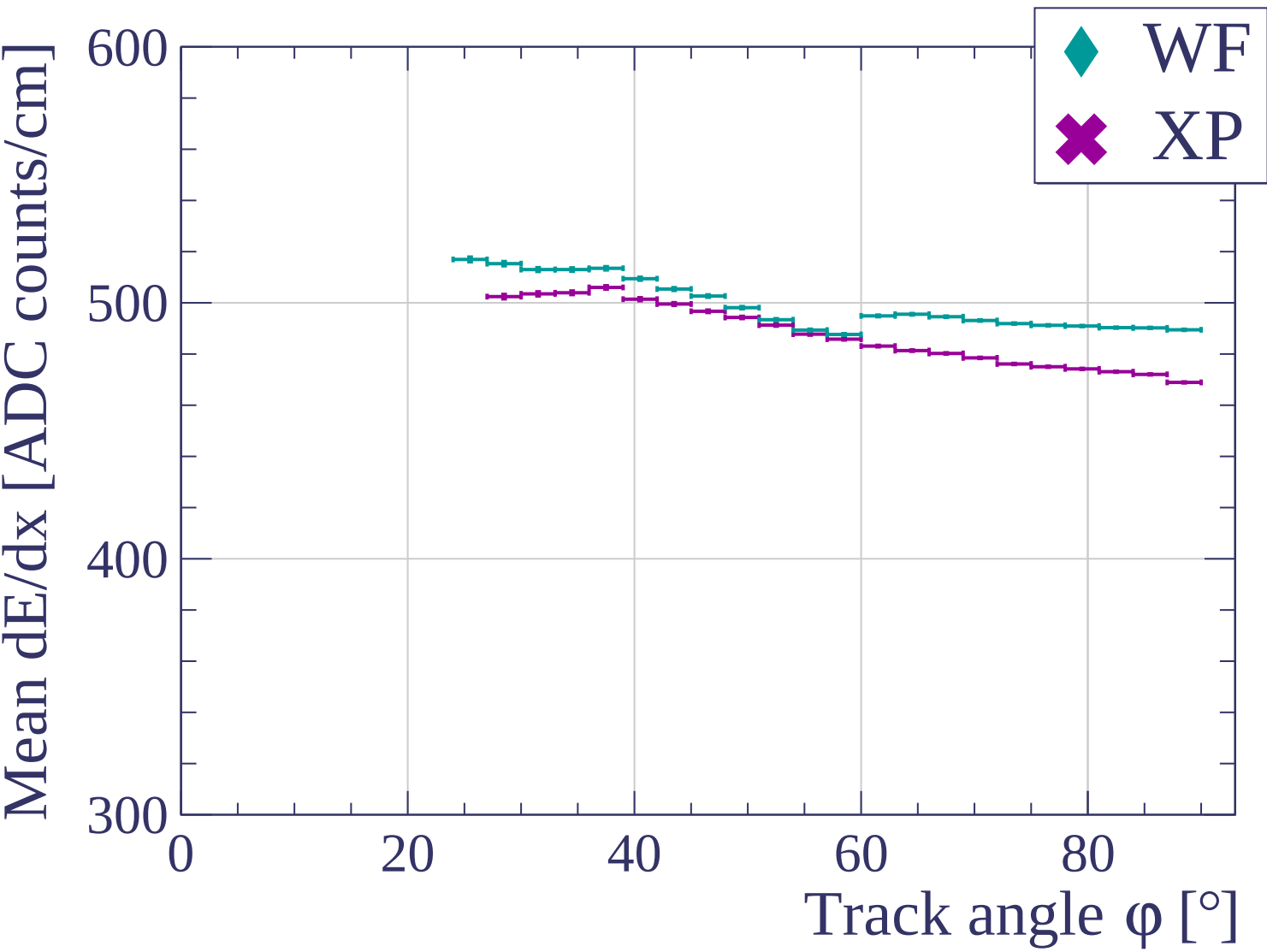


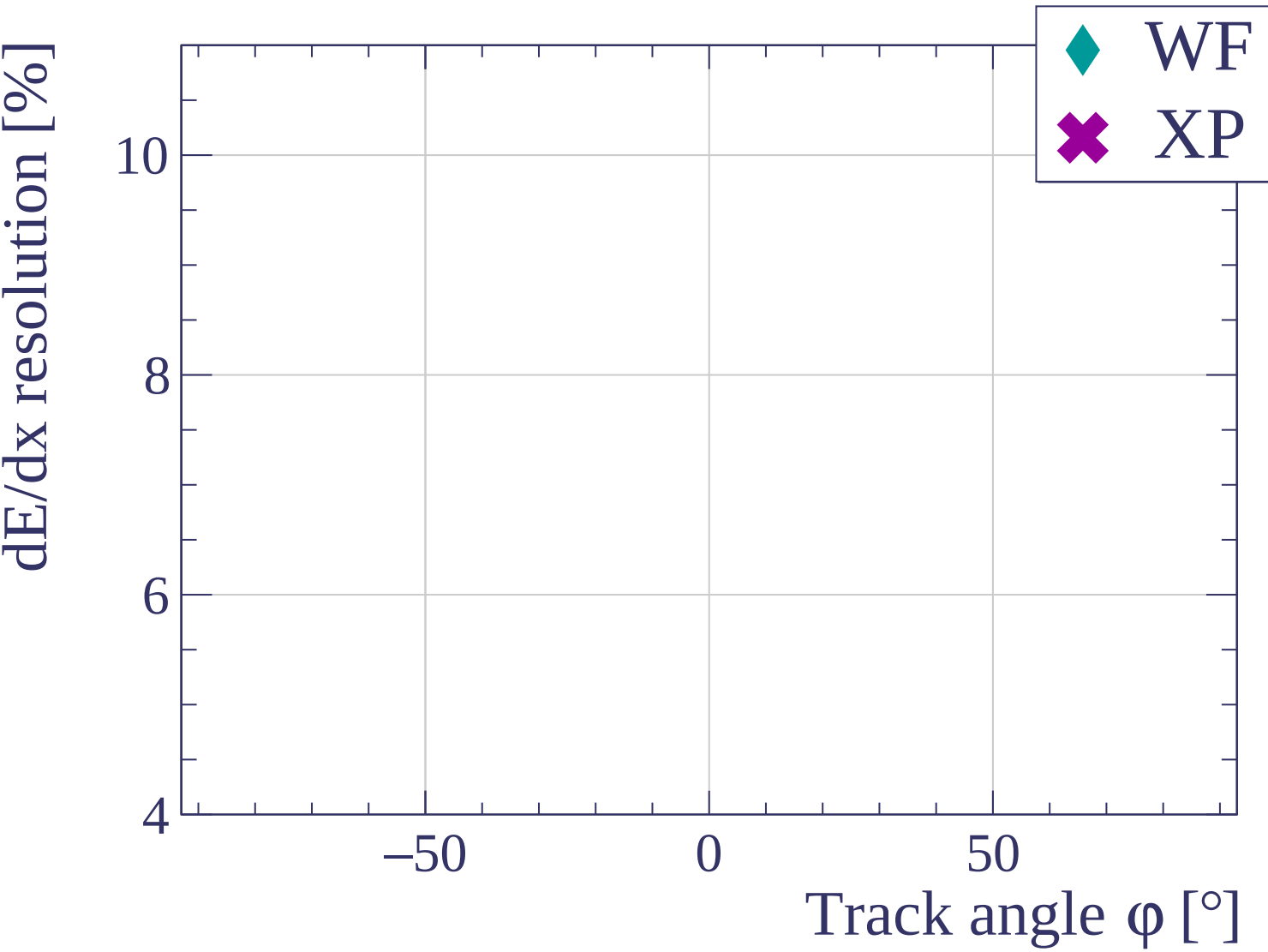


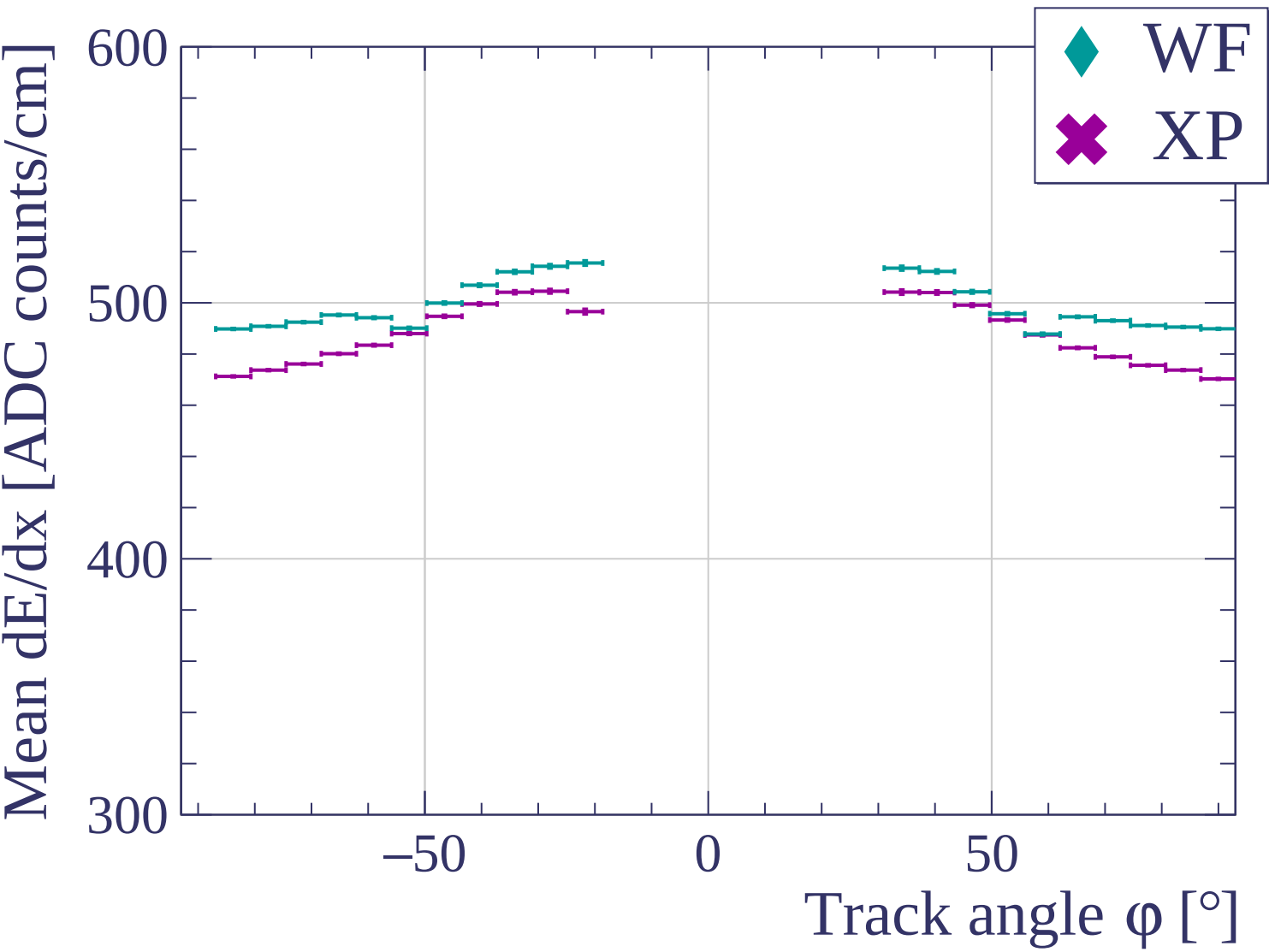


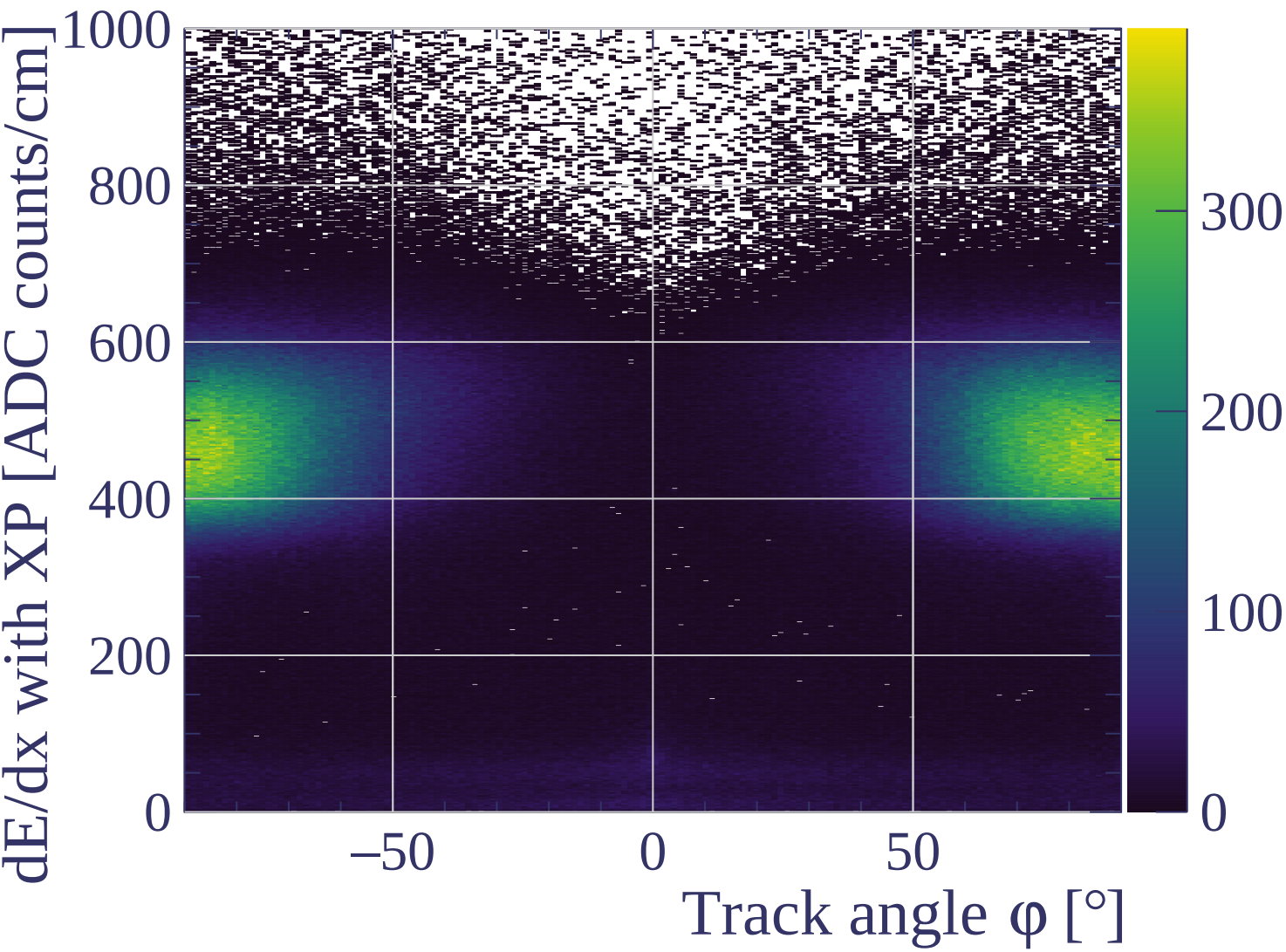




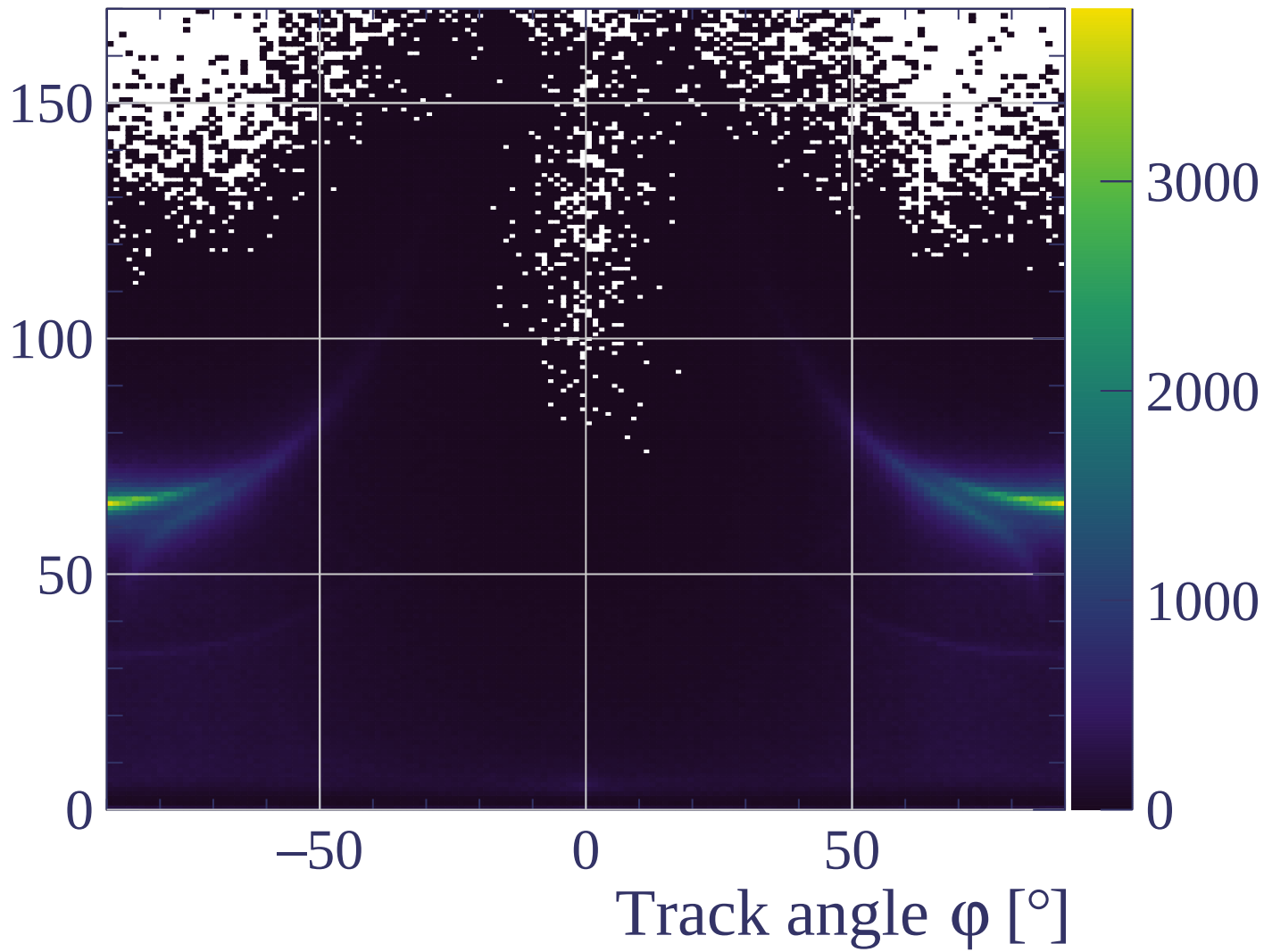


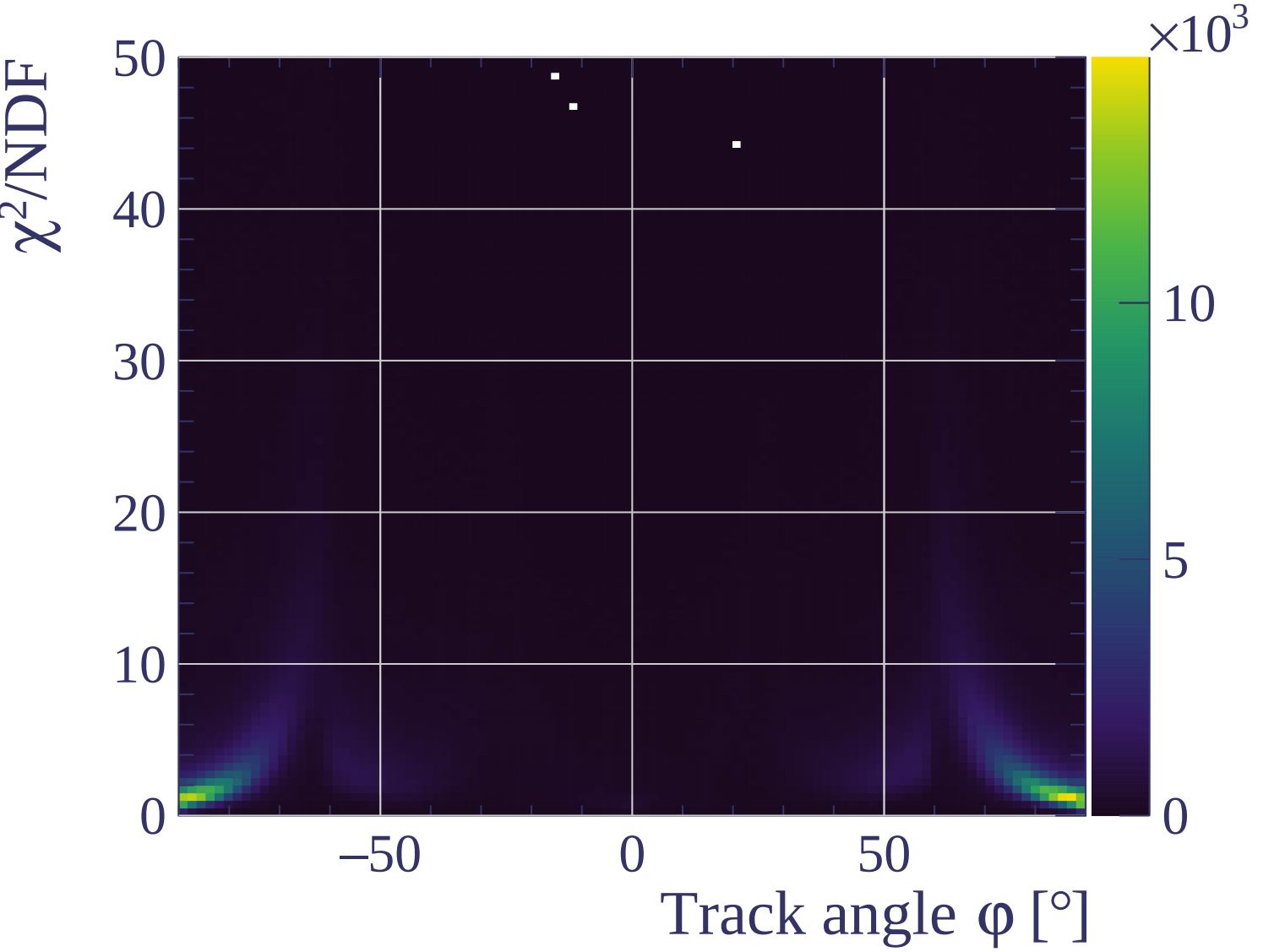


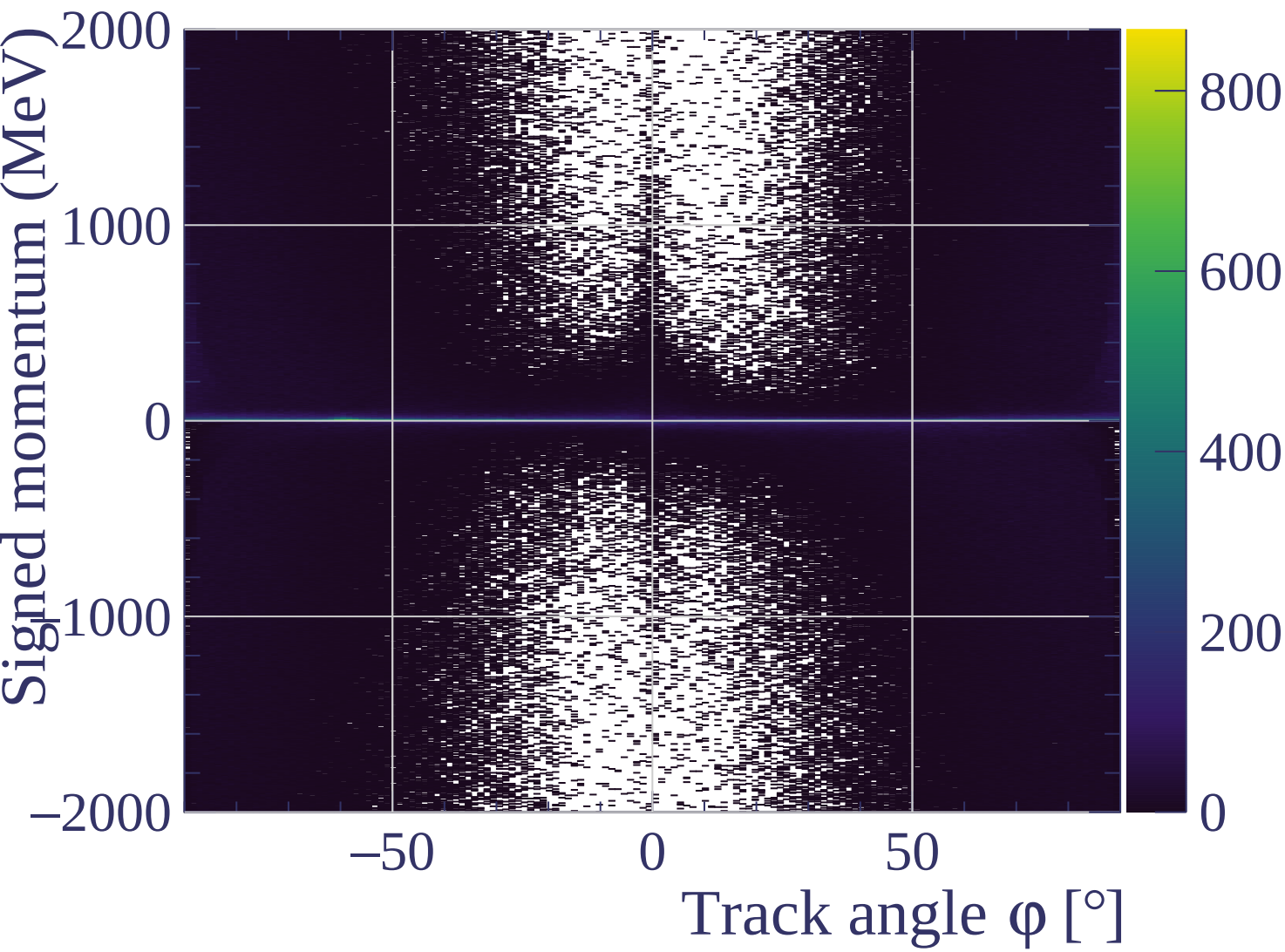


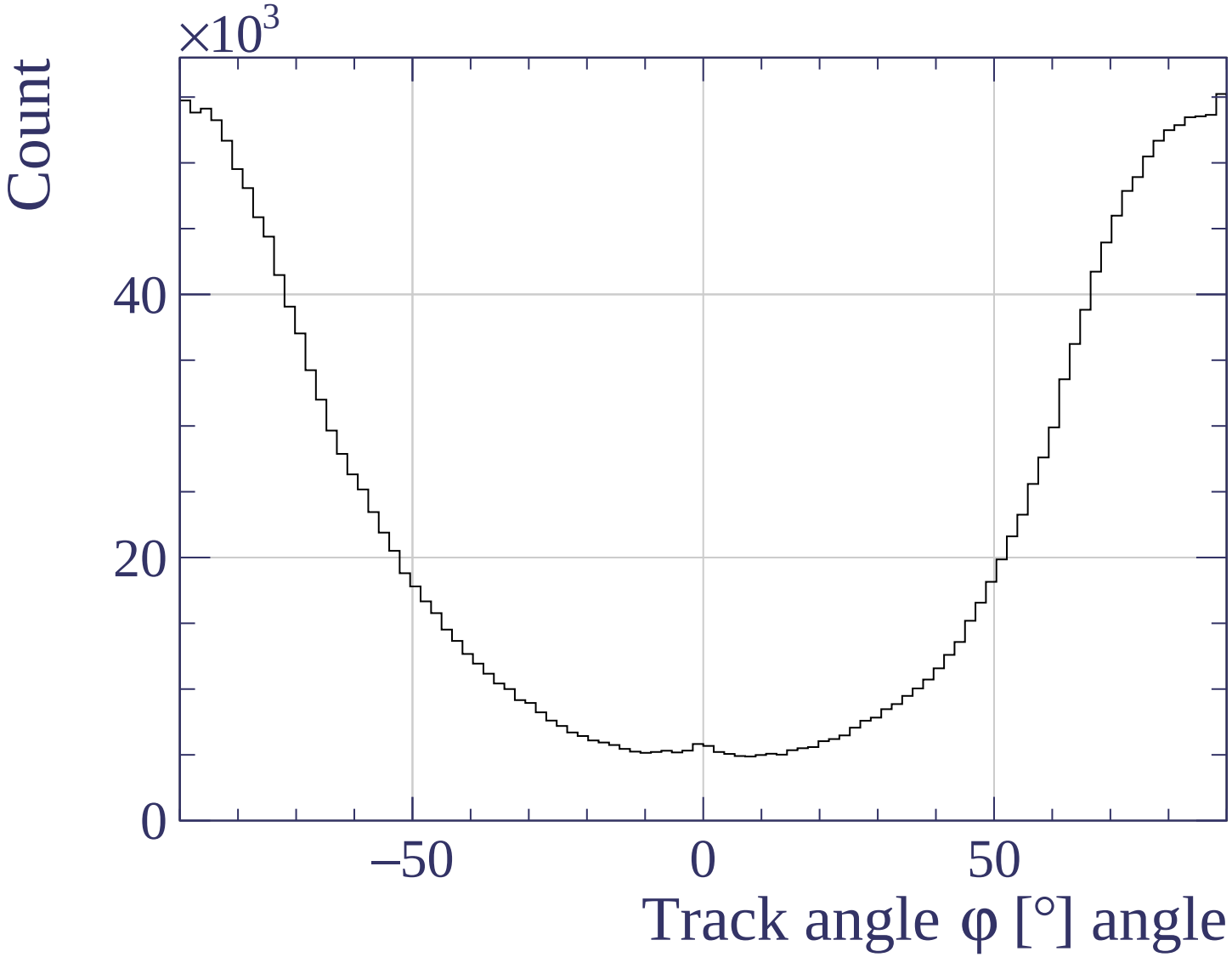


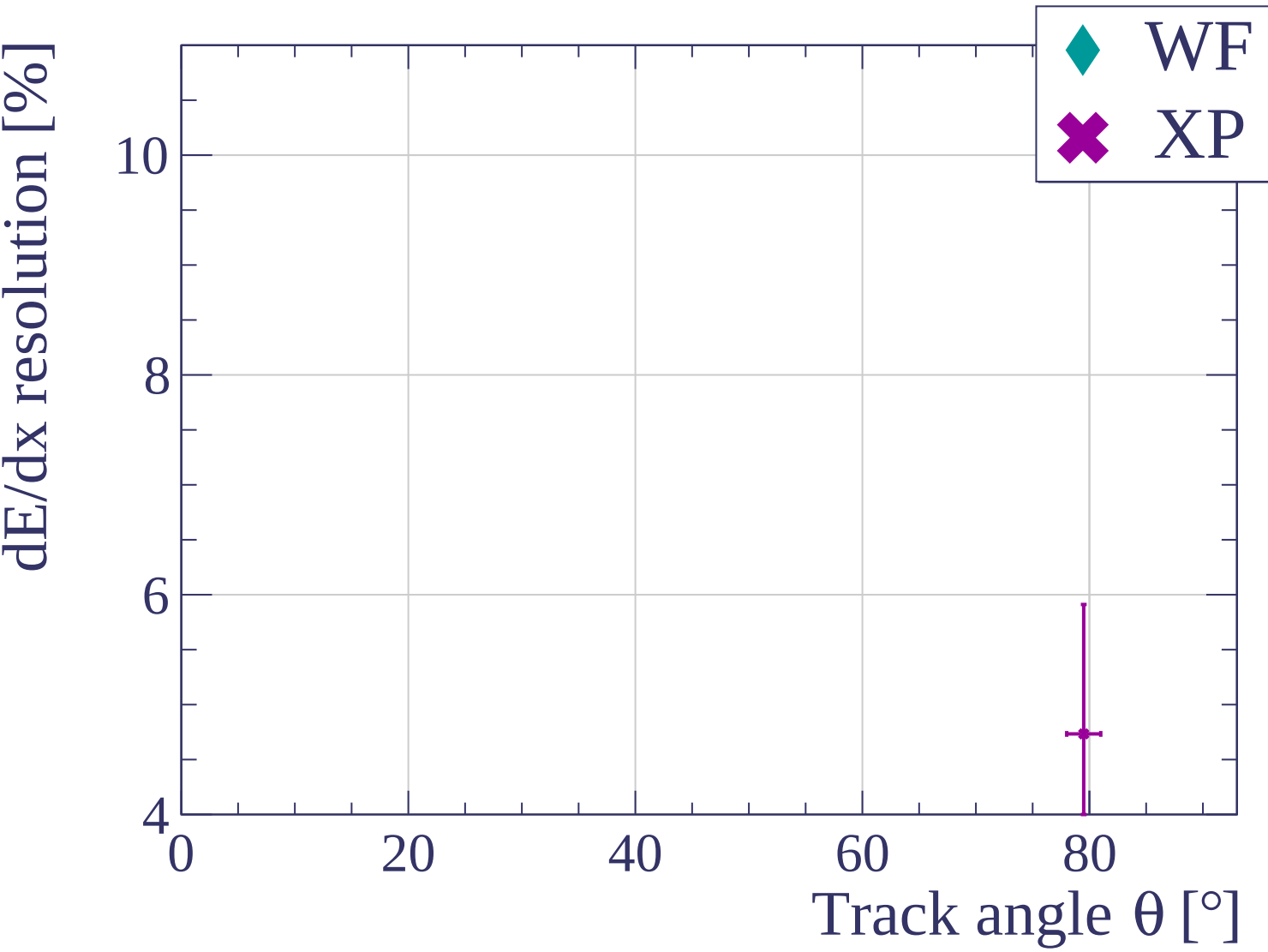
Track length [cm]

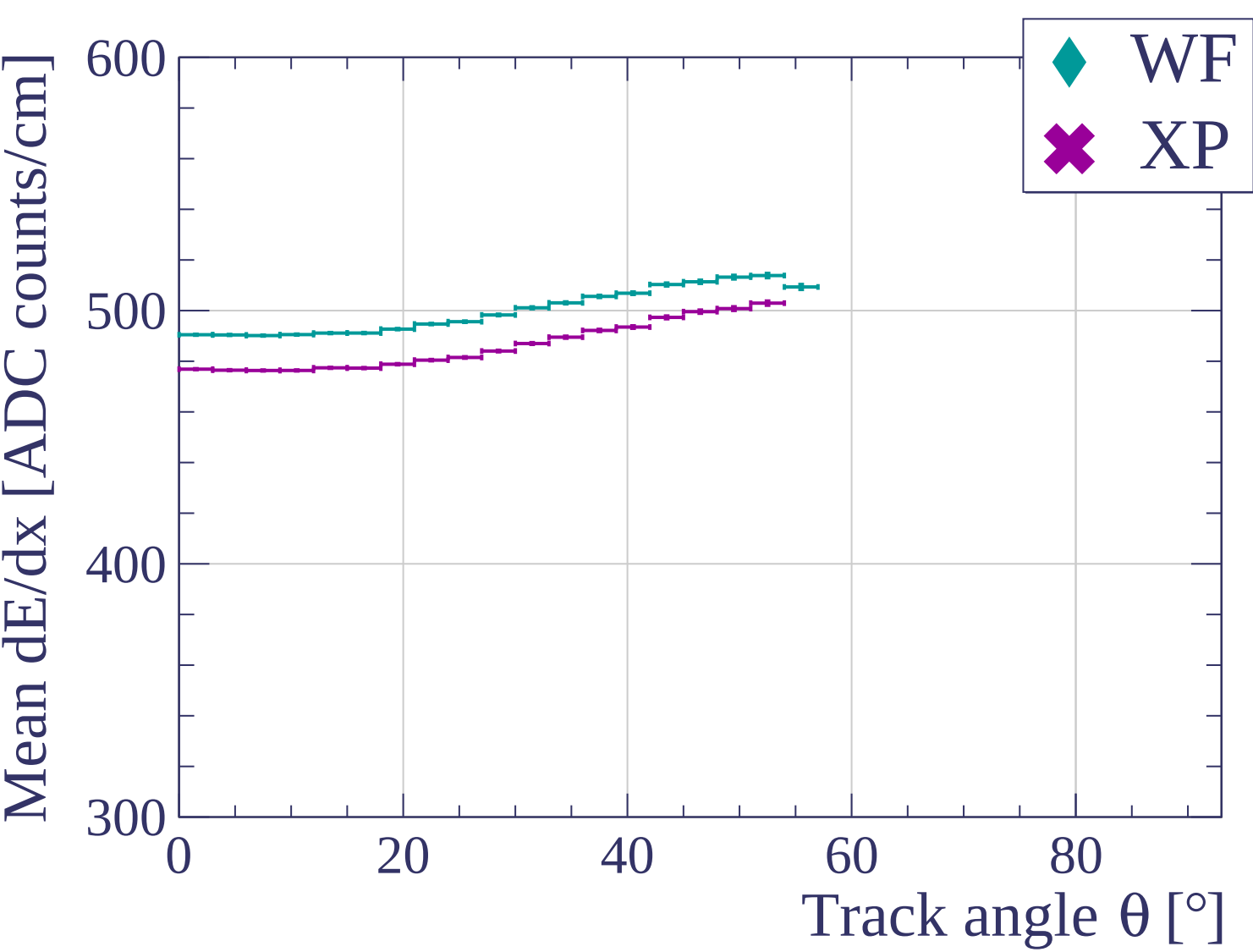


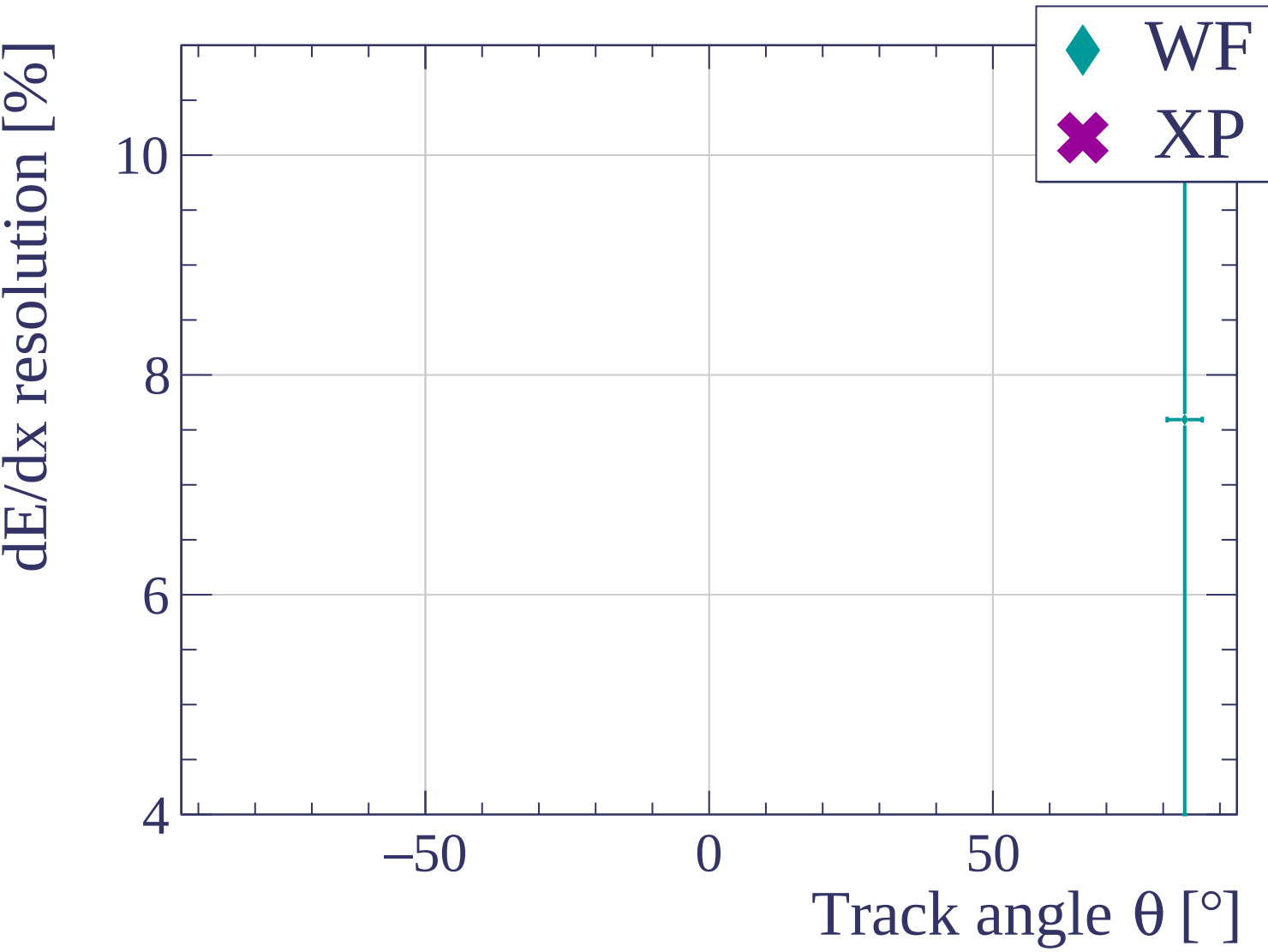


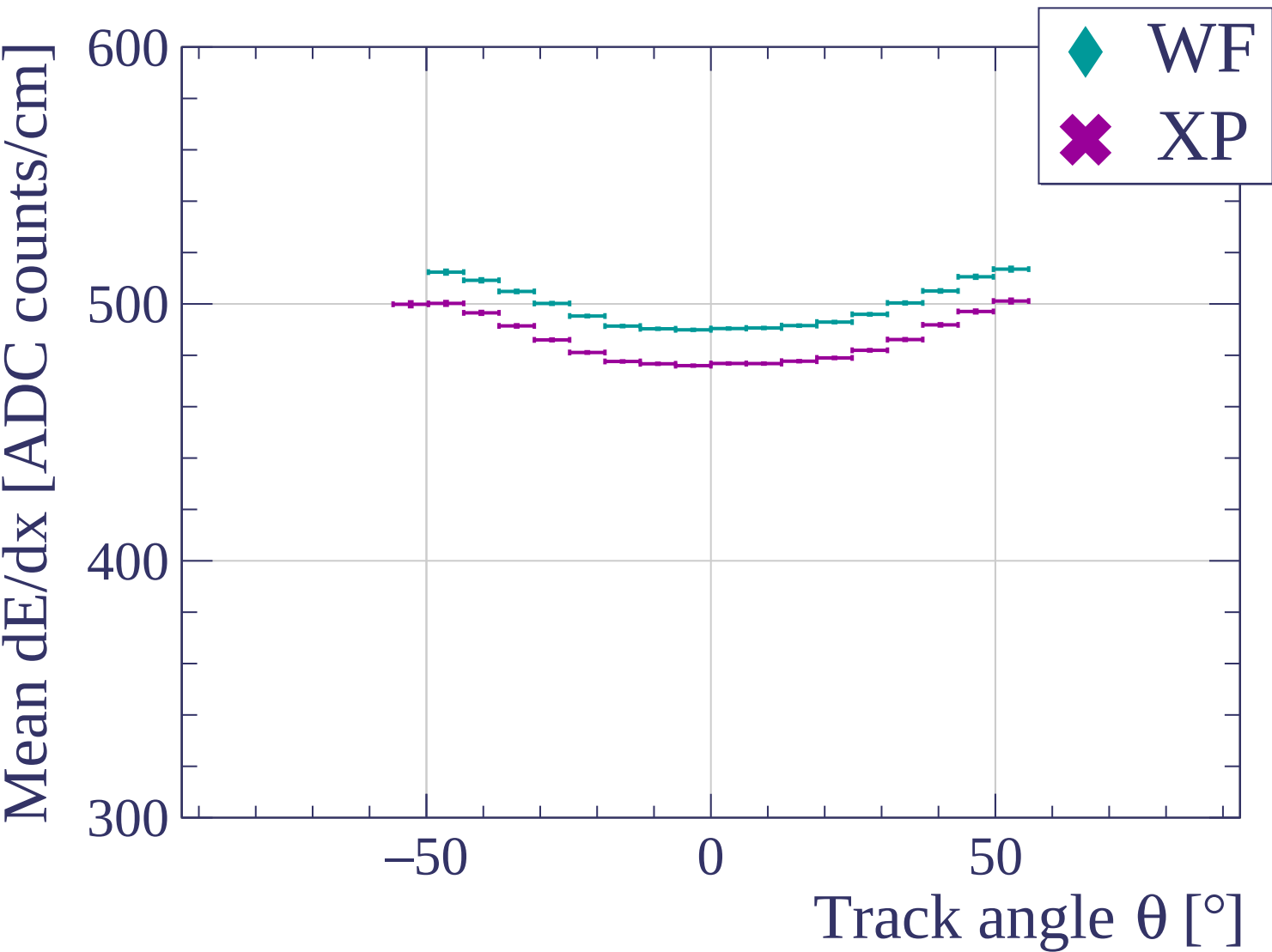


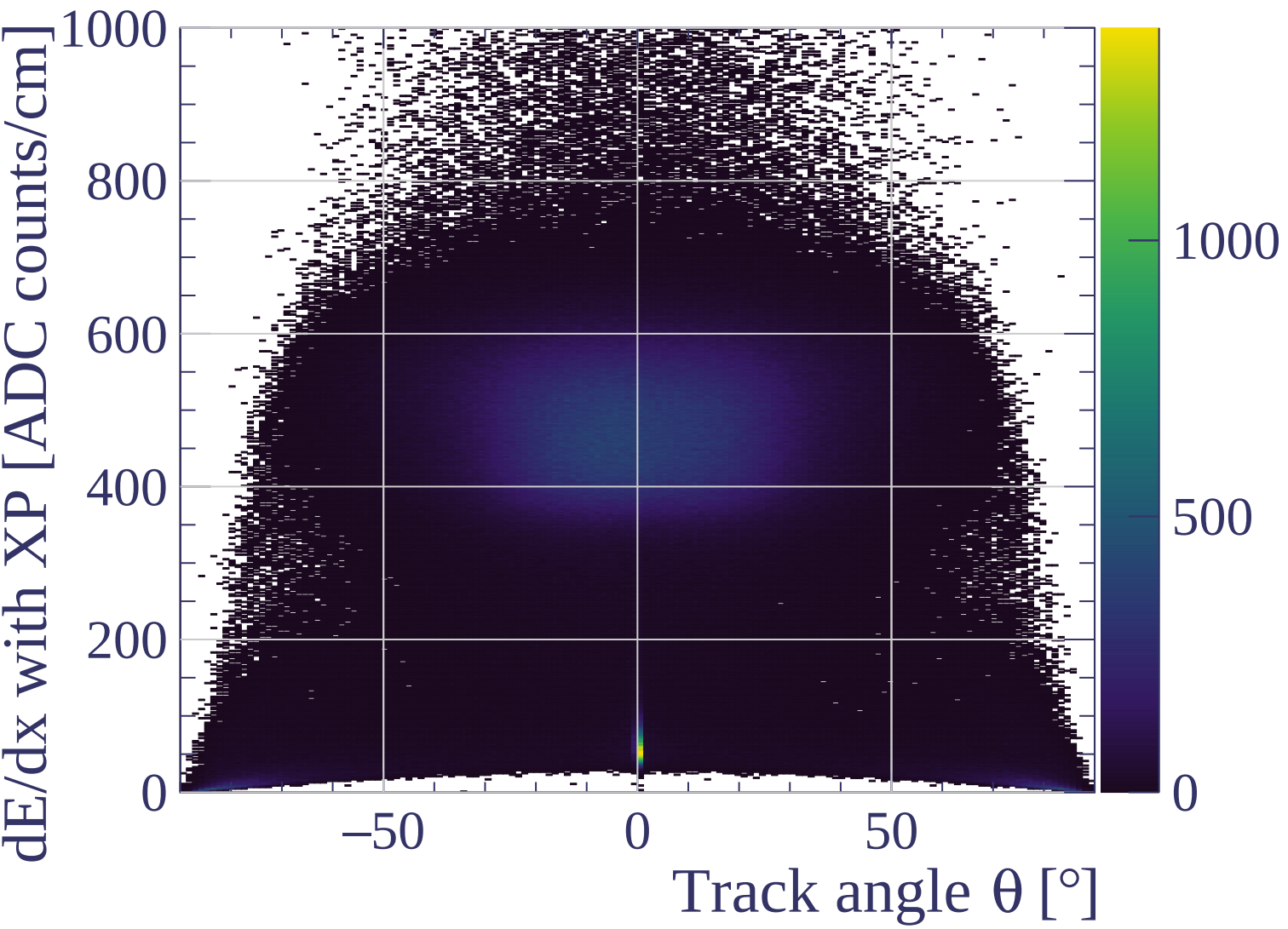




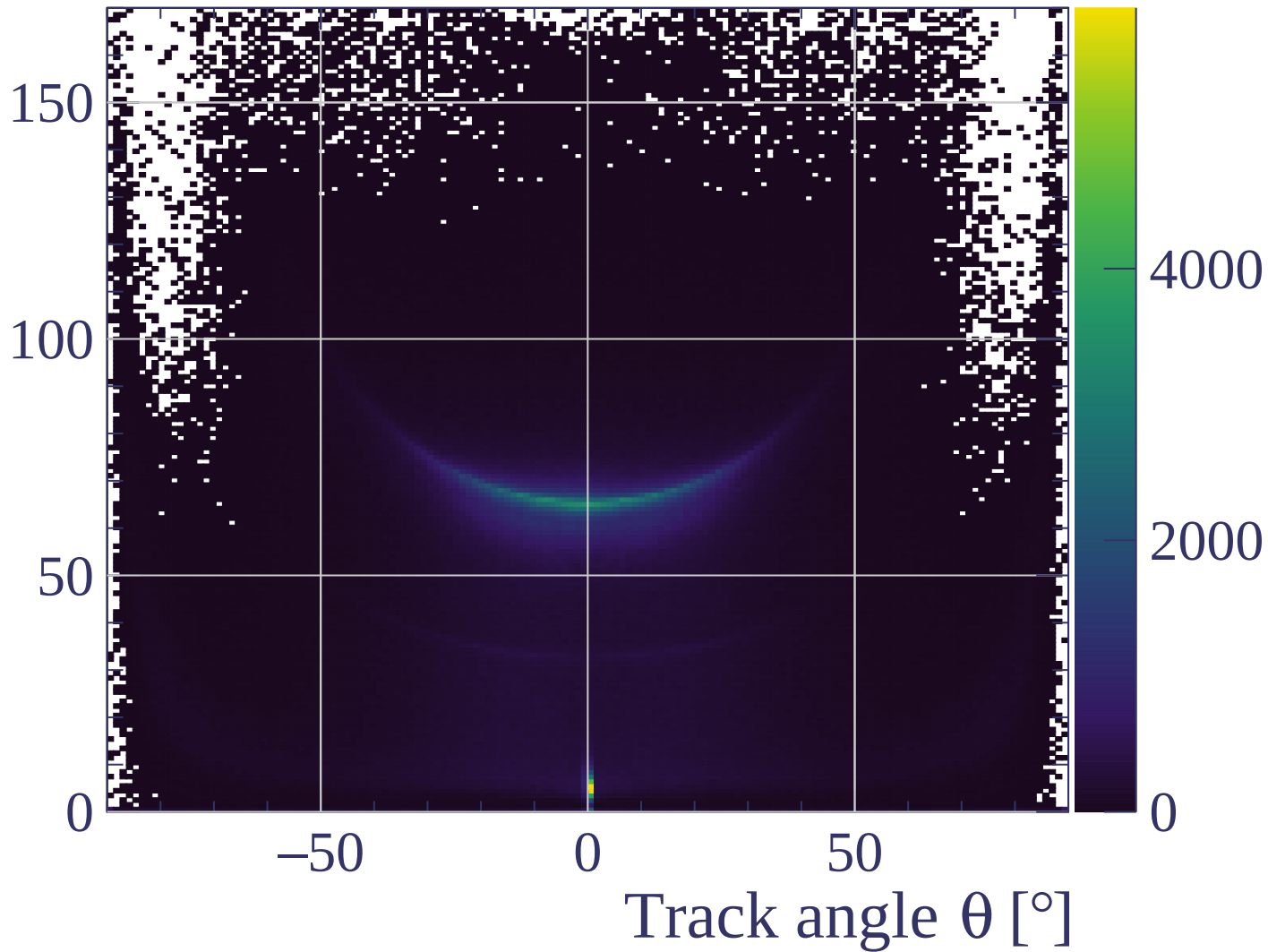


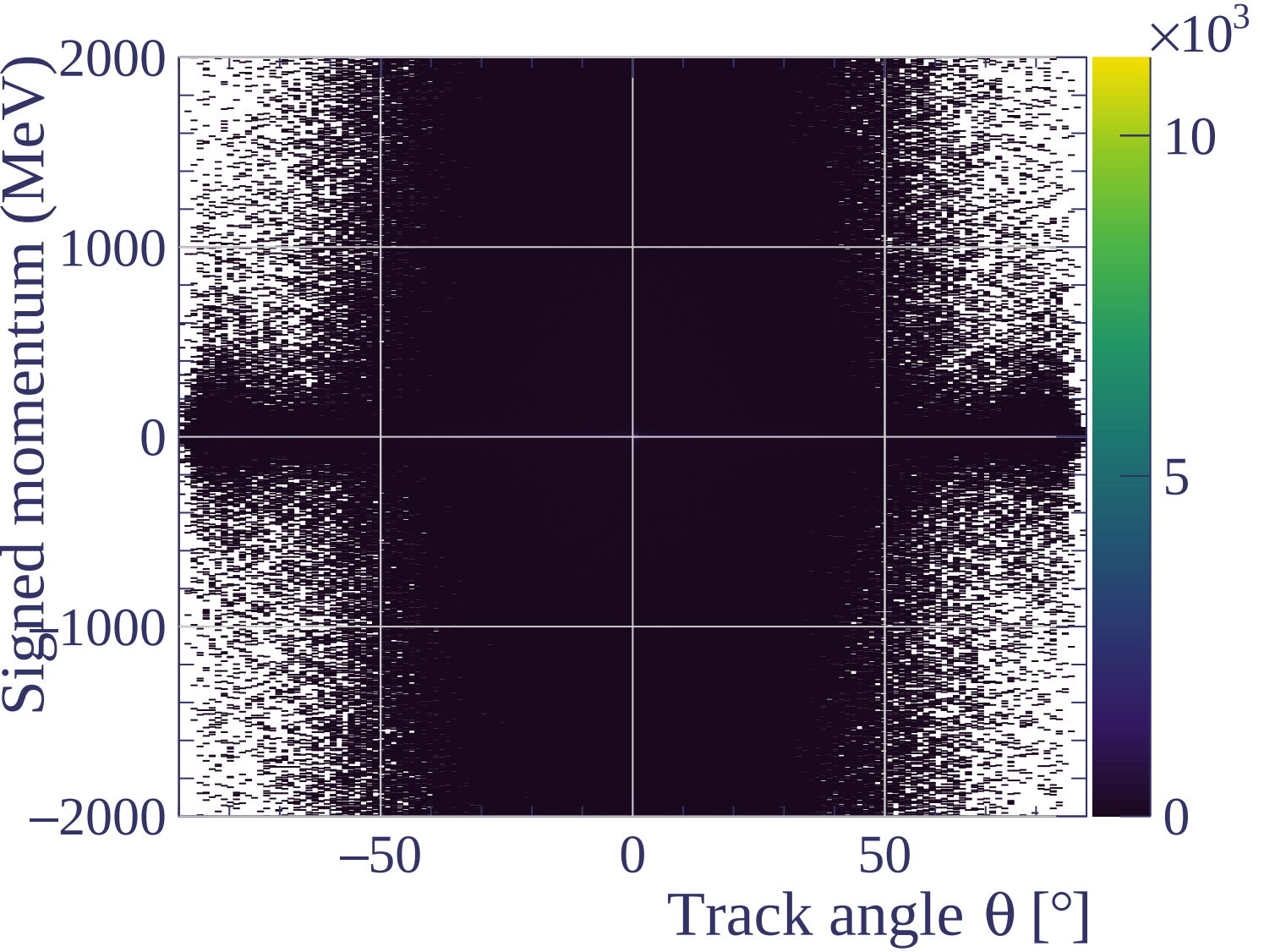


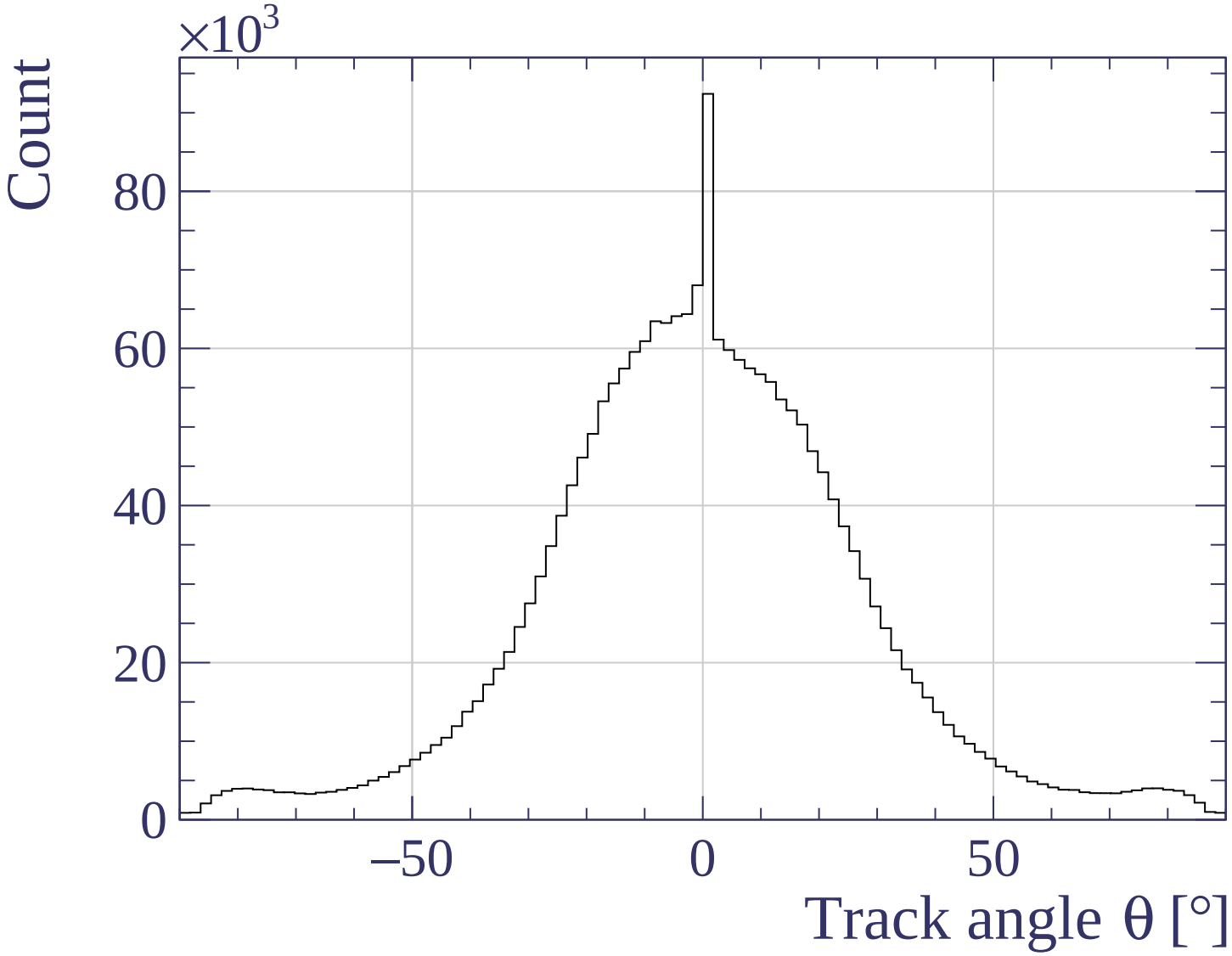




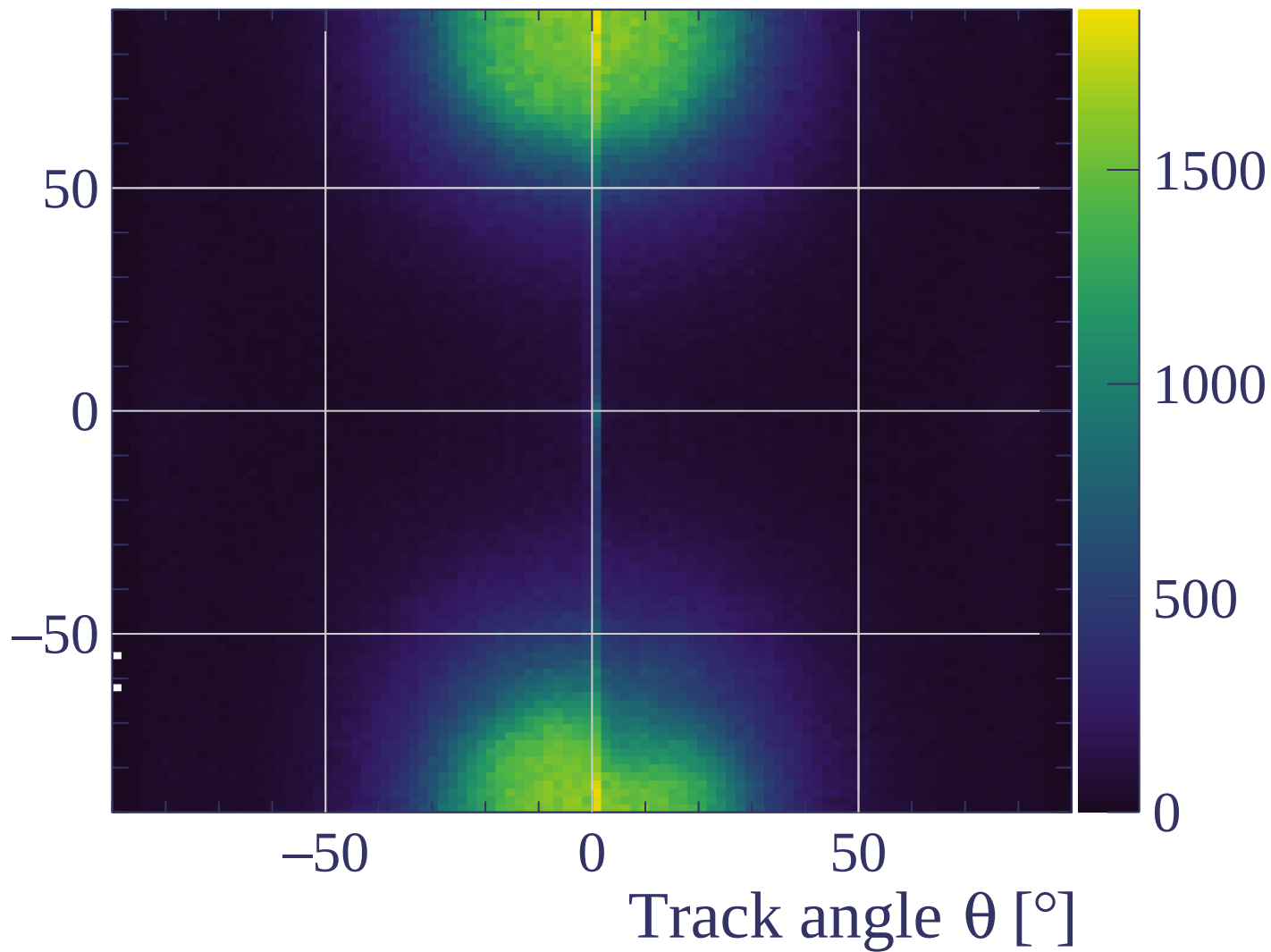
Track length [cm]



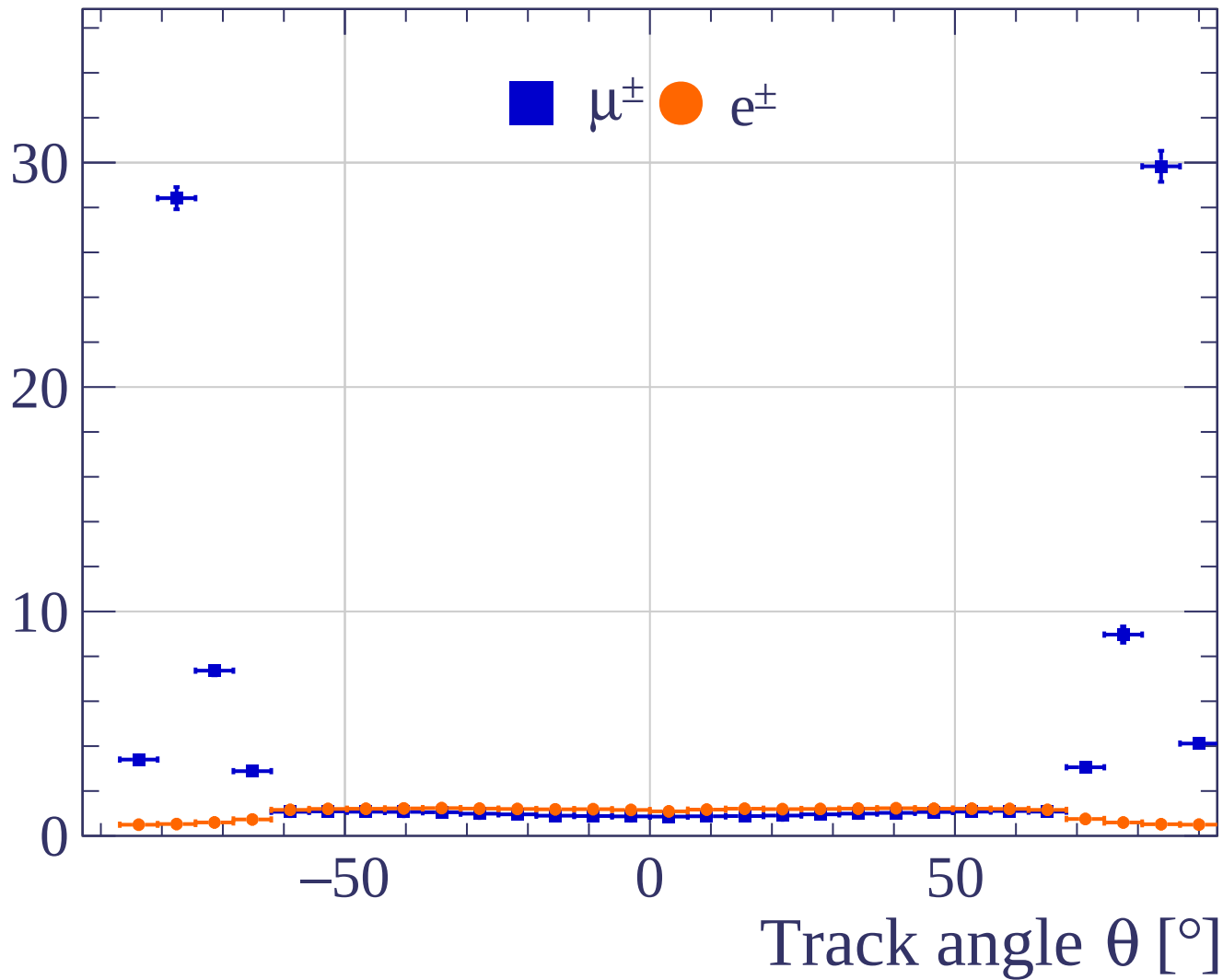




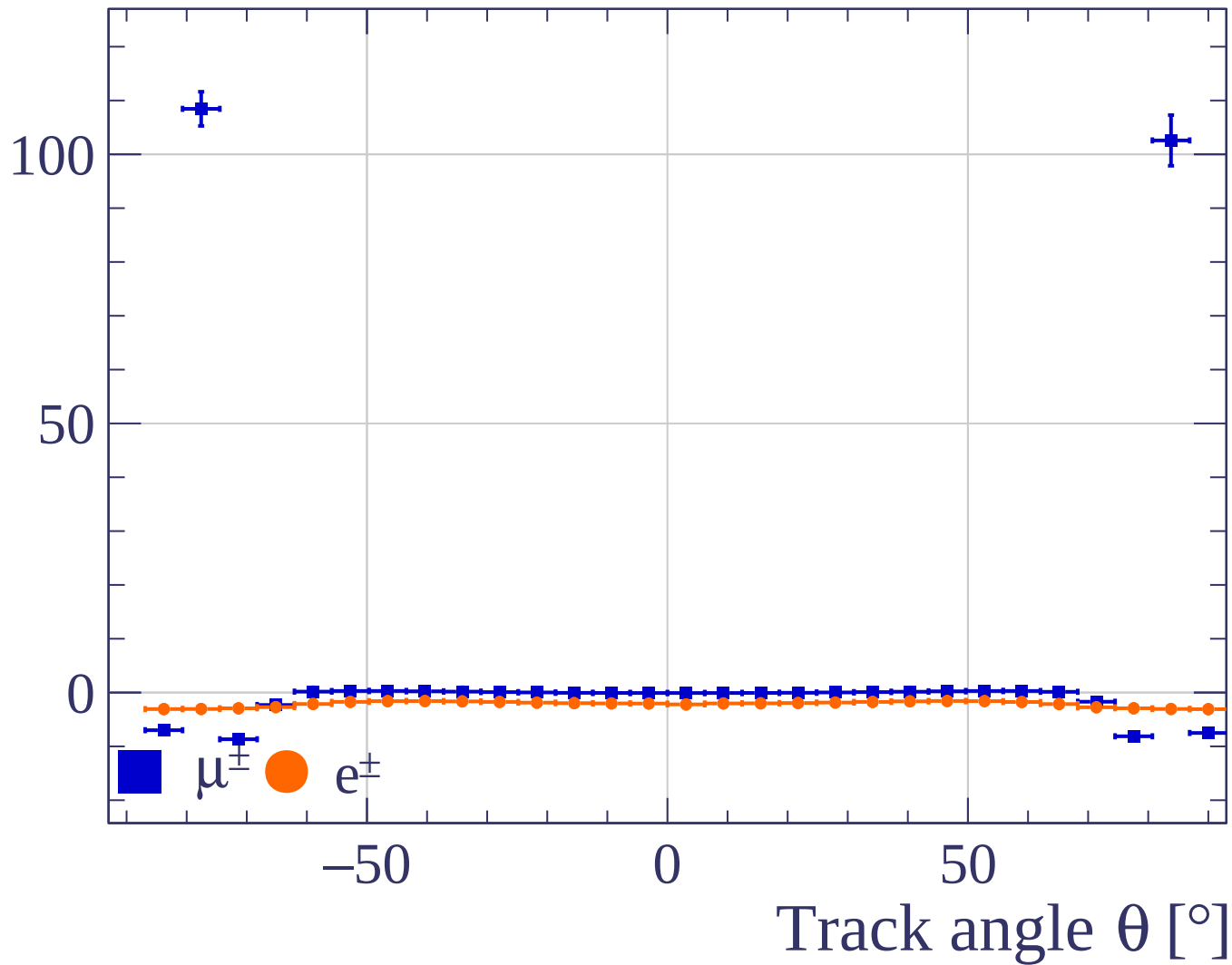
Track angle ϕ [°]



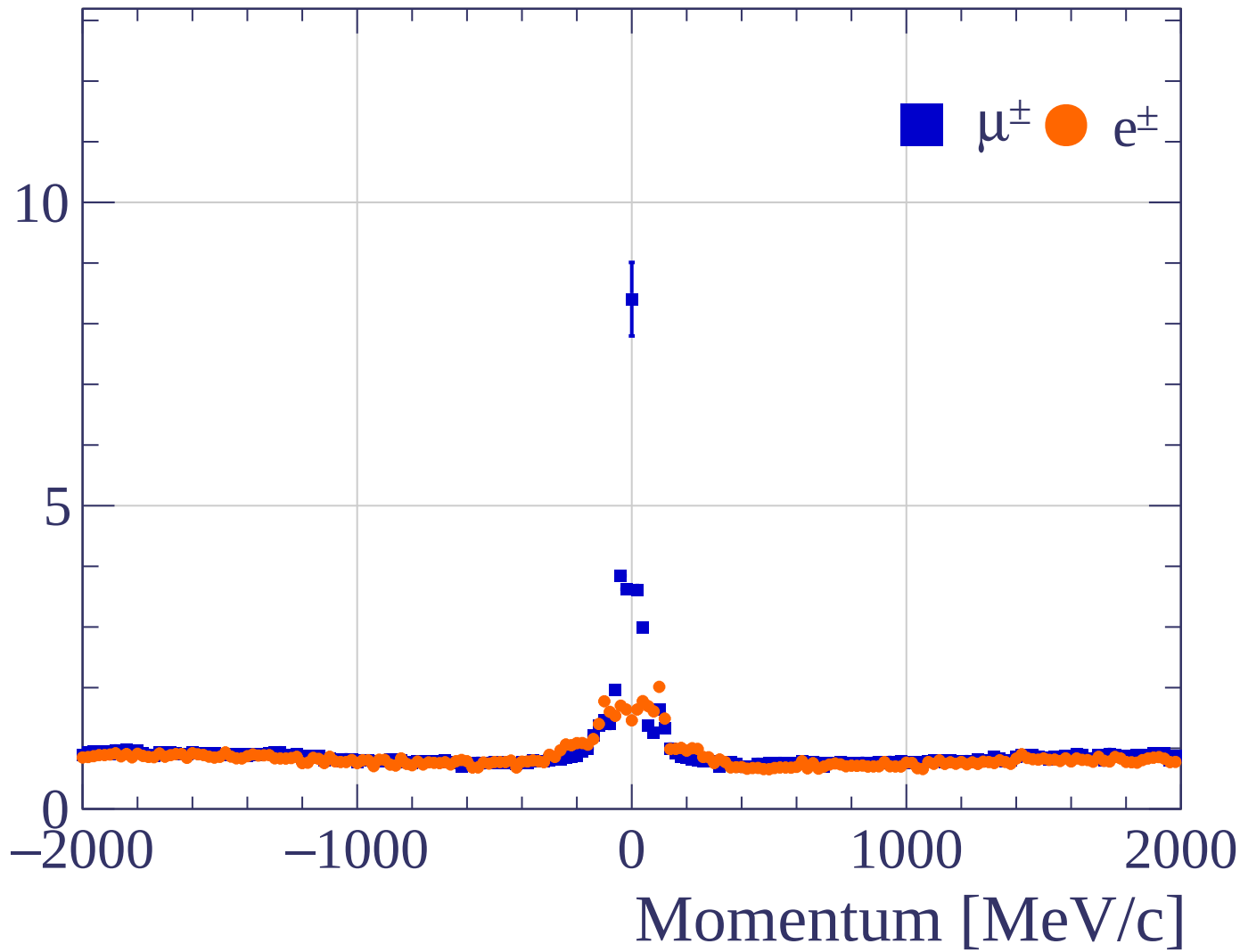
Pulls standard deviation



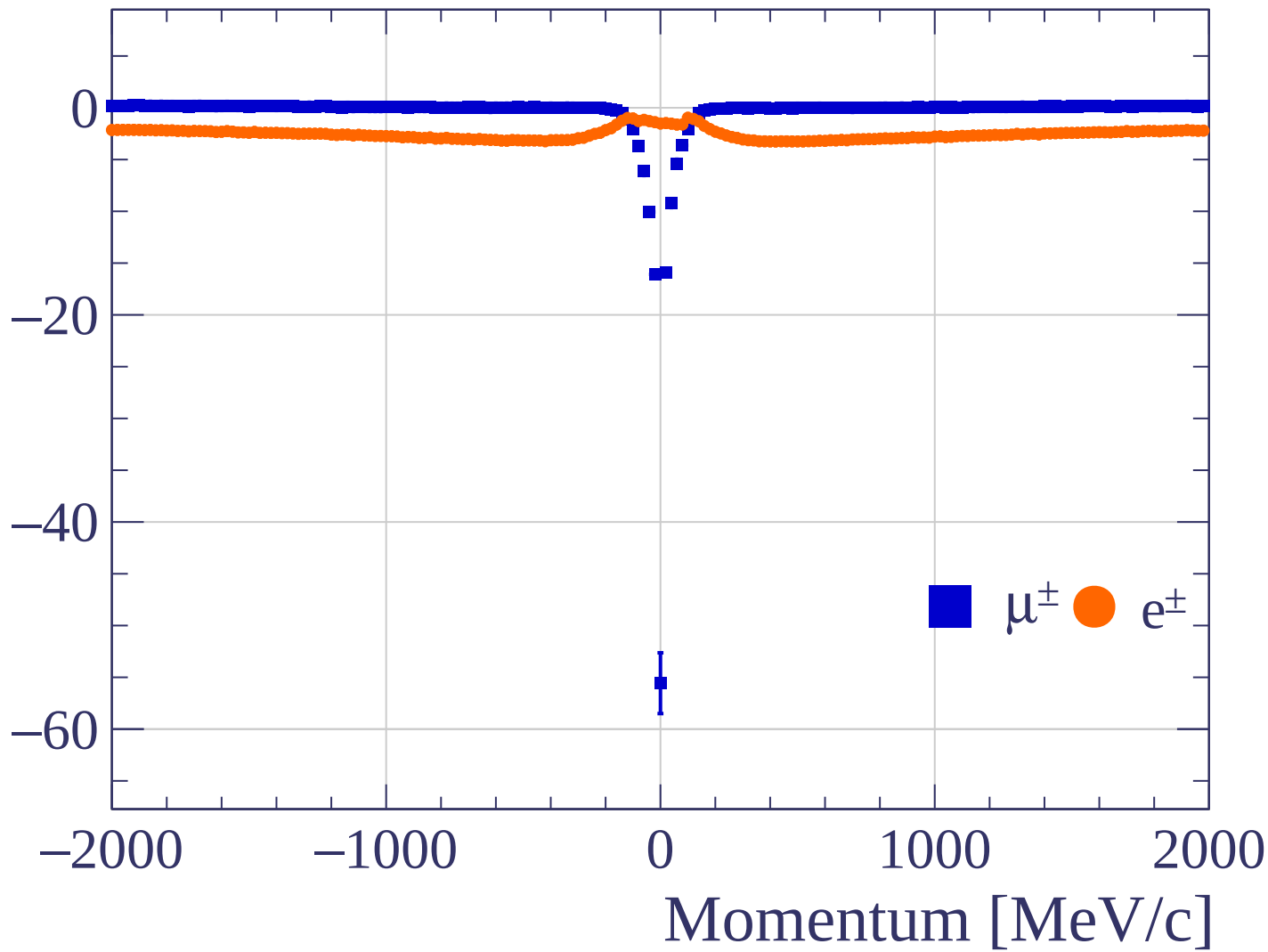
Pulls mean

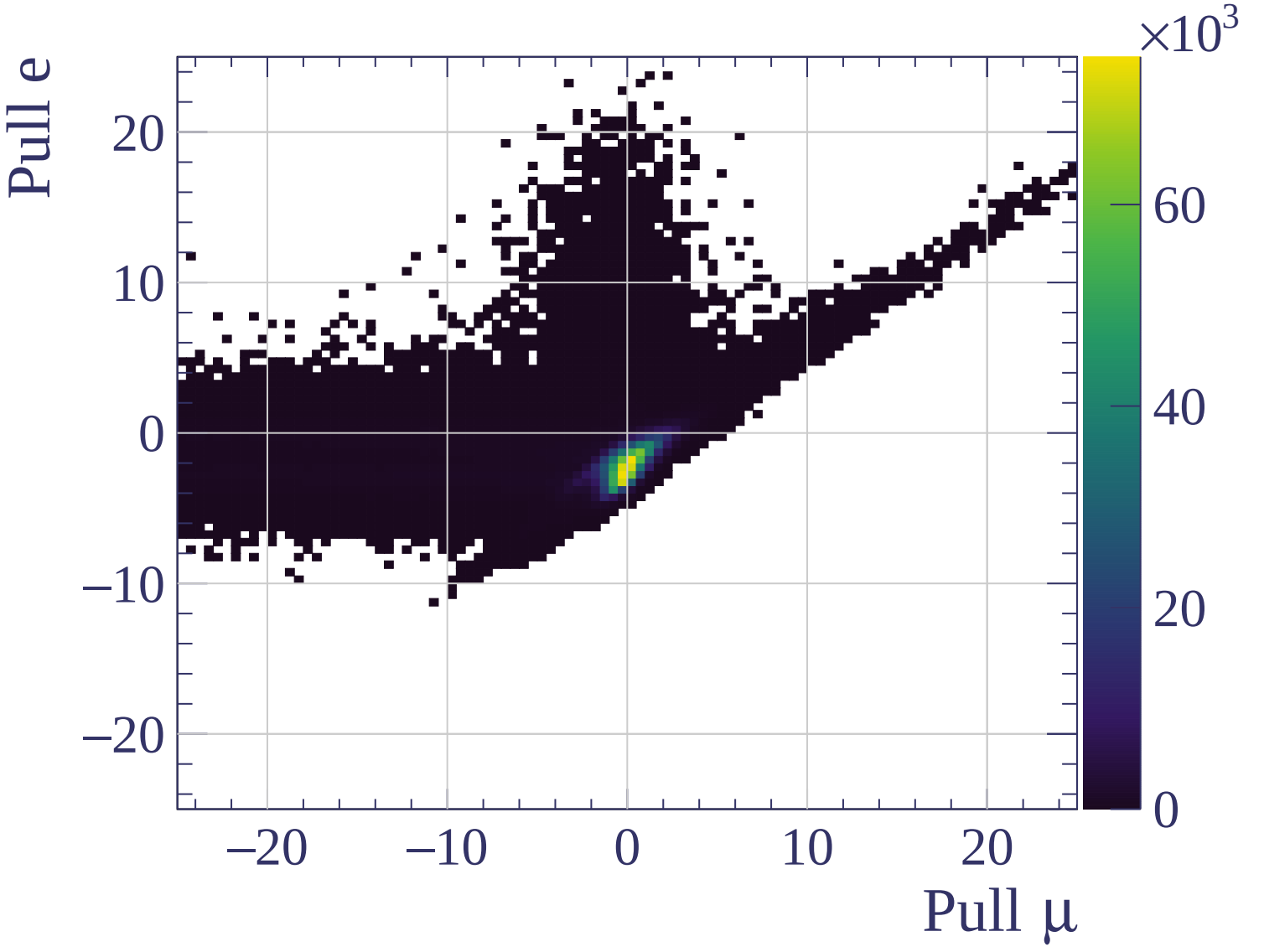


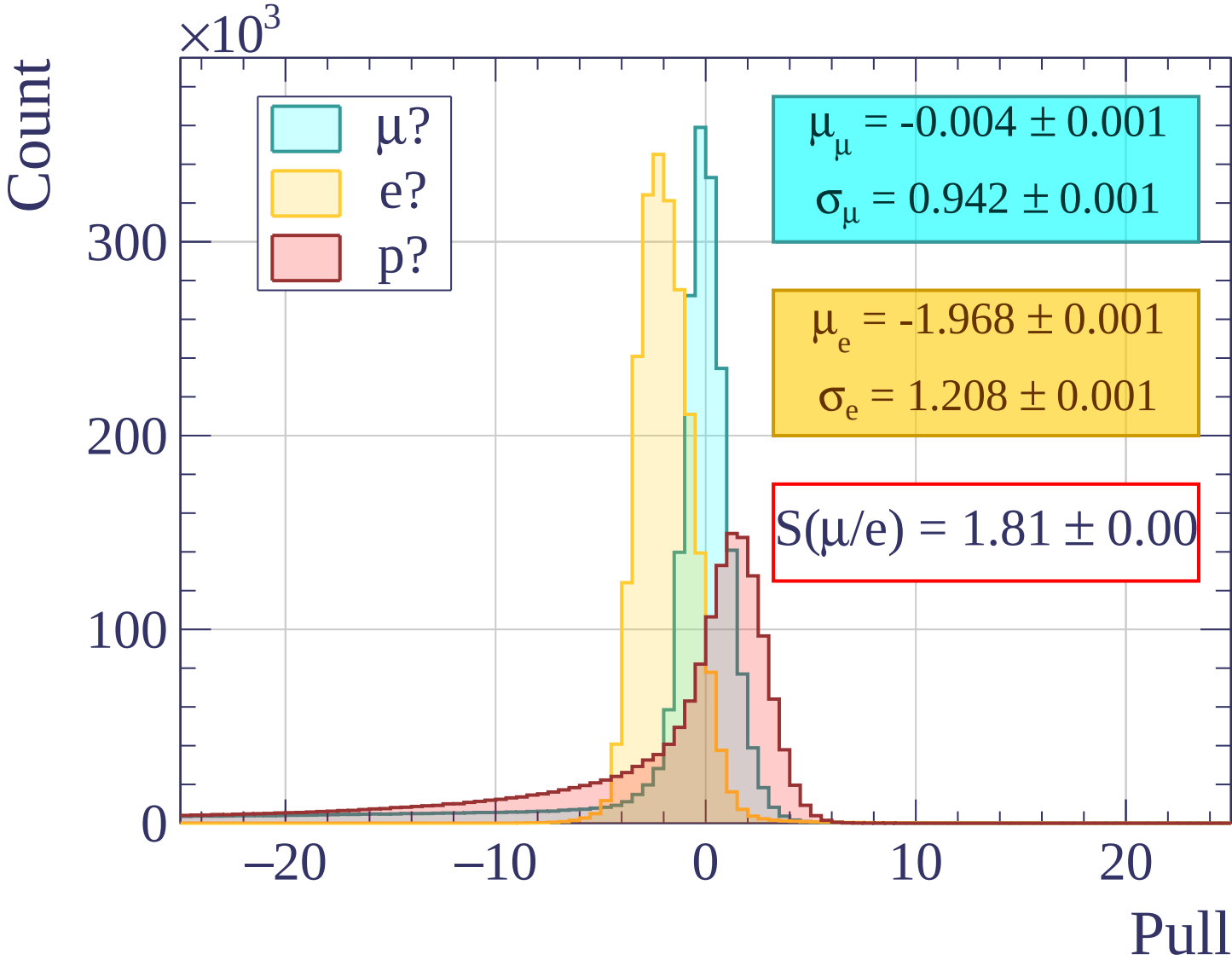
Pulls standard deviation

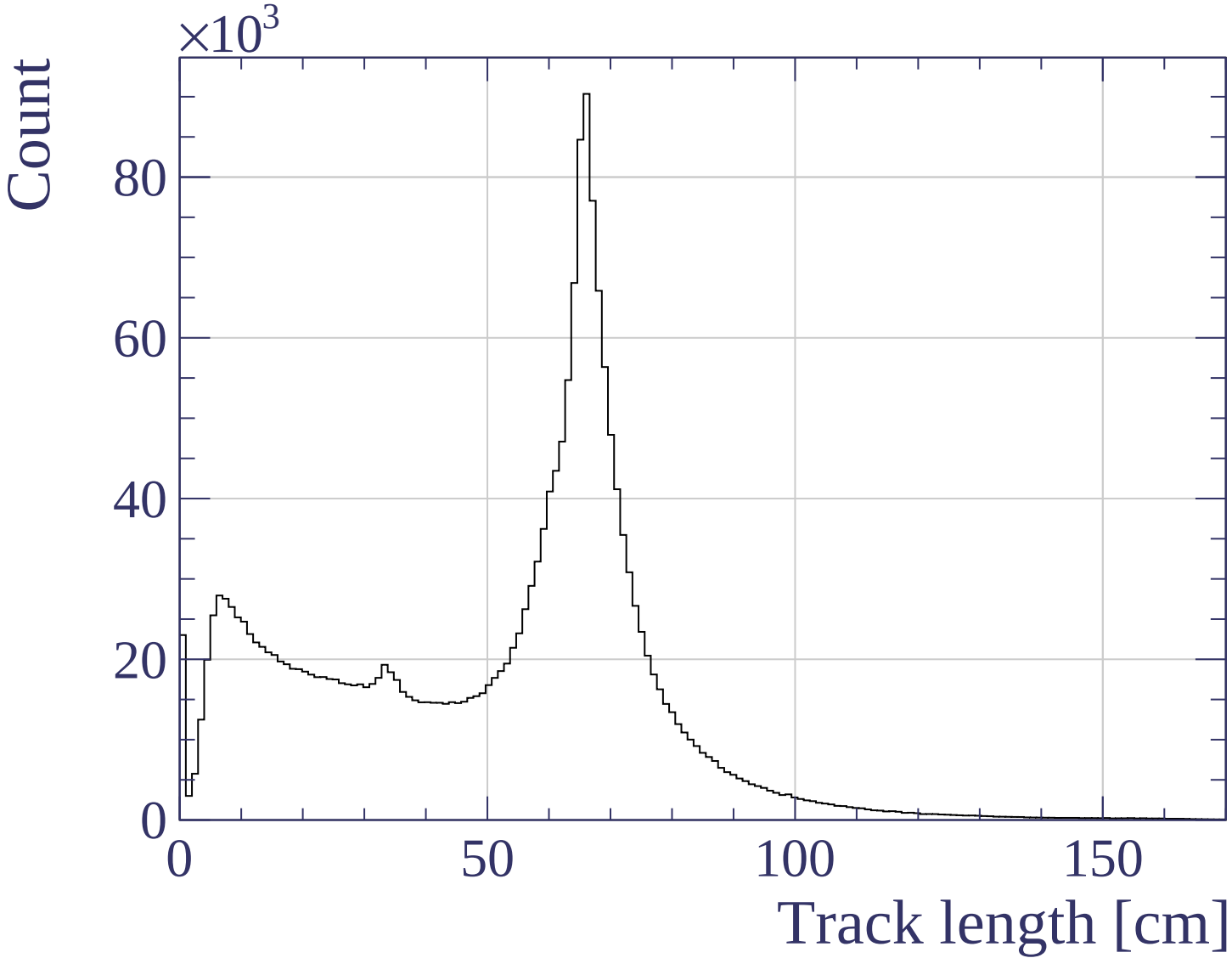


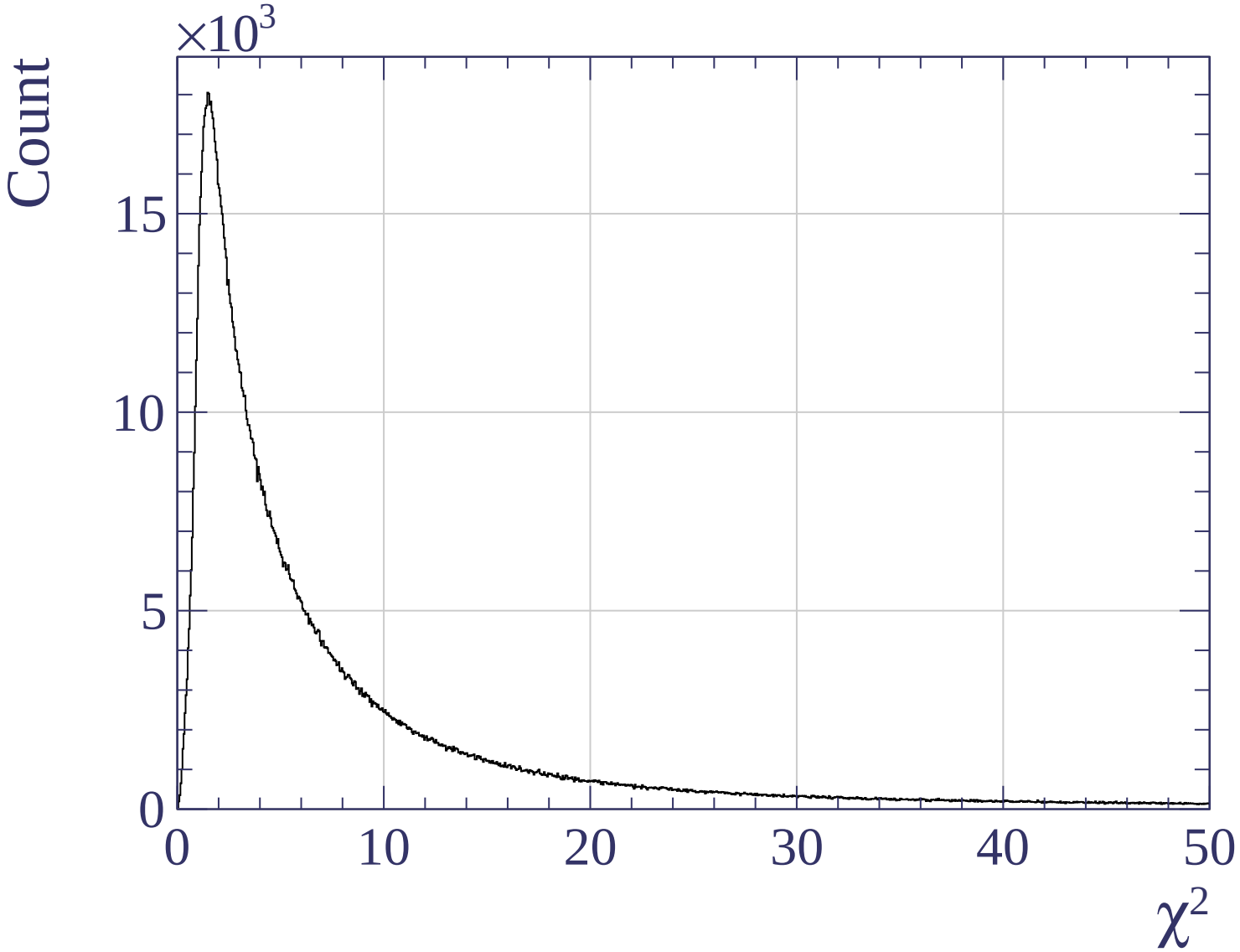
Pulls mean

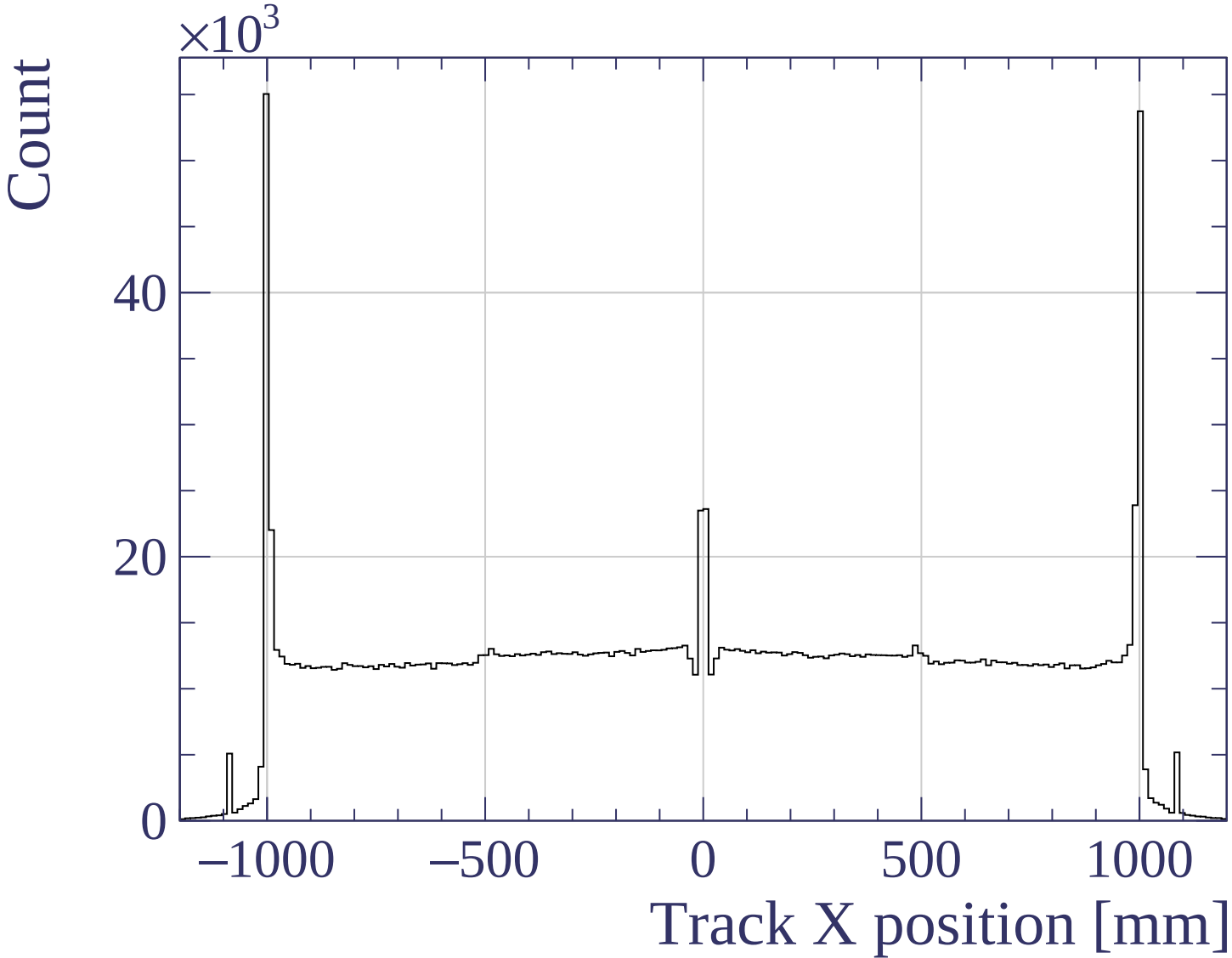


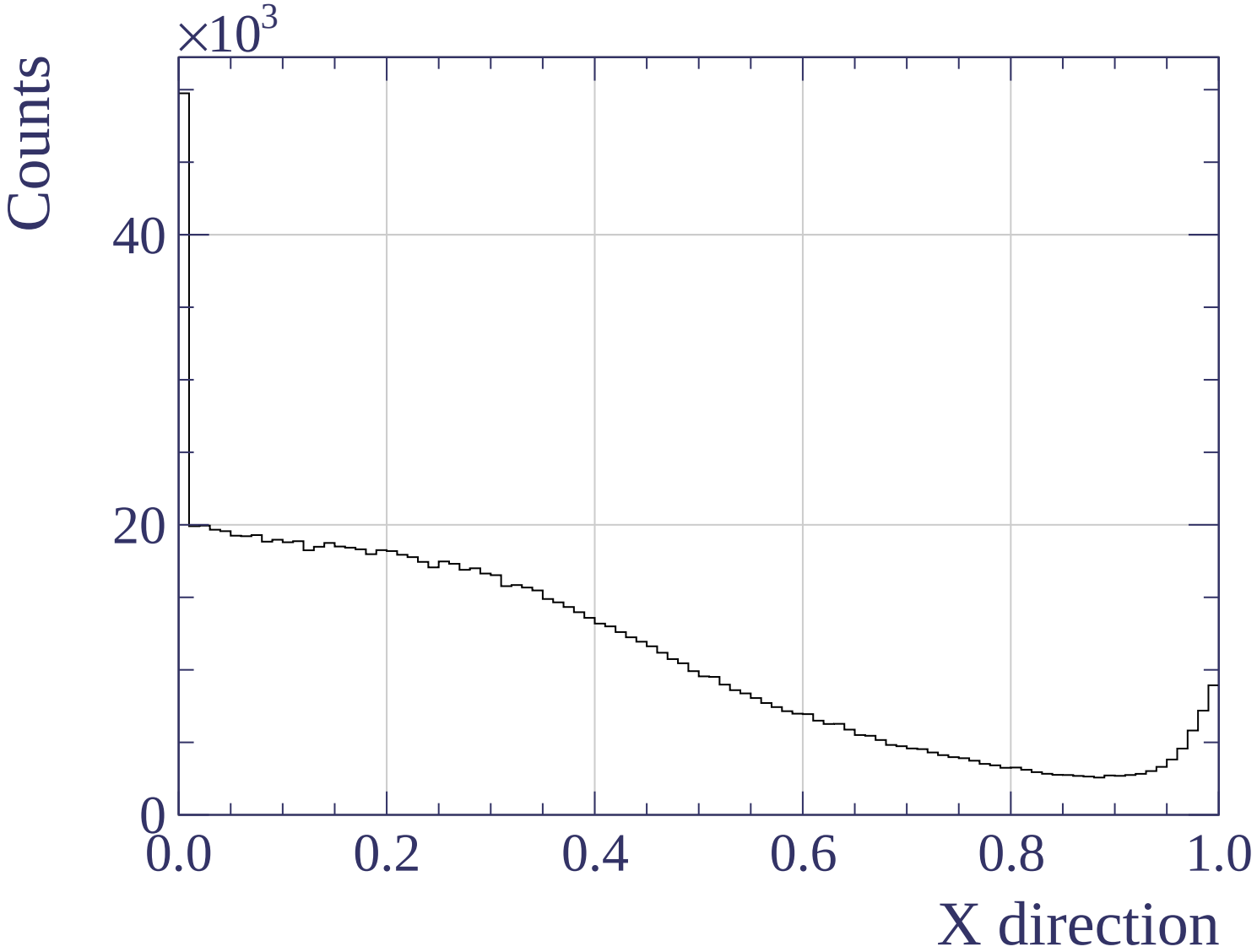


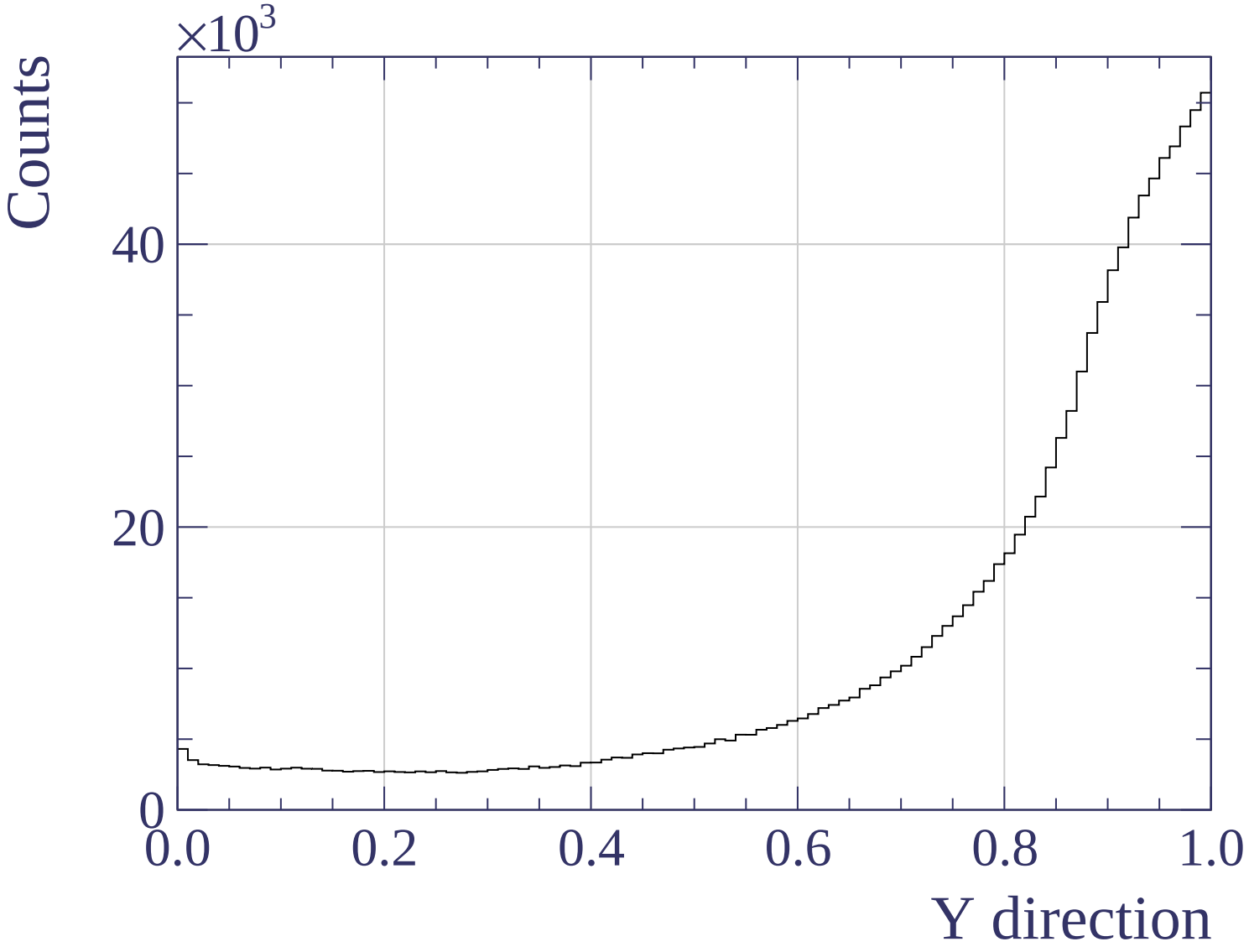


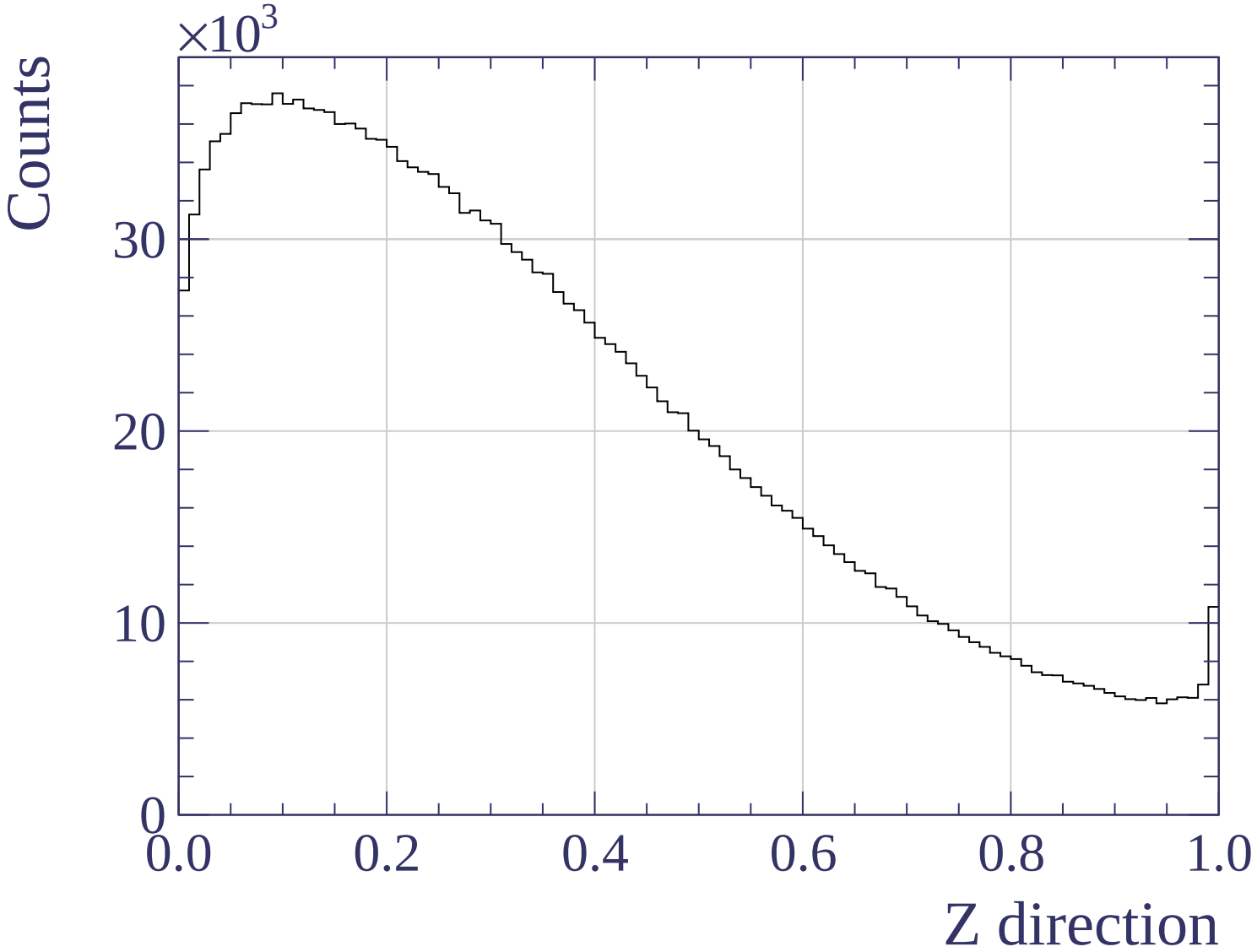


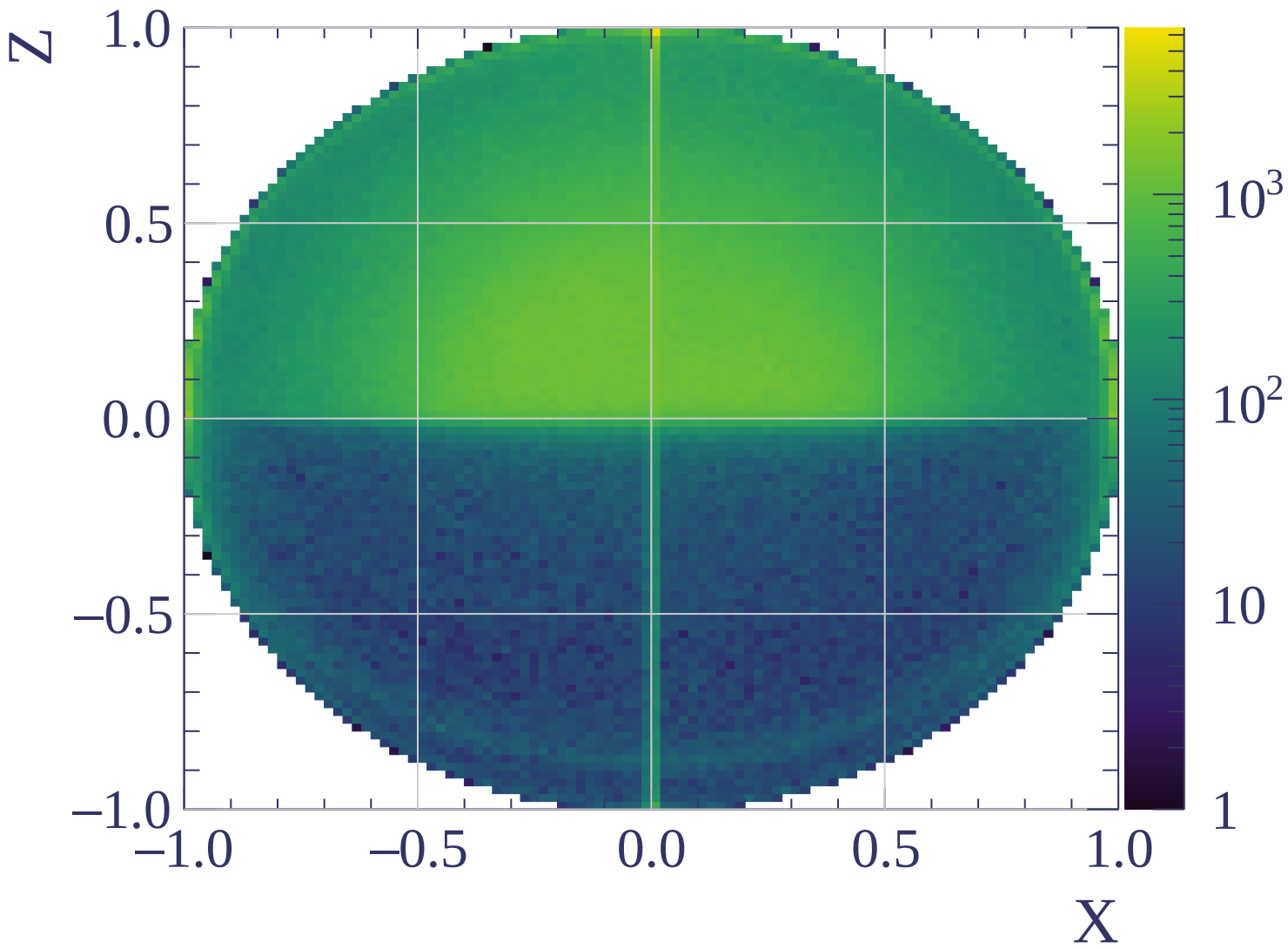


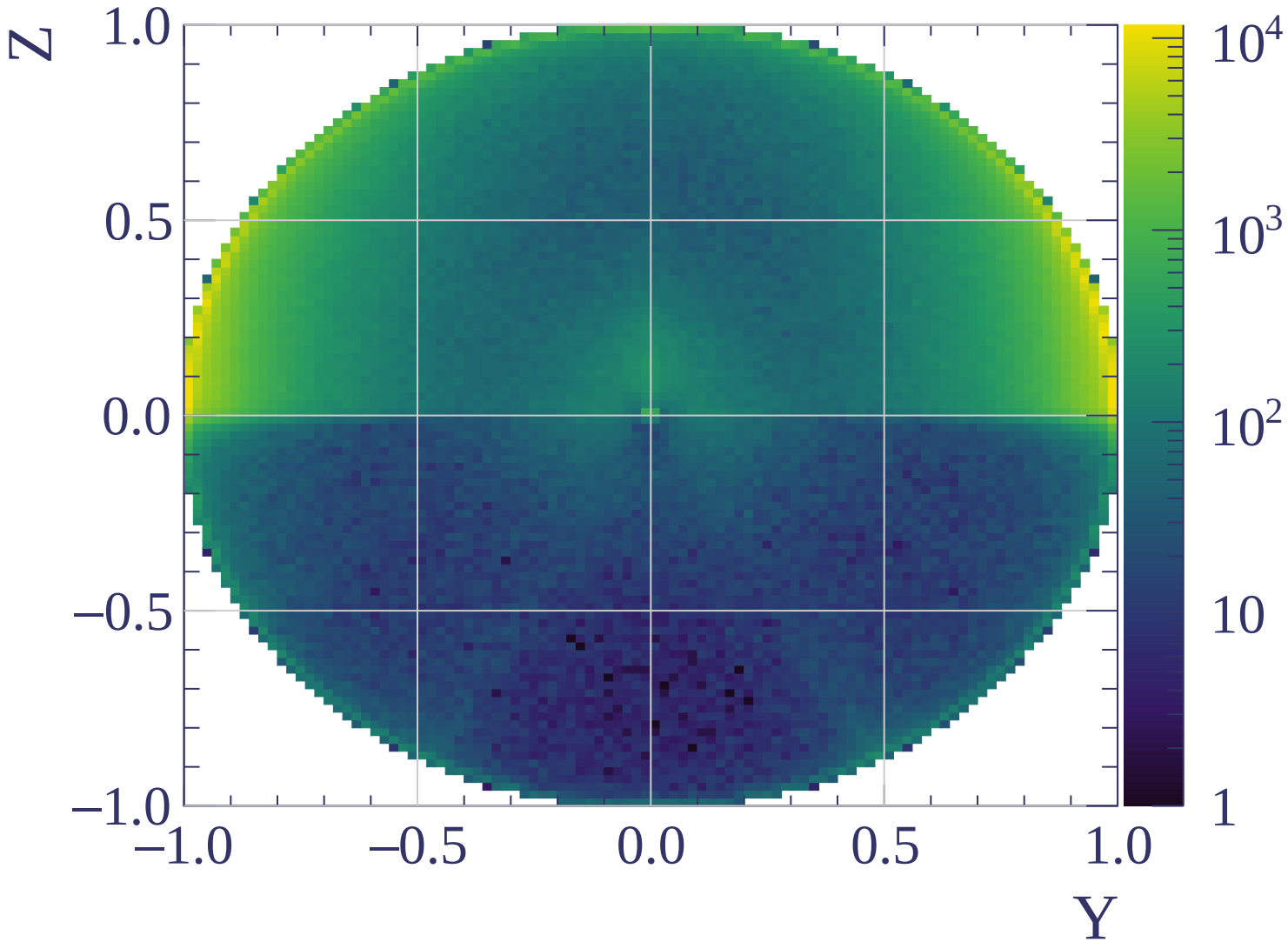


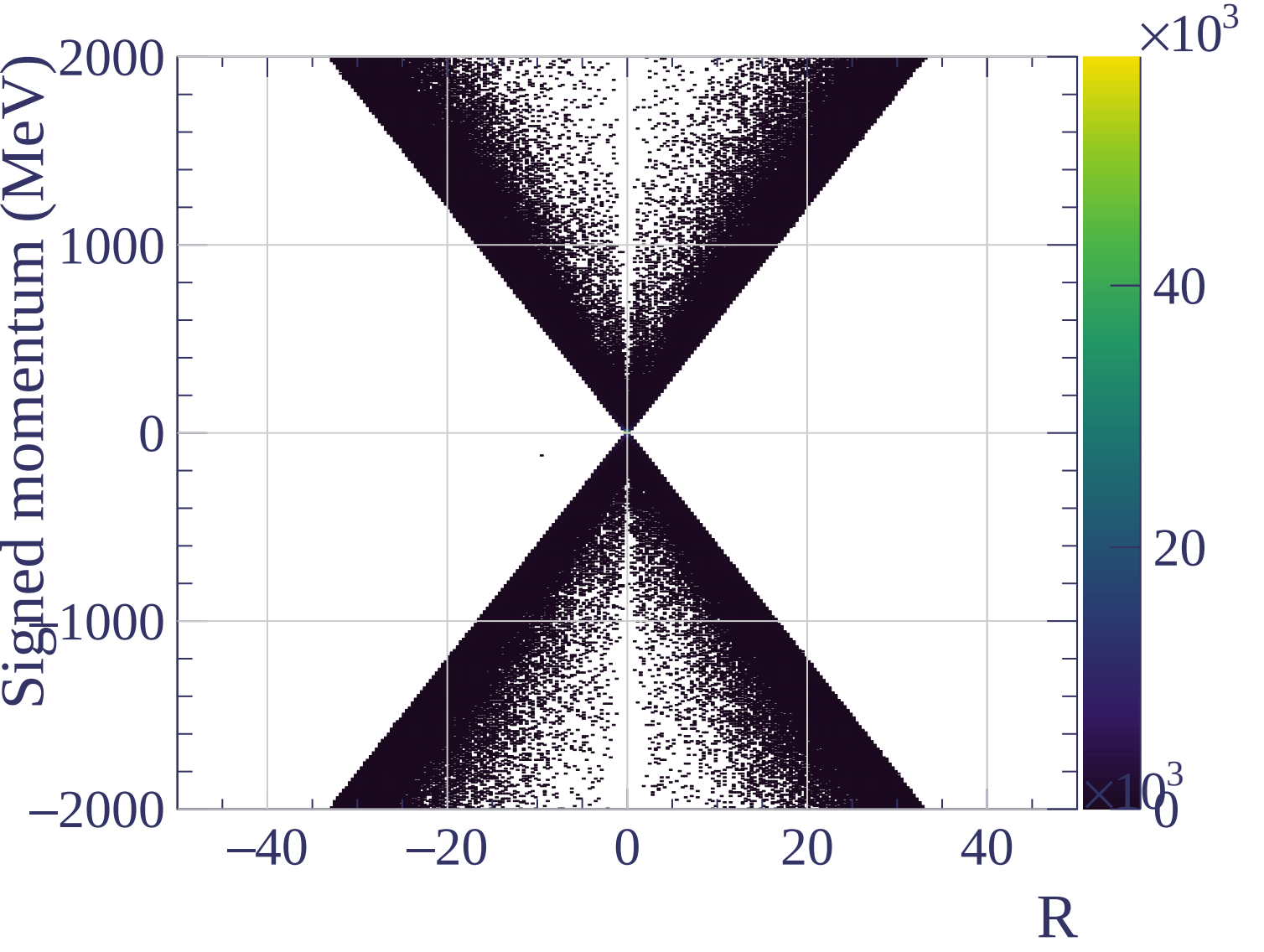


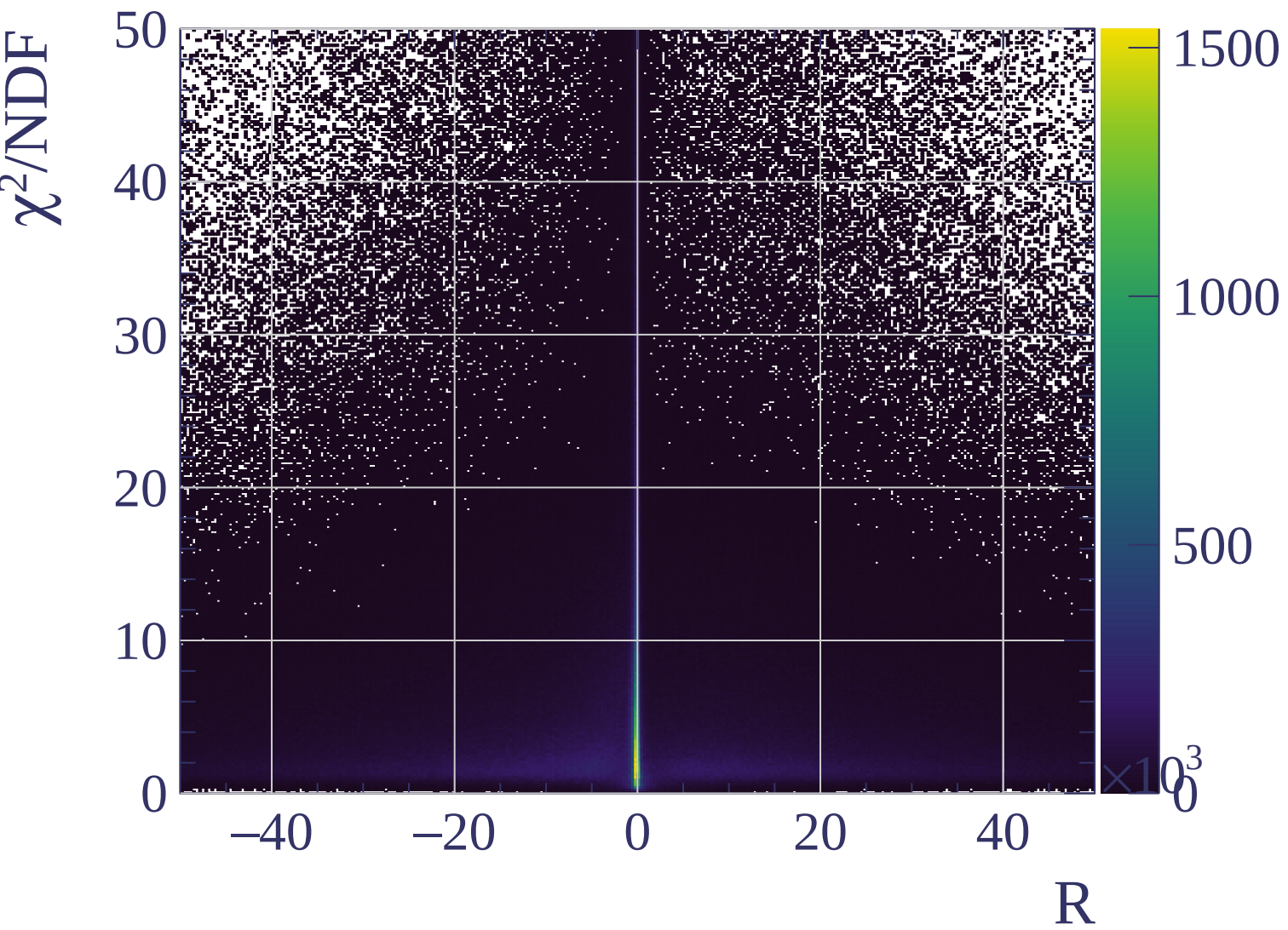












Count

$\times 10^3$

Energy loss | $0 < p < 80$

XP

$$\mu = -488.7 \pm 165531.9$$

$$\sigma = 563.4 \pm 1018.5$$

$$\frac{\sigma}{\mu} = -115.28 \pm 38837.9$$

WF

$$\mu = -1322.8 \pm 916286.9$$

$$\sigma = 1421.9 \pm 1058.3$$

$$\frac{\sigma}{\mu} = -107.49 \pm 744543.6$$

20

10

0

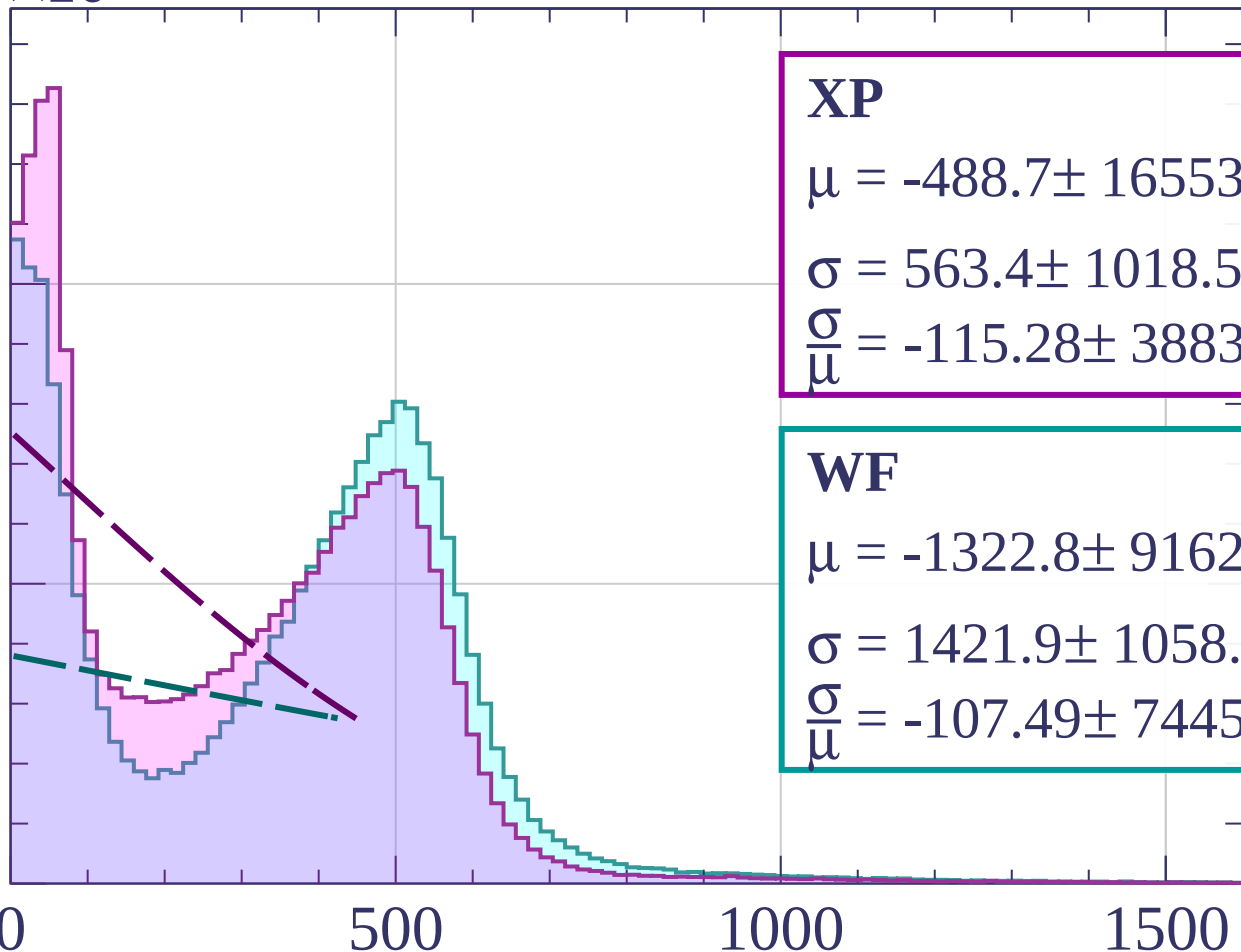
0

500

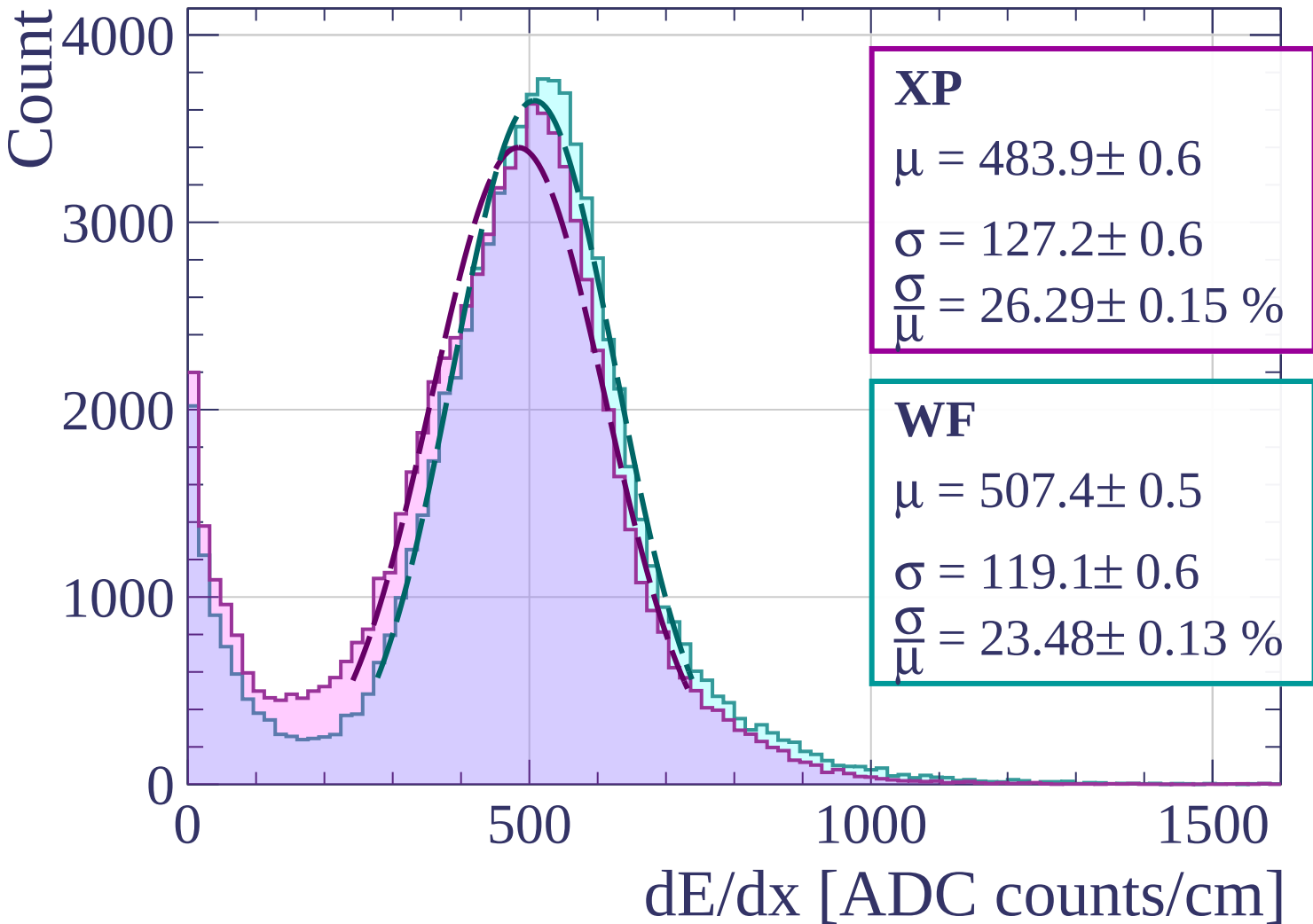
1000

1500

dE/dx [ADC counts/cm]



Energy loss | $80 < p < 160$



Energy loss | $160 < p < 240$

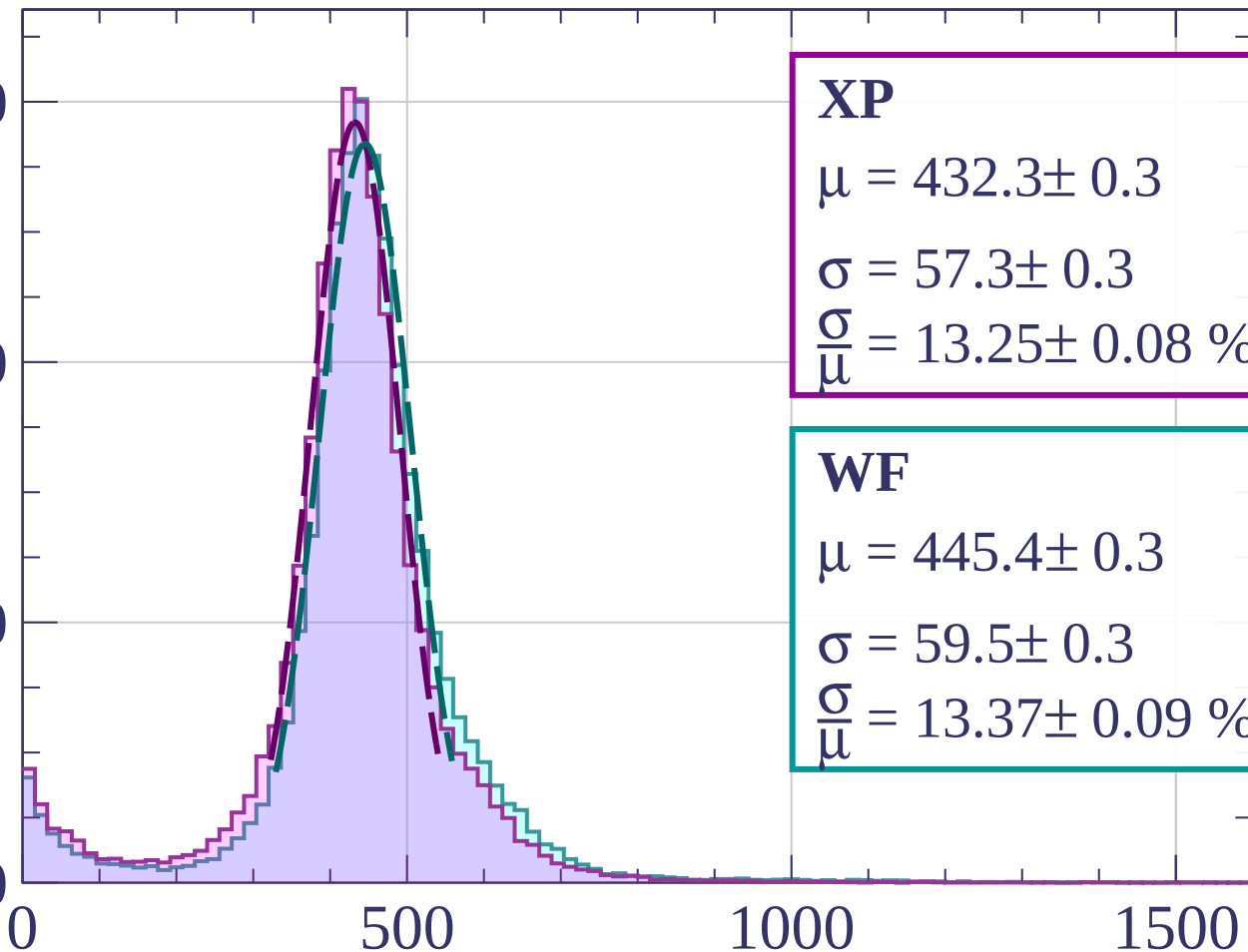
Count

6000

4000

2000

0



XP

$$\mu = 432.3 \pm 0.3$$

$$\sigma = 57.3 \pm 0.3$$

$$\frac{\sigma}{\mu} = 13.25 \pm 0.08 \%$$

WF

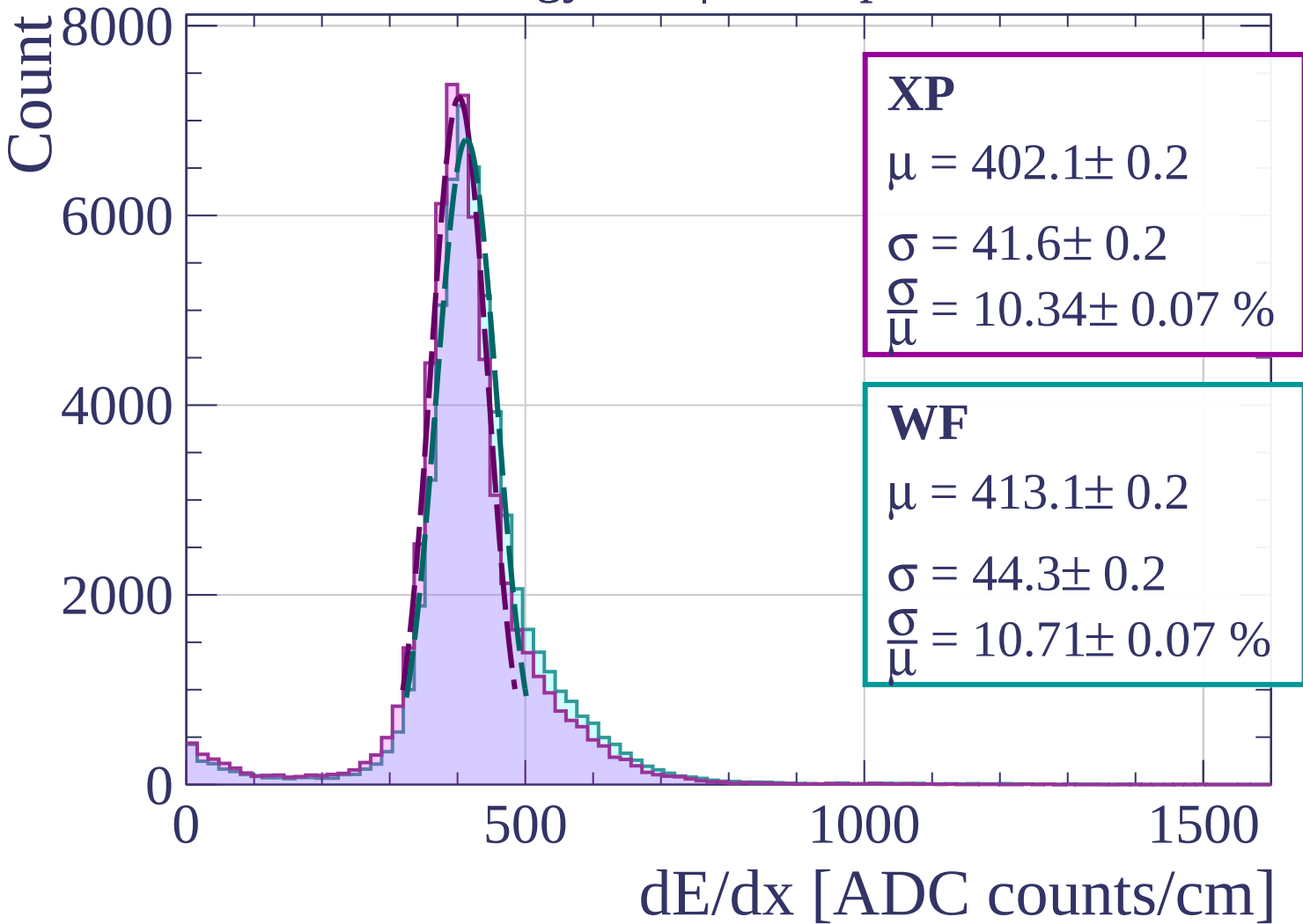
$$\mu = 445.4 \pm 0.3$$

$$\sigma = 59.5 \pm 0.3$$

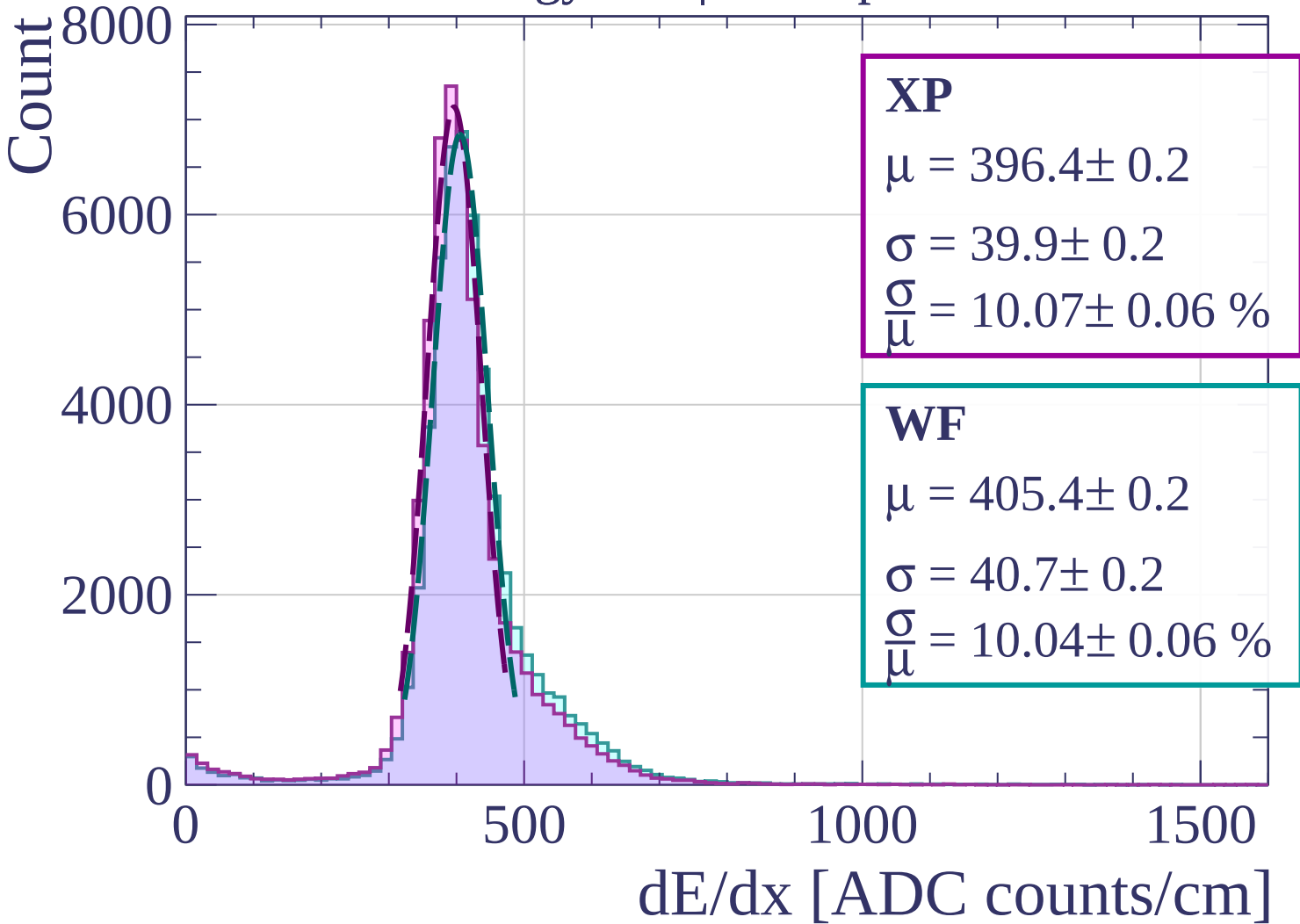
$$\frac{\sigma}{\mu} = 13.37 \pm 0.09 \%$$

dE/dx [ADC counts/cm]

Energy loss | $240 < p < 320$

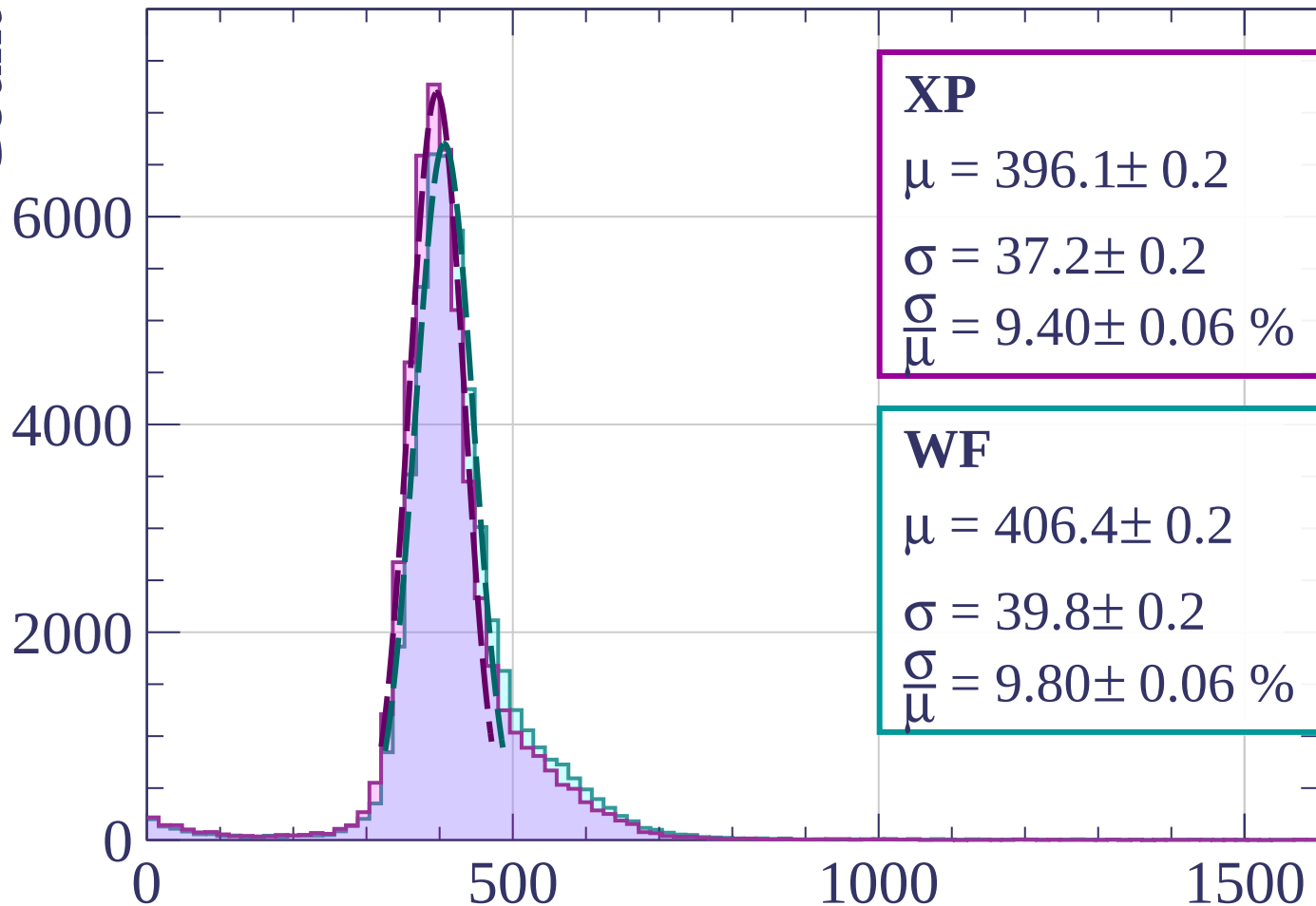


Energy loss | $320 < p < 400$



Energy loss | $400 < p < 480$

Count



XP

$$\mu = 396.1 \pm 0.2$$

$$\sigma = 37.2 \pm 0.2$$

$$\frac{\sigma}{\mu} = 9.40 \pm 0.06 \%$$

WF

$$\mu = 406.4 \pm 0.2$$

$$\sigma = 39.8 \pm 0.2$$

$$\frac{\sigma}{\mu} = 9.80 \pm 0.06 \%$$

dE/dx [ADC counts/cm]

Energy loss | $480 < p < 560$

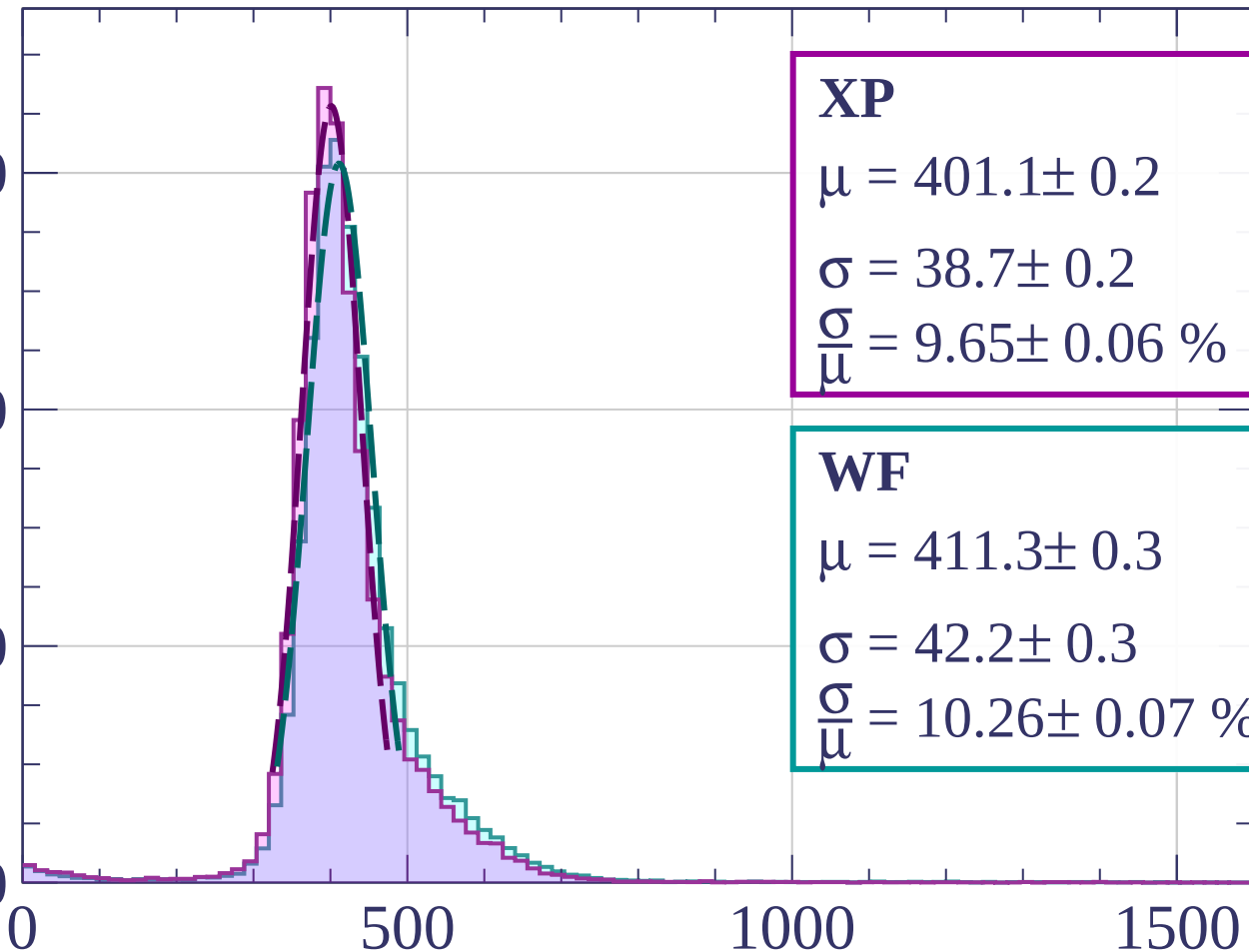
Count

6000

4000

2000

0



XP

$$\mu = 401.1 \pm 0.2$$

$$\sigma = 38.7 \pm 0.2$$

$$\frac{\sigma}{\mu} = 9.65 \pm 0.06 \%$$

WF

$$\mu = 411.3 \pm 0.3$$

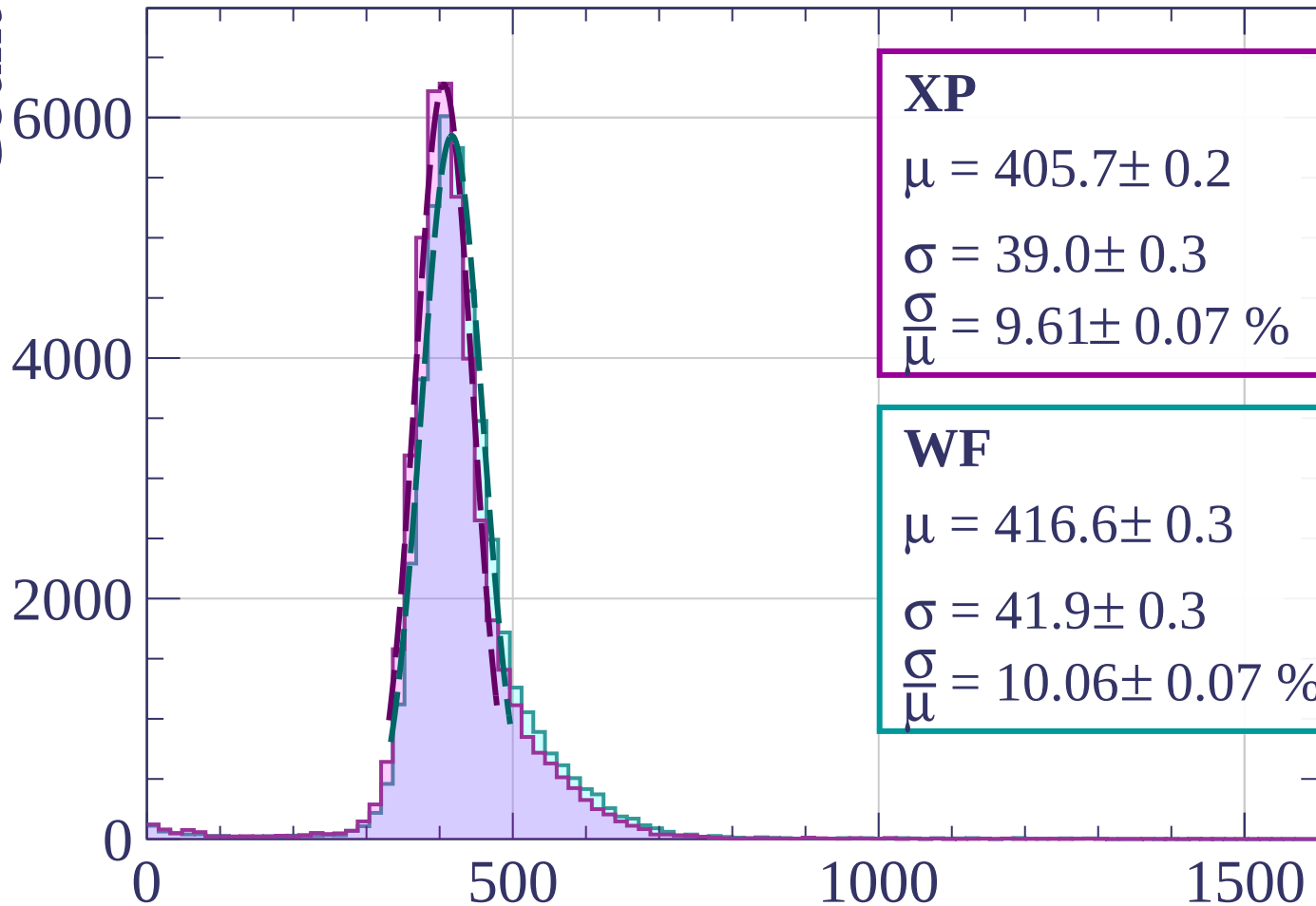
$$\sigma = 42.2 \pm 0.3$$

$$\frac{\sigma}{\mu} = 10.26 \pm 0.07 \%$$

dE/dx [ADC counts/cm]

Energy loss | $560 < p < 640$

Count



XP

$$\mu = 405.7 \pm 0.2$$

$$\sigma = 39.0 \pm 0.3$$

$$\frac{\sigma}{\mu} = 9.61 \pm 0.07 \%$$

WF

$$\mu = 416.6 \pm 0.3$$

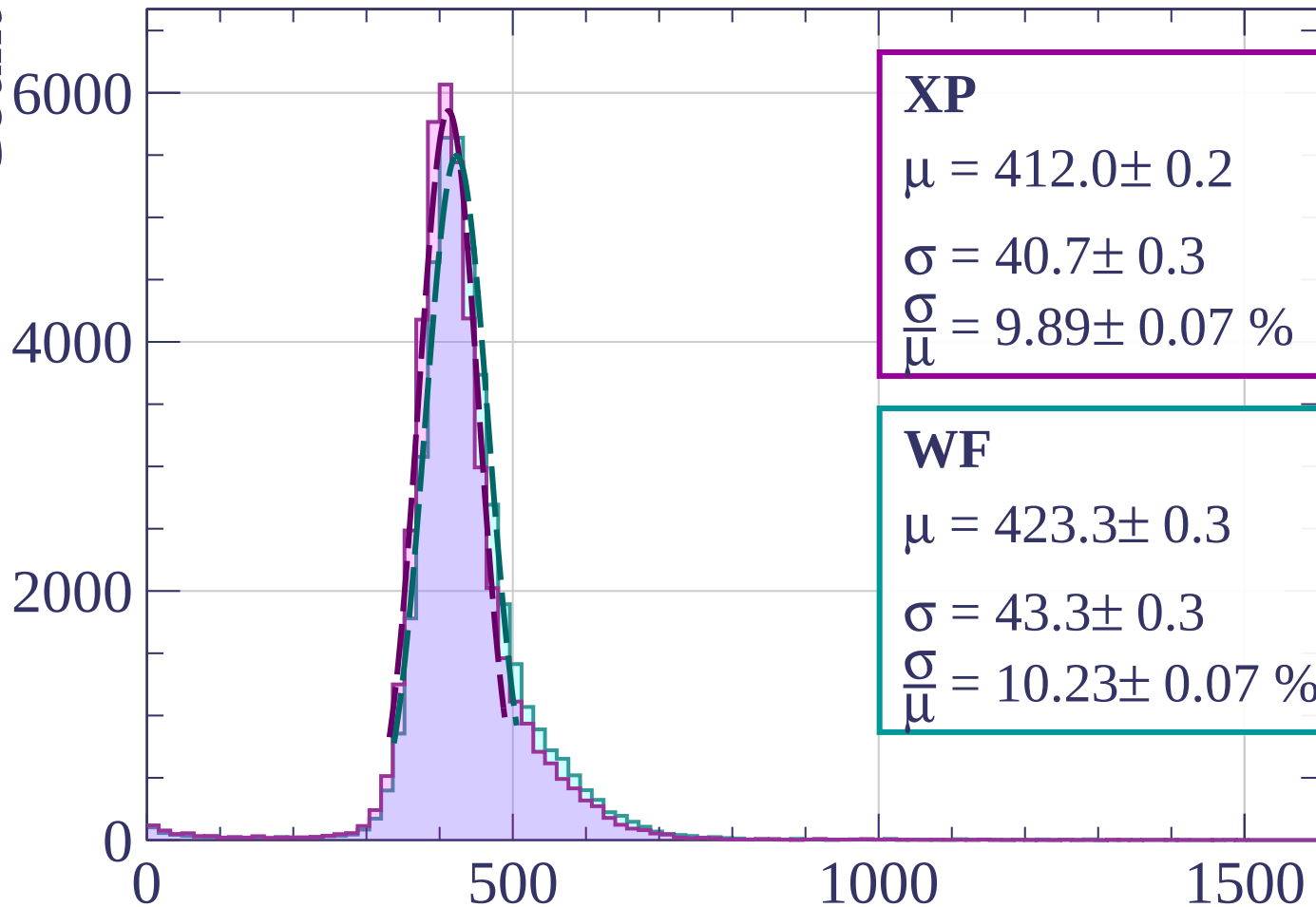
$$\sigma = 41.9 \pm 0.3$$

$$\frac{\sigma}{\mu} = 10.06 \pm 0.07 \%$$

dE/dx [ADC counts/cm]

Energy loss | $640 < p < 720$

Count



XP

$$\mu = 412.0 \pm 0.2$$

$$\sigma = 40.7 \pm 0.3$$

$$\frac{\sigma}{\mu} = 9.89 \pm 0.07 \%$$

WF

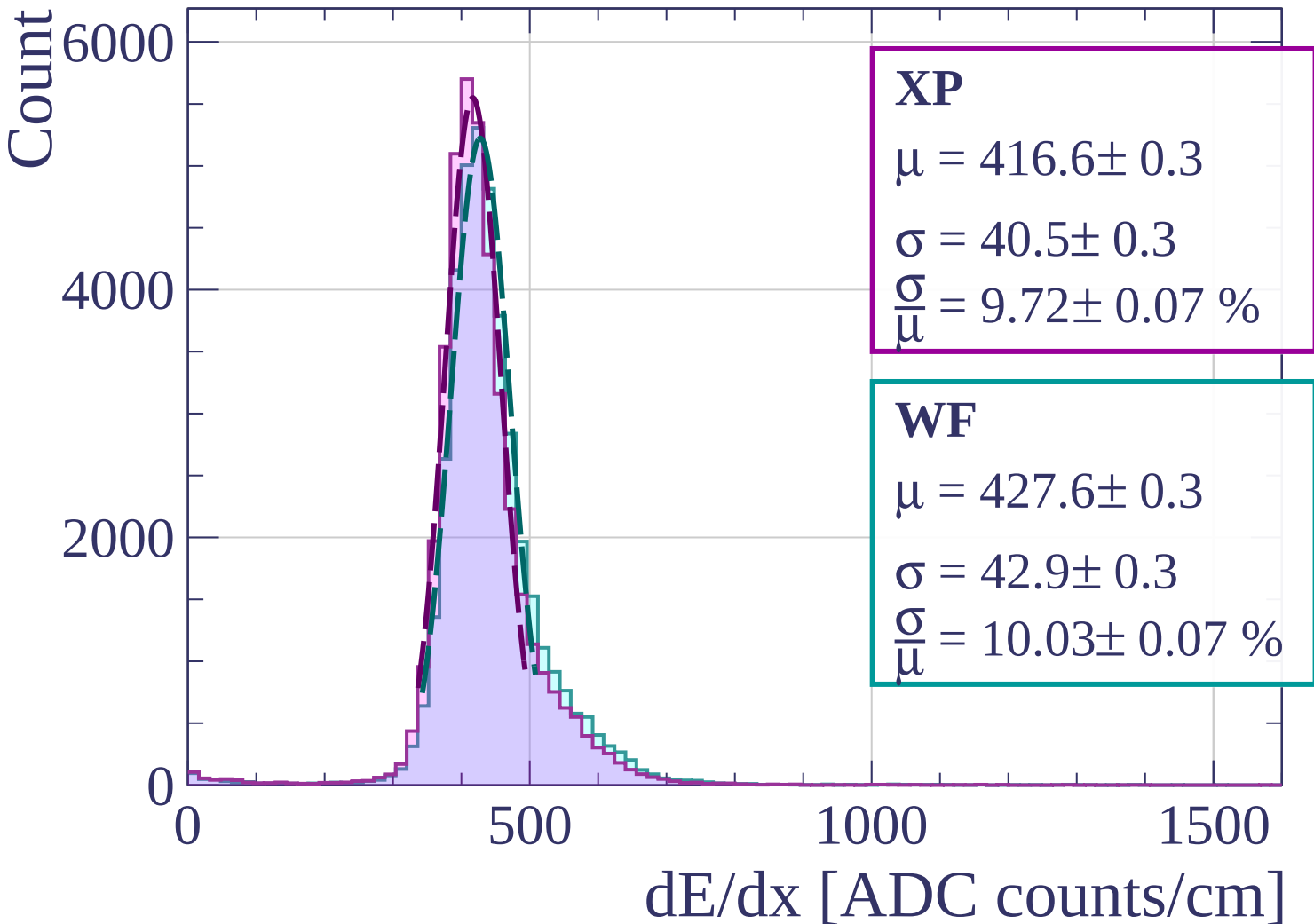
$$\mu = 423.3 \pm 0.3$$

$$\sigma = 43.3 \pm 0.3$$

$$\frac{\sigma}{\mu} = 10.23 \pm 0.07 \%$$

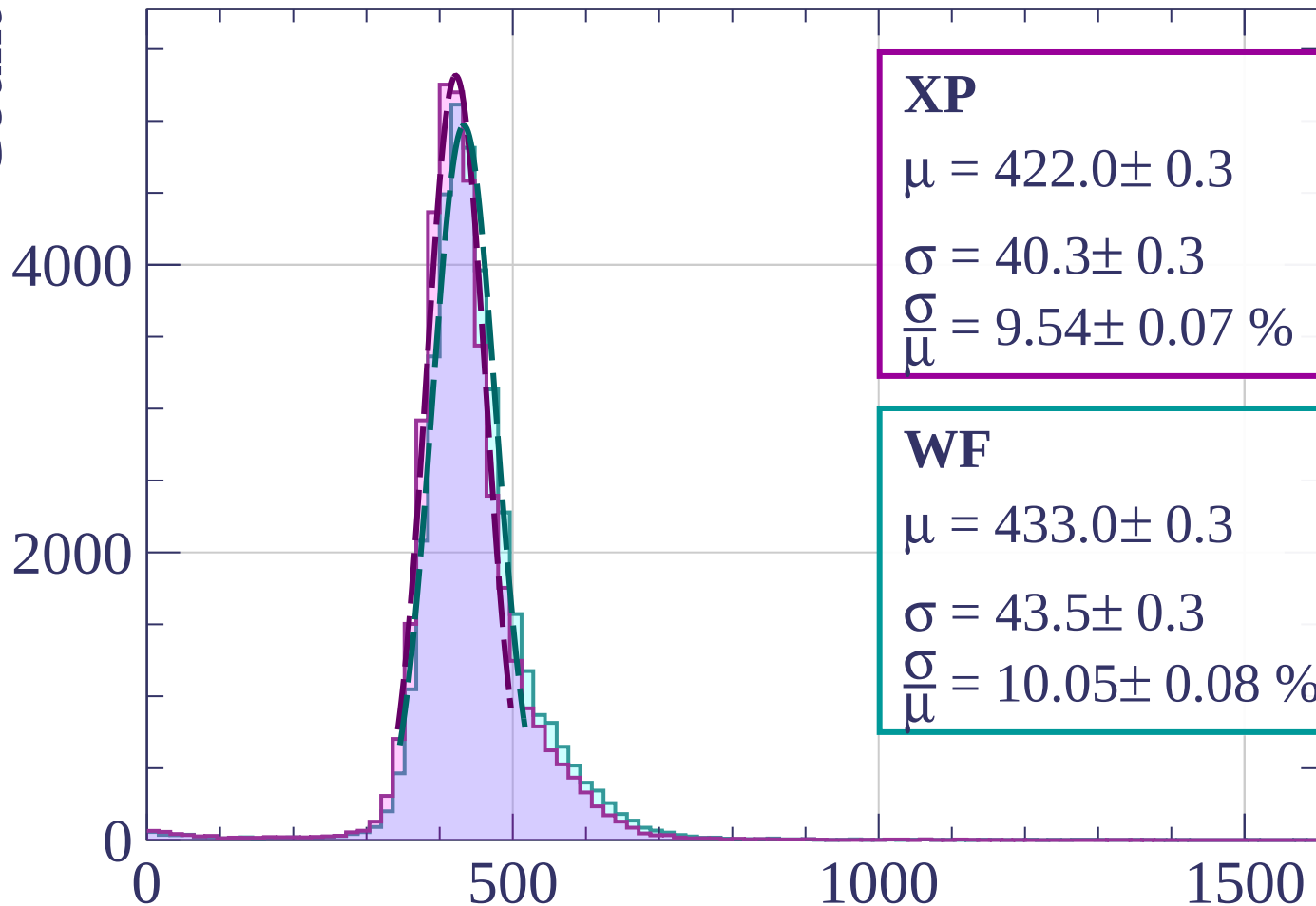
dE/dx [ADC counts/cm]

Energy loss | $720 < p < 800$



Energy loss | $800 < p < 880$

Count



XP

$$\mu = 422.0 \pm 0.3$$

$$\sigma = 40.3 \pm 0.3$$

$$\frac{\sigma}{\mu} = 9.54 \pm 0.07 \%$$

WF

$$\mu = 433.0 \pm 0.3$$

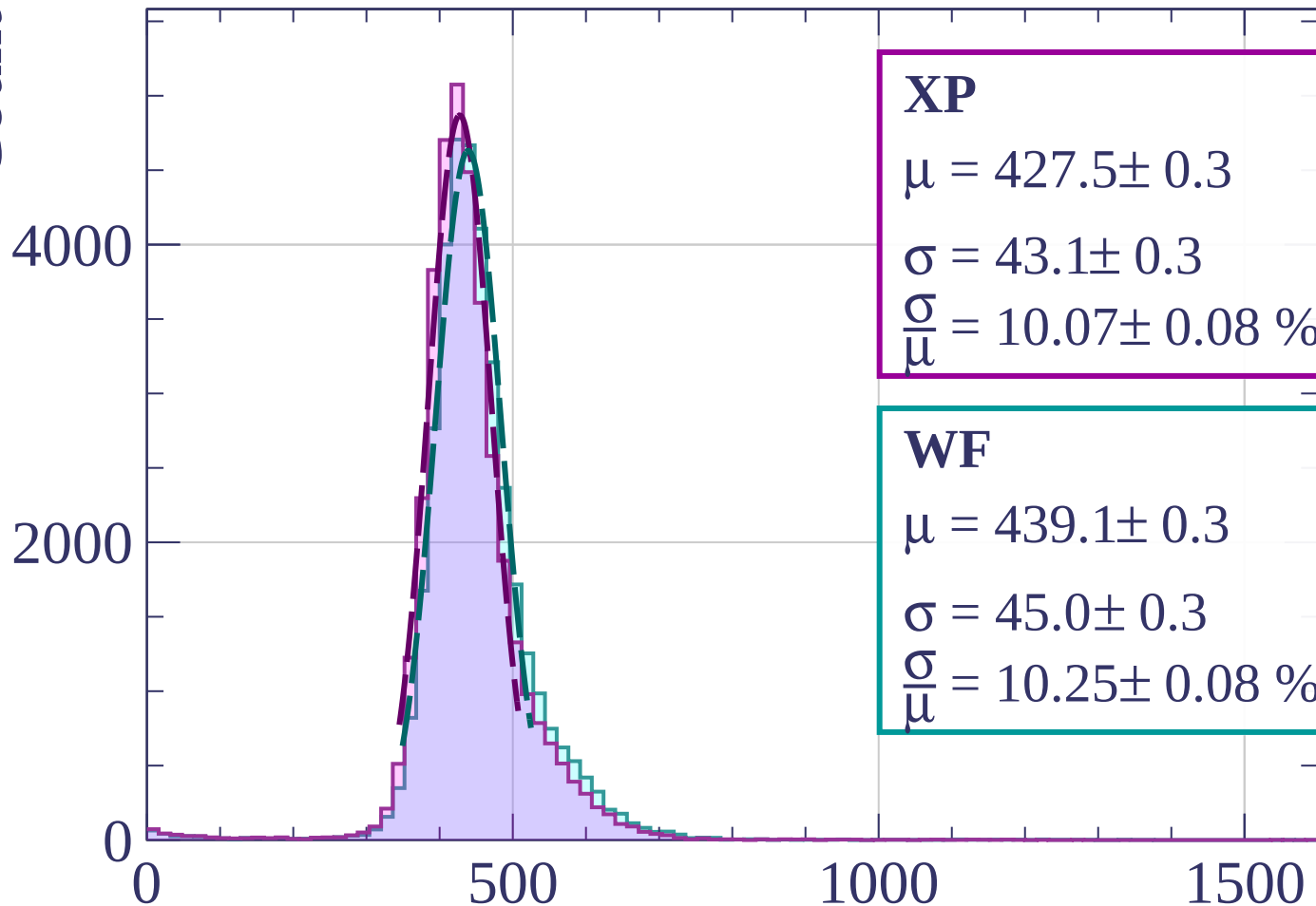
$$\sigma = 43.5 \pm 0.3$$

$$\frac{\sigma}{\mu} = 10.05 \pm 0.08 \%$$

dE/dx [ADC counts/cm]

Energy loss | $880 < p < 960$

Count



XP

$$\mu = 427.5 \pm 0.3$$

$$\sigma = 43.1 \pm 0.3$$

$$\frac{\sigma}{\mu} = 10.07 \pm 0.08 \%$$

WF

$$\mu = 439.1 \pm 0.3$$

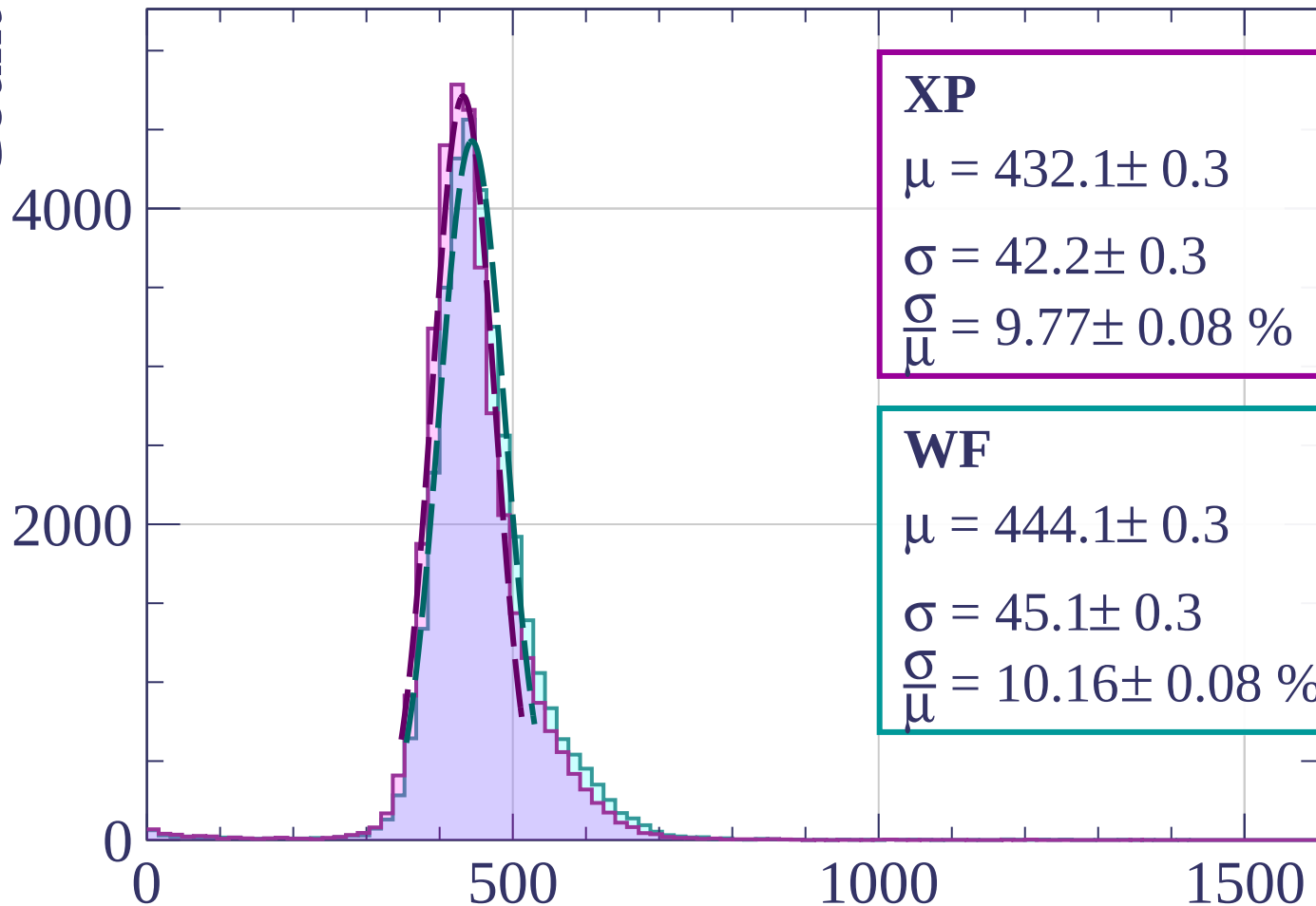
$$\sigma = 45.0 \pm 0.3$$

$$\frac{\sigma}{\mu} = 10.25 \pm 0.08 \%$$

dE/dx [ADC counts/cm]

Energy loss | $960 < p < 1040$

Count



XP

$$\mu = 432.1 \pm 0.3$$

$$\sigma = 42.2 \pm 0.3$$

$$\frac{\sigma}{\mu} = 9.77 \pm 0.08 \%$$

WF

$$\mu = 444.1 \pm 0.3$$

$$\sigma = 45.1 \pm 0.3$$

$$\frac{\sigma}{\mu} = 10.16 \pm 0.08 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1040 < p < 1120$

Count

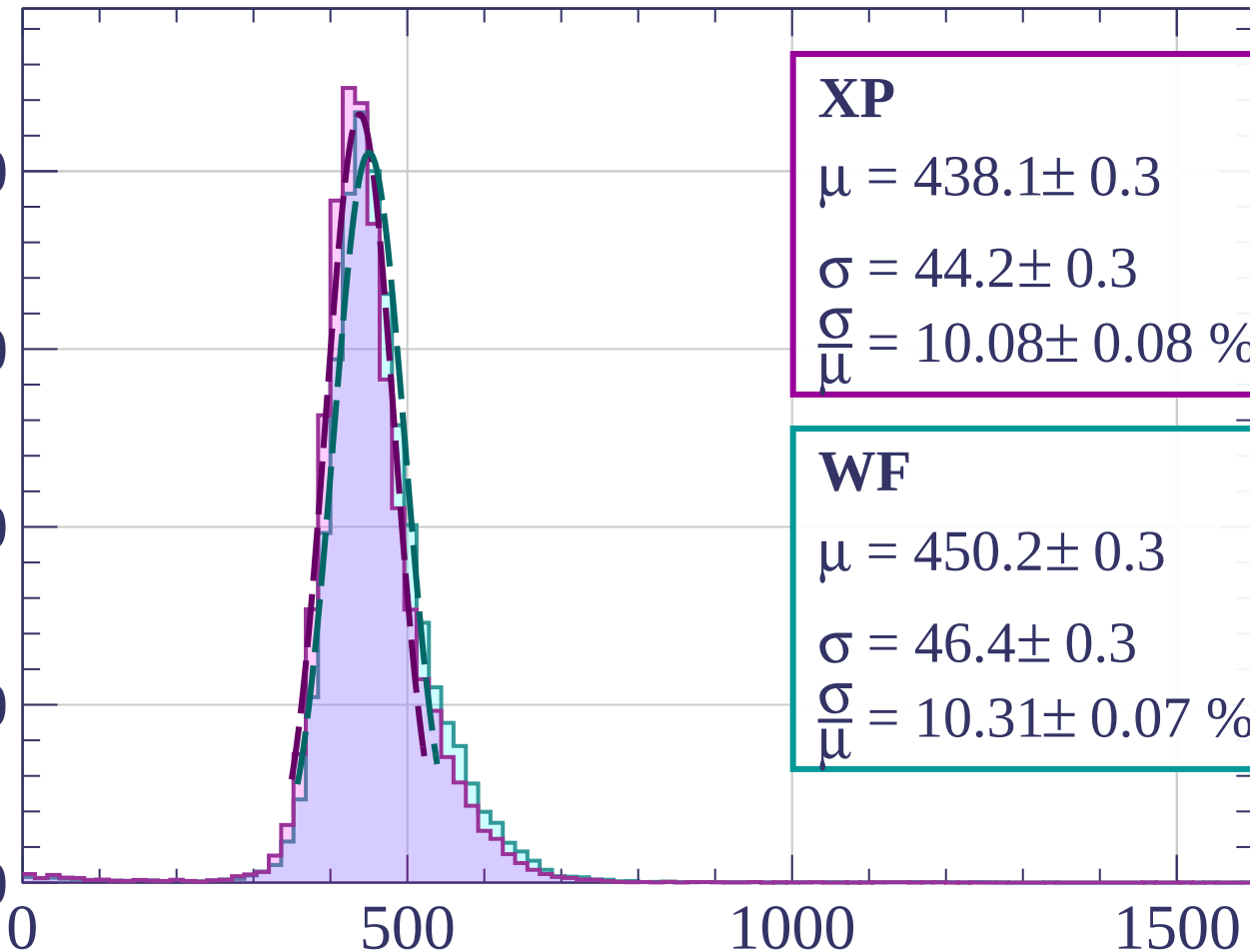
4000

3000

2000

1000

0



XP

$$\mu = 438.1 \pm 0.3$$

$$\sigma = 44.2 \pm 0.3$$

$$\frac{\sigma}{\mu} = 10.08 \pm 0.08 \%$$

WF

$$\mu = 450.2 \pm 0.3$$

$$\sigma = 46.4 \pm 0.3$$

$$\frac{\sigma}{\mu} = 10.31 \pm 0.07 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1120 < p < 1200$

Count

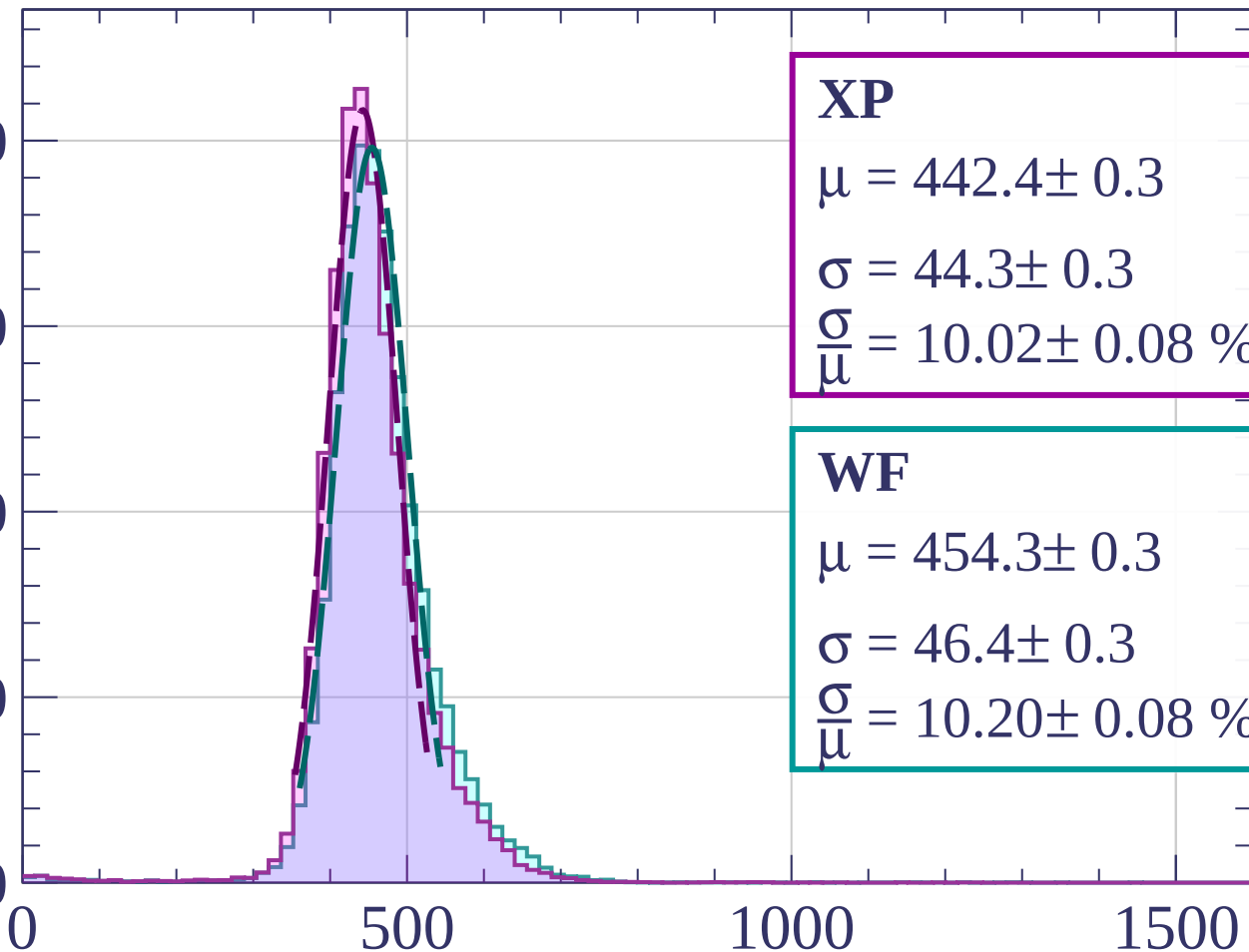
4000

3000

2000

1000

0



XP

$$\mu = 442.4 \pm 0.3$$

$$\sigma = 44.3 \pm 0.3$$

$$\frac{\sigma}{\mu} = 10.02 \pm 0.08 \%$$

WF

$$\mu = 454.3 \pm 0.3$$

$$\sigma = 46.4 \pm 0.3$$

$$\frac{\sigma}{\mu} = 10.20 \pm 0.08 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1200 < p < 1280$

Count

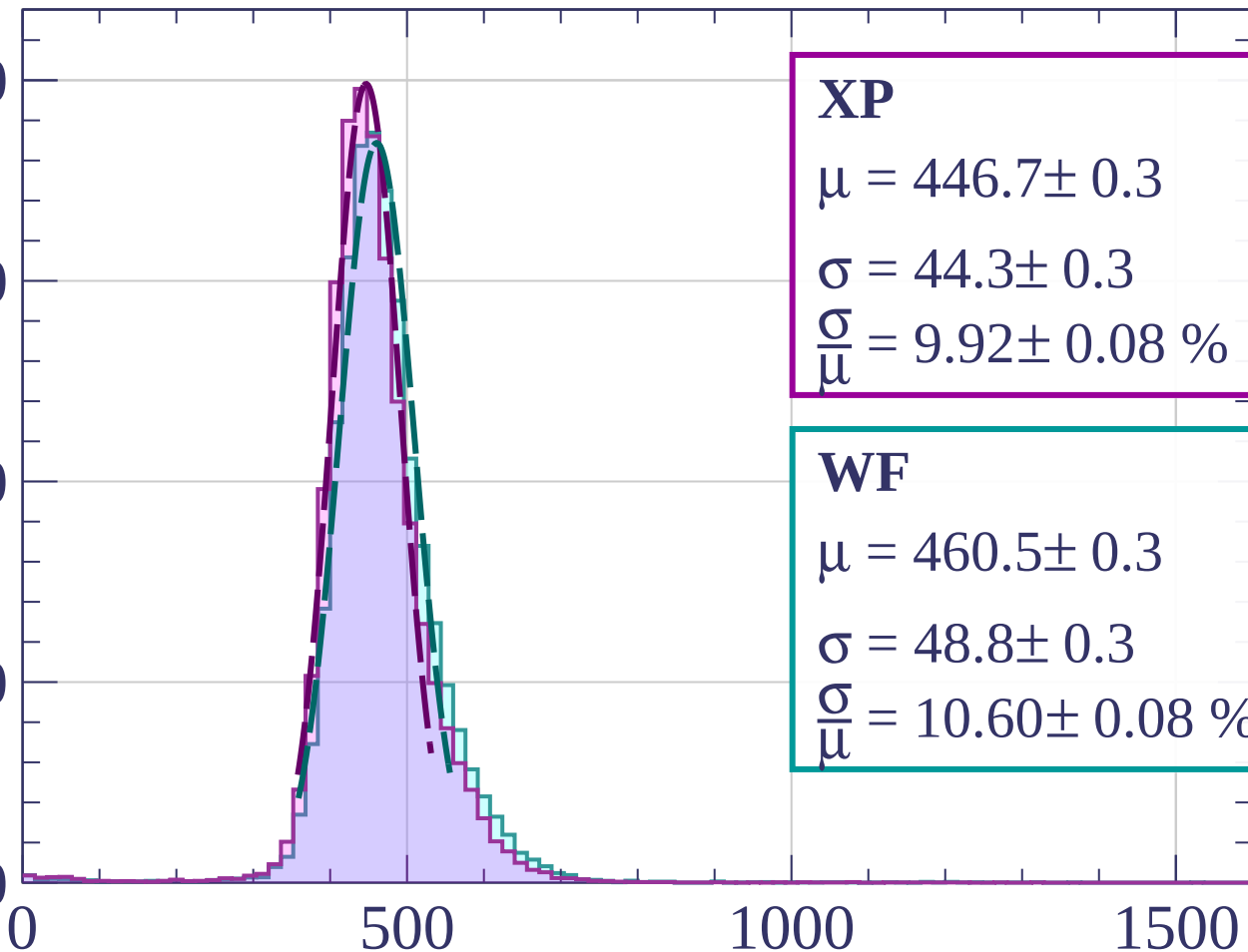
4000

3000

2000

1000

0



XP

$$\mu = 446.7 \pm 0.3$$

$$\sigma = 44.3 \pm 0.3$$

$$\frac{\sigma}{\mu} = 9.92 \pm 0.08 \%$$

WF

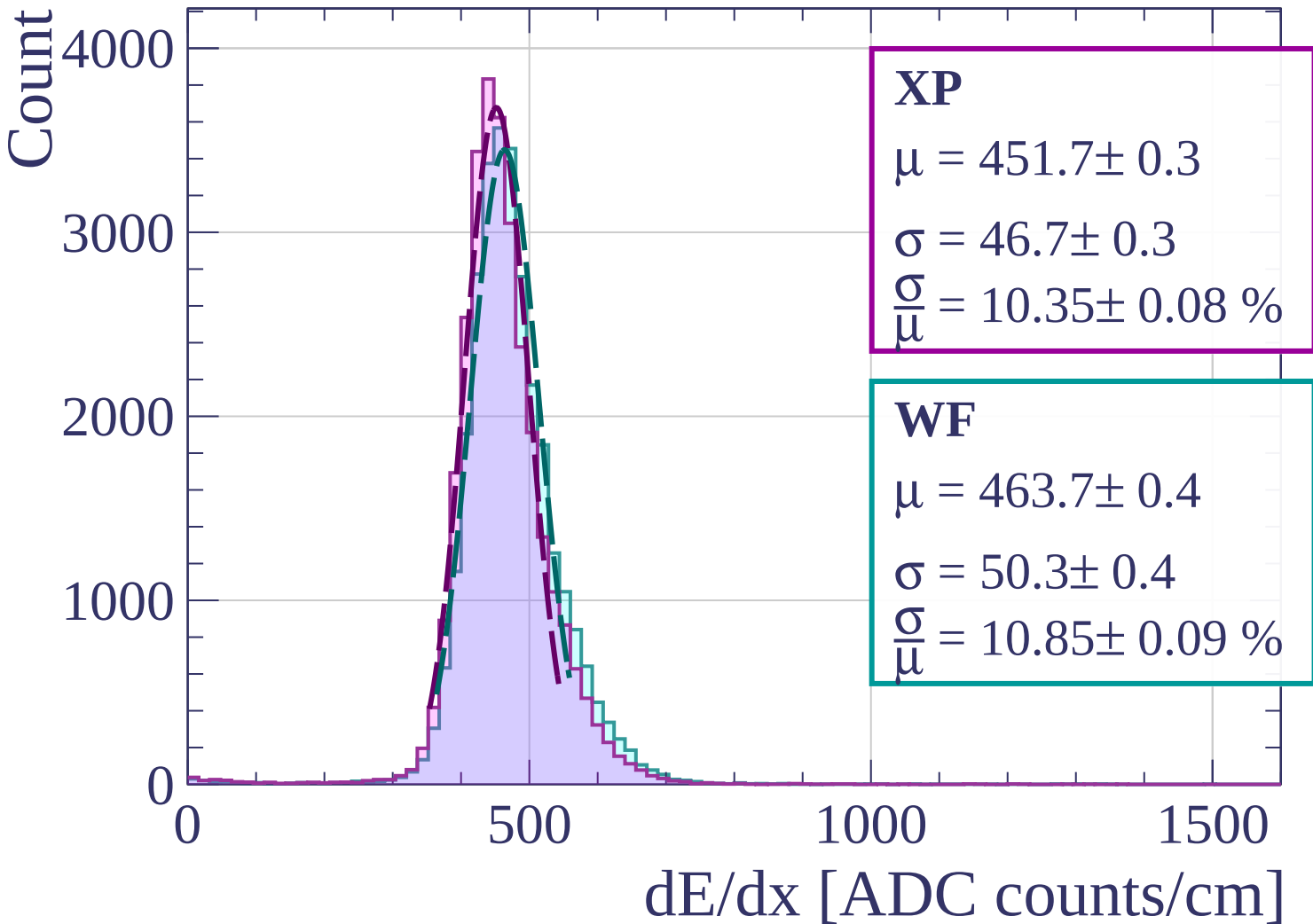
$$\mu = 460.5 \pm 0.3$$

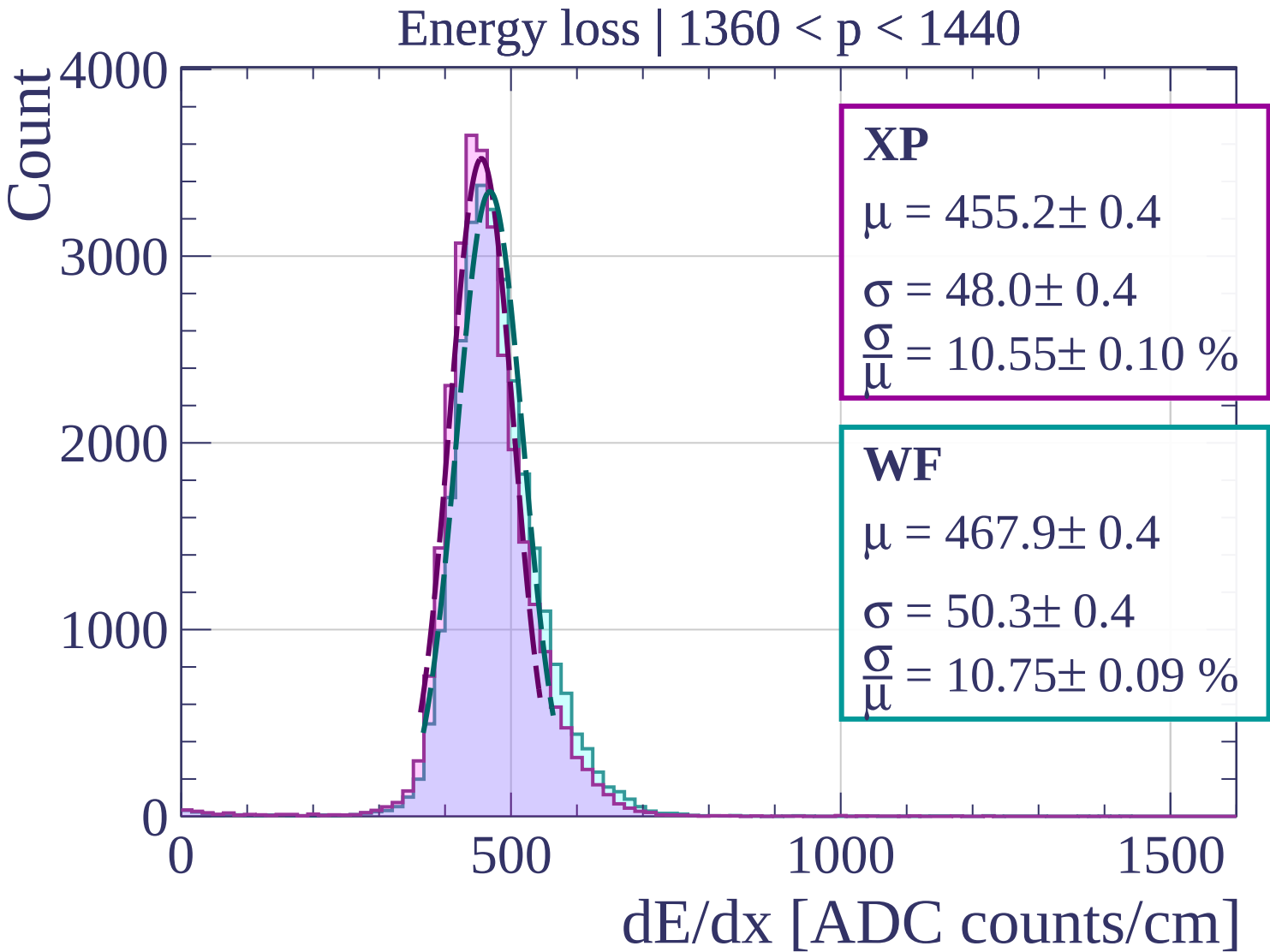
$$\sigma = 48.8 \pm 0.3$$

$$\frac{\sigma}{\mu} = 10.60 \pm 0.08 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1280 < p < 1360$





Energy loss | $1440 < p < 1520$

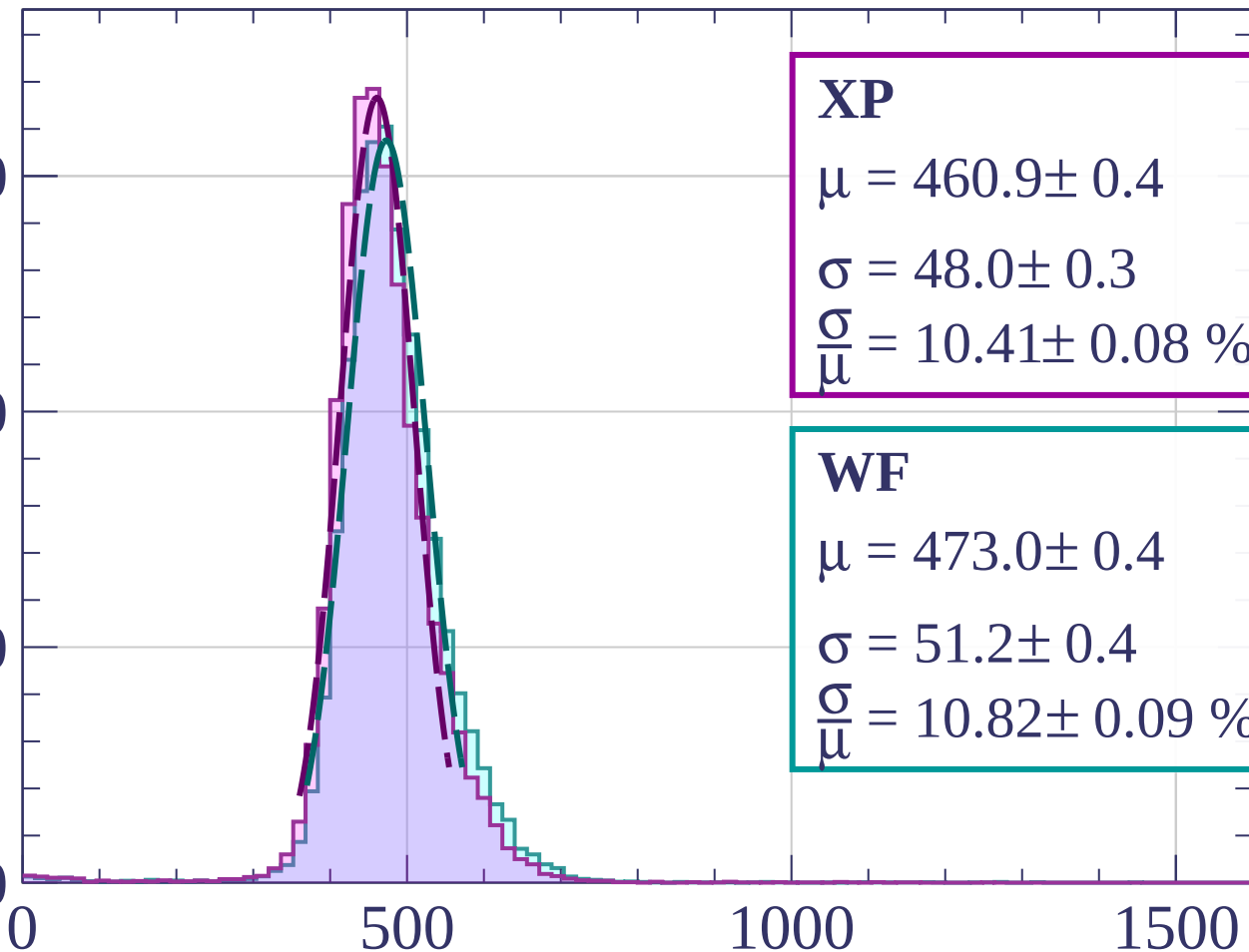
Count

3000

2000

1000

0



XP

$$\mu = 460.9 \pm 0.4$$

$$\sigma = 48.0 \pm 0.3$$

$$\frac{\sigma}{\mu} = 10.41 \pm 0.08 \%$$

WF

$$\mu = 473.0 \pm 0.4$$

$$\sigma = 51.2 \pm 0.4$$

$$\frac{\sigma}{\mu} = 10.82 \pm 0.09 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1520 < p < 1600$

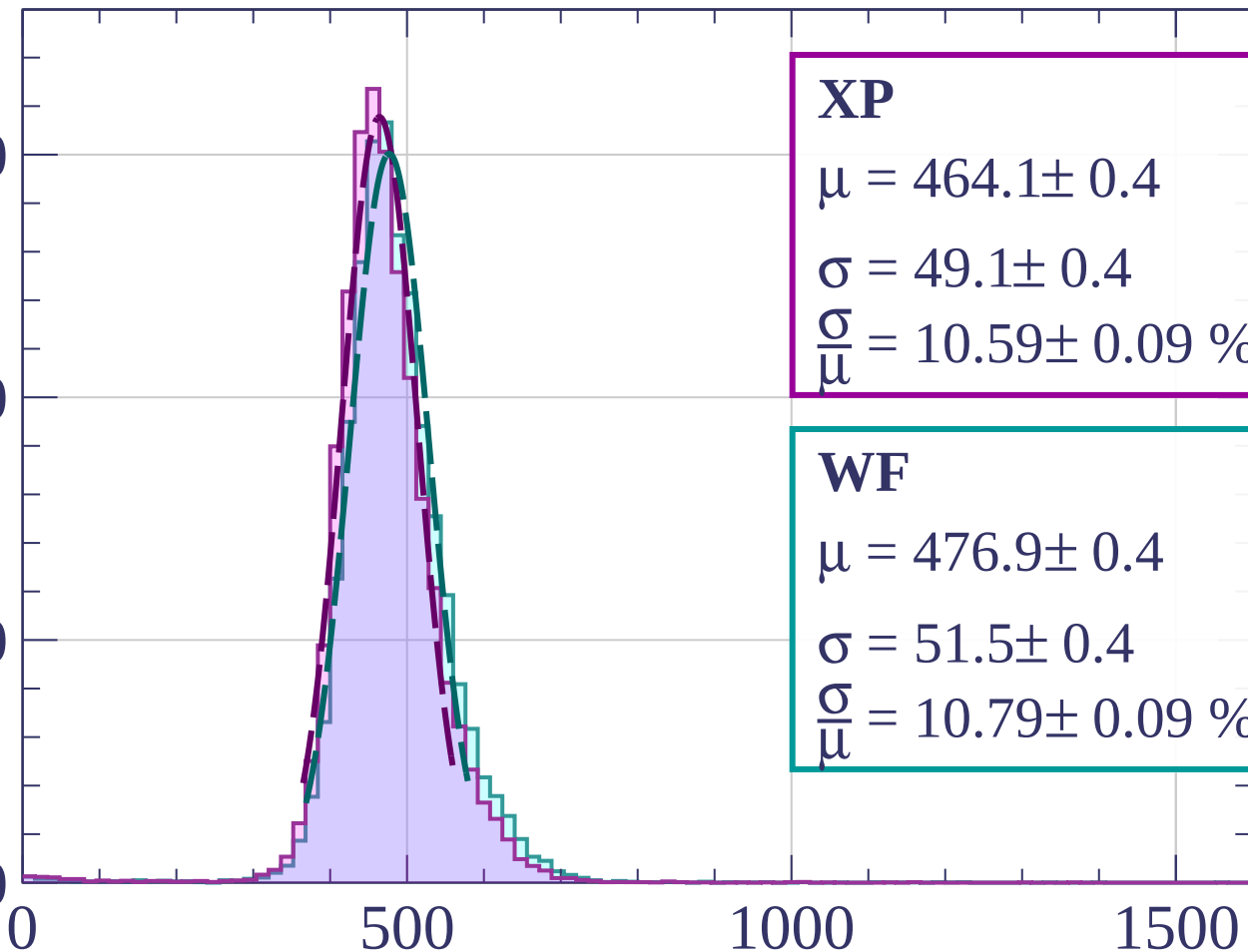
Count

3000

2000

1000

0



XP

$$\mu = 464.1 \pm 0.4$$

$$\sigma = 49.1 \pm 0.4$$

$$\frac{\sigma}{\mu} = 10.59 \pm 0.09 \%$$

WF

$$\mu = 476.9 \pm 0.4$$

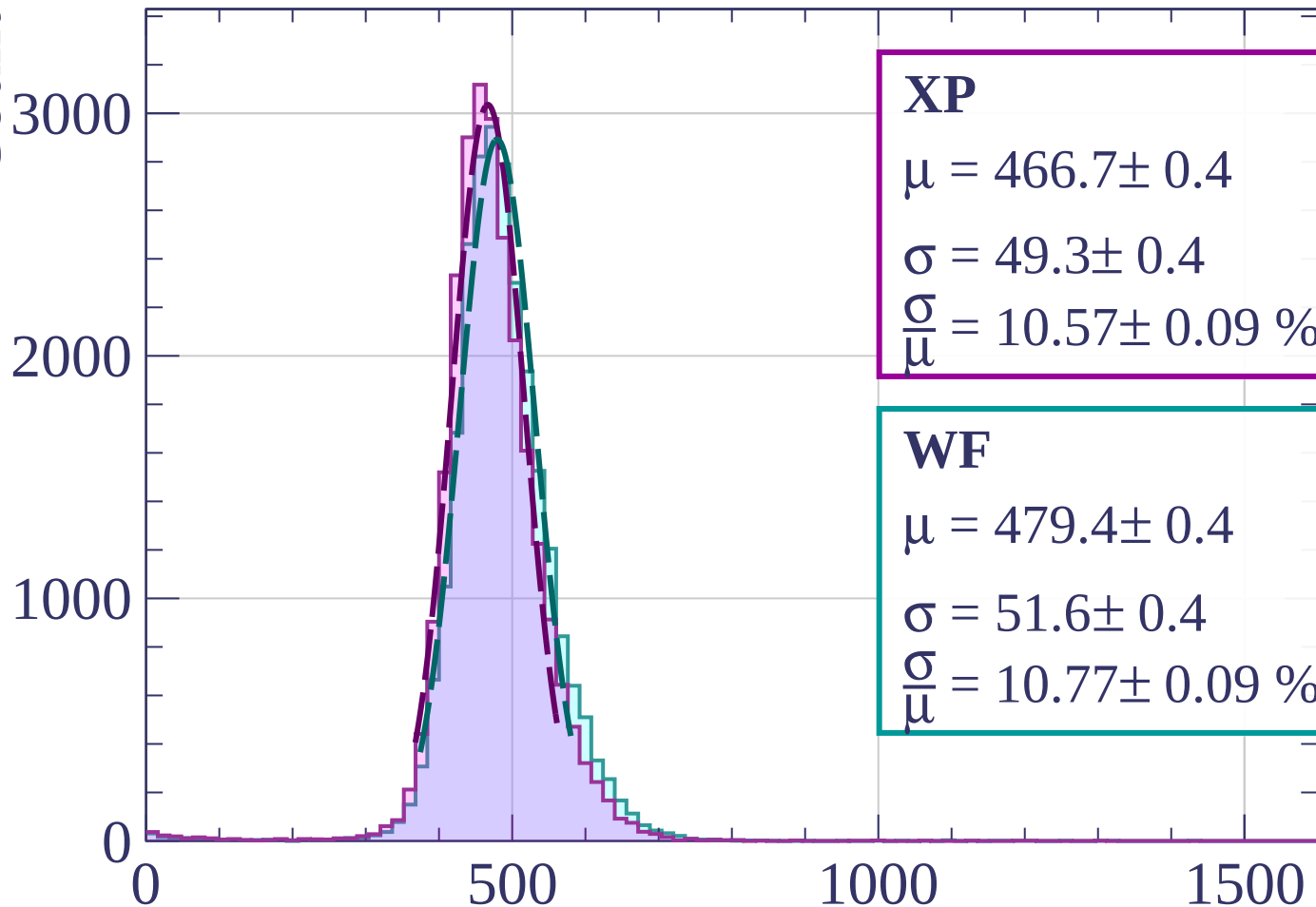
$$\sigma = 51.5 \pm 0.4$$

$$\frac{\sigma}{\mu} = 10.79 \pm 0.09 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1600 < p < 1680$

Count



XP

$$\mu = 466.7 \pm 0.4$$

$$\sigma = 49.3 \pm 0.4$$

$$\frac{\sigma}{\mu} = 10.57 \pm 0.09 \%$$

WF

$$\mu = 479.4 \pm 0.4$$

$$\sigma = 51.6 \pm 0.4$$

$$\frac{\sigma}{\mu} = 10.77 \pm 0.09 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1680 < p < 1760$

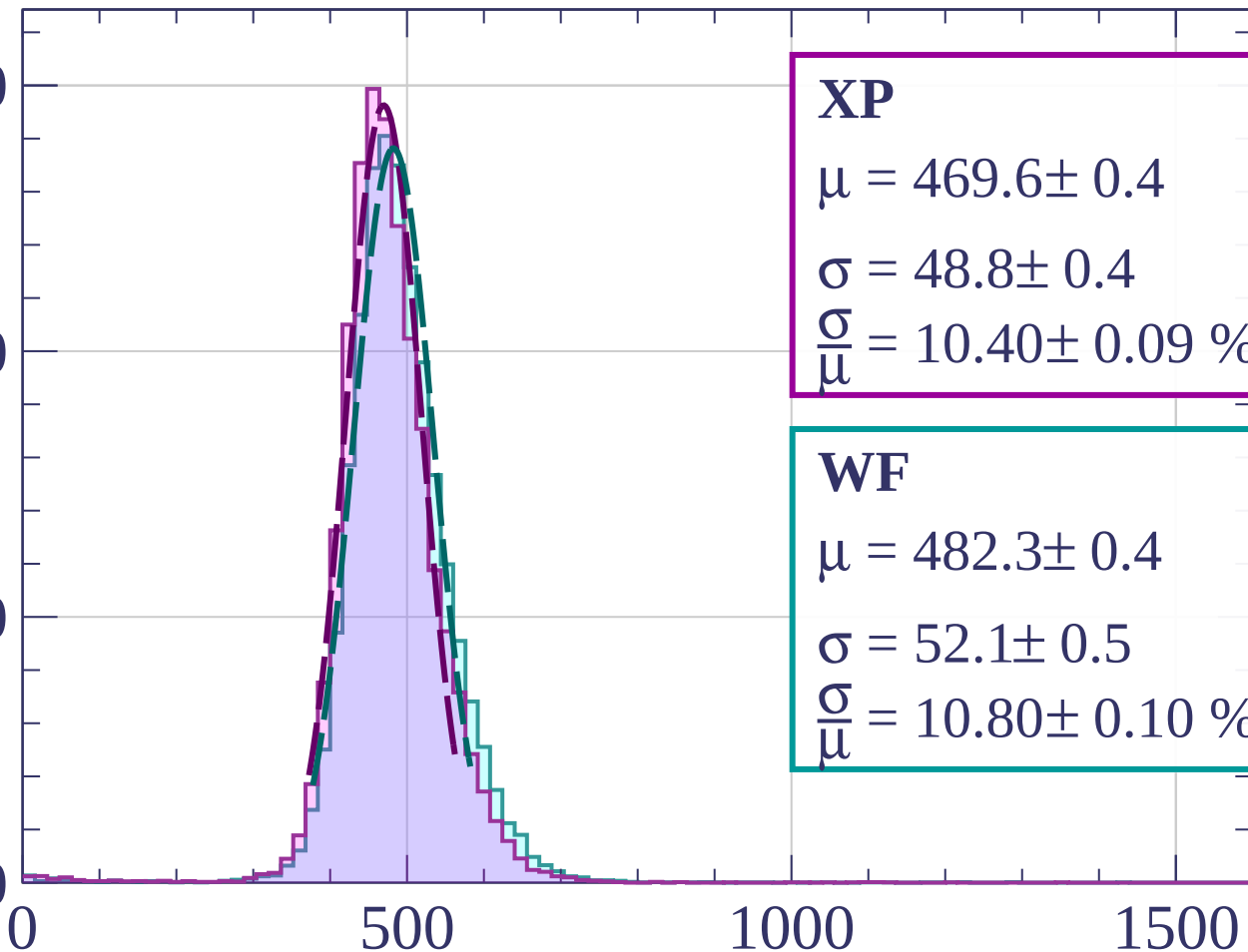
Count

3000

2000

1000

0



XP

$$\mu = 469.6 \pm 0.4$$

$$\sigma = 48.8 \pm 0.4$$

$$\frac{\sigma}{\mu} = 10.40 \pm 0.09 \%$$

WF

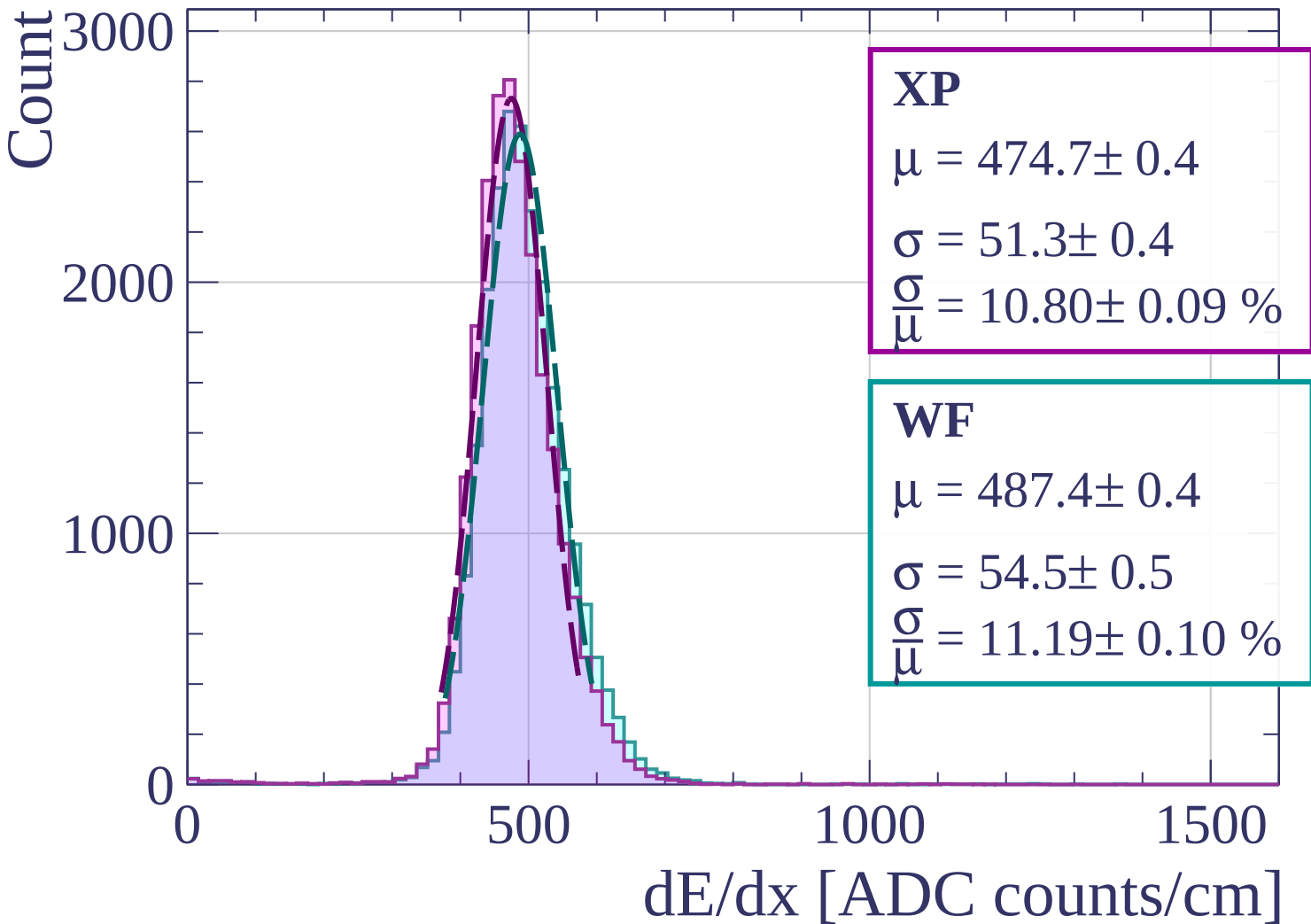
$$\mu = 482.3 \pm 0.4$$

$$\sigma = 52.1 \pm 0.5$$

$$\frac{\sigma}{\mu} = 10.80 \pm 0.10 \%$$

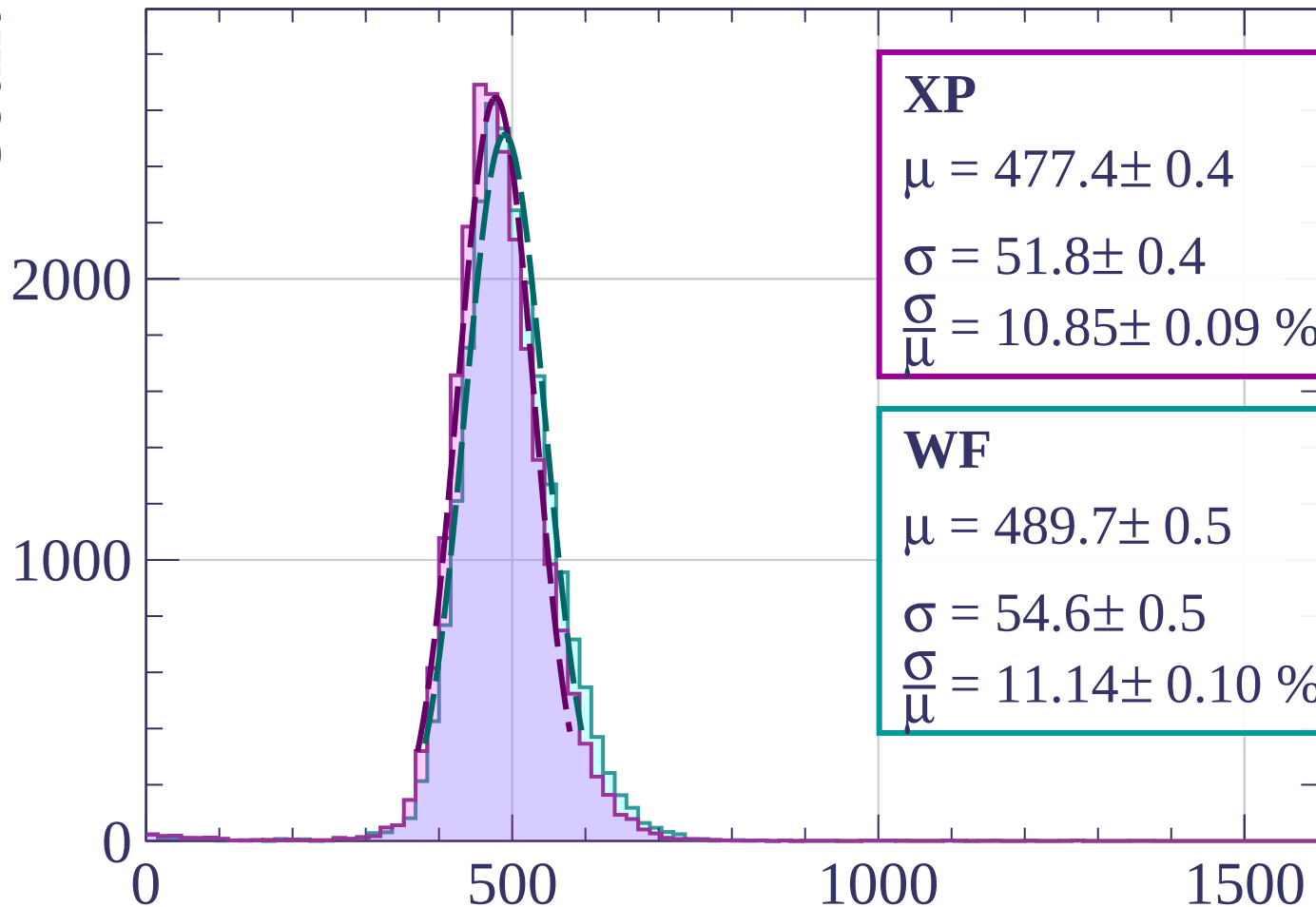
dE/dx [ADC counts/cm]

Energy loss | $1760 < p < 1840$



Energy loss | $1840 < p < 1920$

Count



XP

$$\mu = 477.4 \pm 0.4$$

$$\sigma = 51.8 \pm 0.4$$

$$\frac{\sigma}{\mu} = 10.85 \pm 0.09 \%$$

WF

$$\mu = 489.7 \pm 0.5$$

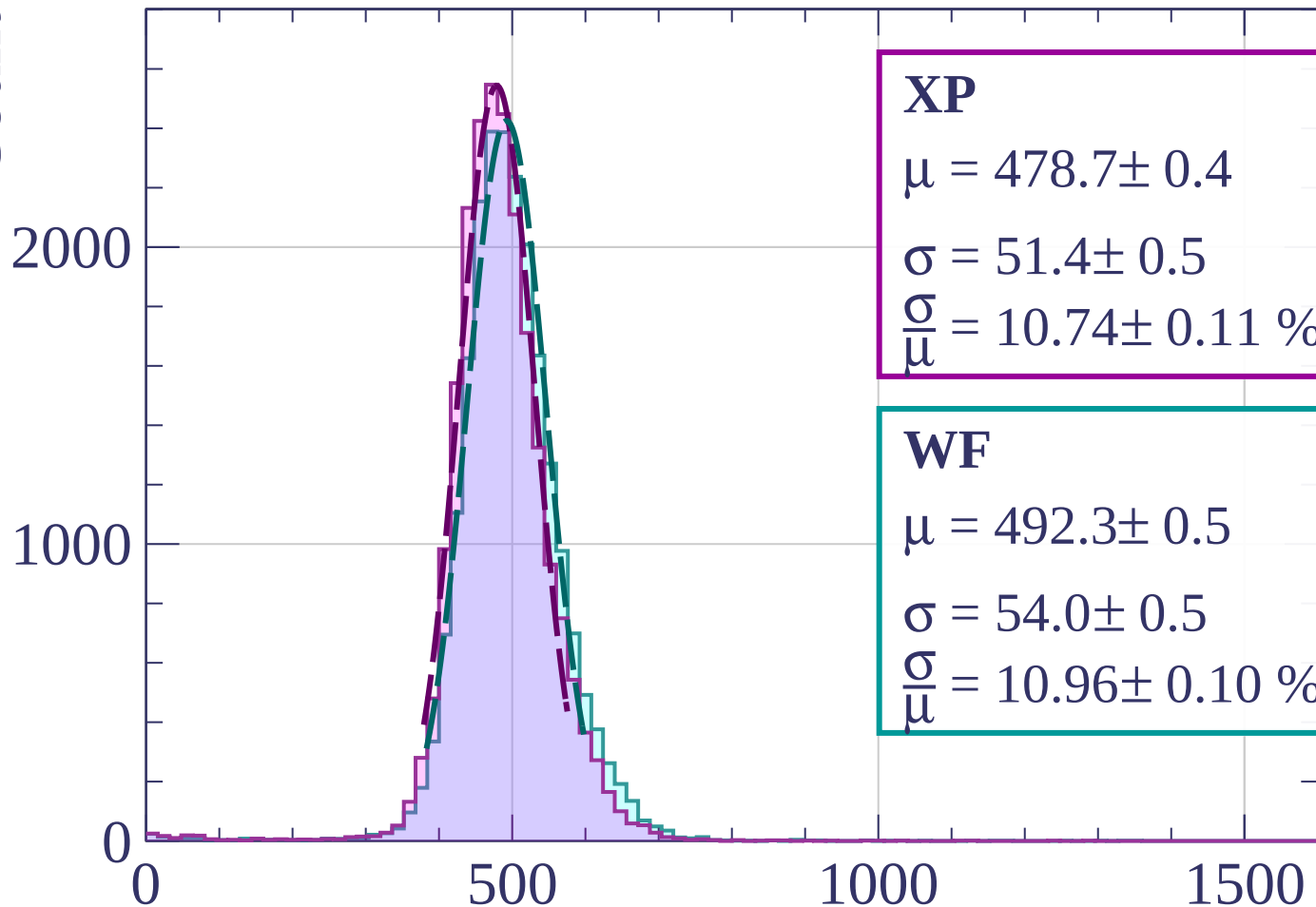
$$\sigma = 54.6 \pm 0.5$$

$$\frac{\sigma}{\mu} = 11.14 \pm 0.10 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1920 < p < 2000$

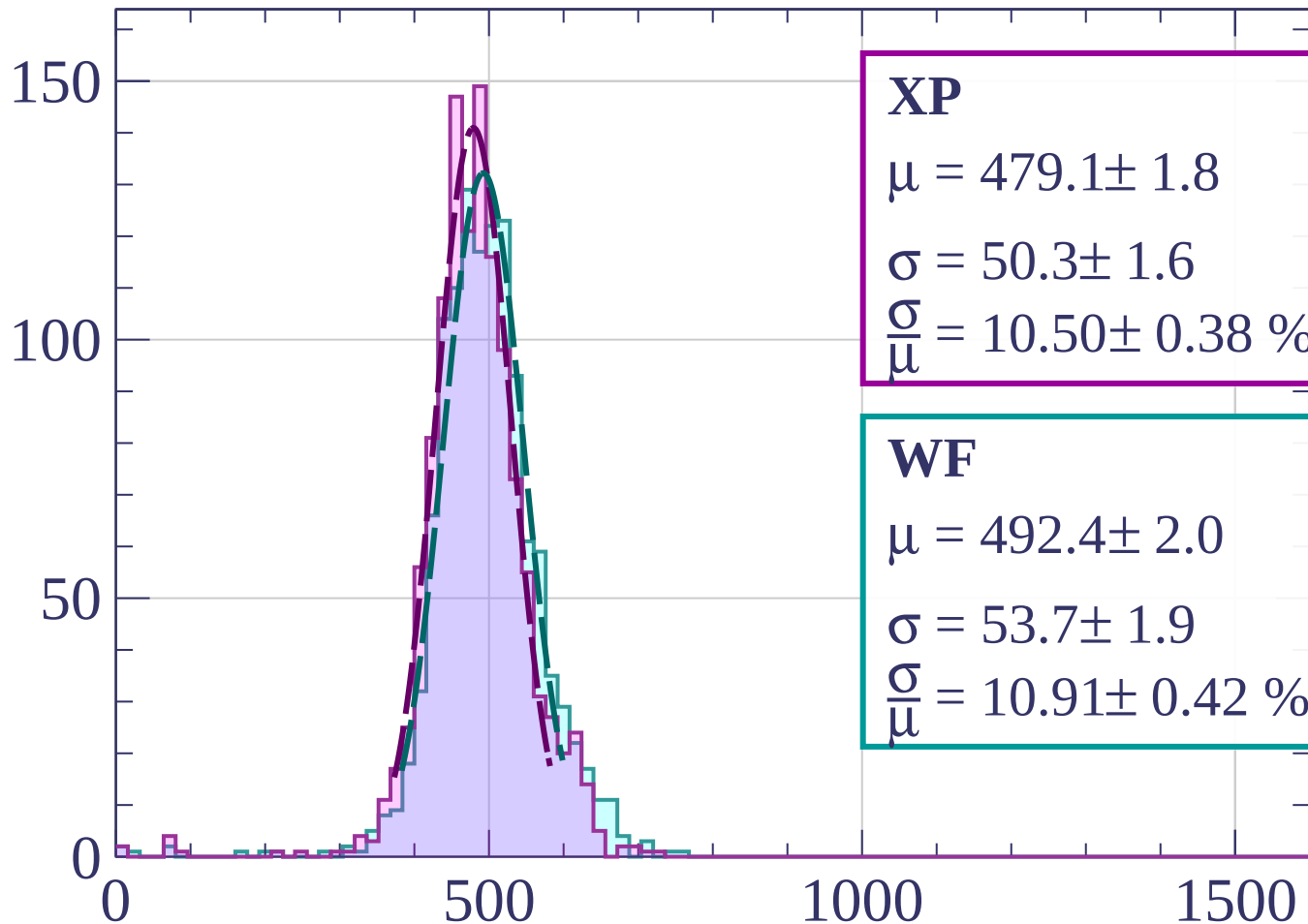
Count



dE/dx [ADC counts/cm]

Energy loss | $-2000 < p < -1980$

Count



XP

$$\mu = 479.1 \pm 1.8$$

$$\sigma = 50.3 \pm 1.6$$

$$\frac{\sigma}{\mu} = 10.50 \pm 0.38 \%$$

WF

$$\mu = 492.4 \pm 2.0$$

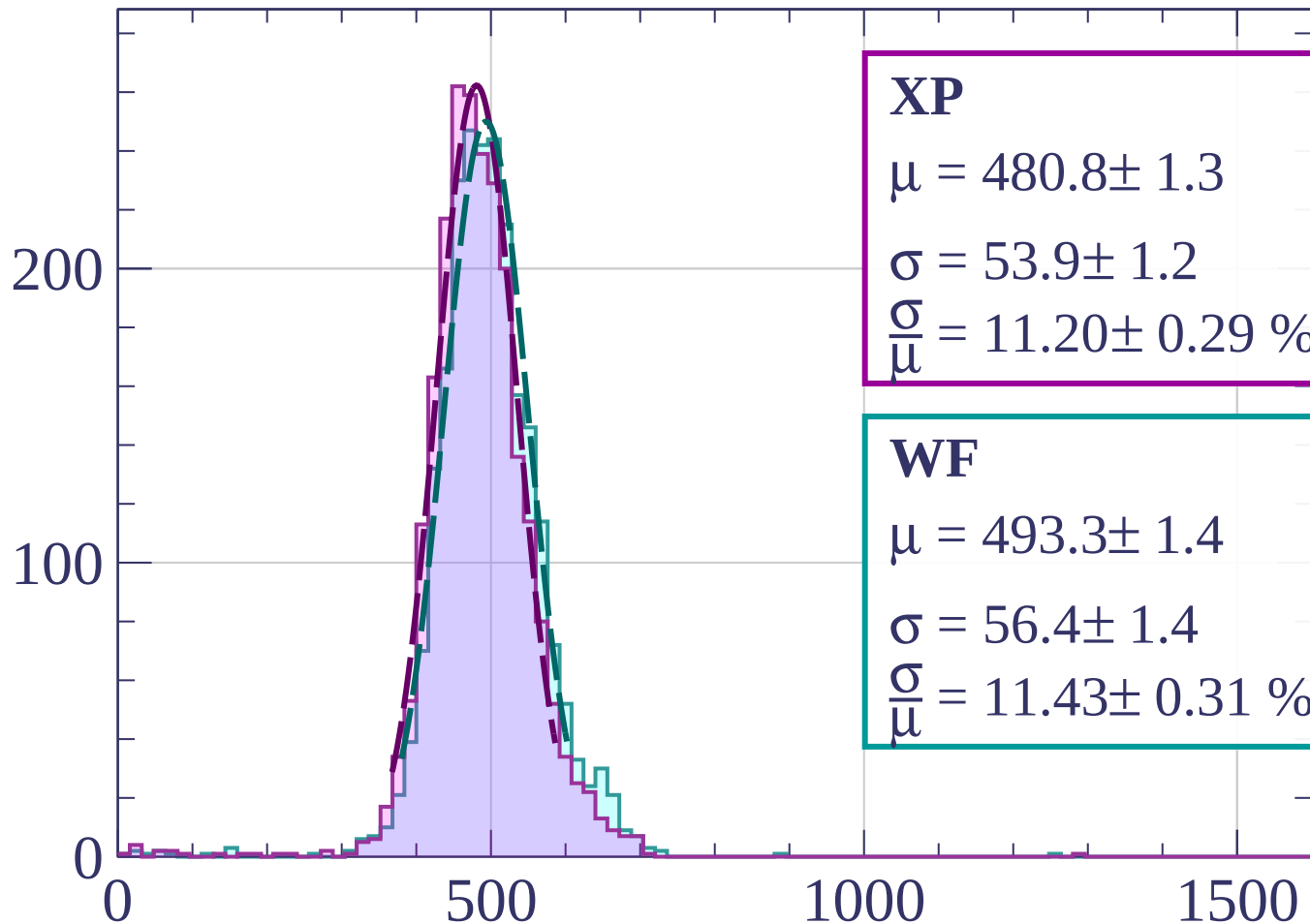
$$\sigma = 53.7 \pm 1.9$$

$$\frac{\sigma}{\mu} = 10.91 \pm 0.42 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1980 < p < -1960$

Count



XP

$$\mu = 480.8 \pm 1.3$$

$$\sigma = 53.9 \pm 1.2$$

$$\frac{\sigma}{\mu} = 11.20 \pm 0.29 \%$$

WF

$$\mu = 493.3 \pm 1.4$$

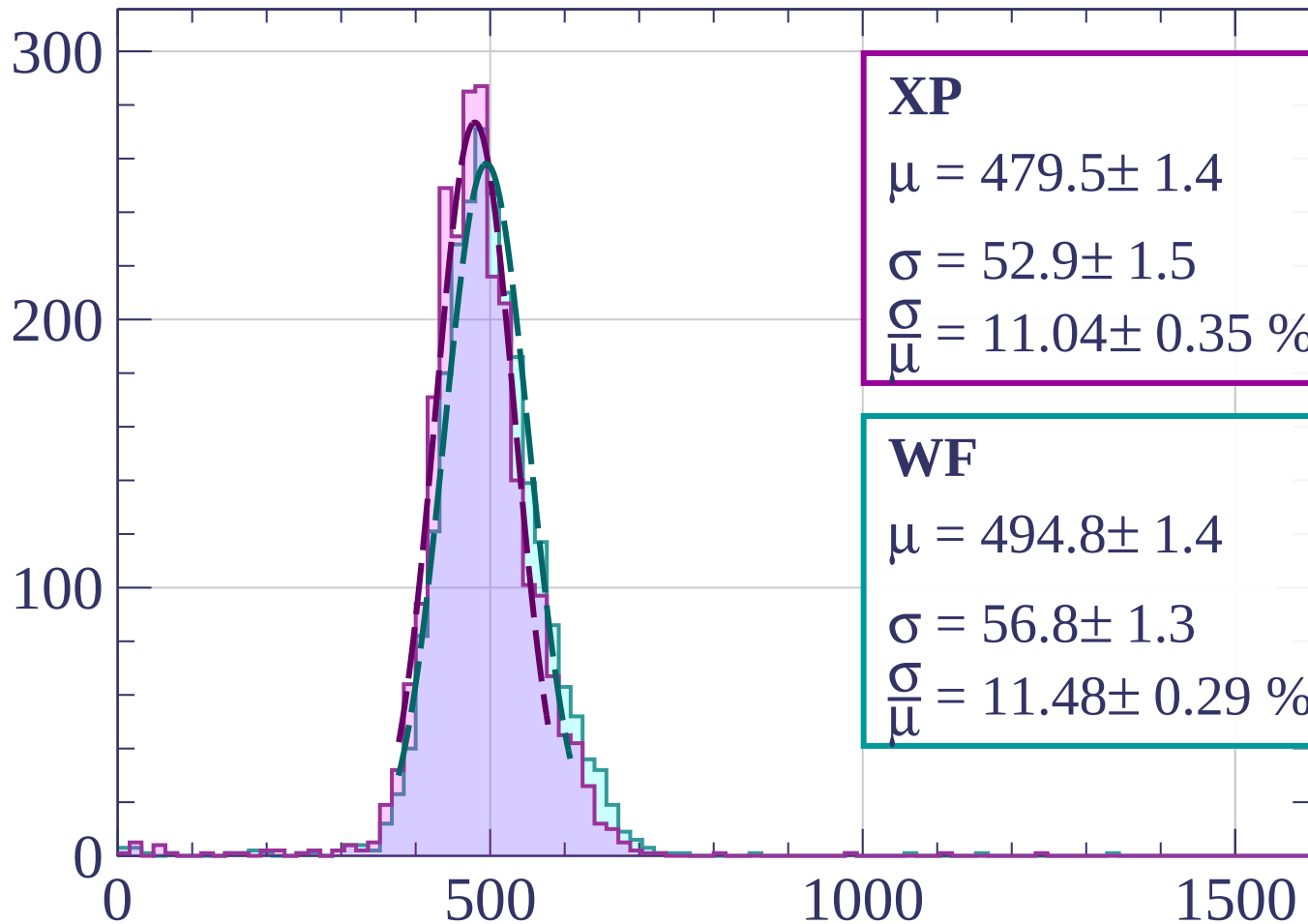
$$\sigma = 56.4 \pm 1.4$$

$$\frac{\sigma}{\mu} = 11.43 \pm 0.31 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1960 < p < -1940$

Count



XP

$$\mu = 479.5 \pm 1.4$$

$$\sigma = 52.9 \pm 1.5$$

$$\frac{\sigma}{\mu} = 11.04 \pm 0.35 \%$$

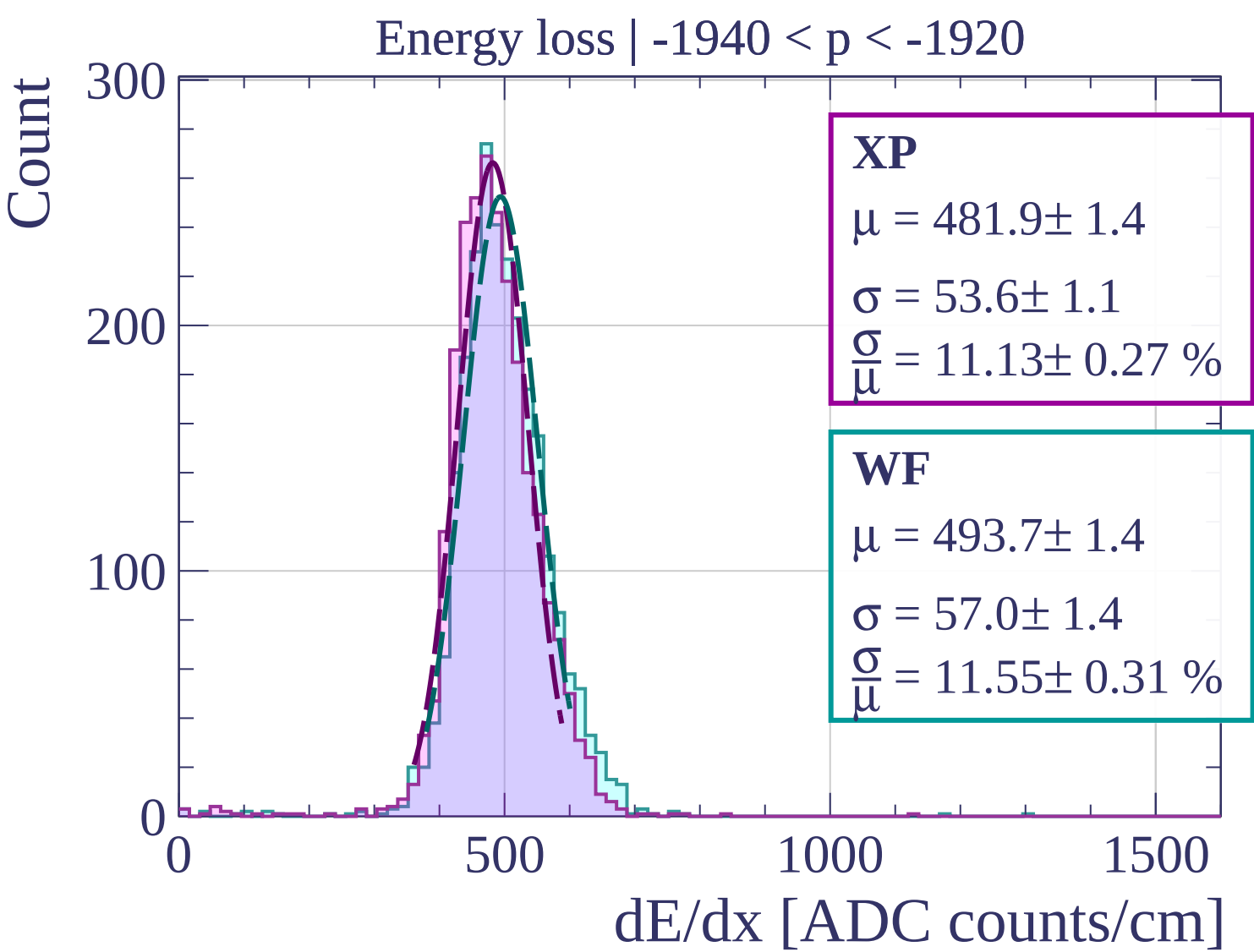
WF

$$\mu = 494.8 \pm 1.4$$

$$\sigma = 56.8 \pm 1.3$$

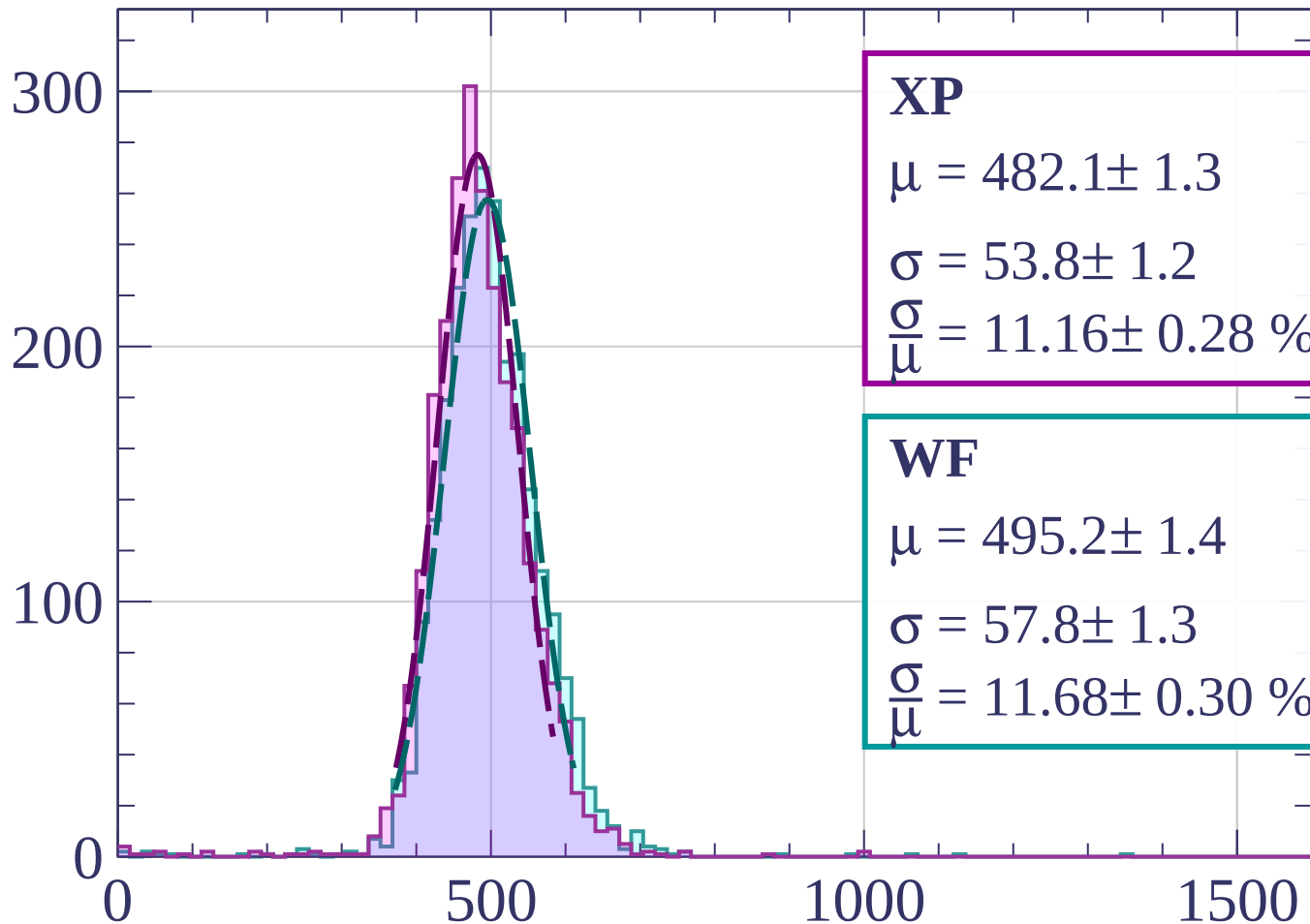
$$\frac{\sigma}{\mu} = 11.48 \pm 0.29 \%$$

dE/dx [ADC counts/cm]



Energy loss | $-1920 < p < -1900$

Count



XP

$$\mu = 482.1 \pm 1.3$$

$$\sigma = 53.8 \pm 1.2$$

$$\frac{\sigma}{\mu} = 11.16 \pm 0.28 \%$$

WF

$$\mu = 495.2 \pm 1.4$$

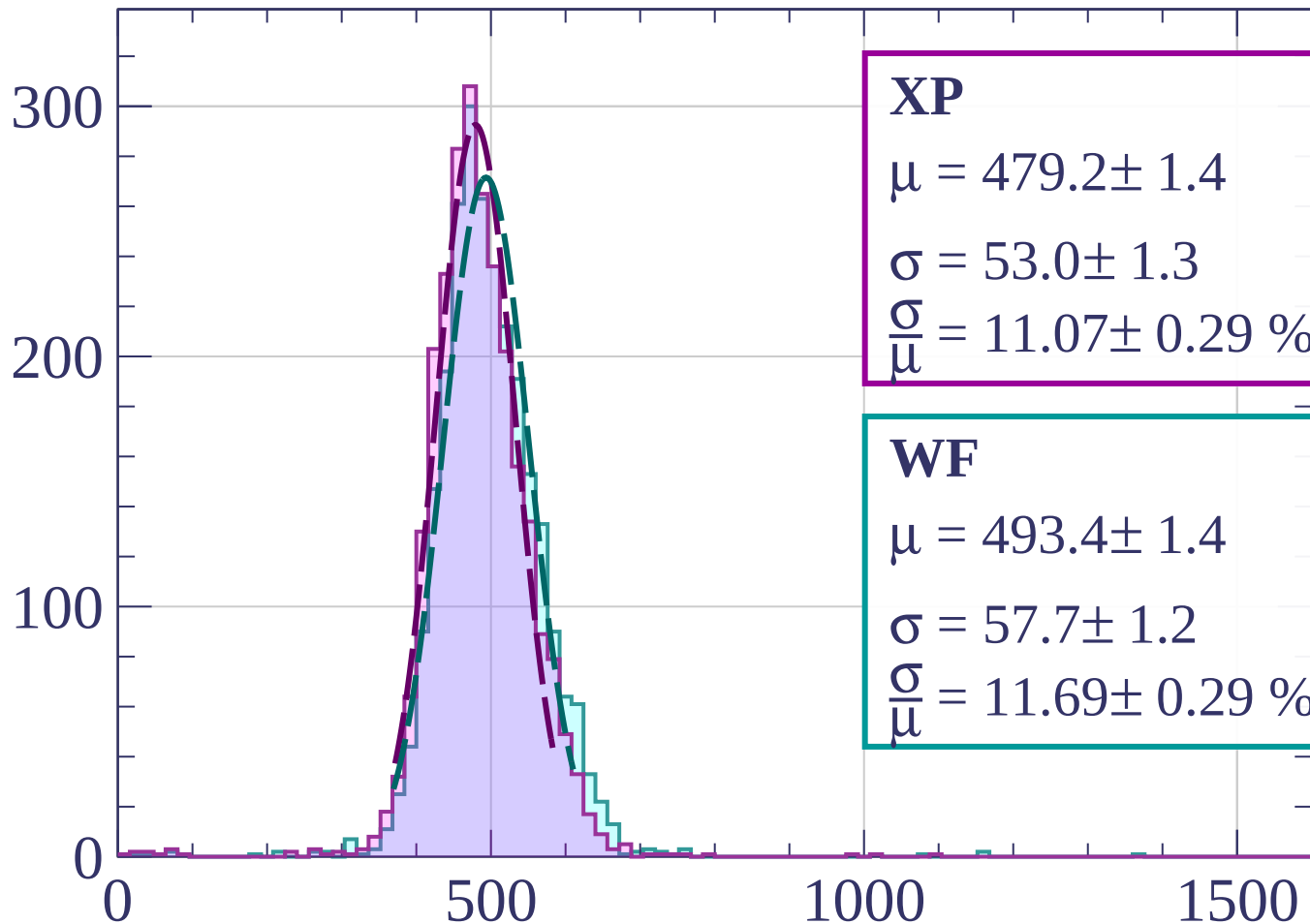
$$\sigma = 57.8 \pm 1.3$$

$$\frac{\sigma}{\mu} = 11.68 \pm 0.30 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1900 < p < -1880$

Count



XP

$$\mu = 479.2 \pm 1.4$$

$$\sigma = 53.0 \pm 1.3$$

$$\frac{\sigma}{\mu} = 11.07 \pm 0.29 \%$$

WF

$$\mu = 493.4 \pm 1.4$$

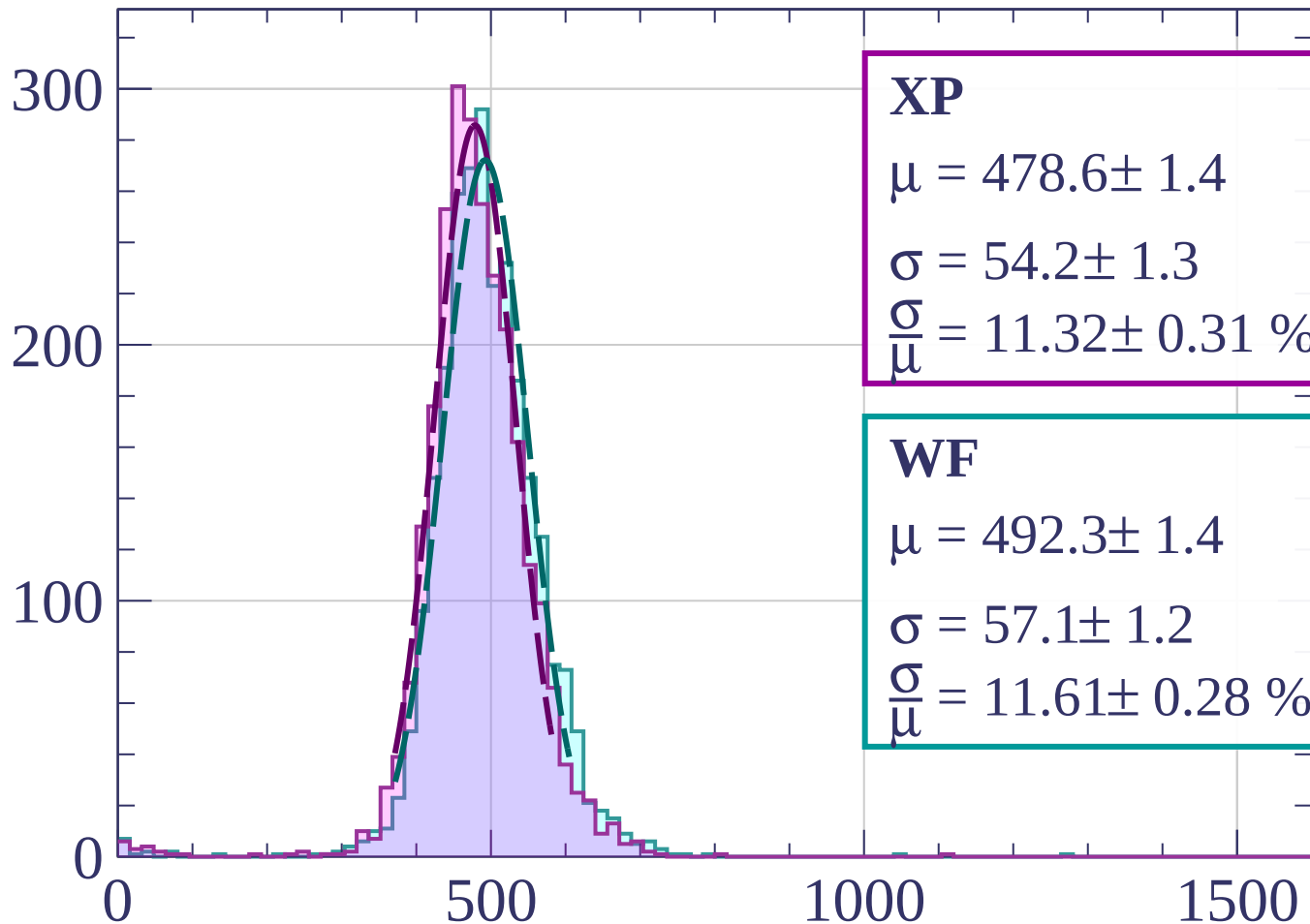
$$\sigma = 57.7 \pm 1.2$$

$$\frac{\sigma}{\mu} = 11.69 \pm 0.29 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1880 < p < -1860$

Count



XP

$$\mu = 478.6 \pm 1.4$$

$$\sigma = 54.2 \pm 1.3$$

$$\frac{\sigma}{\mu} = 11.32 \pm 0.31 \%$$

WF

$$\mu = 492.3 \pm 1.4$$

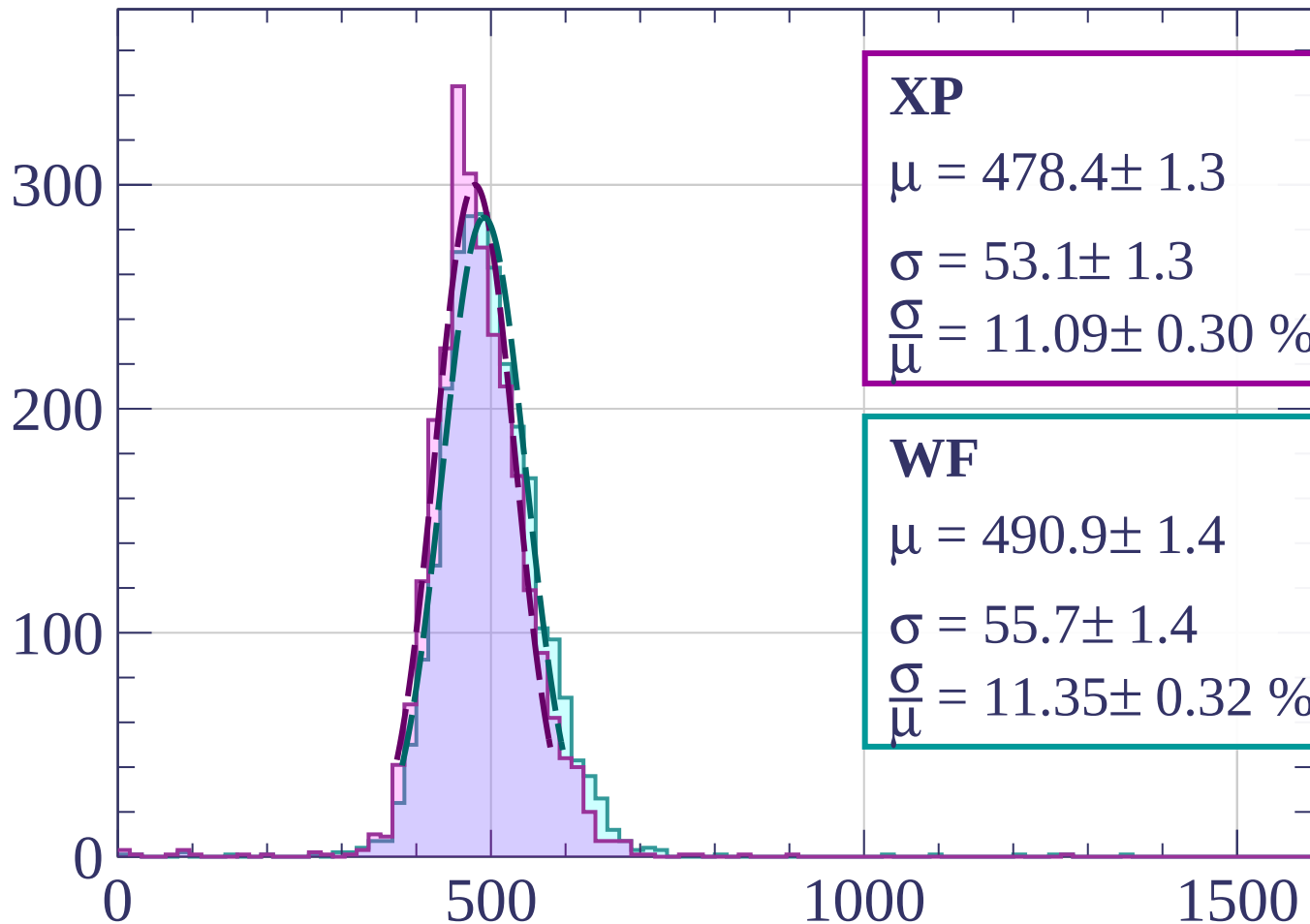
$$\sigma = 57.1 \pm 1.2$$

$$\frac{\sigma}{\mu} = 11.61 \pm 0.28 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1860 < p < -1840$

Count



XP

$$\mu = 478.4 \pm 1.3$$

$$\sigma = 53.1 \pm 1.3$$

$$\frac{\sigma}{\mu} = 11.09 \pm 0.30 \%$$

WF

$$\mu = 490.9 \pm 1.4$$

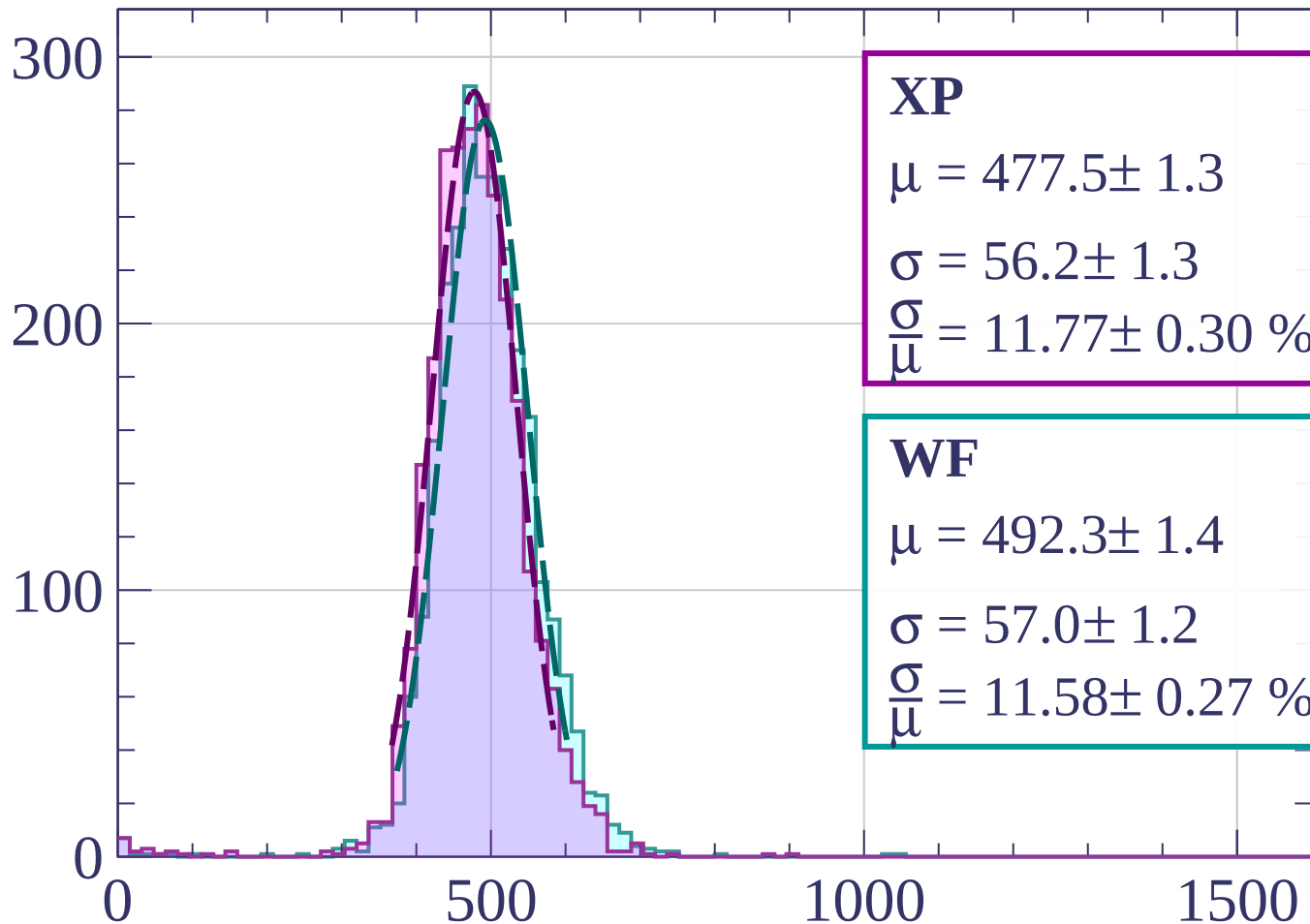
$$\sigma = 55.7 \pm 1.4$$

$$\frac{\sigma}{\mu} = 11.35 \pm 0.32 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1840 < p < -1820$

Count



XP

$$\mu = 477.5 \pm 1.3$$

$$\sigma = 56.2 \pm 1.3$$

$$\frac{\sigma}{\mu} = 11.77 \pm 0.30 \%$$

WF

$$\mu = 492.3 \pm 1.4$$

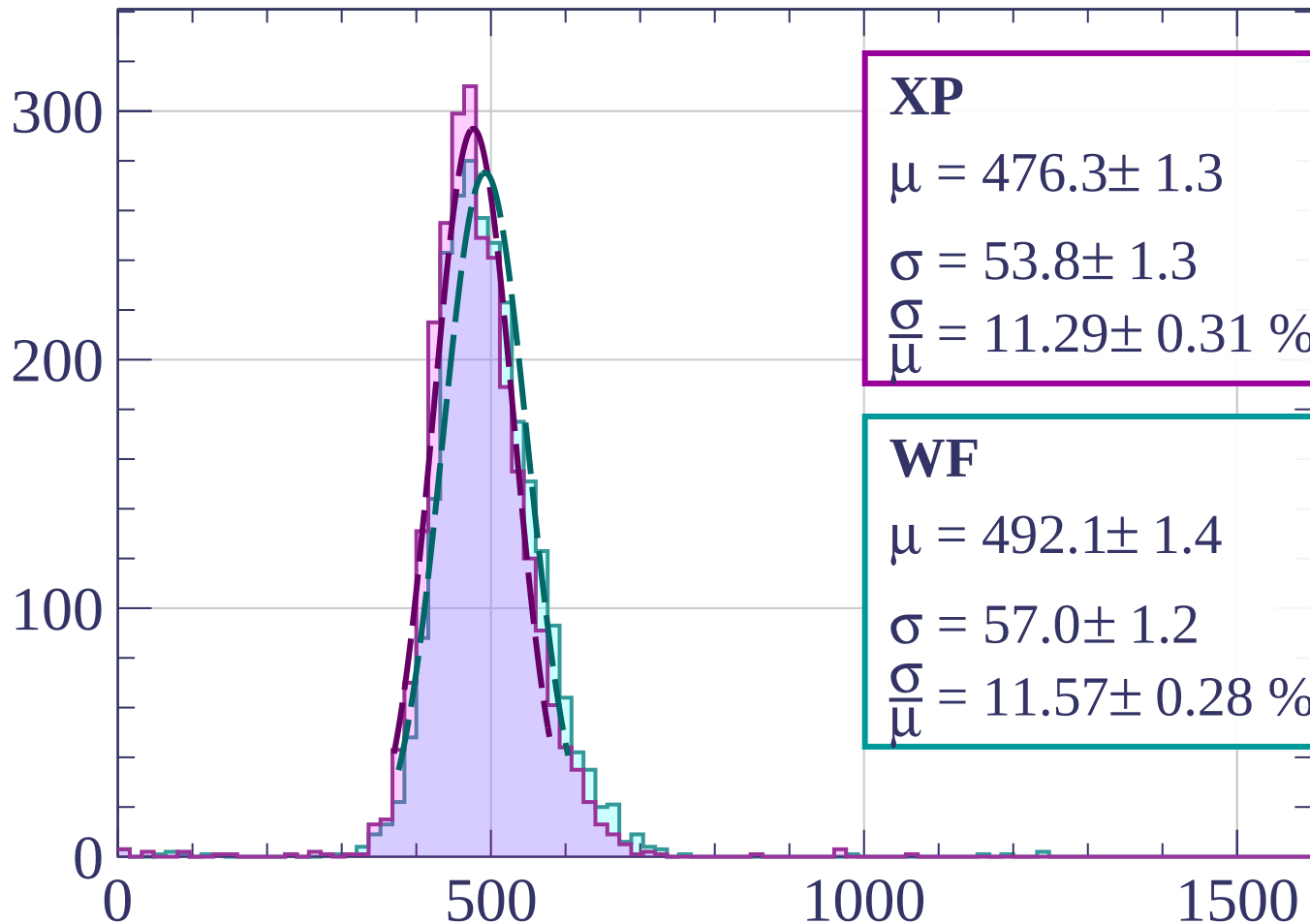
$$\sigma = 57.0 \pm 1.2$$

$$\frac{\sigma}{\mu} = 11.58 \pm 0.27 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1820 < p < -1800$

Count



XP

$$\mu = 476.3 \pm 1.3$$

$$\sigma = 53.8 \pm 1.3$$

$$\frac{\sigma}{\mu} = 11.29 \pm 0.31 \%$$

WF

$$\mu = 492.1 \pm 1.4$$

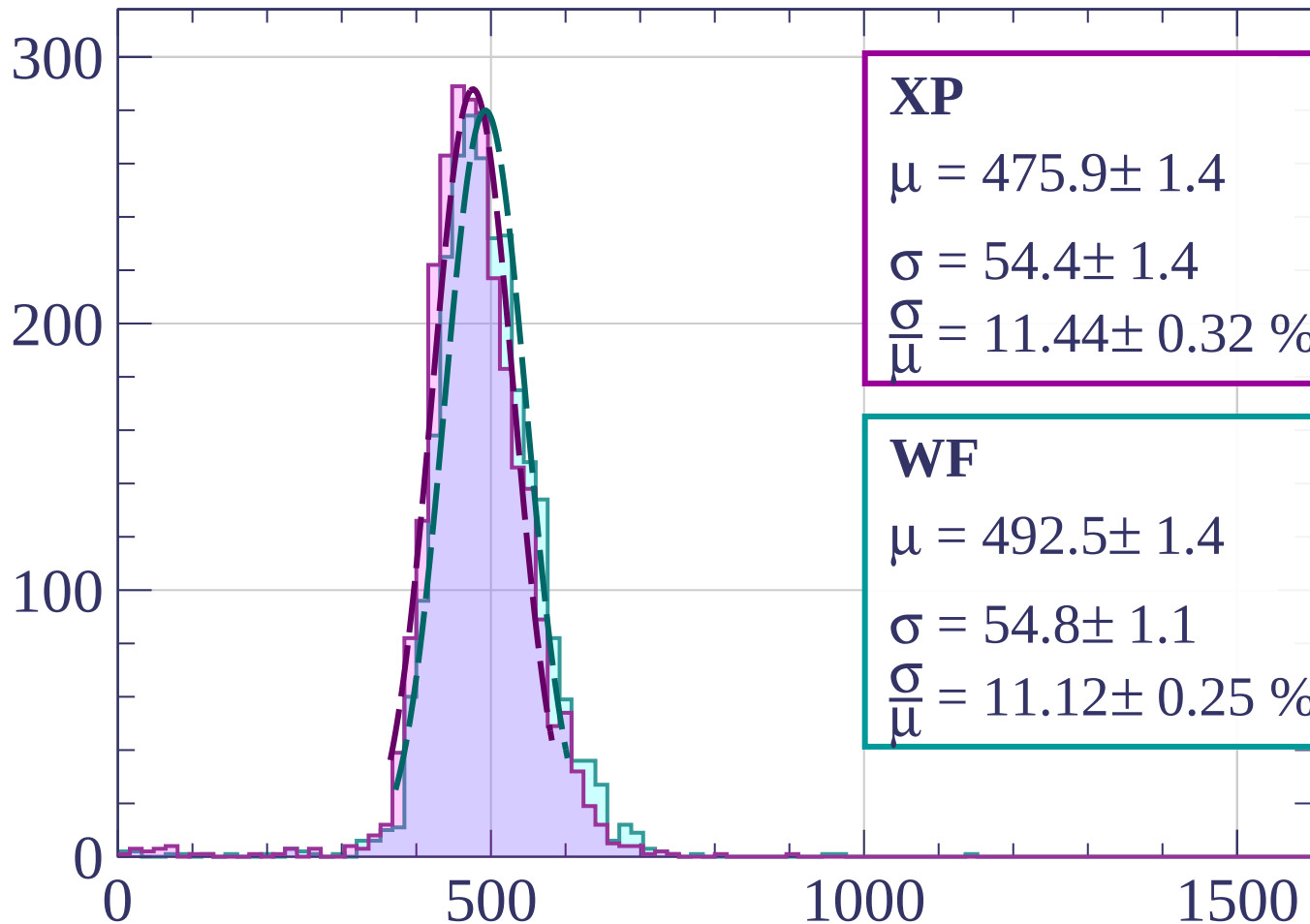
$$\sigma = 57.0 \pm 1.2$$

$$\frac{\sigma}{\mu} = 11.57 \pm 0.28 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1800 < p < -1780$

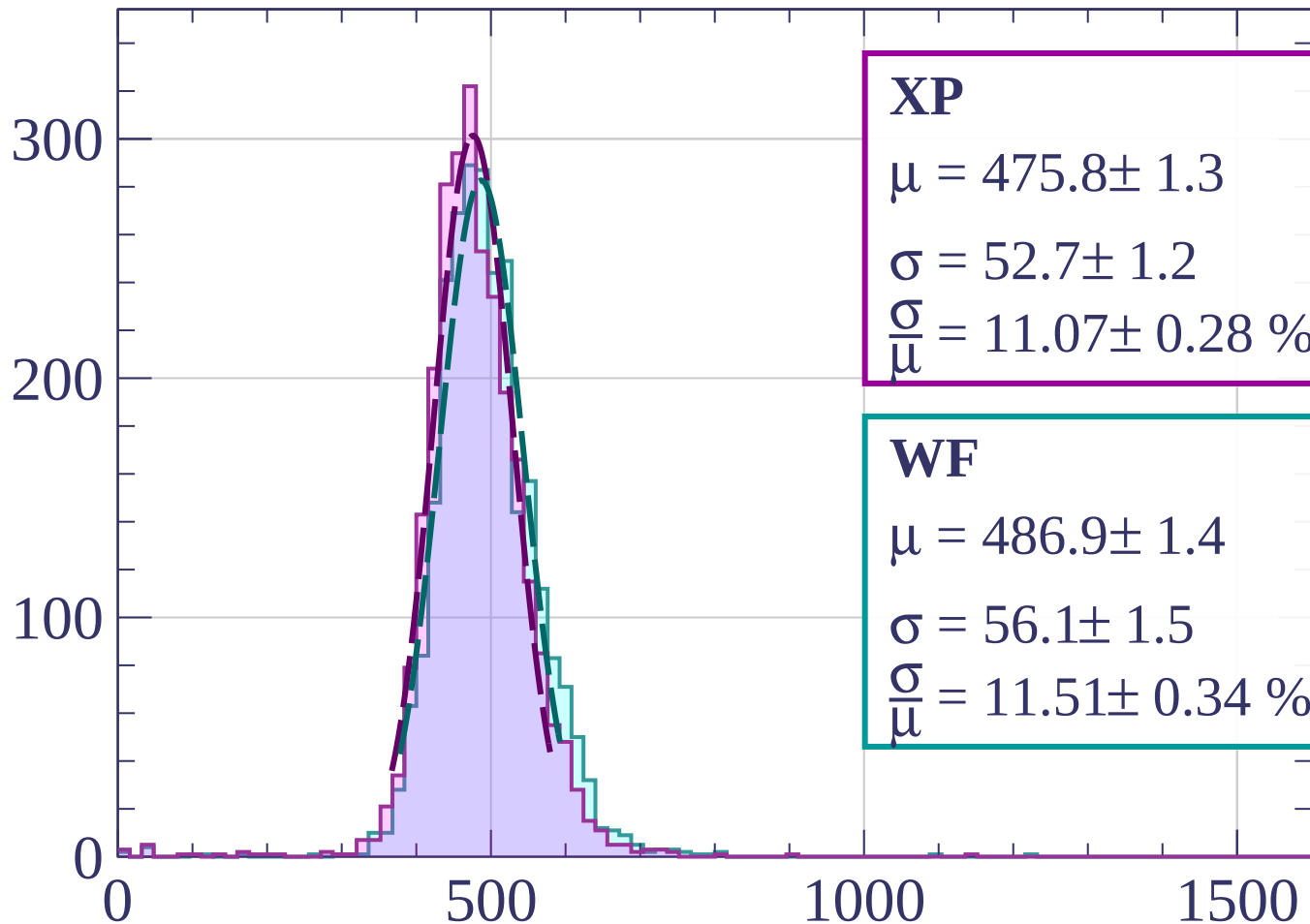
Count



dE/dx [ADC counts/cm]

Energy loss | $-1780 < p < -1760$

Count



XP

$$\mu = 475.8 \pm 1.3$$

$$\sigma = 52.7 \pm 1.2$$

$$\frac{\sigma}{\mu} = 11.07 \pm 0.28 \%$$

WF

$$\mu = 486.9 \pm 1.4$$

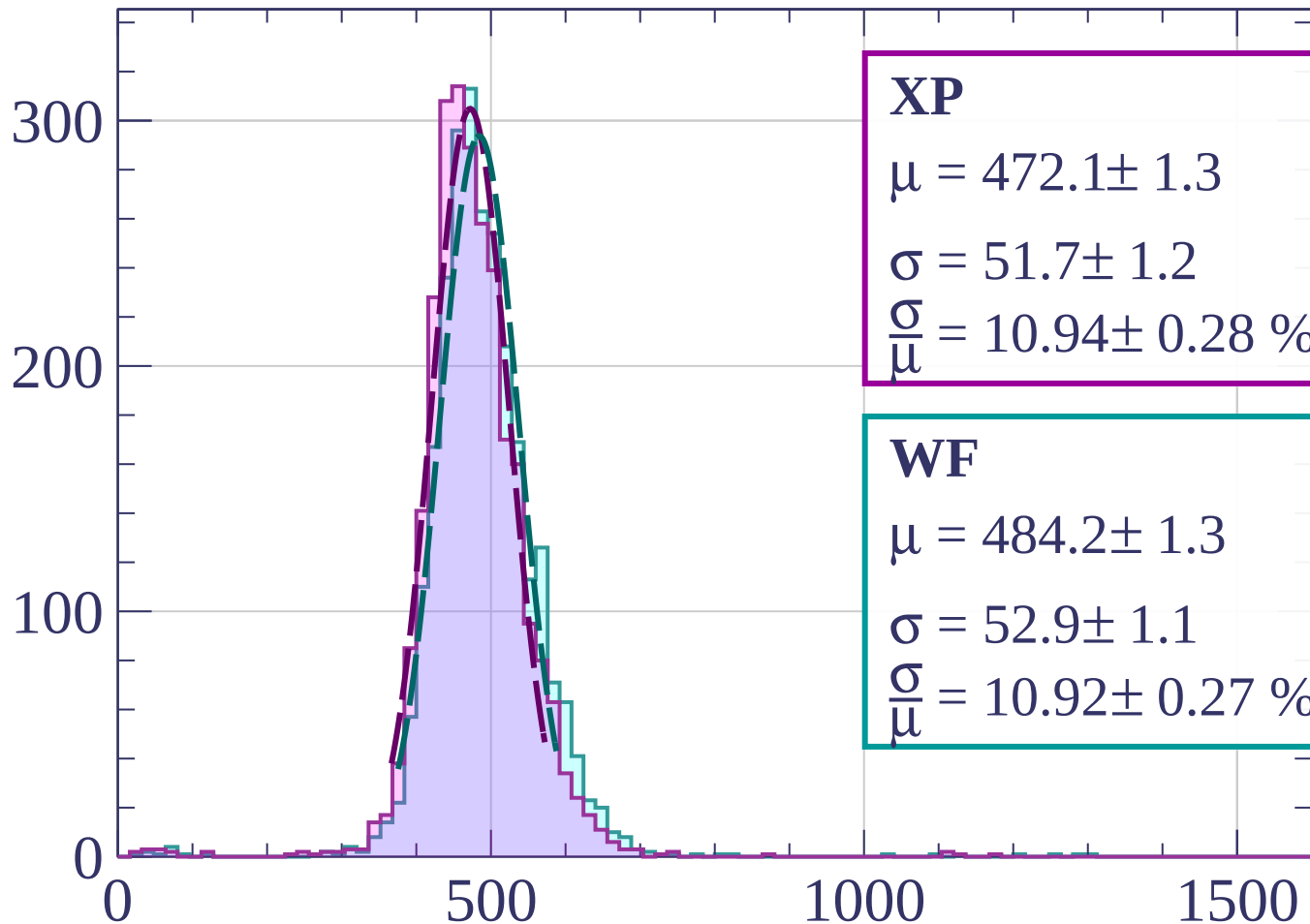
$$\sigma = 56.1 \pm 1.5$$

$$\frac{\sigma}{\mu} = 11.51 \pm 0.34 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1760 < p < -1740$

Count



XP

$$\mu = 472.1 \pm 1.3$$

$$\sigma = 51.7 \pm 1.2$$

$$\frac{\sigma}{\mu} = 10.94 \pm 0.28 \%$$

WF

$$\mu = 484.2 \pm 1.3$$

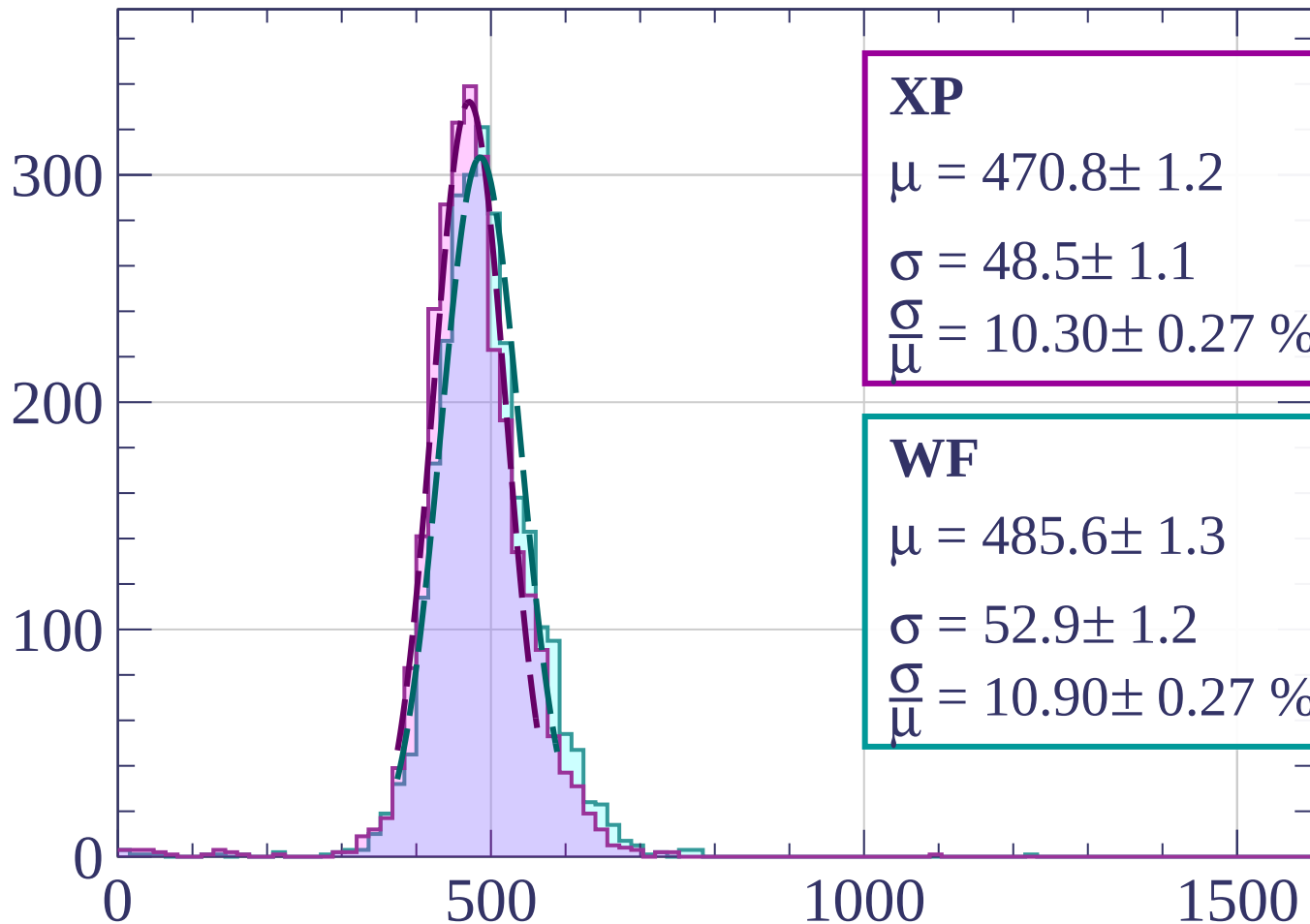
$$\sigma = 52.9 \pm 1.1$$

$$\frac{\sigma}{\mu} = 10.92 \pm 0.27 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1740 < p < -1720$

Count



XP

$$\mu = 470.8 \pm 1.2$$

$$\sigma = 48.5 \pm 1.1$$

$$\frac{\sigma}{\mu} = 10.30 \pm 0.27 \%$$

WF

$$\mu = 485.6 \pm 1.3$$

$$\sigma = 52.9 \pm 1.2$$

$$\frac{\sigma}{\mu} = 10.90 \pm 0.27 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1720 < p < -1700$

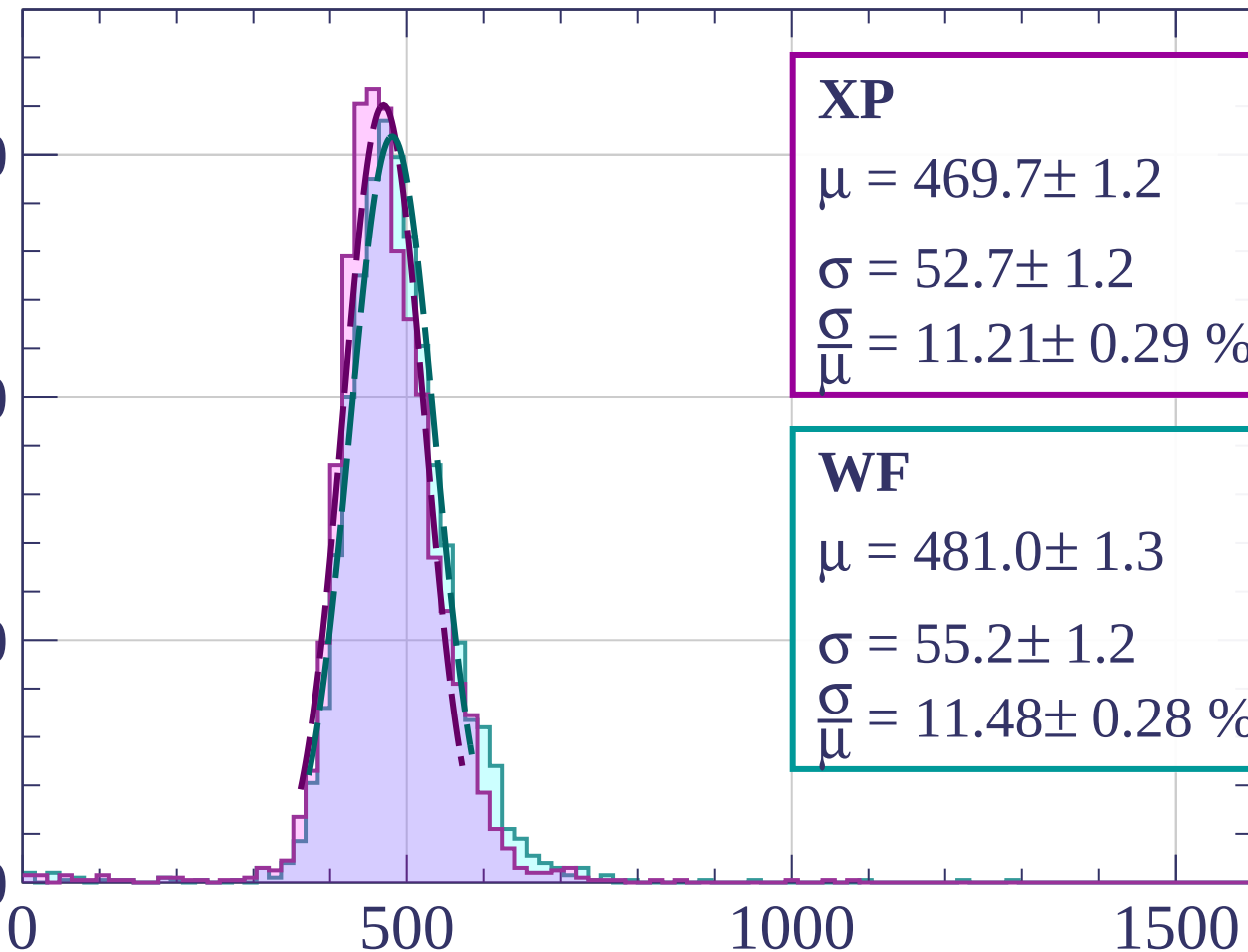
Count

300

200

100

0



XP

$$\mu = 469.7 \pm 1.2$$

$$\sigma = 52.7 \pm 1.2$$

$$\frac{\sigma}{\mu} = 11.21 \pm 0.29 \%$$

WF

$$\mu = 481.0 \pm 1.3$$

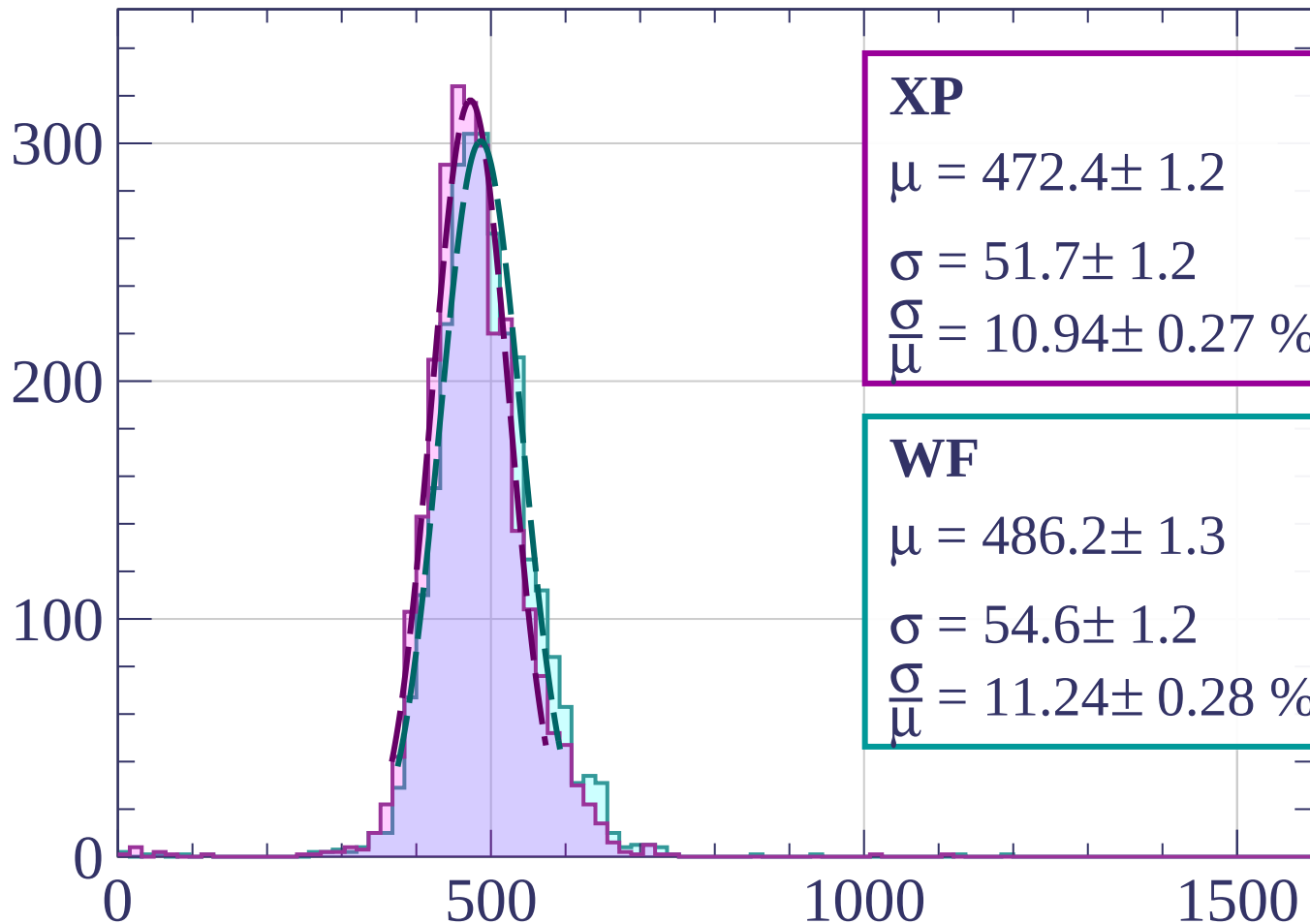
$$\sigma = 55.2 \pm 1.2$$

$$\frac{\sigma}{\mu} = 11.48 \pm 0.28 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1700 < p < -1680$

Count



XP

$$\mu = 472.4 \pm 1.2$$

$$\sigma = 51.7 \pm 1.2$$

$$\frac{\sigma}{\mu} = 10.94 \pm 0.27 \%$$

WF

$$\mu = 486.2 \pm 1.3$$

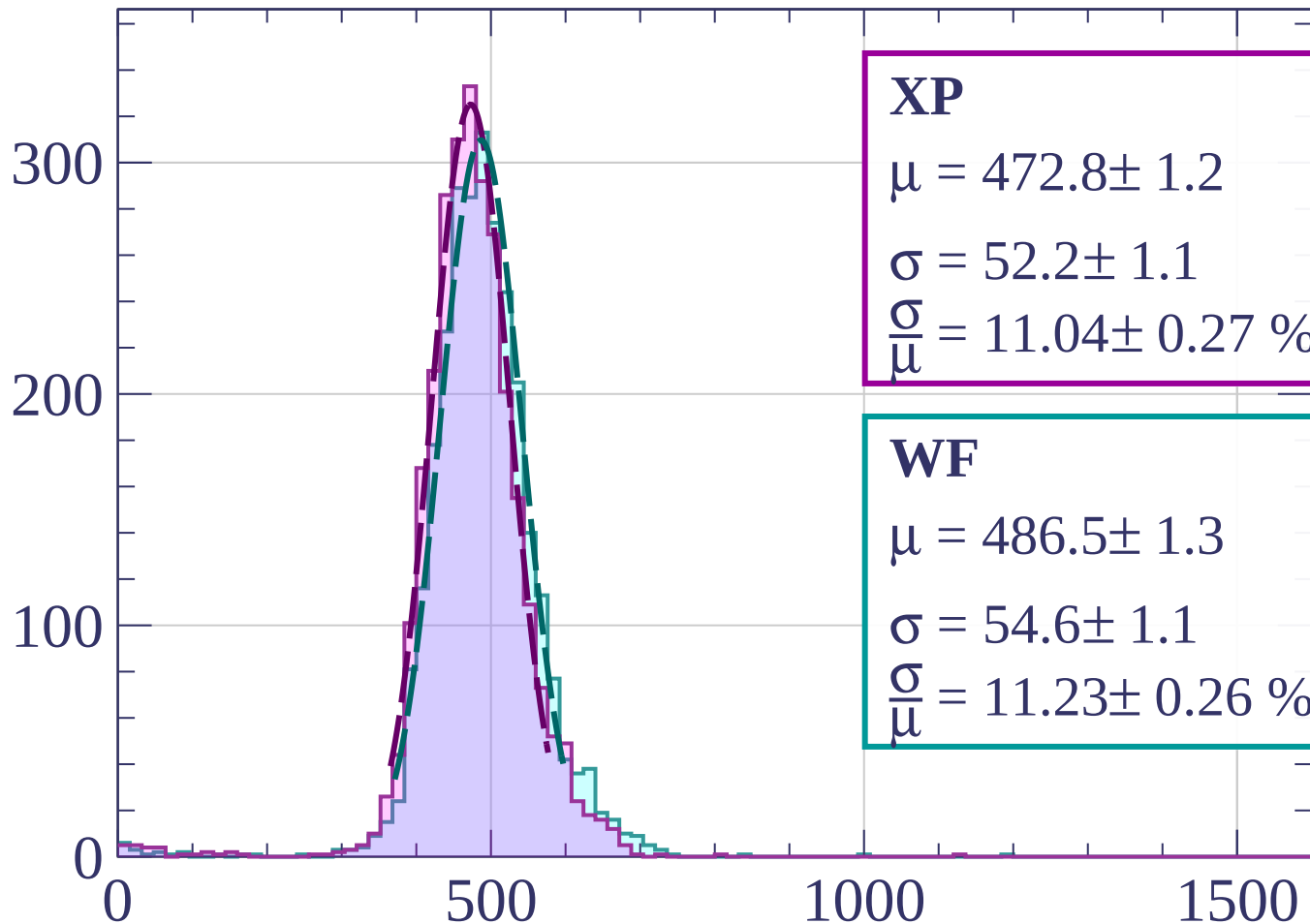
$$\sigma = 54.6 \pm 1.2$$

$$\frac{\sigma}{\mu} = 11.24 \pm 0.28 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1680 < p < -1660$

Count



XP

$$\mu = 472.8 \pm 1.2$$

$$\sigma = 52.2 \pm 1.1$$

$$\frac{\sigma}{\mu} = 11.04 \pm 0.27 \%$$

WF

$$\mu = 486.5 \pm 1.3$$

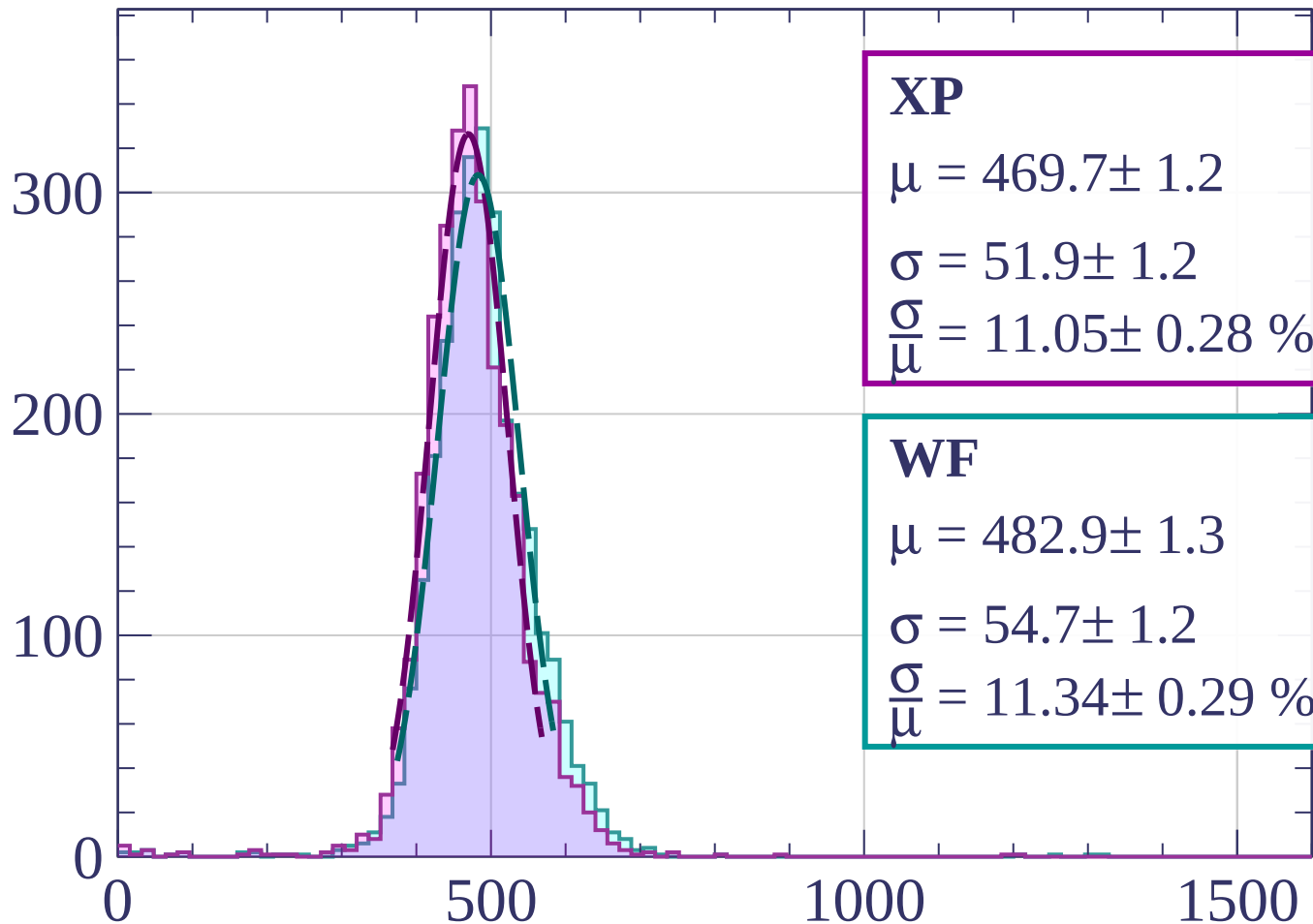
$$\sigma = 54.6 \pm 1.1$$

$$\frac{\sigma}{\mu} = 11.23 \pm 0.26 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1660 < p < -1640$

Count



XP

$$\mu = 469.7 \pm 1.2$$

$$\sigma = 51.9 \pm 1.2$$

$$\frac{\sigma}{\mu} = 11.05 \pm 0.28 \%$$

WF

$$\mu = 482.9 \pm 1.3$$

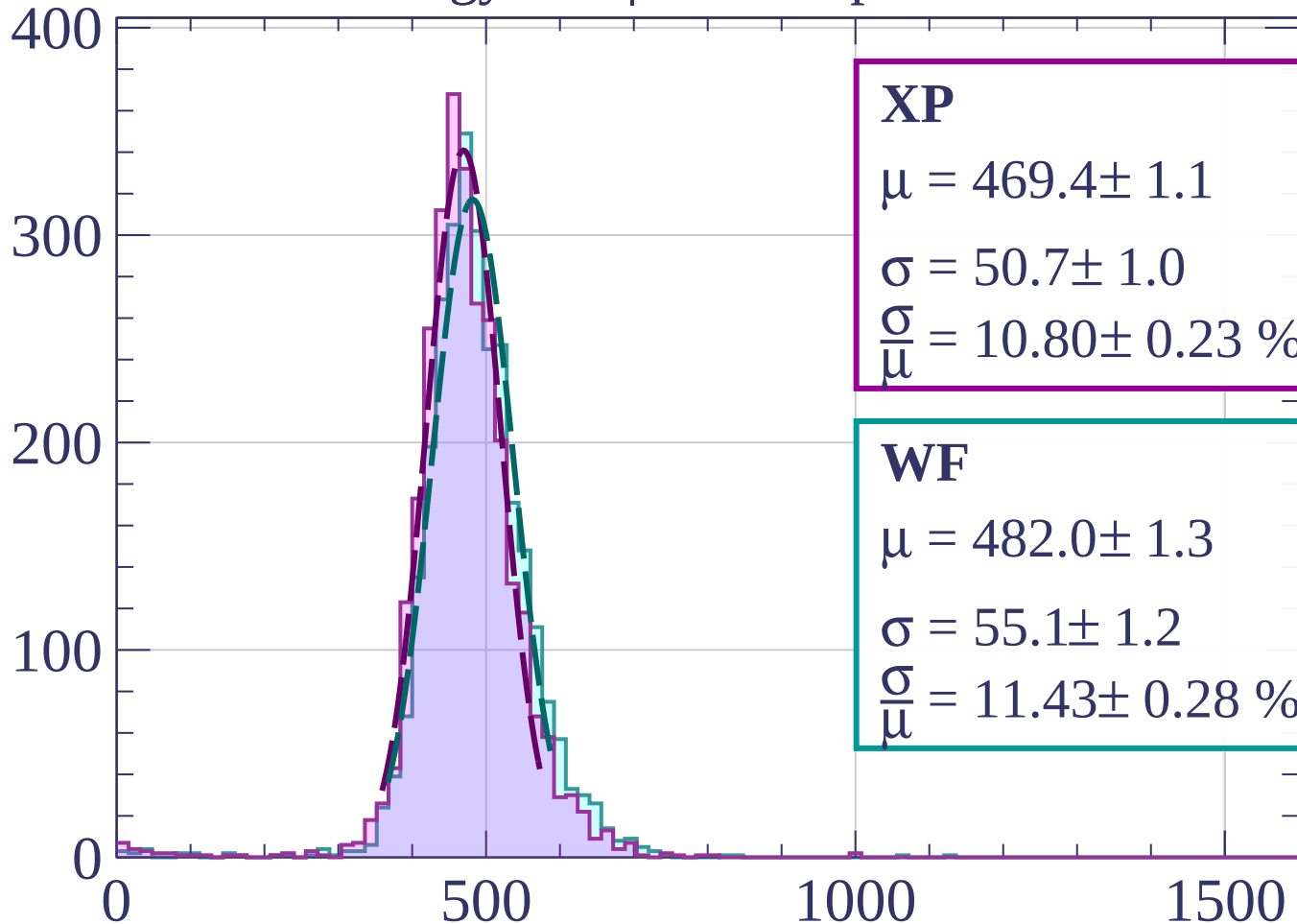
$$\sigma = 54.7 \pm 1.2$$

$$\frac{\sigma}{\mu} = 11.34 \pm 0.29 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1640 < p < -1620$

Count



XP

$$\mu = 469.4 \pm 1.1$$

$$\sigma = 50.7 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.80 \pm 0.23 \%$$

WF

$$\mu = 482.0 \pm 1.3$$

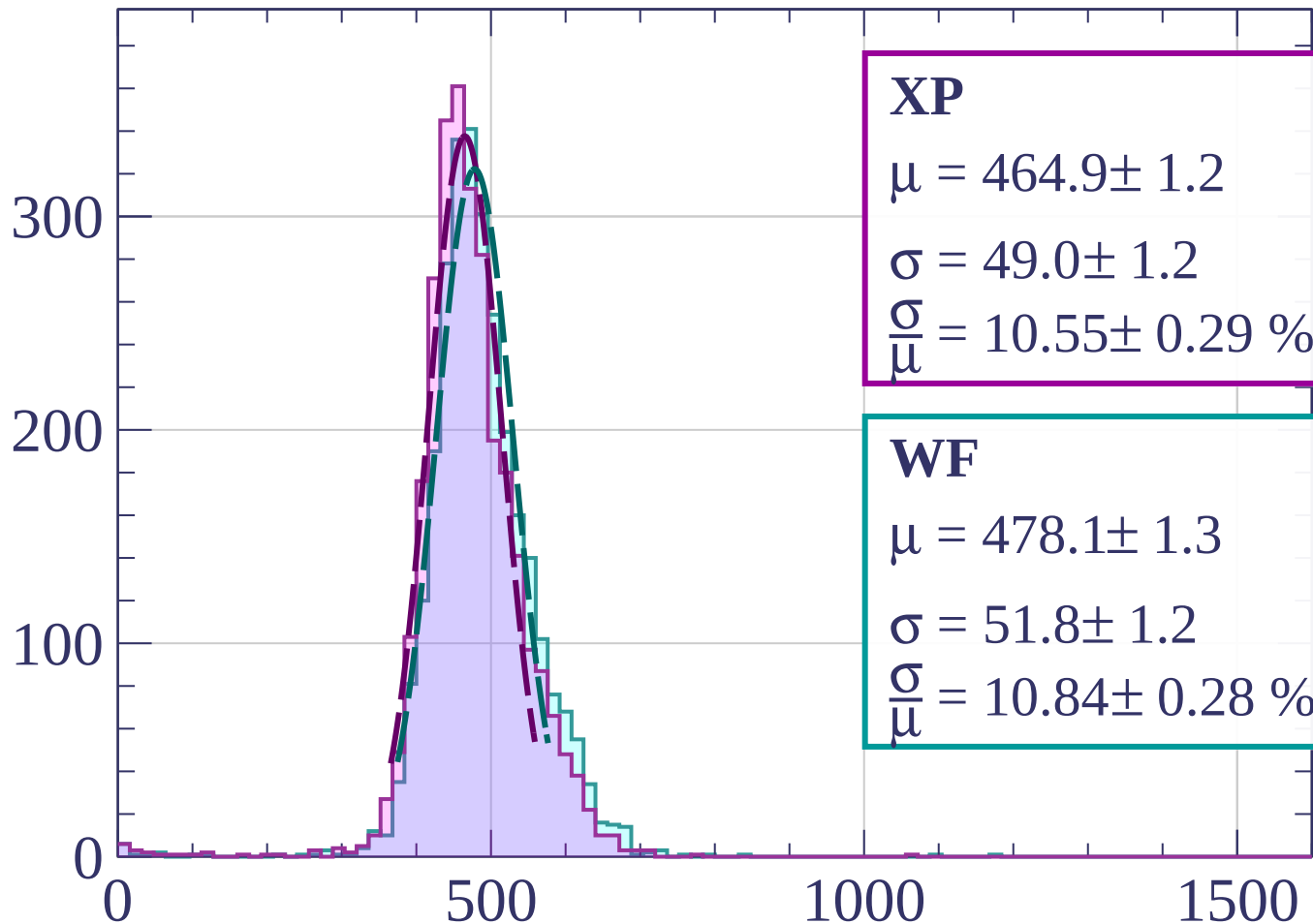
$$\sigma = 55.1 \pm 1.2$$

$$\frac{\sigma}{\mu} = 11.43 \pm 0.28 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1620 < p < -1600$

Count



XP

$$\mu = 464.9 \pm 1.2$$

$$\sigma = 49.0 \pm 1.2$$

$$\frac{\sigma}{\mu} = 10.55 \pm 0.29 \%$$

WF

$$\mu = 478.1 \pm 1.3$$

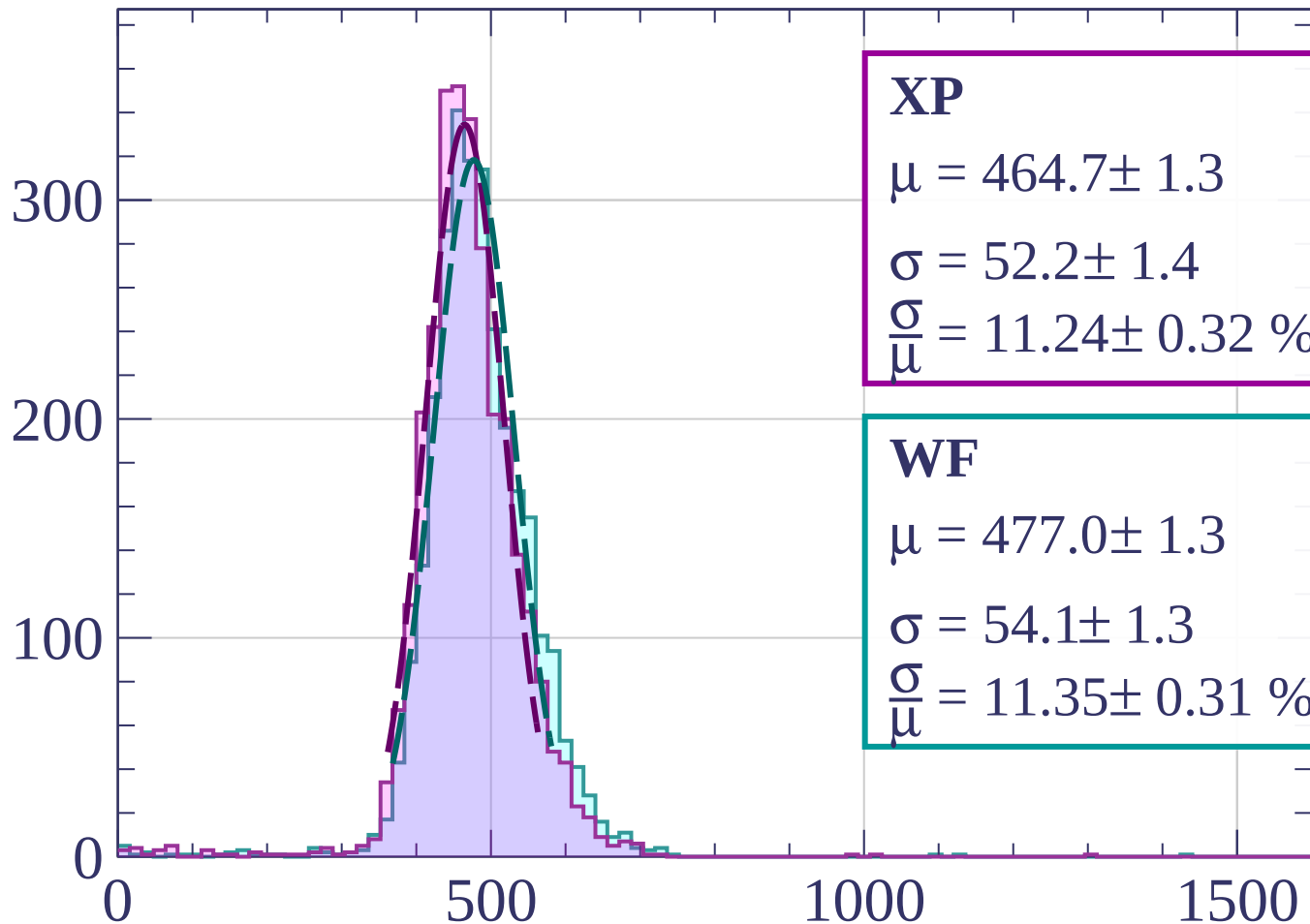
$$\sigma = 51.8 \pm 1.2$$

$$\frac{\sigma}{\mu} = 10.84 \pm 0.28 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1600 < p < -1580$

Count



XP

$$\mu = 464.7 \pm 1.3$$

$$\sigma = 52.2 \pm 1.4$$

$$\frac{\sigma}{\mu} = 11.24 \pm 0.32 \%$$

WF

$$\mu = 477.0 \pm 1.3$$

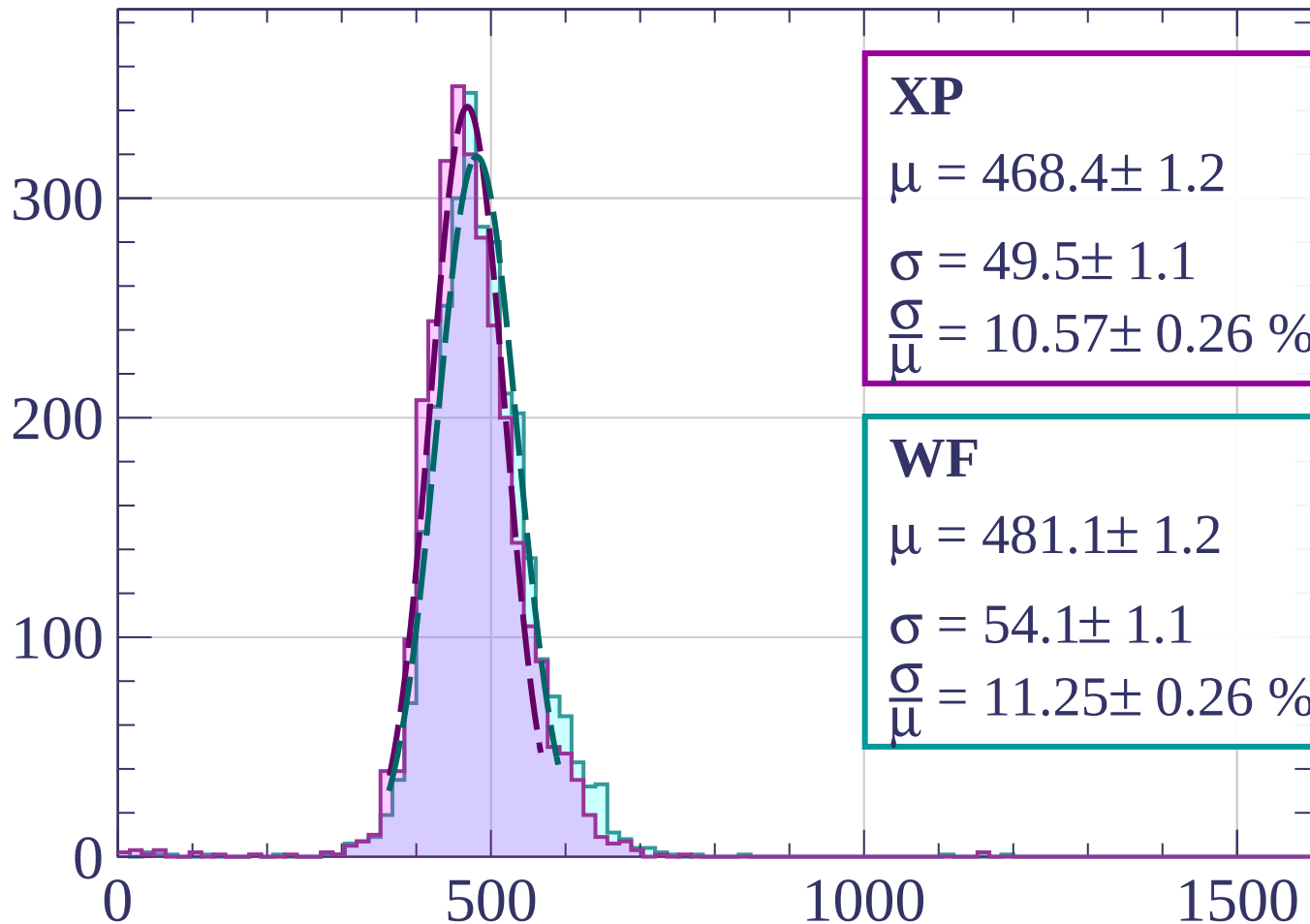
$$\sigma = 54.1 \pm 1.3$$

$$\frac{\sigma}{\mu} = 11.35 \pm 0.31 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1580 < p < -1560$

Count



XP

$$\mu = 468.4 \pm 1.2$$

$$\sigma = 49.5 \pm 1.1$$

$$\frac{\sigma}{\mu} = 10.57 \pm 0.26 \%$$

WF

$$\mu = 481.1 \pm 1.2$$

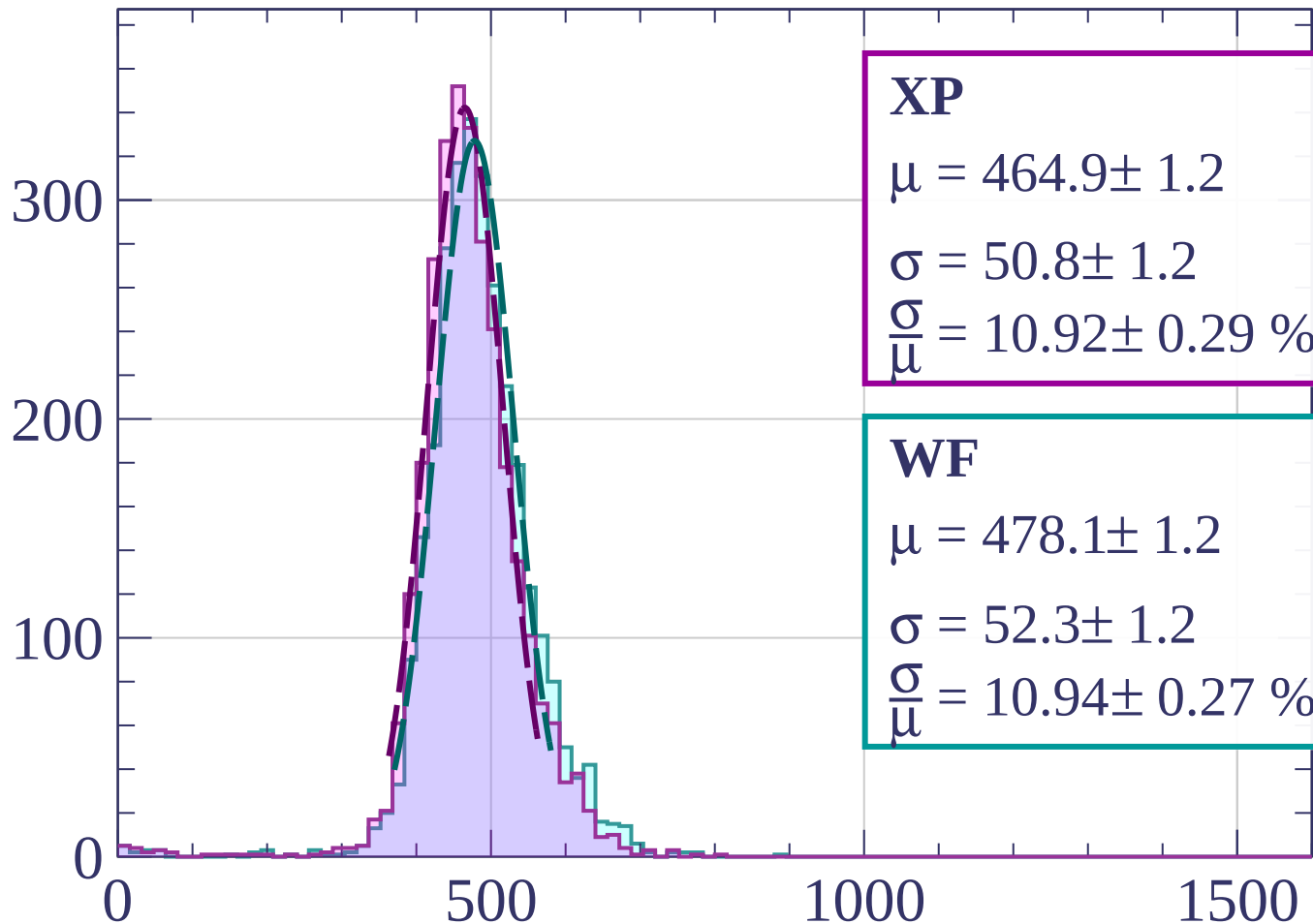
$$\sigma = 54.1 \pm 1.1$$

$$\frac{\sigma}{\mu} = 11.25 \pm 0.26 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1560 < p < -1540$

Count



XP

$$\mu = 464.9 \pm 1.2$$

$$\sigma = 50.8 \pm 1.2$$

$$\frac{\sigma}{\mu} = 10.92 \pm 0.29 \%$$

WF

$$\mu = 478.1 \pm 1.2$$

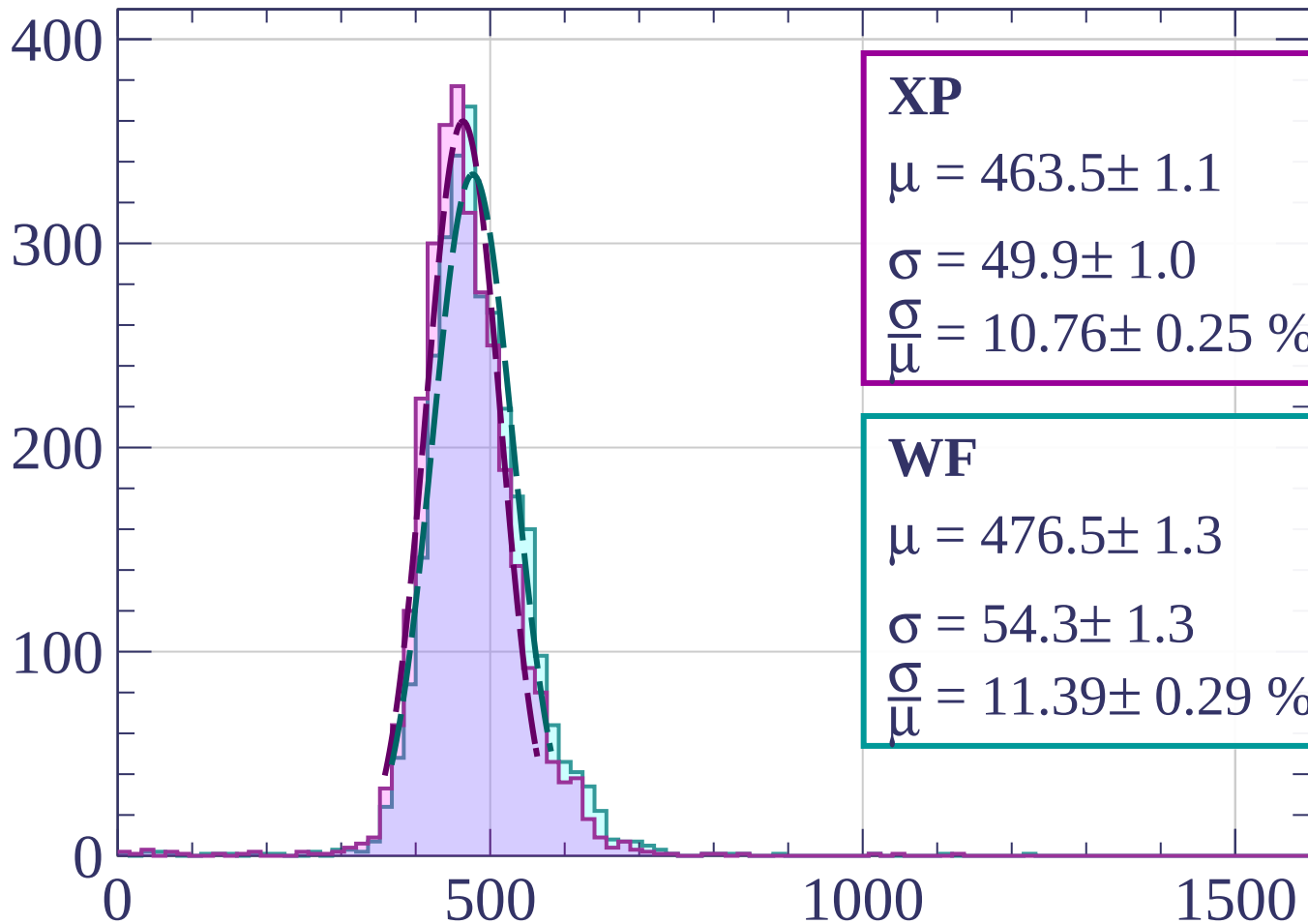
$$\sigma = 52.3 \pm 1.2$$

$$\frac{\sigma}{\mu} = 10.94 \pm 0.27 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1540 < p < -1520$

Count



XP

$$\mu = 463.5 \pm 1.1$$

$$\sigma = 49.9 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.76 \pm 0.25 \%$$

WF

$$\mu = 476.5 \pm 1.3$$

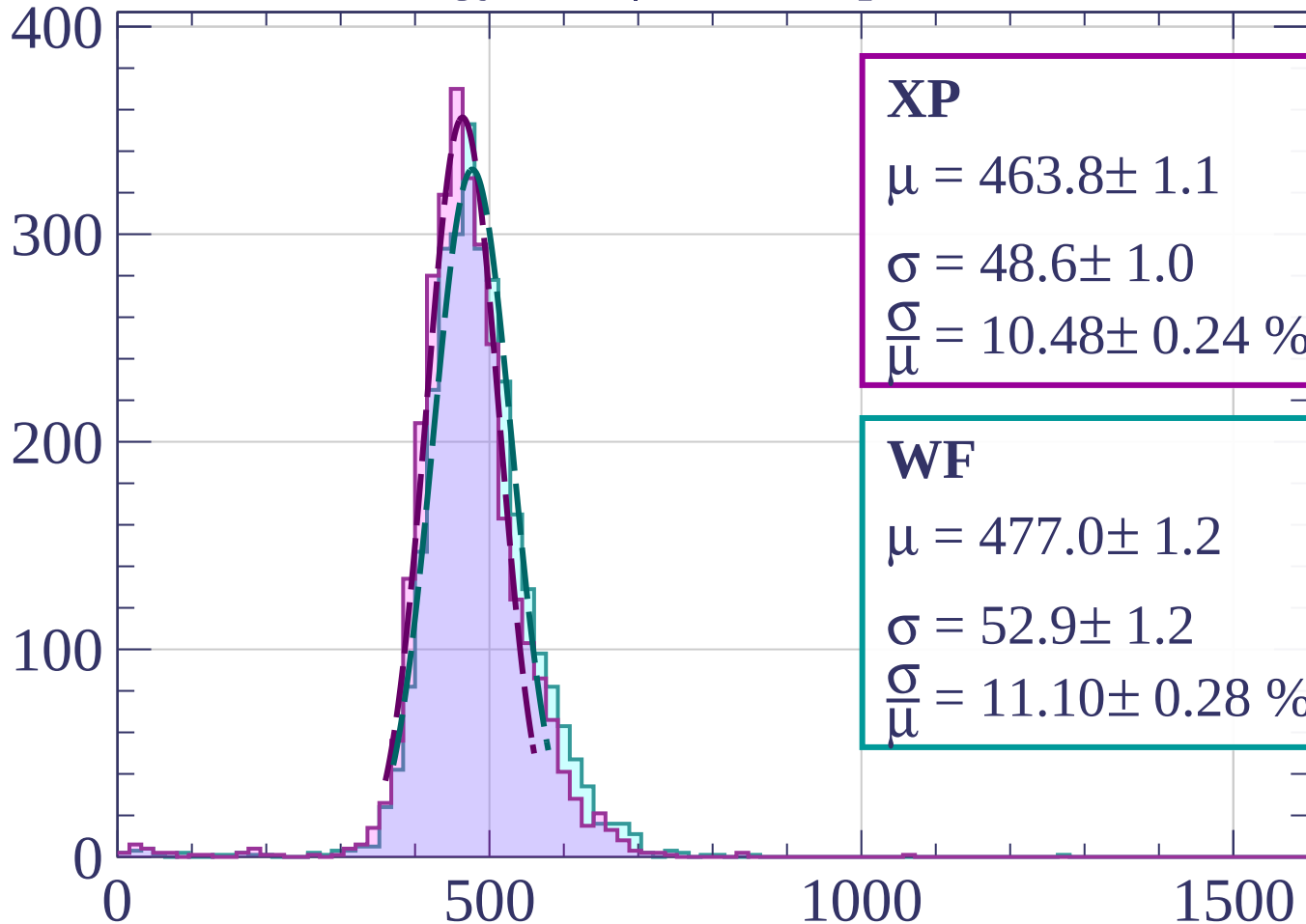
$$\sigma = 54.3 \pm 1.3$$

$$\frac{\sigma}{\mu} = 11.39 \pm 0.29 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1520 < p < -1500$

Count



XP

$$\mu = 463.8 \pm 1.1$$

$$\sigma = 48.6 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.48 \pm 0.24 \%$$

WF

$$\mu = 477.0 \pm 1.2$$

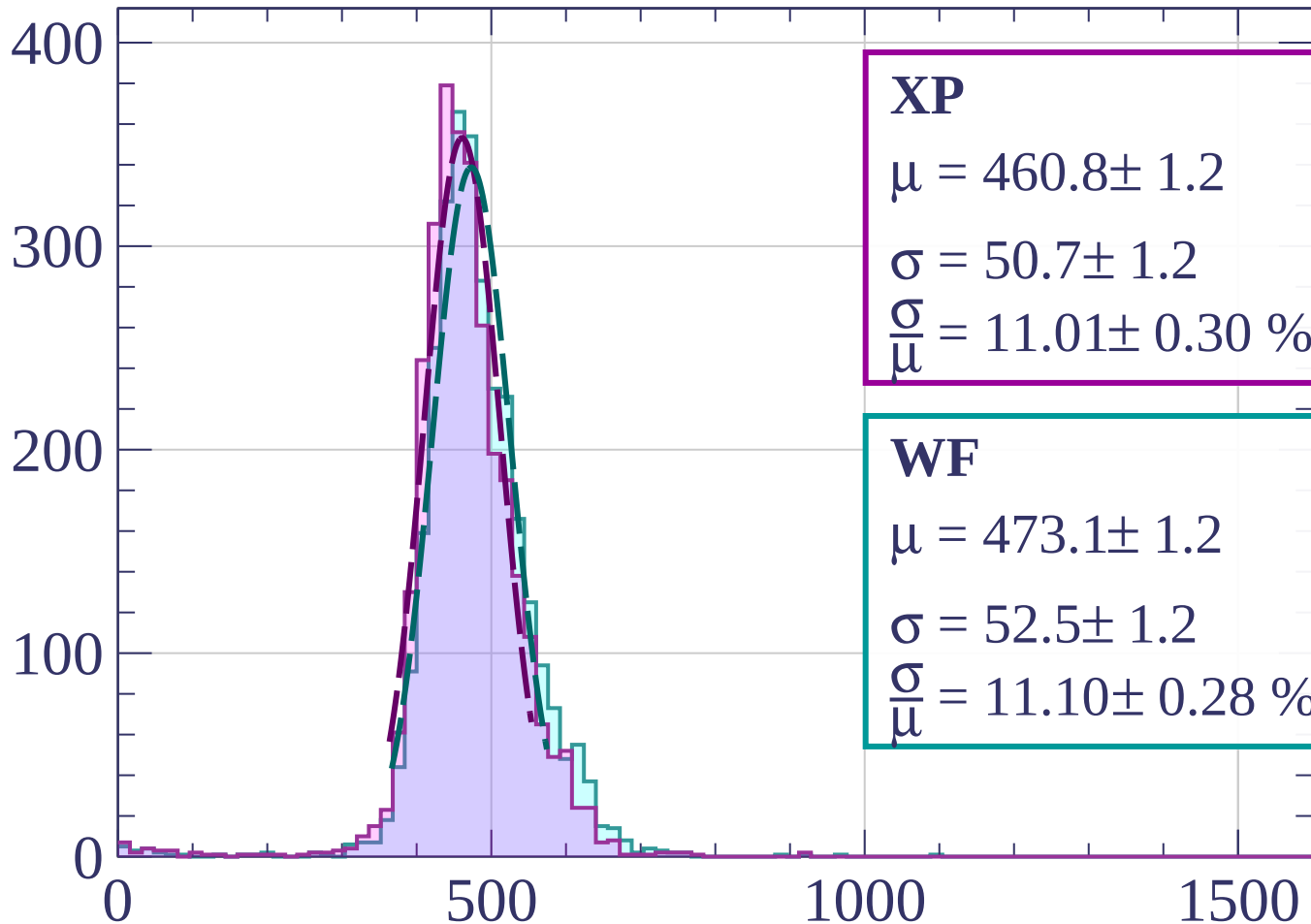
$$\sigma = 52.9 \pm 1.2$$

$$\frac{\sigma}{\mu} = 11.10 \pm 0.28 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1500 < p < -1480$

Count



XP

$$\mu = 460.8 \pm 1.2$$

$$\sigma = 50.7 \pm 1.2$$

$$\frac{\sigma}{\mu} = 11.01 \pm 0.30 \%$$

WF

$$\mu = 473.1 \pm 1.2$$

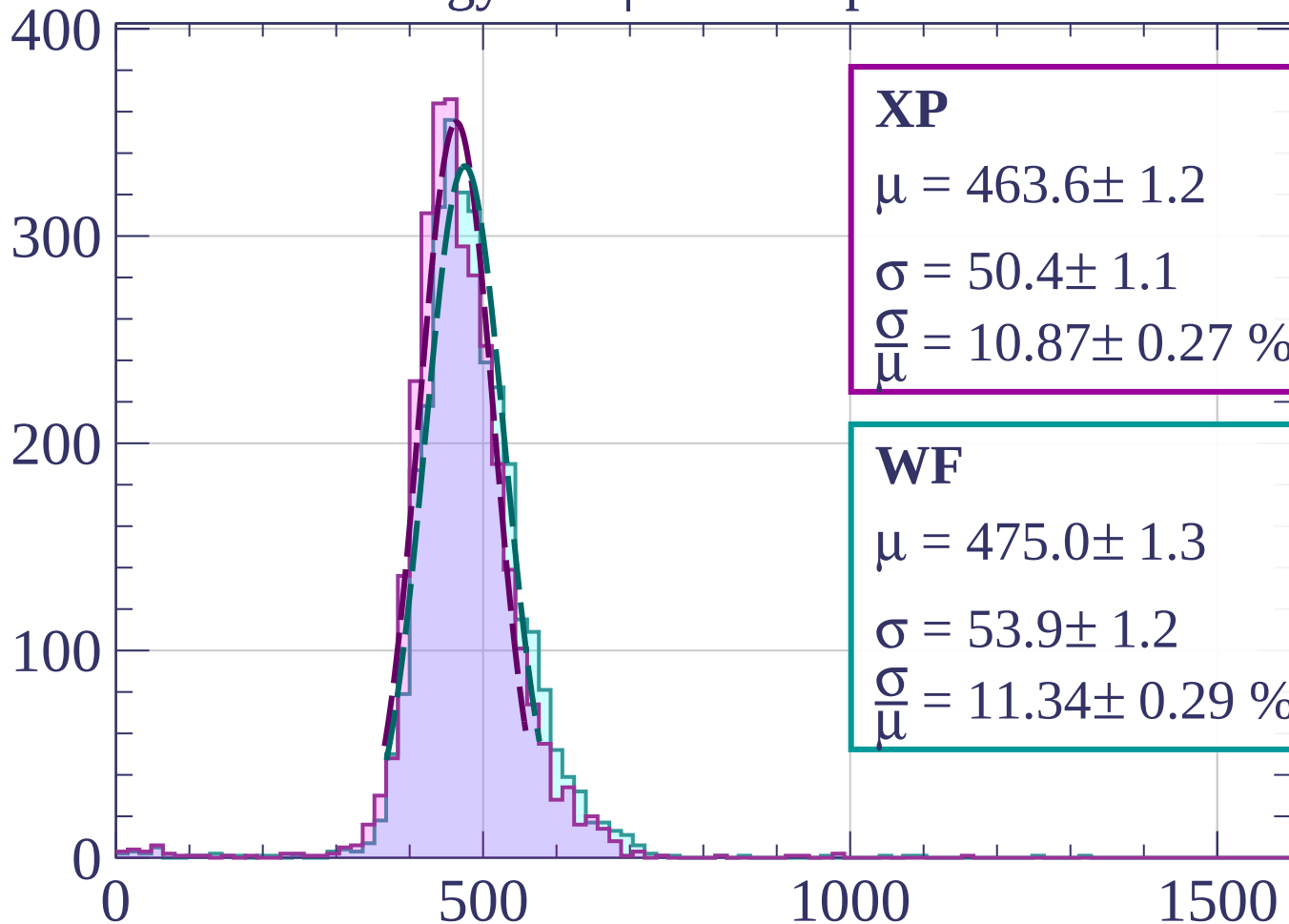
$$\sigma = 52.5 \pm 1.2$$

$$\frac{\sigma}{\mu} = 11.10 \pm 0.28 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1480 < p < -1460$

Count



XP

$$\mu = 463.6 \pm 1.2$$

$$\sigma = 50.4 \pm 1.1$$

$$\frac{\sigma}{\mu} = 10.87 \pm 0.27 \%$$

WF

$$\mu = 475.0 \pm 1.3$$

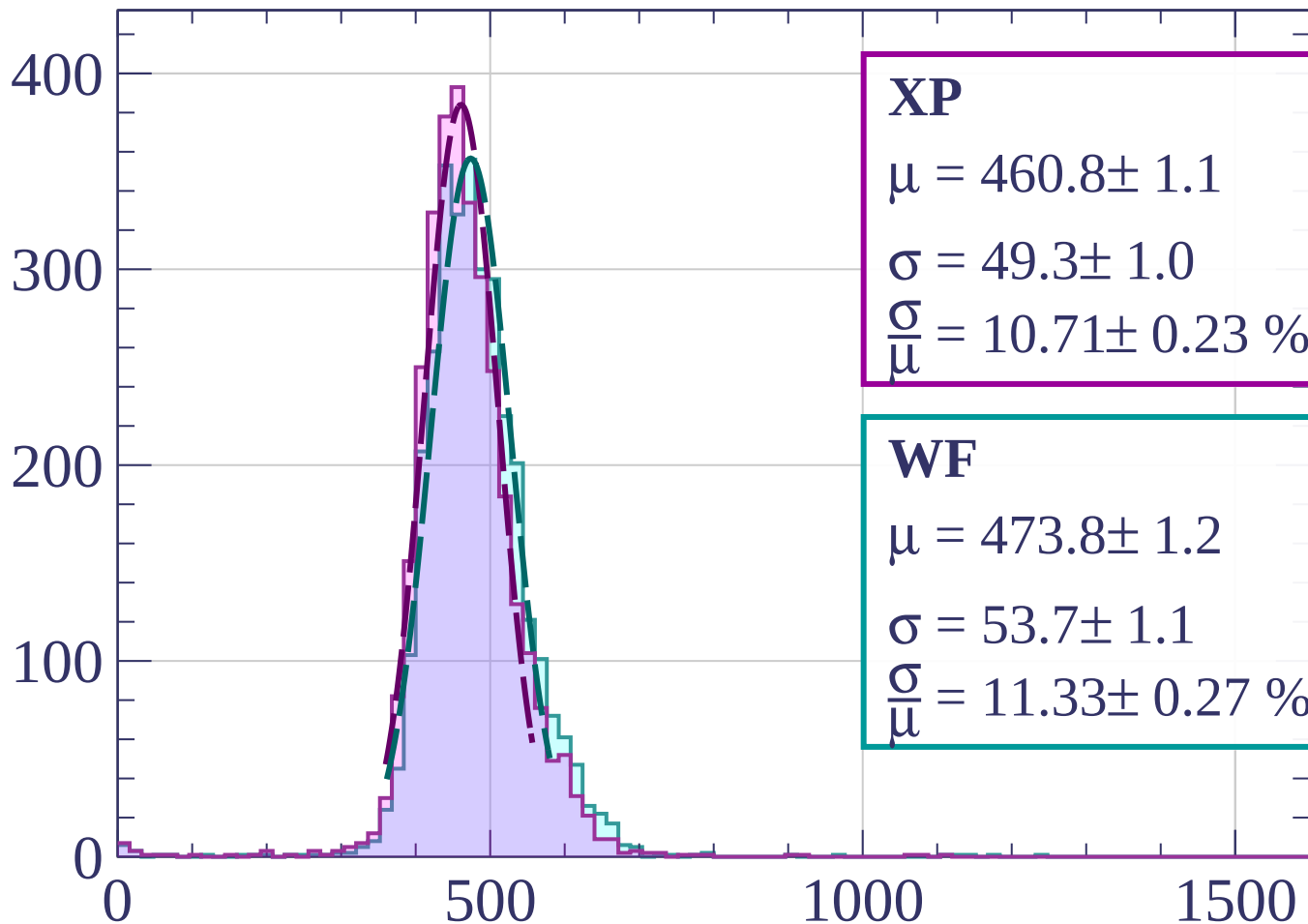
$$\sigma = 53.9 \pm 1.2$$

$$\frac{\sigma}{\mu} = 11.34 \pm 0.29 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1460 < p < -1440$

Count



XP

$$\mu = 460.8 \pm 1.1$$

$$\sigma = 49.3 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.71 \pm 0.23 \%$$

WF

$$\mu = 473.8 \pm 1.2$$

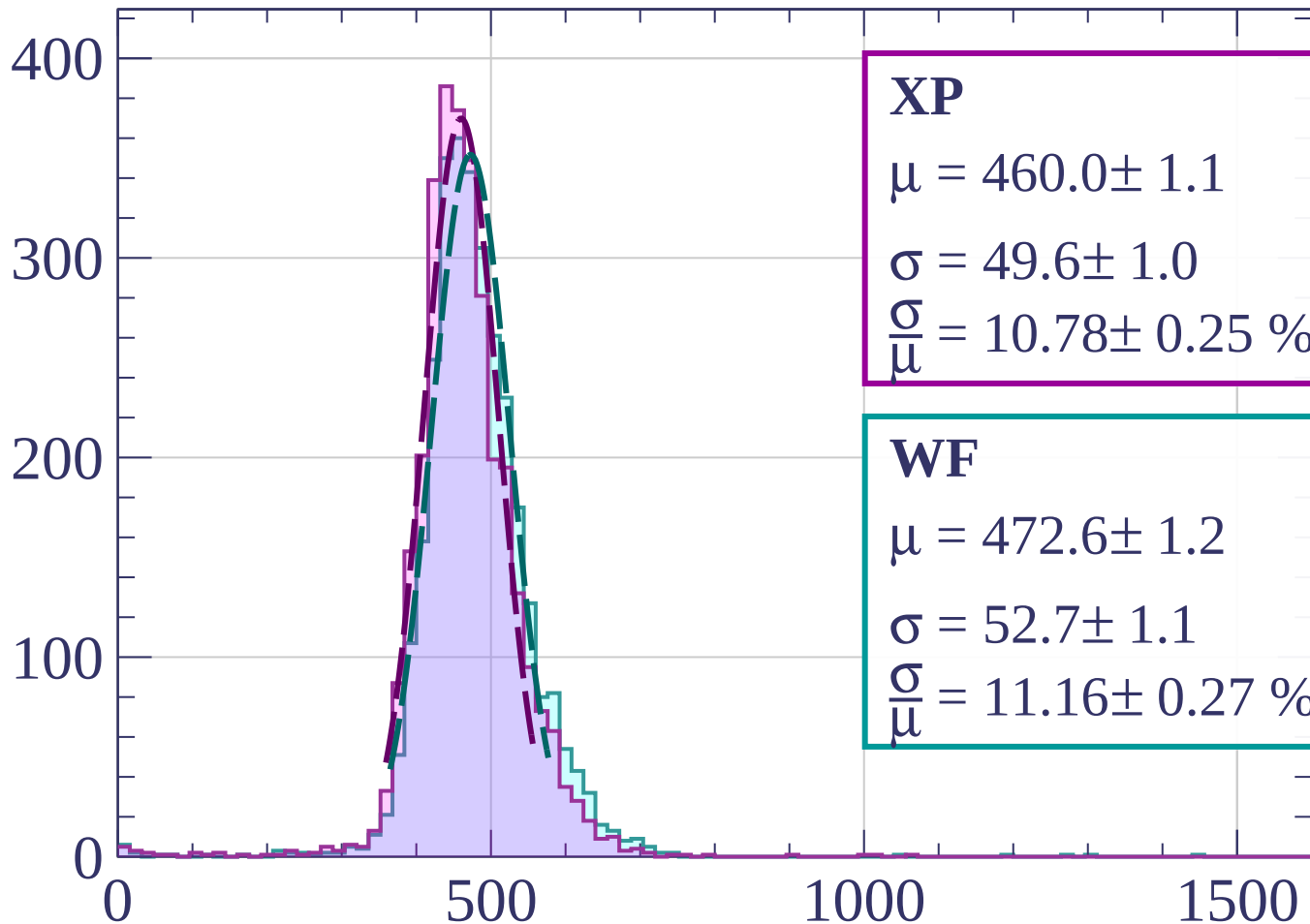
$$\sigma = 53.7 \pm 1.1$$

$$\frac{\sigma}{\mu} = 11.33 \pm 0.27 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1440 < p < -1420$

Count



XP

$$\mu = 460.0 \pm 1.1$$

$$\sigma = 49.6 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.78 \pm 0.25 \%$$

WF

$$\mu = 472.6 \pm 1.2$$

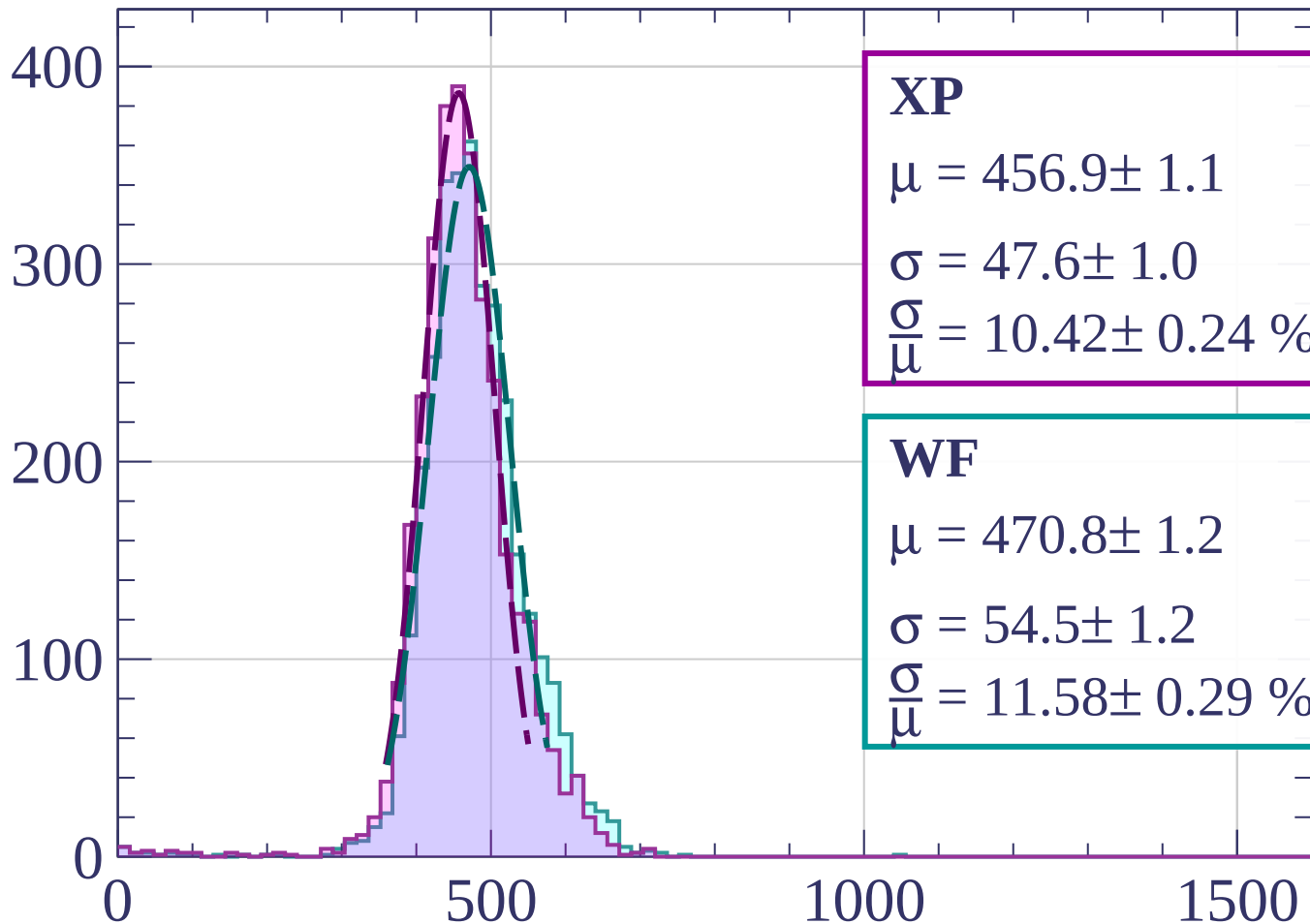
$$\sigma = 52.7 \pm 1.1$$

$$\frac{\sigma}{\mu} = 11.16 \pm 0.27 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1420 < p < -1400$

Count



XP

$$\mu = 456.9 \pm 1.1$$

$$\sigma = 47.6 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.42 \pm 0.24 \%$$

WF

$$\mu = 470.8 \pm 1.2$$

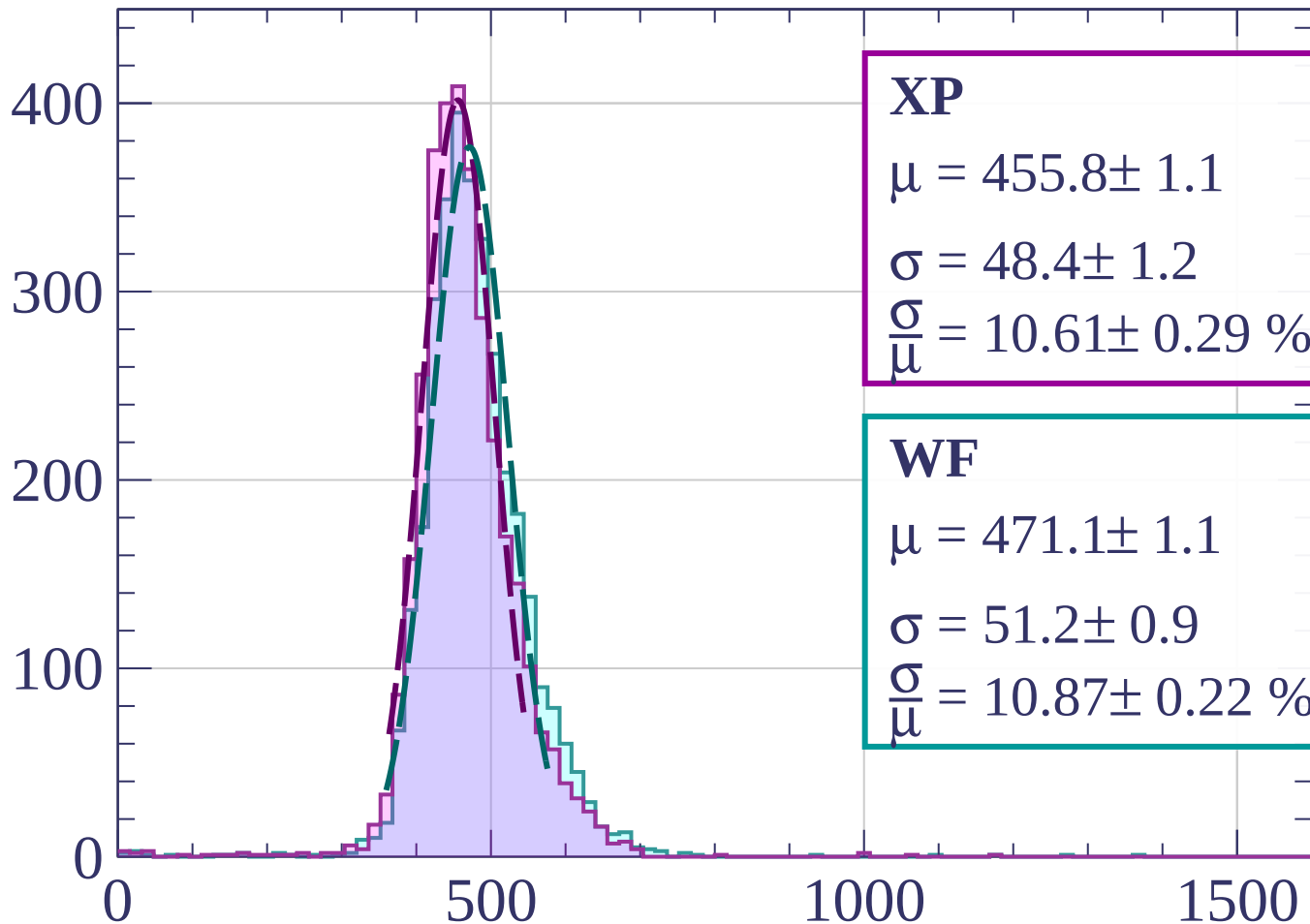
$$\sigma = 54.5 \pm 1.2$$

$$\frac{\sigma}{\mu} = 11.58 \pm 0.29 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1400 < p < -1380$

Count



XP

$$\mu = 455.8 \pm 1.1$$

$$\sigma = 48.4 \pm 1.2$$

$$\frac{\sigma}{\mu} = 10.61 \pm 0.29 \%$$

WF

$$\mu = 471.1 \pm 1.1$$

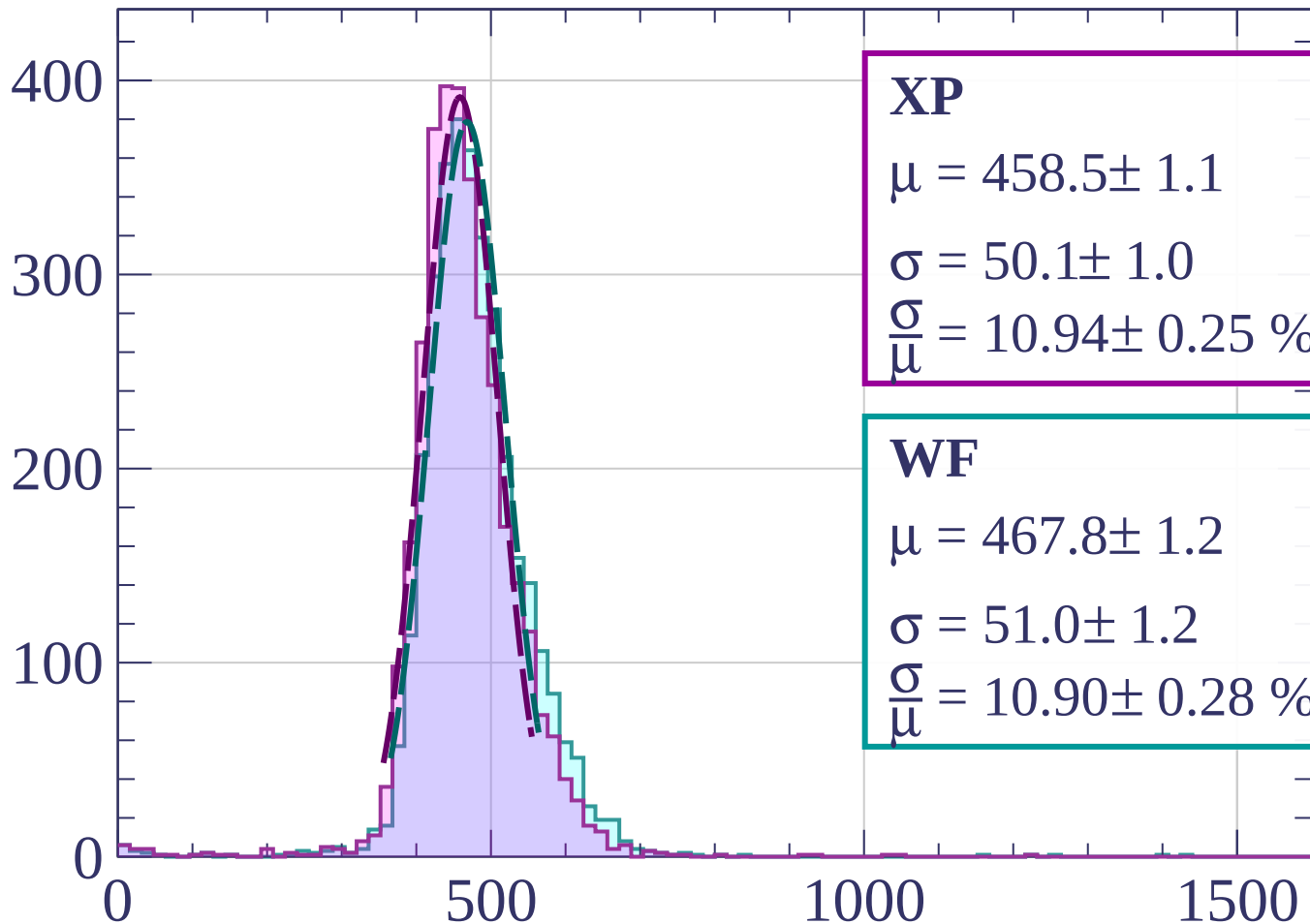
$$\sigma = 51.2 \pm 0.9$$

$$\frac{\sigma}{\mu} = 10.87 \pm 0.22 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1380 < p < -1360$

Count



XP

$$\mu = 458.5 \pm 1.1$$

$$\sigma = 50.1 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.94 \pm 0.25 \%$$

WF

$$\mu = 467.8 \pm 1.2$$

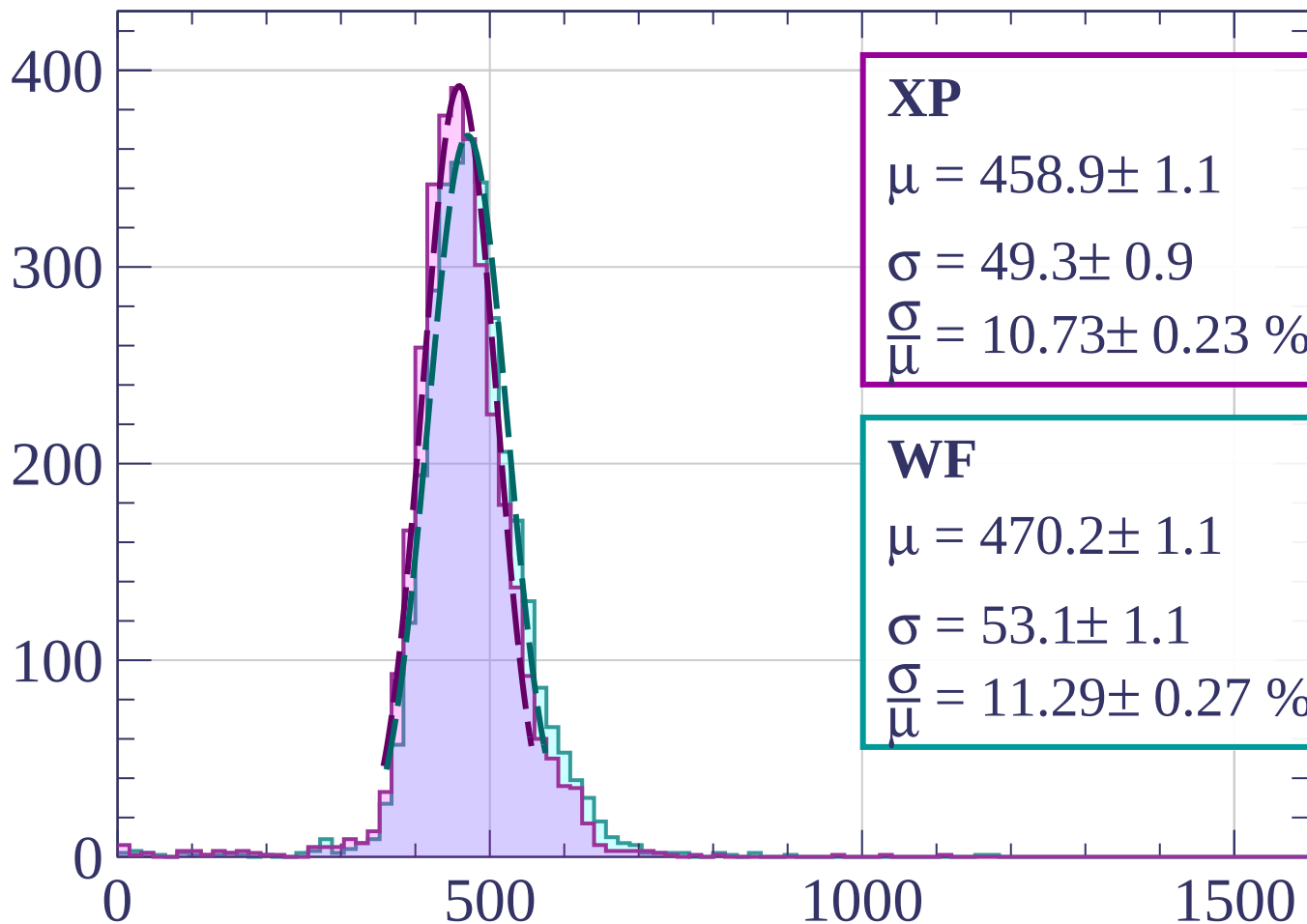
$$\sigma = 51.0 \pm 1.2$$

$$\frac{\sigma}{\mu} = 10.90 \pm 0.28 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1360 < p < -1340$

Count



XP

$$\mu = 458.9 \pm 1.1$$

$$\sigma = 49.3 \pm 0.9$$

$$\frac{\sigma}{\mu} = 10.73 \pm 0.23 \%$$

WF

$$\mu = 470.2 \pm 1.1$$

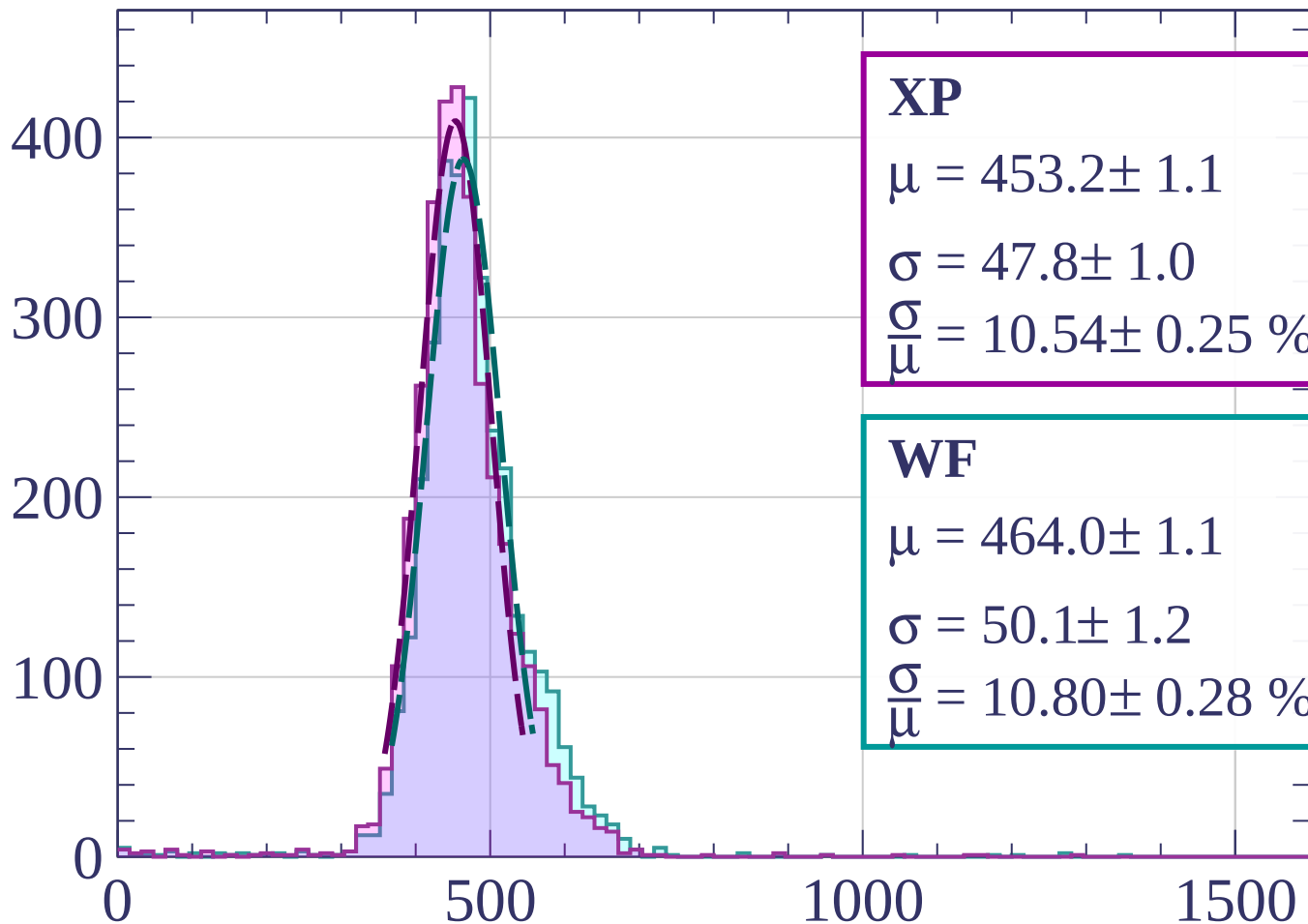
$$\sigma = 53.1 \pm 1.1$$

$$\frac{\sigma}{\mu} = 11.29 \pm 0.27 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1340 < p < -1320$

Count



XP

$$\mu = 453.2 \pm 1.1$$

$$\sigma = 47.8 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.54 \pm 0.25 \%$$

WF

$$\mu = 464.0 \pm 1.1$$

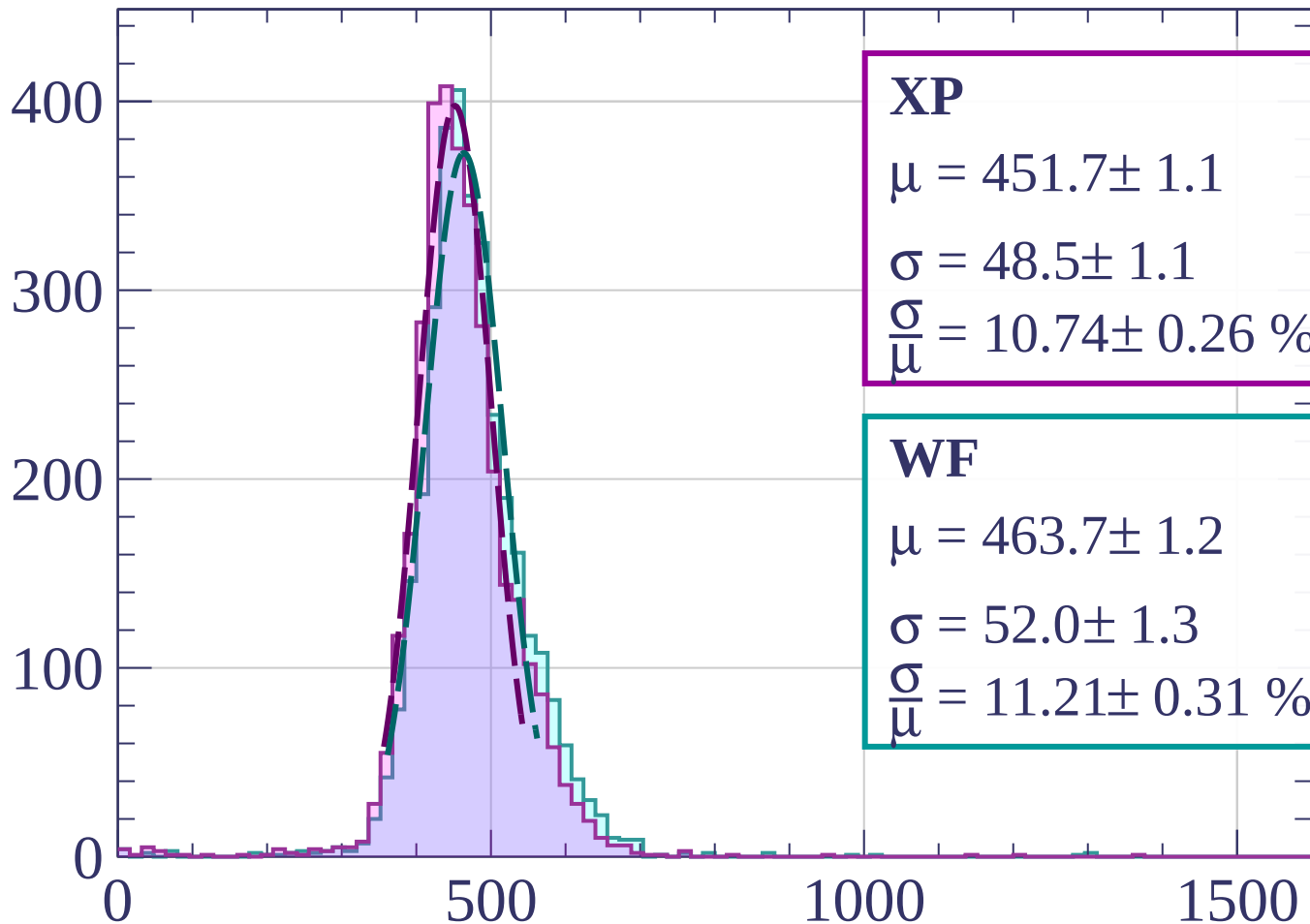
$$\sigma = 50.1 \pm 1.2$$

$$\frac{\sigma}{\mu} = 10.80 \pm 0.28 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1320 < p < -1300$

Count



XP

$$\mu = 451.7 \pm 1.1$$

$$\sigma = 48.5 \pm 1.1$$

$$\frac{\sigma}{\mu} = 10.74 \pm 0.26 \%$$

WF

$$\mu = 463.7 \pm 1.2$$

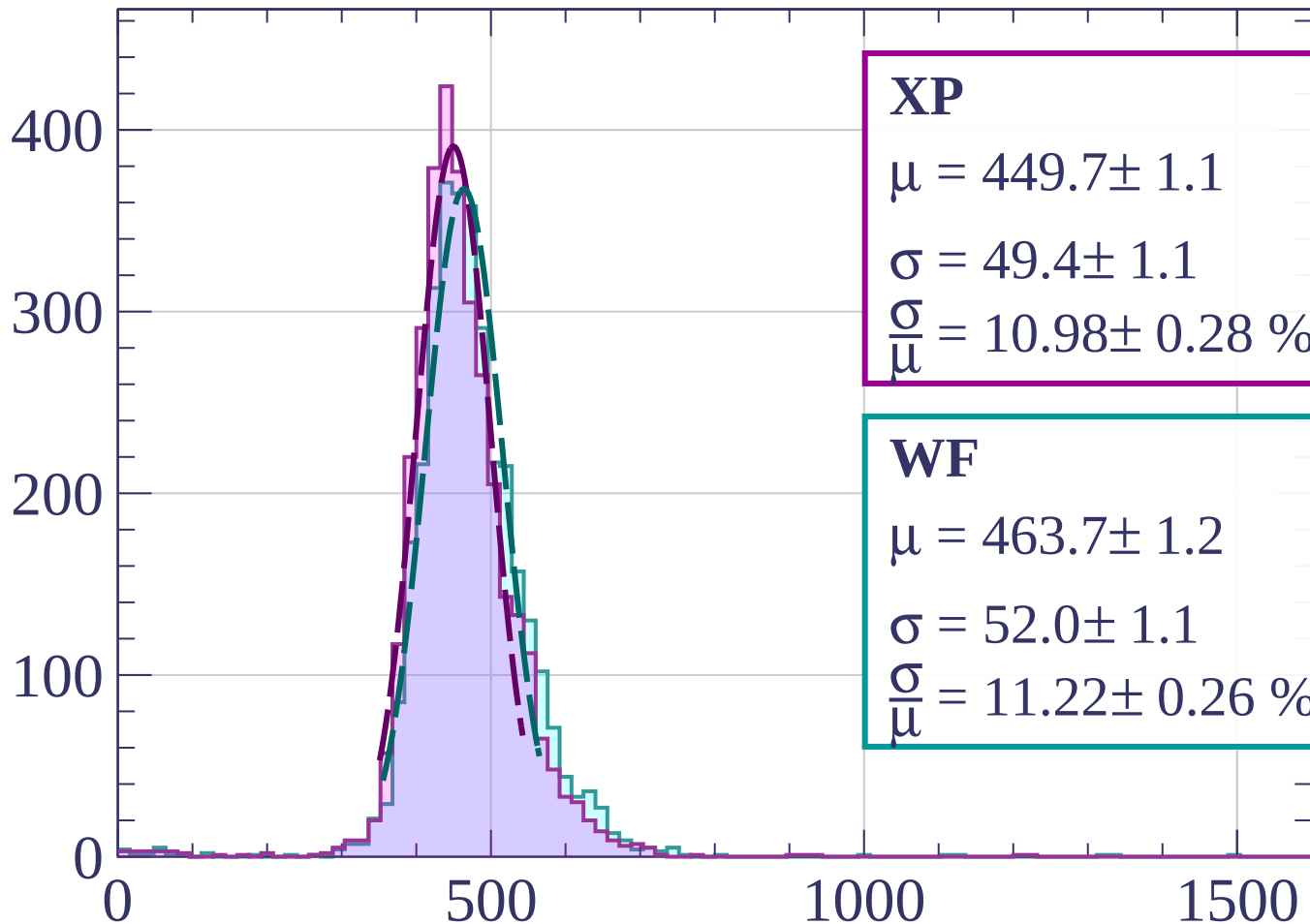
$$\sigma = 52.0 \pm 1.3$$

$$\frac{\sigma}{\mu} = 11.21 \pm 0.31 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1300 < p < -1280$

Count



XP

$$\mu = 449.7 \pm 1.1$$

$$\sigma = 49.4 \pm 1.1$$

$$\frac{\sigma}{\mu} = 10.98 \pm 0.28 \%$$

WF

$$\mu = 463.7 \pm 1.2$$

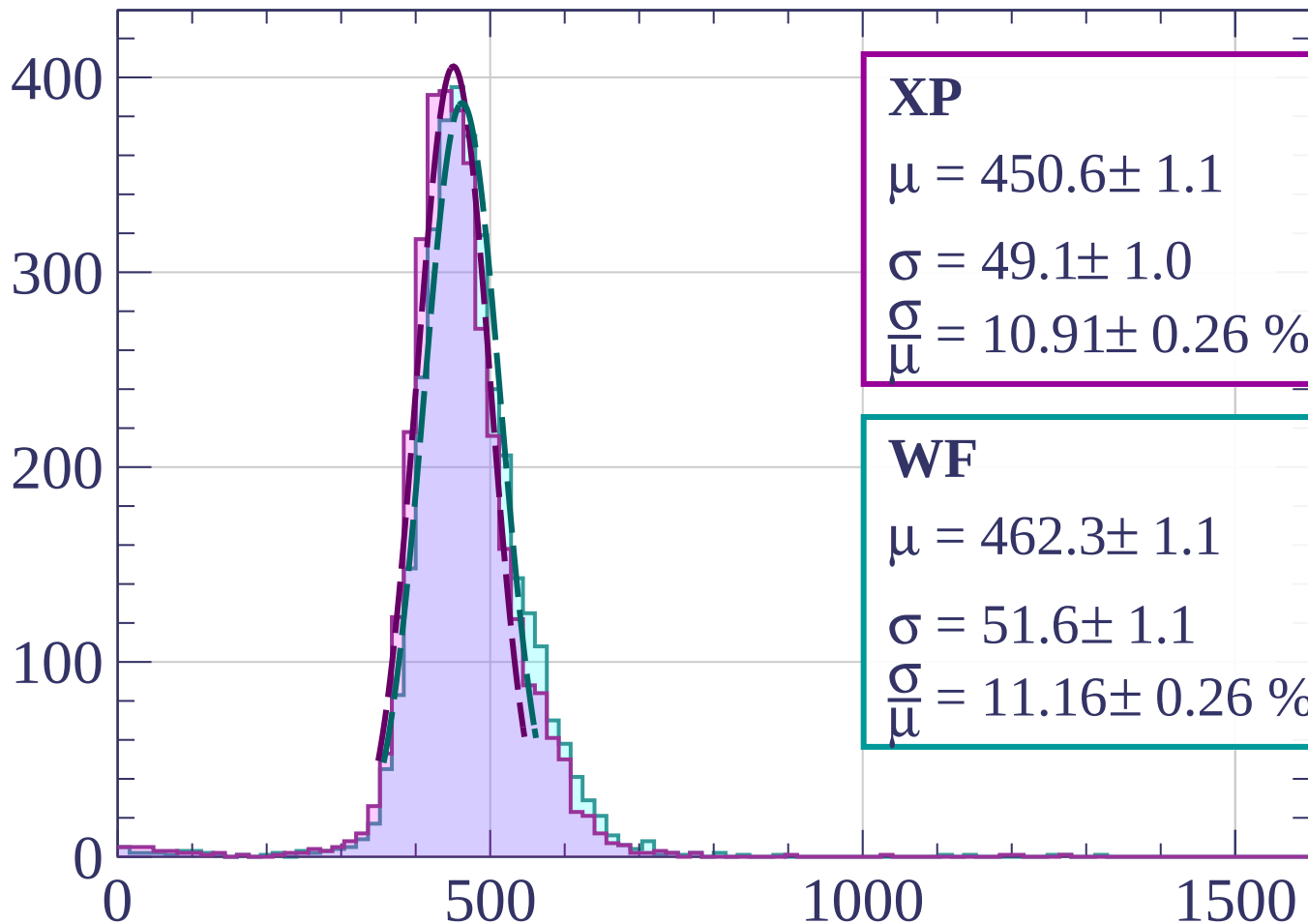
$$\sigma = 52.0 \pm 1.1$$

$$\frac{\sigma}{\mu} = 11.22 \pm 0.26 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1280 < p < -1260$

Count



XP

$$\mu = 450.6 \pm 1.1$$

$$\sigma = 49.1 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.91 \pm 0.26 \%$$

WF

$$\mu = 462.3 \pm 1.1$$

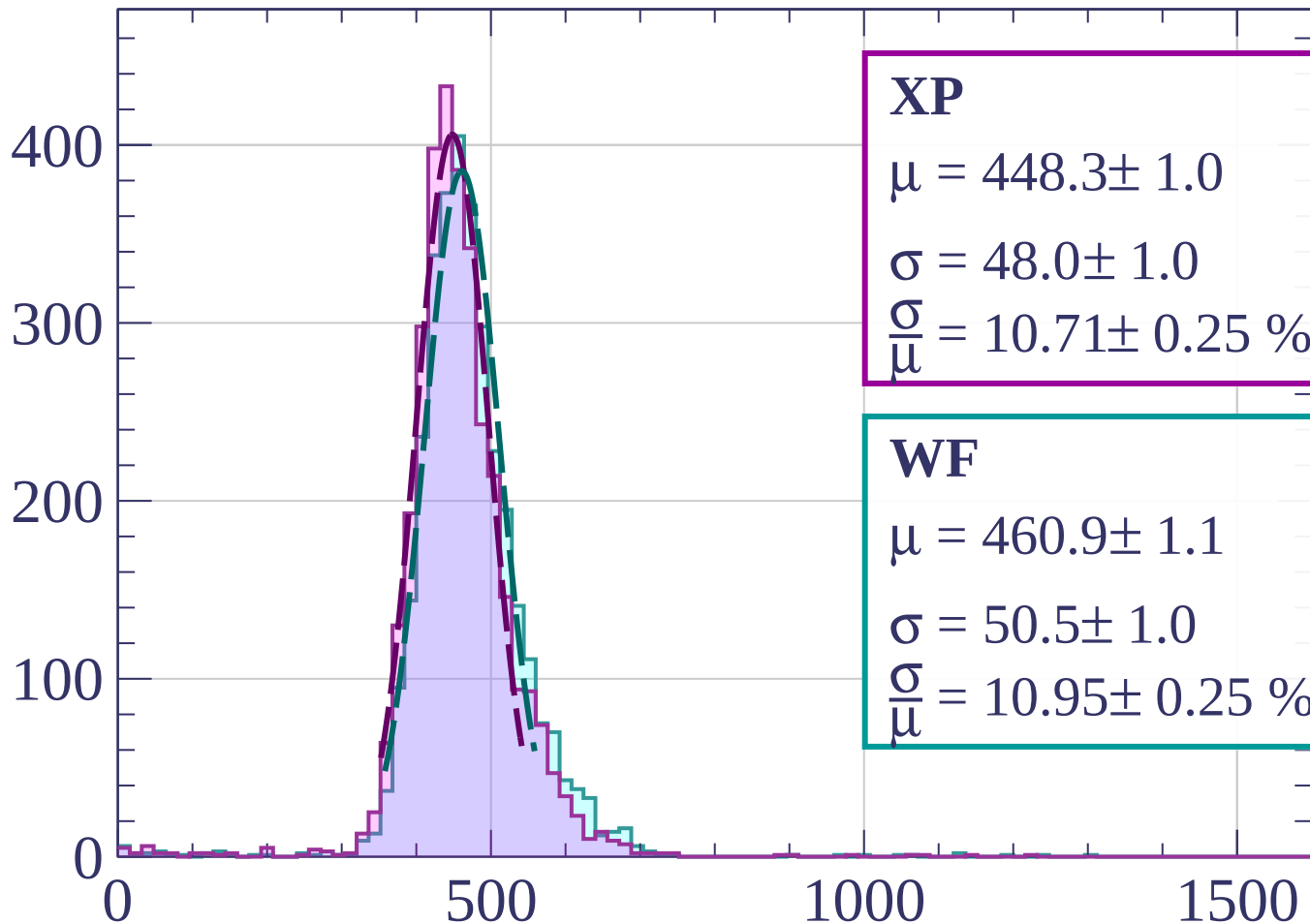
$$\sigma = 51.6 \pm 1.1$$

$$\frac{\sigma}{\mu} = 11.16 \pm 0.26 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1260 < p < -1240$

Count



XP

$$\mu = 448.3 \pm 1.0$$

$$\sigma = 48.0 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.71 \pm 0.25 \%$$

WF

$$\mu = 460.9 \pm 1.1$$

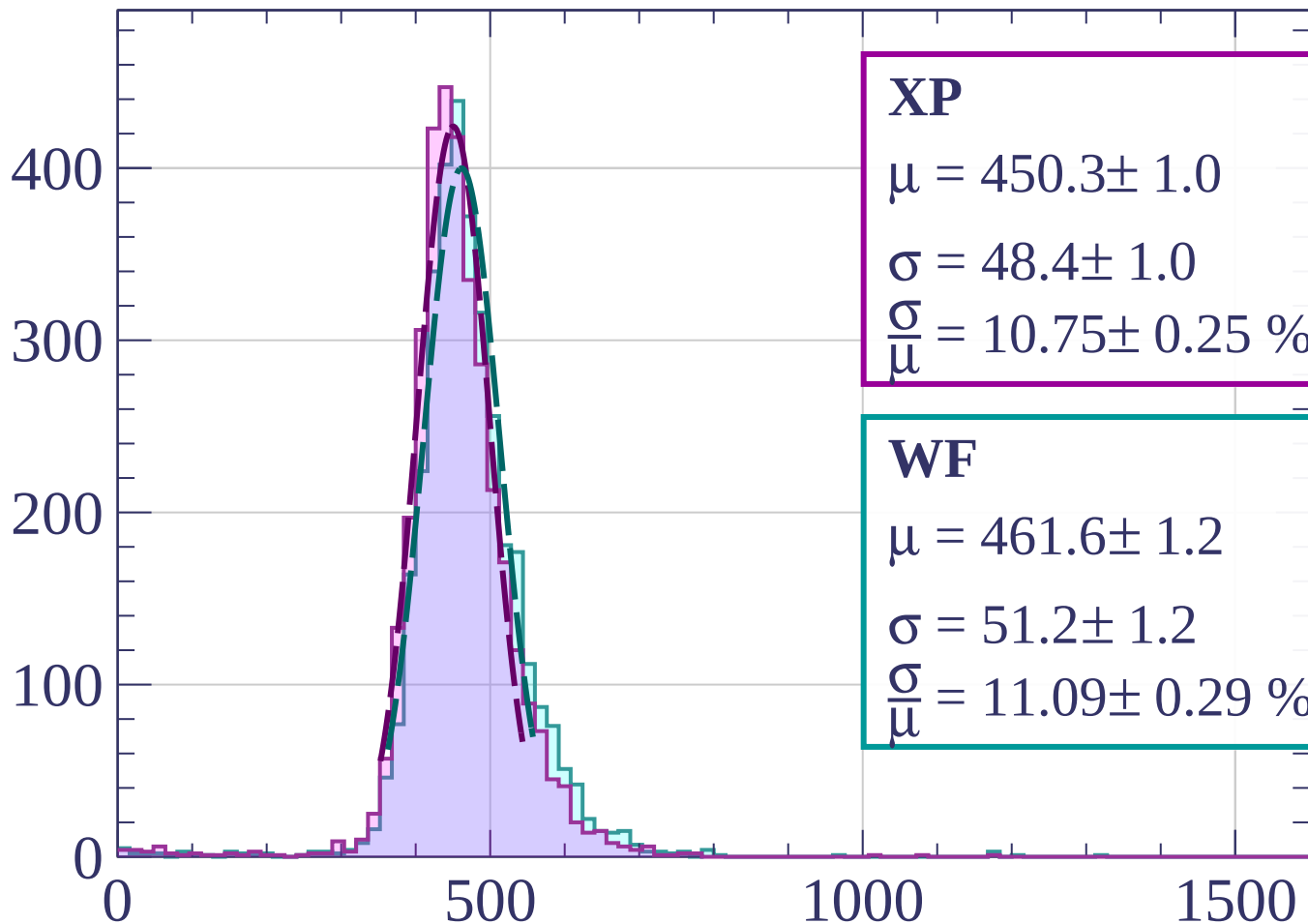
$$\sigma = 50.5 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.95 \pm 0.25 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1240 < p < -1220$

Count



XP

$$\mu = 450.3 \pm 1.0$$

$$\sigma = 48.4 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.75 \pm 0.25 \%$$

WF

$$\mu = 461.6 \pm 1.2$$

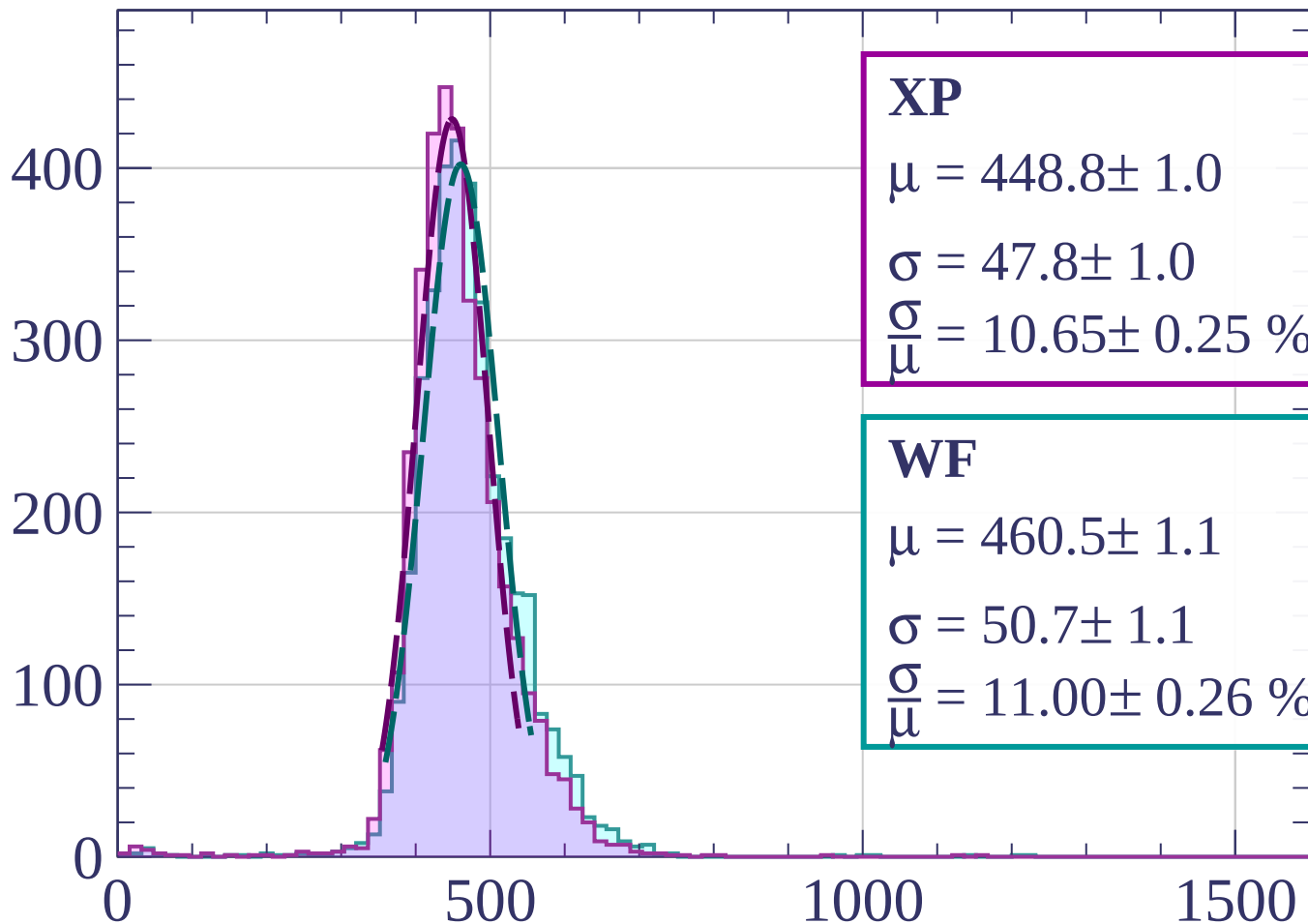
$$\sigma = 51.2 \pm 1.2$$

$$\frac{\sigma}{\mu} = 11.09 \pm 0.29 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1220 < p < -1200$

Count



XP

$$\mu = 448.8 \pm 1.0$$

$$\sigma = 47.8 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.65 \pm 0.25 \%$$

WF

$$\mu = 460.5 \pm 1.1$$

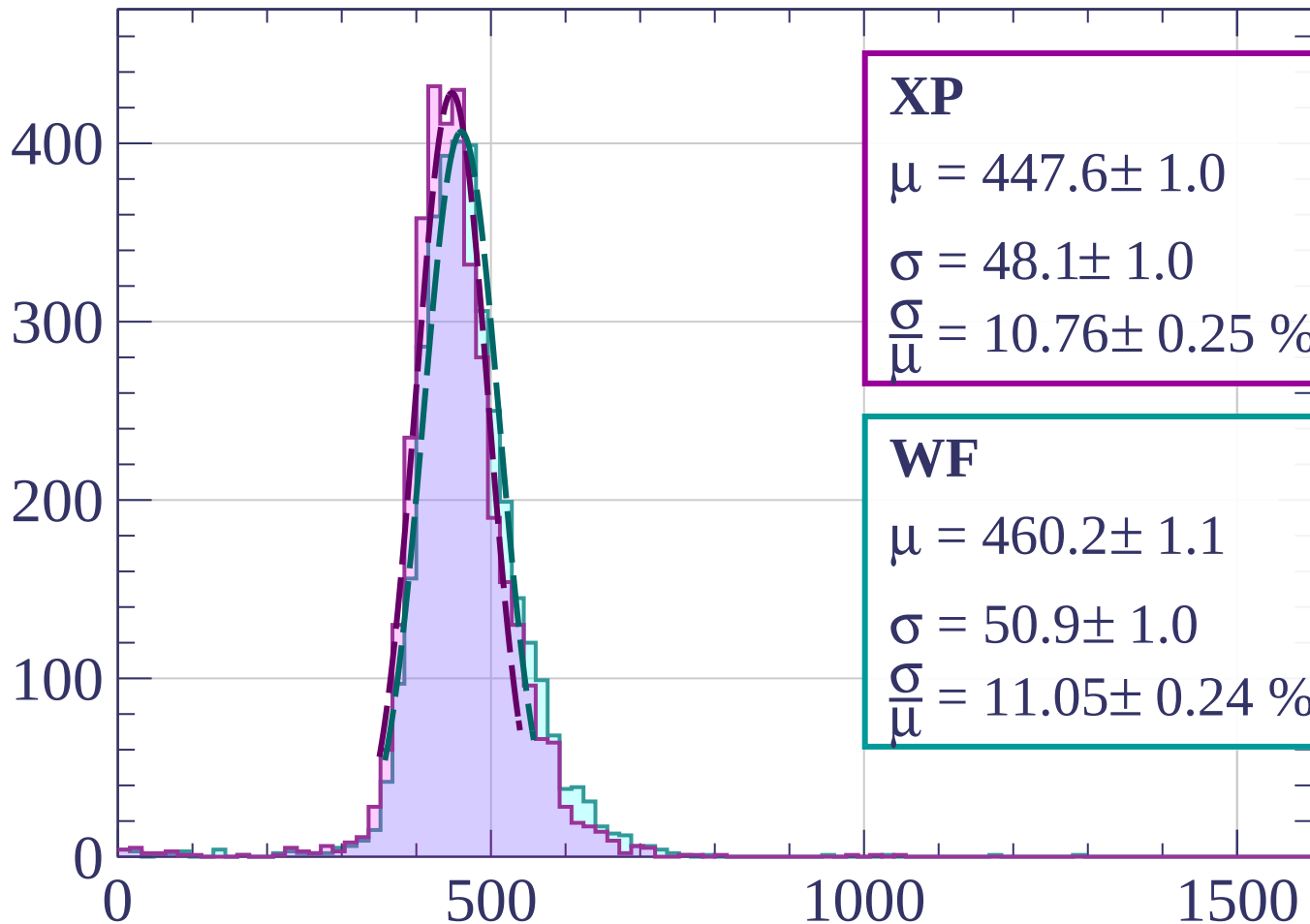
$$\sigma = 50.7 \pm 1.1$$

$$\frac{\sigma}{\mu} = 11.00 \pm 0.26 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1200 < p < -1180$

Count



XP

$$\mu = 447.6 \pm 1.0$$

$$\sigma = 48.1 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.76 \pm 0.25 \%$$

WF

$$\mu = 460.2 \pm 1.1$$

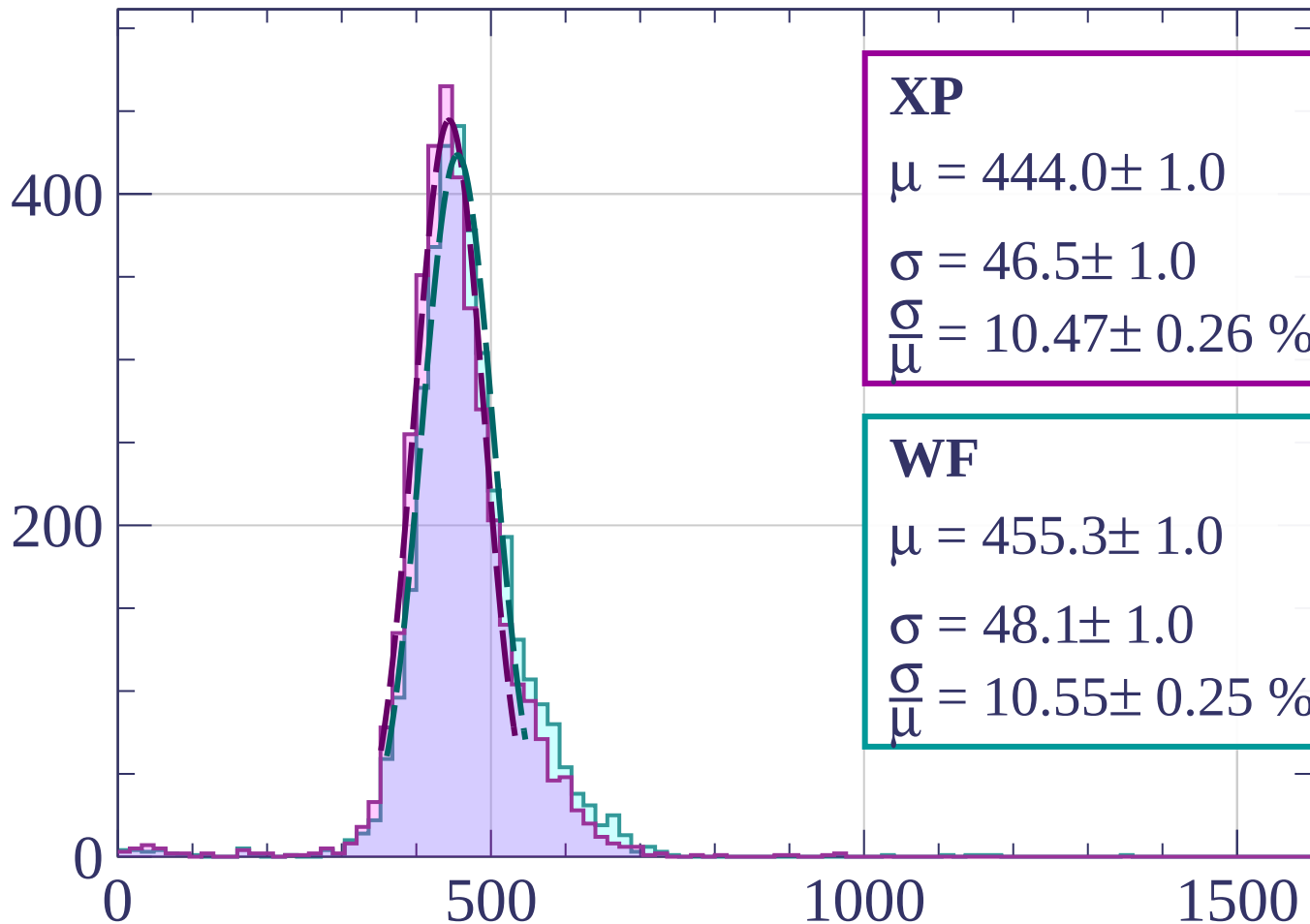
$$\sigma = 50.9 \pm 1.0$$

$$\frac{\sigma}{\mu} = 11.05 \pm 0.24 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1180 < p < -1160$

Count



XP

$$\mu = 444.0 \pm 1.0$$

$$\sigma = 46.5 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.47 \pm 0.26 \%$$

WF

$$\mu = 455.3 \pm 1.0$$

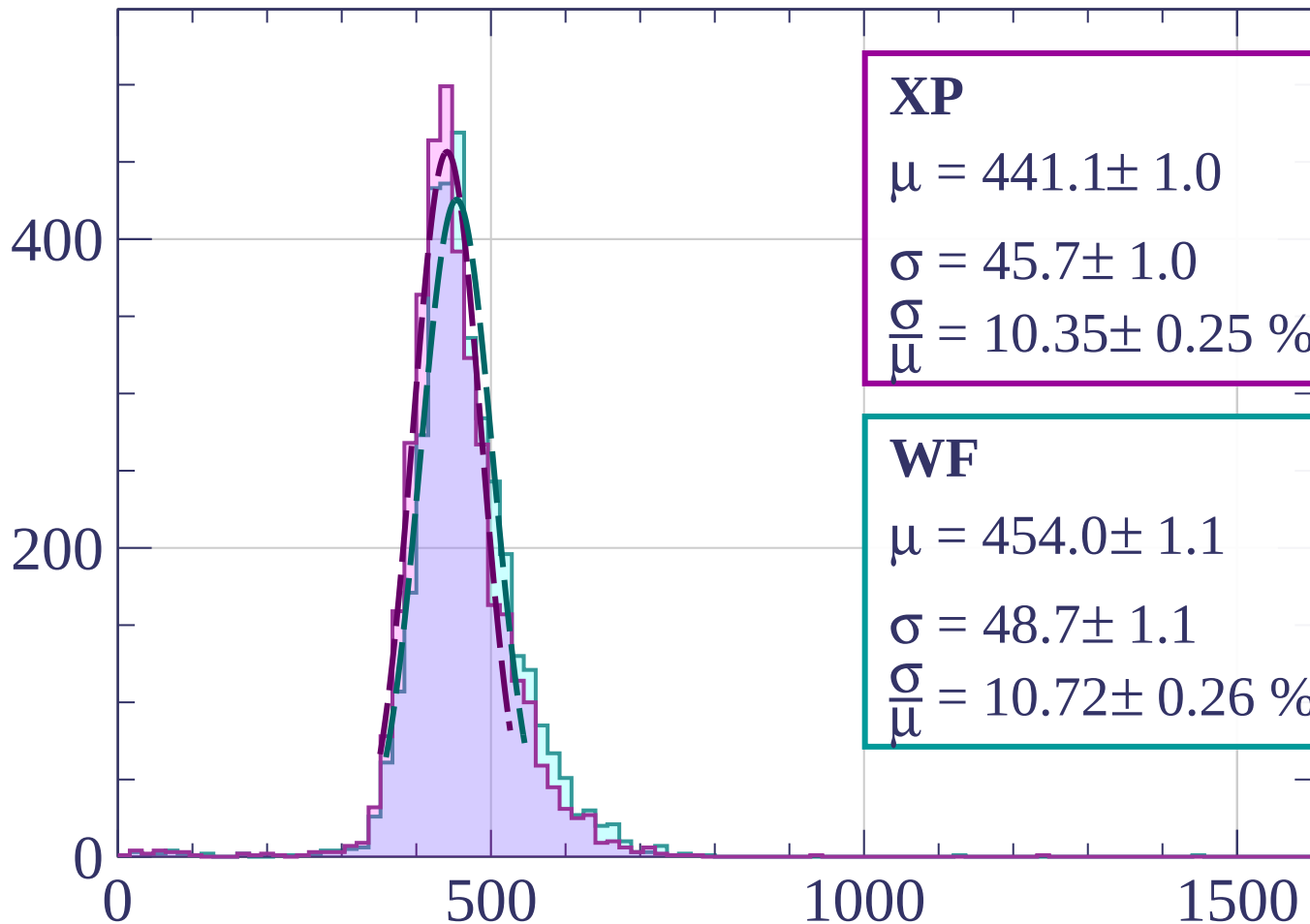
$$\sigma = 48.1 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.55 \pm 0.25 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1160 < p < -1140$

Count



XP

$$\mu = 441.1 \pm 1.0$$

$$\sigma = 45.7 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.35 \pm 0.25 \%$$

WF

$$\mu = 454.0 \pm 1.1$$

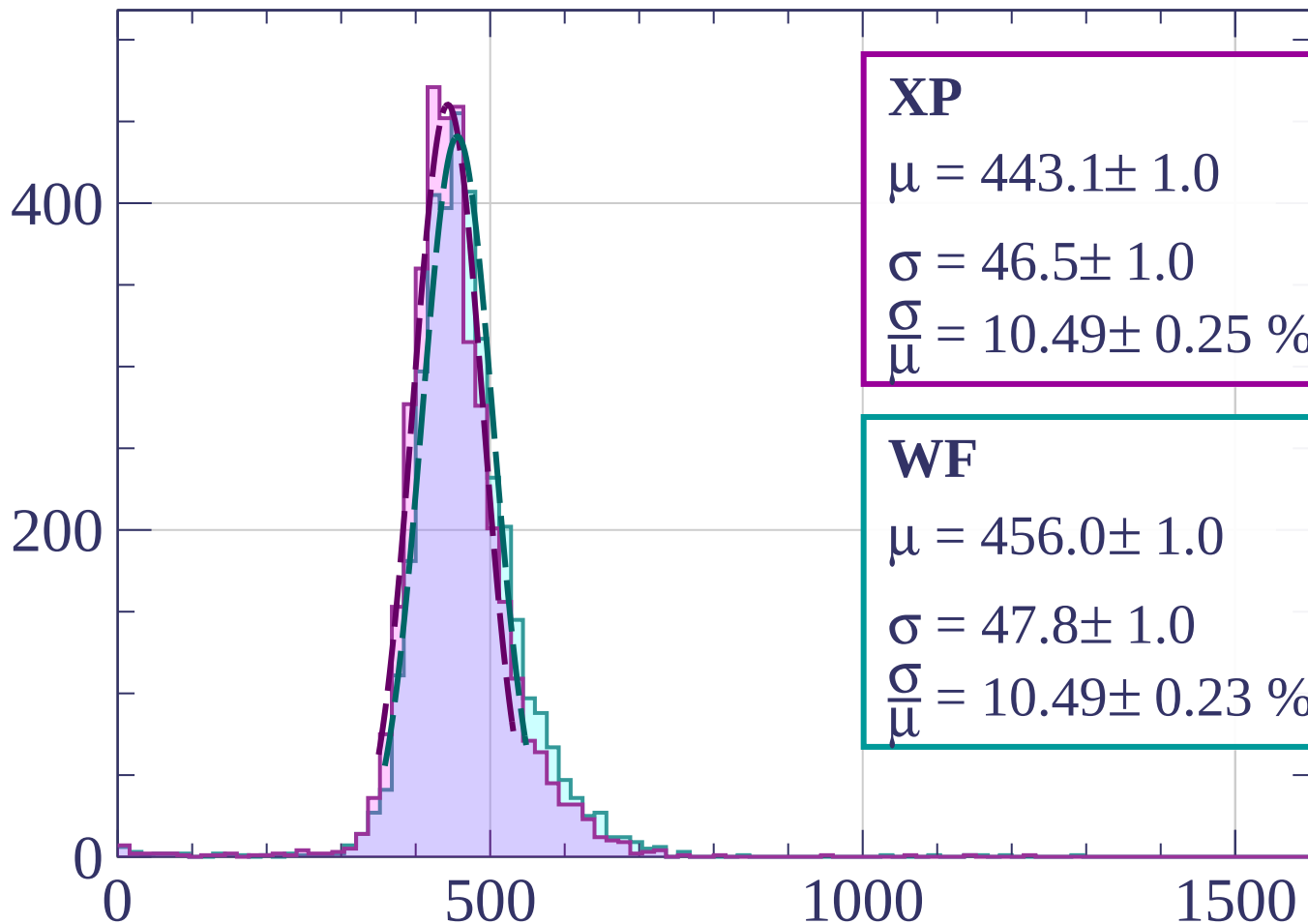
$$\sigma = 48.7 \pm 1.1$$

$$\frac{\sigma}{\mu} = 10.72 \pm 0.26 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1140 < p < -1120$

Count



XP

$$\mu = 443.1 \pm 1.0$$

$$\sigma = 46.5 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.49 \pm 0.25 \%$$

WF

$$\mu = 456.0 \pm 1.0$$

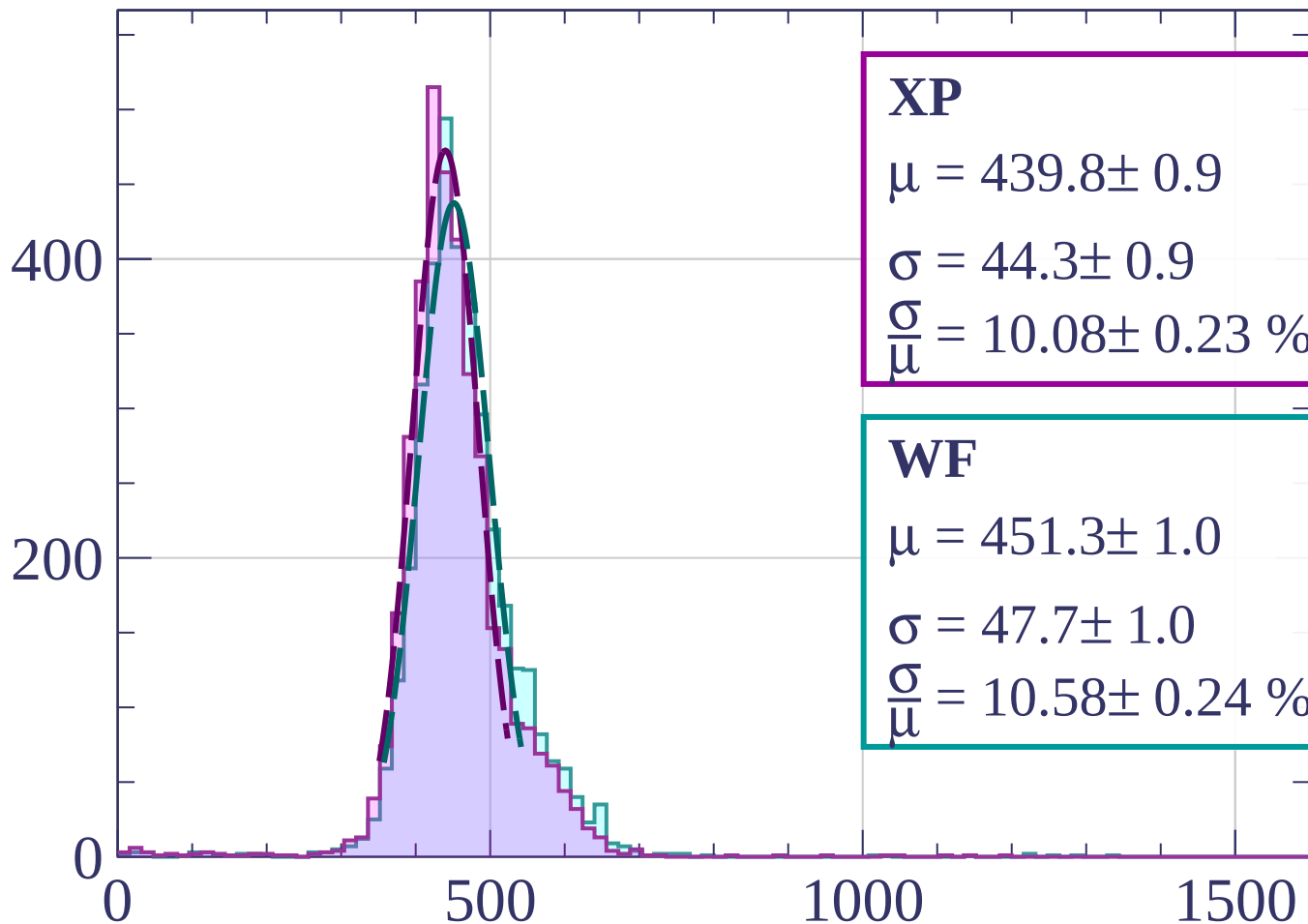
$$\sigma = 47.8 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.49 \pm 0.23 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1120 < p < -1100$

Count



XP

$$\mu = 439.8 \pm 0.9$$

$$\sigma = 44.3 \pm 0.9$$

$$\frac{\sigma}{\mu} = 10.08 \pm 0.23 \%$$

WF

$$\mu = 451.3 \pm 1.0$$

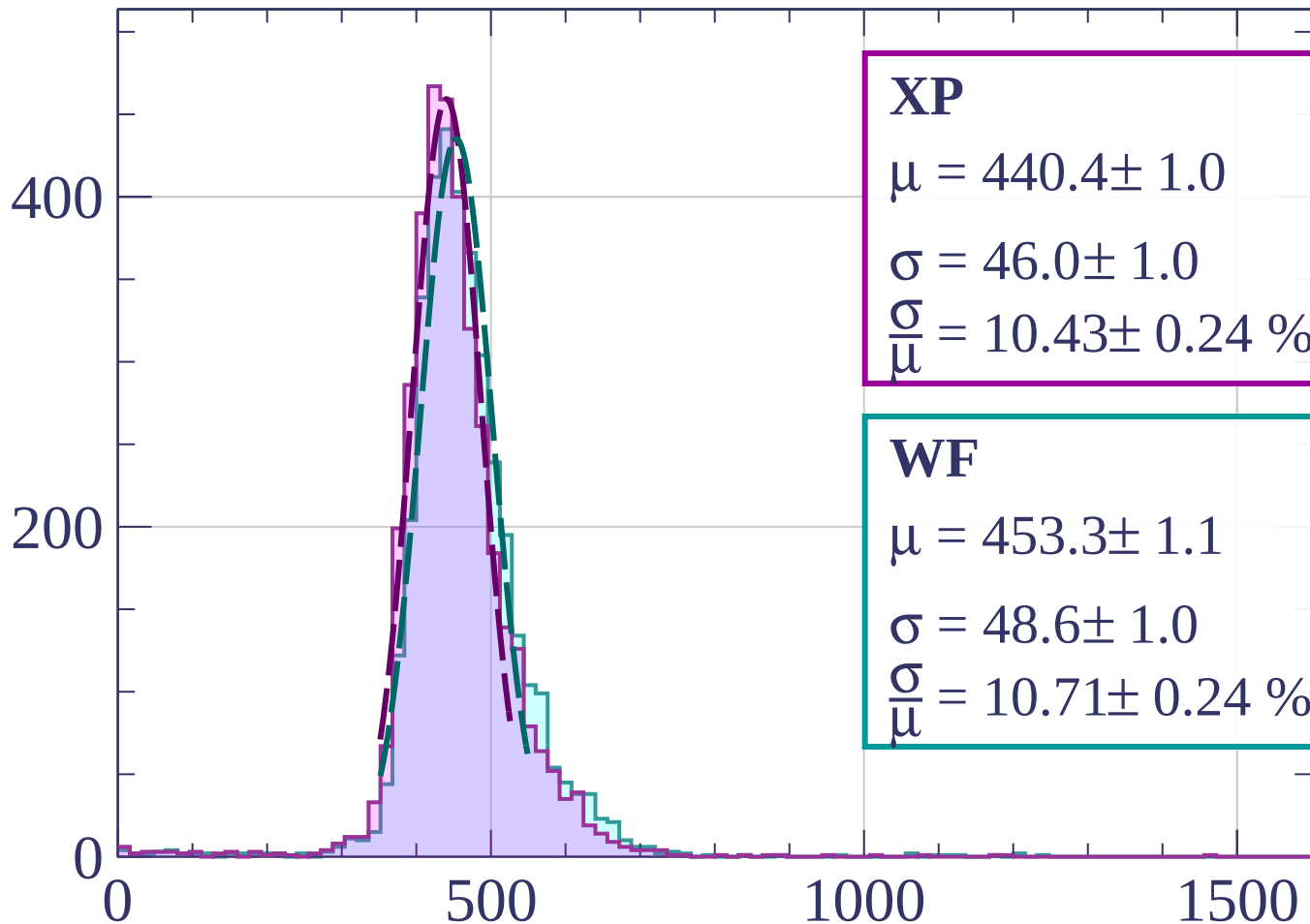
$$\sigma = 47.7 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.58 \pm 0.24 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1100 < p < -1080$

Count



XP

$$\mu = 440.4 \pm 1.0$$

$$\sigma = 46.0 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.43 \pm 0.24 \%$$

WF

$$\mu = 453.3 \pm 1.1$$

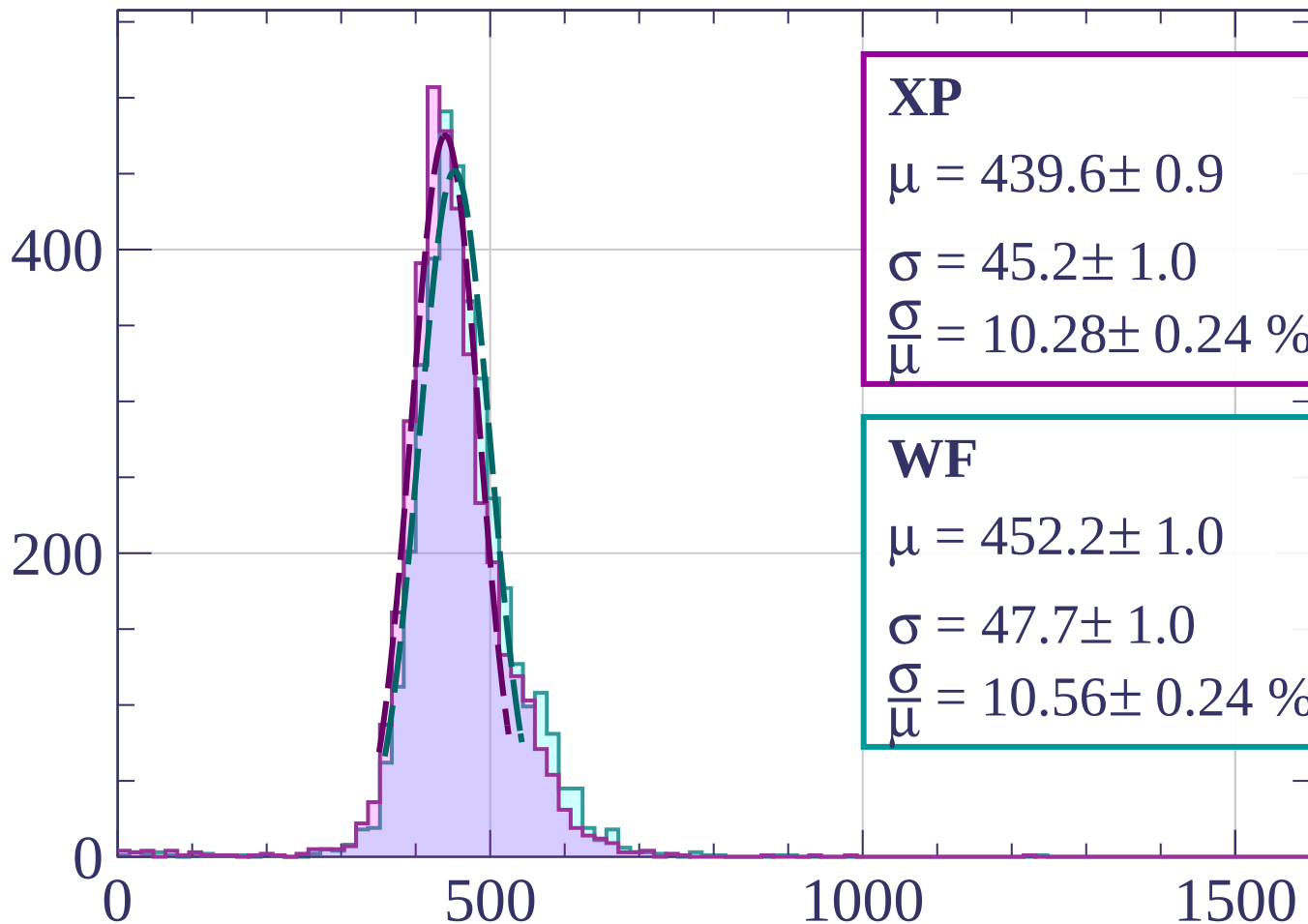
$$\sigma = 48.6 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.71 \pm 0.24 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1080 < p < -1060$

Count



XP

$$\mu = 439.6 \pm 0.9$$

$$\sigma = 45.2 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.28 \pm 0.24 \%$$

WF

$$\mu = 452.2 \pm 1.0$$

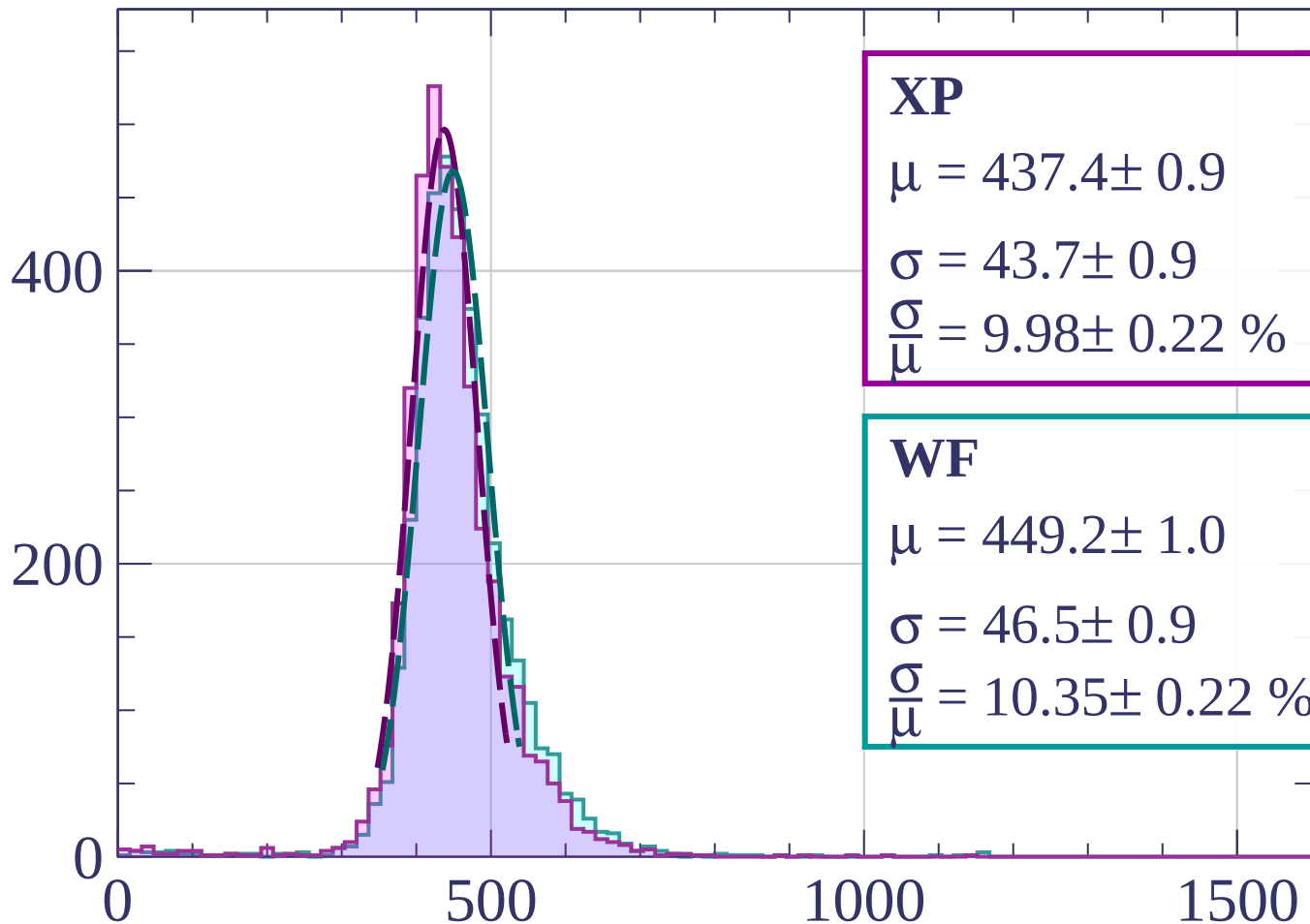
$$\sigma = 47.7 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.56 \pm 0.24 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1060 < p < -1040$

Count



XP

$$\mu = 437.4 \pm 0.9$$

$$\sigma = 43.7 \pm 0.9$$

$$\frac{\sigma}{\mu} = 9.98 \pm 0.22 \%$$

WF

$$\mu = 449.2 \pm 1.0$$

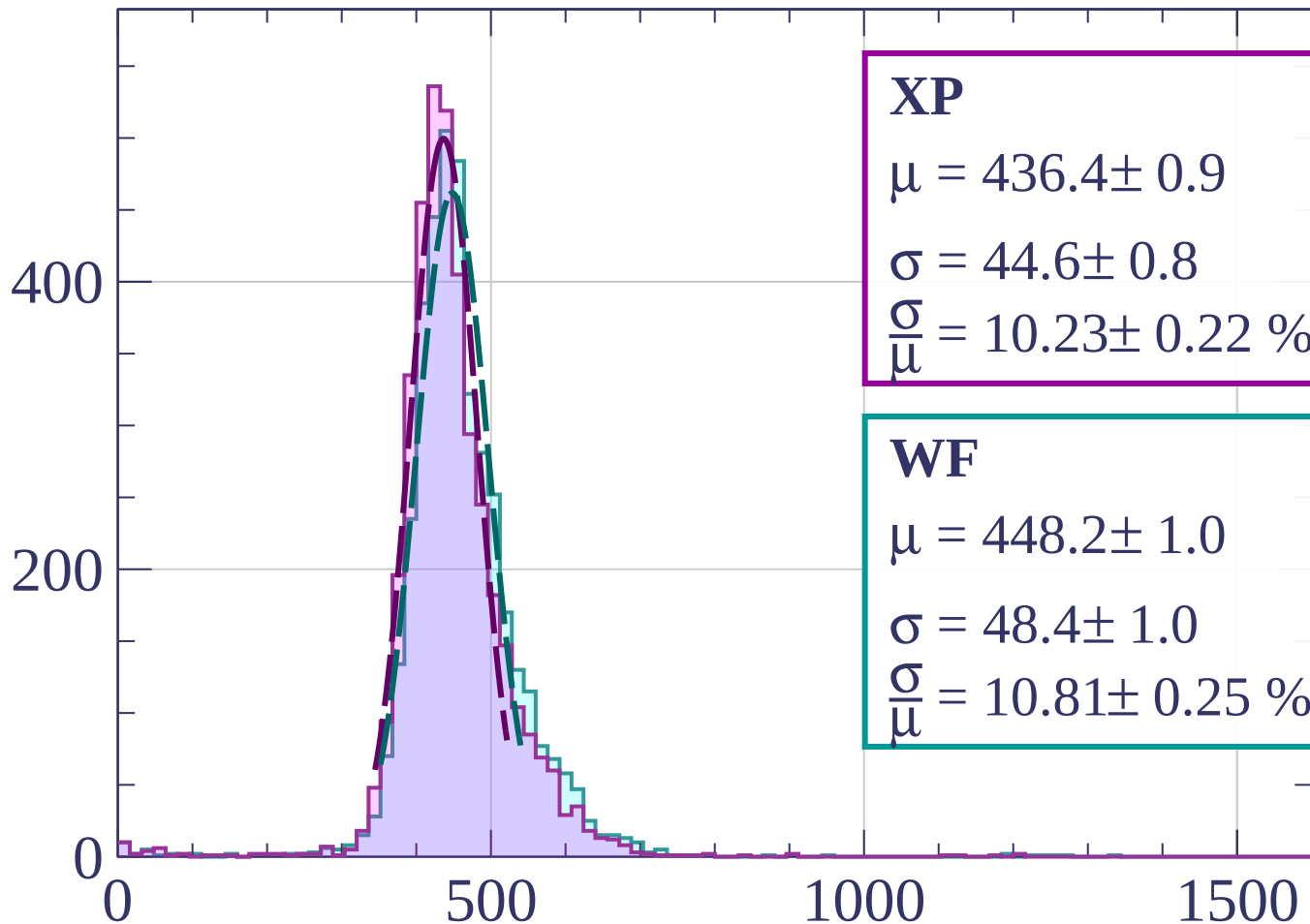
$$\sigma = 46.5 \pm 0.9$$

$$\frac{\sigma}{\mu} = 10.35 \pm 0.22 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1040 < p < -1020$

Count



XP

$$\mu = 436.4 \pm 0.9$$

$$\sigma = 44.6 \pm 0.8$$

$$\frac{\sigma}{\mu} = 10.23 \pm 0.22 \%$$

WF

$$\mu = 448.2 \pm 1.0$$

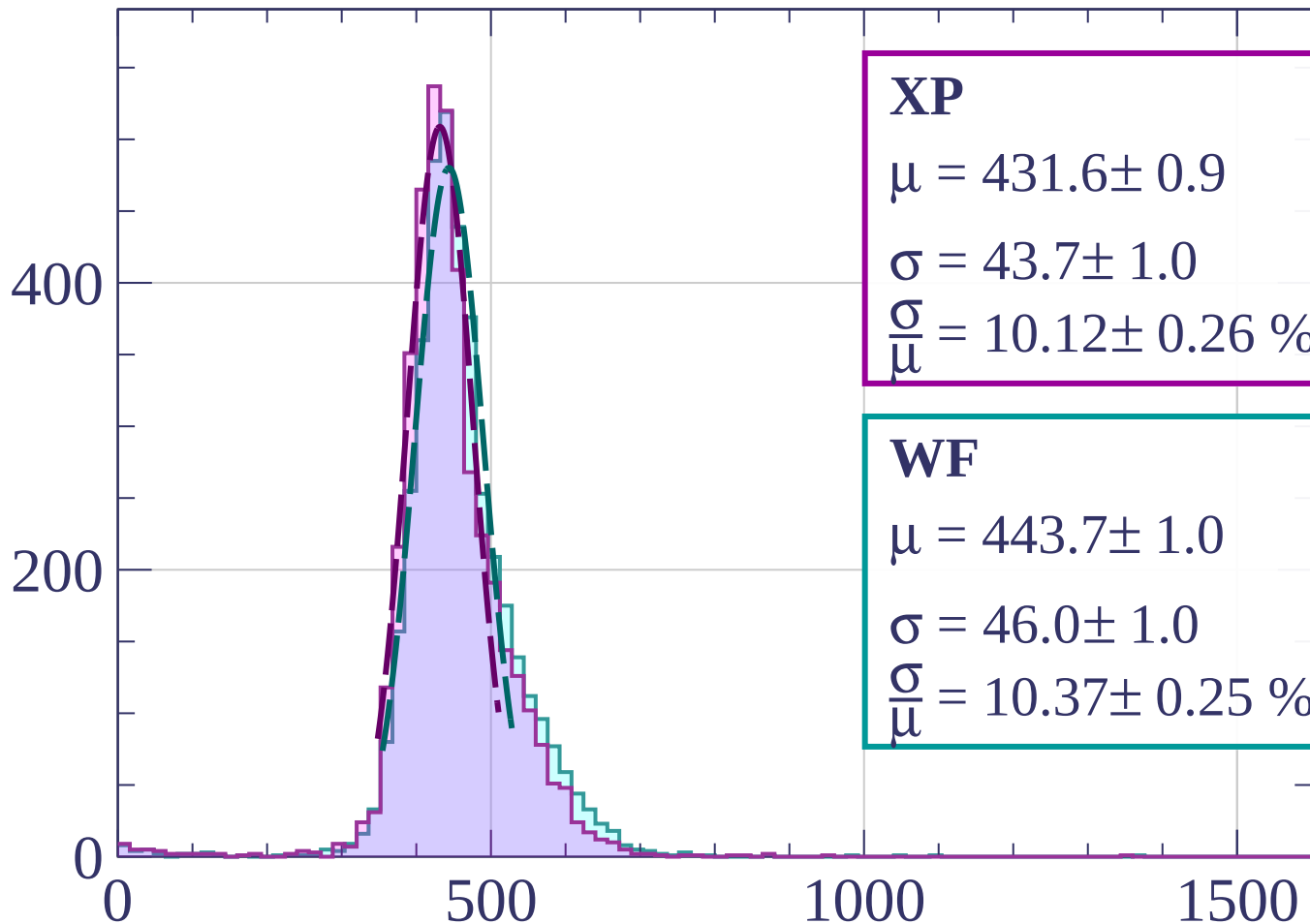
$$\sigma = 48.4 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.81 \pm 0.25 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1020 < p < -1000$

Count



XP

$$\mu = 431.6 \pm 0.9$$

$$\sigma = 43.7 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.12 \pm 0.26 \%$$

WF

$$\mu = 443.7 \pm 1.0$$

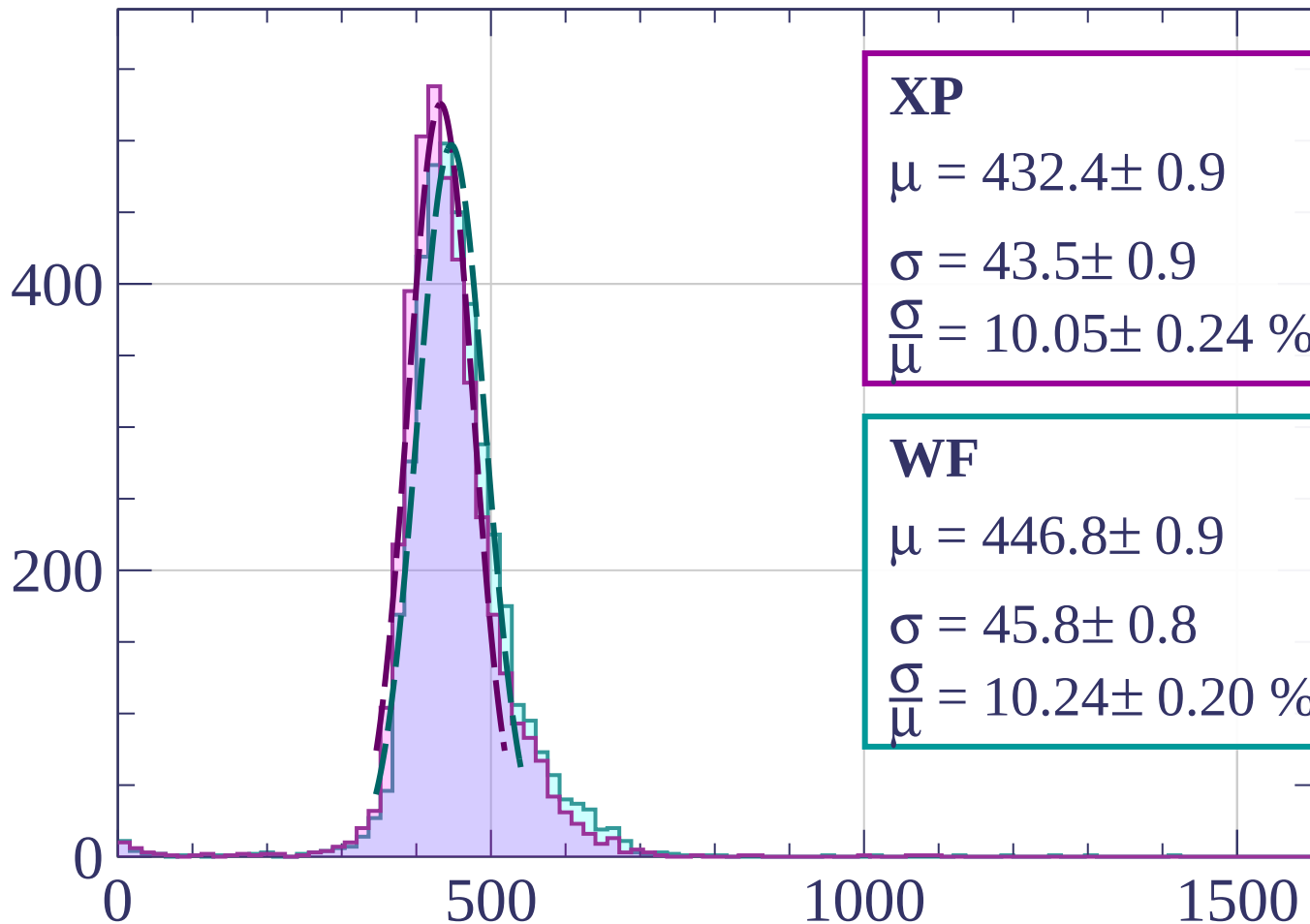
$$\sigma = 46.0 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.37 \pm 0.25 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-1000 < p < -980$

Count



XP

$$\mu = 432.4 \pm 0.9$$

$$\sigma = 43.5 \pm 0.9$$

$$\frac{\sigma}{\mu} = 10.05 \pm 0.24 \%$$

WF

$$\mu = 446.8 \pm 0.9$$

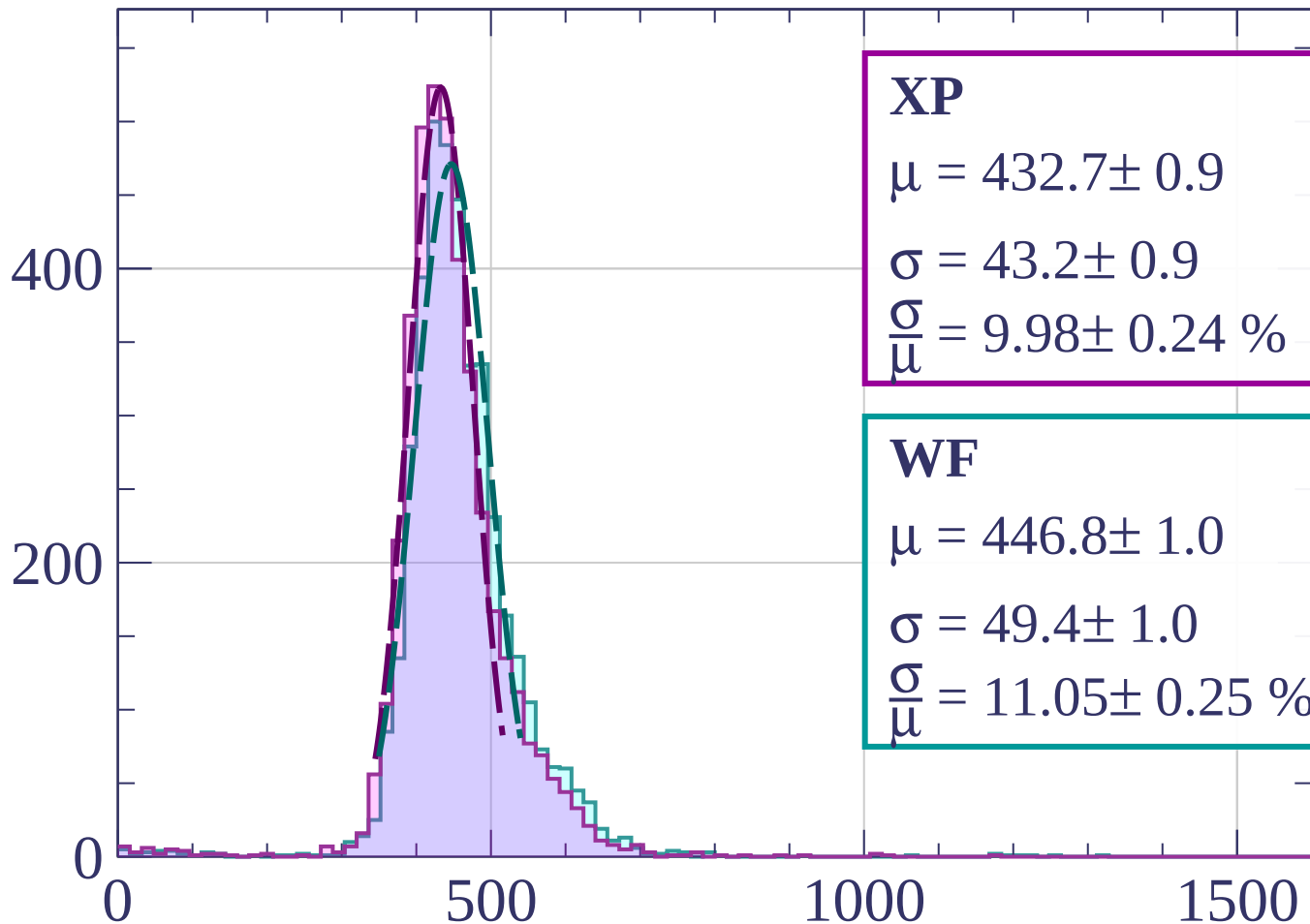
$$\sigma = 45.8 \pm 0.8$$

$$\frac{\sigma}{\mu} = 10.24 \pm 0.20 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-980 < p < -960$

Count



XP

$$\mu = 432.7 \pm 0.9$$

$$\sigma = 43.2 \pm 0.9$$

$$\frac{\sigma}{\mu} = 9.98 \pm 0.24 \%$$

WF

$$\mu = 446.8 \pm 1.0$$

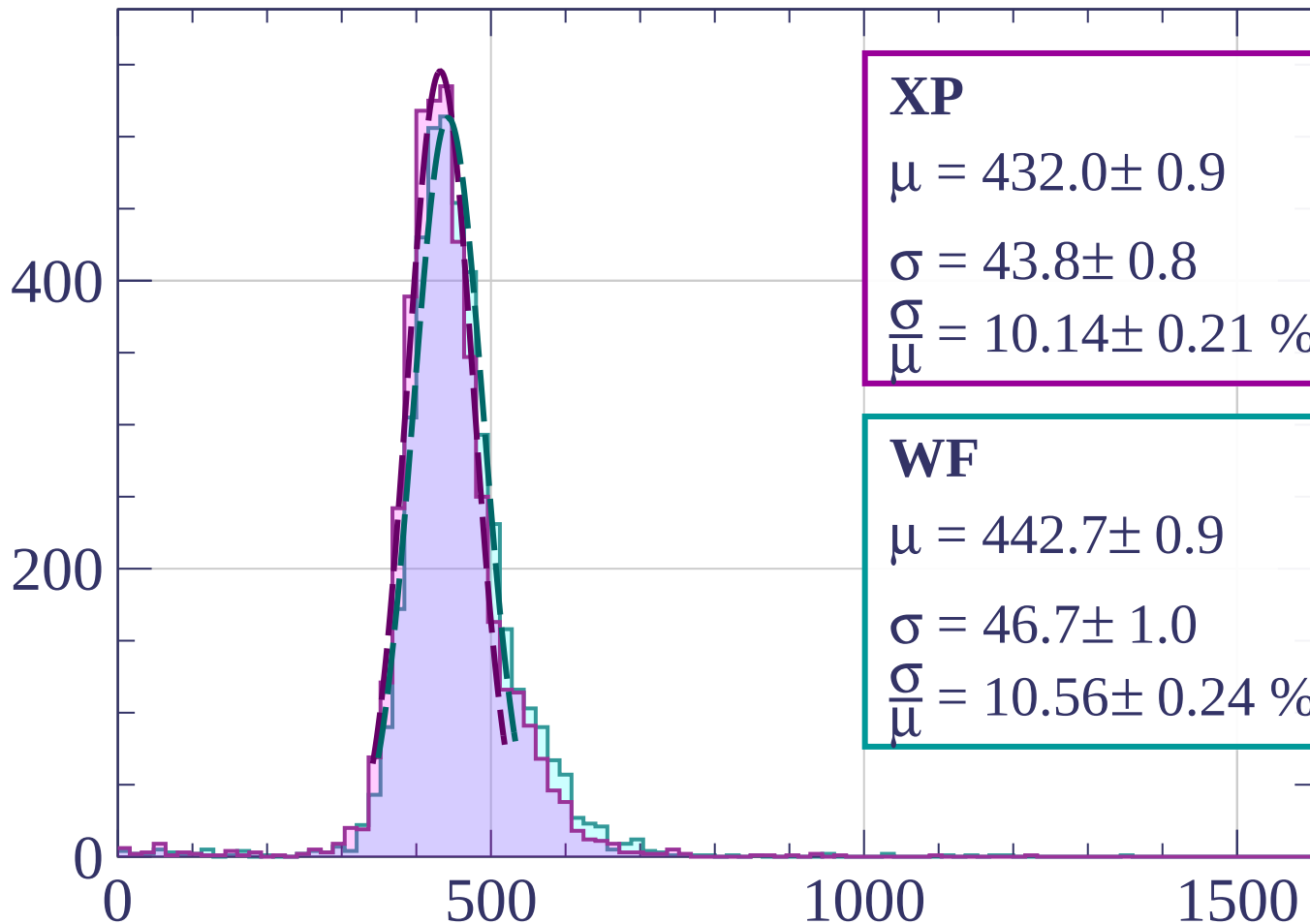
$$\sigma = 49.4 \pm 1.0$$

$$\frac{\sigma}{\mu} = 11.05 \pm 0.25 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-960 < p < -940$

Count



XP

$$\mu = 432.0 \pm 0.9$$

$$\sigma = 43.8 \pm 0.8$$

$$\frac{\sigma}{\mu} = 10.14 \pm 0.21 \%$$

WF

$$\mu = 442.7 \pm 0.9$$

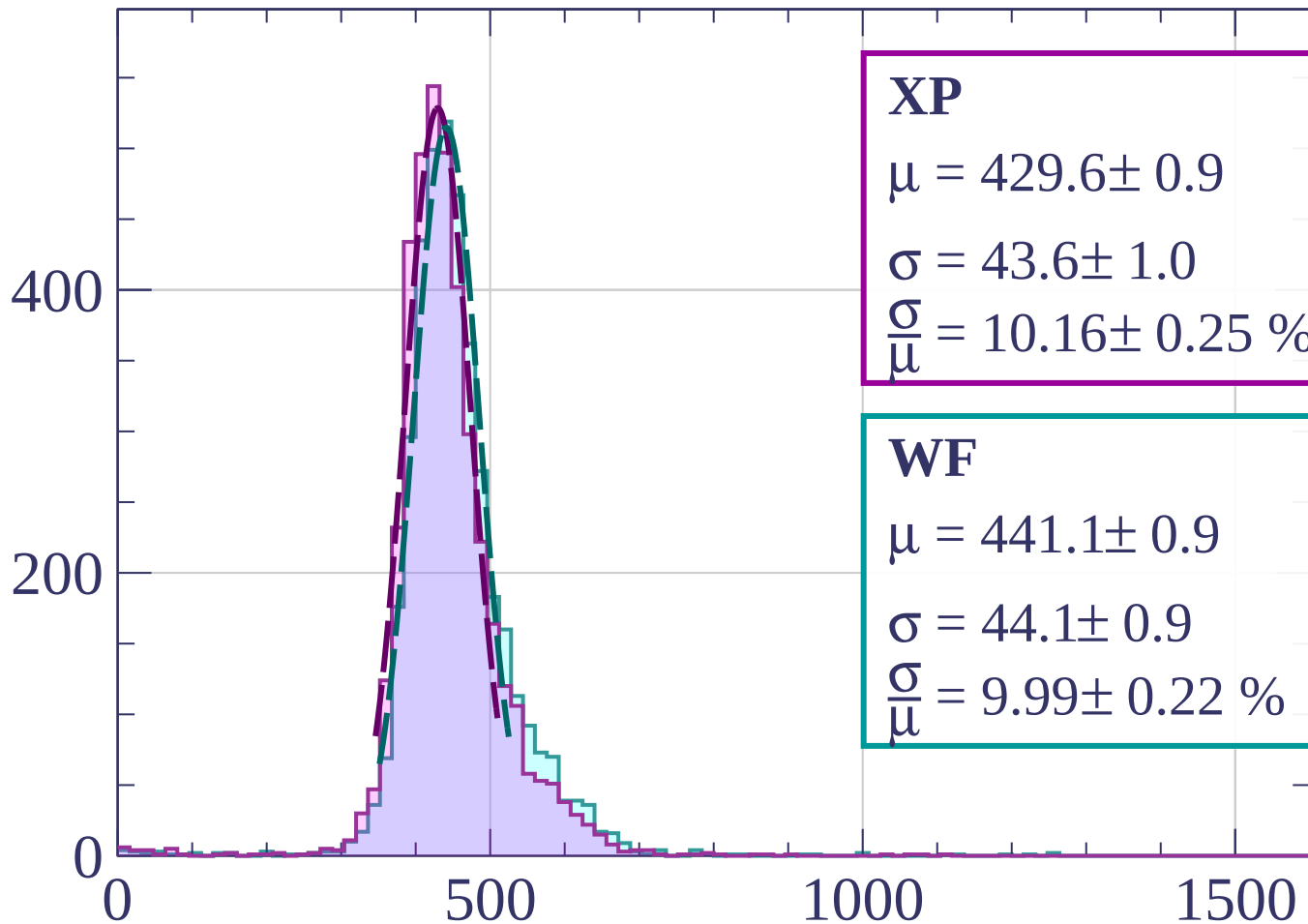
$$\sigma = 46.7 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.56 \pm 0.24 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-940 < p < -920$

Count



XP

$$\mu = 429.6 \pm 0.9$$

$$\sigma = 43.6 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.16 \pm 0.25 \%$$

WF

$$\mu = 441.1 \pm 0.9$$

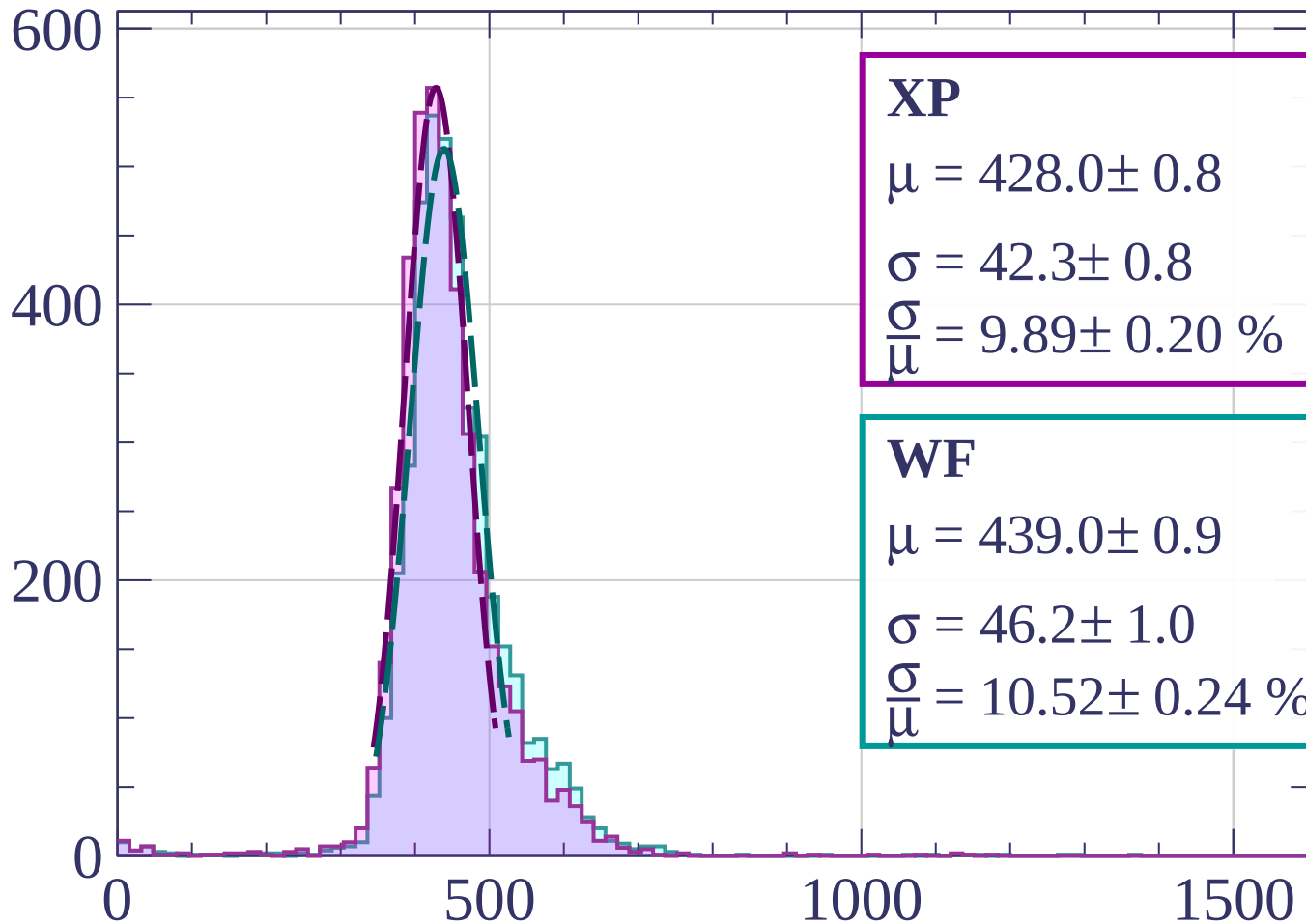
$$\sigma = 44.1 \pm 0.9$$

$$\frac{\sigma}{\mu} = 9.99 \pm 0.22 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-920 < p < -900$

Count



XP

$$\mu = 428.0 \pm 0.8$$

$$\sigma = 42.3 \pm 0.8$$

$$\frac{\sigma}{\mu} = 9.89 \pm 0.20 \%$$

WF

$$\mu = 439.0 \pm 0.9$$

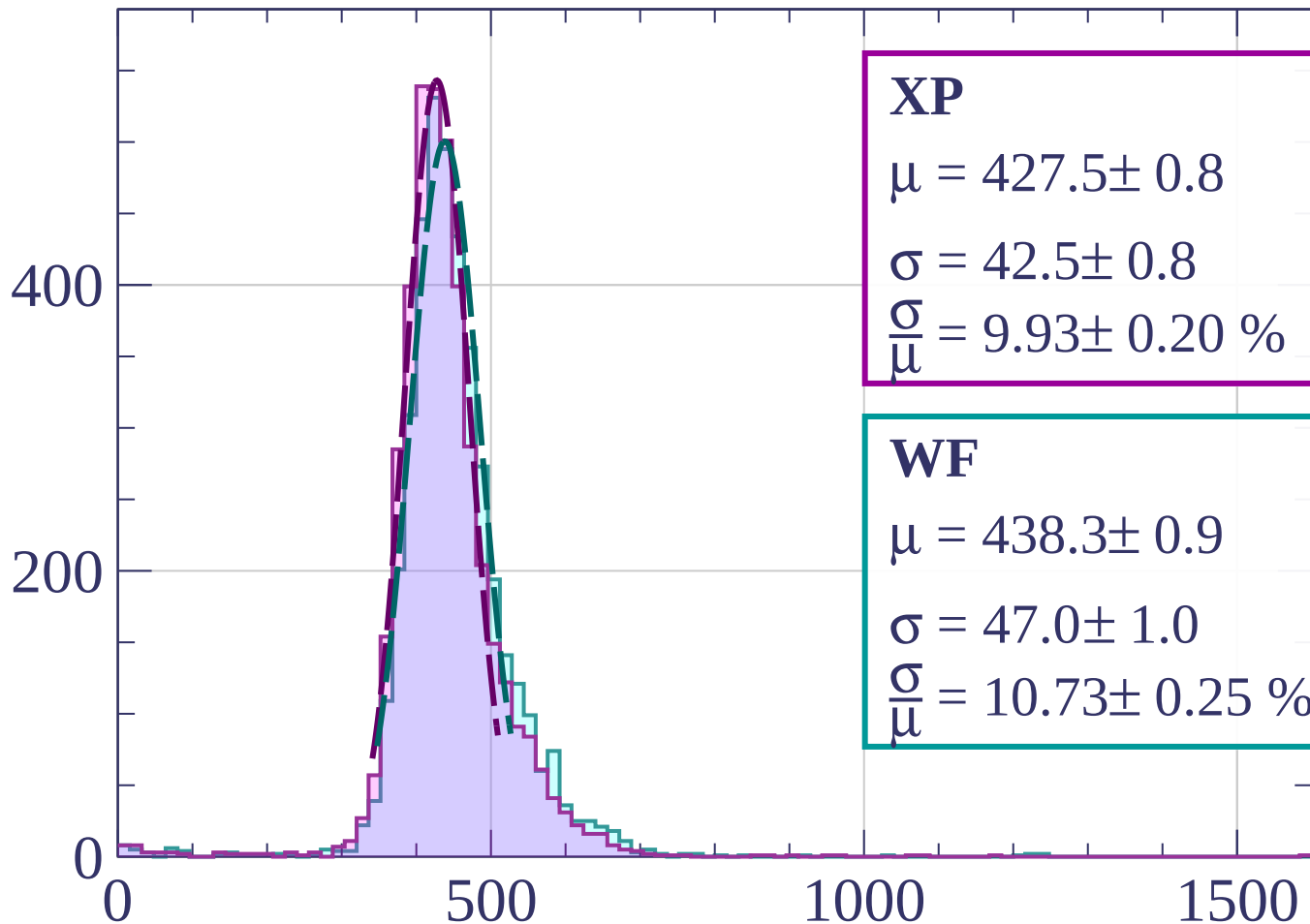
$$\sigma = 46.2 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.52 \pm 0.24 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-900 < p < -880$

Count



XP

$$\mu = 427.5 \pm 0.8$$

$$\sigma = 42.5 \pm 0.8$$

$$\frac{\sigma}{\mu} = 9.93 \pm 0.20 \%$$

WF

$$\mu = 438.3 \pm 0.9$$

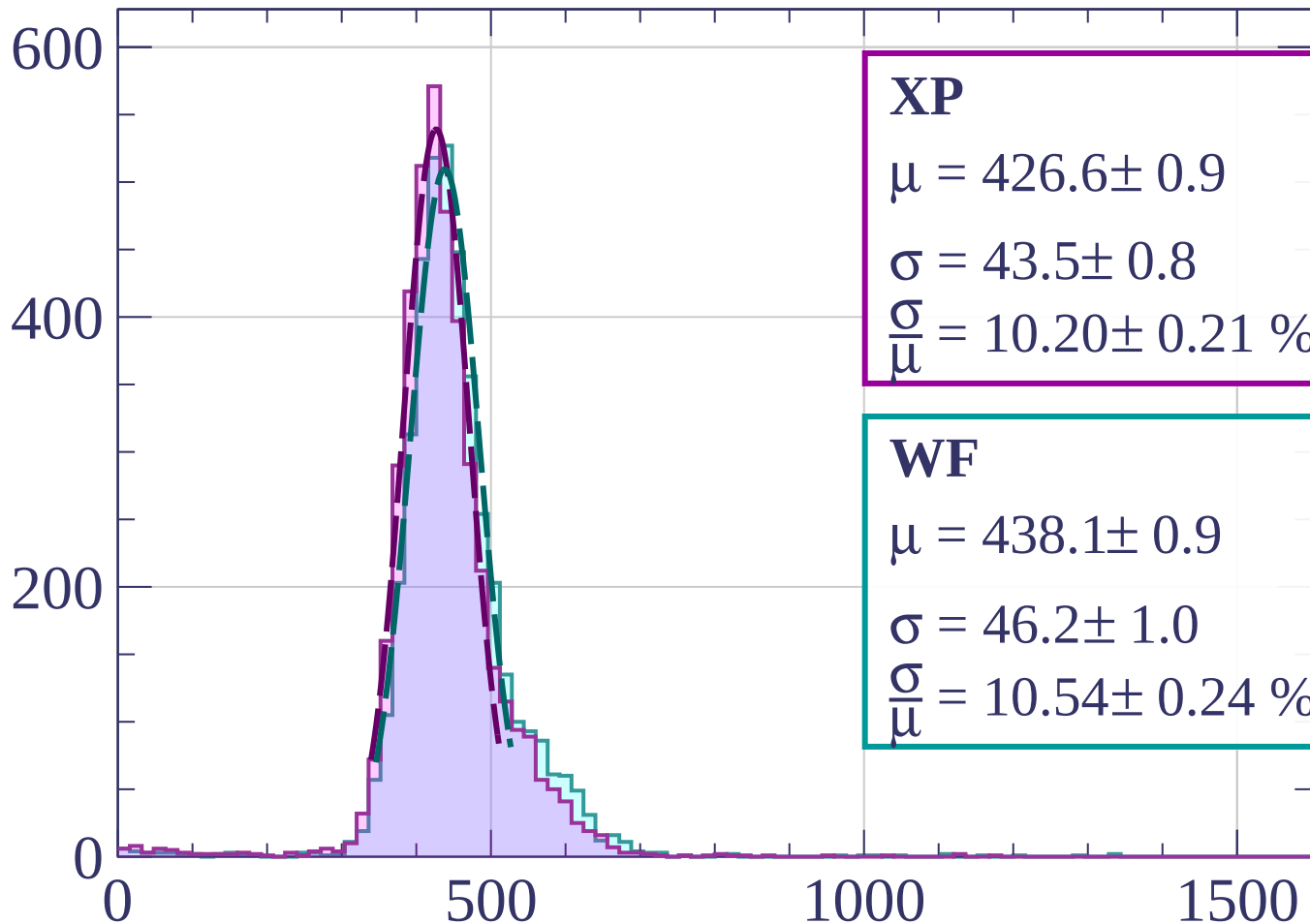
$$\sigma = 47.0 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.73 \pm 0.25 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-880 < p < -860$

Count



XP

$$\mu = 426.6 \pm 0.9$$

$$\sigma = 43.5 \pm 0.8$$

$$\frac{\sigma}{\mu} = 10.20 \pm 0.21 \%$$

WF

$$\mu = 438.1 \pm 0.9$$

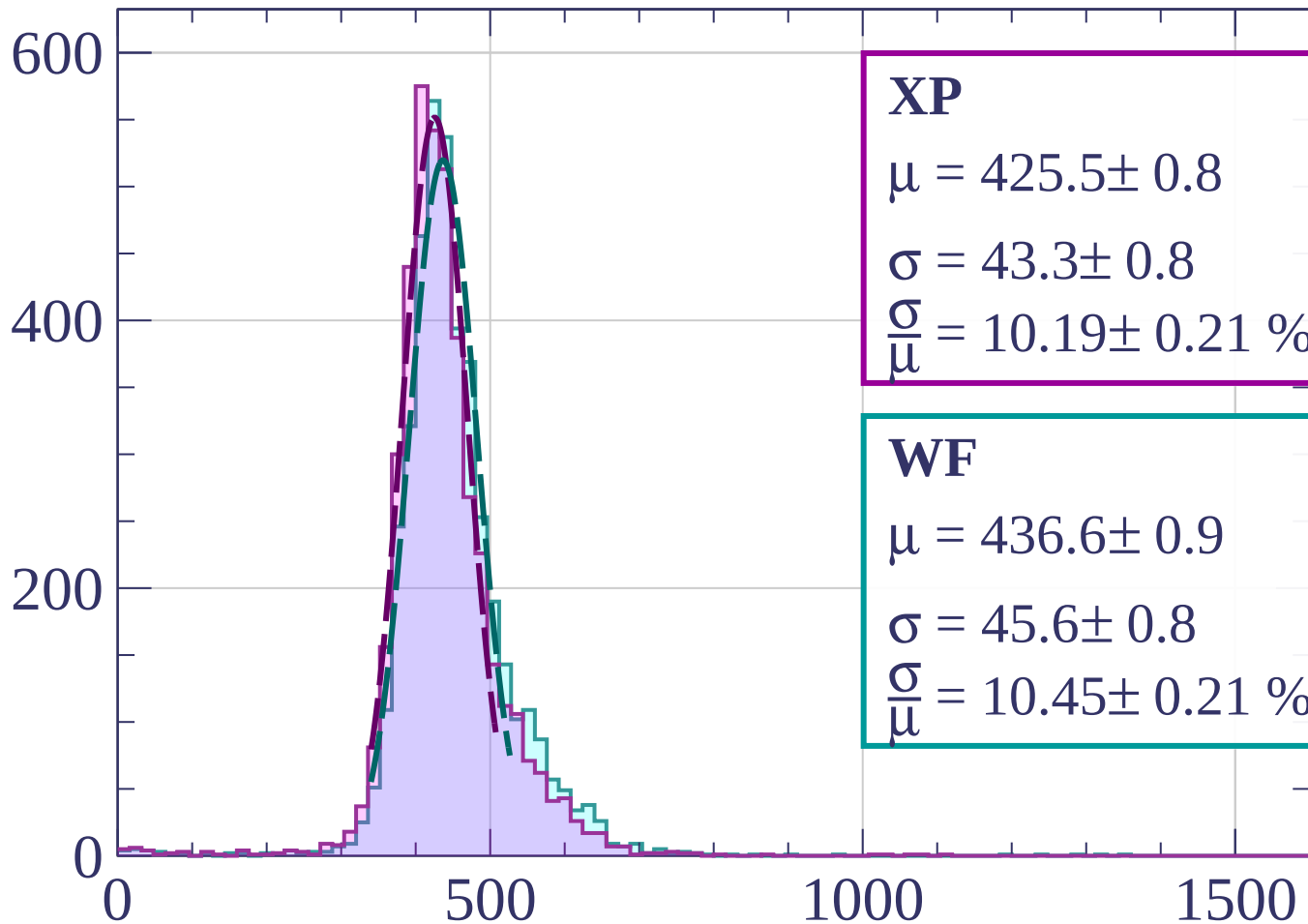
$$\sigma = 46.2 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.54 \pm 0.24 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-860 < p < -840$

Count



XP

$$\mu = 425.5 \pm 0.8$$

$$\sigma = 43.3 \pm 0.8$$

$$\frac{\sigma}{\mu} = 10.19 \pm 0.21 \%$$

WF

$$\mu = 436.6 \pm 0.9$$

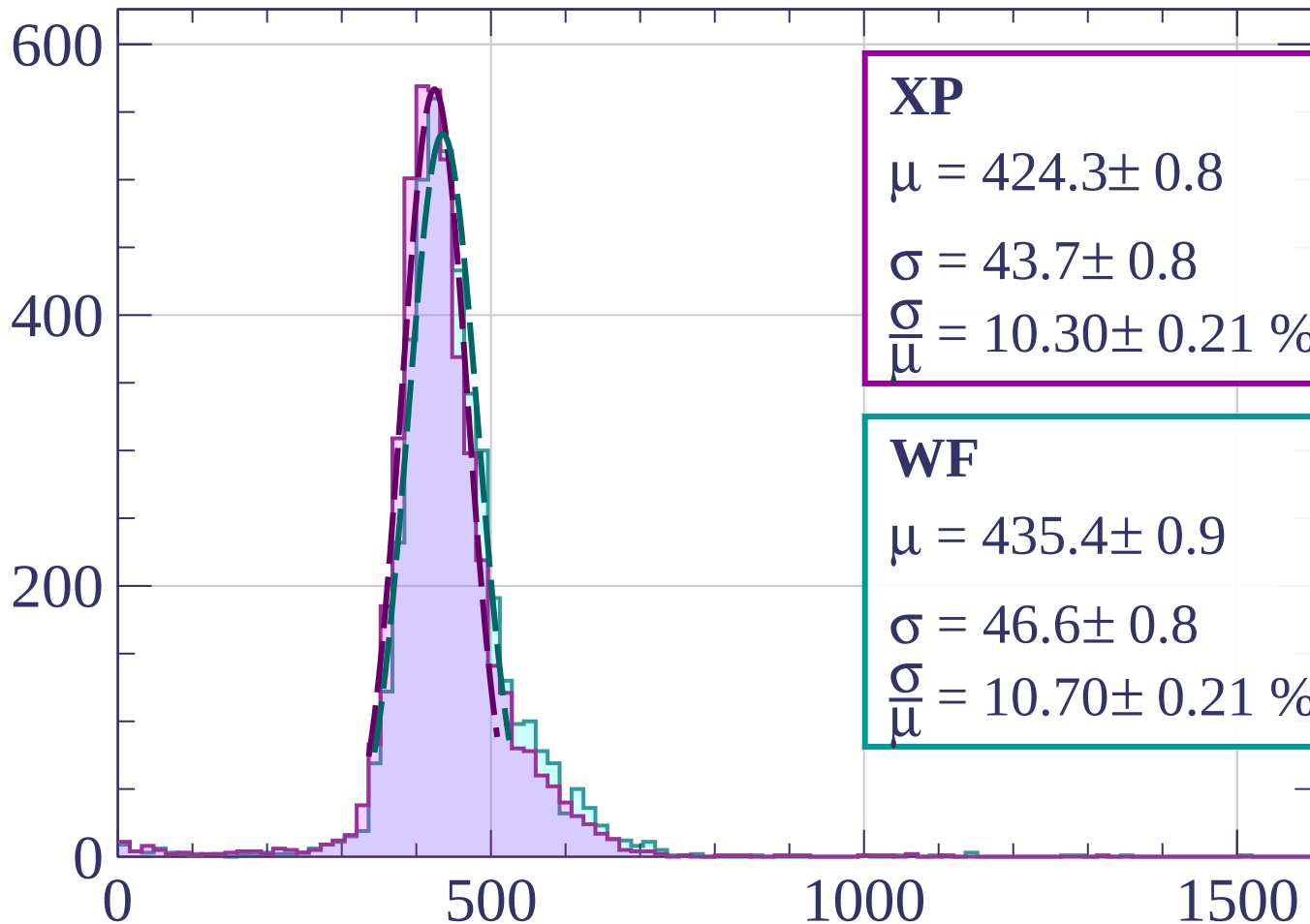
$$\sigma = 45.6 \pm 0.8$$

$$\frac{\sigma}{\mu} = 10.45 \pm 0.21 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-840 < p < -820$

Count



XP

$$\mu = 424.3 \pm 0.8$$

$$\sigma = 43.7 \pm 0.8$$

$$\frac{\sigma}{\mu} = 10.30 \pm 0.21 \%$$

WF

$$\mu = 435.4 \pm 0.9$$

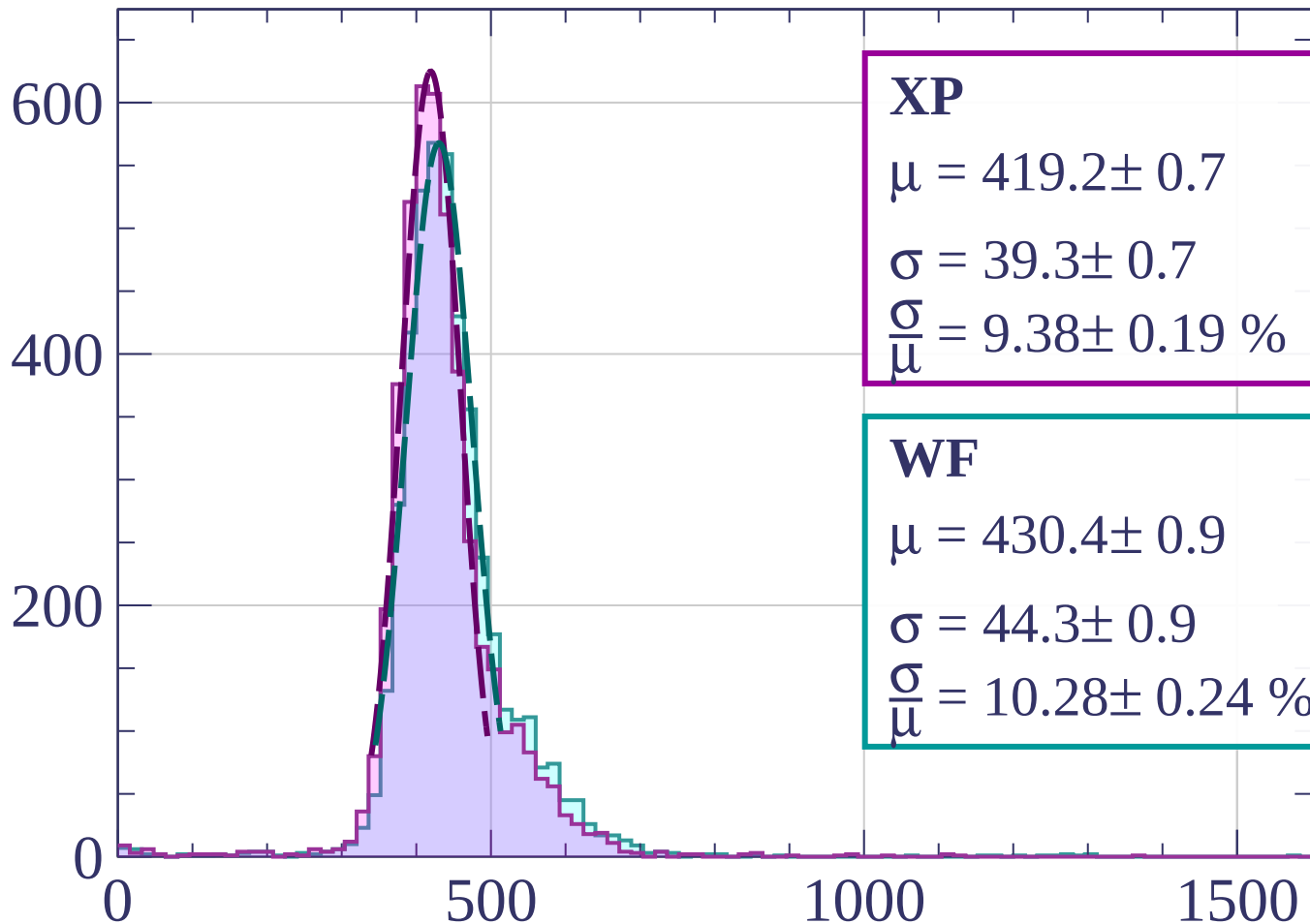
$$\sigma = 46.6 \pm 0.8$$

$$\frac{\sigma}{\mu} = 10.70 \pm 0.21 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-820 < p < -800$

Count



XP

$$\mu = 419.2 \pm 0.7$$

$$\sigma = 39.3 \pm 0.7$$

$$\frac{\sigma}{\mu} = 9.38 \pm 0.19 \%$$

WF

$$\mu = 430.4 \pm 0.9$$

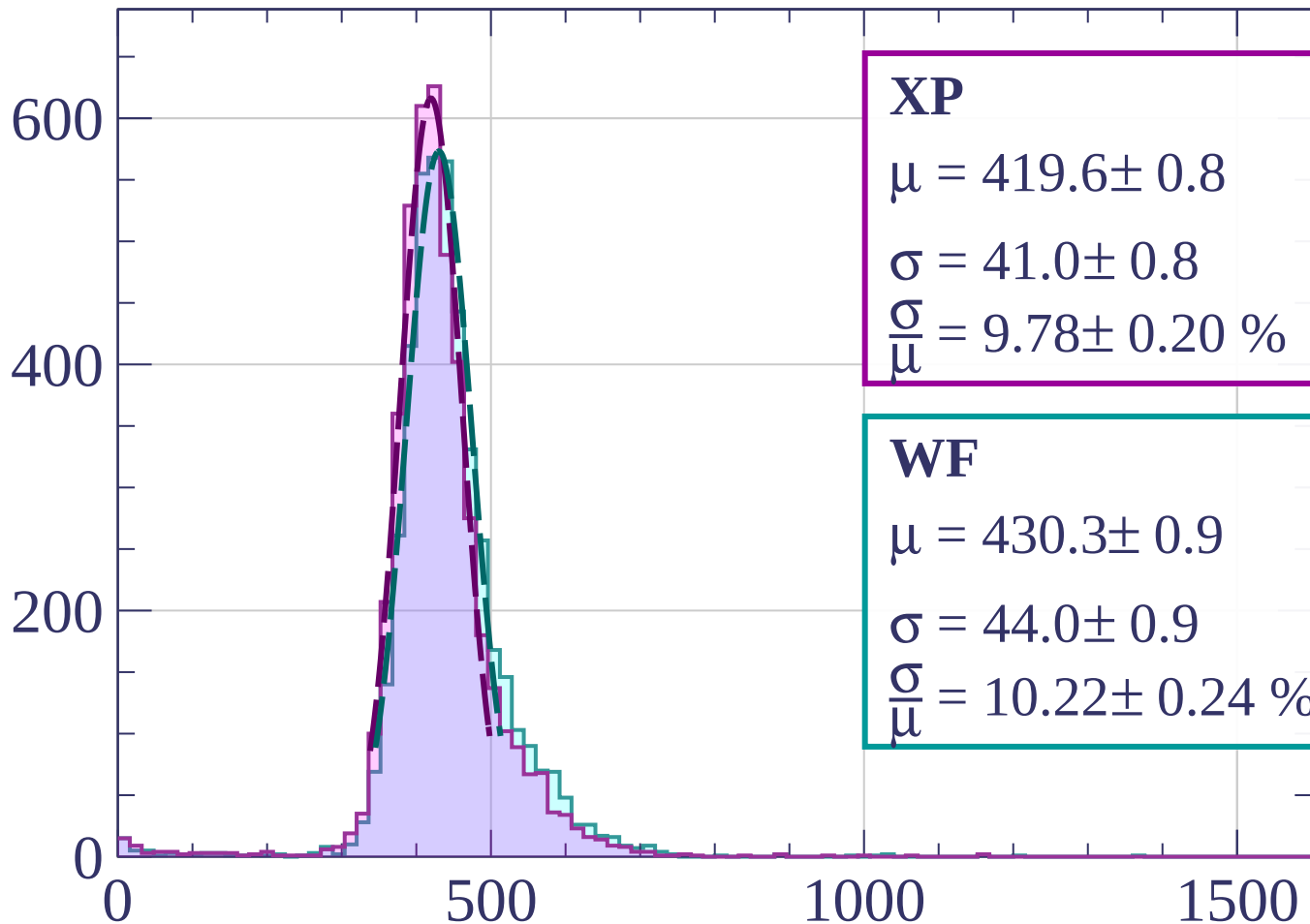
$$\sigma = 44.3 \pm 0.9$$

$$\frac{\sigma}{\mu} = 10.28 \pm 0.24 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-800 < p < -780$

Count



XP

$$\mu = 419.6 \pm 0.8$$

$$\sigma = 41.0 \pm 0.8$$

$$\frac{\sigma}{\mu} = 9.78 \pm 0.20 \%$$

WF

$$\mu = 430.3 \pm 0.9$$

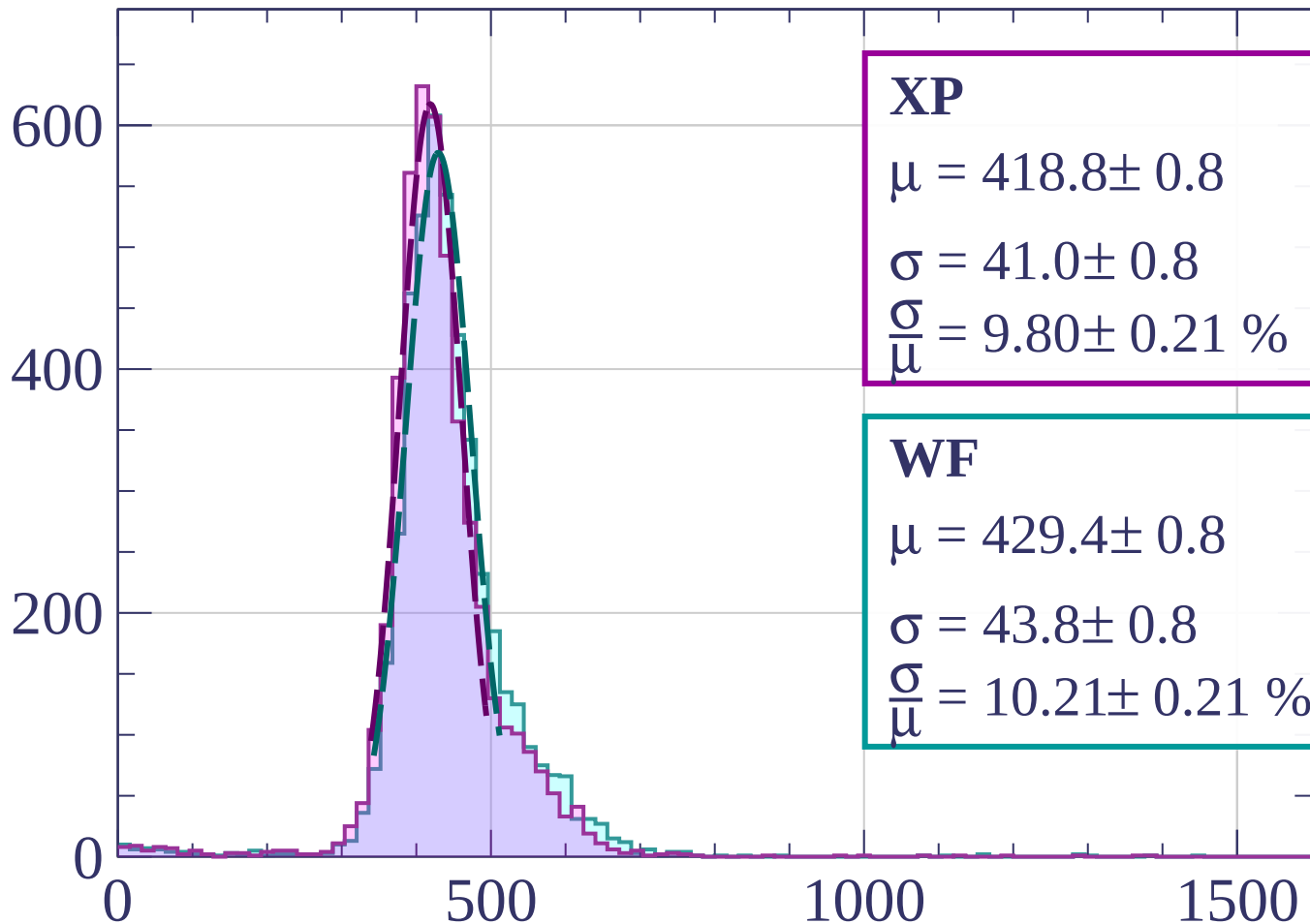
$$\sigma = 44.0 \pm 0.9$$

$$\frac{\sigma}{\mu} = 10.22 \pm 0.24 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-780 < p < -760$

Count



XP

$$\mu = 418.8 \pm 0.8$$

$$\sigma = 41.0 \pm 0.8$$

$$\frac{\sigma}{\mu} = 9.80 \pm 0.21 \%$$

WF

$$\mu = 429.4 \pm 0.8$$

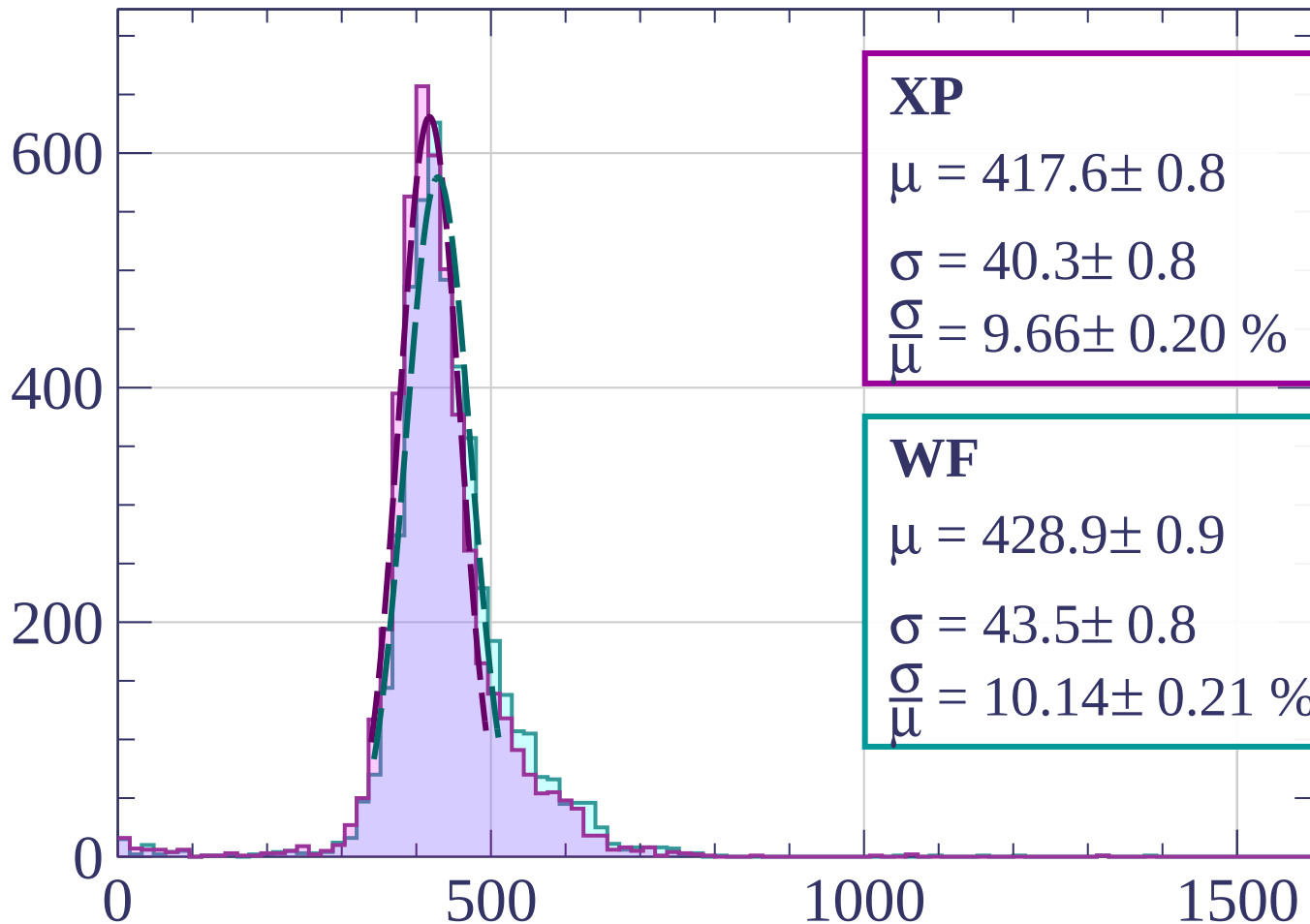
$$\sigma = 43.8 \pm 0.8$$

$$\frac{\sigma}{\mu} = 10.21 \pm 0.21 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-760 < p < -740$

Count



XP

$$\mu = 417.6 \pm 0.8$$

$$\sigma = 40.3 \pm 0.8$$

$$\frac{\sigma}{\mu} = 9.66 \pm 0.20 \%$$

WF

$$\mu = 428.9 \pm 0.9$$

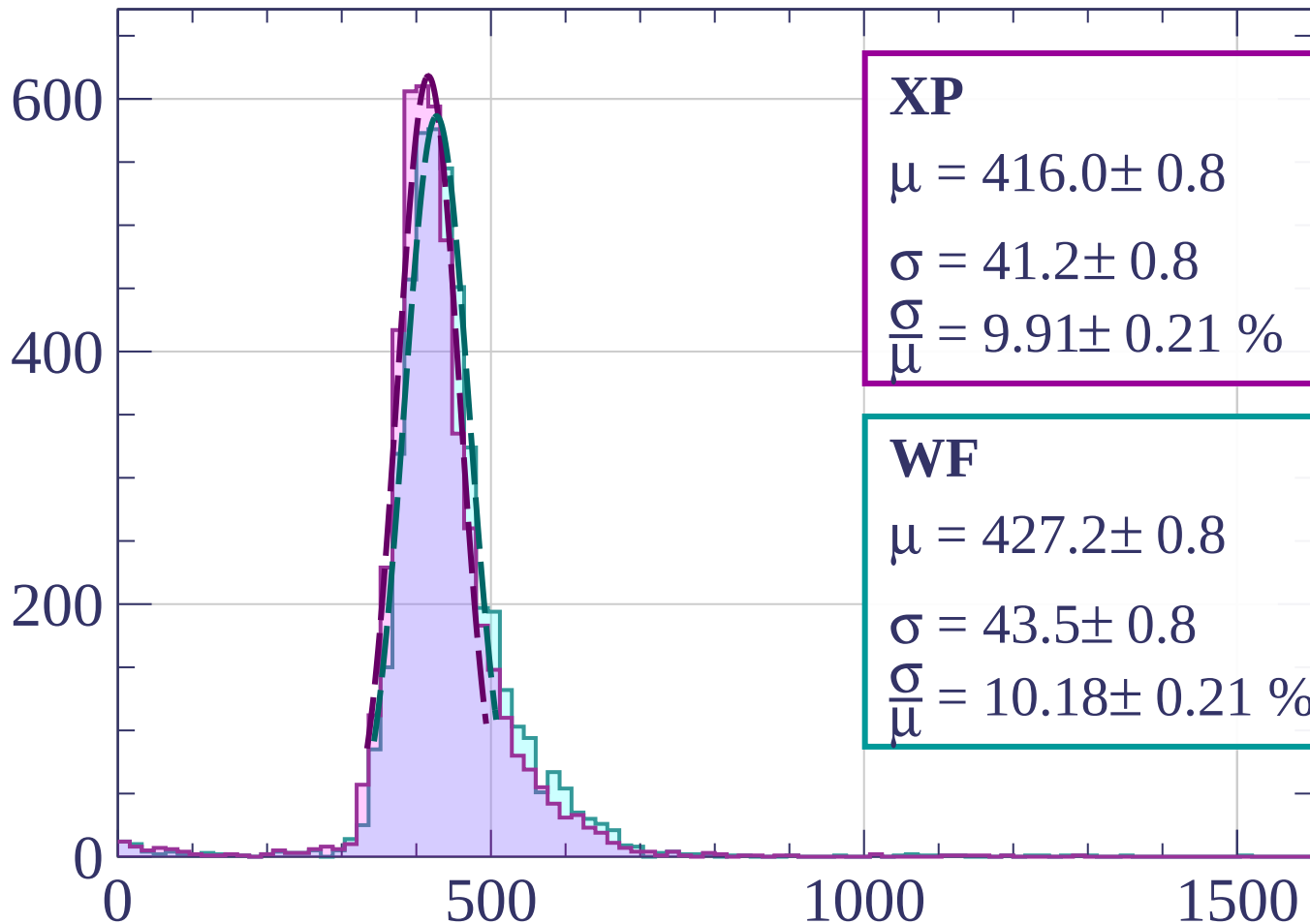
$$\sigma = 43.5 \pm 0.8$$

$$\frac{\sigma}{\mu} = 10.14 \pm 0.21 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-740 < p < -720$

Count



XP

$$\mu = 416.0 \pm 0.8$$

$$\sigma = 41.2 \pm 0.8$$

$$\frac{\sigma}{\mu} = 9.91 \pm 0.21 \%$$

WF

$$\mu = 427.2 \pm 0.8$$

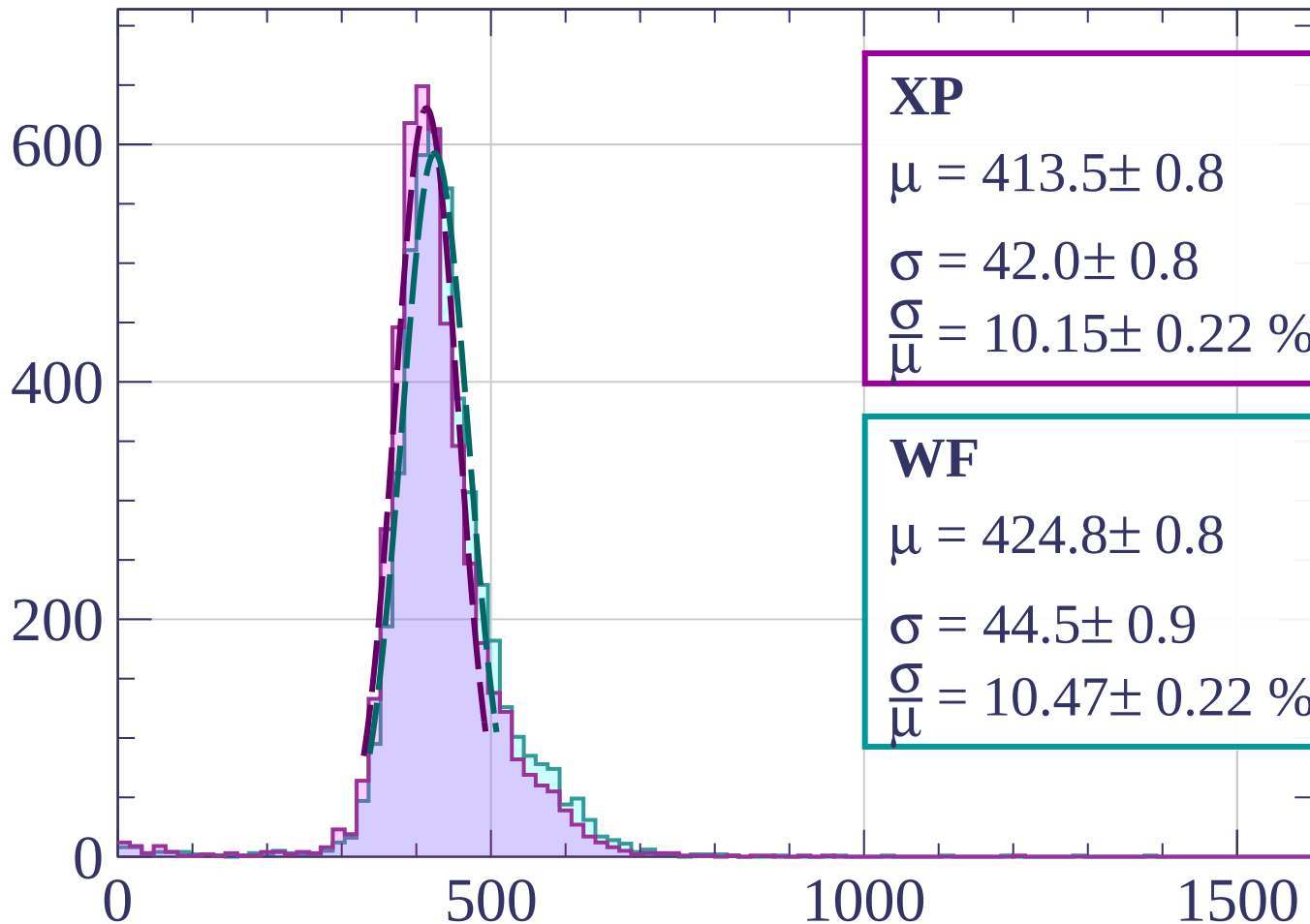
$$\sigma = 43.5 \pm 0.8$$

$$\frac{\sigma}{\mu} = 10.18 \pm 0.21 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-720 < p < -700$

Count



XP

$$\mu = 413.5 \pm 0.8$$

$$\sigma = 42.0 \pm 0.8$$

$$\frac{\sigma}{\mu} = 10.15 \pm 0.22 \%$$

WF

$$\mu = 424.8 \pm 0.8$$

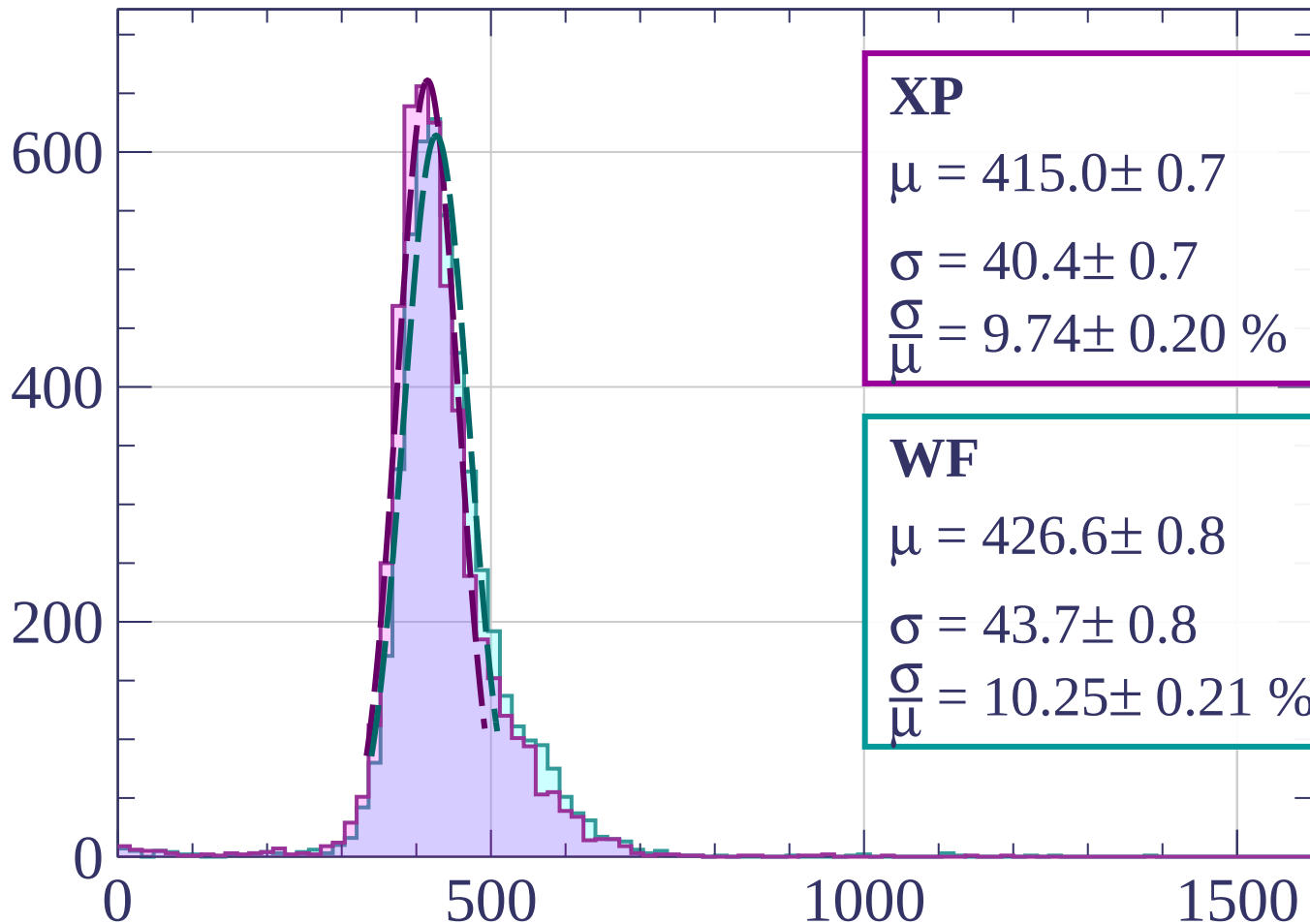
$$\sigma = 44.5 \pm 0.9$$

$$\frac{\sigma}{\mu} = 10.47 \pm 0.22 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-700 < p < -680$

Count



XP

$$\mu = 415.0 \pm 0.7$$

$$\sigma = 40.4 \pm 0.7$$

$$\frac{\sigma}{\mu} = 9.74 \pm 0.20 \%$$

WF

$$\mu = 426.6 \pm 0.8$$

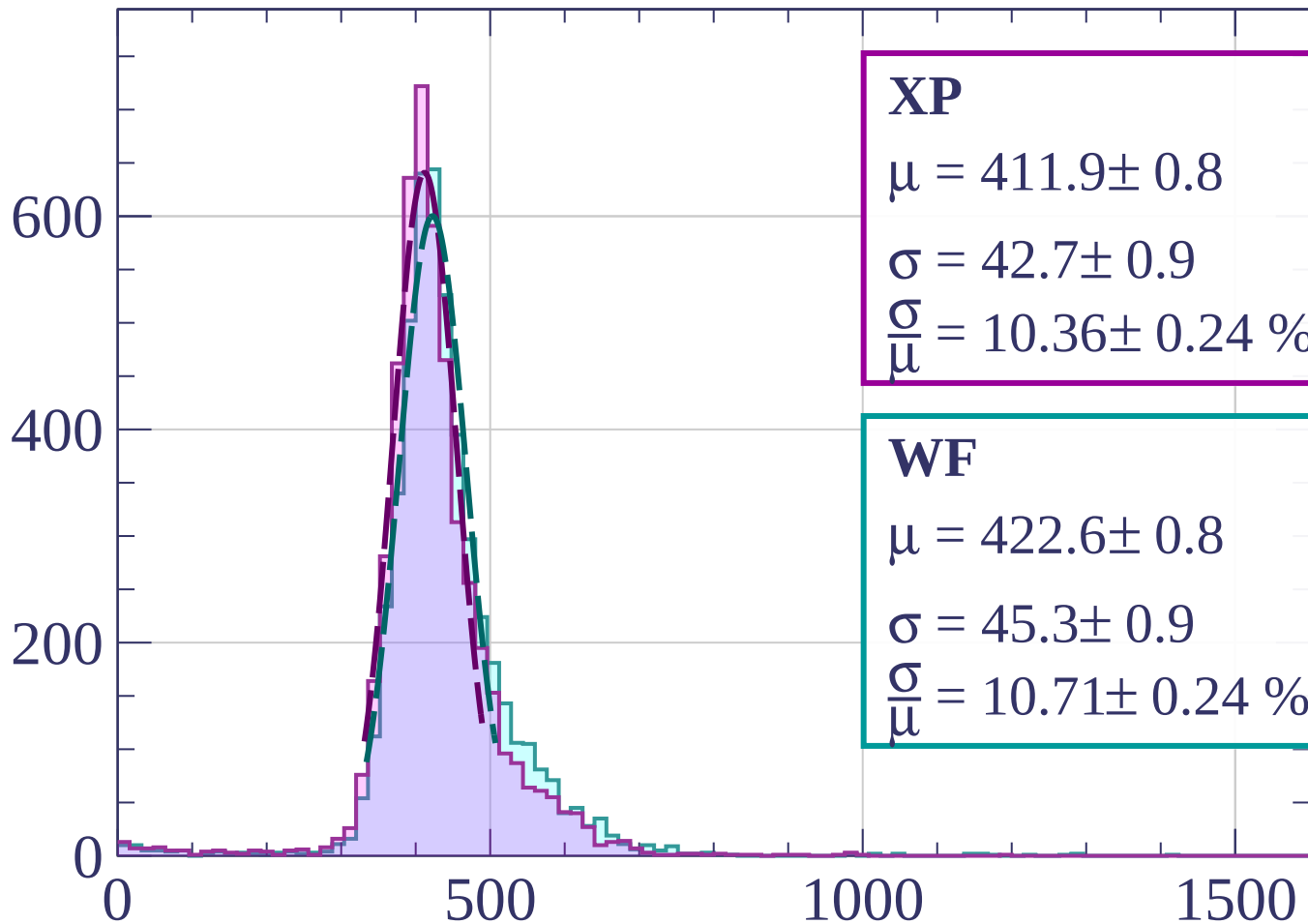
$$\sigma = 43.7 \pm 0.8$$

$$\frac{\sigma}{\mu} = 10.25 \pm 0.21 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-680 < p < -660$

Count



XP

$$\mu = 411.9 \pm 0.8$$

$$\sigma = 42.7 \pm 0.9$$

$$\frac{\sigma}{\mu} = 10.36 \pm 0.24 \%$$

WF

$$\mu = 422.6 \pm 0.8$$

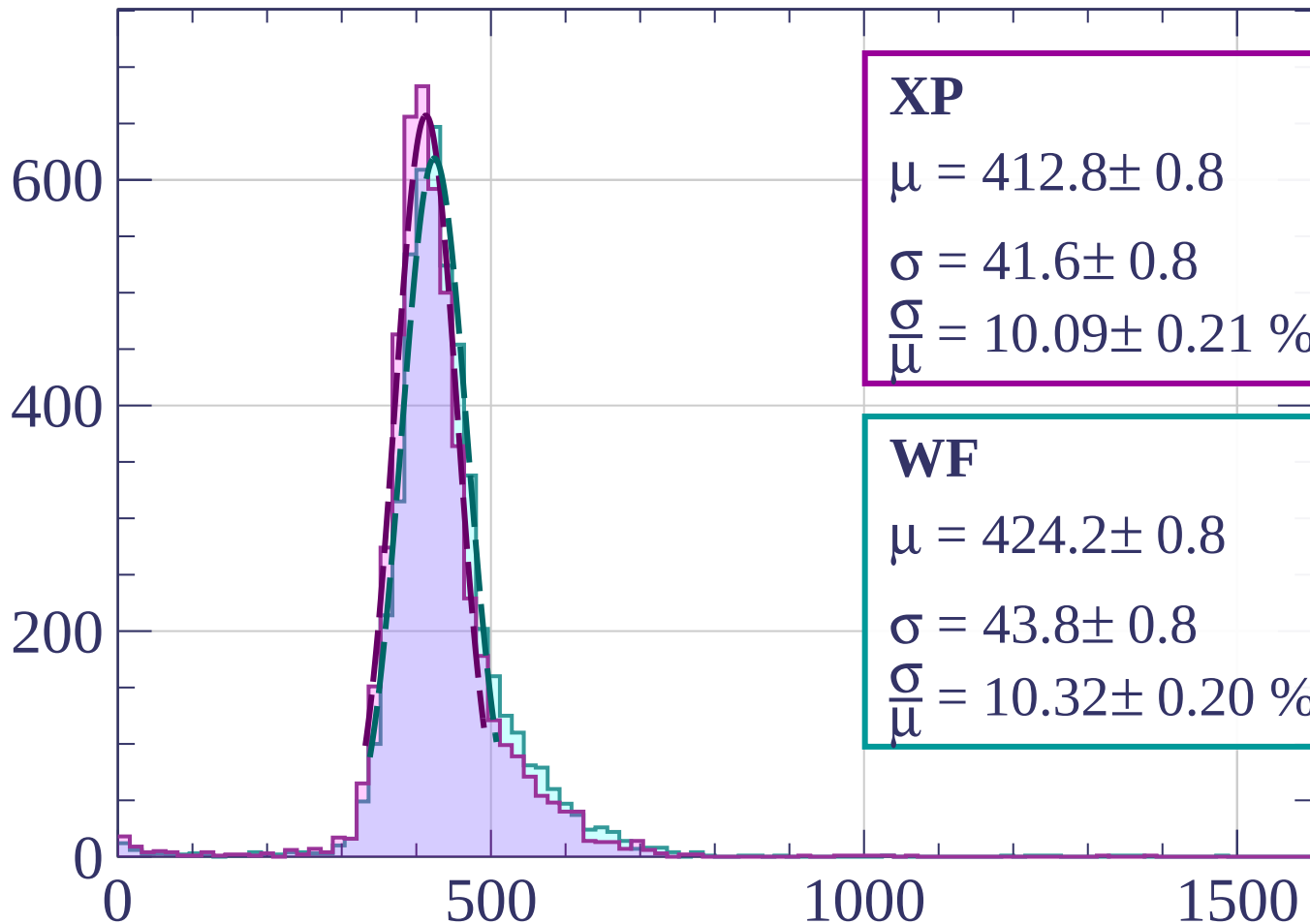
$$\sigma = 45.3 \pm 0.9$$

$$\frac{\sigma}{\mu} = 10.71 \pm 0.24 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-660 < p < -640$

Count



XP

$$\mu = 412.8 \pm 0.8$$

$$\sigma = 41.6 \pm 0.8$$

$$\frac{\sigma}{\mu} = 10.09 \pm 0.21 \%$$

WF

$$\mu = 424.2 \pm 0.8$$

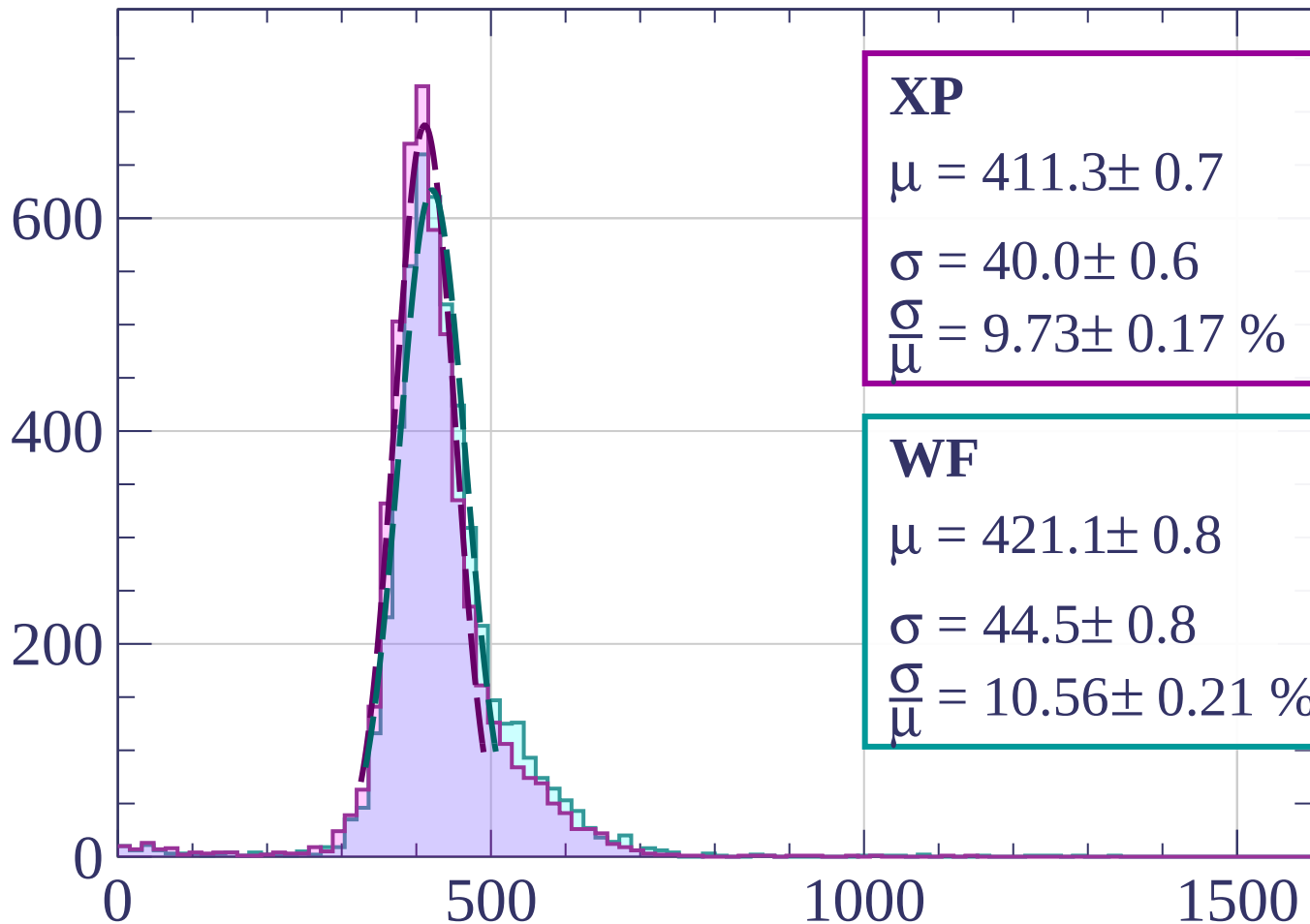
$$\sigma = 43.8 \pm 0.8$$

$$\frac{\sigma}{\mu} = 10.32 \pm 0.20 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-640 < p < -620$

Count



XP

$$\mu = 411.3 \pm 0.7$$

$$\sigma = 40.0 \pm 0.6$$

$$\frac{\sigma}{\mu} = 9.73 \pm 0.17 \%$$

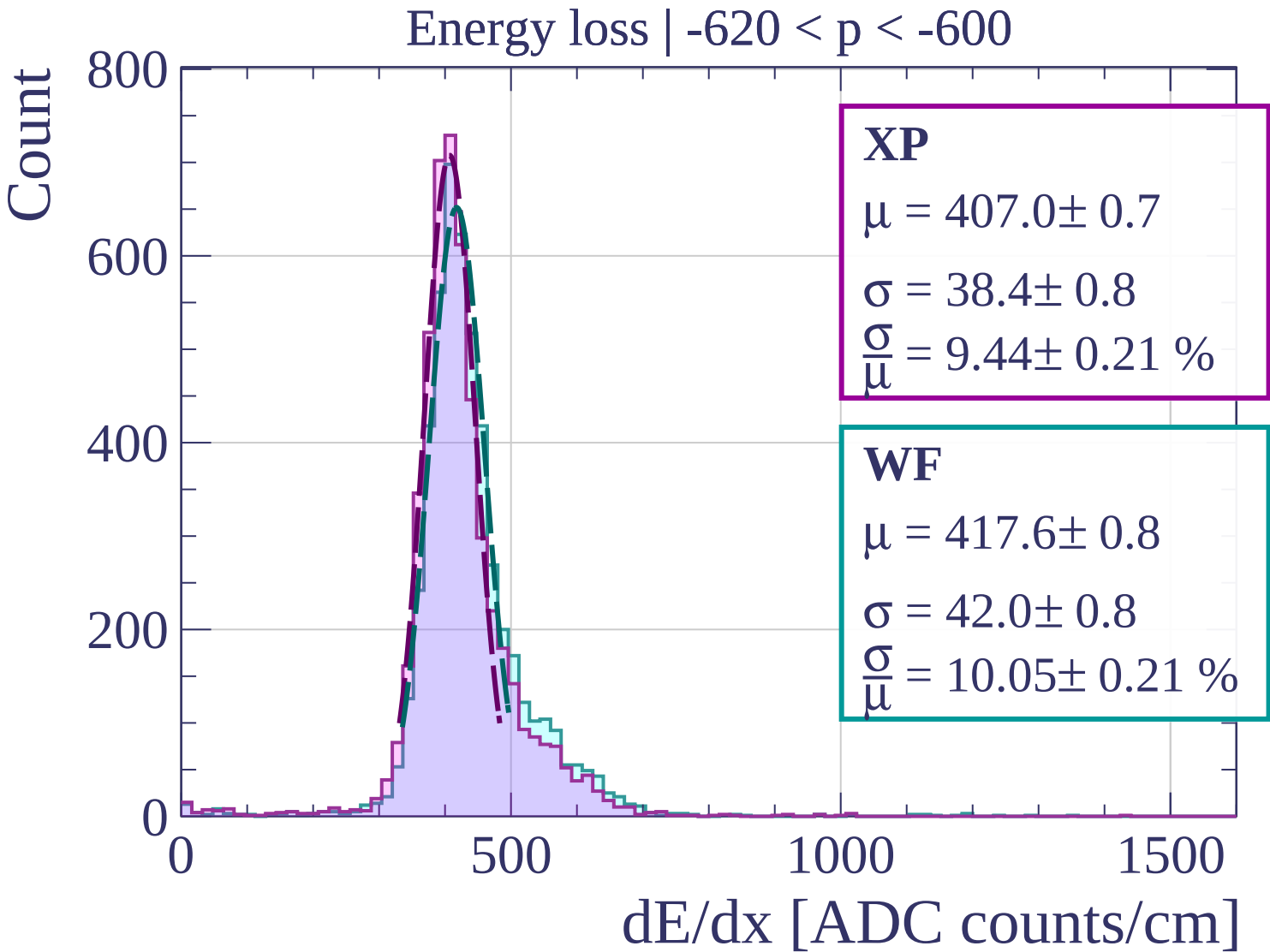
WF

$$\mu = 421.1 \pm 0.8$$

$$\sigma = 44.5 \pm 0.8$$

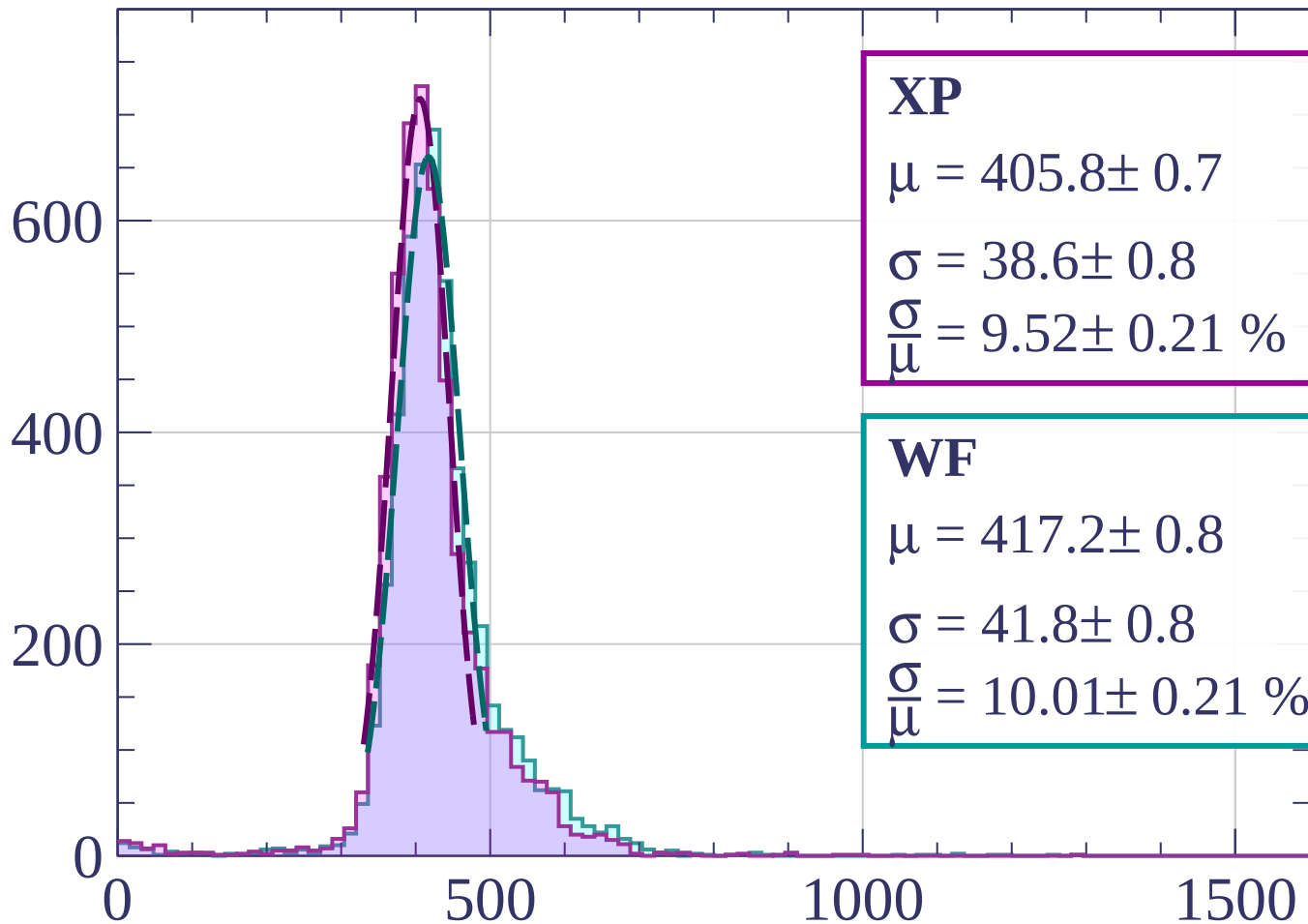
$$\frac{\sigma}{\mu} = 10.56 \pm 0.21 \%$$

dE/dx [ADC counts/cm]



Energy loss | $-600 < p < -580$

Count



XP

$$\mu = 405.8 \pm 0.7$$

$$\sigma = 38.6 \pm 0.8$$

$$\frac{\sigma}{\mu} = 9.52 \pm 0.21 \%$$

WF

$$\mu = 417.2 \pm 0.8$$

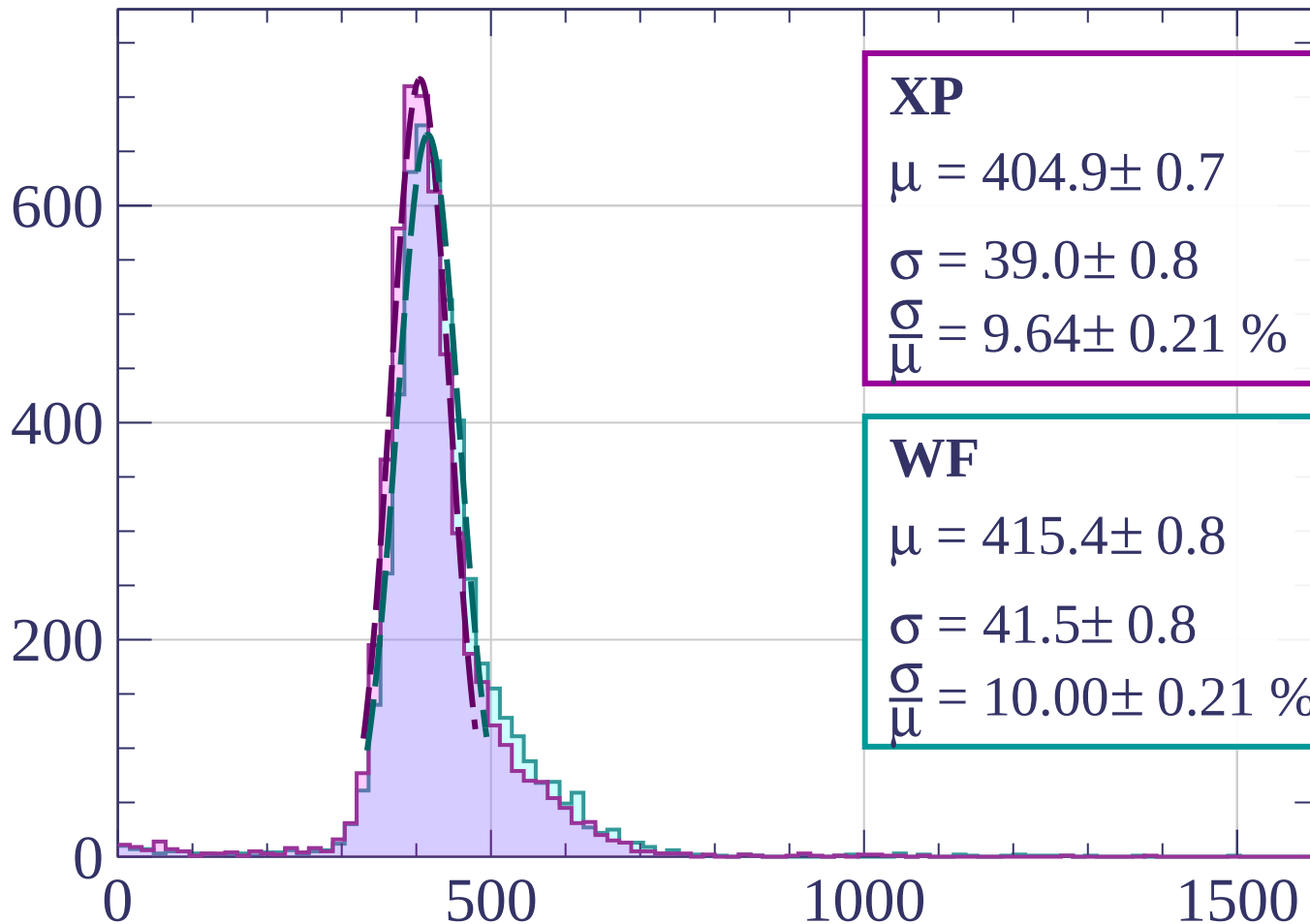
$$\sigma = 41.8 \pm 0.8$$

$$\frac{\sigma}{\mu} = 10.01 \pm 0.21 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-580 < p < -560$

Count



XP

$$\mu = 404.9 \pm 0.7$$

$$\sigma = 39.0 \pm 0.8$$

$$\frac{\sigma}{\mu} = 9.64 \pm 0.21 \%$$

WF

$$\mu = 415.4 \pm 0.8$$

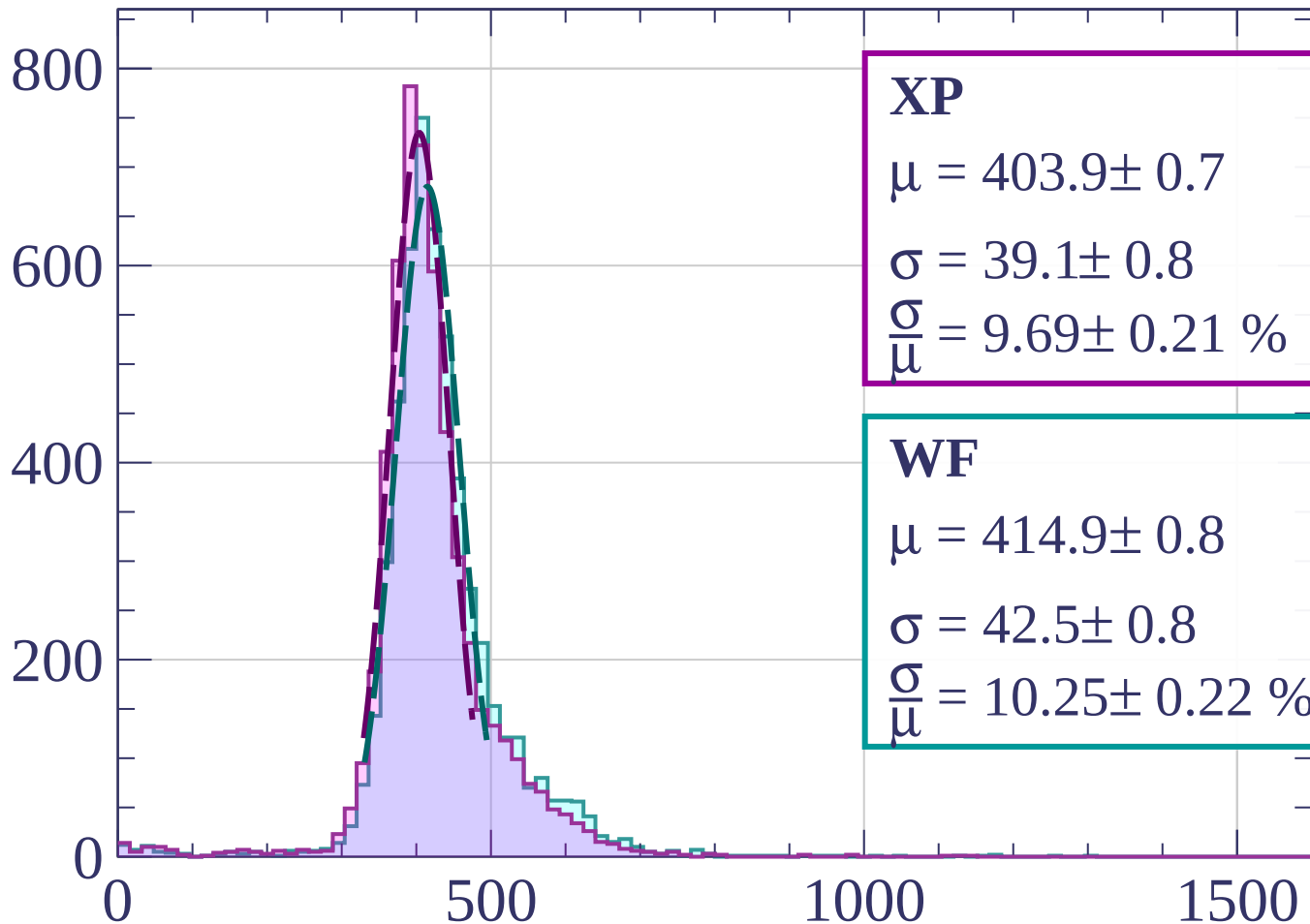
$$\sigma = 41.5 \pm 0.8$$

$$\frac{\sigma}{\mu} = 10.00 \pm 0.21 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-560 < p < -540$

Count



XP

$$\mu = 403.9 \pm 0.7$$

$$\sigma = 39.1 \pm 0.8$$

$$\frac{\sigma}{\mu} = 9.69 \pm 0.21 \%$$

WF

$$\mu = 414.9 \pm 0.8$$

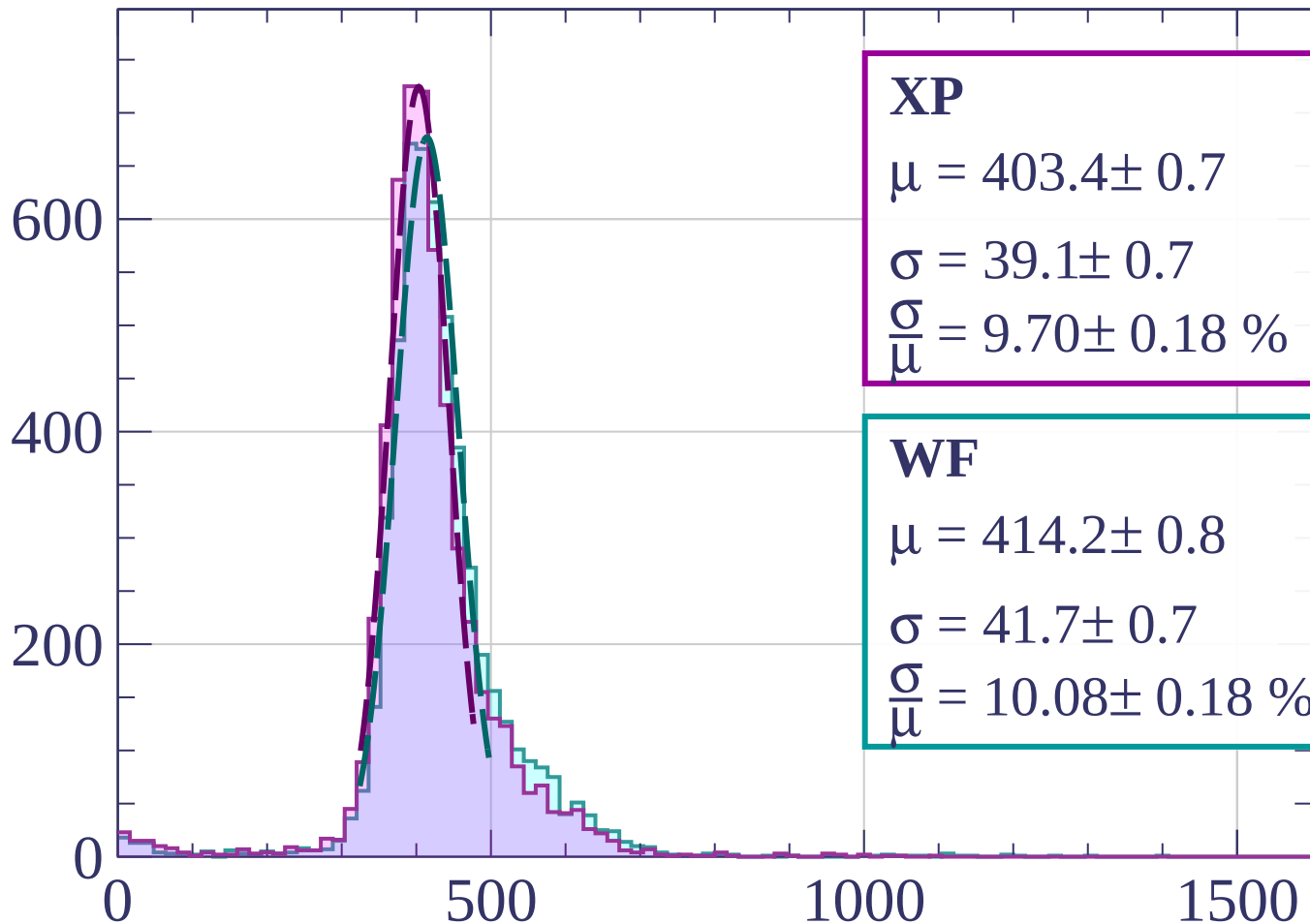
$$\sigma = 42.5 \pm 0.8$$

$$\frac{\sigma}{\mu} = 10.25 \pm 0.22 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-540 < p < -520$

Count



XP

$$\mu = 403.4 \pm 0.7$$

$$\sigma = 39.1 \pm 0.7$$

$$\frac{\sigma}{\mu} = 9.70 \pm 0.18 \%$$

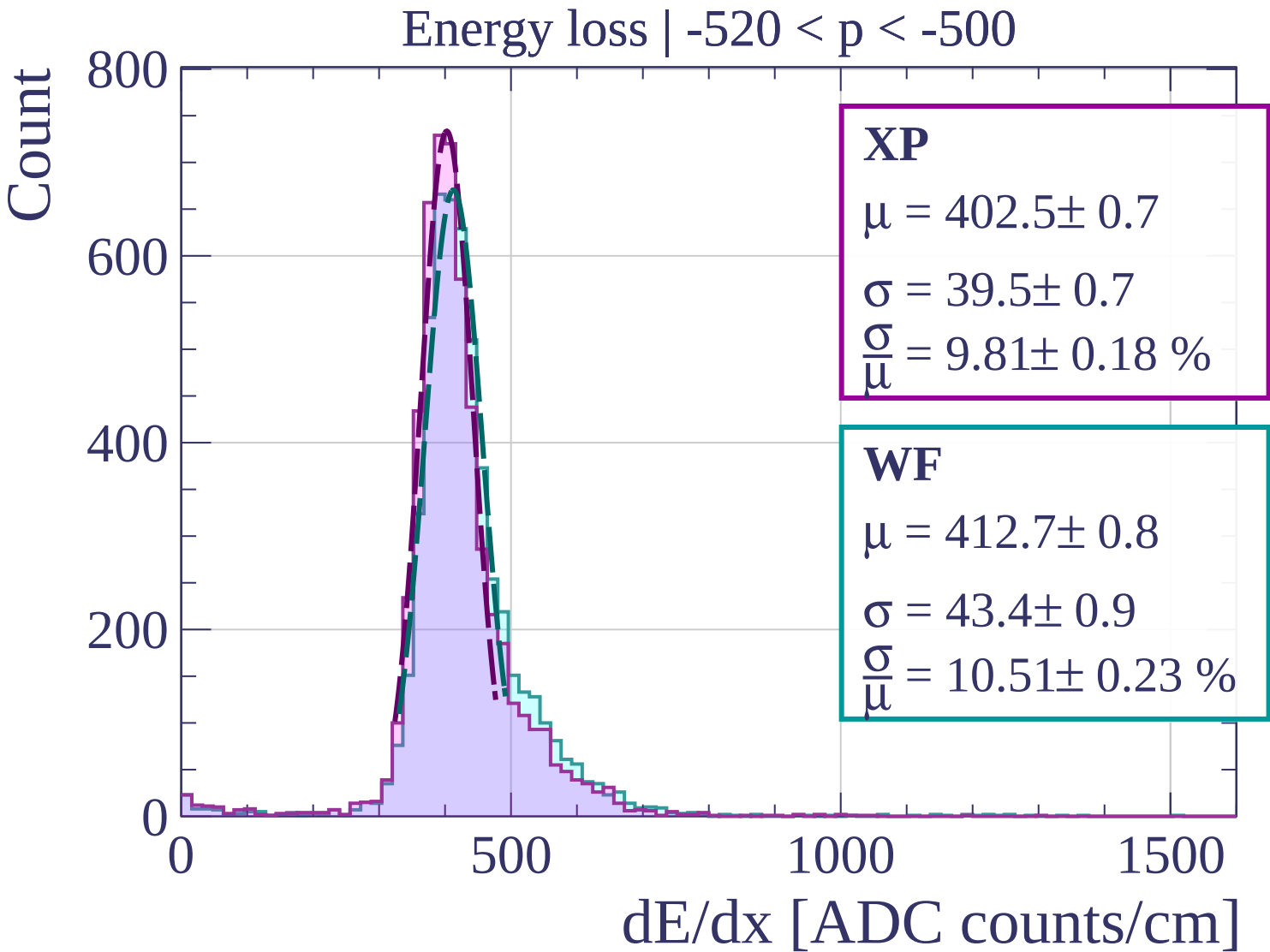
WF

$$\mu = 414.2 \pm 0.8$$

$$\sigma = 41.7 \pm 0.7$$

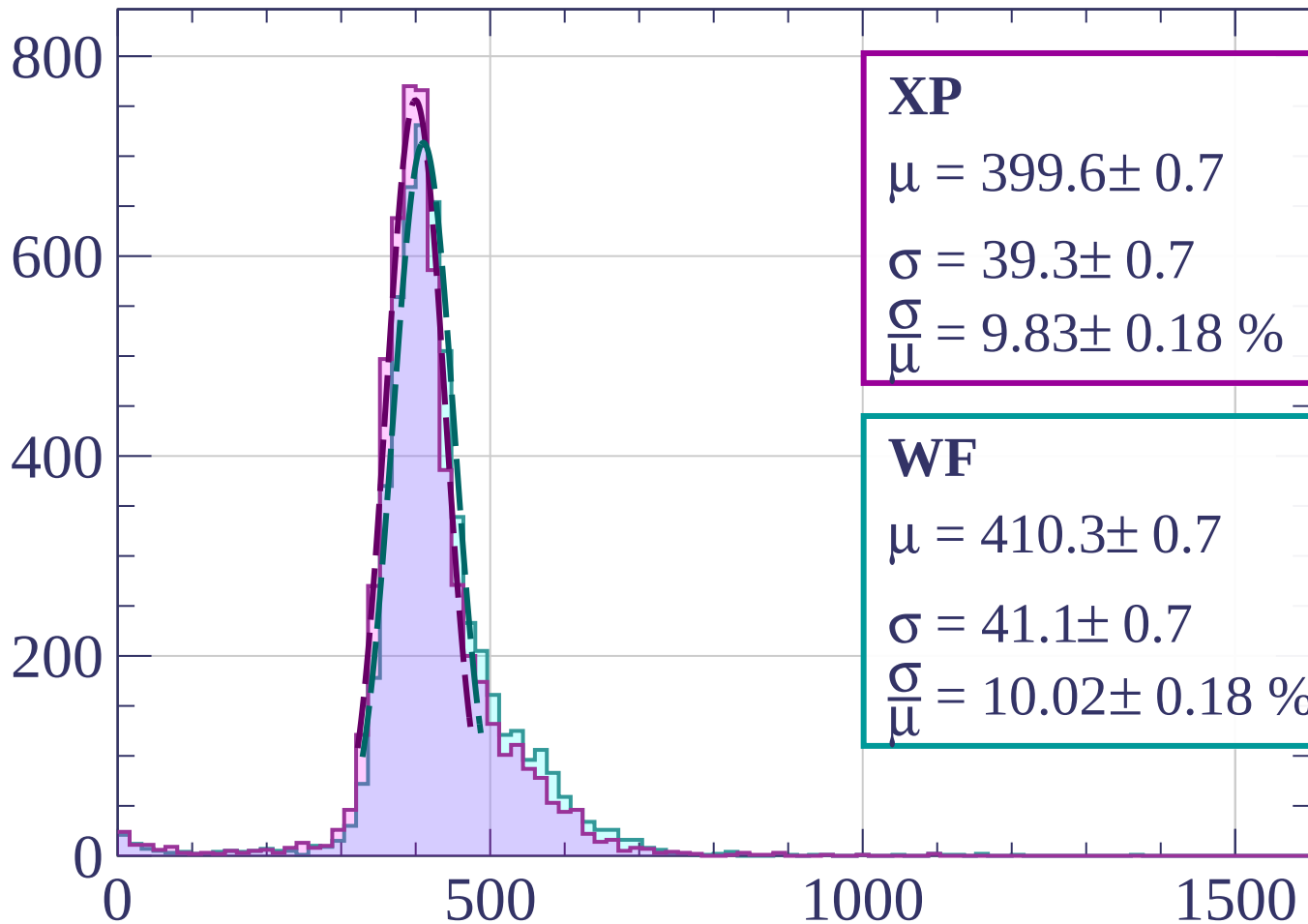
$$\frac{\sigma}{\mu} = 10.08 \pm 0.18 \%$$

dE/dx [ADC counts/cm]



Energy loss | $-500 < p < -480$

Count



XP

$$\mu = 399.6 \pm 0.7$$

$$\sigma = 39.3 \pm 0.7$$

$$\frac{\sigma}{\mu} = 9.83 \pm 0.18 \%$$

WF

$$\mu = 410.3 \pm 0.7$$

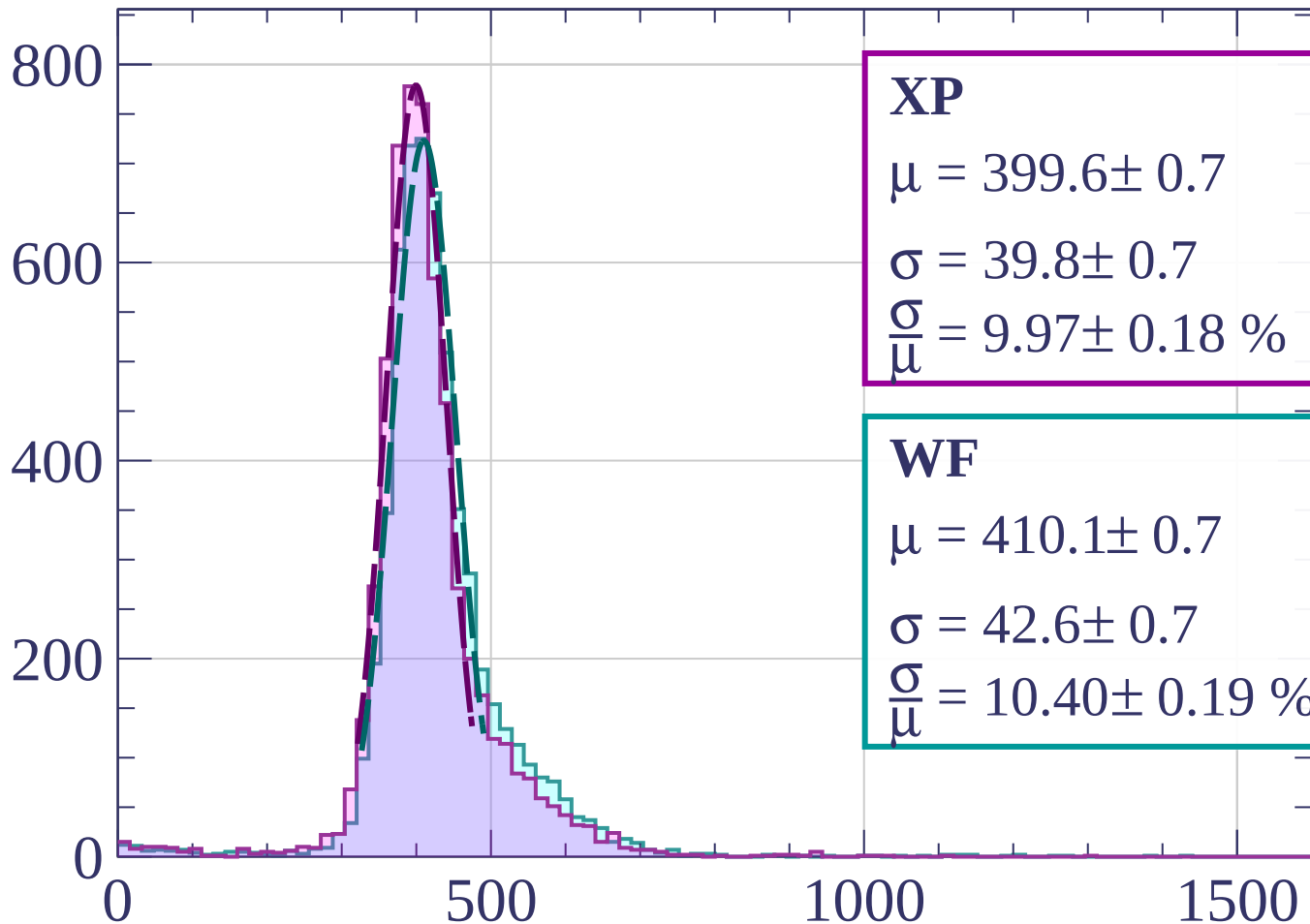
$$\sigma = 41.1 \pm 0.7$$

$$\frac{\sigma}{\mu} = 10.02 \pm 0.18 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-480 < p < -460$

Count



XP

$$\mu = 399.6 \pm 0.7$$

$$\sigma = 39.8 \pm 0.7$$

$$\frac{\sigma}{\mu} = 9.97 \pm 0.18 \%$$

WF

$$\mu = 410.1 \pm 0.7$$

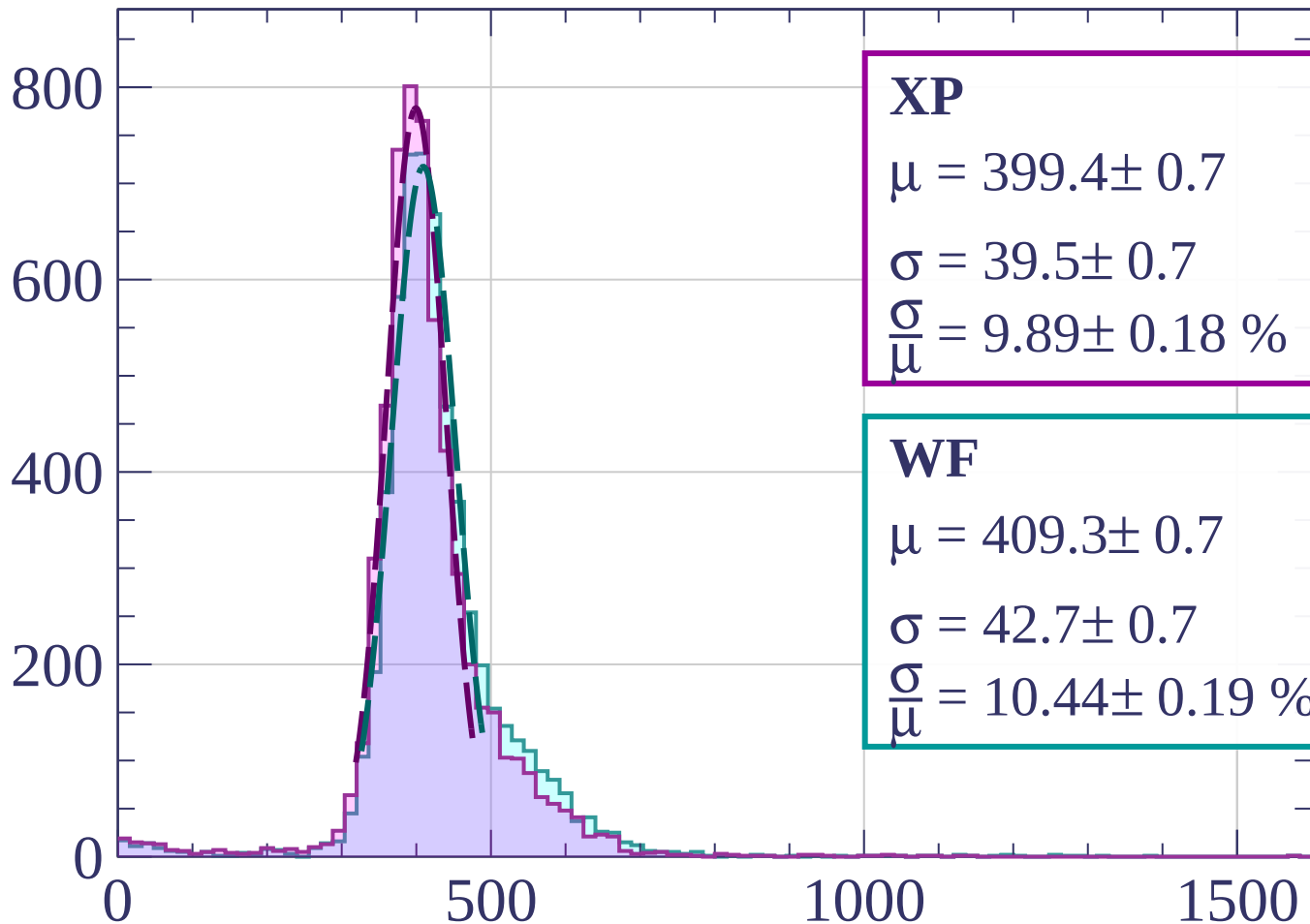
$$\sigma = 42.6 \pm 0.7$$

$$\frac{\sigma}{\mu} = 10.40 \pm 0.19 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-460 < p < -440$

Count



XP

$$\mu = 399.4 \pm 0.7$$

$$\sigma = 39.5 \pm 0.7$$

$$\frac{\sigma}{\mu} = 9.89 \pm 0.18 \%$$

WF

$$\mu = 409.3 \pm 0.7$$

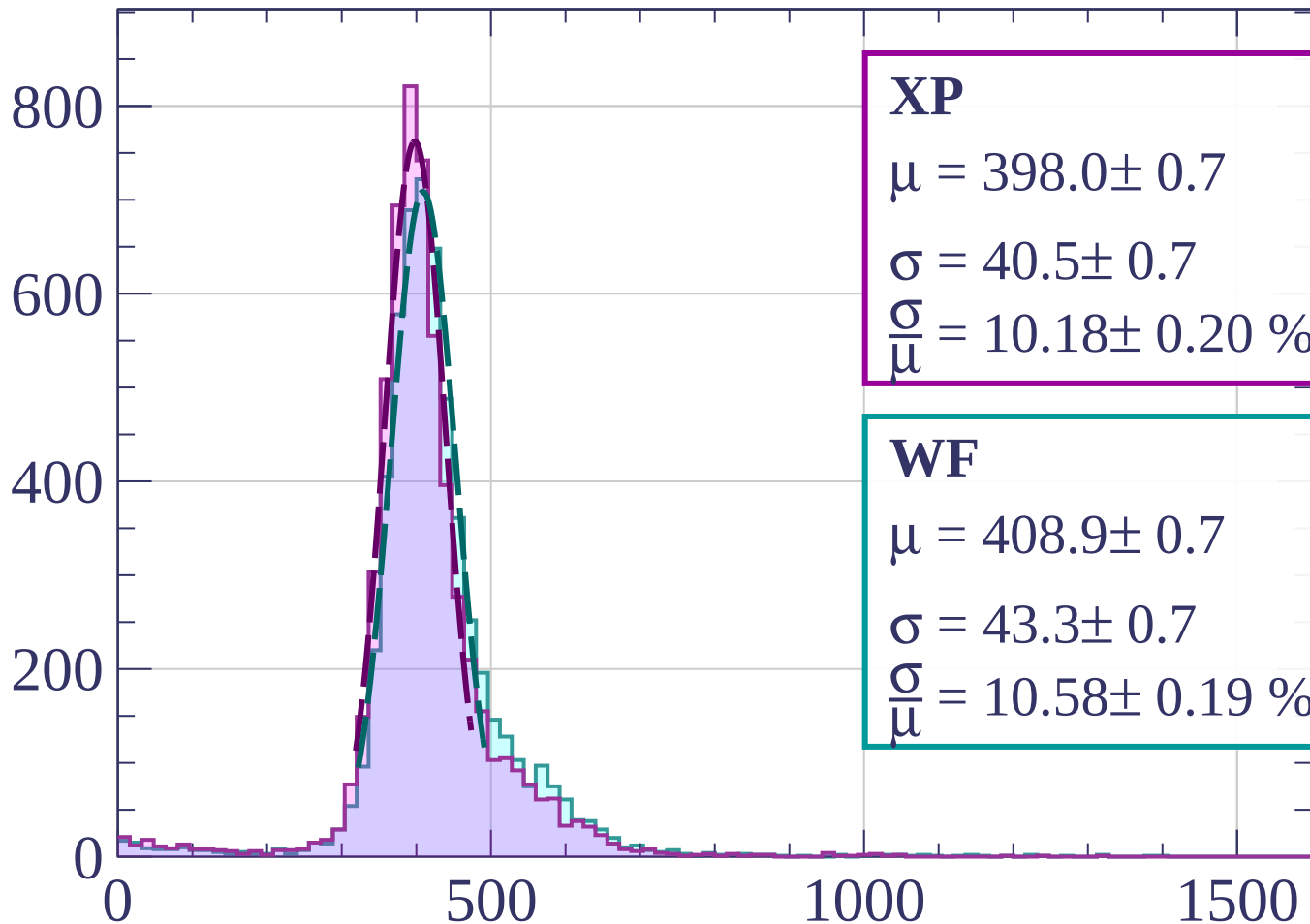
$$\sigma = 42.7 \pm 0.7$$

$$\frac{\sigma}{\mu} = 10.44 \pm 0.19 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-440 < p < -420$

Count



XP

$$\mu = 398.0 \pm 0.7$$

$$\sigma = 40.5 \pm 0.7$$

$$\frac{\sigma}{\mu} = 10.18 \pm 0.20 \%$$

WF

$$\mu = 408.9 \pm 0.7$$

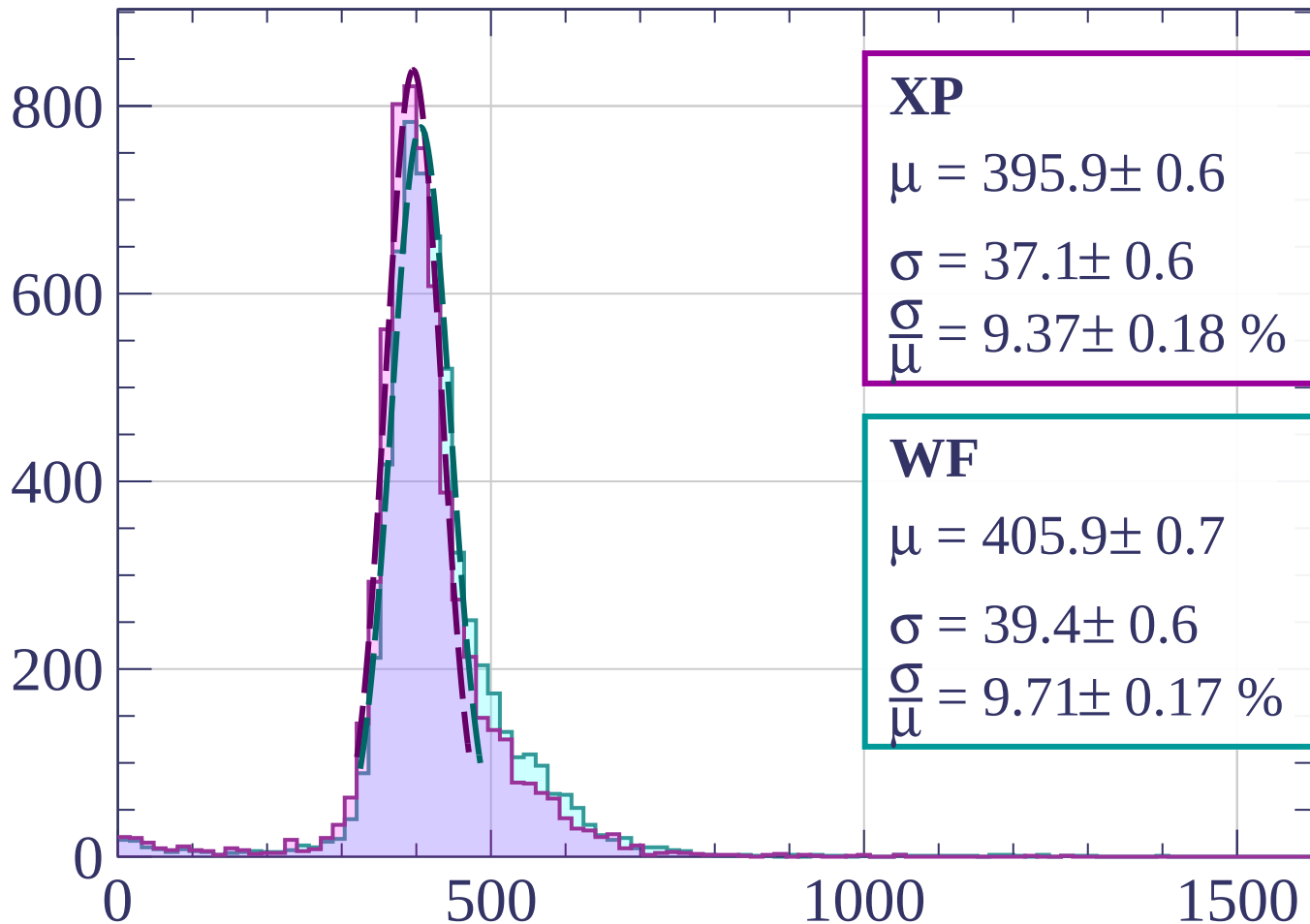
$$\sigma = 43.3 \pm 0.7$$

$$\frac{\sigma}{\mu} = 10.58 \pm 0.19 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-420 < p < -400$

Count



XP

$$\mu = 395.9 \pm 0.6$$

$$\sigma = 37.1 \pm 0.6$$

$$\frac{\sigma}{\mu} = 9.37 \pm 0.18 \%$$

WF

$$\mu = 405.9 \pm 0.7$$

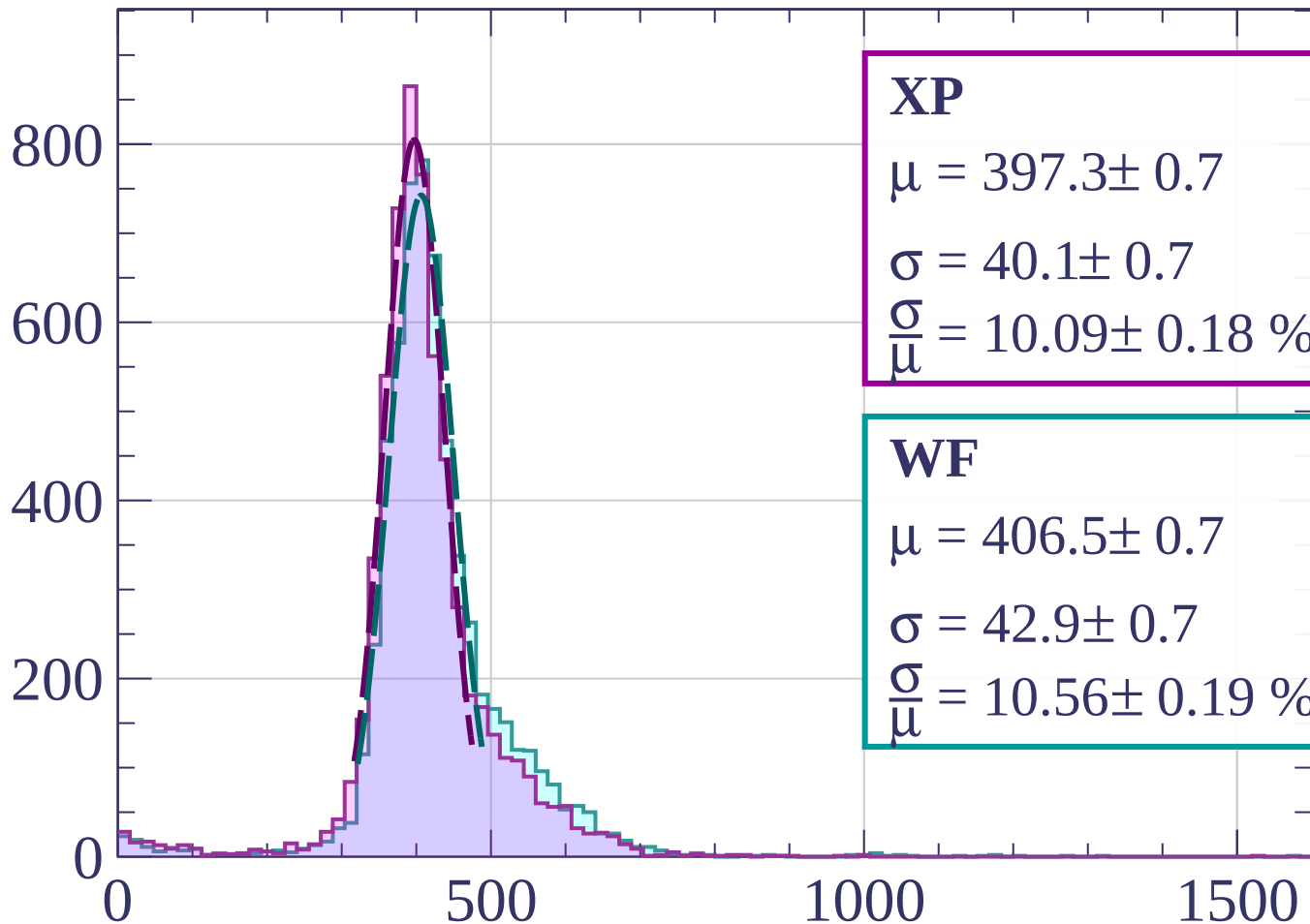
$$\sigma = 39.4 \pm 0.6$$

$$\frac{\sigma}{\mu} = 9.71 \pm 0.17 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-400 < p < -380$

Count



XP

$$\mu = 397.3 \pm 0.7$$

$$\sigma = 40.1 \pm 0.7$$

$$\frac{\sigma}{\mu} = 10.09 \pm 0.18 \%$$

WF

$$\mu = 406.5 \pm 0.7$$

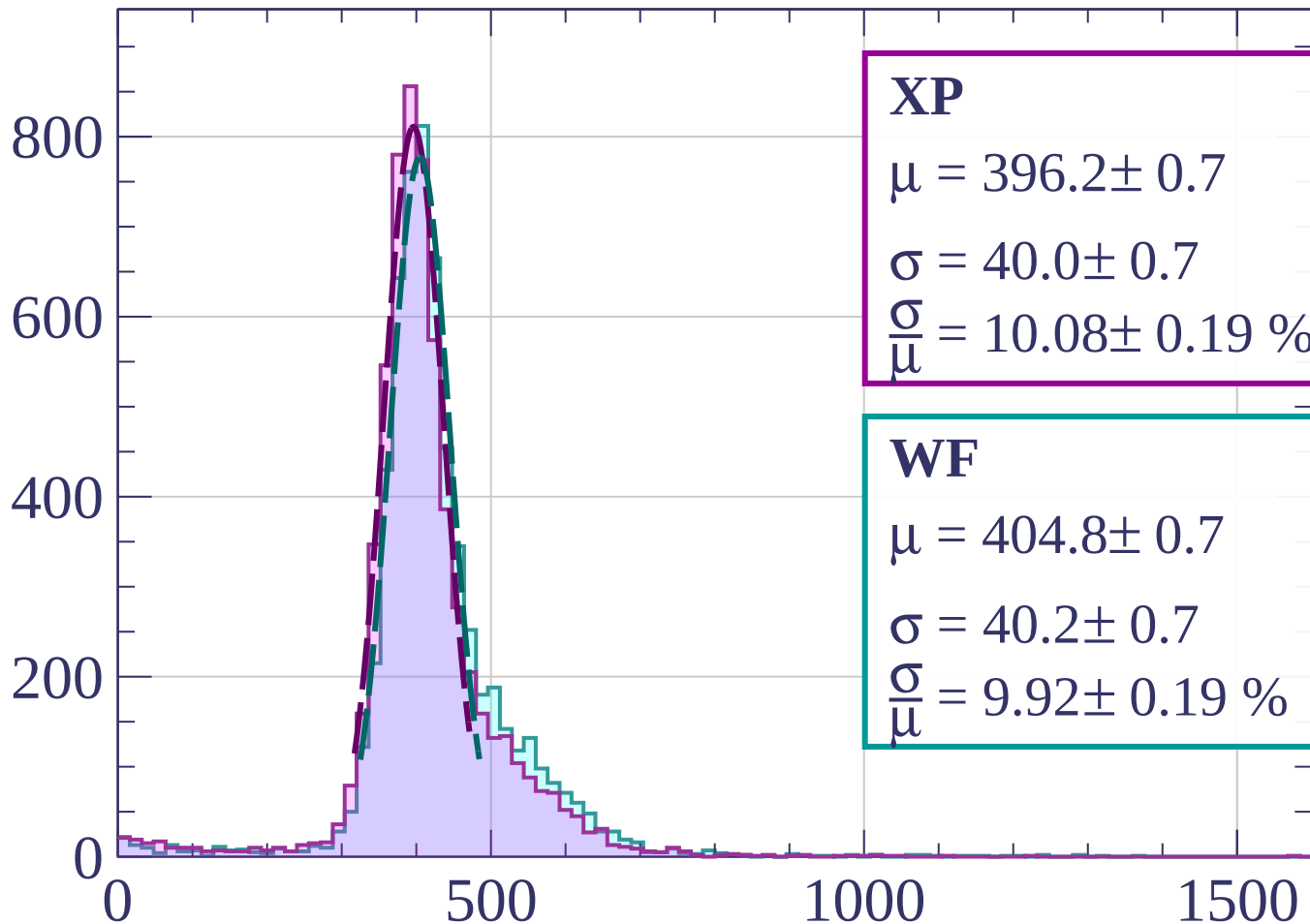
$$\sigma = 42.9 \pm 0.7$$

$$\frac{\sigma}{\mu} = 10.56 \pm 0.19 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-380 < p < -360$

Count



XP

$$\mu = 396.2 \pm 0.7$$

$$\sigma = 40.0 \pm 0.7$$

$$\frac{\sigma}{\mu} = 10.08 \pm 0.19 \%$$

WF

$$\mu = 404.8 \pm 0.7$$

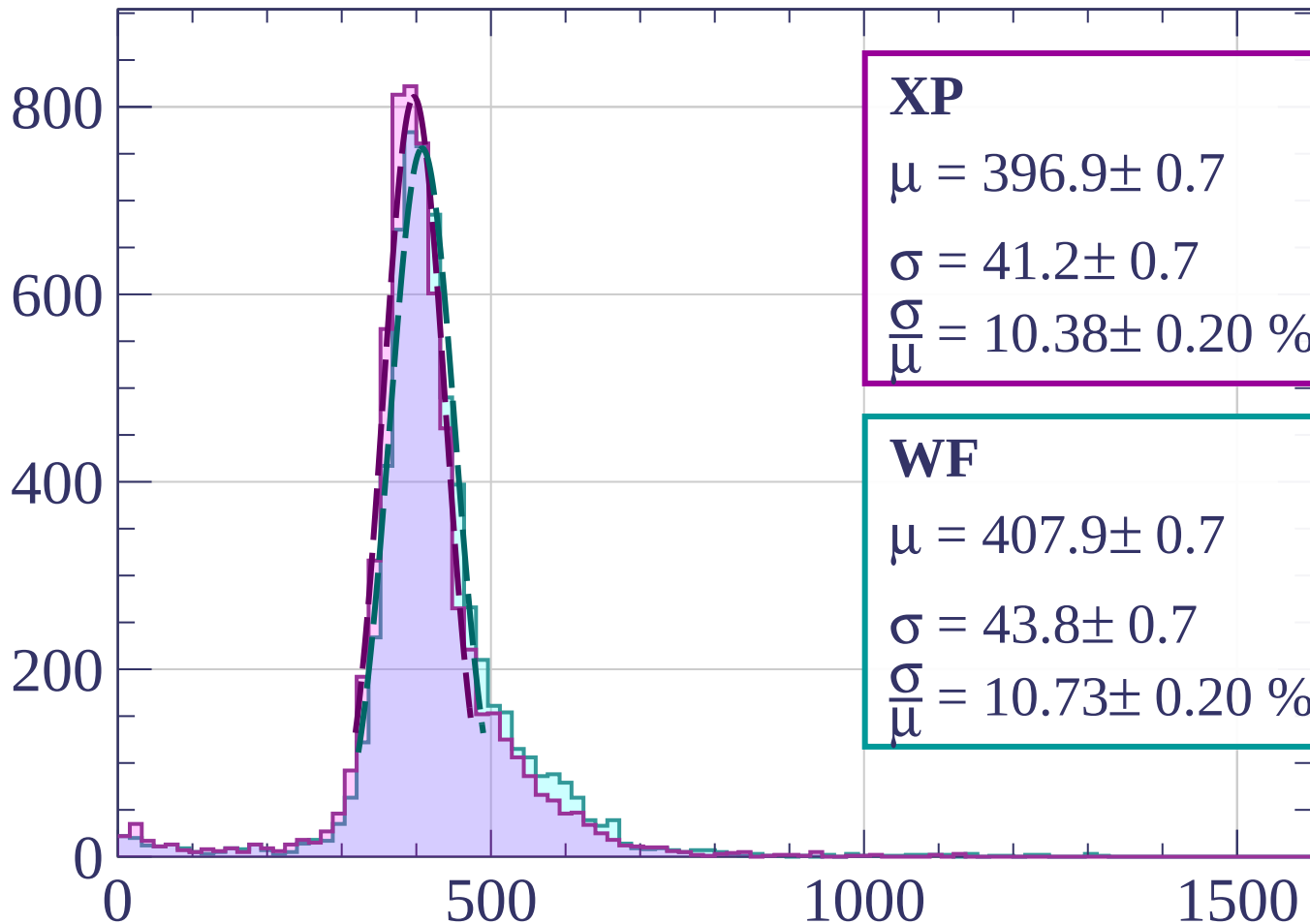
$$\sigma = 40.2 \pm 0.7$$

$$\frac{\sigma}{\mu} = 9.92 \pm 0.19 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-360 < p < -340$

Count



XP

$$\mu = 396.9 \pm 0.7$$

$$\sigma = 41.2 \pm 0.7$$

$$\frac{\sigma}{\mu} = 10.38 \pm 0.20 \%$$

WF

$$\mu = 407.9 \pm 0.7$$

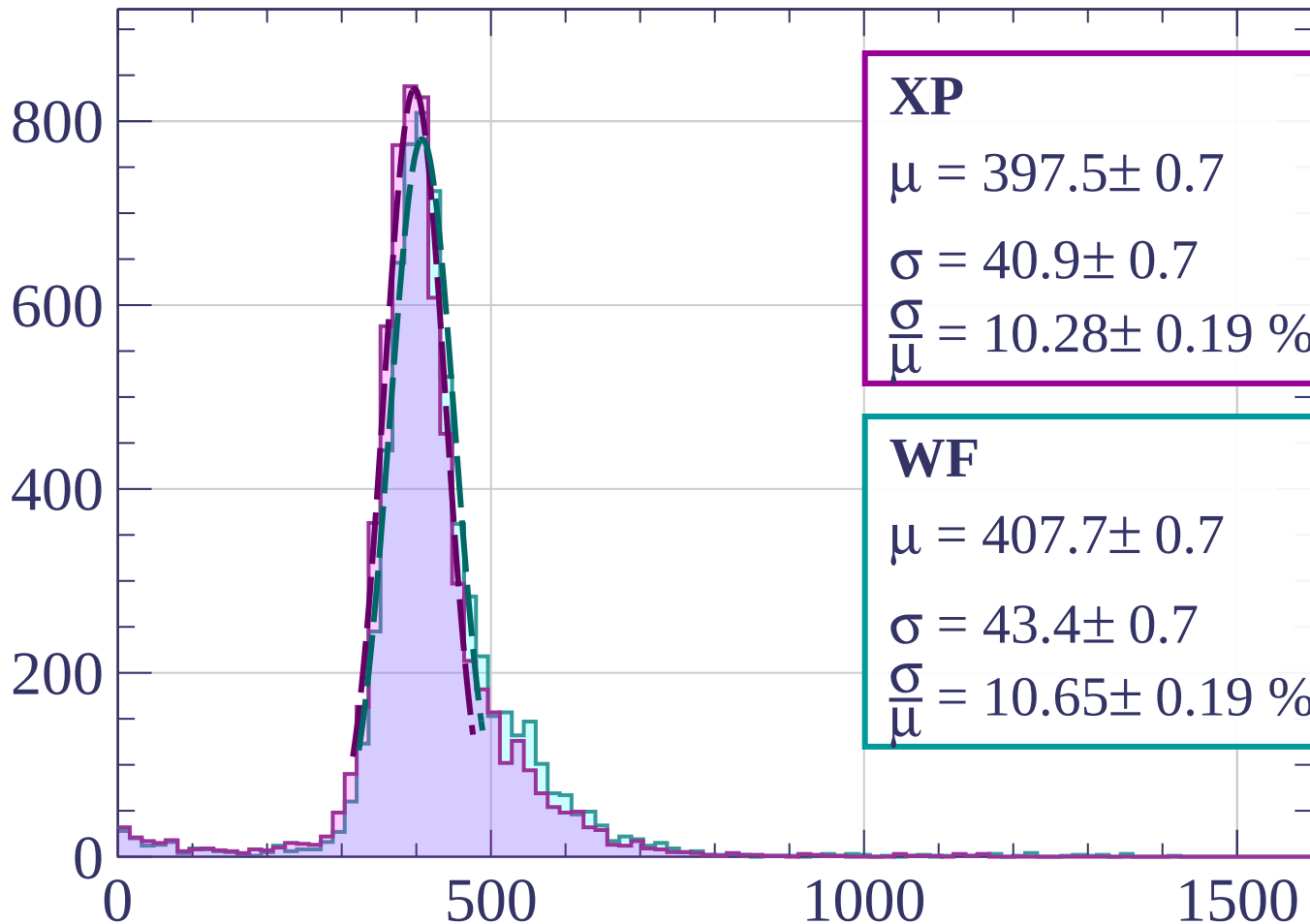
$$\sigma = 43.8 \pm 0.7$$

$$\frac{\sigma}{\mu} = 10.73 \pm 0.20 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-340 < p < -320$

Count



XP

$$\mu = 397.5 \pm 0.7$$

$$\sigma = 40.9 \pm 0.7$$

$$\frac{\sigma}{\mu} = 10.28 \pm 0.19 \%$$

WF

$$\mu = 407.7 \pm 0.7$$

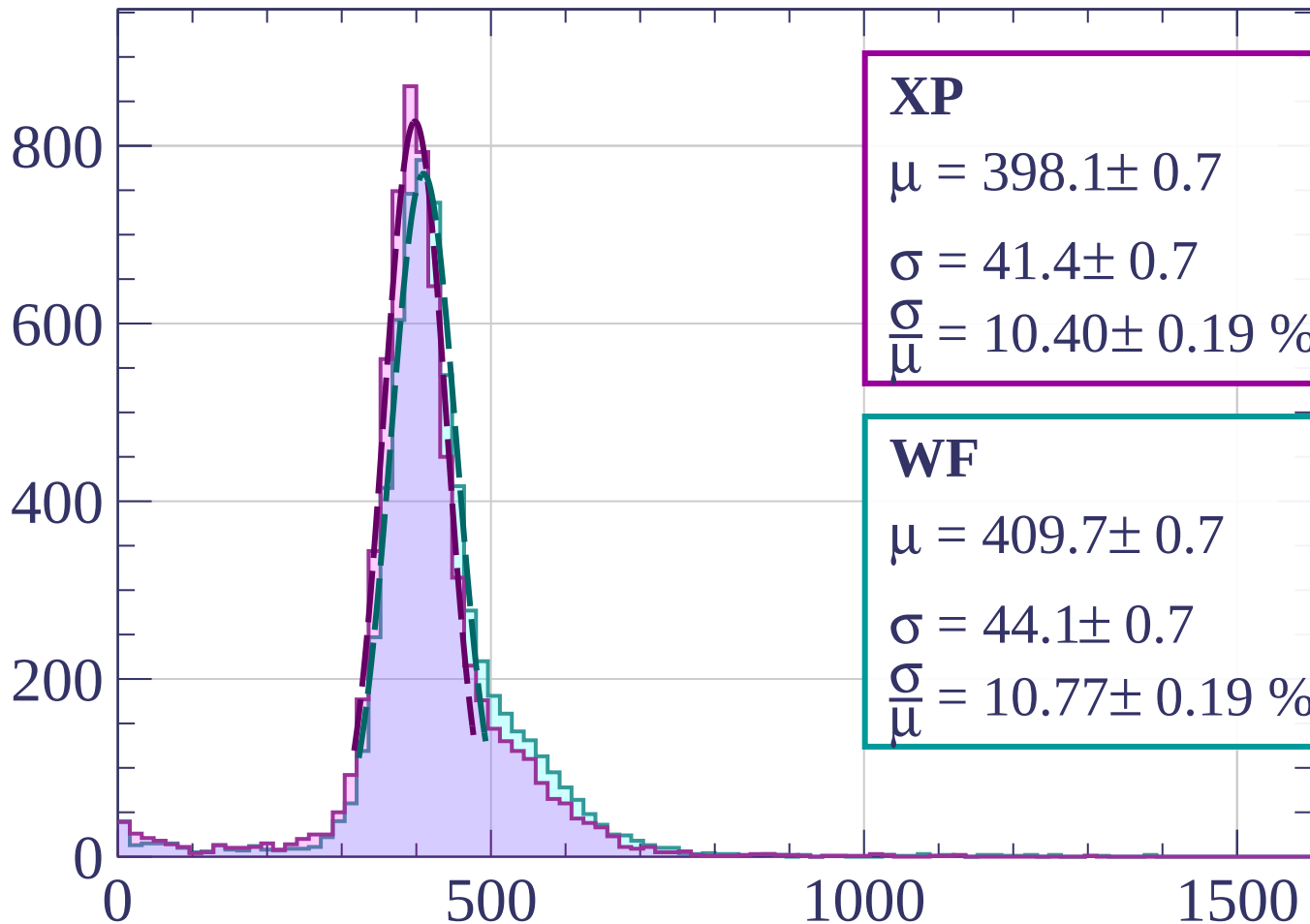
$$\sigma = 43.4 \pm 0.7$$

$$\frac{\sigma}{\mu} = 10.65 \pm 0.19 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-320 < p < -300$

Count



XP

$$\mu = 398.1 \pm 0.7$$

$$\sigma = 41.4 \pm 0.7$$

$$\frac{\sigma}{\mu} = 10.40 \pm 0.19 \%$$

WF

$$\mu = 409.7 \pm 0.7$$

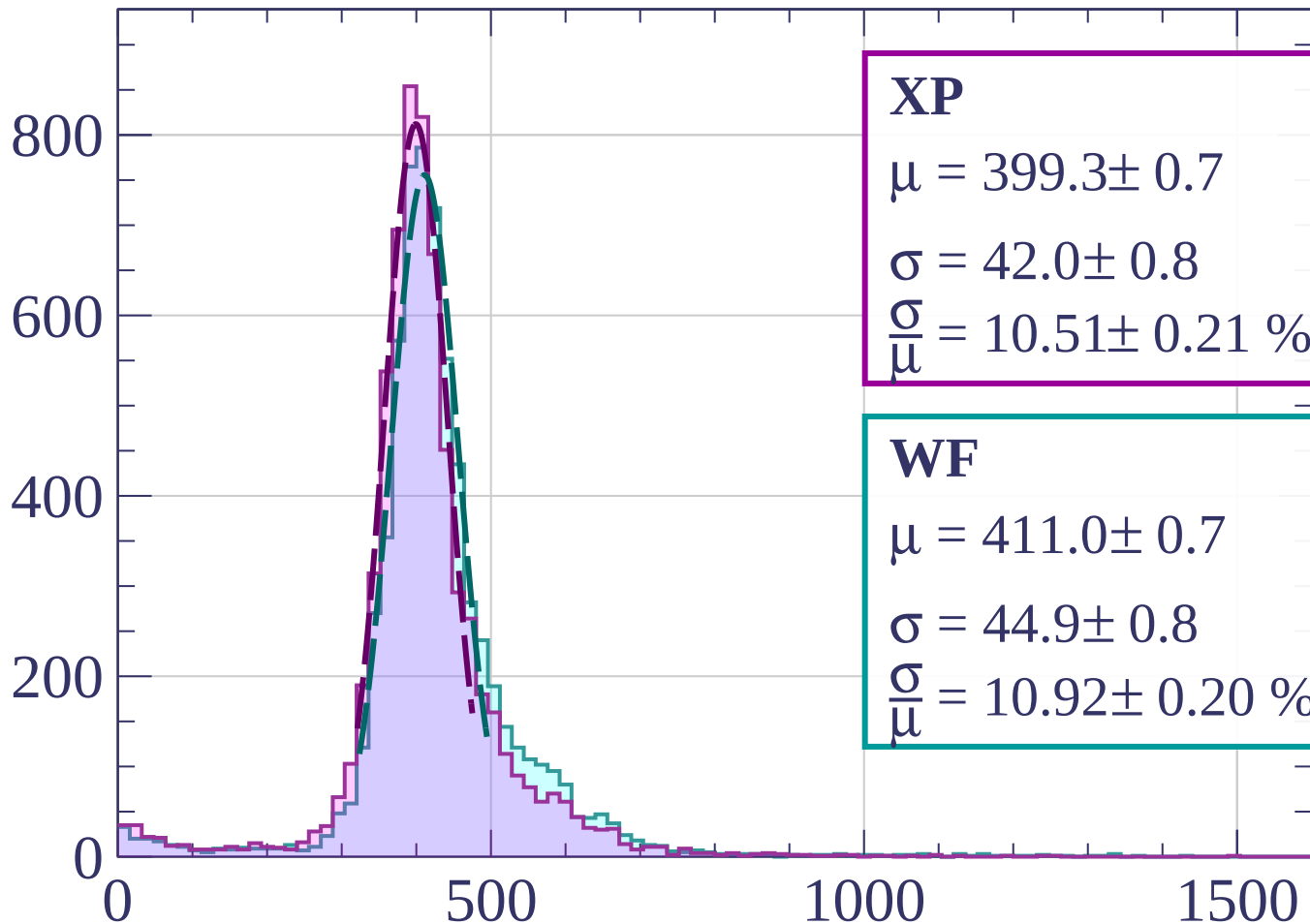
$$\sigma = 44.1 \pm 0.7$$

$$\frac{\sigma}{\mu} = 10.77 \pm 0.19 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-300 < p < -280$

Count



XP

$$\mu = 399.3 \pm 0.7$$

$$\sigma = 42.0 \pm 0.8$$

$$\frac{\sigma}{\mu} = 10.51 \pm 0.21 \%$$

WF

$$\mu = 411.0 \pm 0.7$$

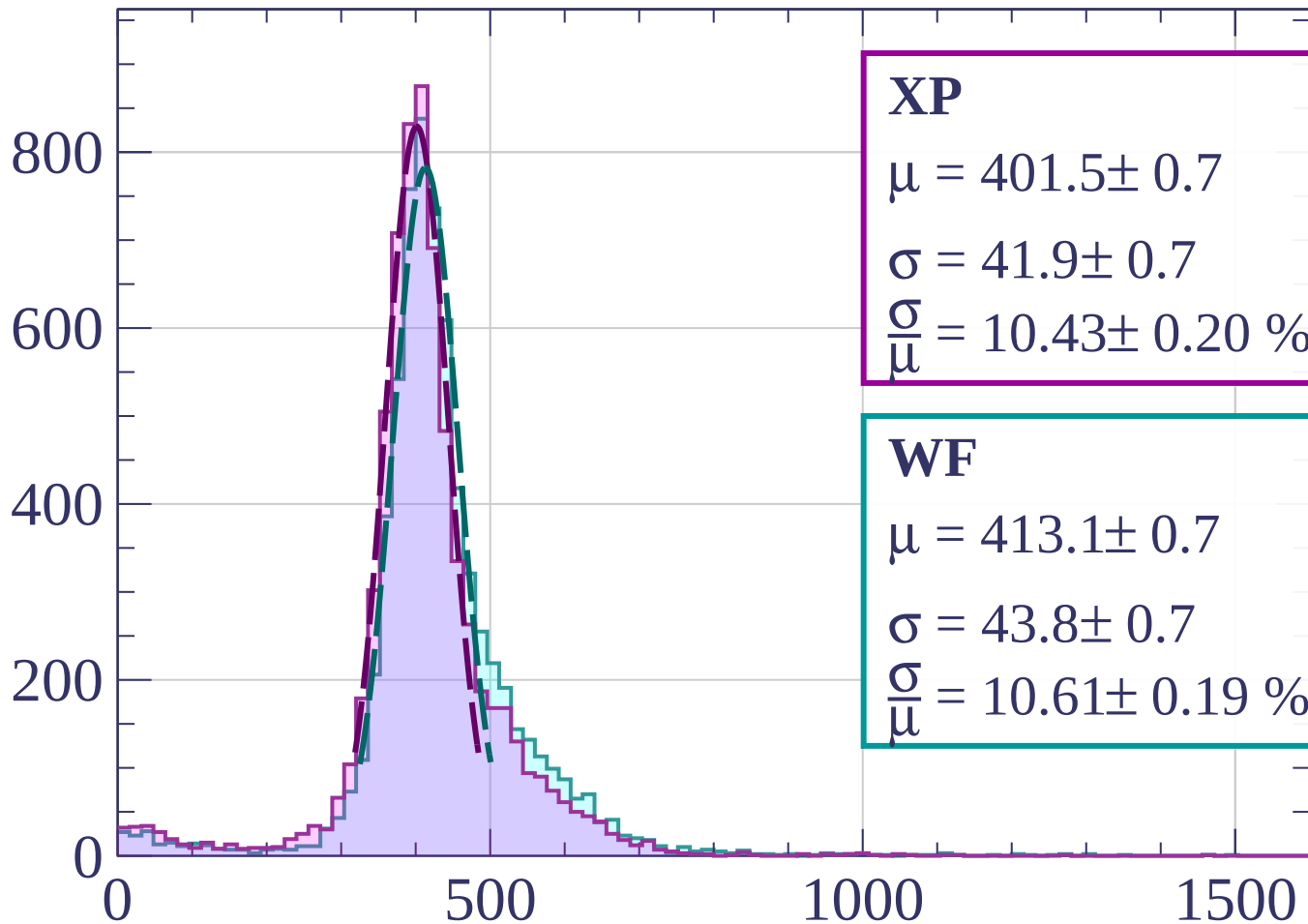
$$\sigma = 44.9 \pm 0.8$$

$$\frac{\sigma}{\mu} = 10.92 \pm 0.20 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-280 < p < -260$

Count



XP

$$\mu = 401.5 \pm 0.7$$

$$\sigma = 41.9 \pm 0.7$$

$$\frac{\sigma}{\mu} = 10.43 \pm 0.20 \%$$

WF

$$\mu = 413.1 \pm 0.7$$

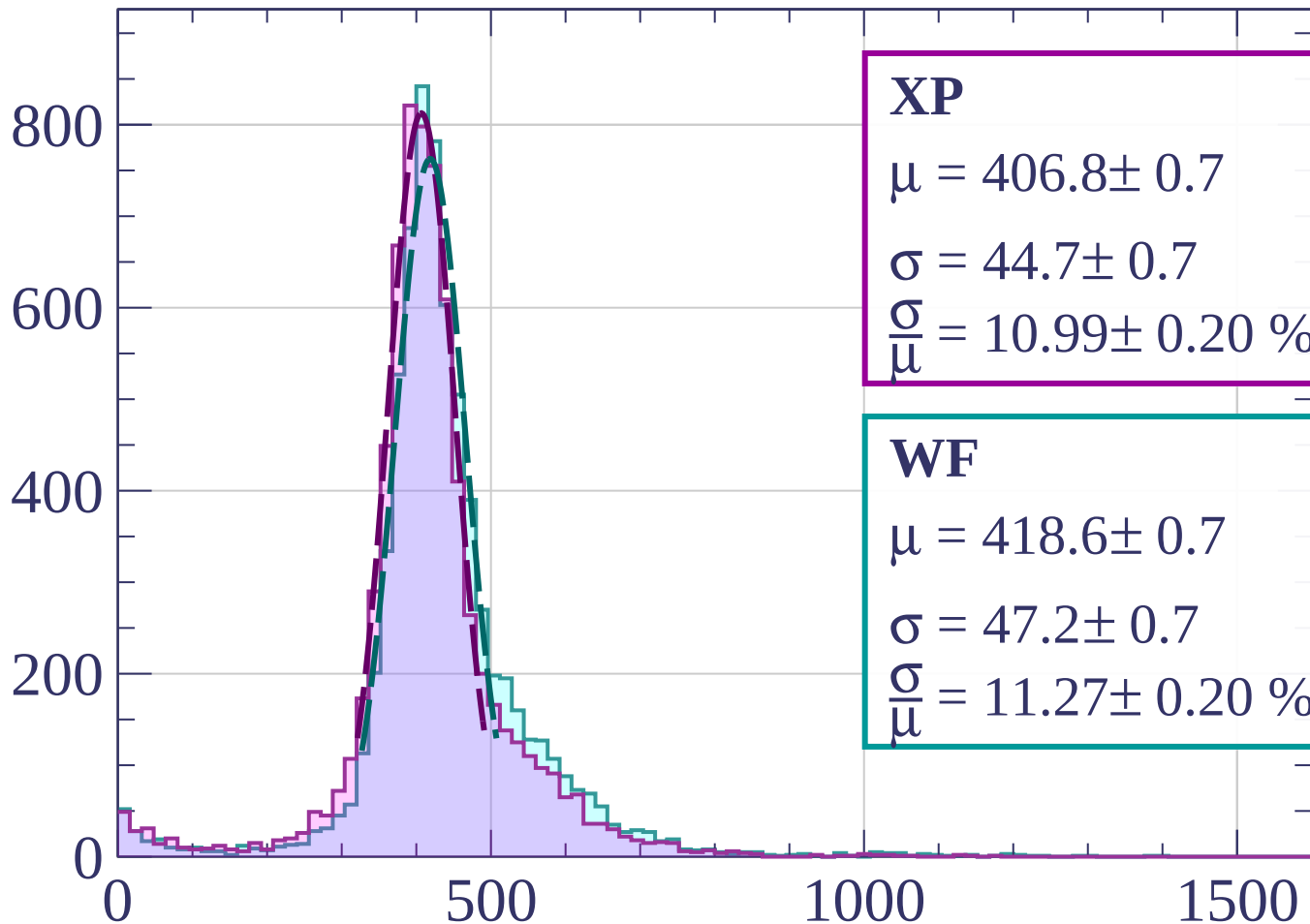
$$\sigma = 43.8 \pm 0.7$$

$$\frac{\sigma}{\mu} = 10.61 \pm 0.19 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-260 < p < -240$

Count



XP

$$\mu = 406.8 \pm 0.7$$

$$\sigma = 44.7 \pm 0.7$$

$$\frac{\sigma}{\mu} = 10.99 \pm 0.20 \%$$

WF

$$\mu = 418.6 \pm 0.7$$

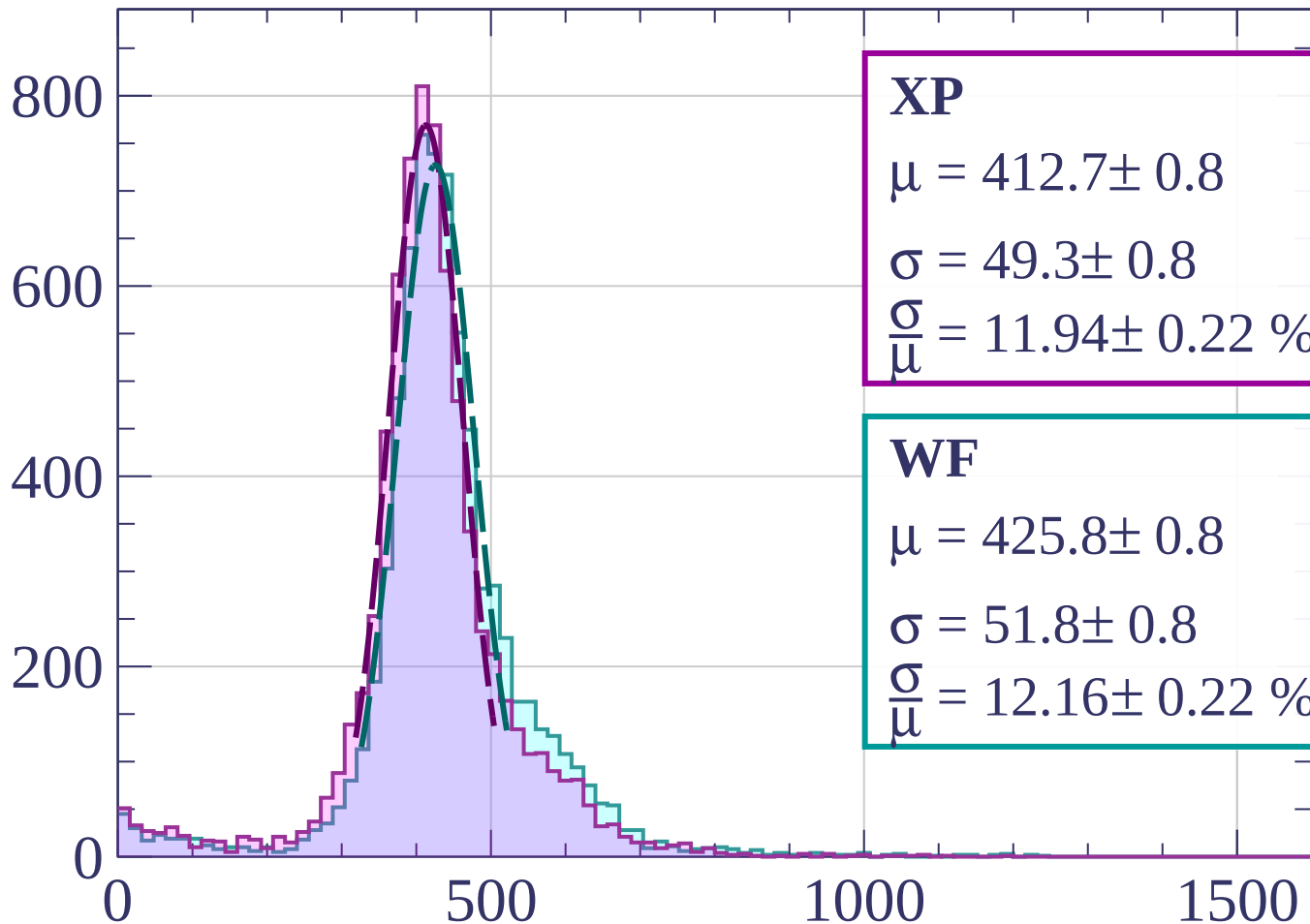
$$\sigma = 47.2 \pm 0.7$$

$$\frac{\sigma}{\mu} = 11.27 \pm 0.20 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-240 < p < -220$

Count



XP

$$\mu = 412.7 \pm 0.8$$

$$\sigma = 49.3 \pm 0.8$$

$$\frac{\sigma}{\mu} = 11.94 \pm 0.22 \%$$

WF

$$\mu = 425.8 \pm 0.8$$

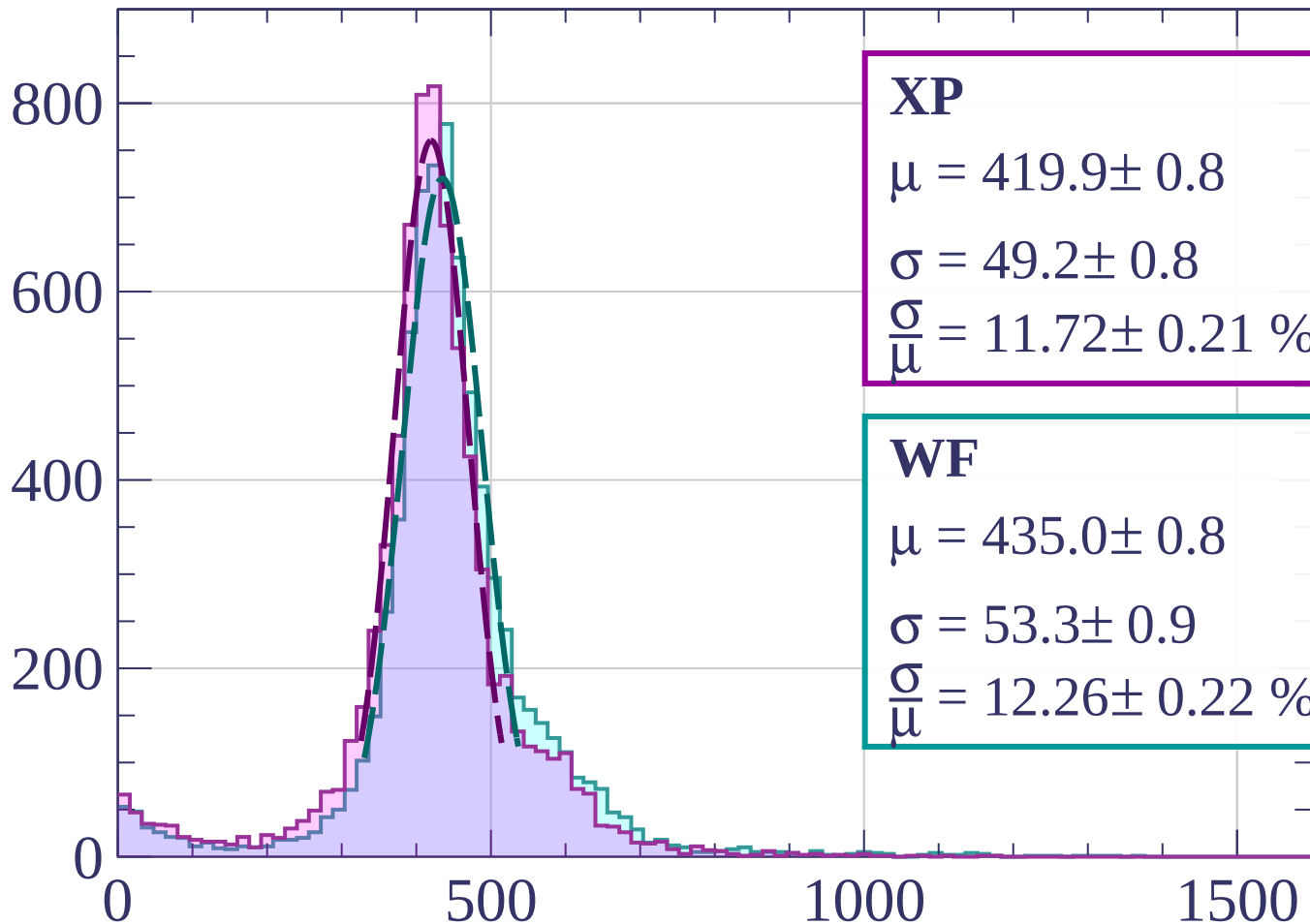
$$\sigma = 51.8 \pm 0.8$$

$$\frac{\sigma}{\mu} = 12.16 \pm 0.22 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-220 < p < -200$

Count



XP

$$\mu = 419.9 \pm 0.8$$

$$\sigma = 49.2 \pm 0.8$$

$$\frac{\sigma}{\mu} = 11.72 \pm 0.21 \%$$

WF

$$\mu = 435.0 \pm 0.8$$

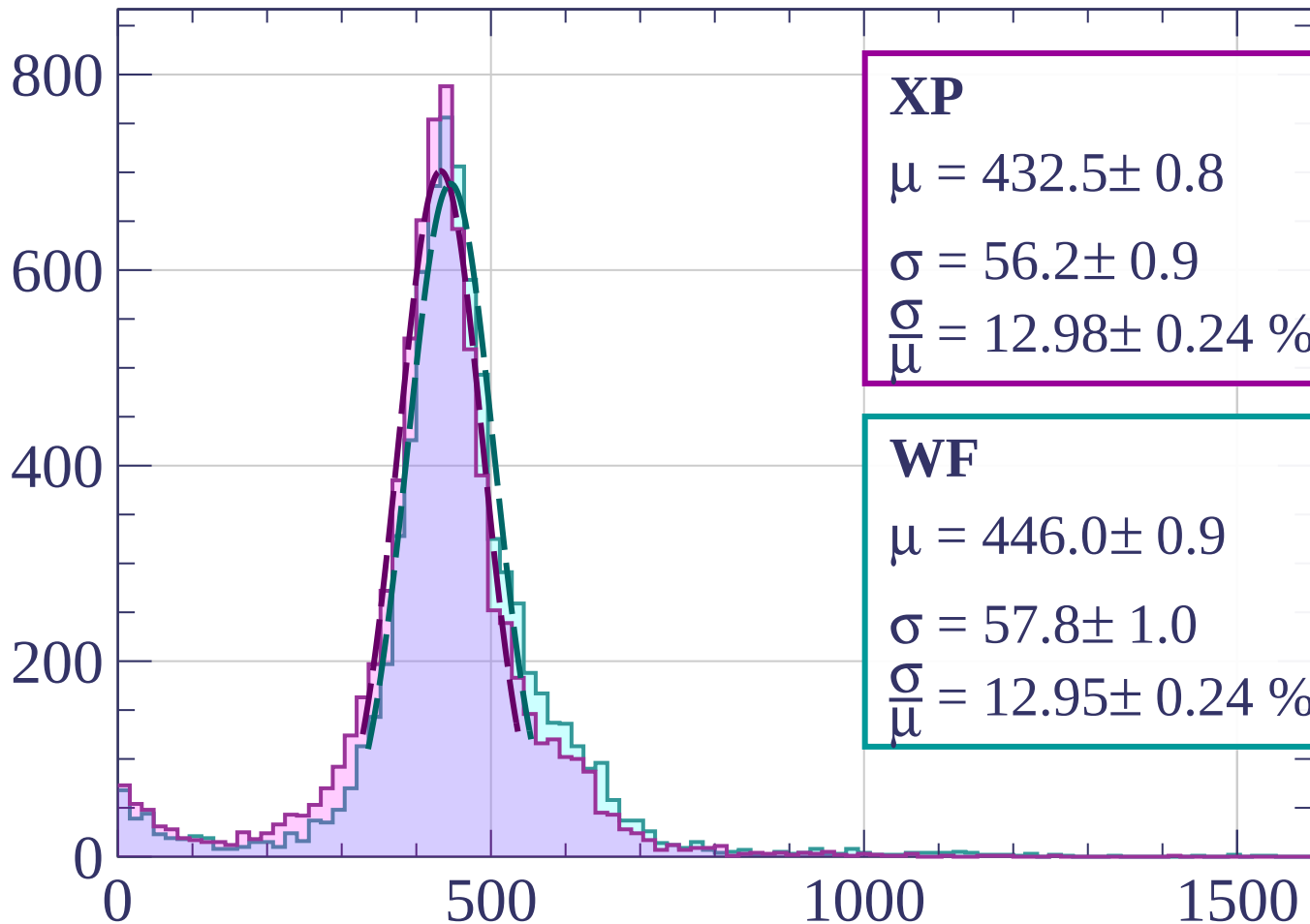
$$\sigma = 53.3 \pm 0.9$$

$$\frac{\sigma}{\mu} = 12.26 \pm 0.22 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-200 < p < -180$

Count



XP

$$\mu = 432.5 \pm 0.8$$

$$\sigma = 56.2 \pm 0.9$$

$$\frac{\sigma}{\mu} = 12.98 \pm 0.24 \%$$

WF

$$\mu = 446.0 \pm 0.9$$

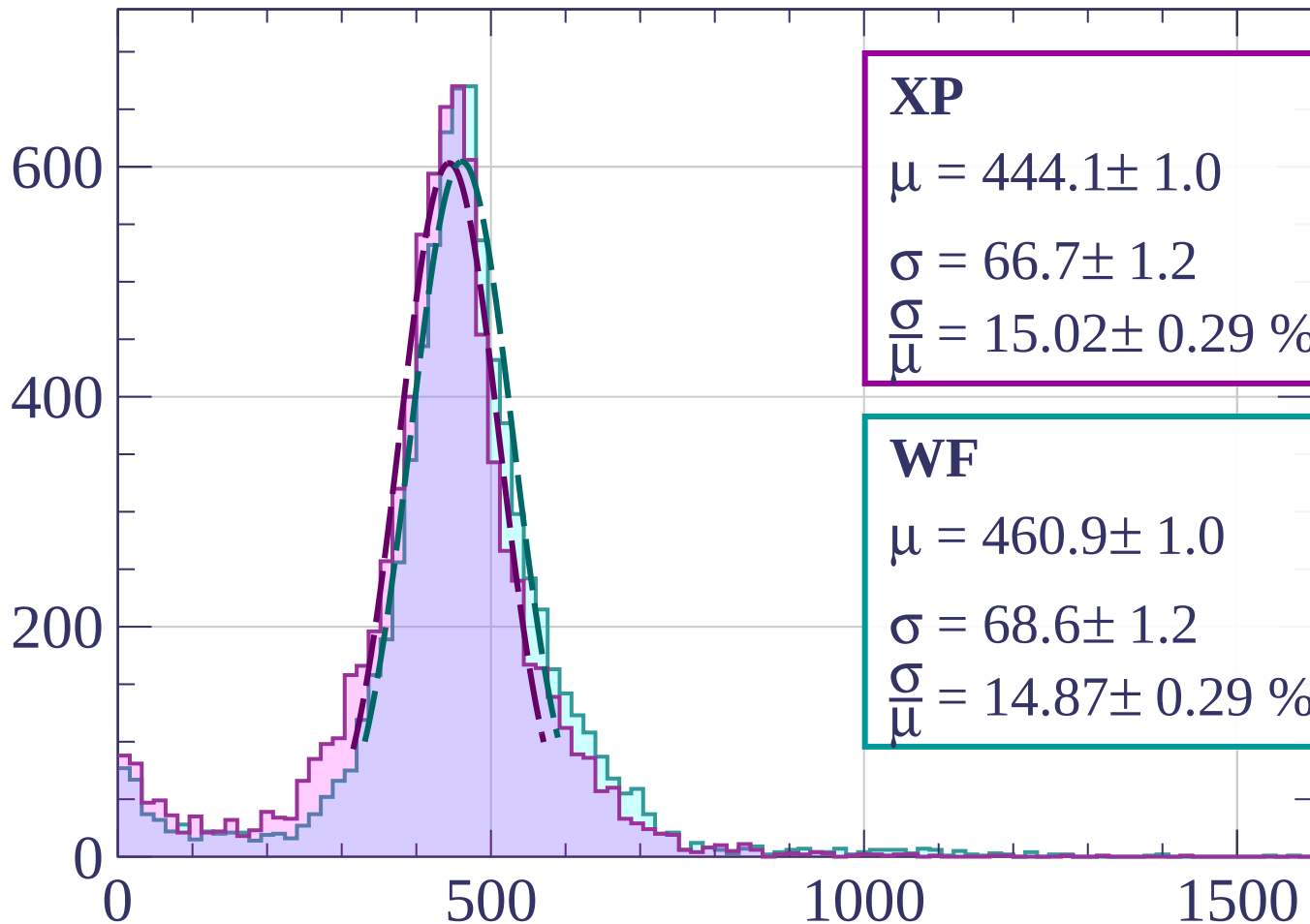
$$\sigma = 57.8 \pm 1.0$$

$$\frac{\sigma}{\mu} = 12.95 \pm 0.24 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-180 < p < -160$

Count



XP

$$\mu = 444.1 \pm 1.0$$

$$\sigma = 66.7 \pm 1.2$$

$$\frac{\sigma}{\mu} = 15.02 \pm 0.29 \%$$

WF

$$\mu = 460.9 \pm 1.0$$

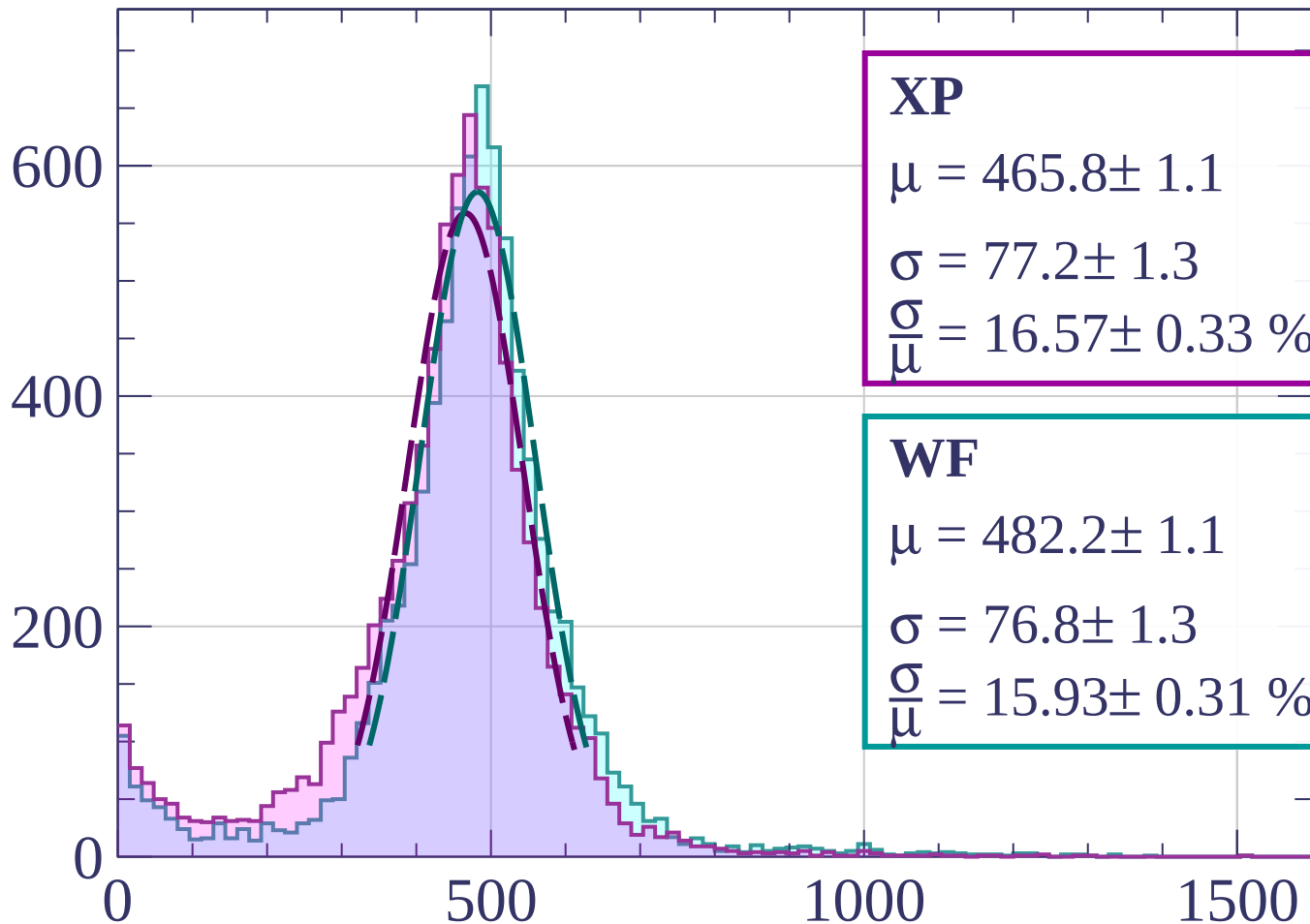
$$\sigma = 68.6 \pm 1.2$$

$$\frac{\sigma}{\mu} = 14.87 \pm 0.29 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-160 < p < -140$

Count



XP

$$\mu = 465.8 \pm 1.1$$

$$\sigma = 77.2 \pm 1.3$$

$$\frac{\sigma}{\mu} = 16.57 \pm 0.33 \%$$

WF

$$\mu = 482.2 \pm 1.1$$

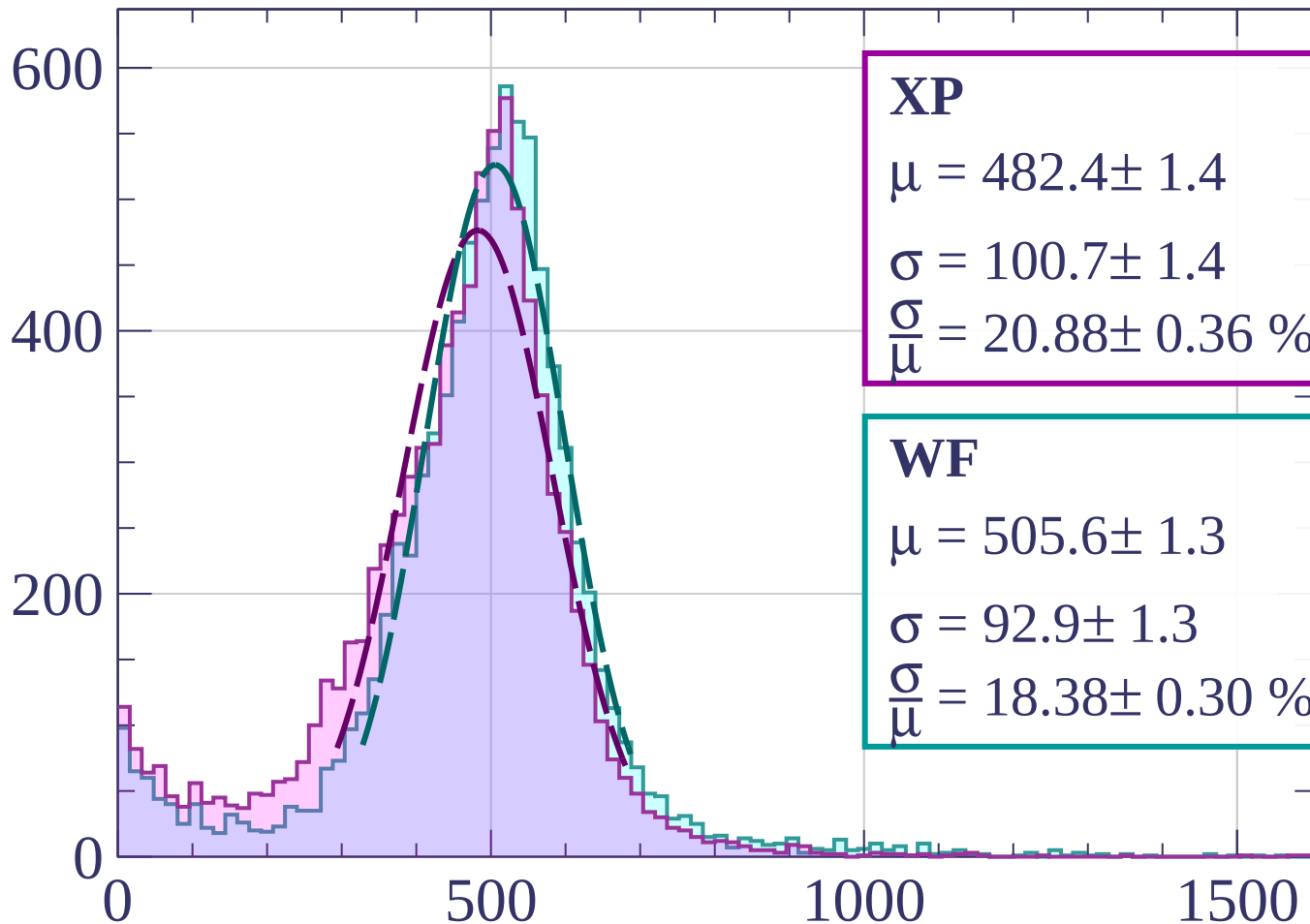
$$\sigma = 76.8 \pm 1.3$$

$$\frac{\sigma}{\mu} = 15.93 \pm 0.31 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-140 < p < -120$

Count



XP

$$\mu = 482.4 \pm 1.4$$

$$\sigma = 100.7 \pm 1.4$$

$$\frac{\sigma}{\mu} = 20.88 \pm 0.36 \%$$

WF

$$\mu = 505.6 \pm 1.3$$

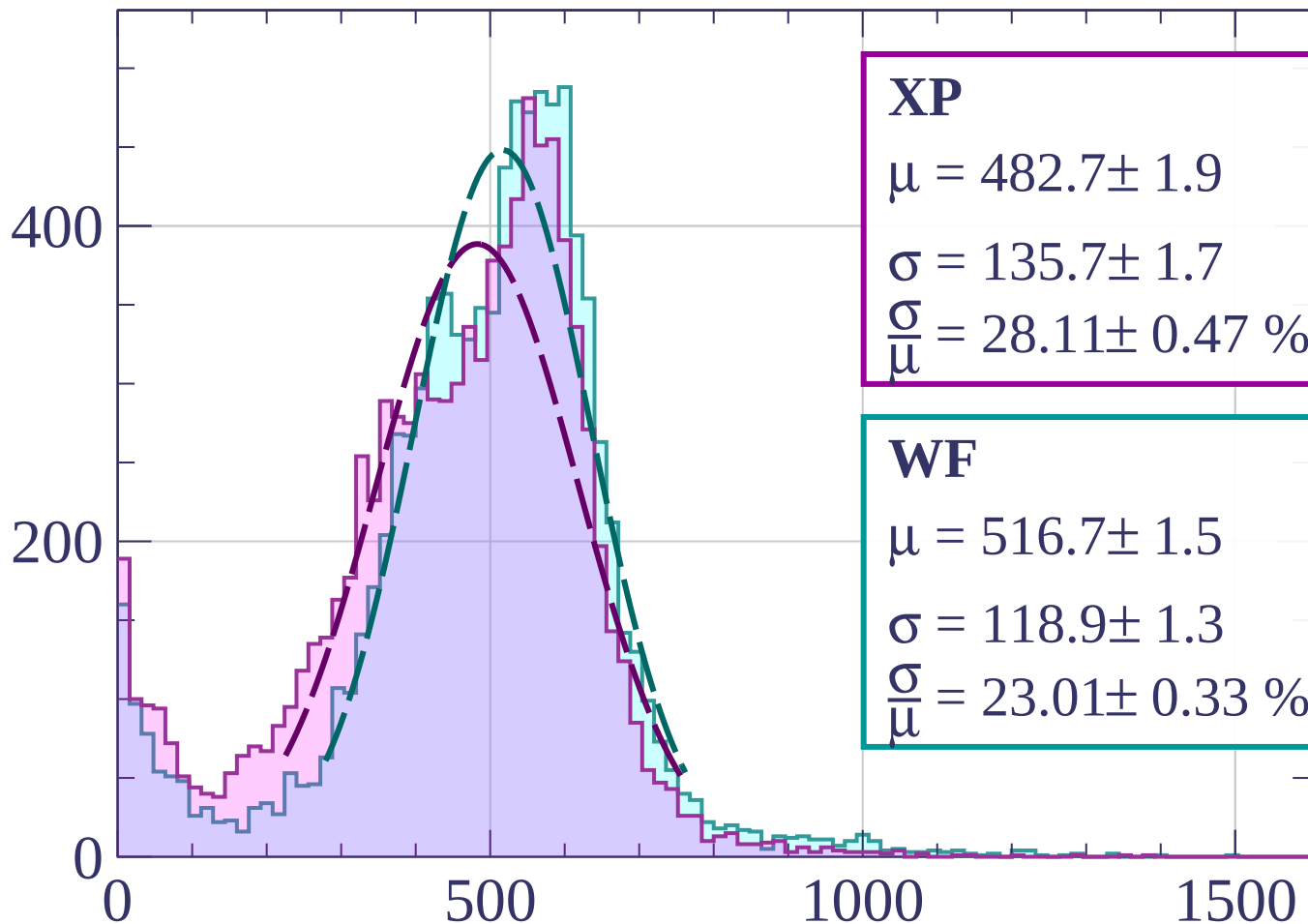
$$\sigma = 92.9 \pm 1.3$$

$$\frac{\sigma}{\mu} = 18.38 \pm 0.30 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-120 < p < -100$

Count



XP

$$\mu = 482.7 \pm 1.9$$

$$\sigma = 135.7 \pm 1.7$$

$$\frac{\sigma}{\mu} = 28.11 \pm 0.47 \%$$

WF

$$\mu = 516.7 \pm 1.5$$

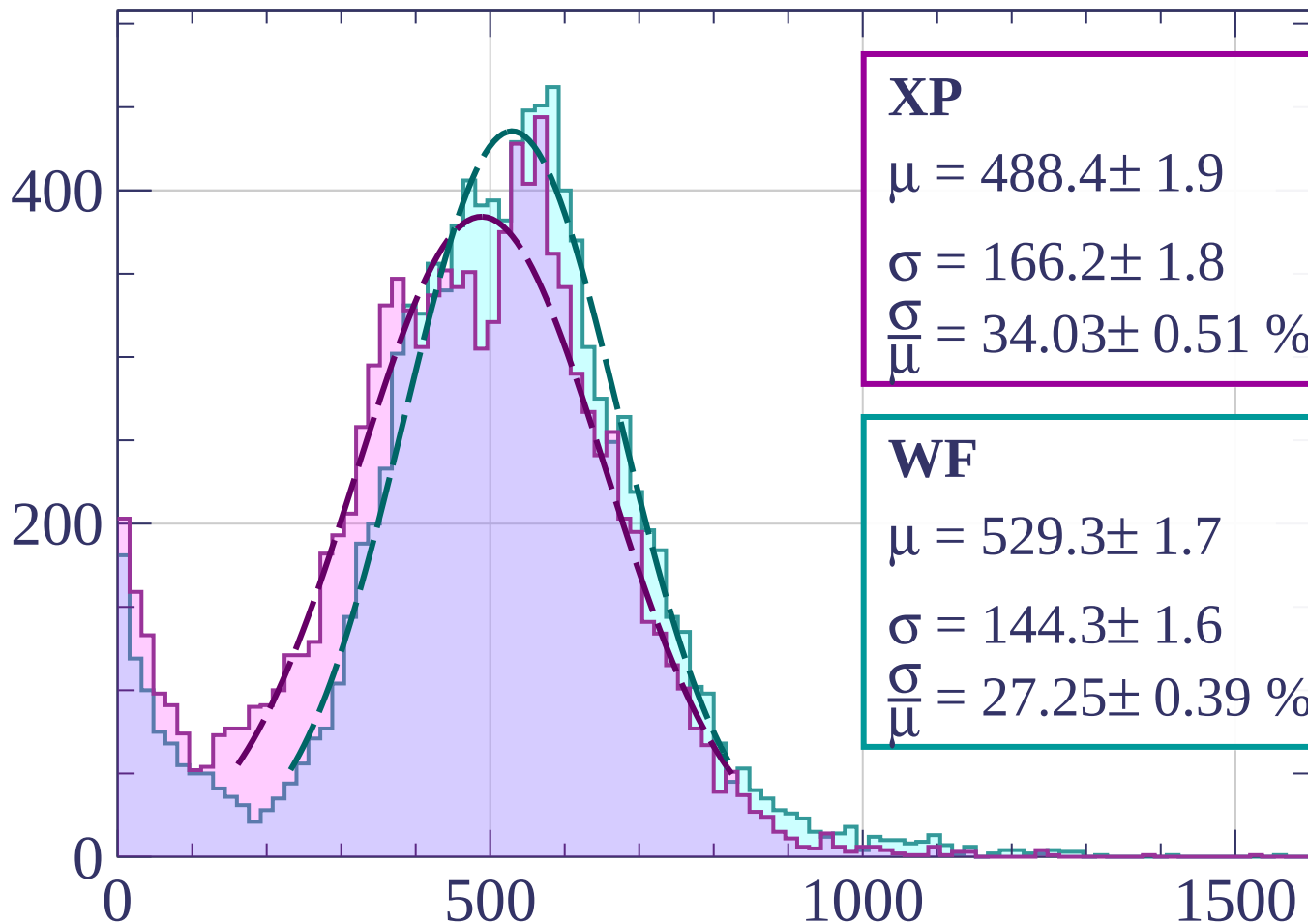
$$\sigma = 118.9 \pm 1.3$$

$$\frac{\sigma}{\mu} = 23.01 \pm 0.33 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-100 < p < -80$

Count



XP

$$\mu = 488.4 \pm 1.9$$

$$\sigma = 166.2 \pm 1.8$$

$$\frac{\sigma}{\mu} = 34.03 \pm 0.51 \%$$

WF

$$\mu = 529.3 \pm 1.7$$

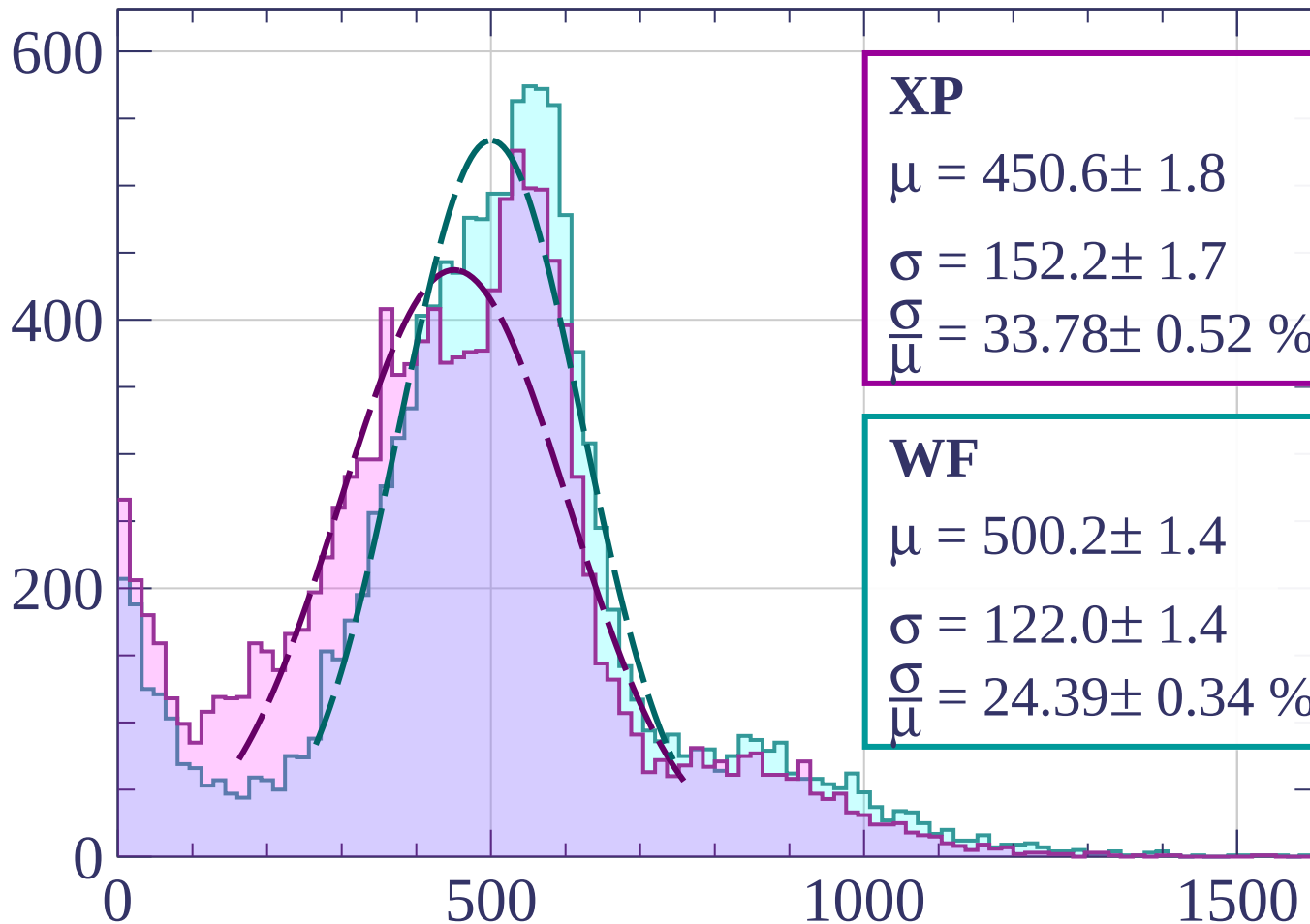
$$\sigma = 144.3 \pm 1.6$$

$$\frac{\sigma}{\mu} = 27.25 \pm 0.39 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-80 < p < -60$

Count



XP

$$\mu = 450.6 \pm 1.8$$

$$\sigma = 152.2 \pm 1.7$$

$$\frac{\sigma}{\mu} = 33.78 \pm 0.52 \%$$

WF

$$\mu = 500.2 \pm 1.4$$

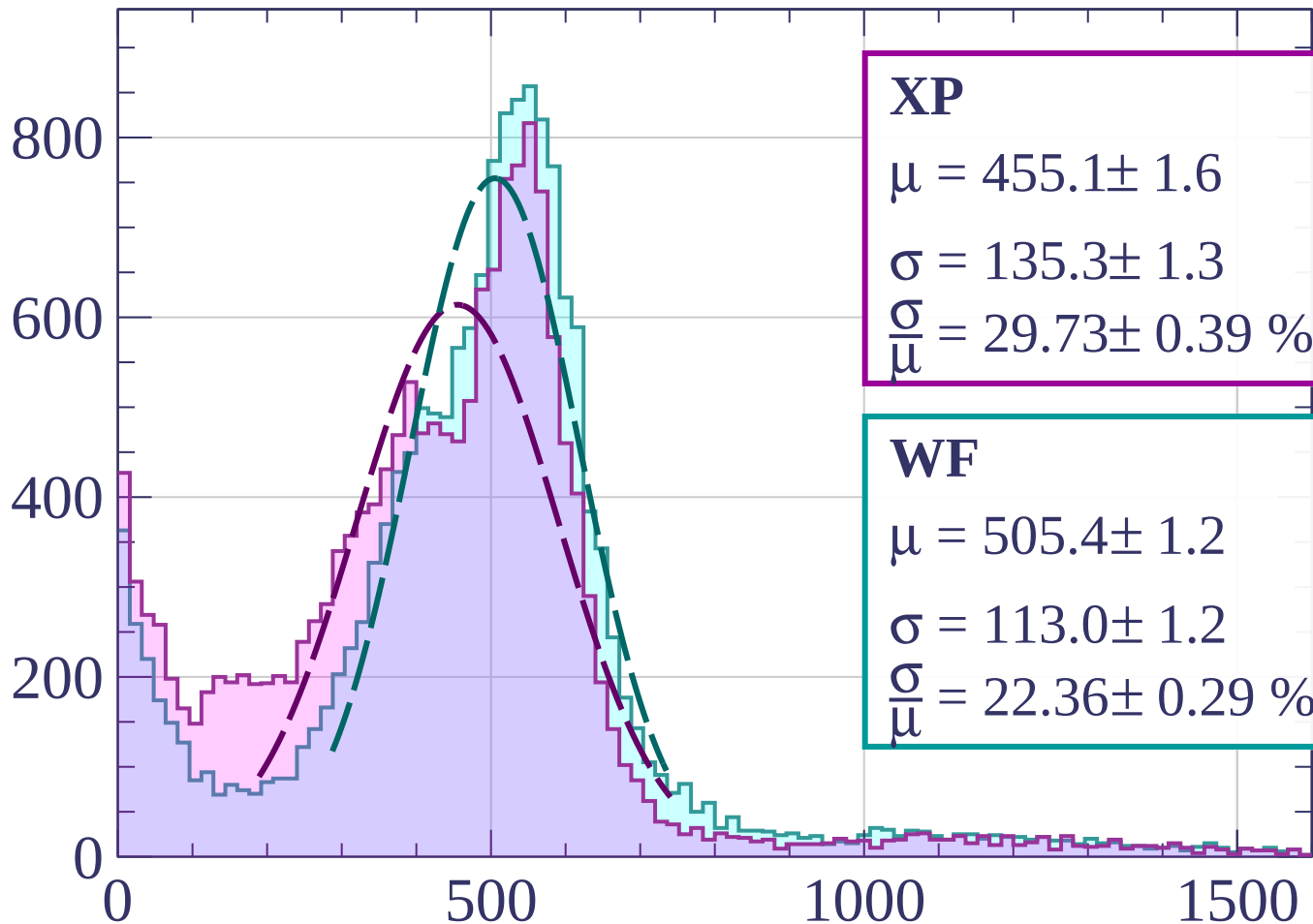
$$\sigma = 122.0 \pm 1.4$$

$$\frac{\sigma}{\mu} = 24.39 \pm 0.34 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-60 < p < -40$

Count



XP

$$\mu = 455.1 \pm 1.6$$

$$\sigma = 135.3 \pm 1.3$$

$$\frac{\sigma}{\mu} = 29.73 \pm 0.39 \%$$

WF

$$\mu = 505.4 \pm 1.2$$

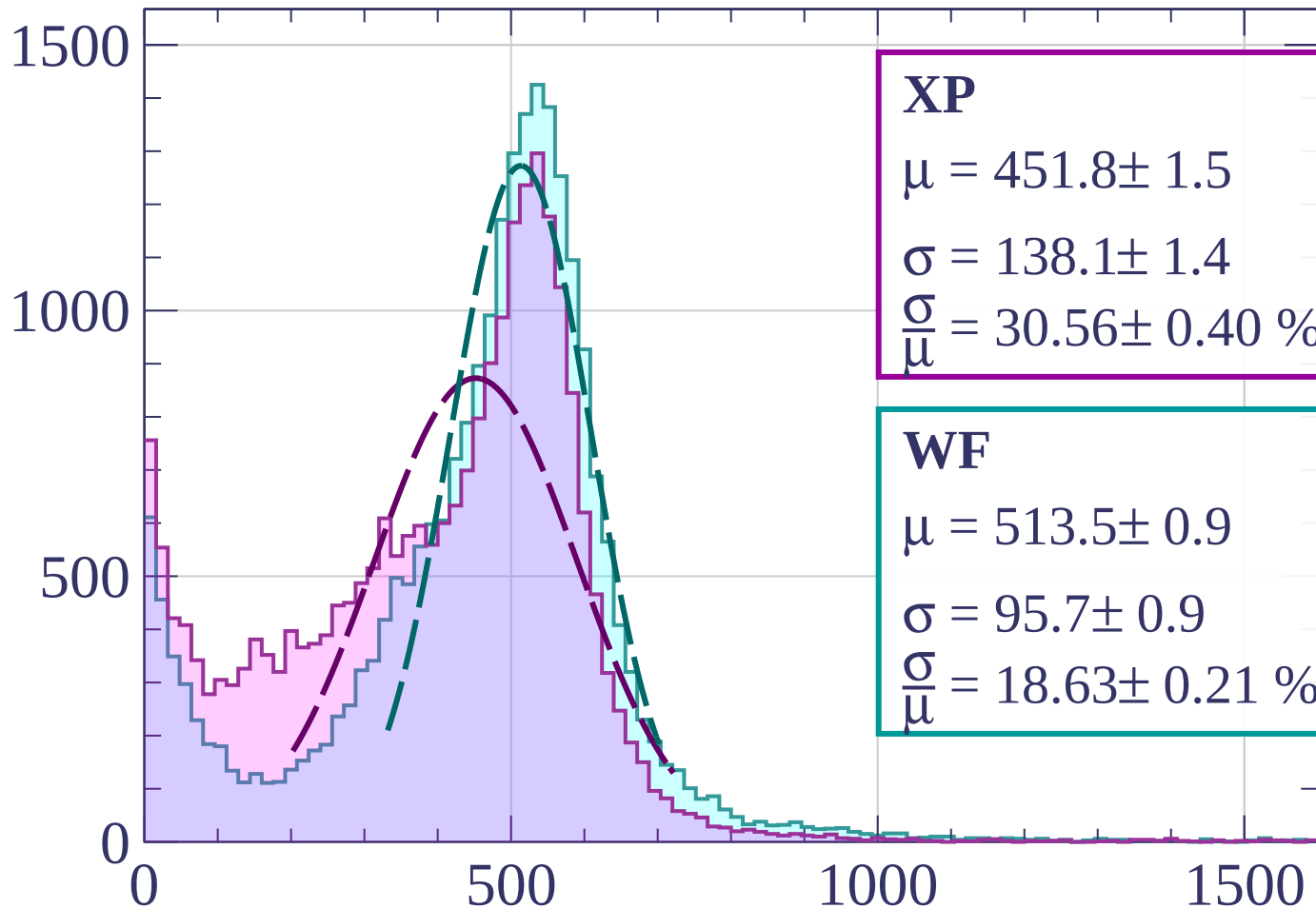
$$\sigma = 113.0 \pm 1.2$$

$$\frac{\sigma}{\mu} = 22.36 \pm 0.29 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-40 < p < -20$

Count



XP

$$\mu = 451.8 \pm 1.5$$

$$\sigma = 138.1 \pm 1.4$$

$$\frac{\sigma}{\mu} = 30.56 \pm 0.40 \%$$

WF

$$\mu = 513.5 \pm 0.9$$

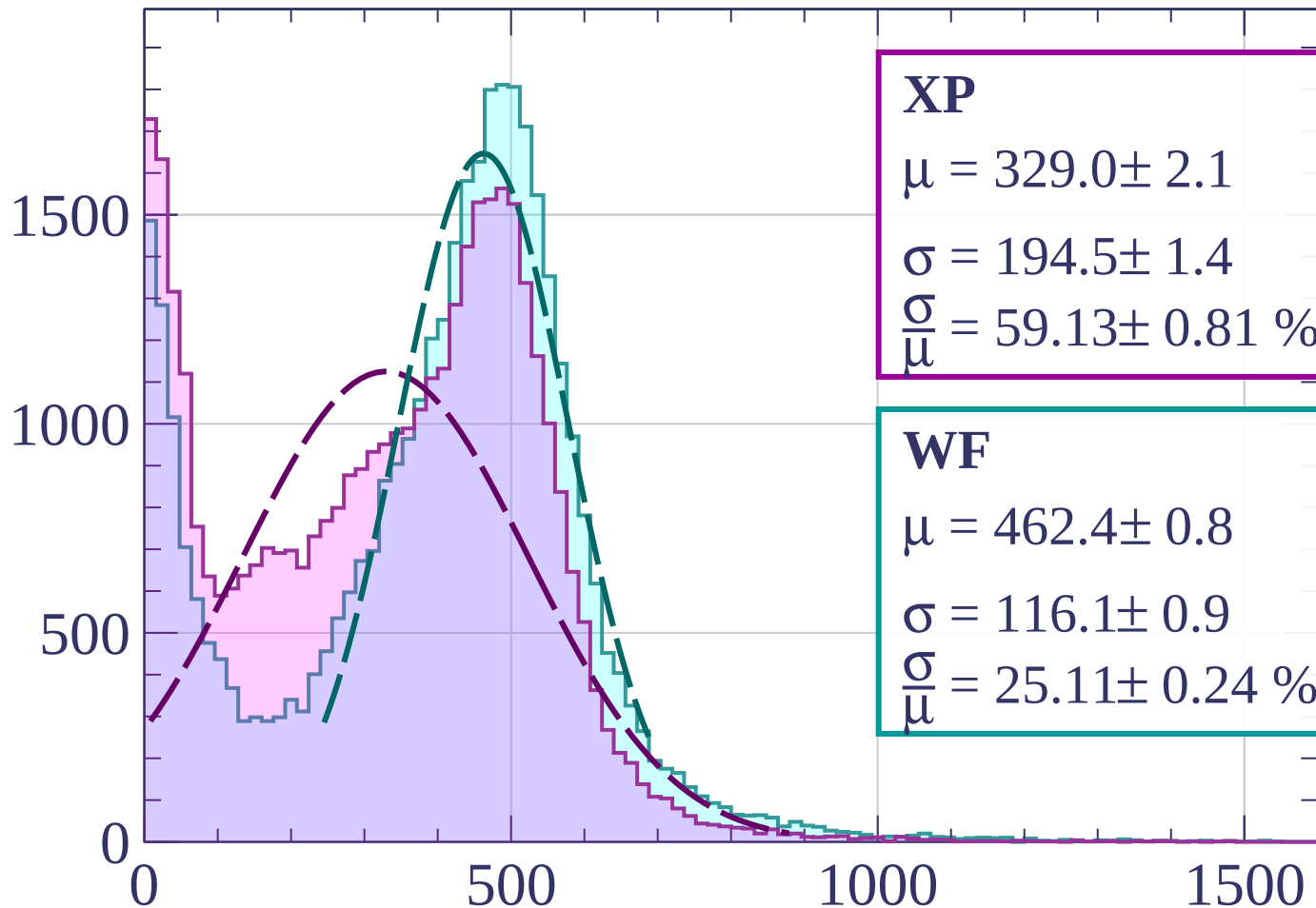
$$\sigma = 95.7 \pm 0.9$$

$$\frac{\sigma}{\mu} = 18.63 \pm 0.21 \%$$

dE/dx [ADC counts/cm]

Energy loss | $-20 < p < 0$

Count



XP

$$\mu = 329.0 \pm 2.1$$

$$\sigma = 194.5 \pm 1.4$$

$$\frac{\sigma}{\mu} = 59.13 \pm 0.81 \%$$

WF

$$\mu = 462.4 \pm 0.8$$

$$\sigma = 116.1 \pm 0.9$$

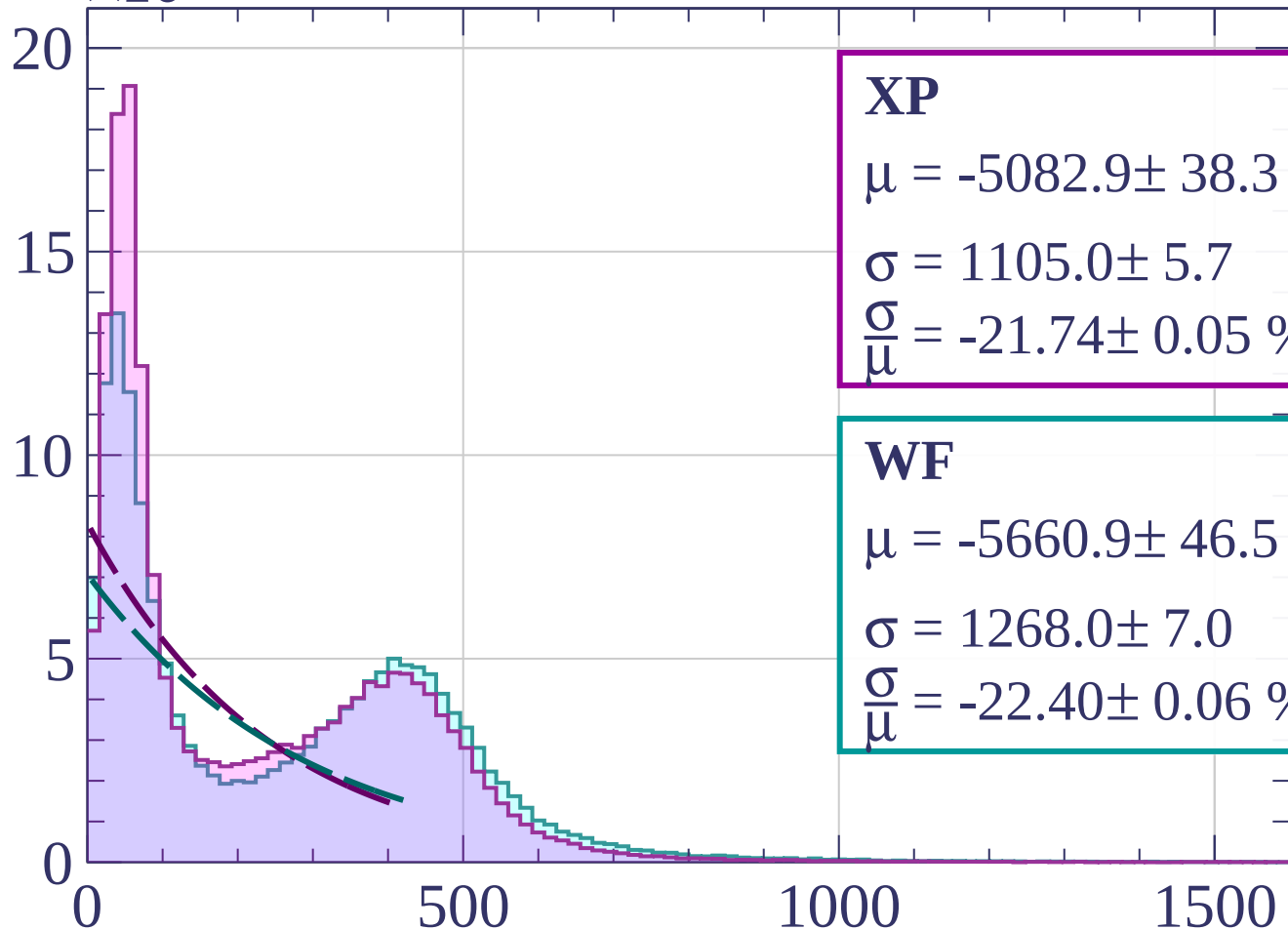
$$\frac{\sigma}{\mu} = 25.11 \pm 0.24 \%$$

dE/dx [ADC counts/cm]

Count

$\times 10^3$

Energy loss | $0 < p < 20$



XP

$$\mu = -5082.9 \pm 38.3$$

$$\sigma = 1105.0 \pm 5.7$$

$$\frac{\sigma}{\mu} = -21.74 \pm 0.05 \%$$

WF

$$\mu = -5660.9 \pm 46.5$$

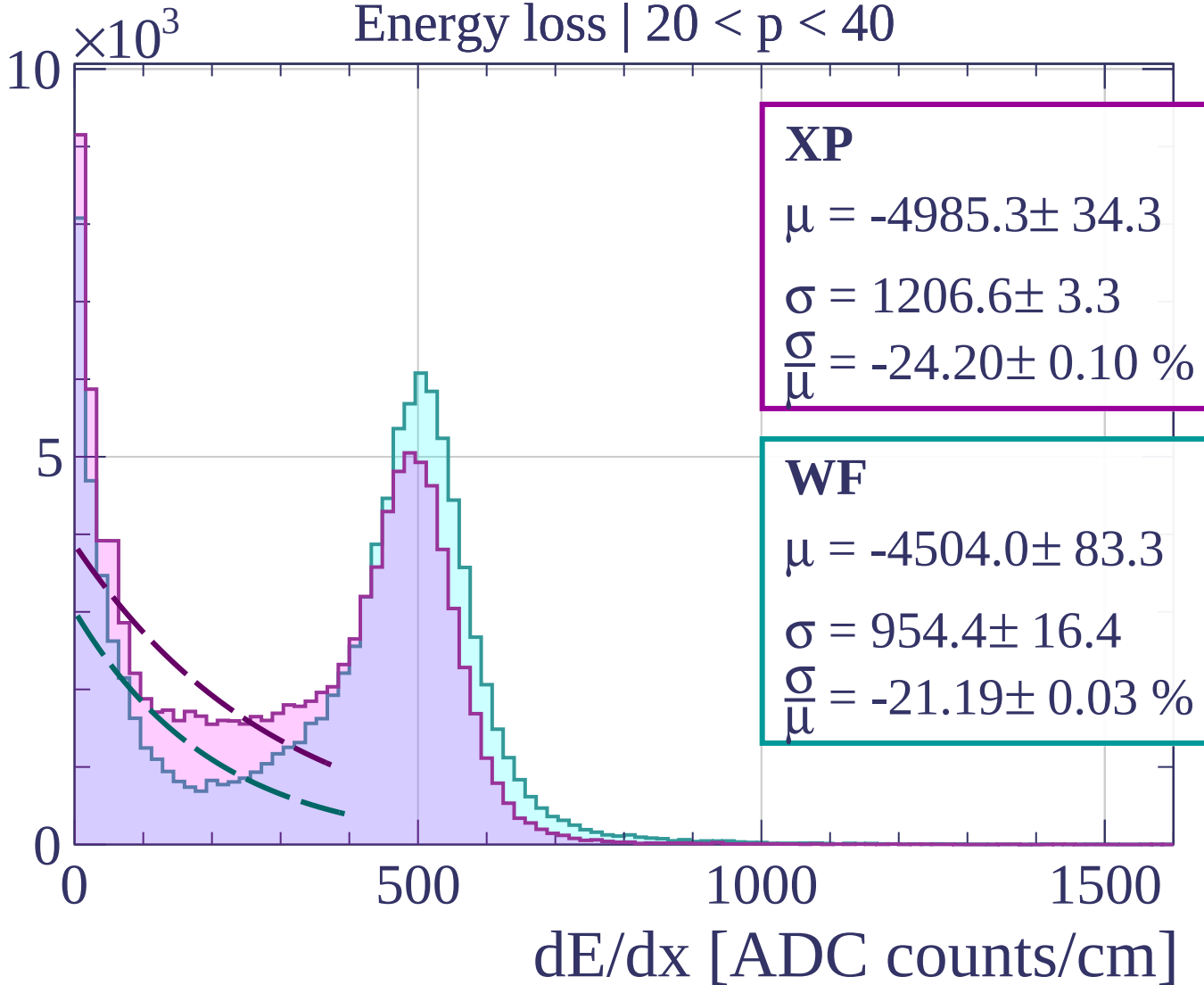
$$\sigma = 1268.0 \pm 7.0$$

$$\frac{\sigma}{\mu} = -22.40 \pm 0.06 \%$$

dE/dx [ADC counts/cm]

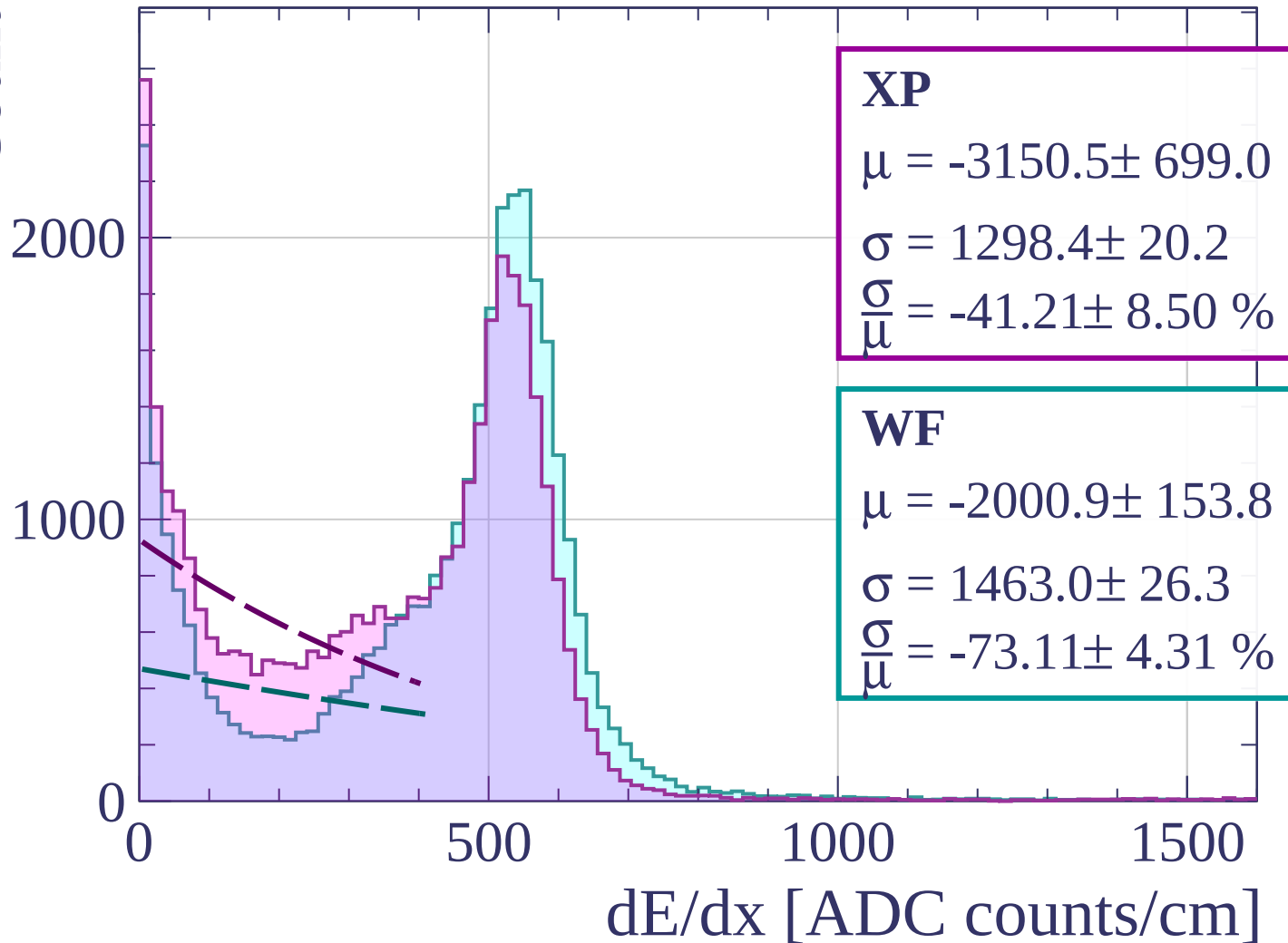
Count

Energy loss | $20 < p < 40$



Energy loss | $40 < p < 60$

Count



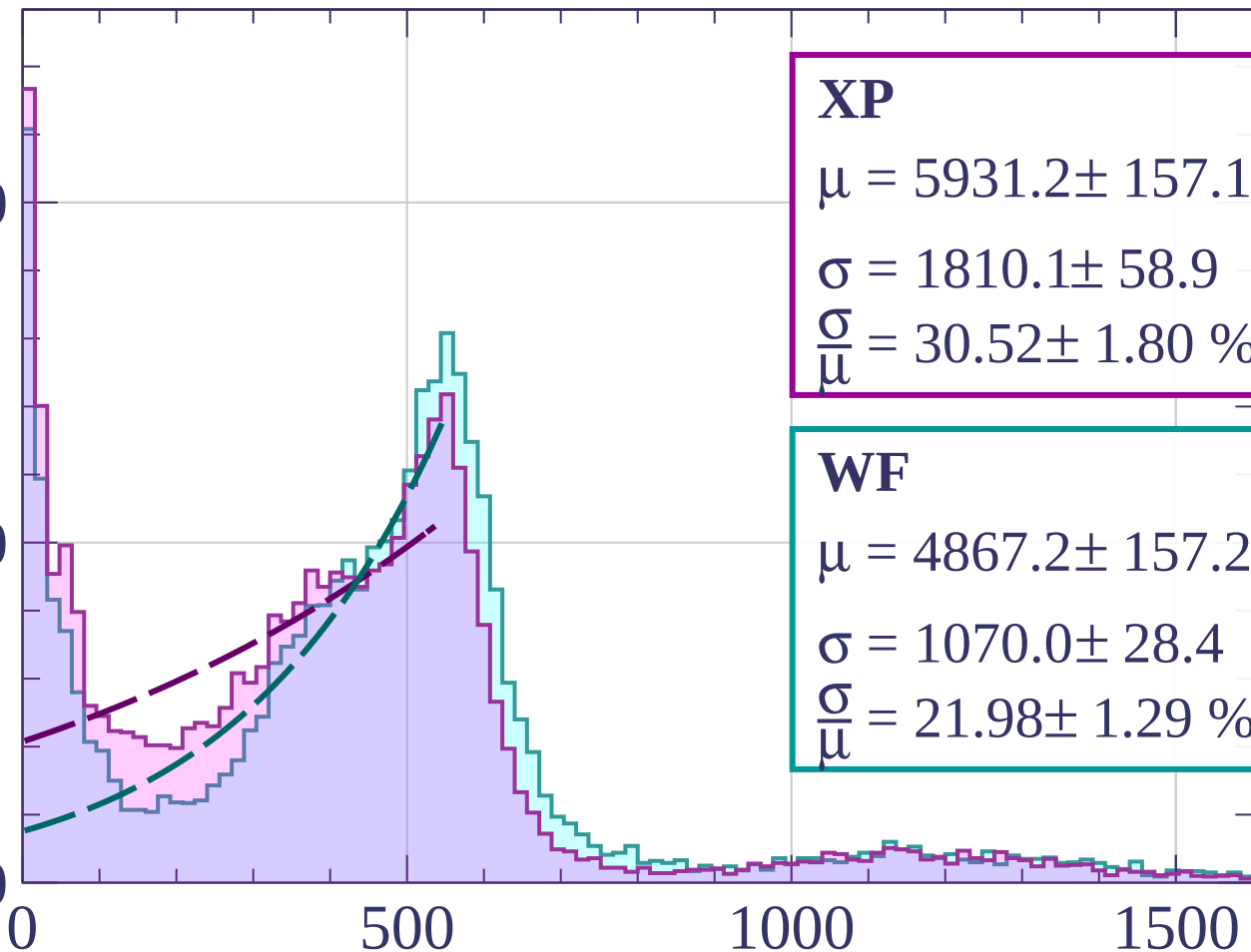
Energy loss | $60 < p < 80$

Count

1000

500

0



XP

$$\mu = 5931.2 \pm 157.1$$

$$\sigma = 1810.1 \pm 58.9$$

$$\frac{\sigma}{\mu} = 30.52 \pm 1.80 \%$$

WF

$$\mu = 4867.2 \pm 157.2$$

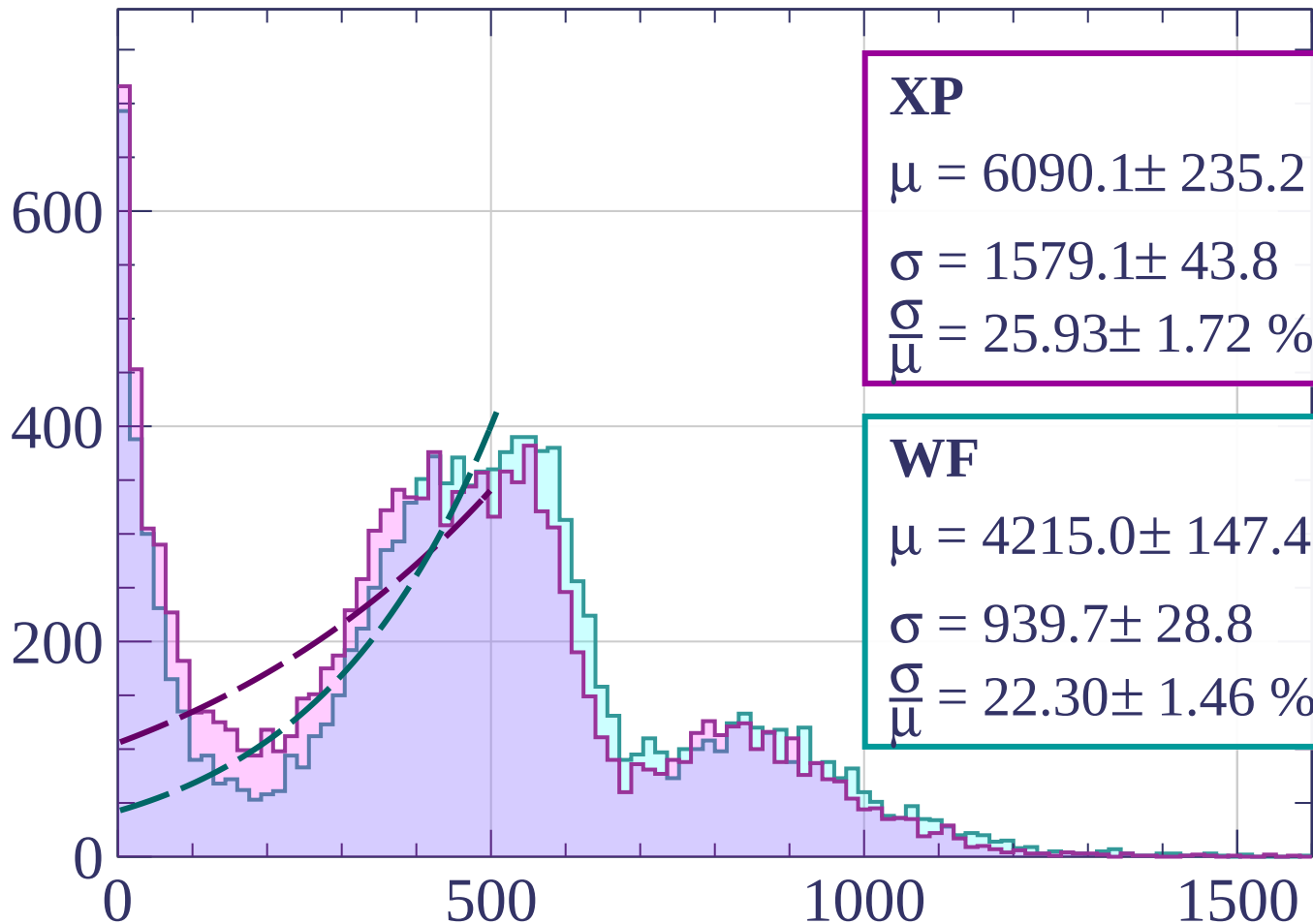
$$\sigma = 1070.0 \pm 28.4$$

$$\frac{\sigma}{\mu} = 21.98 \pm 1.29 \%$$

dE/dx [ADC counts/cm]

Energy loss | $80 < p < 100$

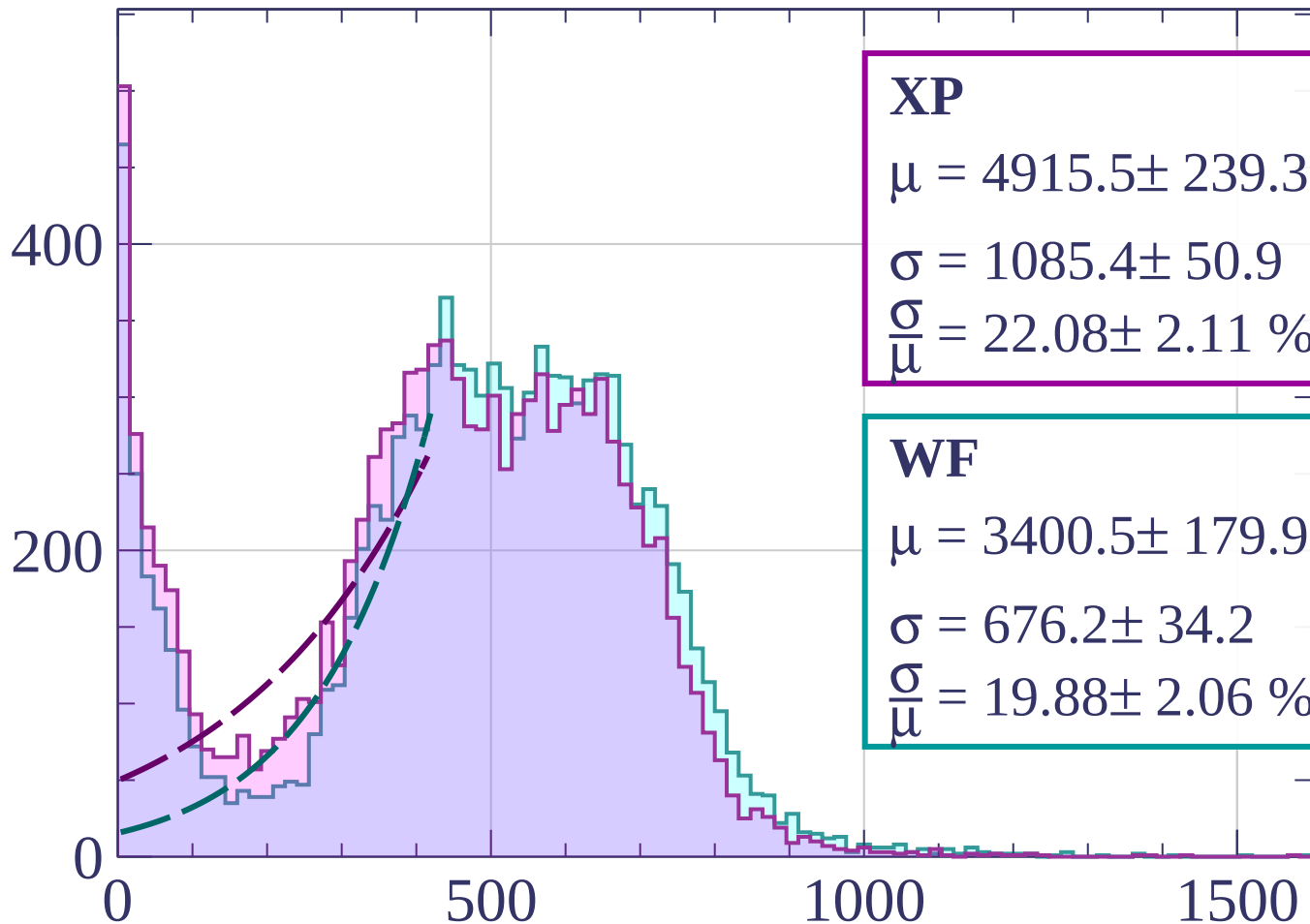
Count



dE/dx [ADC counts/cm]

Energy loss | $100 < p < 120$

Count



XP

$$\mu = 4915.5 \pm 239.3$$

$$\sigma = 1085.4 \pm 50.9$$

$$\frac{\sigma}{\mu} = 22.08 \pm 2.11 \%$$

WF

$$\mu = 3400.5 \pm 179.9$$

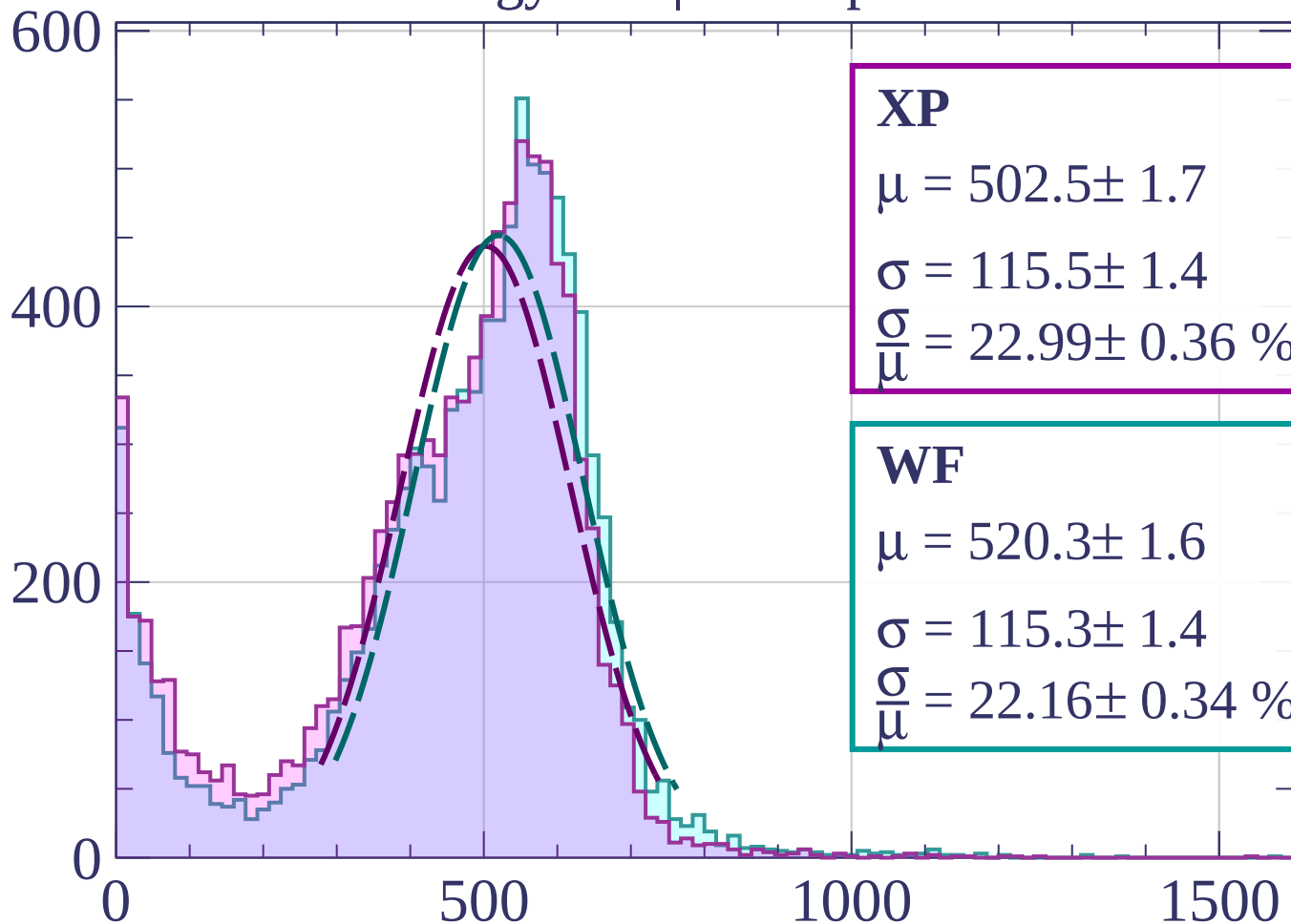
$$\sigma = 676.2 \pm 34.2$$

$$\frac{\sigma}{\mu} = 19.88 \pm 2.06 \%$$

dE/dx [ADC counts/cm]

Energy loss | $120 < p < 140$

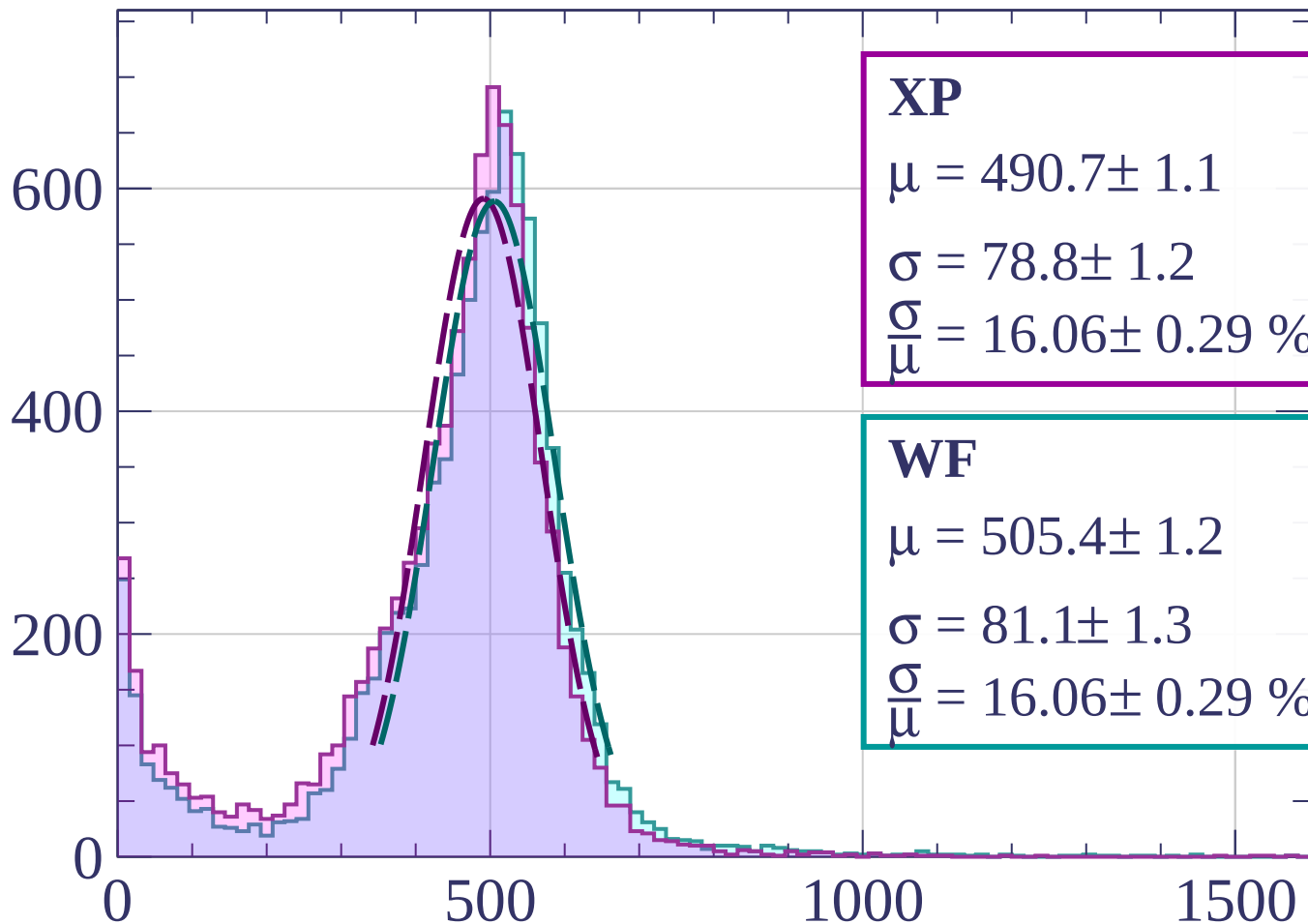
Count



dE/dx [ADC counts/cm]

Energy loss | $140 < p < 160$

Count



XP

$$\mu = 490.7 \pm 1.1$$

$$\sigma = 78.8 \pm 1.2$$

$$\frac{\sigma}{\mu} = 16.06 \pm 0.29 \%$$

WF

$$\mu = 505.4 \pm 1.2$$

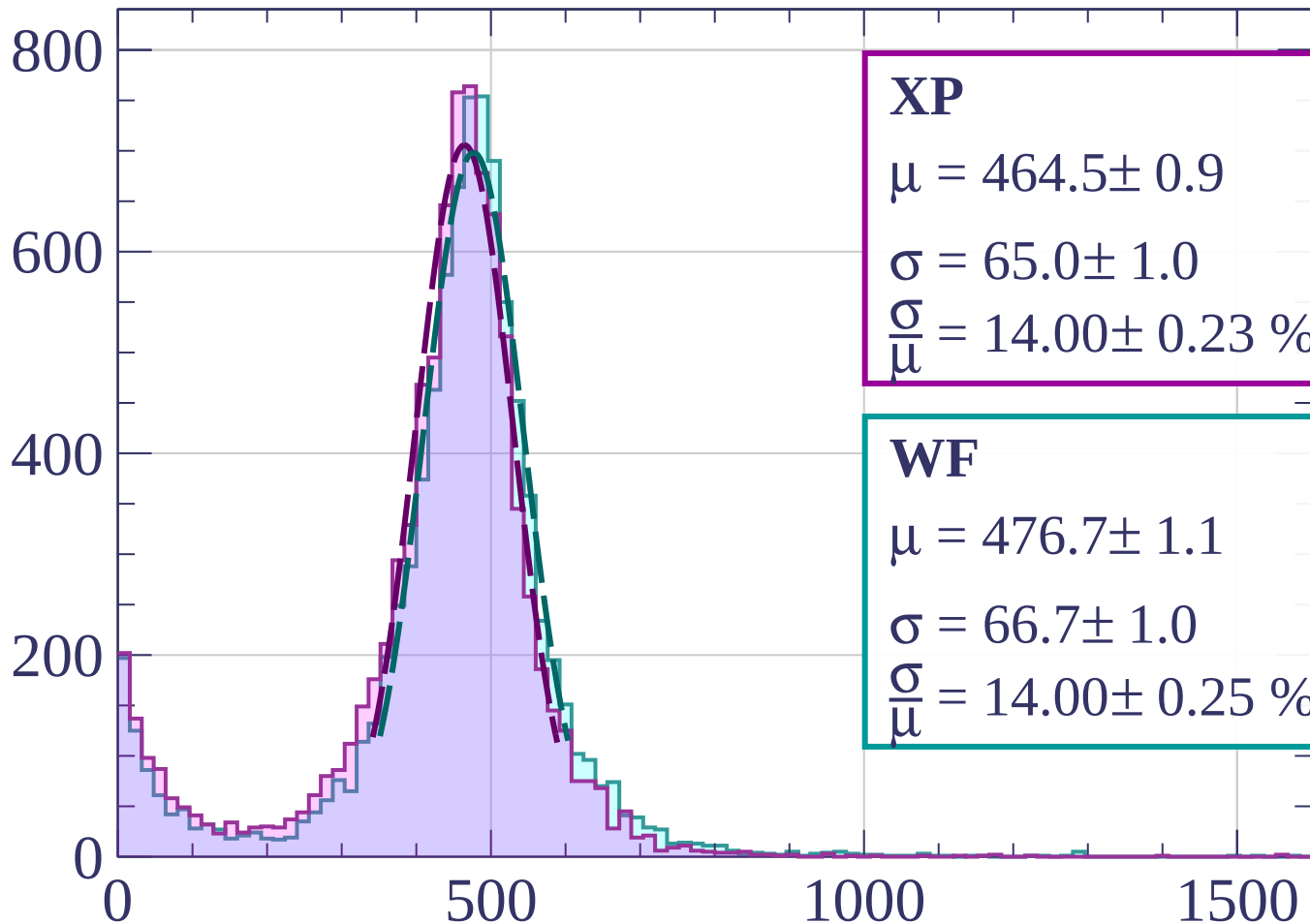
$$\sigma = 81.1 \pm 1.3$$

$$\frac{\sigma}{\mu} = 16.06 \pm 0.29 \%$$

dE/dx [ADC counts/cm]

Energy loss | $160 < p < 180$

Count



XP

$$\mu = 464.5 \pm 0.9$$

$$\sigma = 65.0 \pm 1.0$$

$$\frac{\sigma}{\mu} = 14.00 \pm 0.23 \%$$

WF

$$\mu = 476.7 \pm 1.1$$

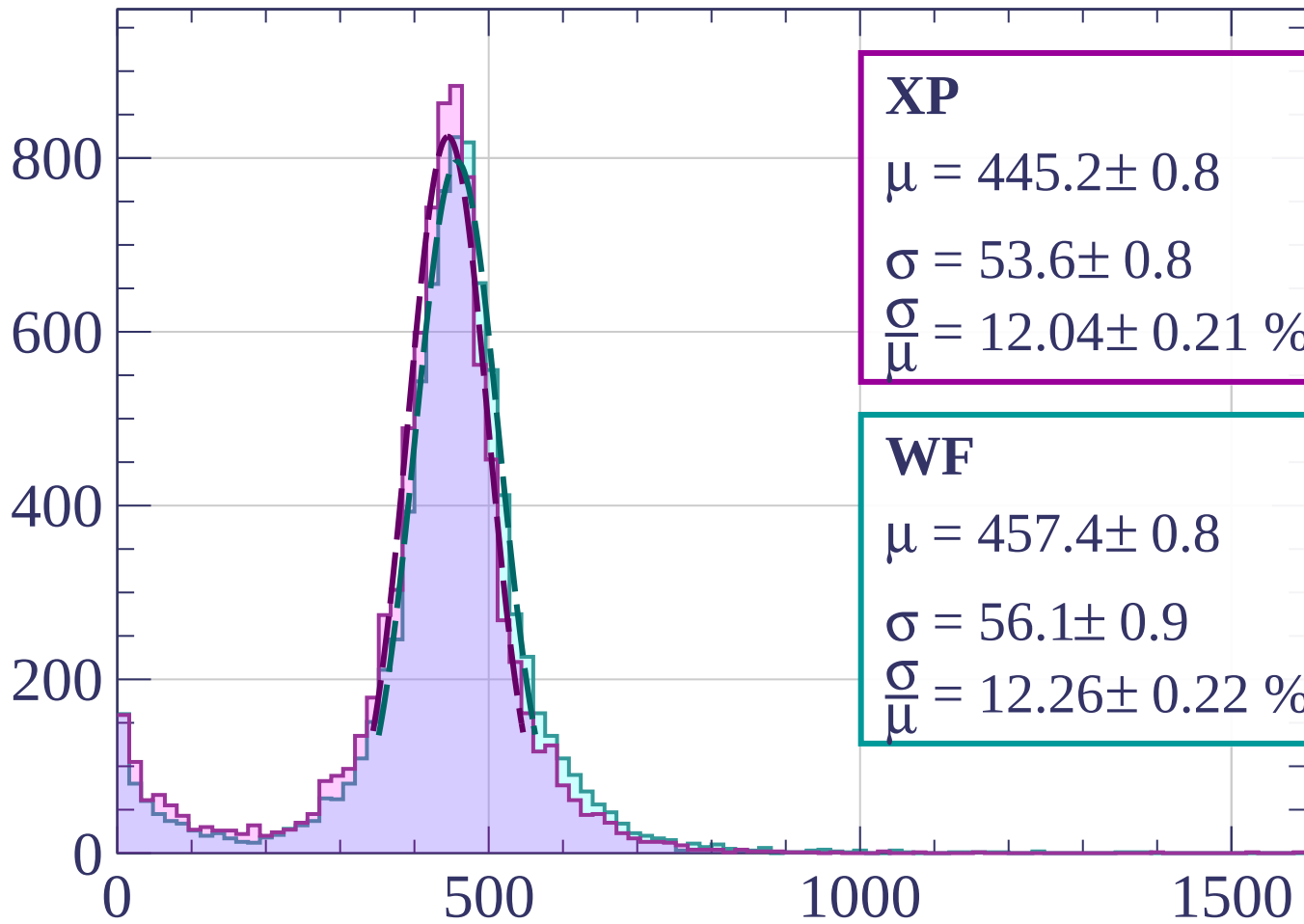
$$\sigma = 66.7 \pm 1.0$$

$$\frac{\sigma}{\mu} = 14.00 \pm 0.25 \%$$

dE/dx [ADC counts/cm]

Energy loss | $180 < p < 200$

Count



XP

$$\mu = 445.2 \pm 0.8$$

$$\sigma = 53.6 \pm 0.8$$

$$\frac{\sigma}{\mu} = 12.04 \pm 0.21 \%$$

WF

$$\mu = 457.4 \pm 0.8$$

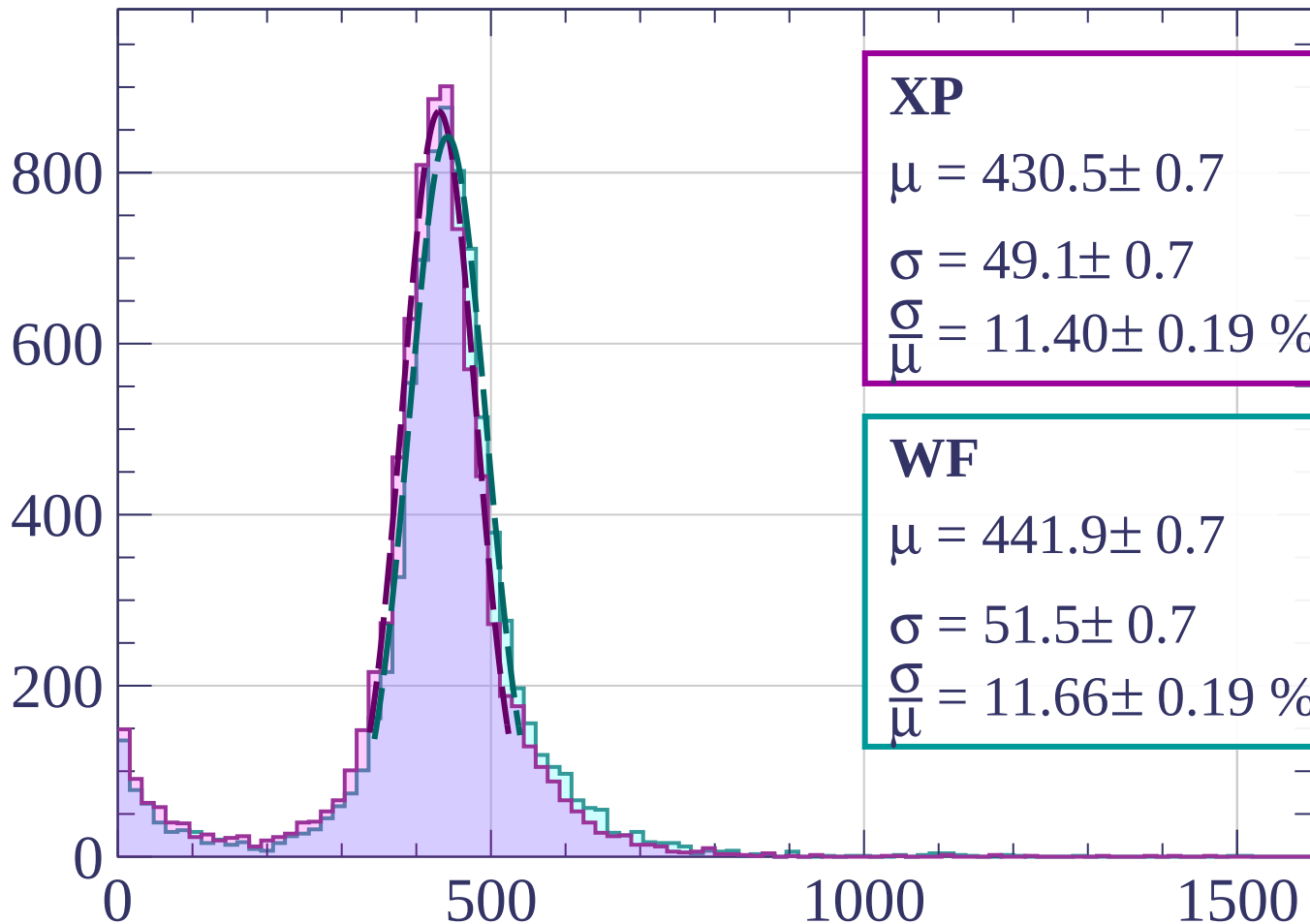
$$\sigma = 56.1 \pm 0.9$$

$$\frac{\sigma}{\mu} = 12.26 \pm 0.22 \%$$

dE/dx [ADC counts/cm]

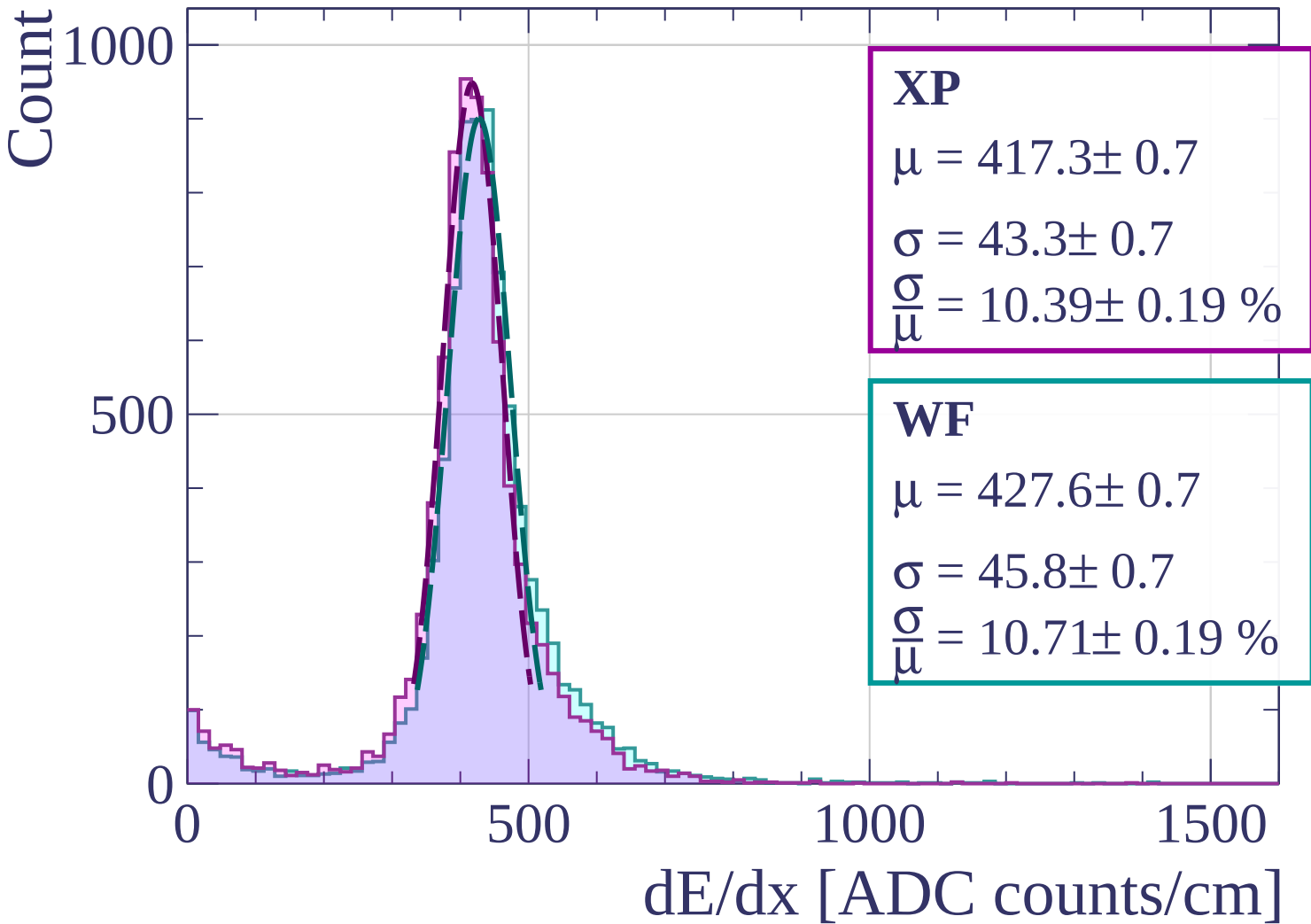
Energy loss | $200 < p < 220$

Count



dE/dx [ADC counts/cm]

Energy loss | $220 < p < 240$



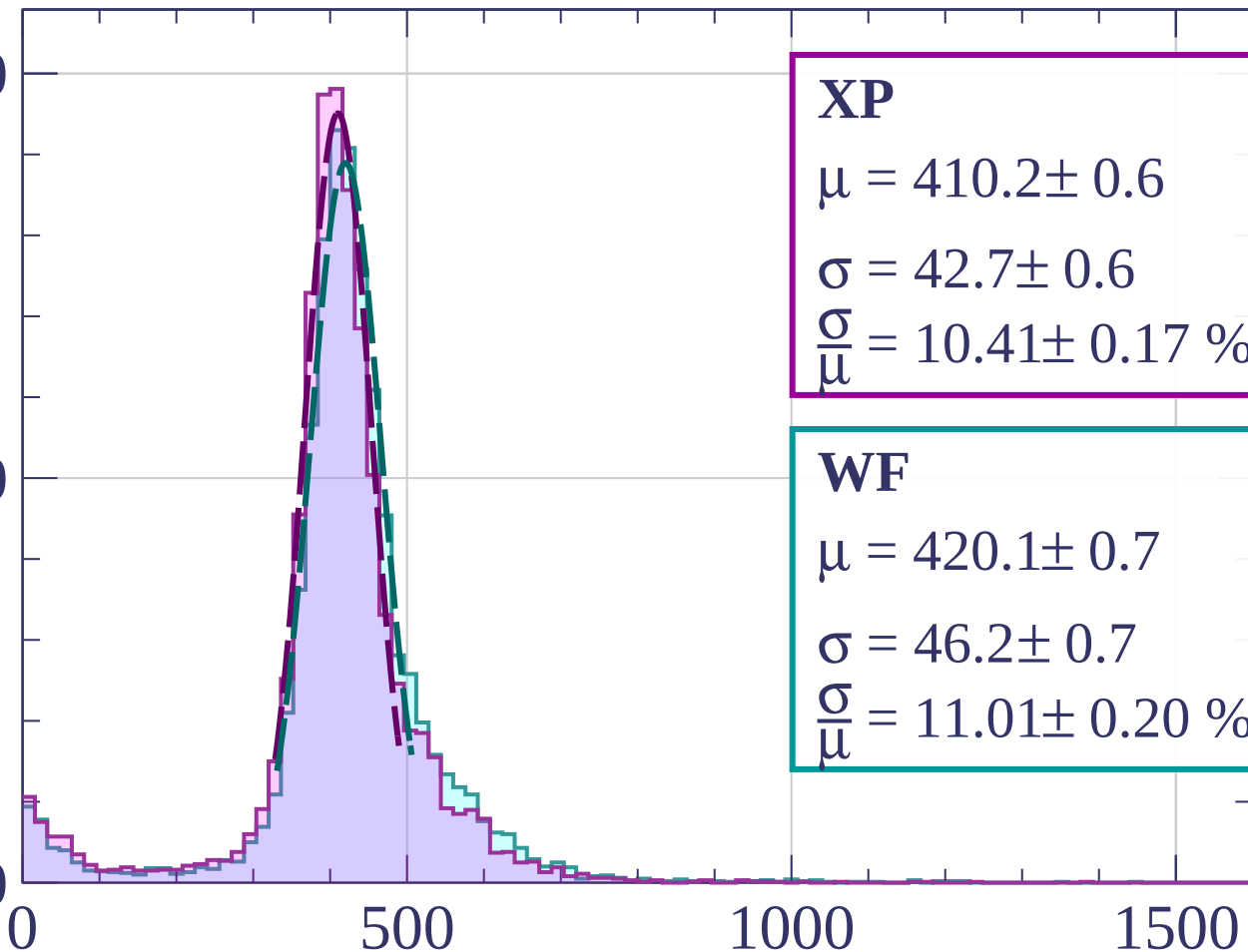
Energy loss | $240 < p < 260$

Count

1000

500

0



XP

$$\mu = 410.2 \pm 0.6$$

$$\sigma = 42.7 \pm 0.6$$

$$\frac{\sigma}{\mu} = 10.41 \pm 0.17 \%$$

WF

$$\mu = 420.1 \pm 0.7$$

$$\sigma = 46.2 \pm 0.7$$

$$\frac{\sigma}{\mu} = 11.01 \pm 0.20 \%$$

dE/dx [ADC counts/cm]

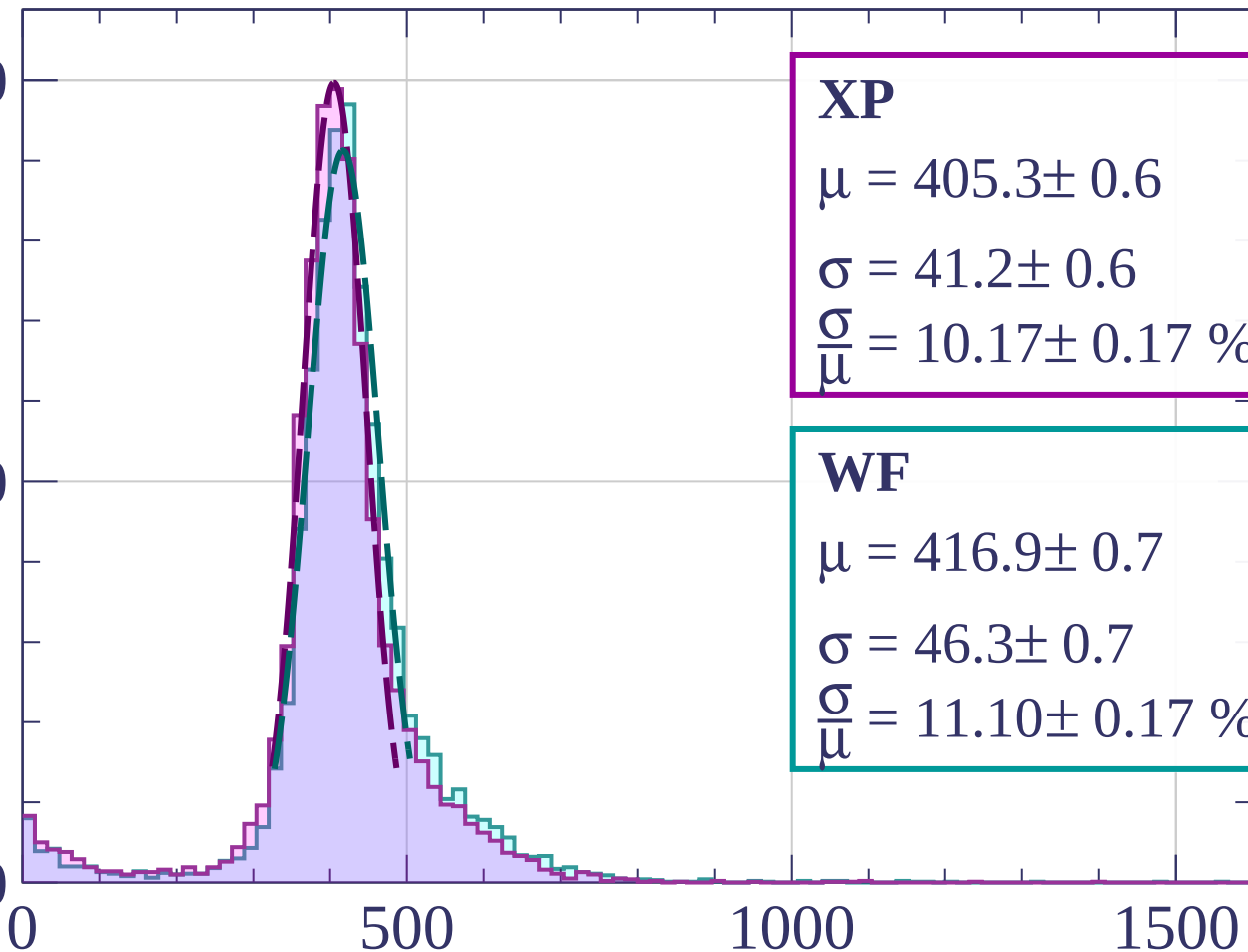
Energy loss | $260 < p < 280$

Count

1000

500

0



XP

$$\mu = 405.3 \pm 0.6$$

$$\sigma = 41.2 \pm 0.6$$

$$\frac{\sigma}{\mu} = 10.17 \pm 0.17 \%$$

WF

$$\mu = 416.9 \pm 0.7$$

$$\sigma = 46.3 \pm 0.7$$

$$\frac{\sigma}{\mu} = 11.10 \pm 0.17 \%$$

dE/dx [ADC counts/cm]

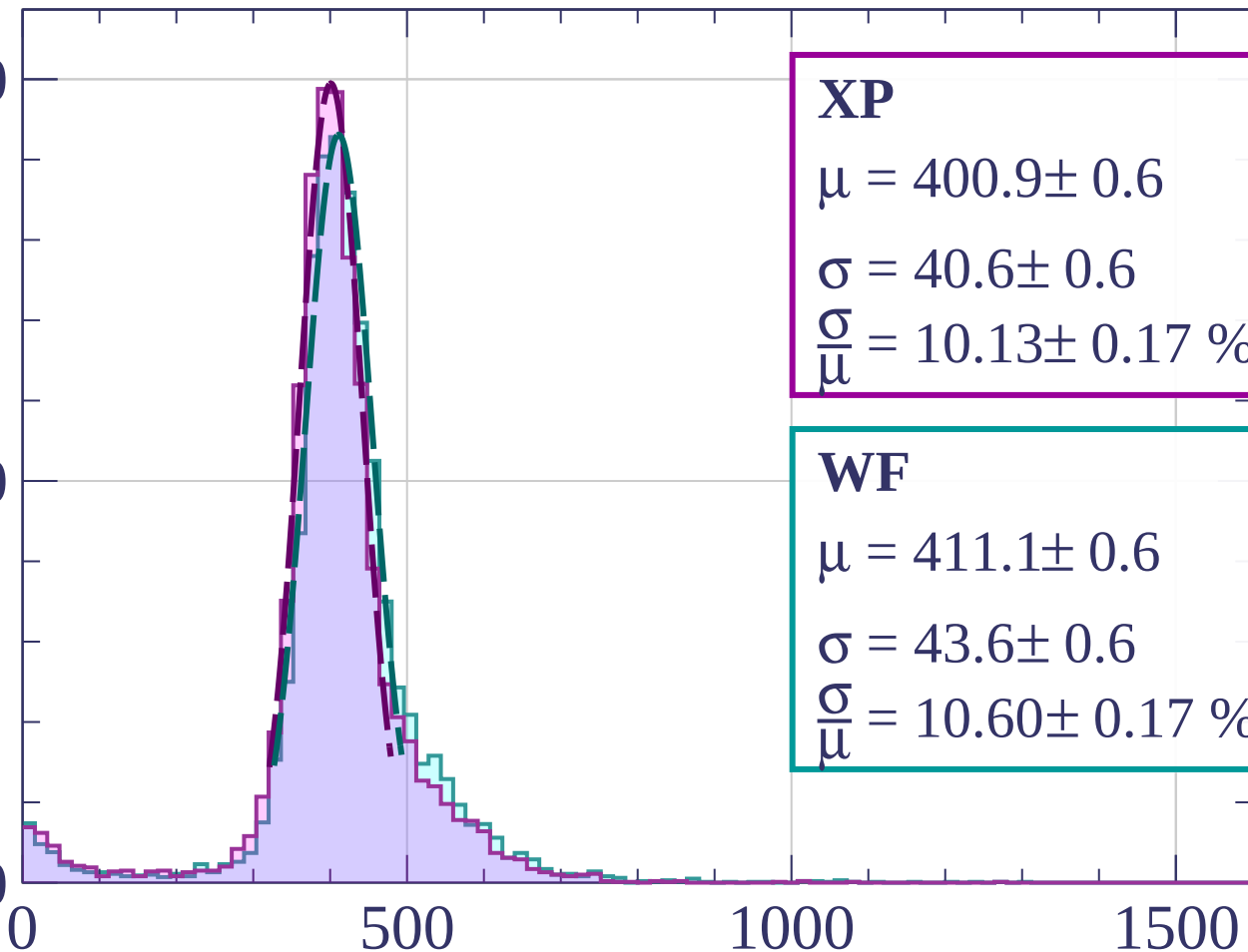
Energy loss | $280 < p < 300$

Count

1000

500

0



XP

$$\mu = 400.9 \pm 0.6$$

$$\sigma = 40.6 \pm 0.6$$

$$\frac{\sigma}{\mu} = 10.13 \pm 0.17 \%$$

WF

$$\mu = 411.1 \pm 0.6$$

$$\sigma = 43.6 \pm 0.6$$

$$\frac{\sigma}{\mu} = 10.60 \pm 0.17 \%$$

dE/dx [ADC counts/cm]

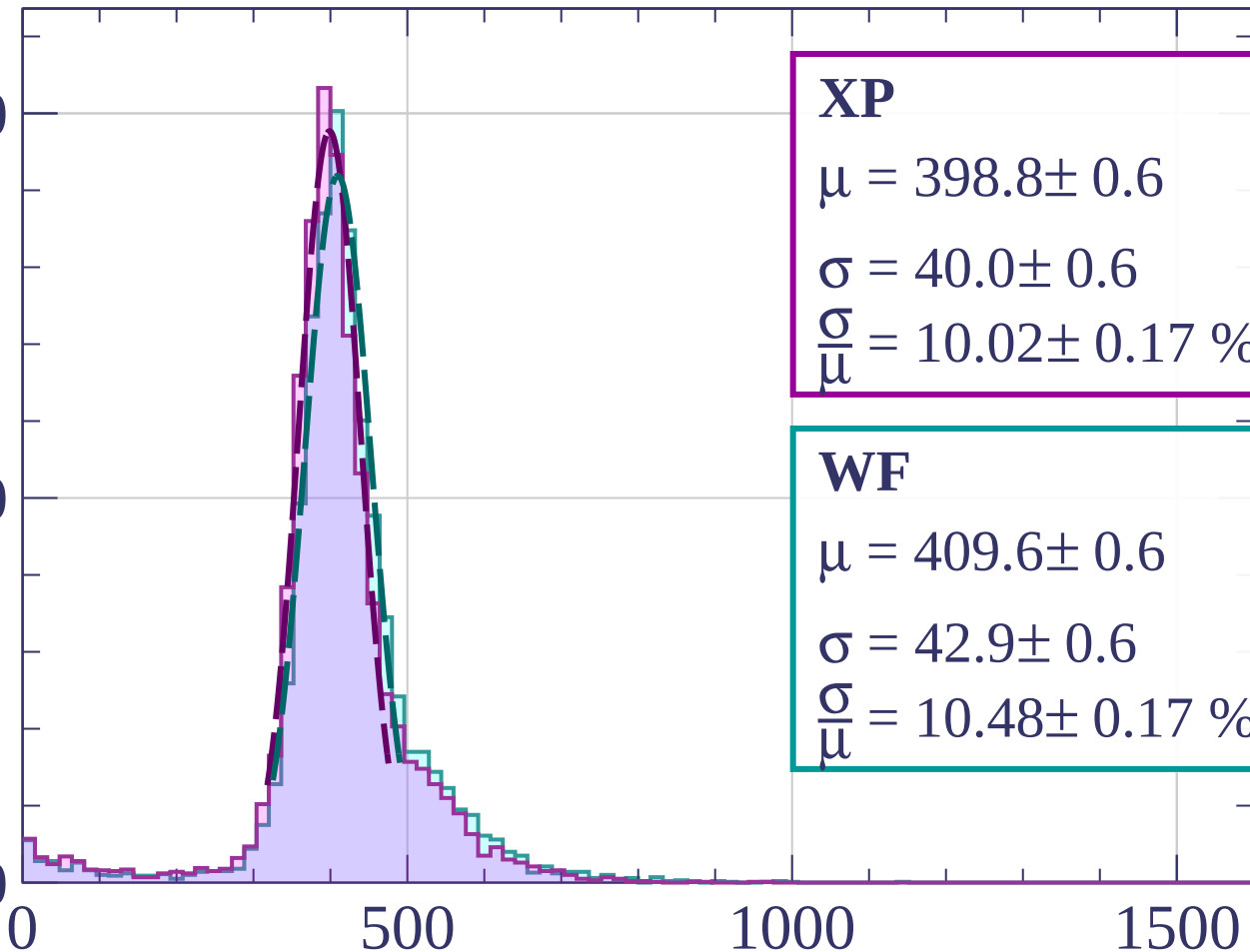
Energy loss | $300 < p < 320$

Count

1000

500

0



dE/dx [ADC counts/cm]

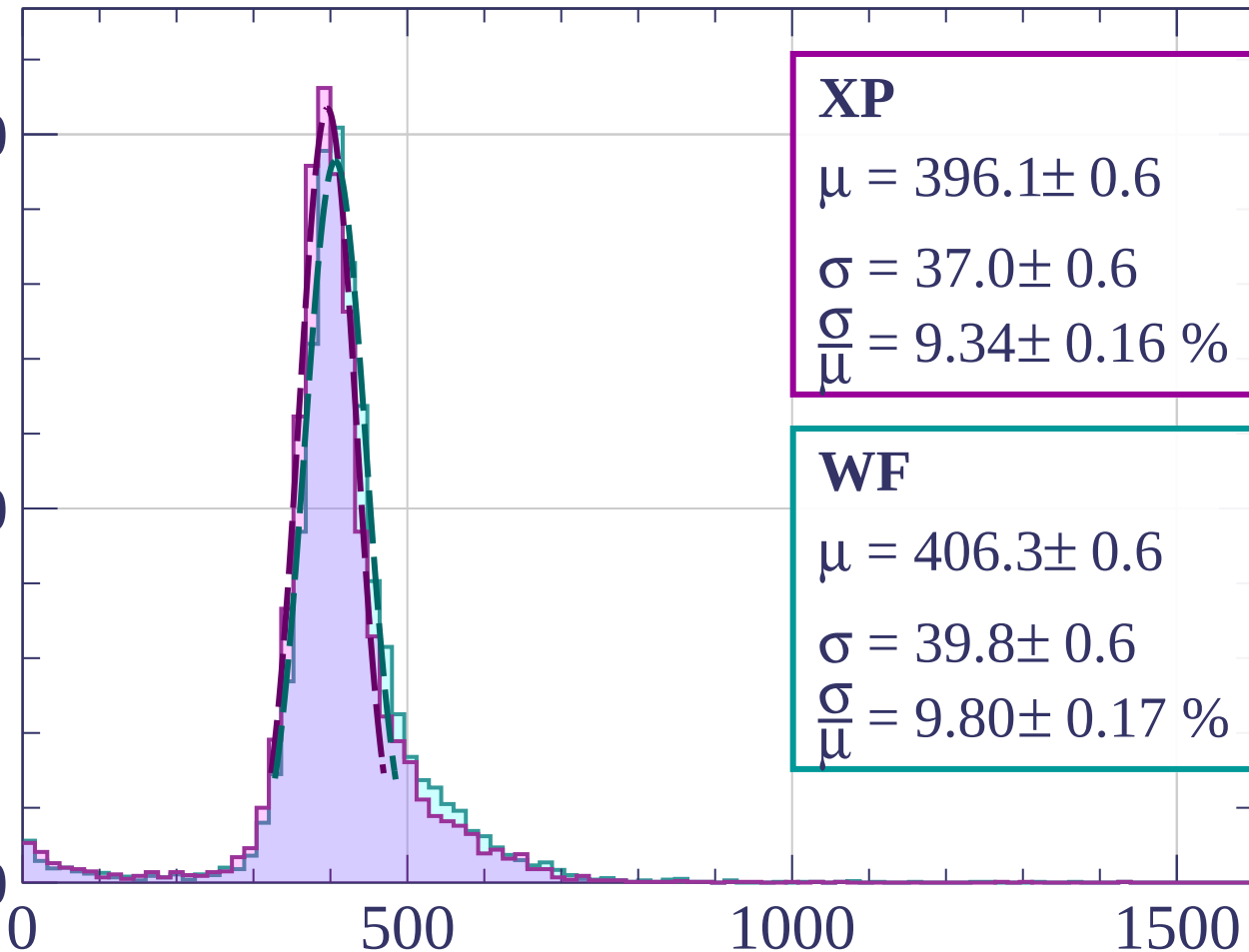
Energy loss | $320 < p < 340$

Count

1000

500

0



XP

$$\mu = 396.1 \pm 0.6$$

$$\sigma = 37.0 \pm 0.6$$

$$\frac{\sigma}{\mu} = 9.34 \pm 0.16 \%$$

WF

$$\mu = 406.3 \pm 0.6$$

$$\sigma = 39.8 \pm 0.6$$

$$\frac{\sigma}{\mu} = 9.80 \pm 0.17 \%$$

dE/dx [ADC counts/cm]

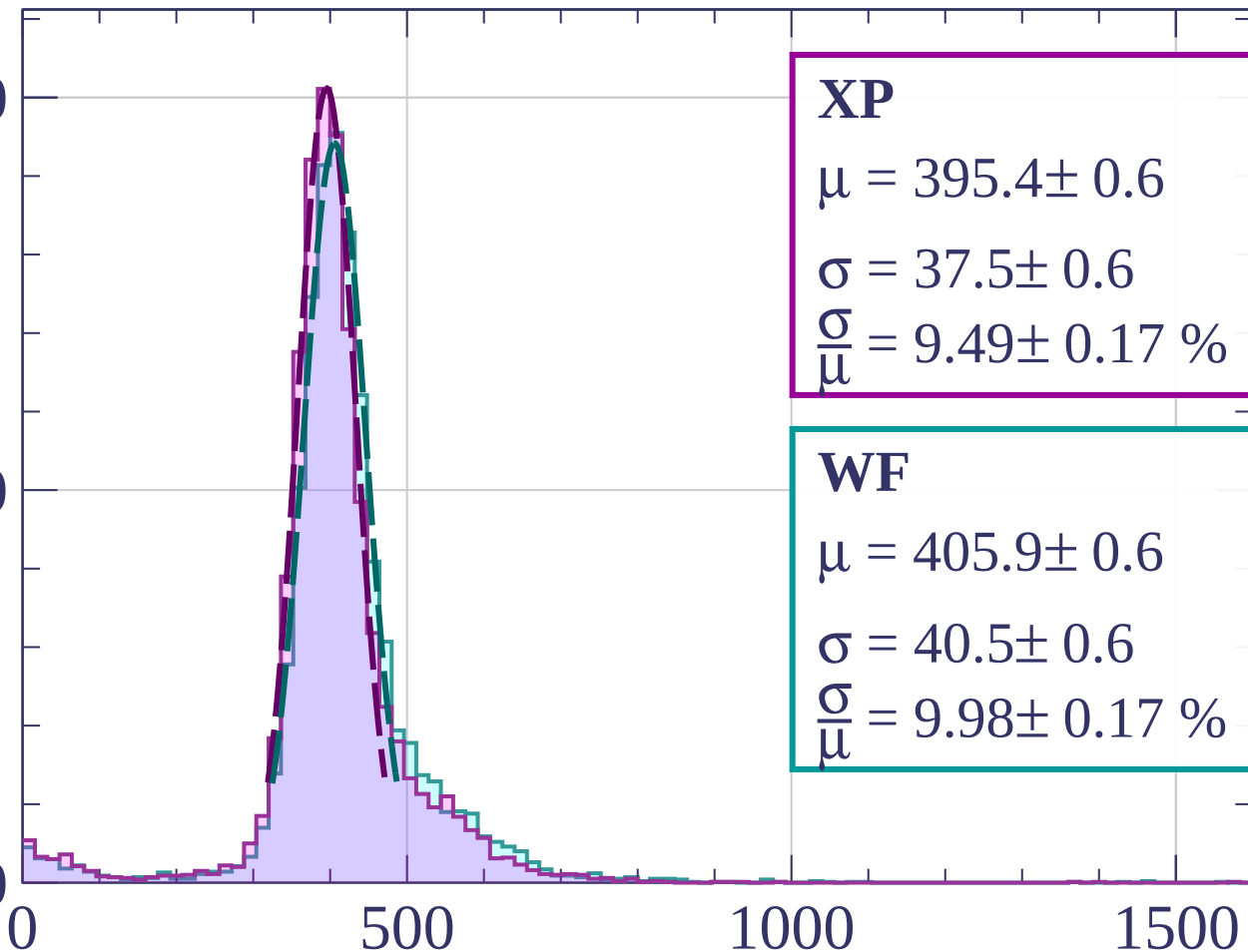
Energy loss | $340 < p < 360$

Count

1000

500

0



XP

$$\mu = 395.4 \pm 0.6$$

$$\sigma = 37.5 \pm 0.6$$

$$\frac{\sigma}{\mu} = 9.49 \pm 0.17 \%$$

WF

$$\mu = 405.9 \pm 0.6$$

$$\sigma = 40.5 \pm 0.6$$

$$\frac{\sigma}{\mu} = 9.98 \pm 0.17 \%$$

dE/dx [ADC counts/cm]

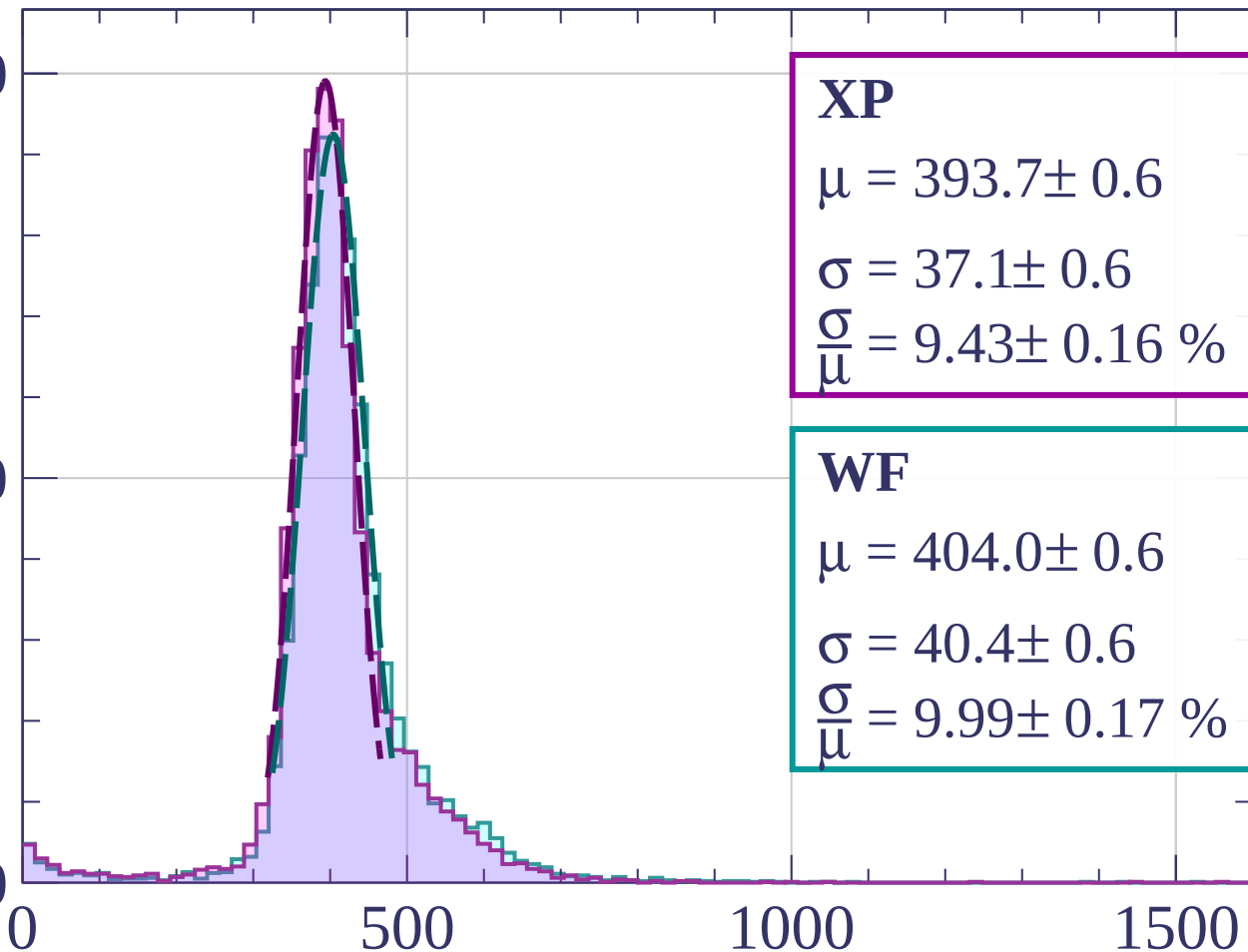
Energy loss | $360 < p < 380$

Count

1000

500

0



XP

$$\mu = 393.7 \pm 0.6$$

$$\sigma = 37.1 \pm 0.6$$

$$\frac{\sigma}{\mu} = 9.43 \pm 0.16 \%$$

WF

$$\mu = 404.0 \pm 0.6$$

$$\sigma = 40.4 \pm 0.6$$

$$\frac{\sigma}{\mu} = 9.99 \pm 0.17 \%$$

dE/dx [ADC counts/cm]

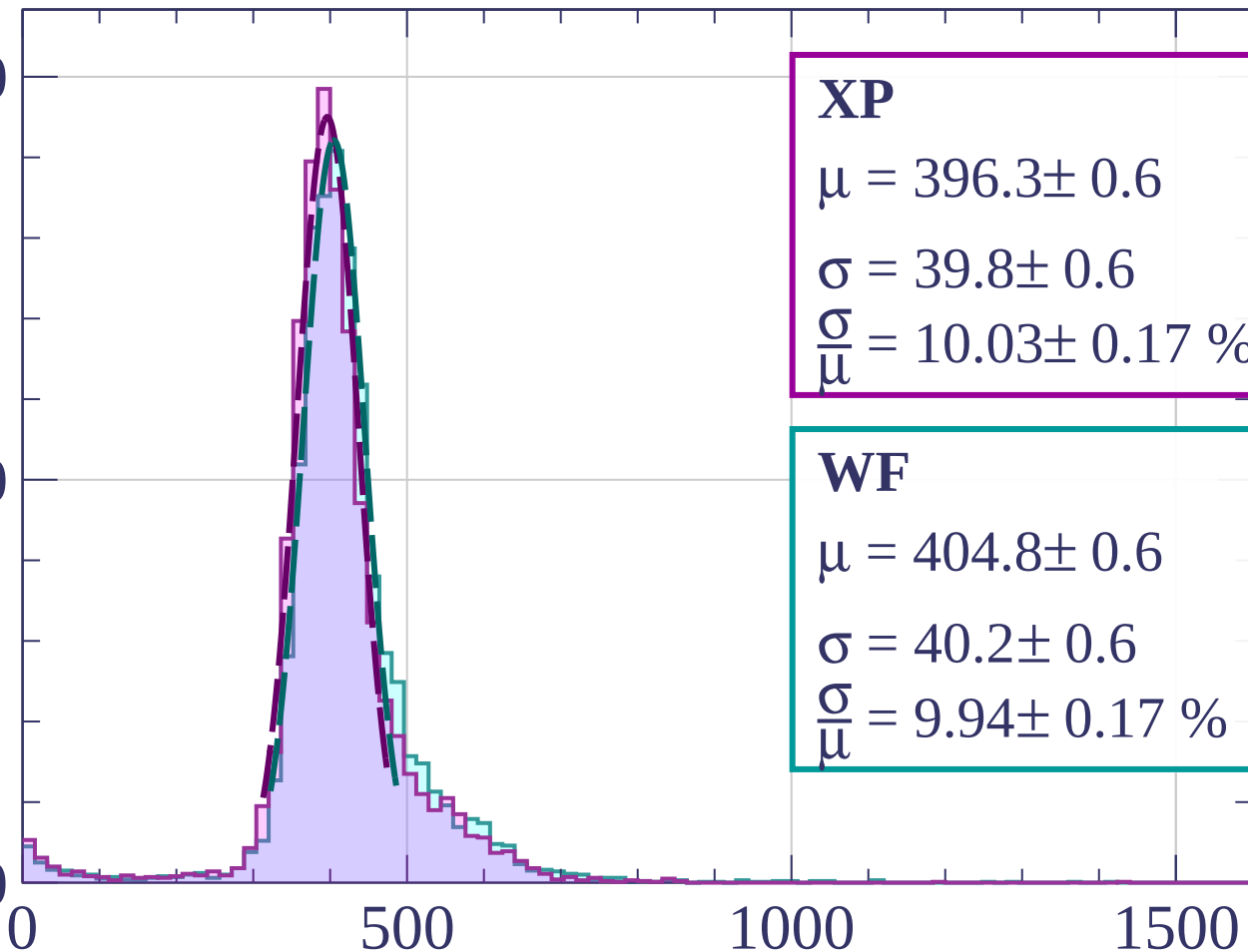
Energy loss | $380 < p < 400$

Count

1000

500

0



XP

$$\mu = 396.3 \pm 0.6$$

$$\sigma = 39.8 \pm 0.6$$

$$\frac{\sigma}{\mu} = 10.03 \pm 0.17 \%$$

WF

$$\mu = 404.8 \pm 0.6$$

$$\sigma = 40.2 \pm 0.6$$

$$\frac{\sigma}{\mu} = 9.94 \pm 0.17 \%$$

dE/dx [ADC counts/cm]

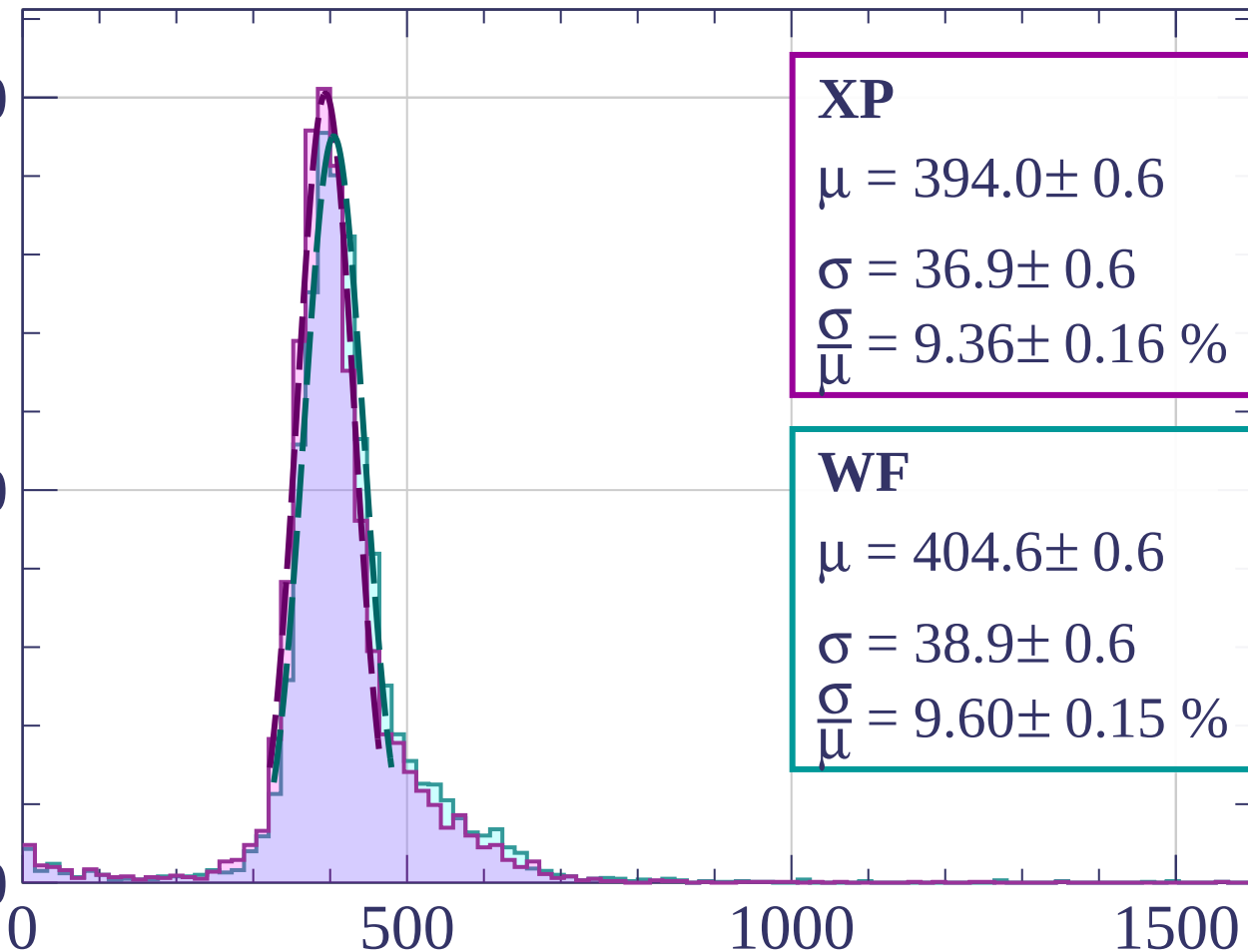
Energy loss | $400 < p < 420$

Count

1000

500

0



XP

$$\mu = 394.0 \pm 0.6$$

$$\sigma = 36.9 \pm 0.6$$

$$\frac{\sigma}{\mu} = 9.36 \pm 0.16 \%$$

WF

$$\mu = 404.6 \pm 0.6$$

$$\sigma = 38.9 \pm 0.6$$

$$\frac{\sigma}{\mu} = 9.60 \pm 0.15 \%$$

dE/dx [ADC counts/cm]

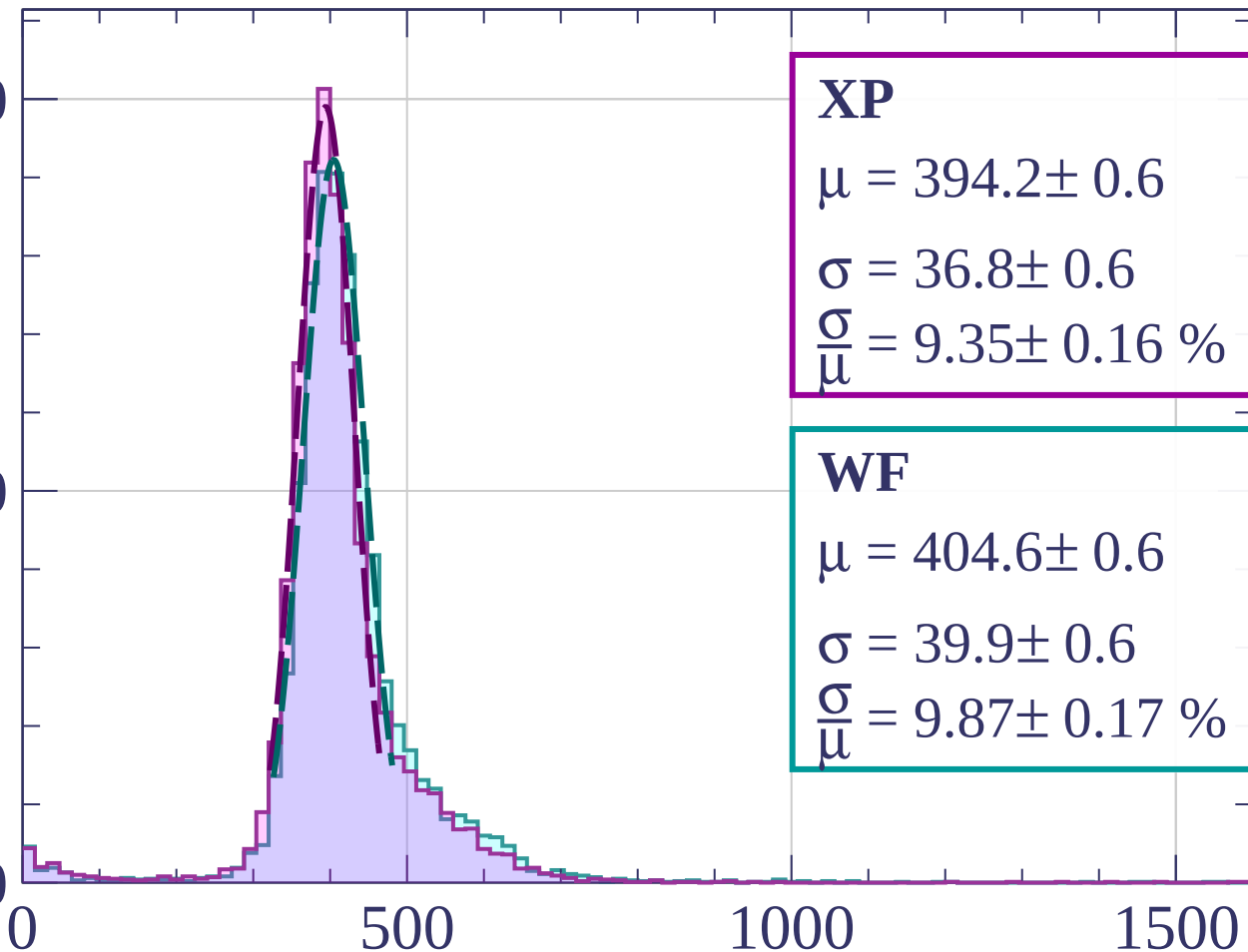
Energy loss | $420 < p < 440$

Count

1000

500

0



XP

$$\mu = 394.2 \pm 0.6$$

$$\sigma = 36.8 \pm 0.6$$

$$\frac{\sigma}{\mu} = 9.35 \pm 0.16 \%$$

WF

$$\mu = 404.6 \pm 0.6$$

$$\sigma = 39.9 \pm 0.6$$

$$\frac{\sigma}{\mu} = 9.87 \pm 0.17 \%$$

dE/dx [ADC counts/cm]

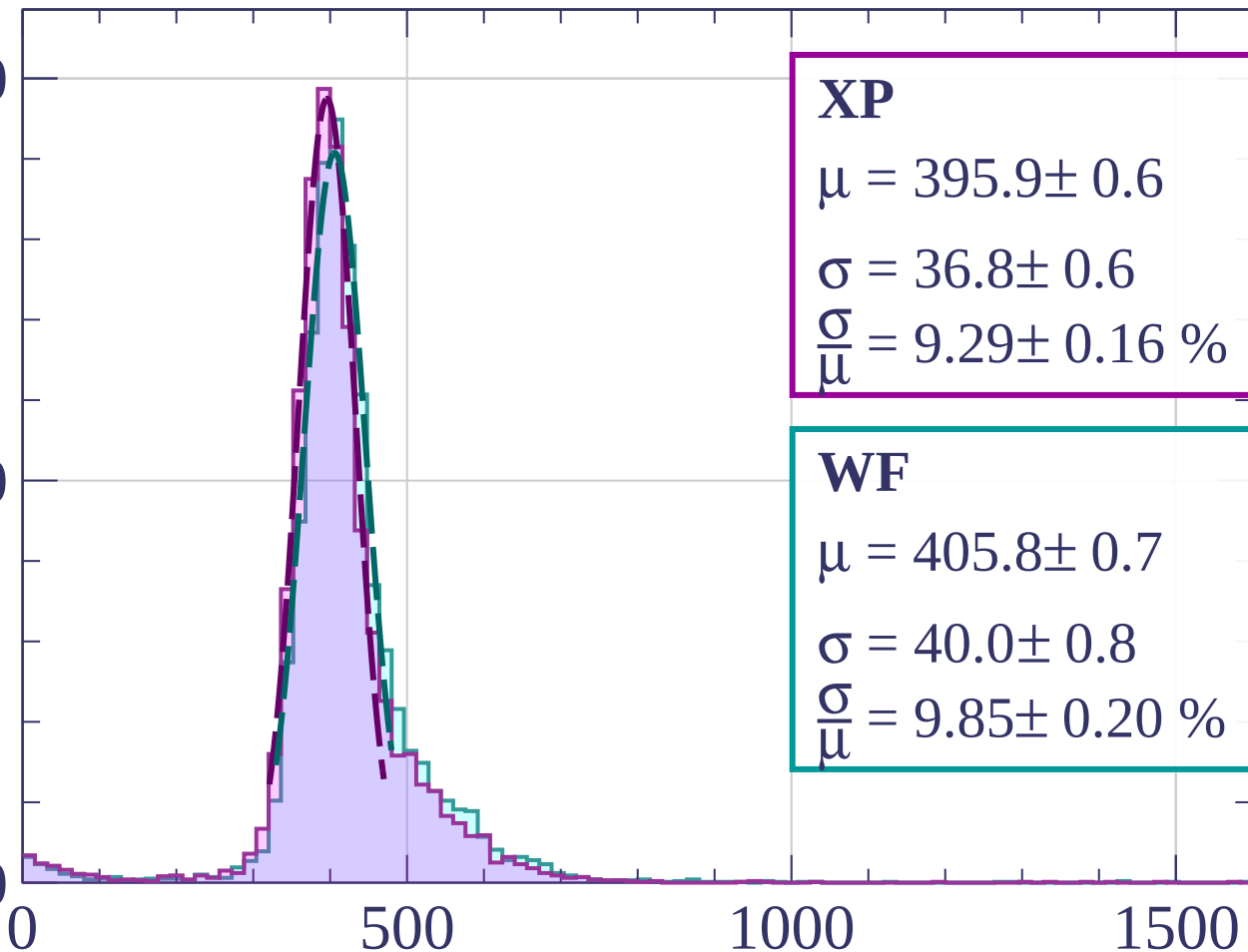
Energy loss | $440 < p < 460$

Count

1000

500

0



XP

$$\mu = 395.9 \pm 0.6$$

$$\sigma = 36.8 \pm 0.6$$

$$\frac{\sigma}{\mu} = 9.29 \pm 0.16 \%$$

WF

$$\mu = 405.8 \pm 0.7$$

$$\sigma = 40.0 \pm 0.8$$

$$\frac{\sigma}{\mu} = 9.85 \pm 0.20 \%$$

dE/dx [ADC counts/cm]

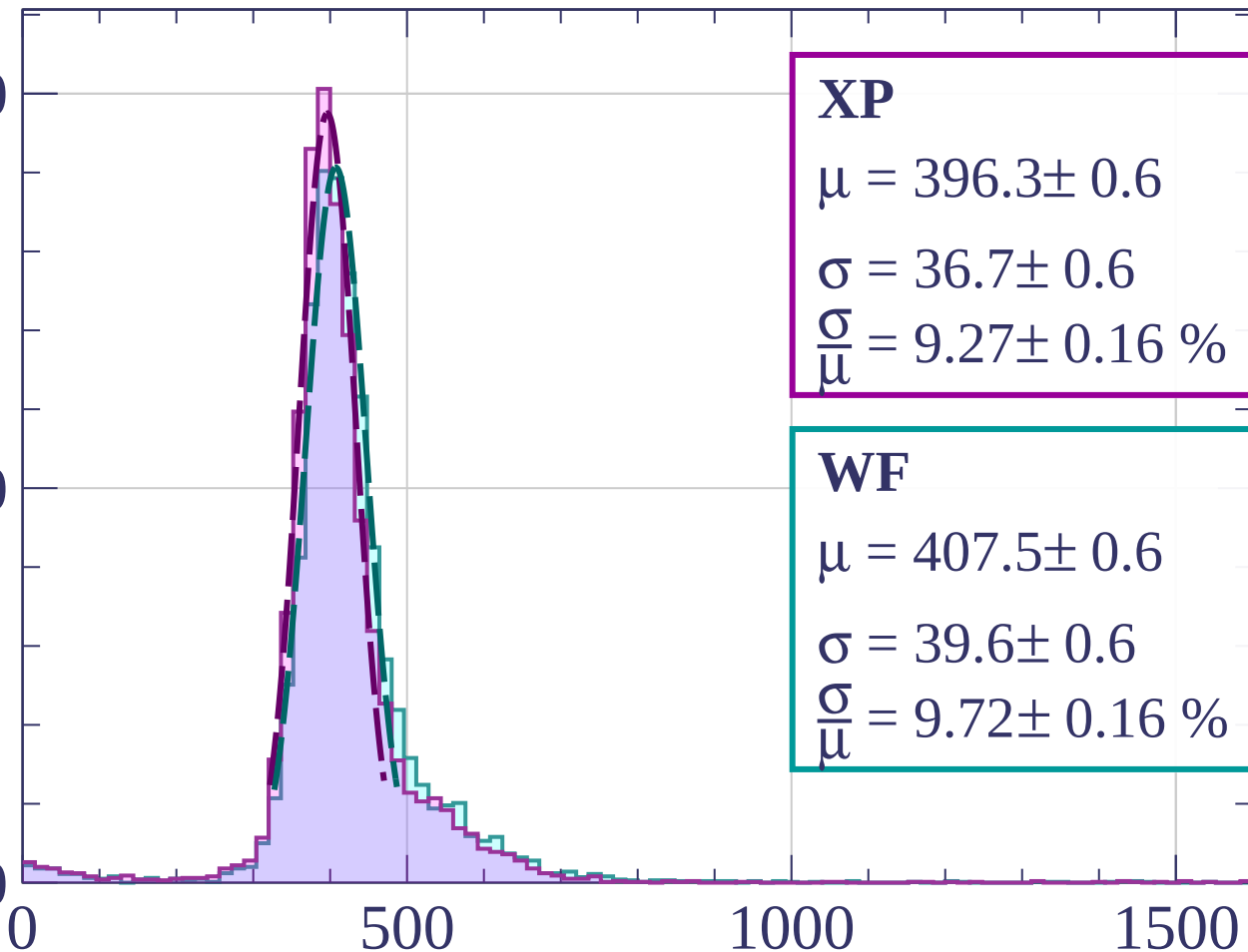
Energy loss | $460 < p < 480$

Count

1000

500

0



XP

$$\mu = 396.3 \pm 0.6$$

$$\sigma = 36.7 \pm 0.6$$

$$\frac{\sigma}{\mu} = 9.27 \pm 0.16 \%$$

WF

$$\mu = 407.5 \pm 0.6$$

$$\sigma = 39.6 \pm 0.6$$

$$\frac{\sigma}{\mu} = 9.72 \pm 0.16 \%$$

dE/dx [ADC counts/cm]

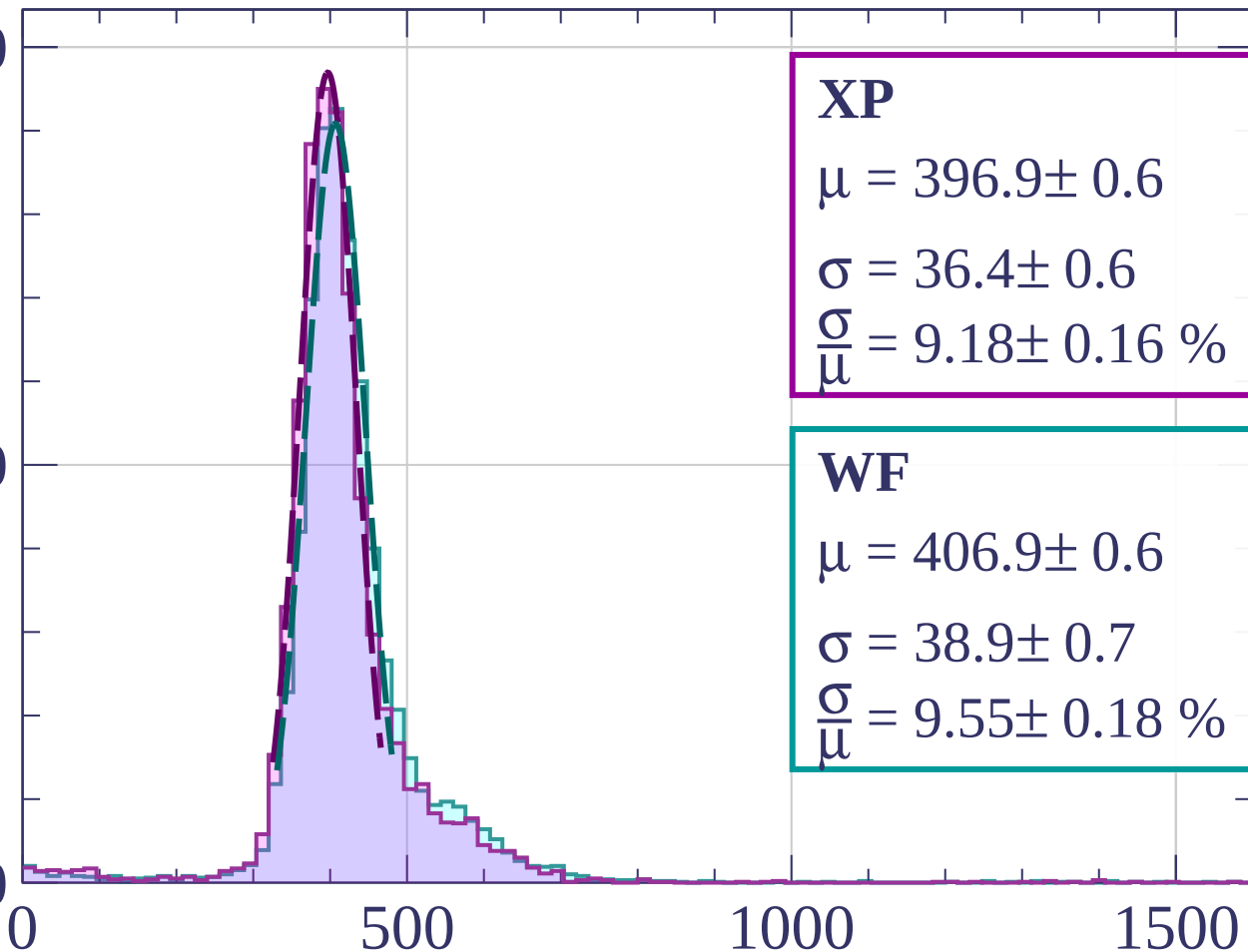
Energy loss | $480 < p < 500$

Count

1000

500

0



XP

$$\mu = 396.9 \pm 0.6$$

$$\sigma = 36.4 \pm 0.6$$

$$\frac{\sigma}{\mu} = 9.18 \pm 0.16 \%$$

WF

$$\mu = 406.9 \pm 0.6$$

$$\sigma = 38.9 \pm 0.7$$

$$\frac{\sigma}{\mu} = 9.55 \pm 0.18 \%$$

dE/dx [ADC counts/cm]

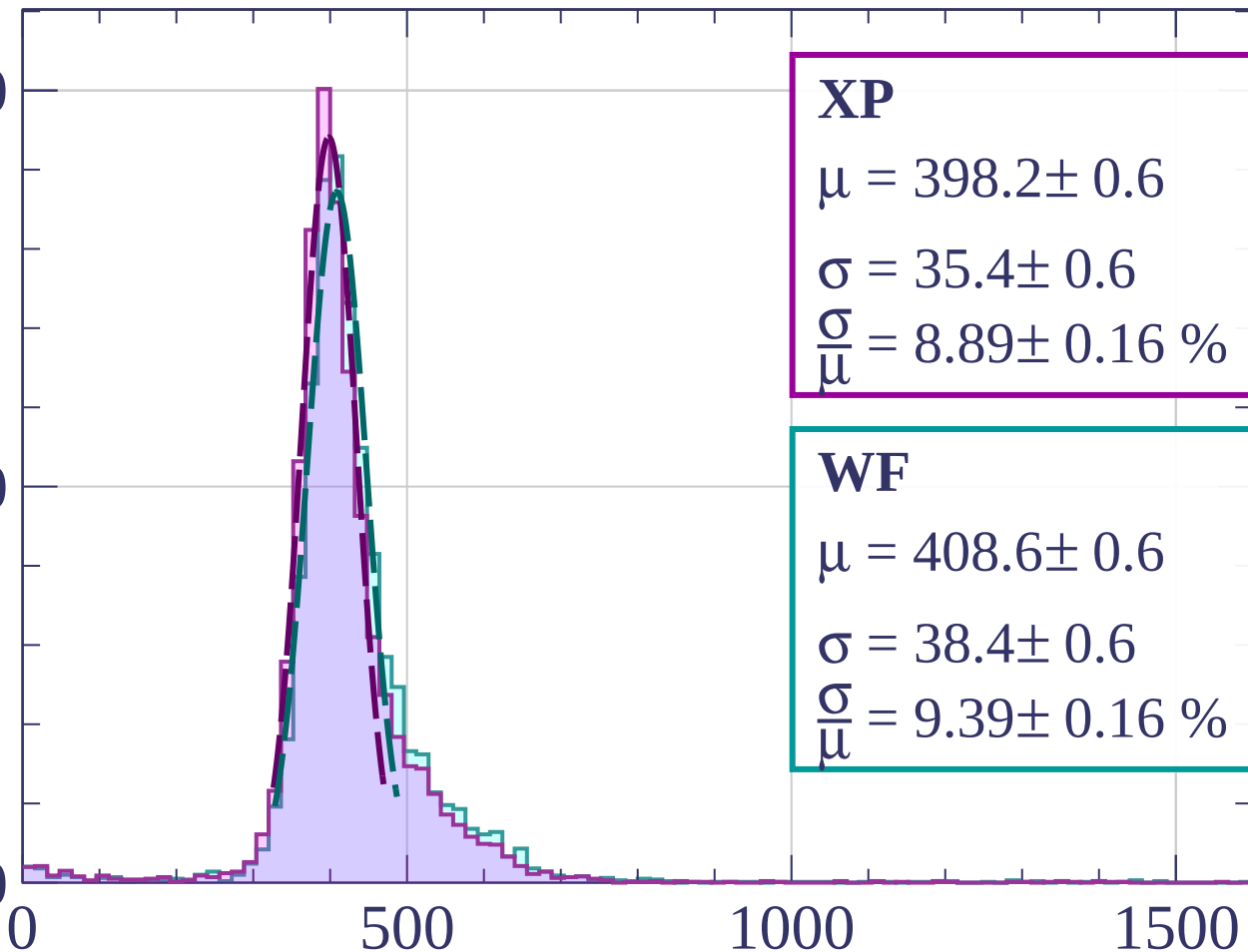
Energy loss | $500 < p < 520$

Count

1000

500

0



XP

$$\mu = 398.2 \pm 0.6$$

$$\sigma = 35.4 \pm 0.6$$

$$\frac{\sigma}{\mu} = 8.89 \pm 0.16 \%$$

WF

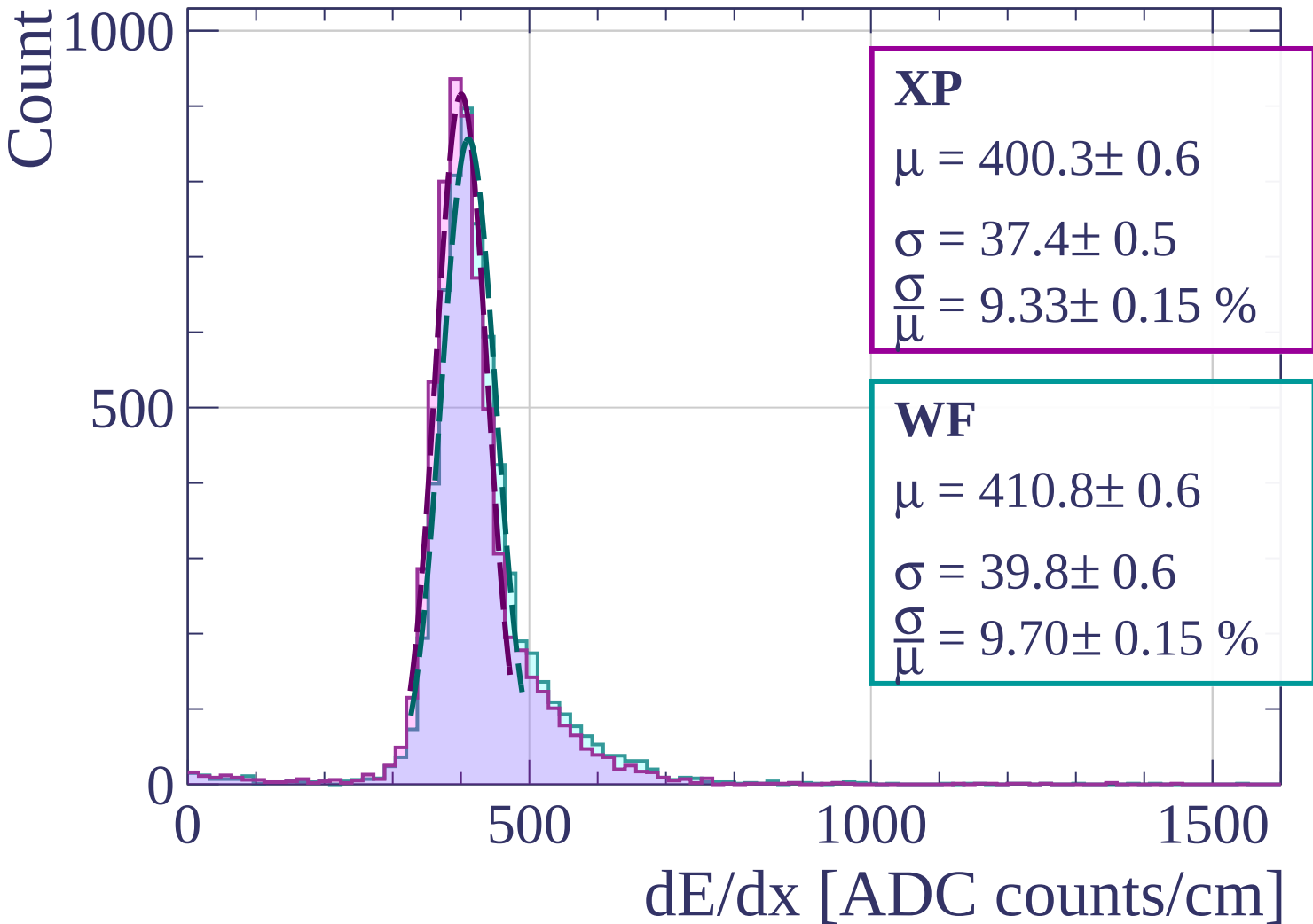
$$\mu = 408.6 \pm 0.6$$

$$\sigma = 38.4 \pm 0.6$$

$$\frac{\sigma}{\mu} = 9.39 \pm 0.16 \%$$

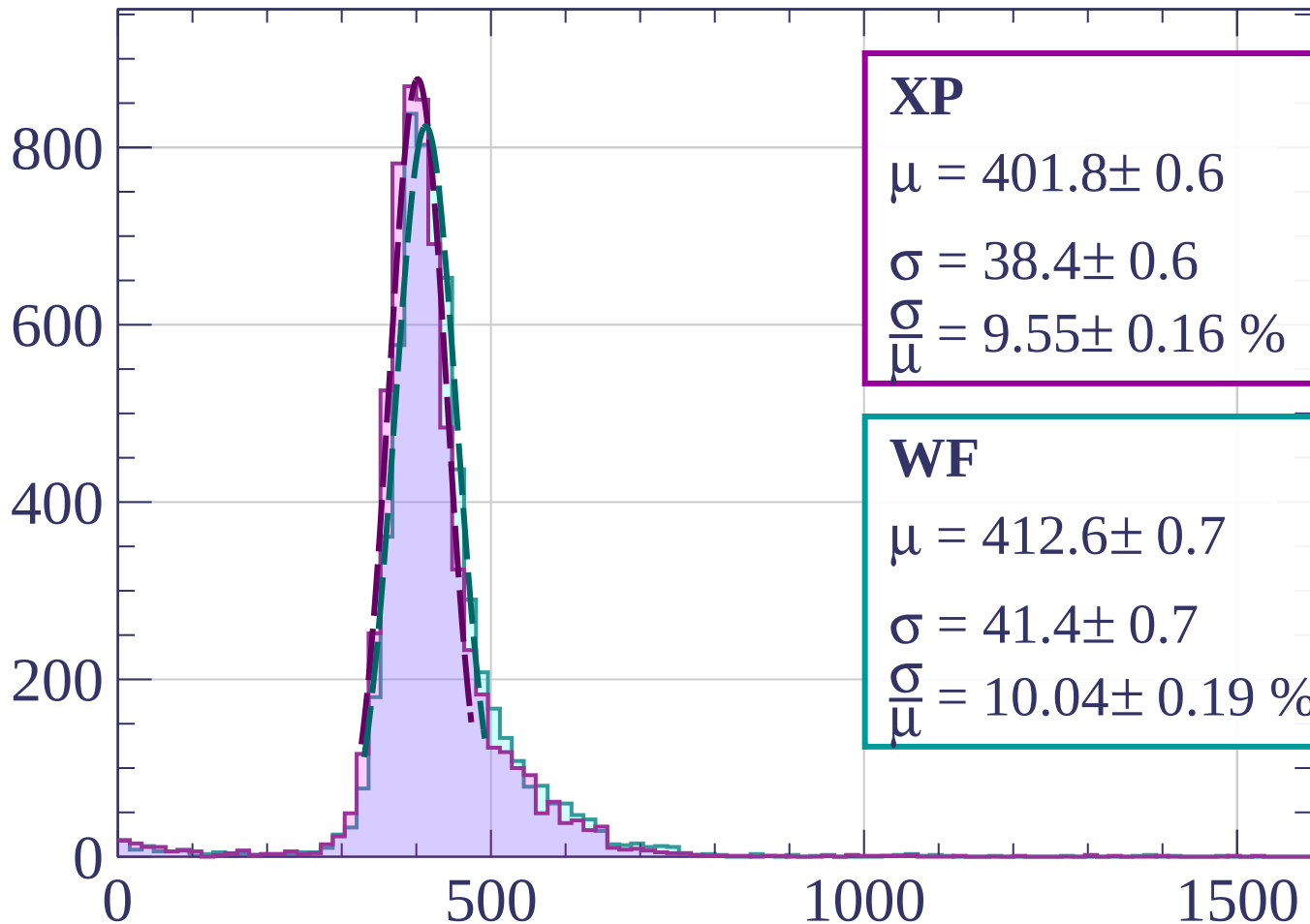
dE/dx [ADC counts/cm]

Energy loss | $520 < p < 540$



Energy loss | $540 < p < 560$

Count



XP

$$\mu = 401.8 \pm 0.6$$

$$\sigma = 38.4 \pm 0.6$$

$$\frac{\sigma}{\mu} = 9.55 \pm 0.16 \%$$

WF

$$\mu = 412.6 \pm 0.7$$

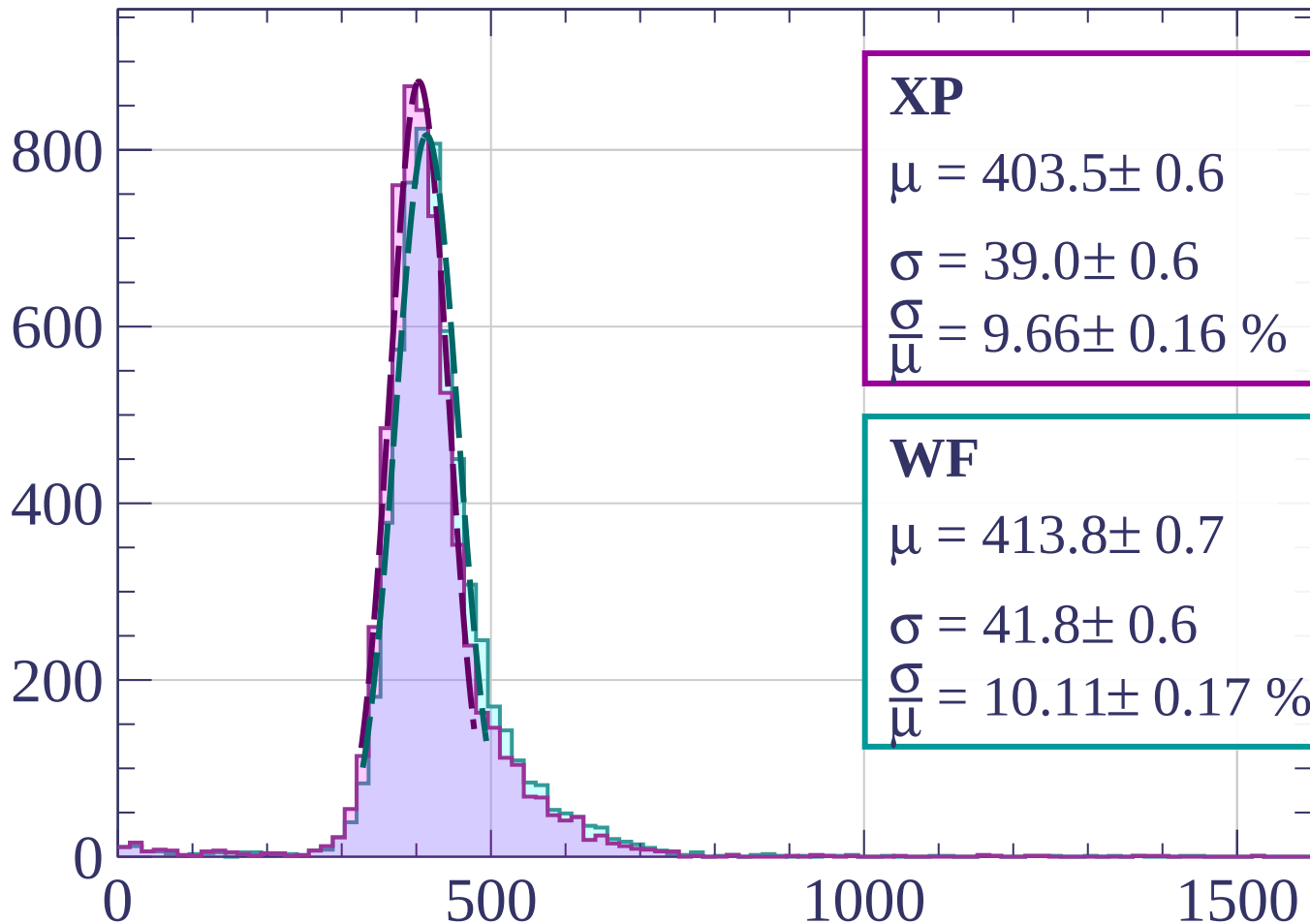
$$\sigma = 41.4 \pm 0.7$$

$$\frac{\sigma}{\mu} = 10.04 \pm 0.19 \%$$

dE/dx [ADC counts/cm]

Energy loss | $560 < p < 580$

Count



XP

$$\mu = 403.5 \pm 0.6$$

$$\sigma = 39.0 \pm 0.6$$

$$\frac{\sigma}{\mu} = 9.66 \pm 0.16 \%$$

WF

$$\mu = 413.8 \pm 0.7$$

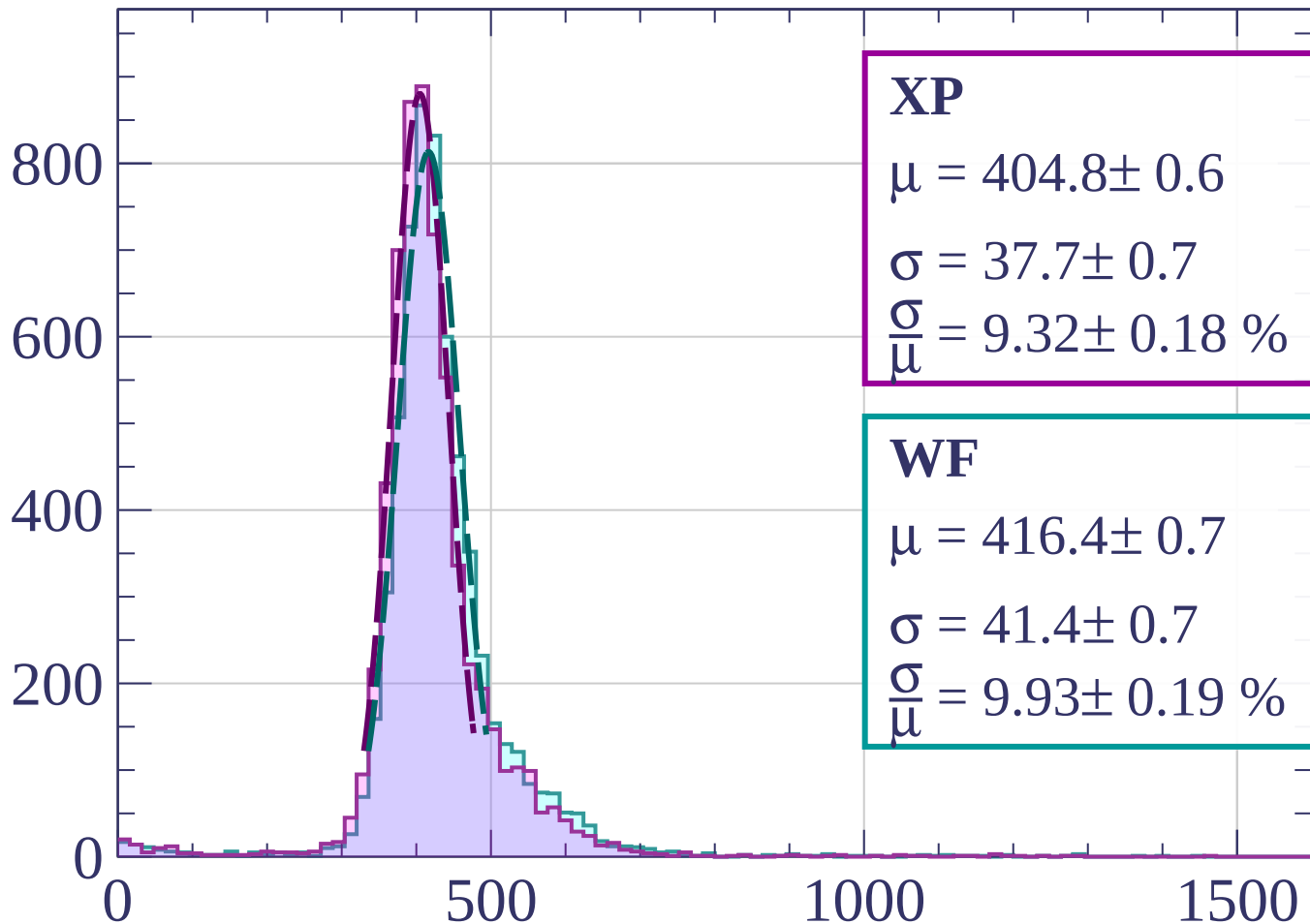
$$\sigma = 41.8 \pm 0.6$$

$$\frac{\sigma}{\mu} = 10.11 \pm 0.17 \%$$

dE/dx [ADC counts/cm]

Energy loss | $580 < p < 600$

Count



XP

$$\mu = 404.8 \pm 0.6$$

$$\sigma = 37.7 \pm 0.7$$

$$\frac{\sigma}{\mu} = 9.32 \pm 0.18 \%$$

WF

$$\mu = 416.4 \pm 0.7$$

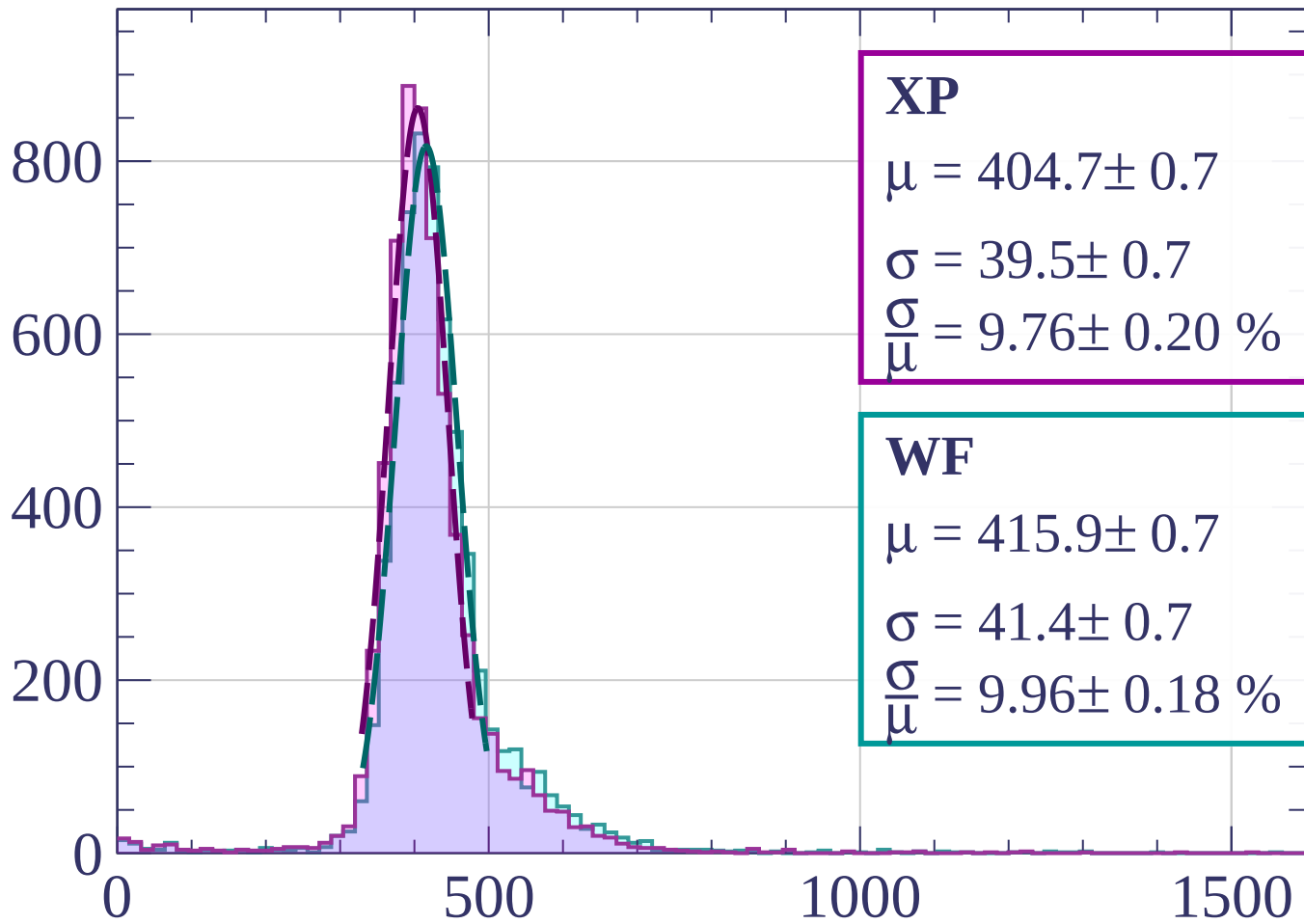
$$\sigma = 41.4 \pm 0.7$$

$$\frac{\sigma}{\mu} = 9.93 \pm 0.19 \%$$

dE/dx [ADC counts/cm]

Energy loss | $600 < p < 620$

Count



XP

$$\mu = 404.7 \pm 0.7$$

$$\sigma = 39.5 \pm 0.7$$

$$\frac{\sigma}{\mu} = 9.76 \pm 0.20 \%$$

WF

$$\mu = 415.9 \pm 0.7$$

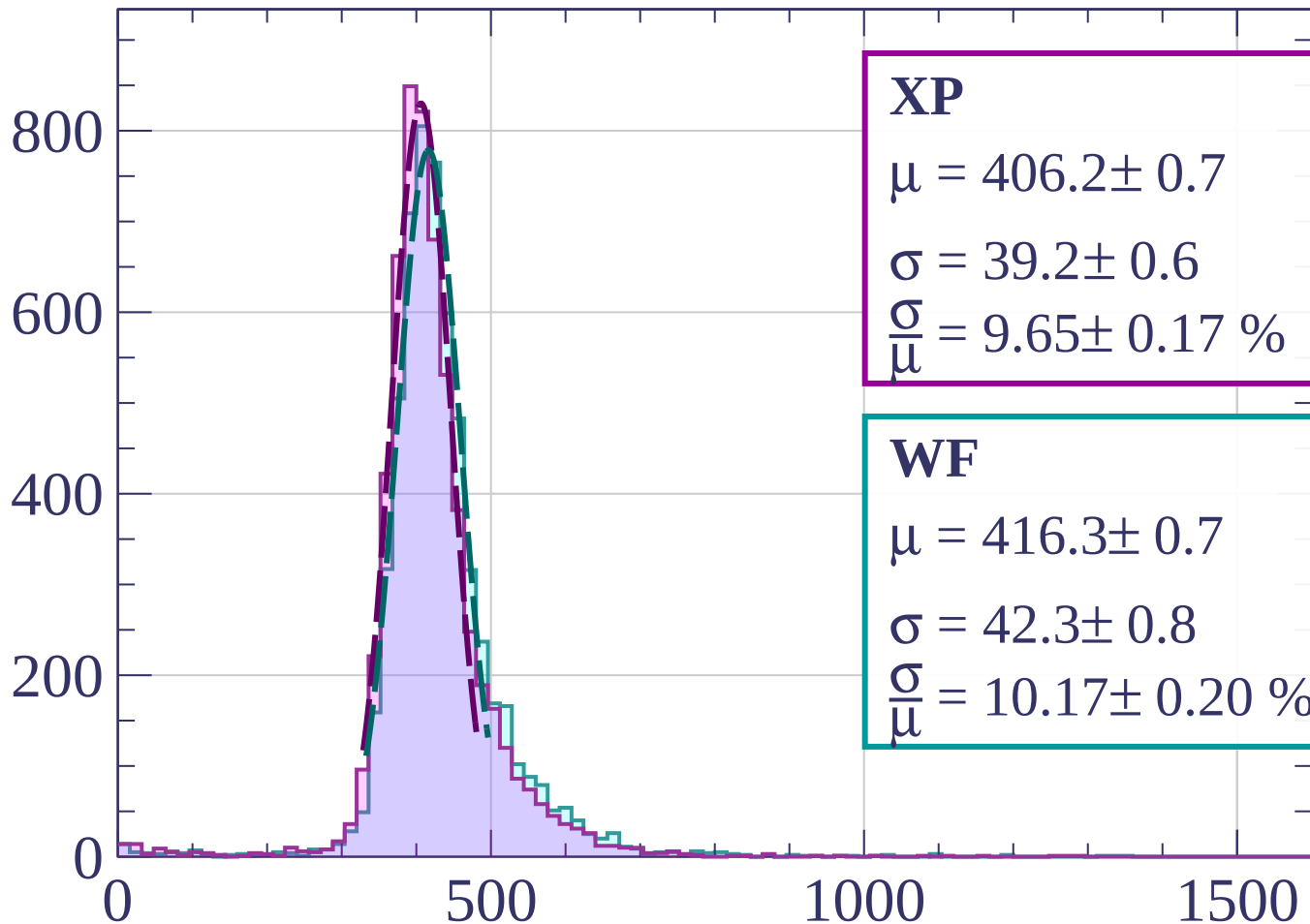
$$\sigma = 41.4 \pm 0.7$$

$$\frac{\sigma}{\mu} = 9.96 \pm 0.18 \%$$

dE/dx [ADC counts/cm]

Energy loss | $620 < p < 640$

Count



XP

$$\mu = 406.2 \pm 0.7$$

$$\sigma = 39.2 \pm 0.6$$

$$\frac{\sigma}{\mu} = 9.65 \pm 0.17 \%$$

WF

$$\mu = 416.3 \pm 0.7$$

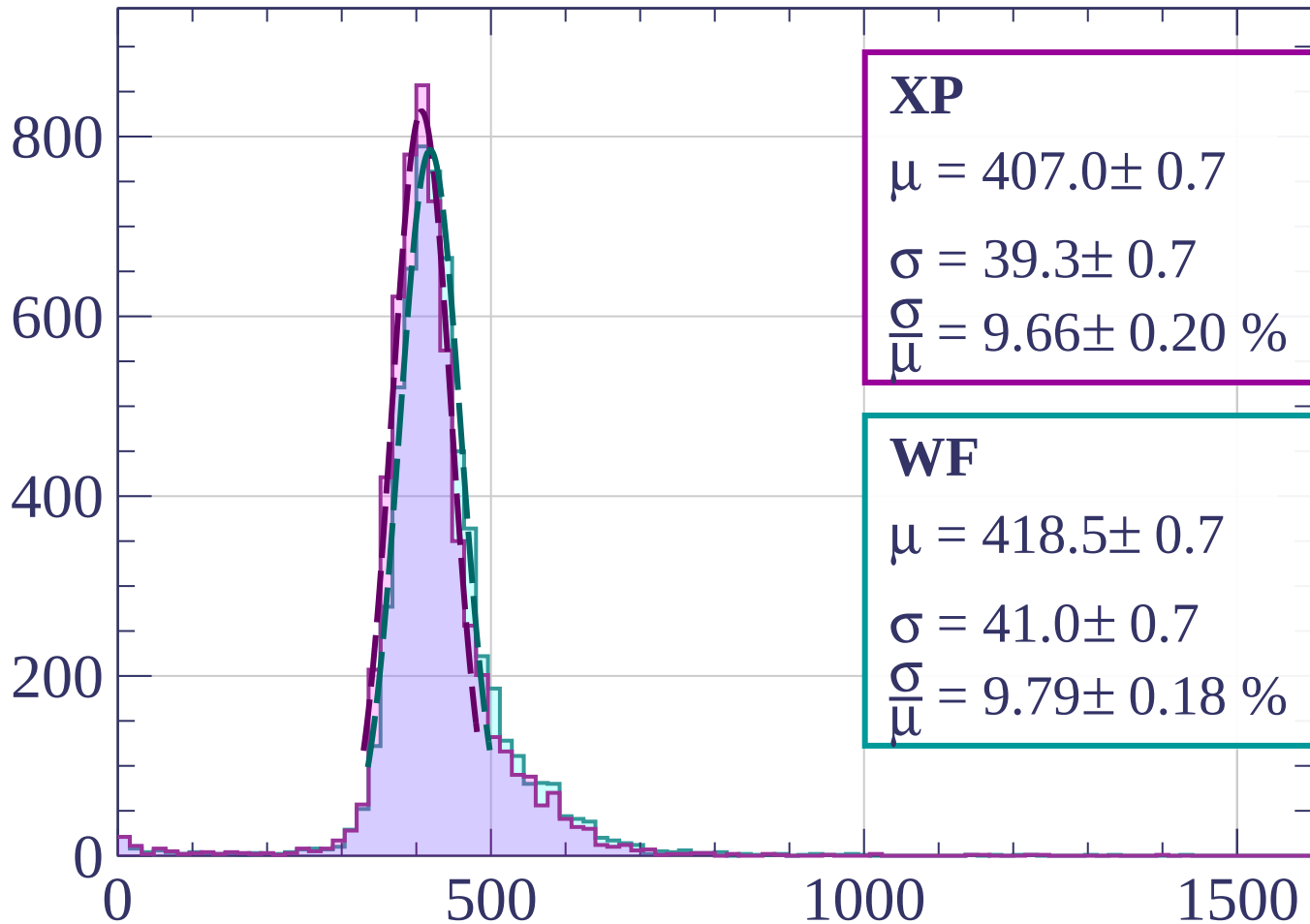
$$\sigma = 42.3 \pm 0.8$$

$$\frac{\sigma}{\mu} = 10.17 \pm 0.20 \%$$

dE/dx [ADC counts/cm]

Energy loss | $640 < p < 660$

Count



XP

$$\mu = 407.0 \pm 0.7$$

$$\sigma = 39.3 \pm 0.7$$

$$\frac{\sigma}{\mu} = 9.66 \pm 0.20 \%$$

WF

$$\mu = 418.5 \pm 0.7$$

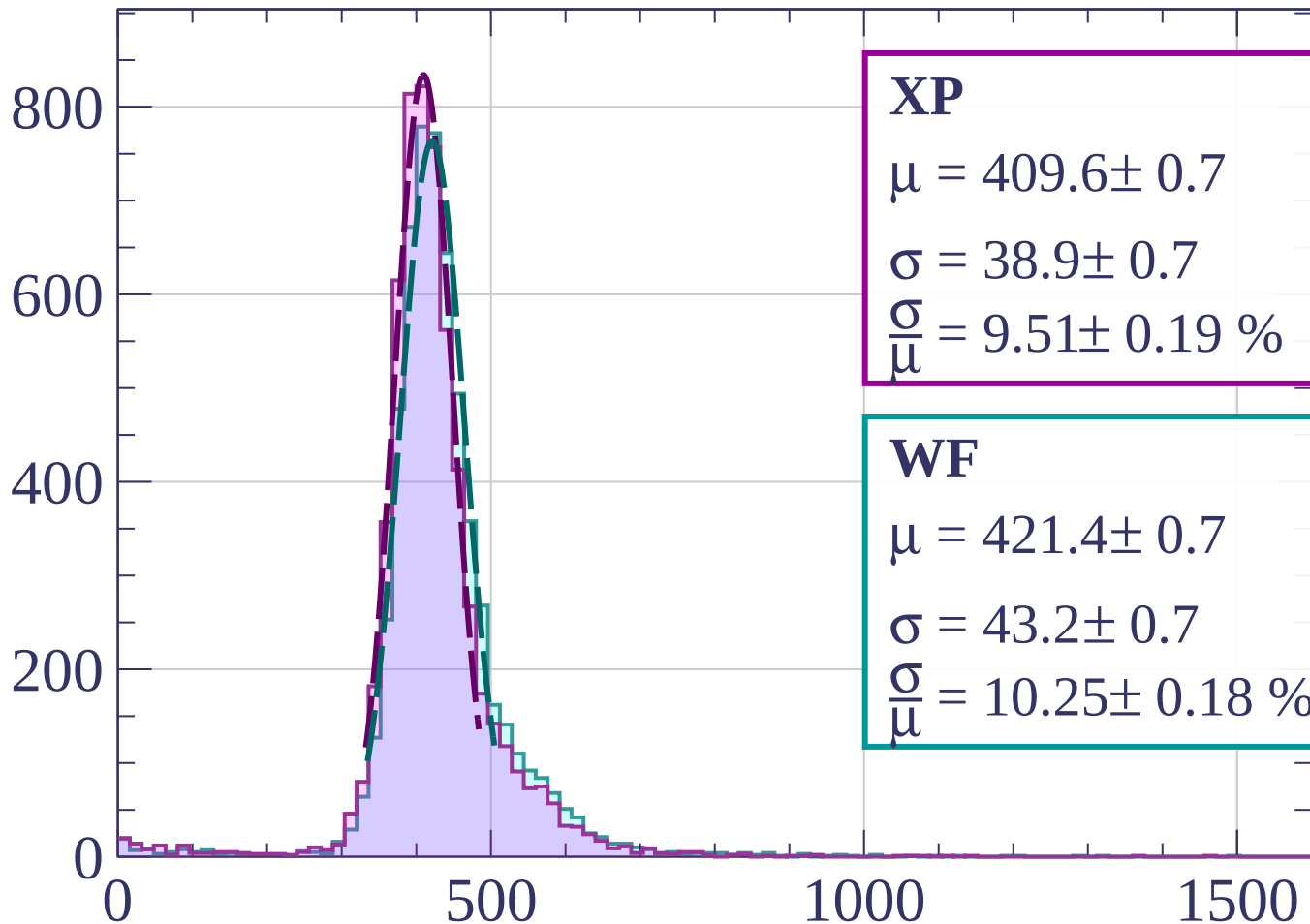
$$\sigma = 41.0 \pm 0.7$$

$$\frac{\sigma}{\mu} = 9.79 \pm 0.18 \%$$

dE/dx [ADC counts/cm]

Energy loss | $660 < p < 680$

Count



XP

$$\mu = 409.6 \pm 0.7$$

$$\sigma = 38.9 \pm 0.7$$

$$\frac{\sigma}{\mu} = 9.51 \pm 0.19 \%$$

WF

$$\mu = 421.4 \pm 0.7$$

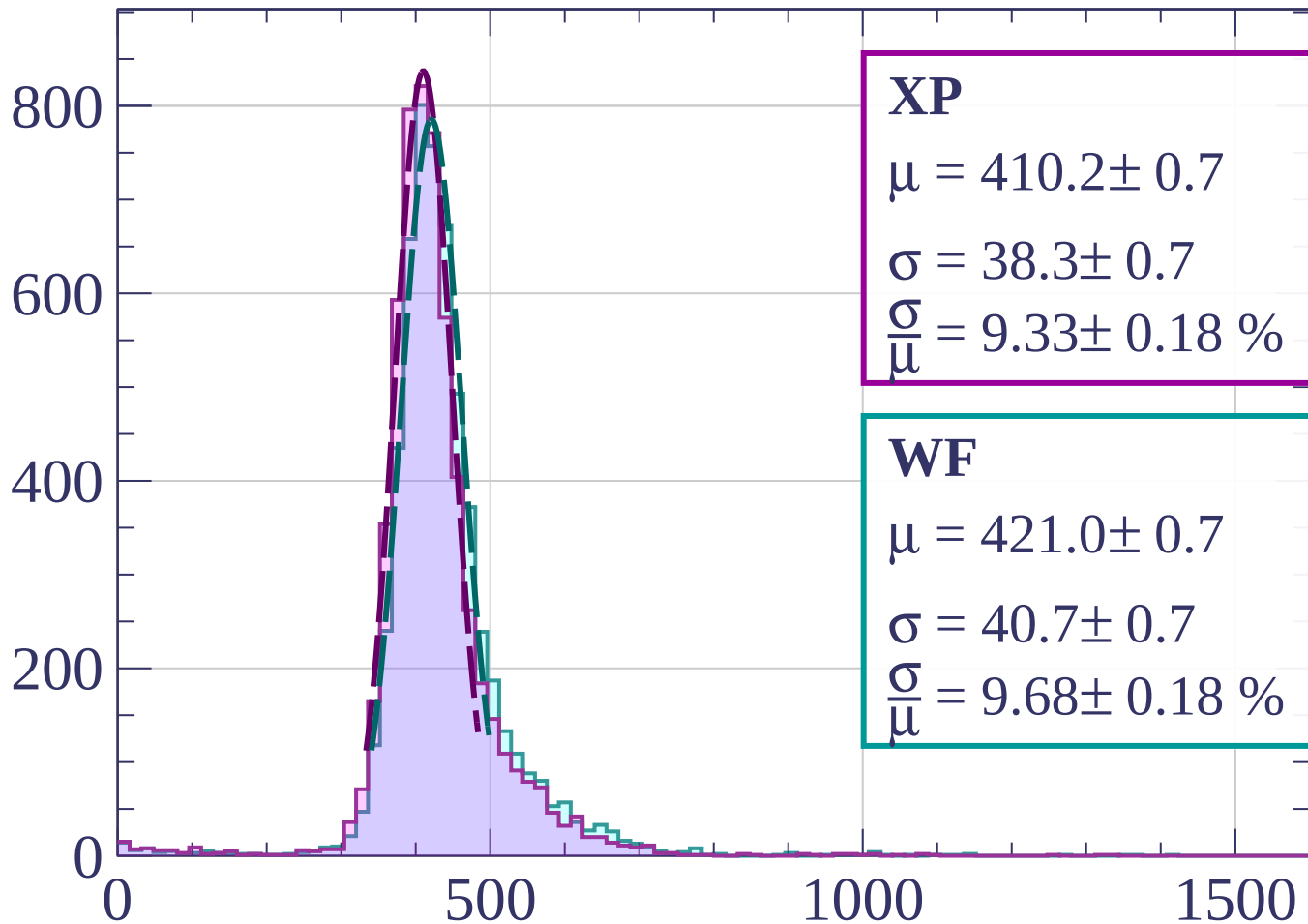
$$\sigma = 43.2 \pm 0.7$$

$$\frac{\sigma}{\mu} = 10.25 \pm 0.18 \%$$

dE/dx [ADC counts/cm]

Energy loss | $680 < p < 700$

Count



XP

$$\mu = 410.2 \pm 0.7$$

$$\sigma = 38.3 \pm 0.7$$

$$\frac{\sigma}{\mu} = 9.33 \pm 0.18 \%$$

WF

$$\mu = 421.0 \pm 0.7$$

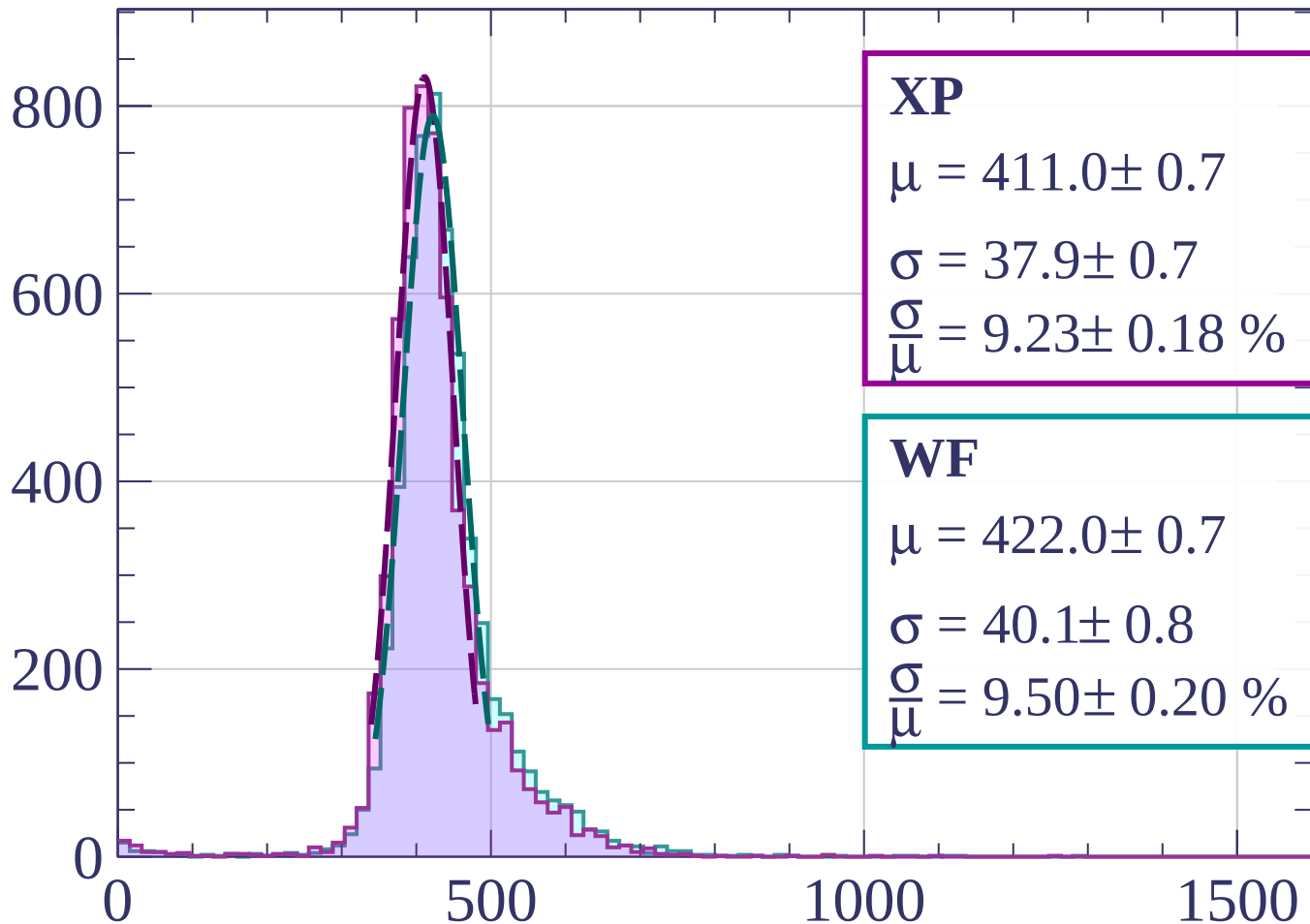
$$\sigma = 40.7 \pm 0.7$$

$$\frac{\sigma}{\mu} = 9.68 \pm 0.18 \%$$

dE/dx [ADC counts/cm]

Energy loss | $700 < p < 720$

Count



XP

$$\mu = 411.0 \pm 0.7$$

$$\sigma = 37.9 \pm 0.7$$

$$\frac{\sigma}{\mu} = 9.23 \pm 0.18 \%$$

WF

$$\mu = 422.0 \pm 0.7$$

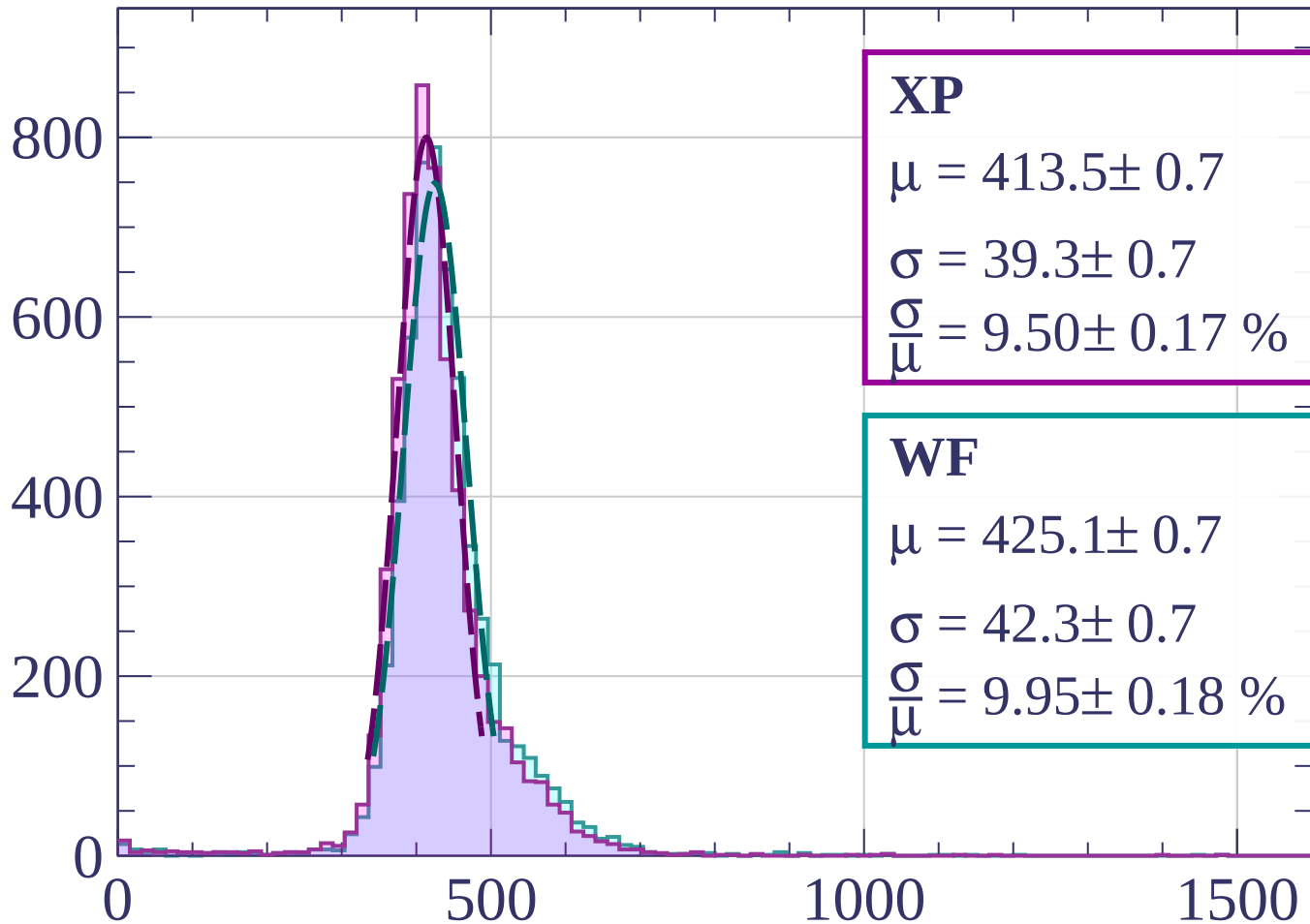
$$\sigma = 40.1 \pm 0.8$$

$$\frac{\sigma}{\mu} = 9.50 \pm 0.20 \%$$

dE/dx [ADC counts/cm]

Energy loss | $720 < p < 740$

Count



XP

$$\mu = 413.5 \pm 0.7$$

$$\sigma = 39.3 \pm 0.7$$

$$\frac{\sigma}{\mu} = 9.50 \pm 0.17 \%$$

WF

$$\mu = 425.1 \pm 0.7$$

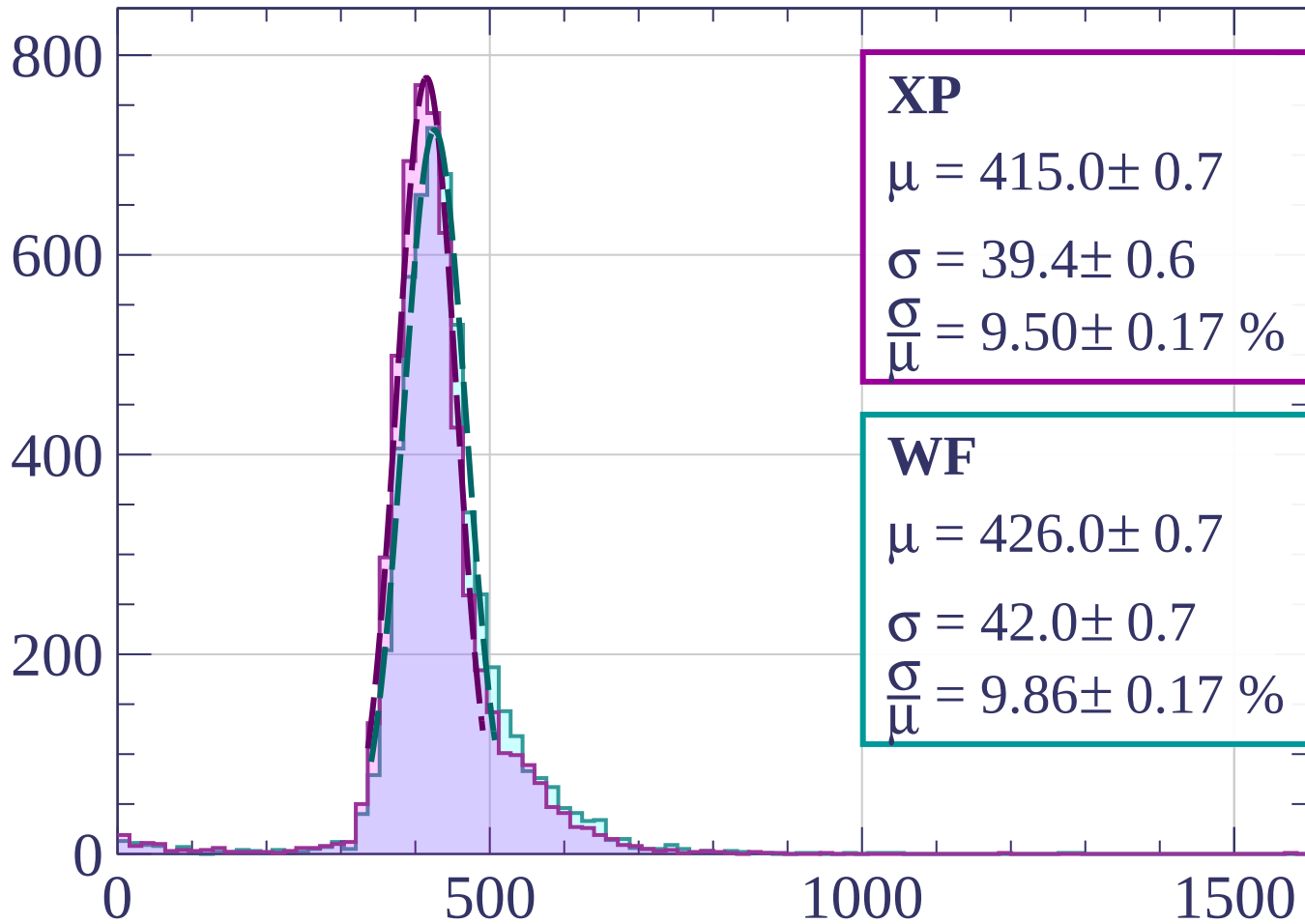
$$\sigma = 42.3 \pm 0.7$$

$$\frac{\sigma}{\mu} = 9.95 \pm 0.18 \%$$

dE/dx [ADC counts/cm]

Energy loss | $740 < p < 760$

Count



XP

$$\mu = 415.0 \pm 0.7$$

$$\sigma = 39.4 \pm 0.6$$

$$\frac{\sigma}{\mu} = 9.50 \pm 0.17 \%$$

WF

$$\mu = 426.0 \pm 0.7$$

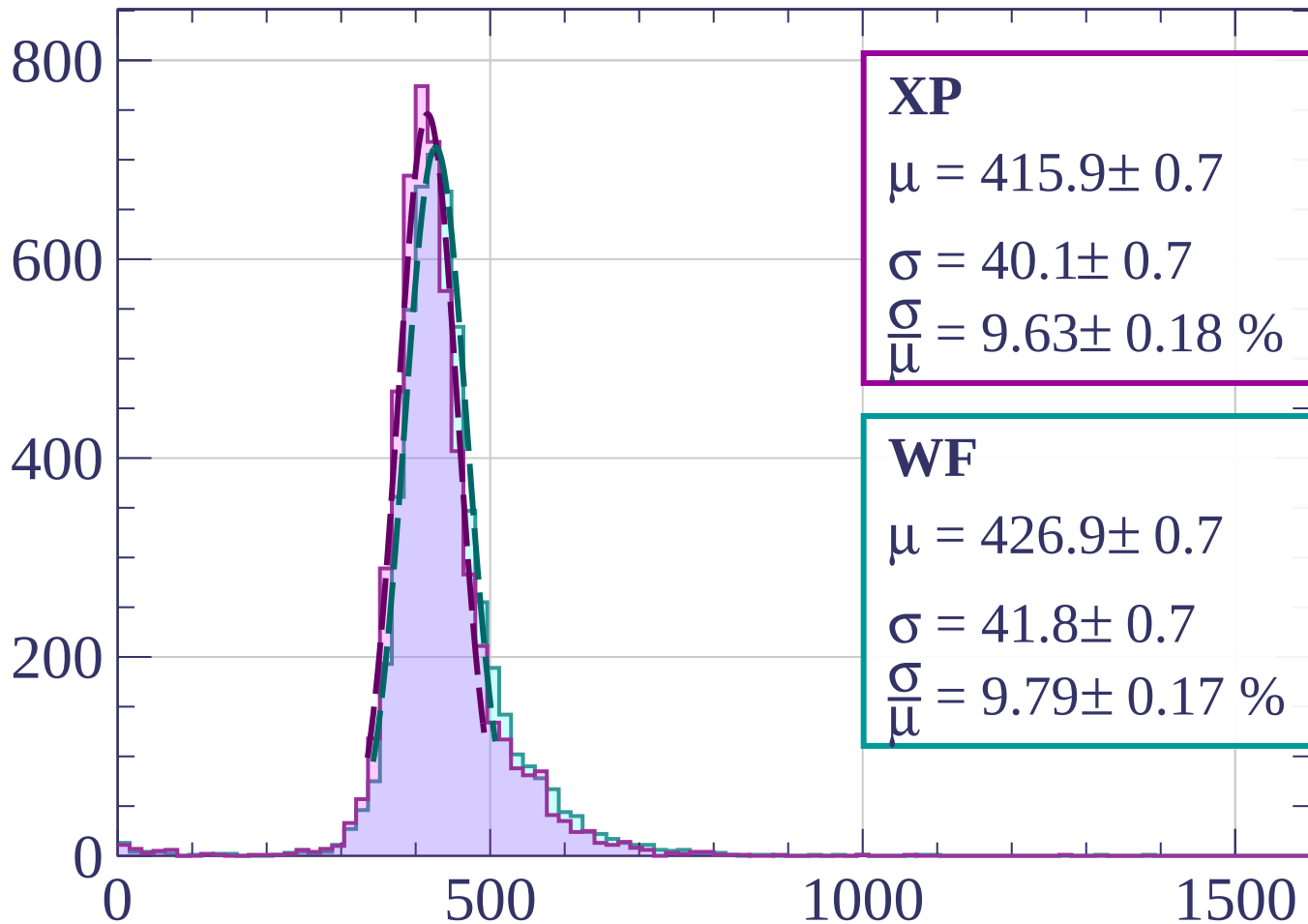
$$\sigma = 42.0 \pm 0.7$$

$$\frac{\sigma}{\mu} = 9.86 \pm 0.17 \%$$

dE/dx [ADC counts/cm]

Energy loss | $760 < p < 780$

Count



XP

$$\mu = 415.9 \pm 0.7$$

$$\sigma = 40.1 \pm 0.7$$

$$\frac{\sigma}{\mu} = 9.63 \pm 0.18 \%$$

WF

$$\mu = 426.9 \pm 0.7$$

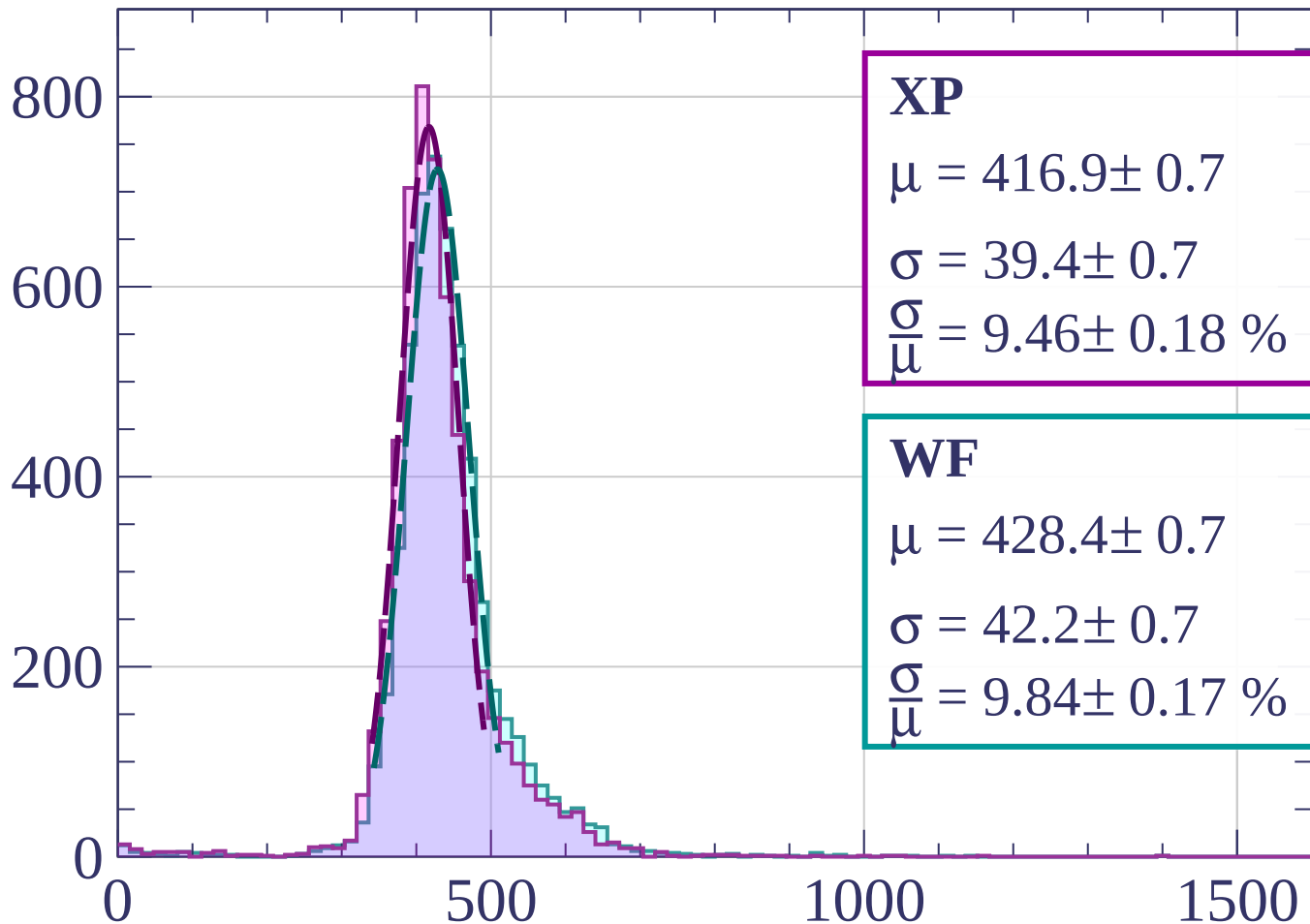
$$\sigma = 41.8 \pm 0.7$$

$$\frac{\sigma}{\mu} = 9.79 \pm 0.17 \%$$

dE/dx [ADC counts/cm]

Energy loss | $780 < p < 800$

Count



XP

$$\mu = 416.9 \pm 0.7$$

$$\sigma = 39.4 \pm 0.7$$

$$\frac{\sigma}{\mu} = 9.46 \pm 0.18 \%$$

WF

$$\mu = 428.4 \pm 0.7$$

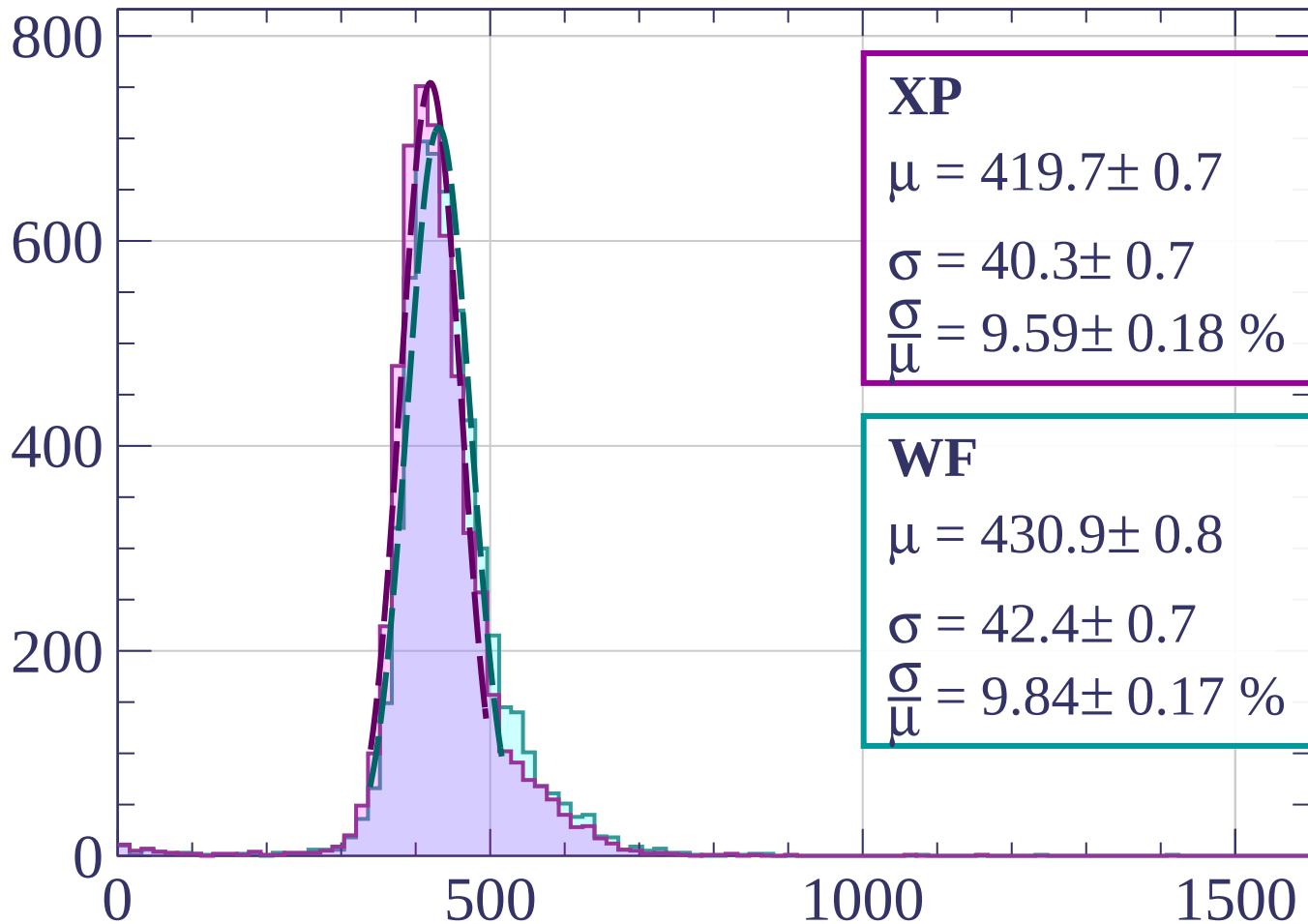
$$\sigma = 42.2 \pm 0.7$$

$$\frac{\sigma}{\mu} = 9.84 \pm 0.17 \%$$

dE/dx [ADC counts/cm]

Energy loss | $800 < p < 820$

Count



XP

$$\mu = 419.7 \pm 0.7$$

$$\sigma = 40.3 \pm 0.7$$

$$\frac{\sigma}{\mu} = 9.59 \pm 0.18 \%$$

WF

$$\mu = 430.9 \pm 0.8$$

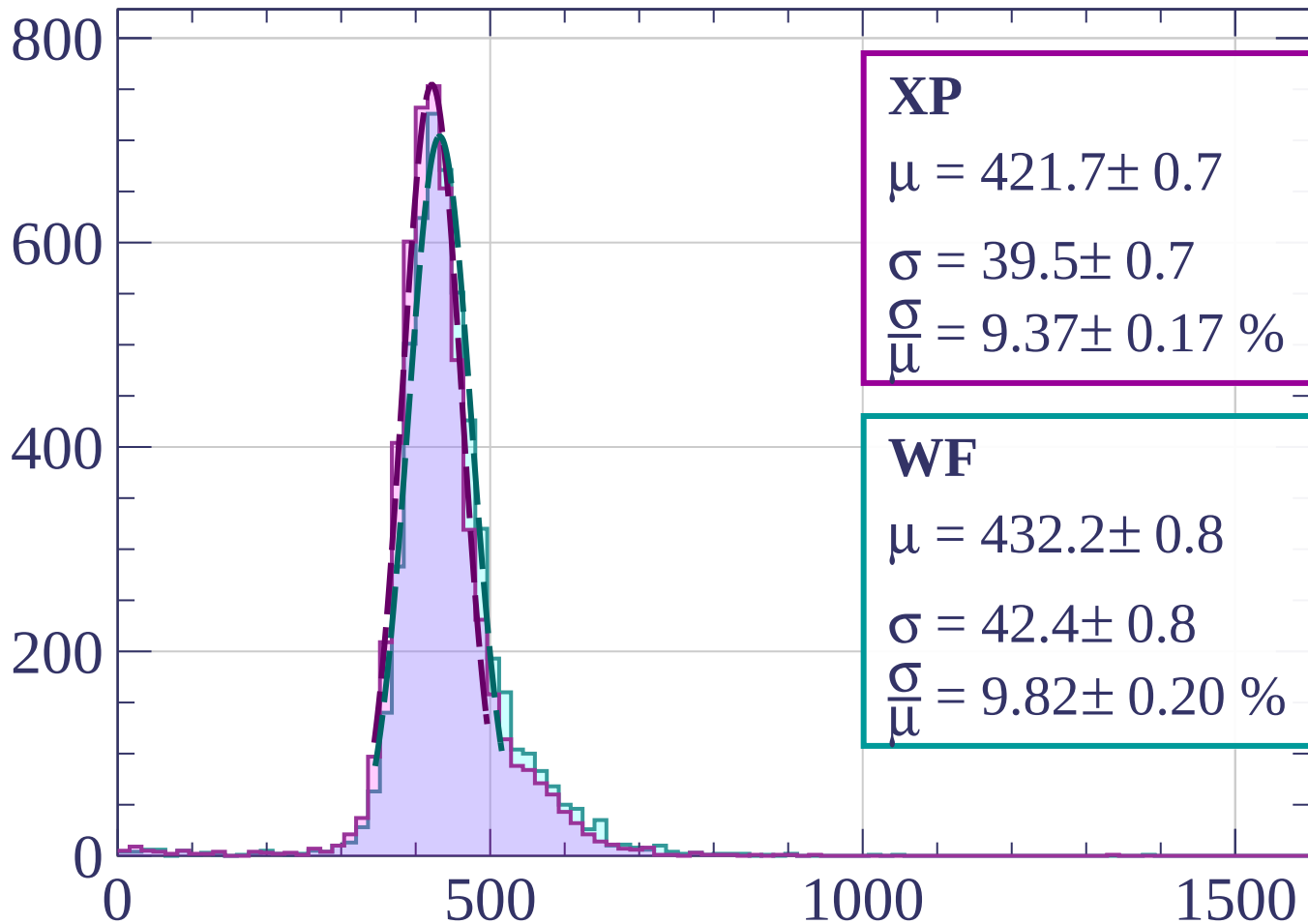
$$\sigma = 42.4 \pm 0.7$$

$$\frac{\sigma}{\mu} = 9.84 \pm 0.17 \%$$

dE/dx [ADC counts/cm]

Energy loss | $820 < p < 840$

Count



XP

$$\mu = 421.7 \pm 0.7$$

$$\sigma = 39.5 \pm 0.7$$

$$\frac{\sigma}{\mu} = 9.37 \pm 0.17 \%$$

WF

$$\mu = 432.2 \pm 0.8$$

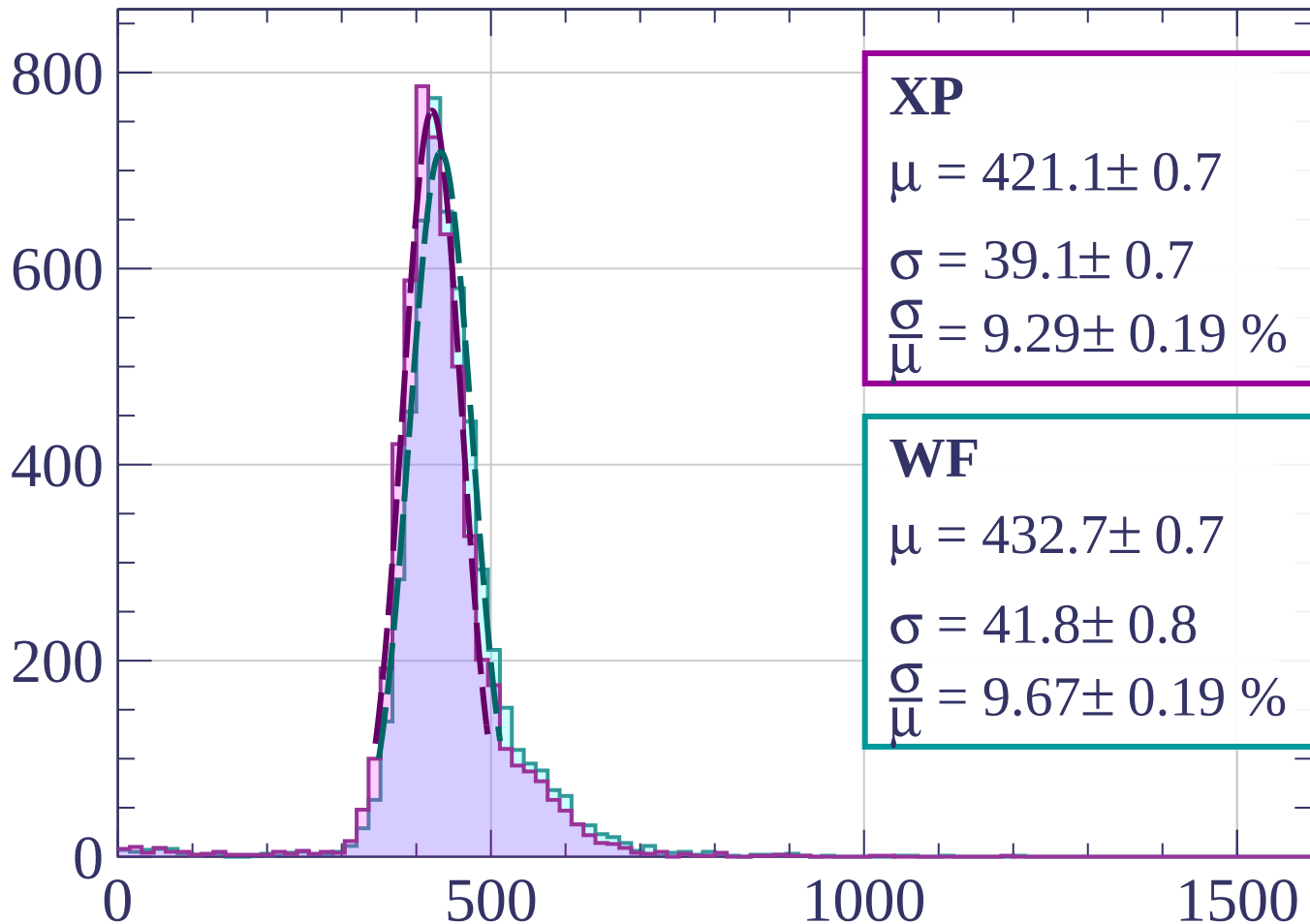
$$\sigma = 42.4 \pm 0.8$$

$$\frac{\sigma}{\mu} = 9.82 \pm 0.20 \%$$

dE/dx [ADC counts/cm]

Energy loss | $840 < p < 860$

Count



XP

$$\mu = 421.1 \pm 0.7$$

$$\sigma = 39.1 \pm 0.7$$

$$\frac{\sigma}{\mu} = 9.29 \pm 0.19 \%$$

WF

$$\mu = 432.7 \pm 0.7$$

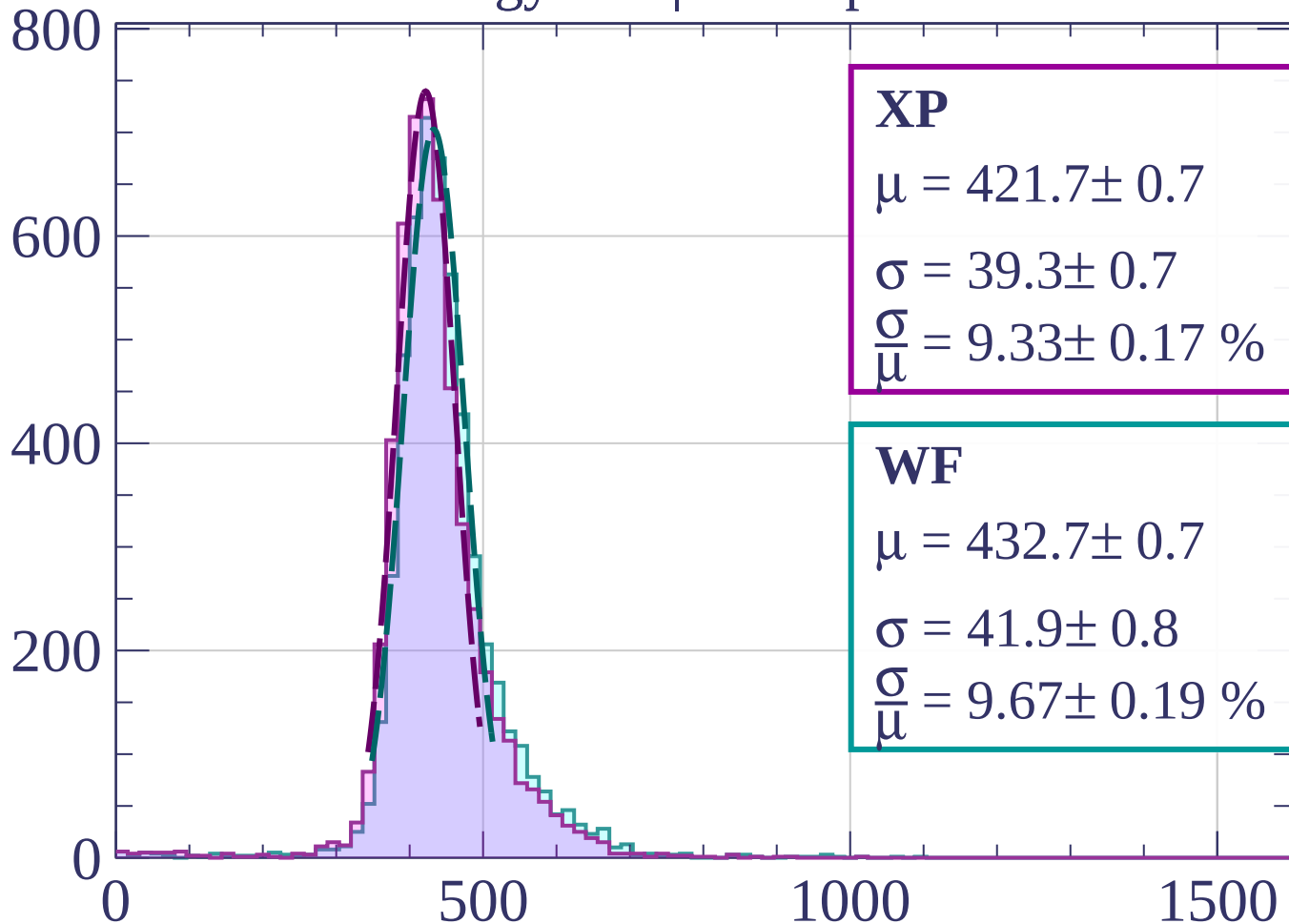
$$\sigma = 41.8 \pm 0.8$$

$$\frac{\sigma}{\mu} = 9.67 \pm 0.19 \%$$

dE/dx [ADC counts/cm]

Energy loss | $860 < p < 880$

Count



XP

$$\mu = 421.7 \pm 0.7$$

$$\sigma = 39.3 \pm 0.7$$

$$\frac{\sigma}{\mu} = 9.33 \pm 0.17 \%$$

WF

$$\mu = 432.7 \pm 0.7$$

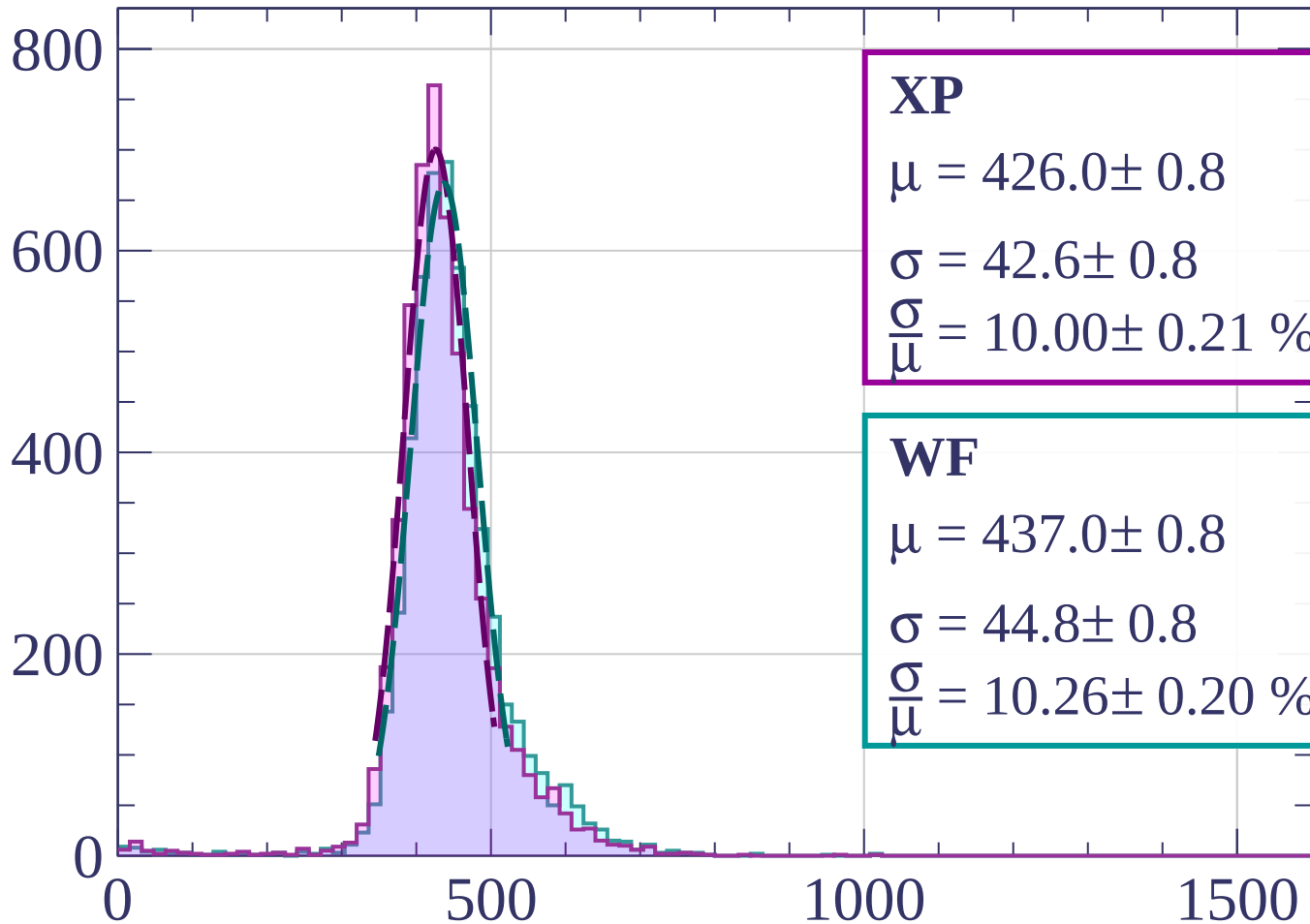
$$\sigma = 41.9 \pm 0.8$$

$$\frac{\sigma}{\mu} = 9.67 \pm 0.19 \%$$

dE/dx [ADC counts/cm]

Energy loss | $880 < p < 900$

Count



XP

$$\mu = 426.0 \pm 0.8$$

$$\sigma = 42.6 \pm 0.8$$

$$\frac{\sigma}{\mu} = 10.00 \pm 0.21 \%$$

WF

$$\mu = 437.0 \pm 0.8$$

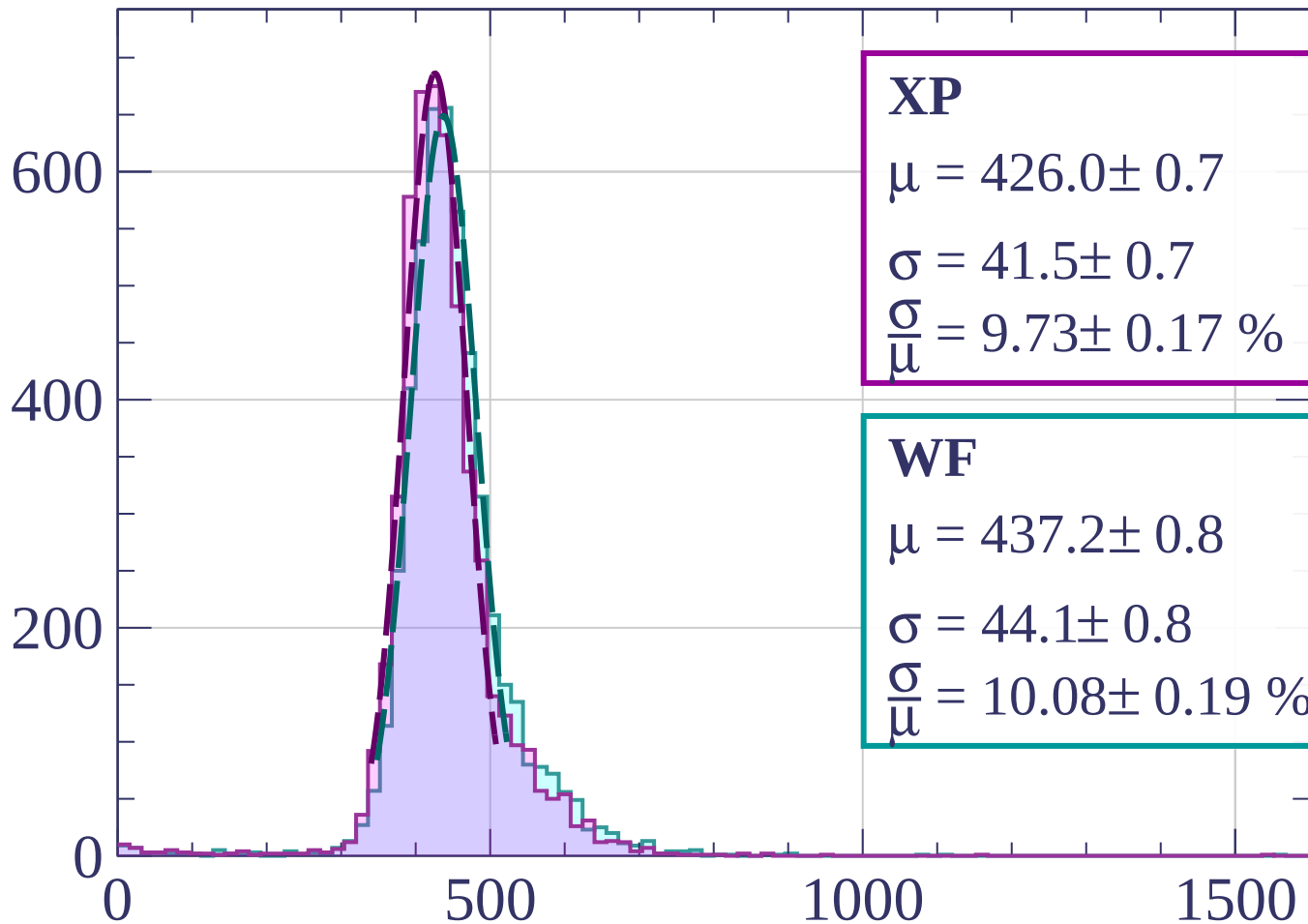
$$\sigma = 44.8 \pm 0.8$$

$$\frac{\sigma}{\mu} = 10.26 \pm 0.20 \%$$

dE/dx [ADC counts/cm]

Energy loss | $900 < p < 920$

Count



XP

$$\mu = 426.0 \pm 0.7$$

$$\sigma = 41.5 \pm 0.7$$

$$\frac{\sigma}{\mu} = 9.73 \pm 0.17 \%$$

WF

$$\mu = 437.2 \pm 0.8$$

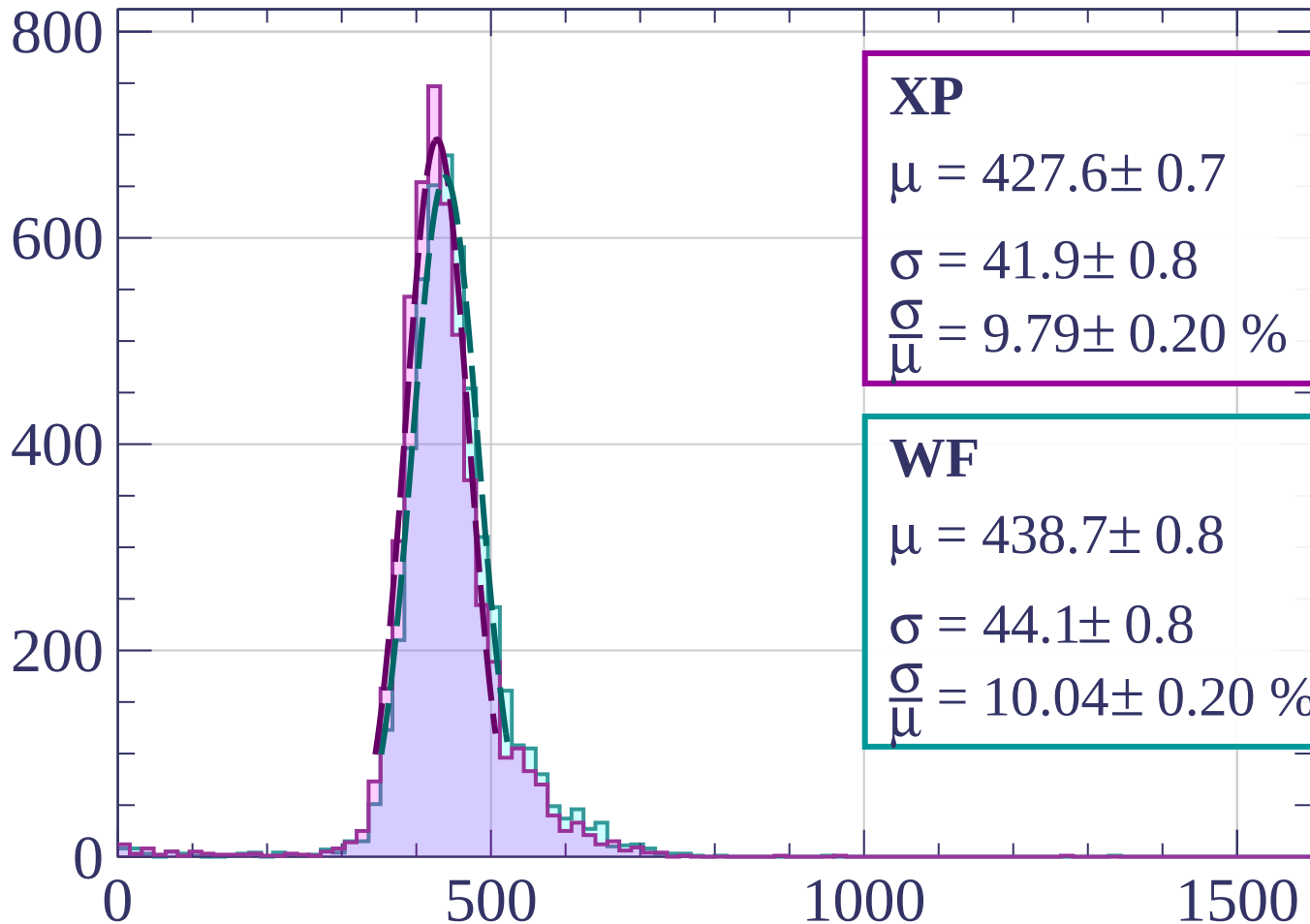
$$\sigma = 44.1 \pm 0.8$$

$$\frac{\sigma}{\mu} = 10.08 \pm 0.19 \%$$

dE/dx [ADC counts/cm]

Energy loss | $920 < p < 940$

Count



XP

$$\mu = 427.6 \pm 0.7$$

$$\sigma = 41.9 \pm 0.8$$

$$\frac{\sigma}{\mu} = 9.79 \pm 0.20 \%$$

WF

$$\mu = 438.7 \pm 0.8$$

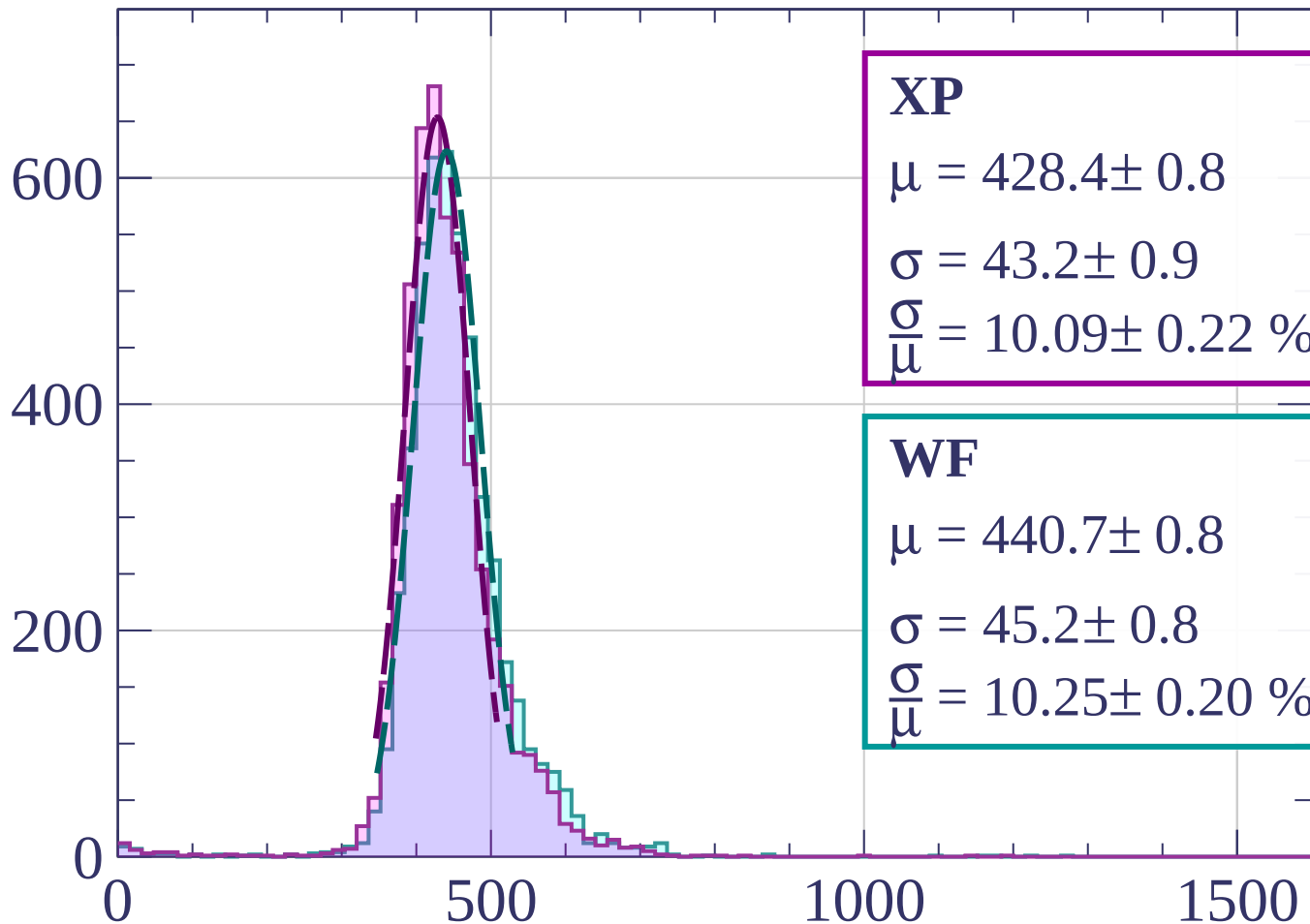
$$\sigma = 44.1 \pm 0.8$$

$$\frac{\sigma}{\mu} = 10.04 \pm 0.20 \%$$

dE/dx [ADC counts/cm]

Energy loss | $940 < p < 960$

Count



XP

$$\mu = 428.4 \pm 0.8$$

$$\sigma = 43.2 \pm 0.9$$

$$\frac{\sigma}{\mu} = 10.09 \pm 0.22 \%$$

WF

$$\mu = 440.7 \pm 0.8$$

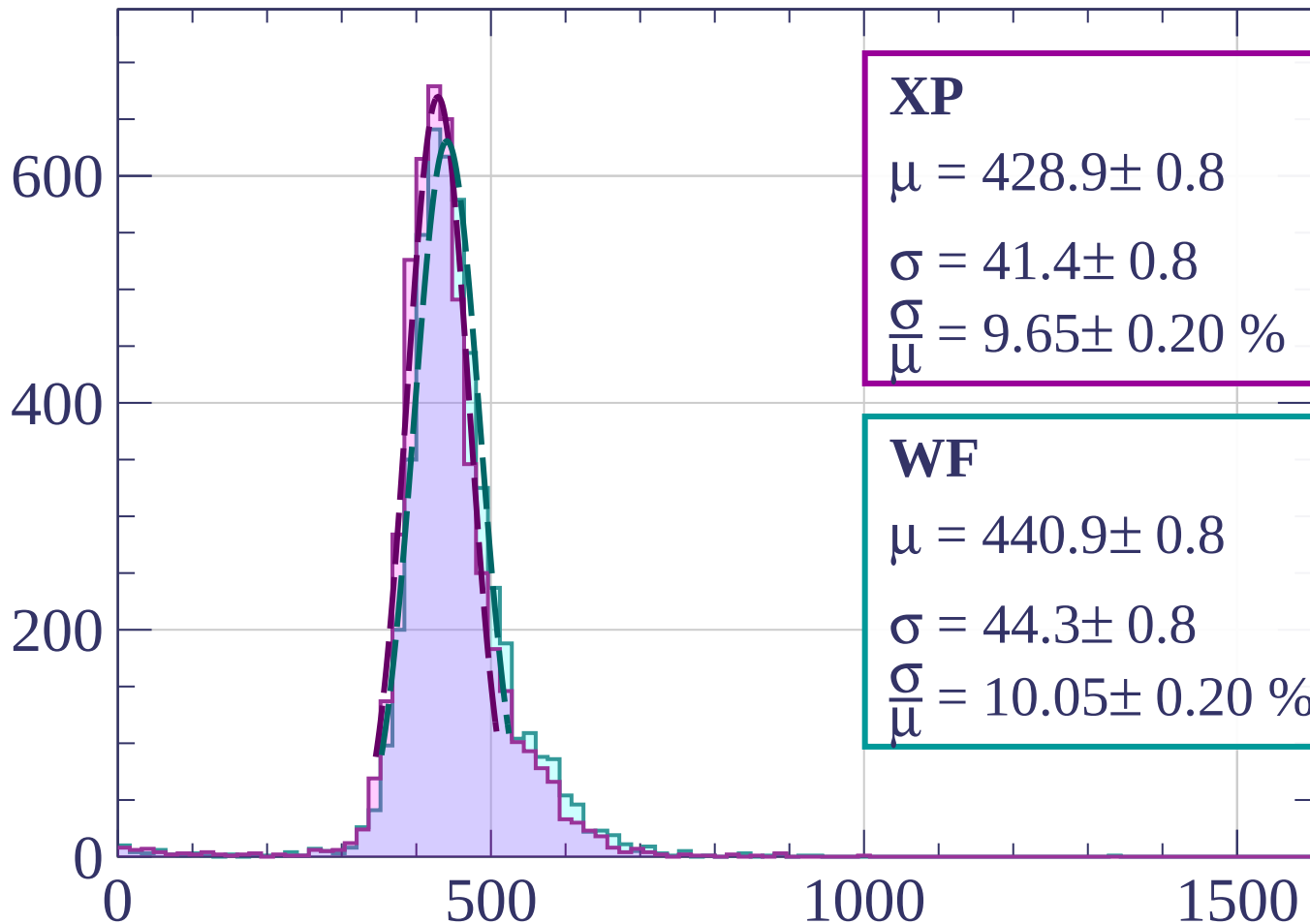
$$\sigma = 45.2 \pm 0.8$$

$$\frac{\sigma}{\mu} = 10.25 \pm 0.20 \%$$

dE/dx [ADC counts/cm]

Energy loss | $960 < p < 980$

Count



XP

$$\mu = 428.9 \pm 0.8$$

$$\sigma = 41.4 \pm 0.8$$

$$\frac{\sigma}{\mu} = 9.65 \pm 0.20 \%$$

WF

$$\mu = 440.9 \pm 0.8$$

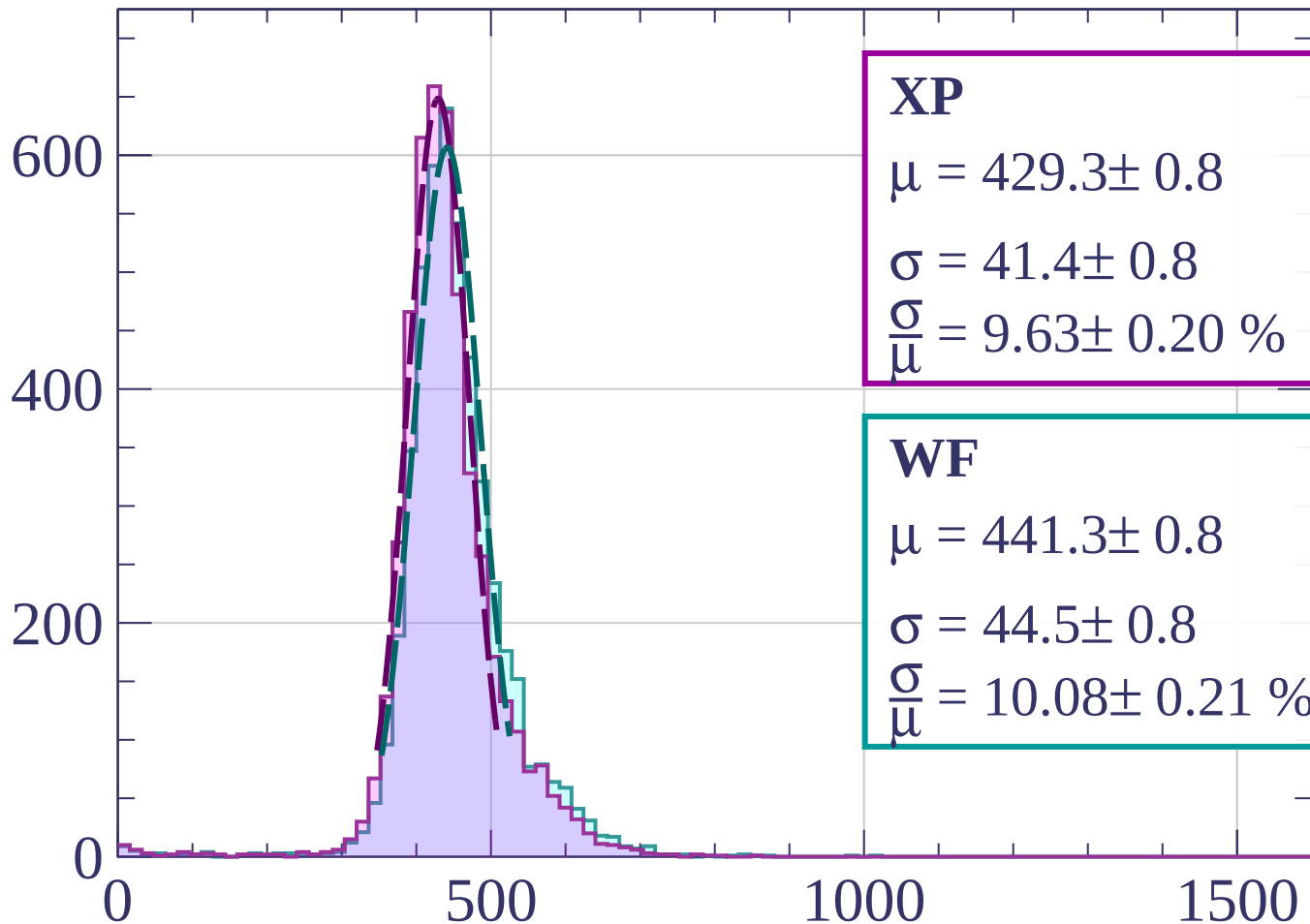
$$\sigma = 44.3 \pm 0.8$$

$$\frac{\sigma}{\mu} = 10.05 \pm 0.20 \%$$

dE/dx [ADC counts/cm]

Energy loss | $980 < p < 1000$

Count



XP

$$\mu = 429.3 \pm 0.8$$

$$\sigma = 41.4 \pm 0.8$$

$$\frac{\sigma}{\mu} = 9.63 \pm 0.20 \%$$

WF

$$\mu = 441.3 \pm 0.8$$

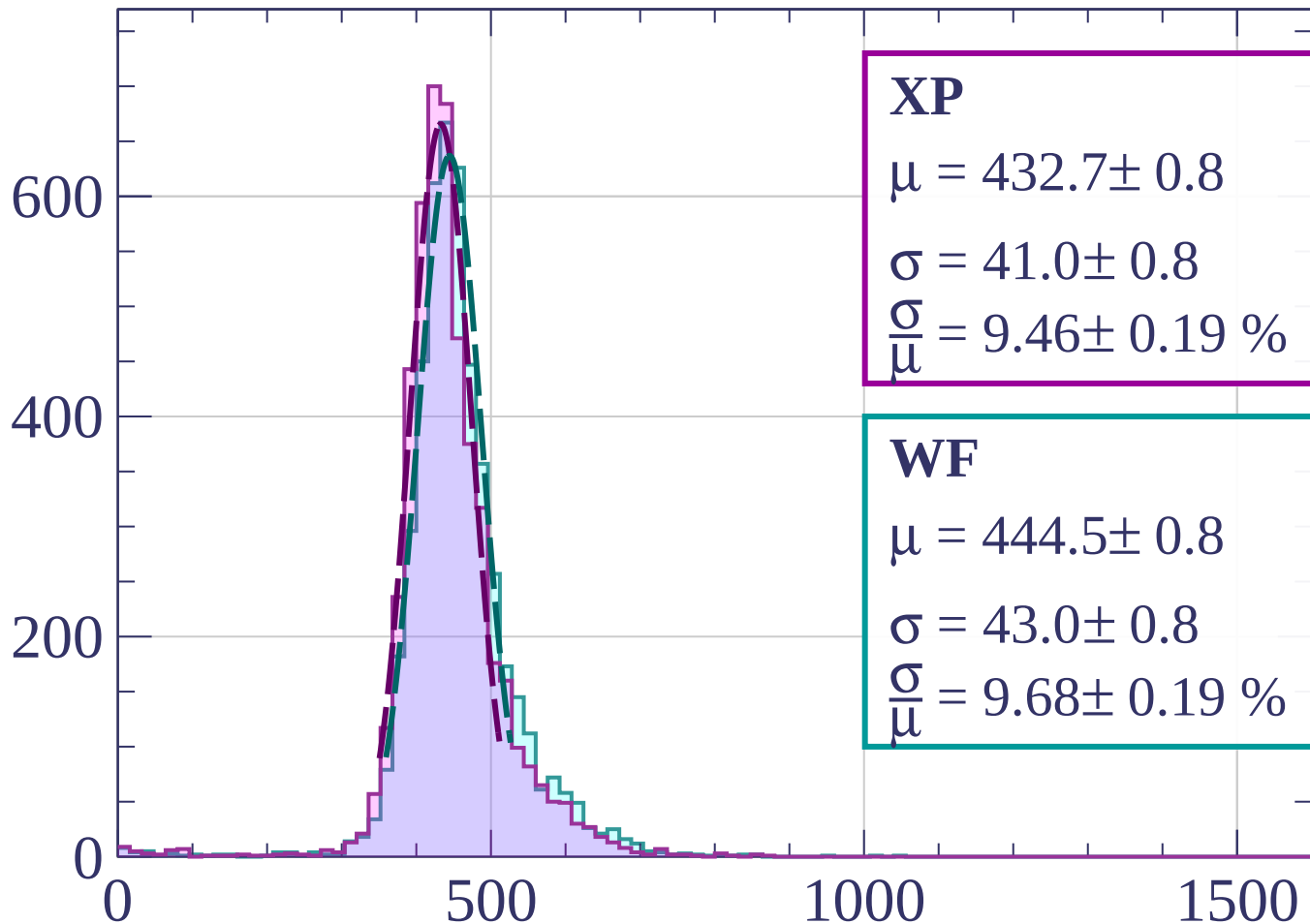
$$\sigma = 44.5 \pm 0.8$$

$$\frac{\sigma}{\mu} = 10.08 \pm 0.21 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1000 < p < 1020$

Count



XP

$$\mu = 432.7 \pm 0.8$$

$$\sigma = 41.0 \pm 0.8$$

$$\frac{\sigma}{\mu} = 9.46 \pm 0.19 \%$$

WF

$$\mu = 444.5 \pm 0.8$$

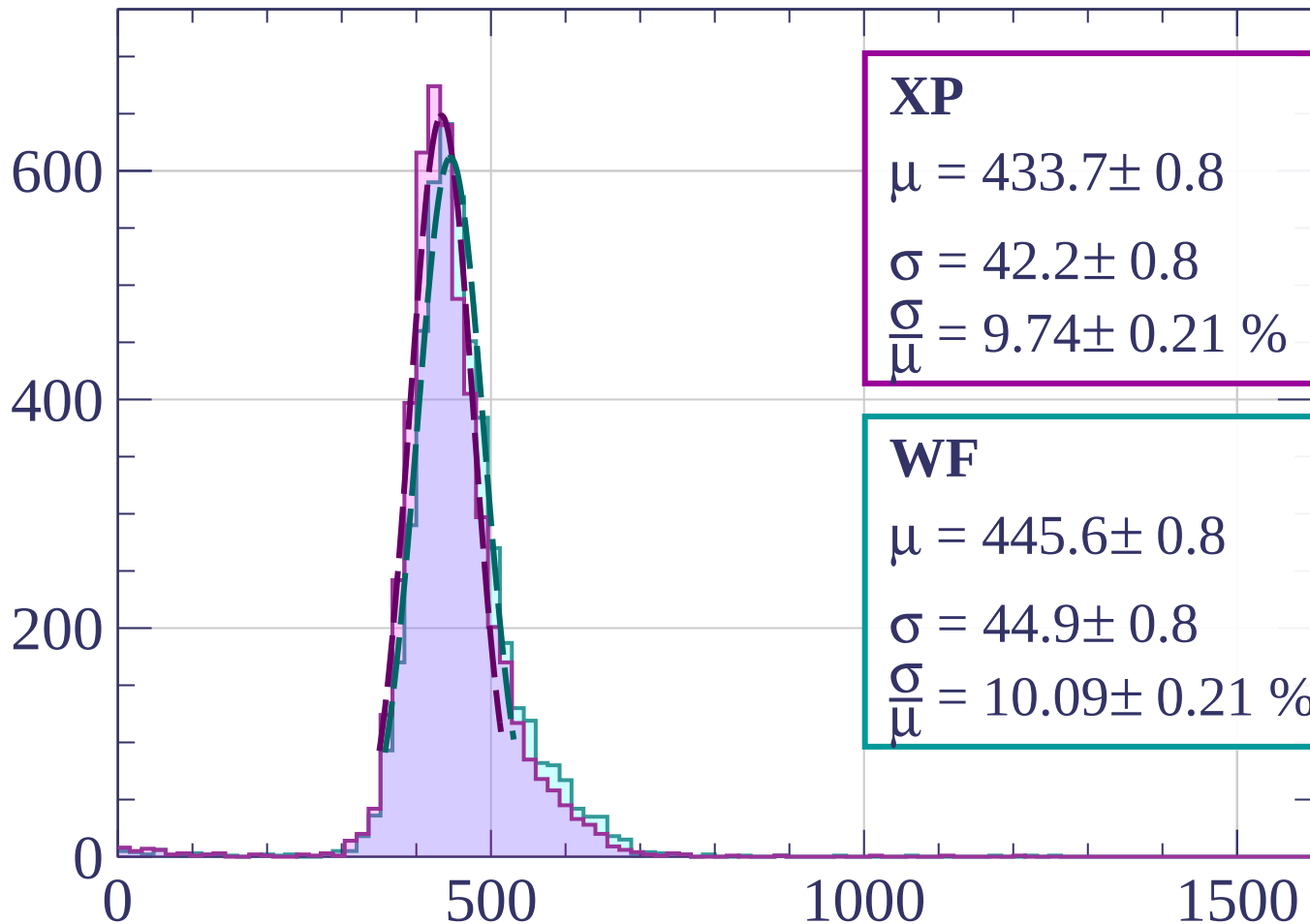
$$\sigma = 43.0 \pm 0.8$$

$$\frac{\sigma}{\mu} = 9.68 \pm 0.19 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1020 < p < 1040$

Count



XP

$$\mu = 433.7 \pm 0.8$$

$$\sigma = 42.2 \pm 0.8$$

$$\frac{\sigma}{\mu} = 9.74 \pm 0.21 \%$$

WF

$$\mu = 445.6 \pm 0.8$$

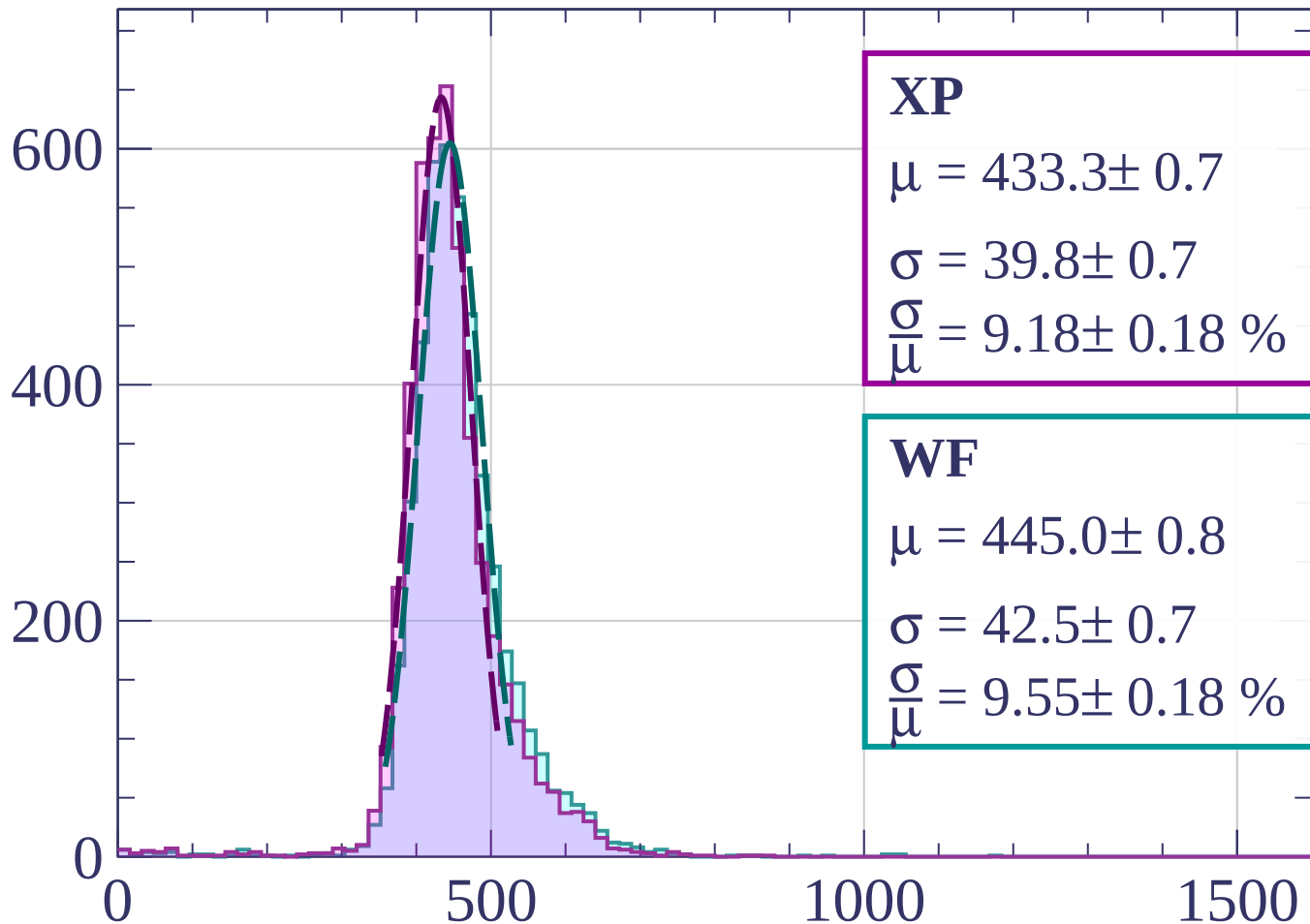
$$\sigma = 44.9 \pm 0.8$$

$$\frac{\sigma}{\mu} = 10.09 \pm 0.21 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1040 < p < 1060$

Count



XP

$$\mu = 433.3 \pm 0.7$$

$$\sigma = 39.8 \pm 0.7$$

$$\frac{\sigma}{\mu} = 9.18 \pm 0.18 \%$$

WF

$$\mu = 445.0 \pm 0.8$$

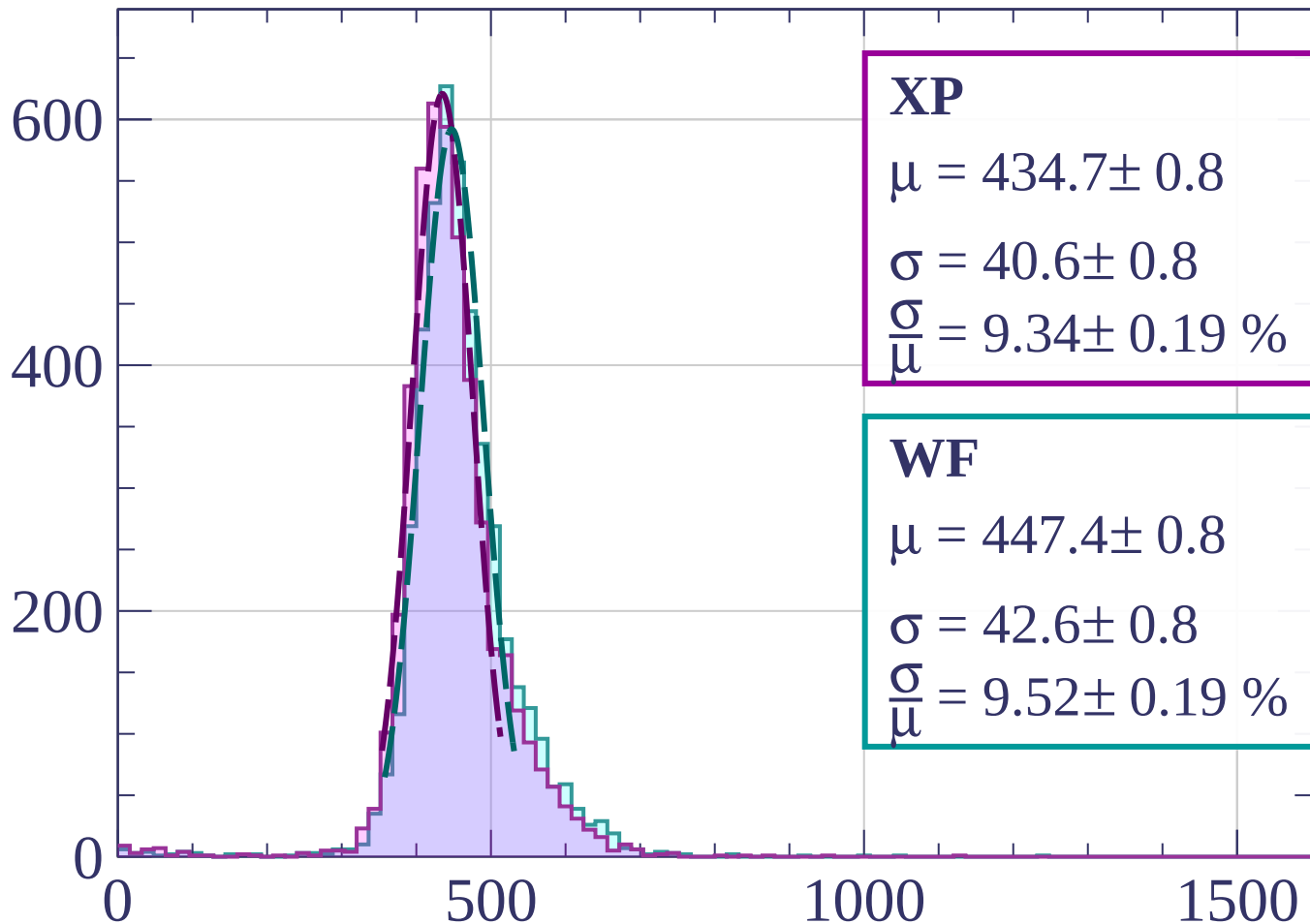
$$\sigma = 42.5 \pm 0.7$$

$$\frac{\sigma}{\mu} = 9.55 \pm 0.18 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1060 < p < 1080$

Count



XP

$$\mu = 434.7 \pm 0.8$$

$$\sigma = 40.6 \pm 0.8$$

$$\frac{\sigma}{\mu} = 9.34 \pm 0.19 \%$$

WF

$$\mu = 447.4 \pm 0.8$$

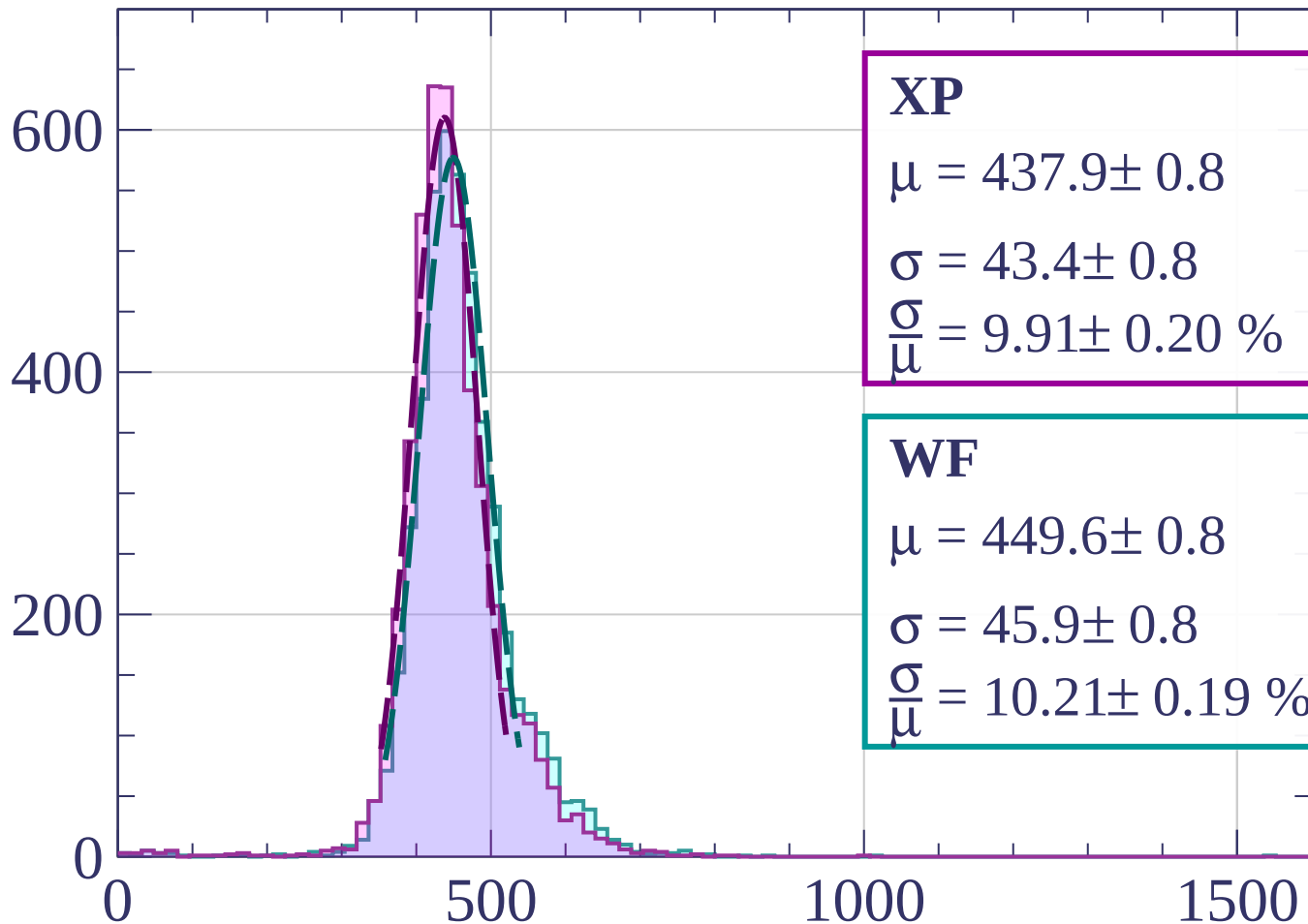
$$\sigma = 42.6 \pm 0.8$$

$$\frac{\sigma}{\mu} = 9.52 \pm 0.19 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1080 < p < 1100$

Count



XP

$$\mu = 437.9 \pm 0.8$$

$$\sigma = 43.4 \pm 0.8$$

$$\frac{\sigma}{\mu} = 9.91 \pm 0.20 \%$$

WF

$$\mu = 449.6 \pm 0.8$$

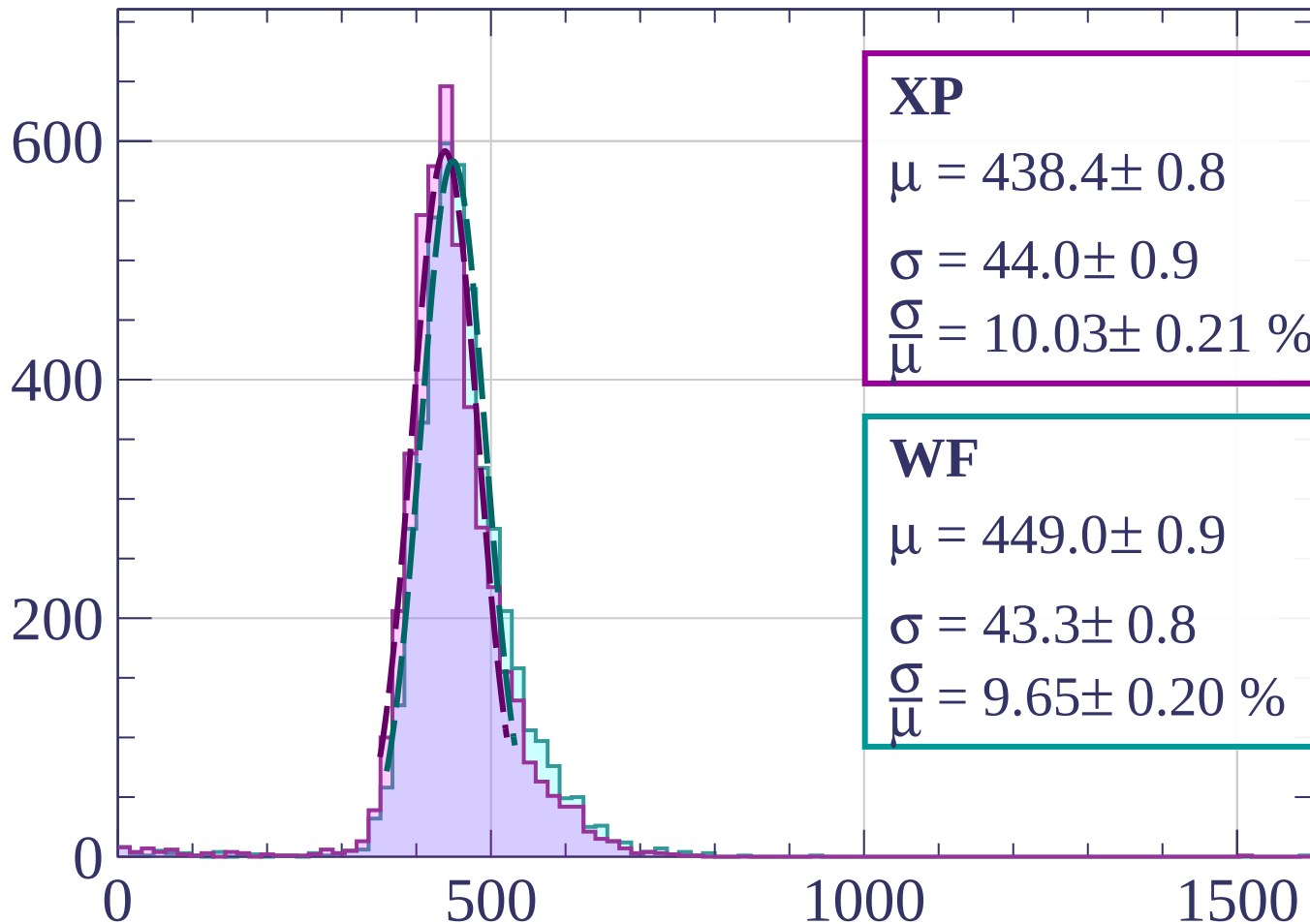
$$\sigma = 45.9 \pm 0.8$$

$$\frac{\sigma}{\mu} = 10.21 \pm 0.19 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1100 < p < 1120$

Count



XP

$$\mu = 438.4 \pm 0.8$$

$$\sigma = 44.0 \pm 0.9$$

$$\frac{\sigma}{\mu} = 10.03 \pm 0.21 \%$$

WF

$$\mu = 449.0 \pm 0.9$$

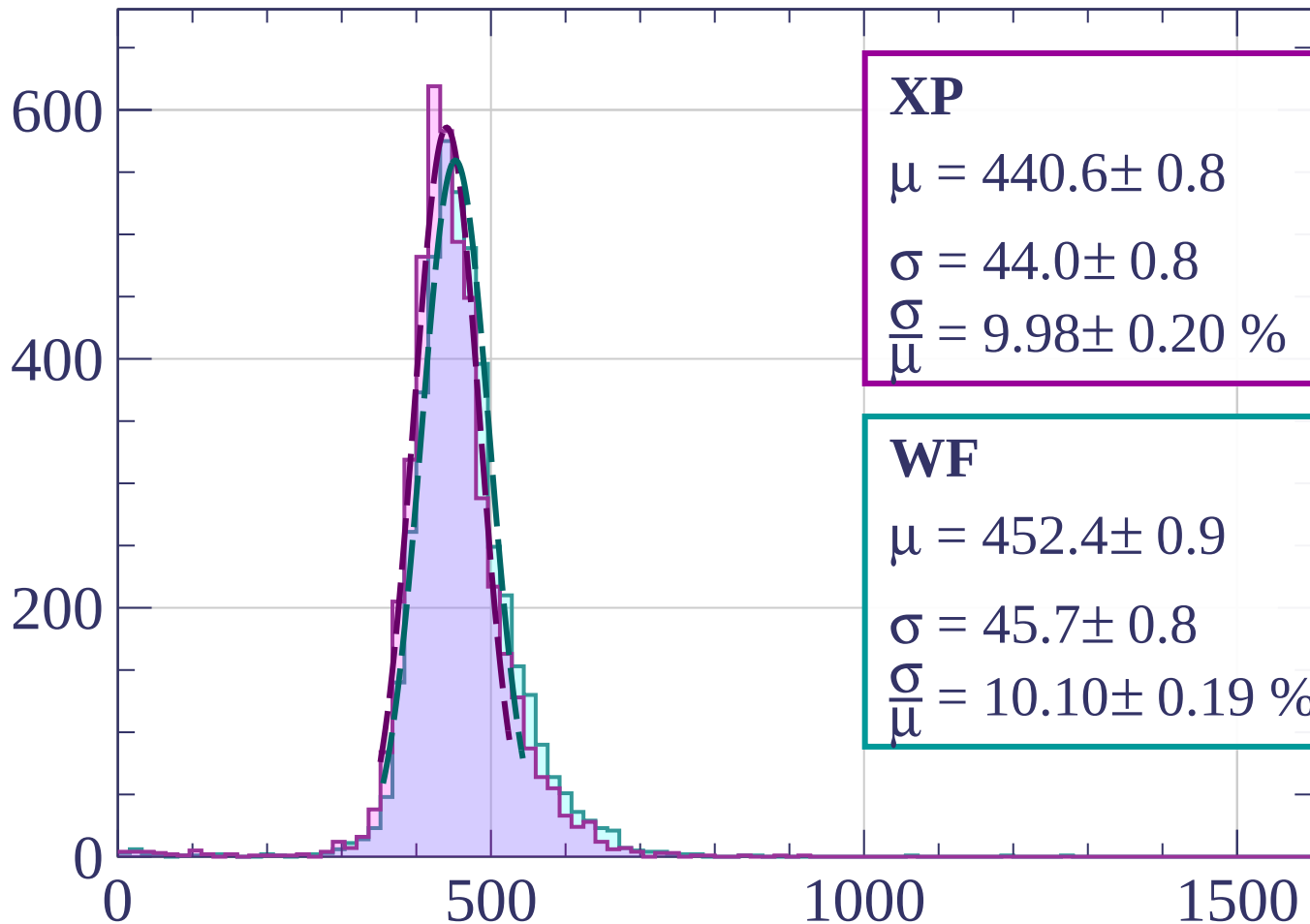
$$\sigma = 43.3 \pm 0.8$$

$$\frac{\sigma}{\mu} = 9.65 \pm 0.20 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1120 < p < 1140$

Count



XP

$$\mu = 440.6 \pm 0.8$$

$$\sigma = 44.0 \pm 0.8$$

$$\frac{\sigma}{\mu} = 9.98 \pm 0.20 \%$$

WF

$$\mu = 452.4 \pm 0.9$$

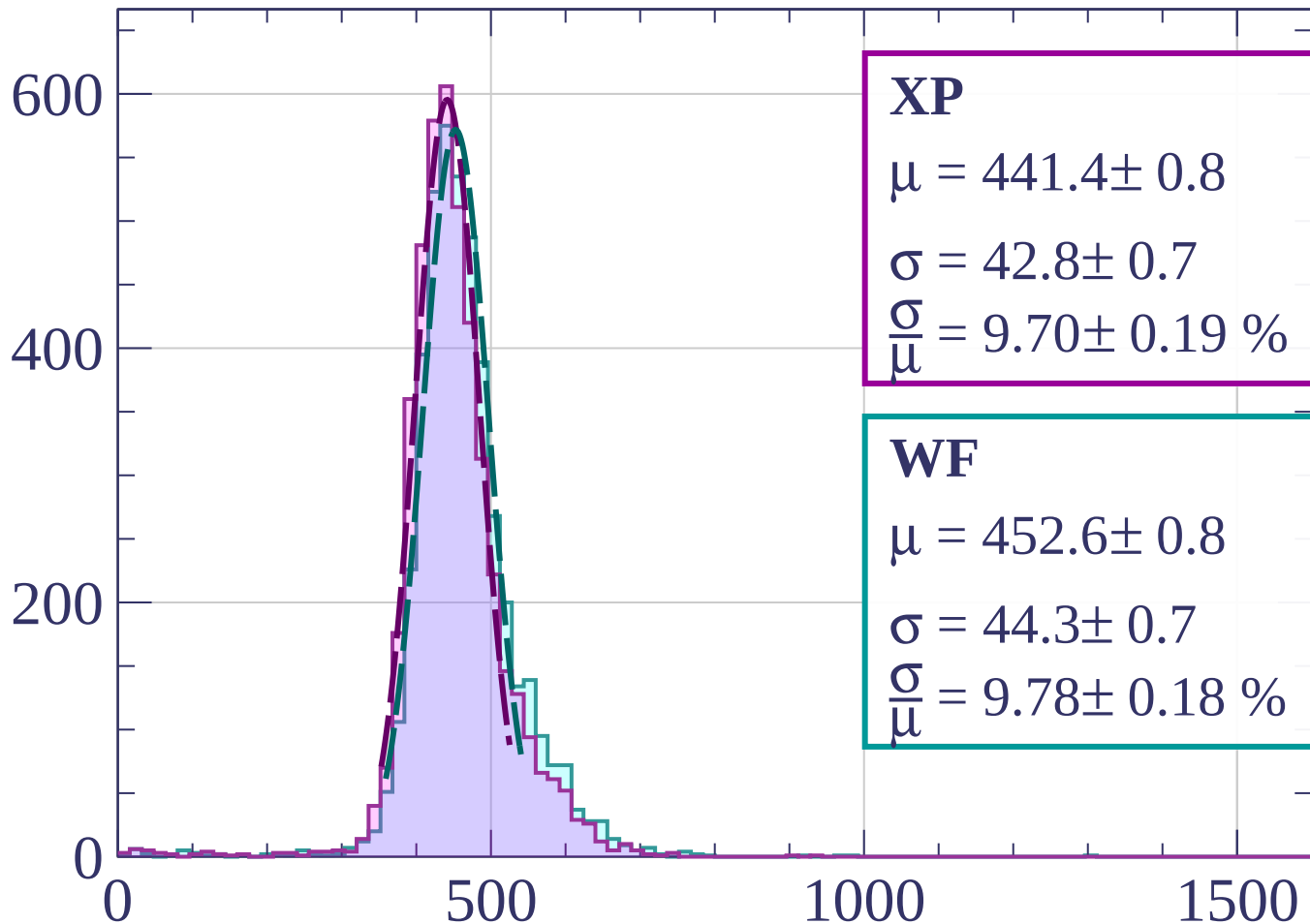
$$\sigma = 45.7 \pm 0.8$$

$$\frac{\sigma}{\mu} = 10.10 \pm 0.19 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1140 < p < 1160$

Count



XP

$$\mu = 441.4 \pm 0.8$$

$$\sigma = 42.8 \pm 0.7$$

$$\frac{\sigma}{\mu} = 9.70 \pm 0.19 \%$$

WF

$$\mu = 452.6 \pm 0.8$$

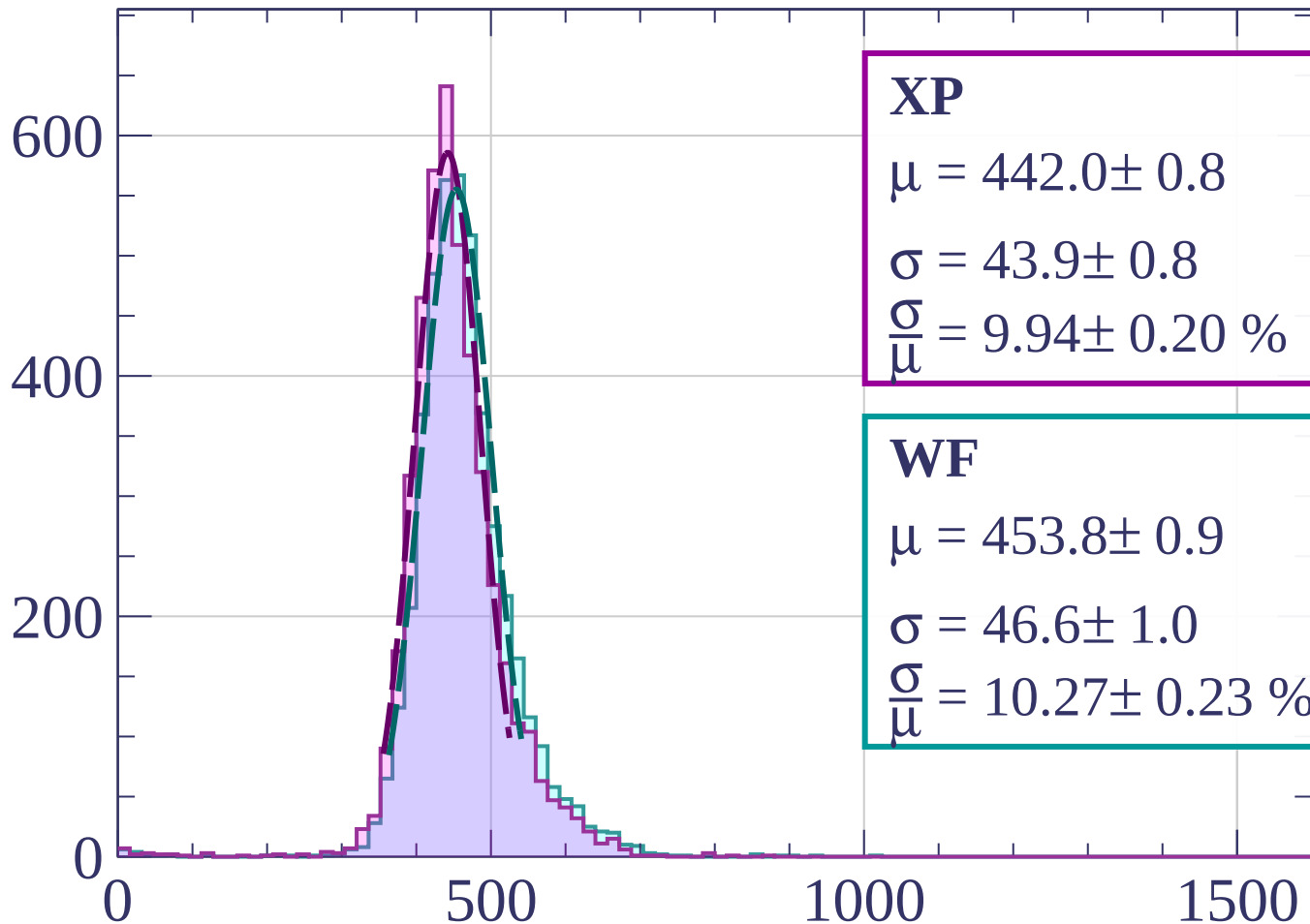
$$\sigma = 44.3 \pm 0.7$$

$$\frac{\sigma}{\mu} = 9.78 \pm 0.18 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1160 < p < 1180$

Count



XP

$$\mu = 442.0 \pm 0.8$$

$$\sigma = 43.9 \pm 0.8$$

$$\frac{\sigma}{\mu} = 9.94 \pm 0.20 \%$$

WF

$$\mu = 453.8 \pm 0.9$$

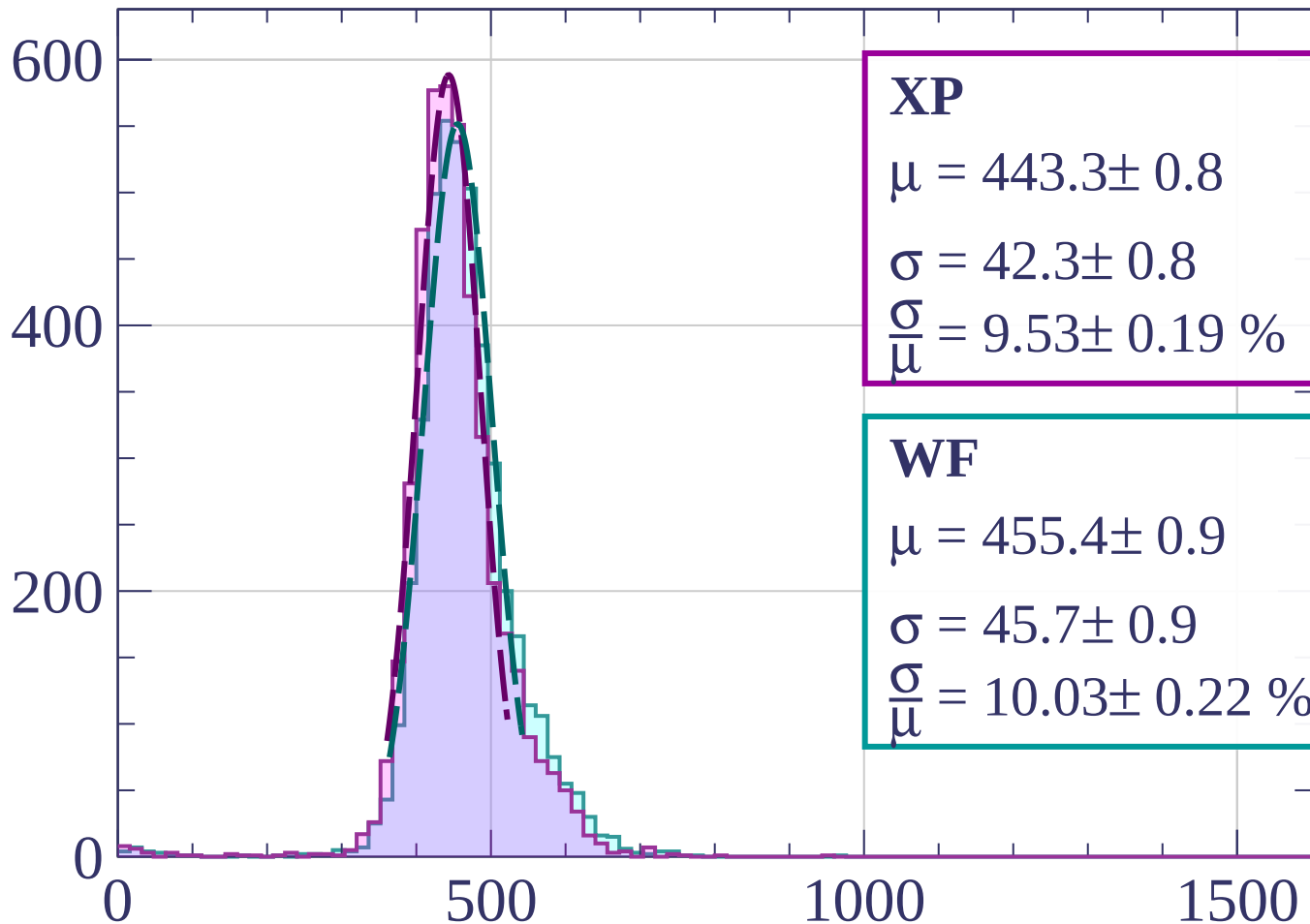
$$\sigma = 46.6 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.27 \pm 0.23 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1180 < p < 1200$

Count



XP

$$\mu = 443.3 \pm 0.8$$

$$\sigma = 42.3 \pm 0.8$$

$$\frac{\sigma}{\mu} = 9.53 \pm 0.19 \%$$

WF

$$\mu = 455.4 \pm 0.9$$

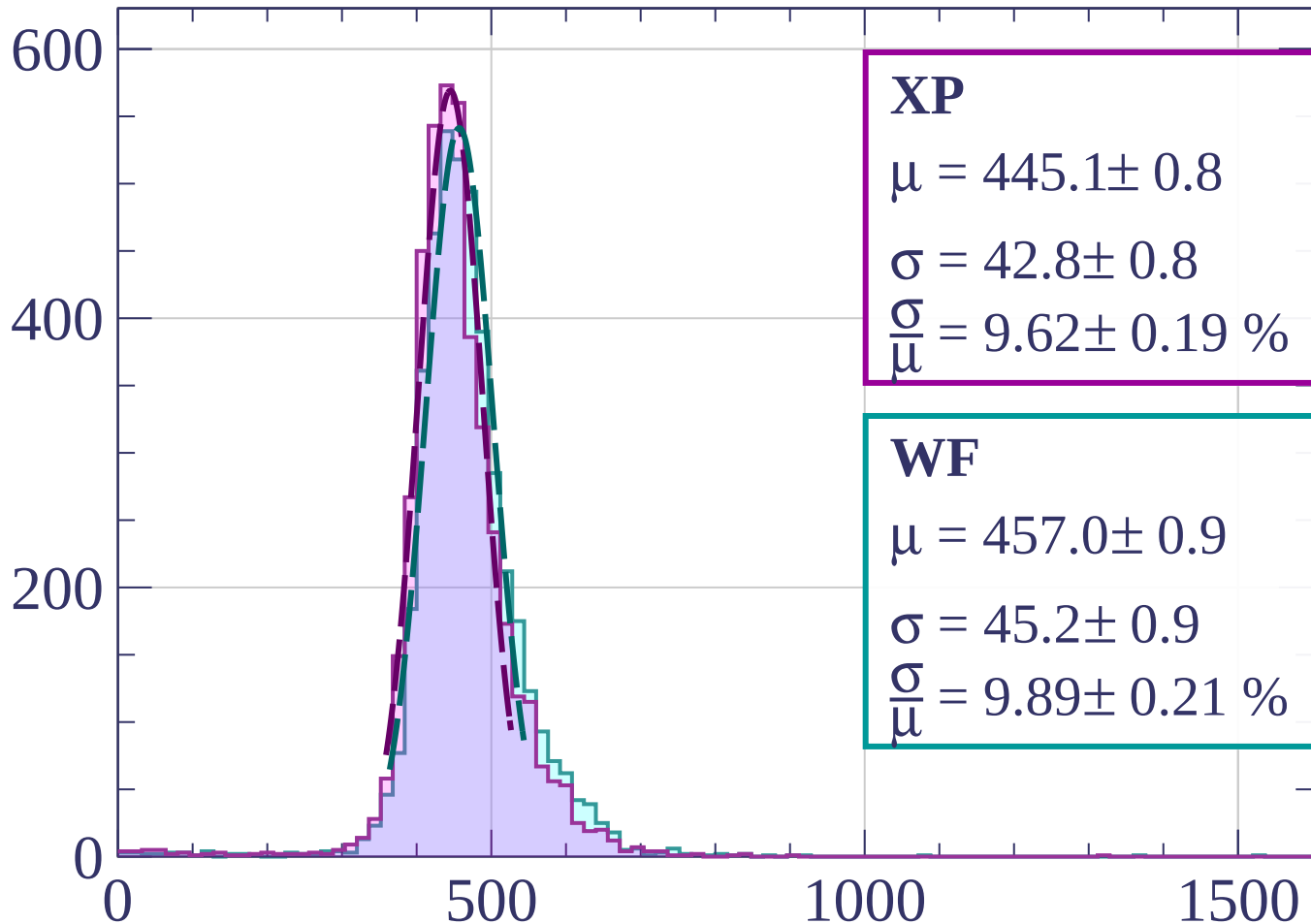
$$\sigma = 45.7 \pm 0.9$$

$$\frac{\sigma}{\mu} = 10.03 \pm 0.22 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1200 < p < 1220$

Count



XP

$$\mu = 445.1 \pm 0.8$$

$$\sigma = 42.8 \pm 0.8$$

$$\frac{\sigma}{\mu} = 9.62 \pm 0.19 \%$$

WF

$$\mu = 457.0 \pm 0.9$$

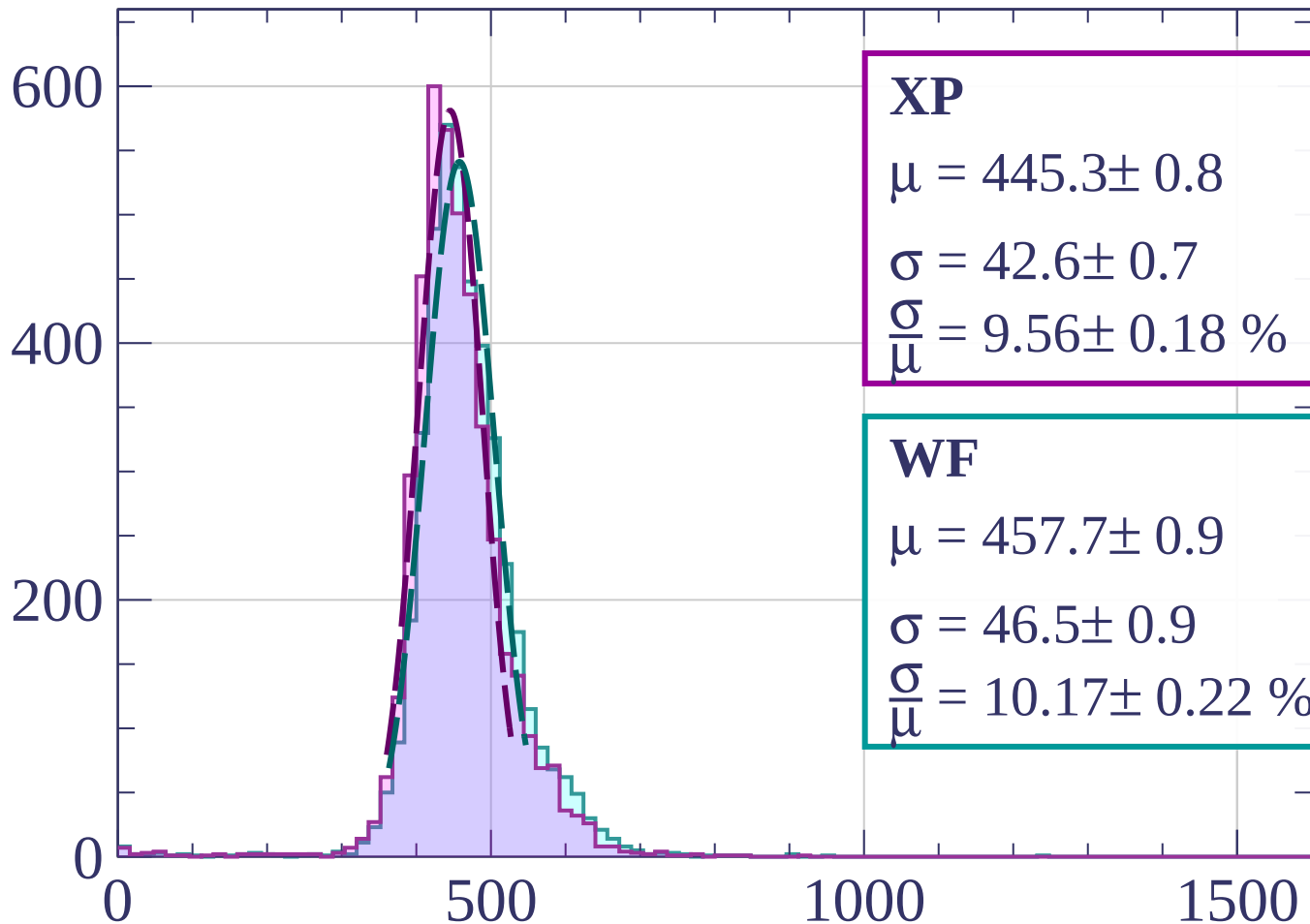
$$\sigma = 45.2 \pm 0.9$$

$$\frac{\sigma}{\mu} = 9.89 \pm 0.21 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1220 < p < 1240$

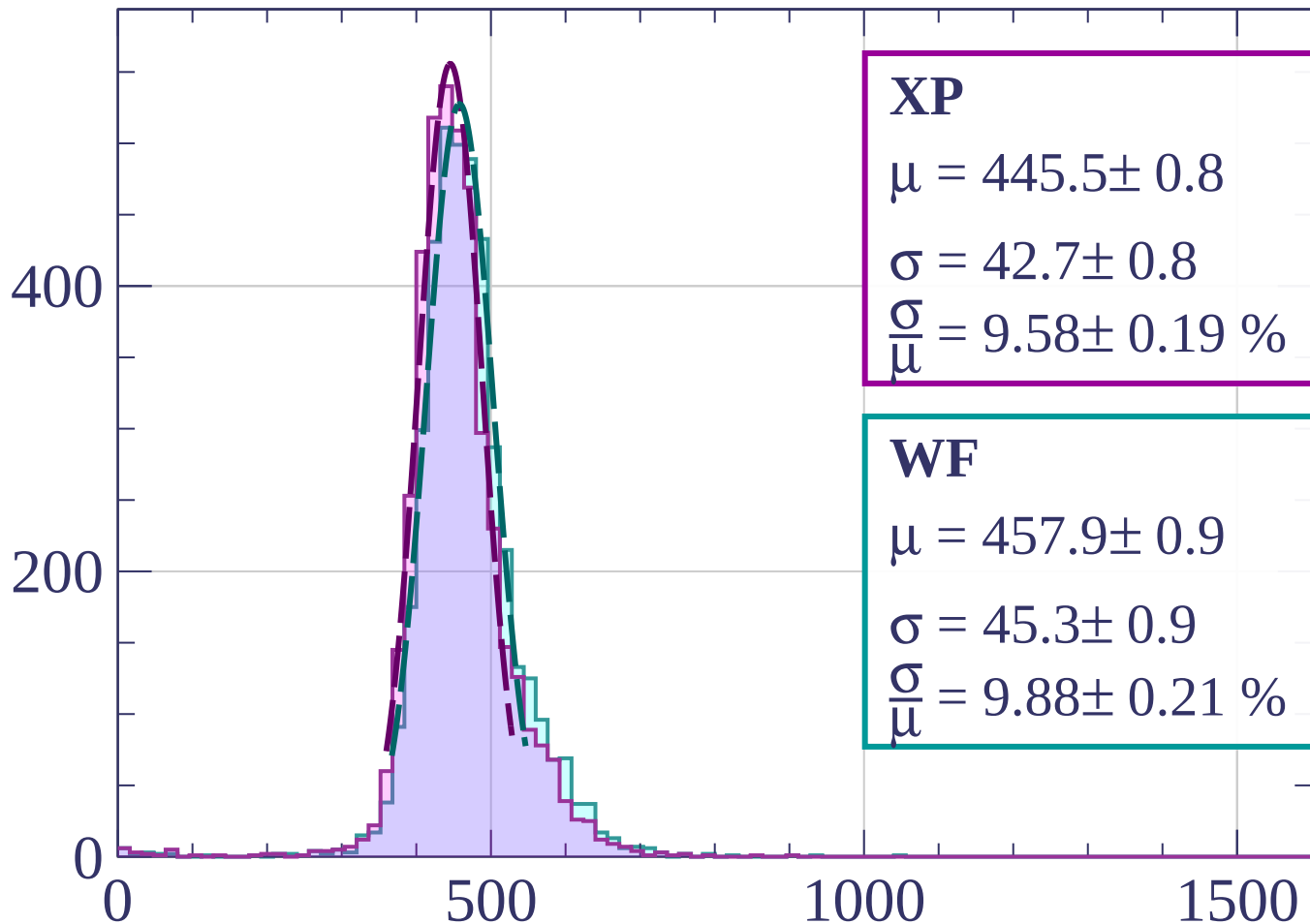
Count



dE/dx [ADC counts/cm]

Energy loss | $1240 < p < 1260$

Count



XP

$$\mu = 445.5 \pm 0.8$$

$$\sigma = 42.7 \pm 0.8$$

$$\frac{\sigma}{\mu} = 9.58 \pm 0.19 \%$$

WF

$$\mu = 457.9 \pm 0.9$$

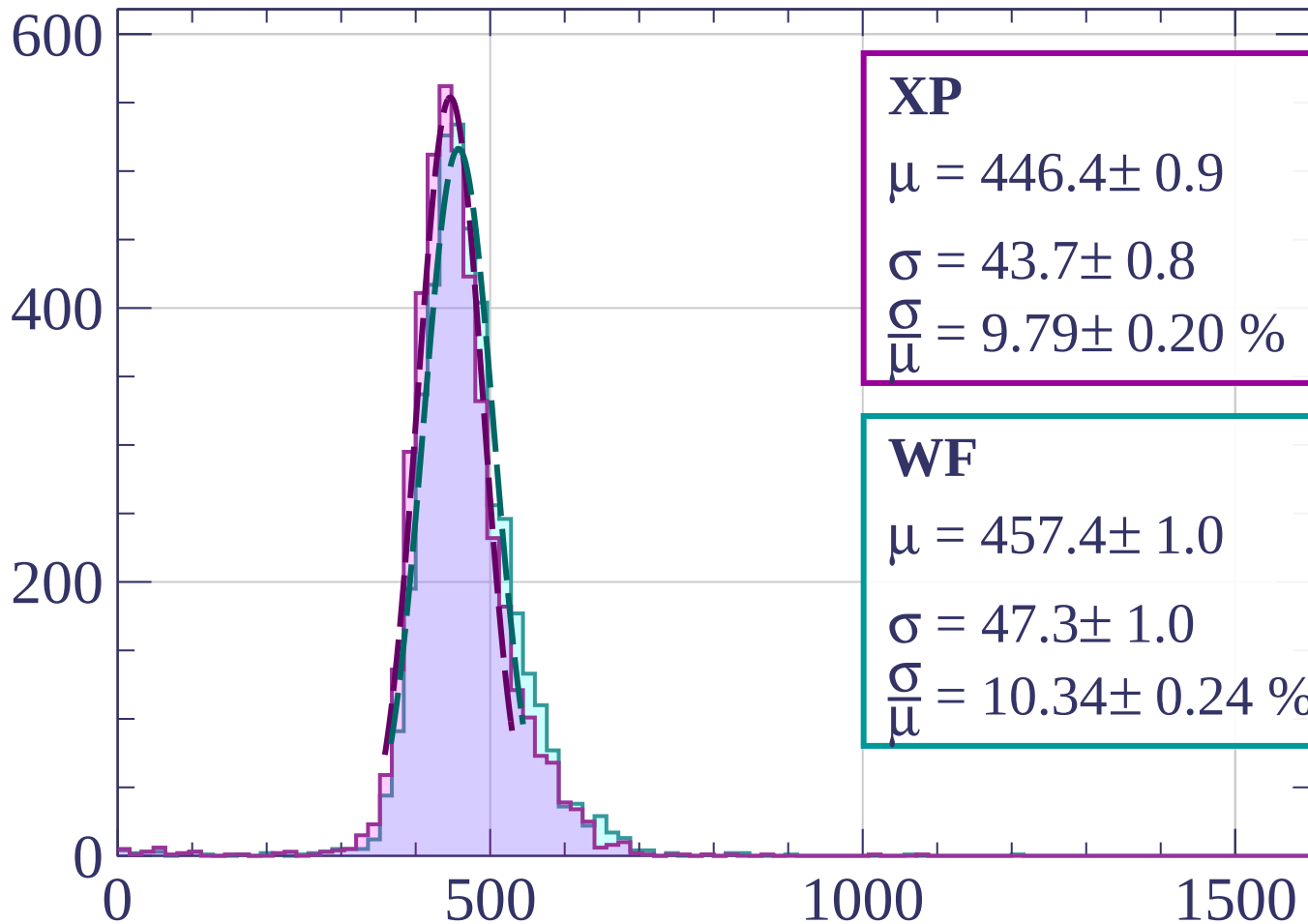
$$\sigma = 45.3 \pm 0.9$$

$$\frac{\sigma}{\mu} = 9.88 \pm 0.21 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1260 < p < 1280$

Count



XP

$$\mu = 446.4 \pm 0.9$$

$$\sigma = 43.7 \pm 0.8$$

$$\frac{\sigma}{\mu} = 9.79 \pm 0.20 \%$$

WF

$$\mu = 457.4 \pm 1.0$$

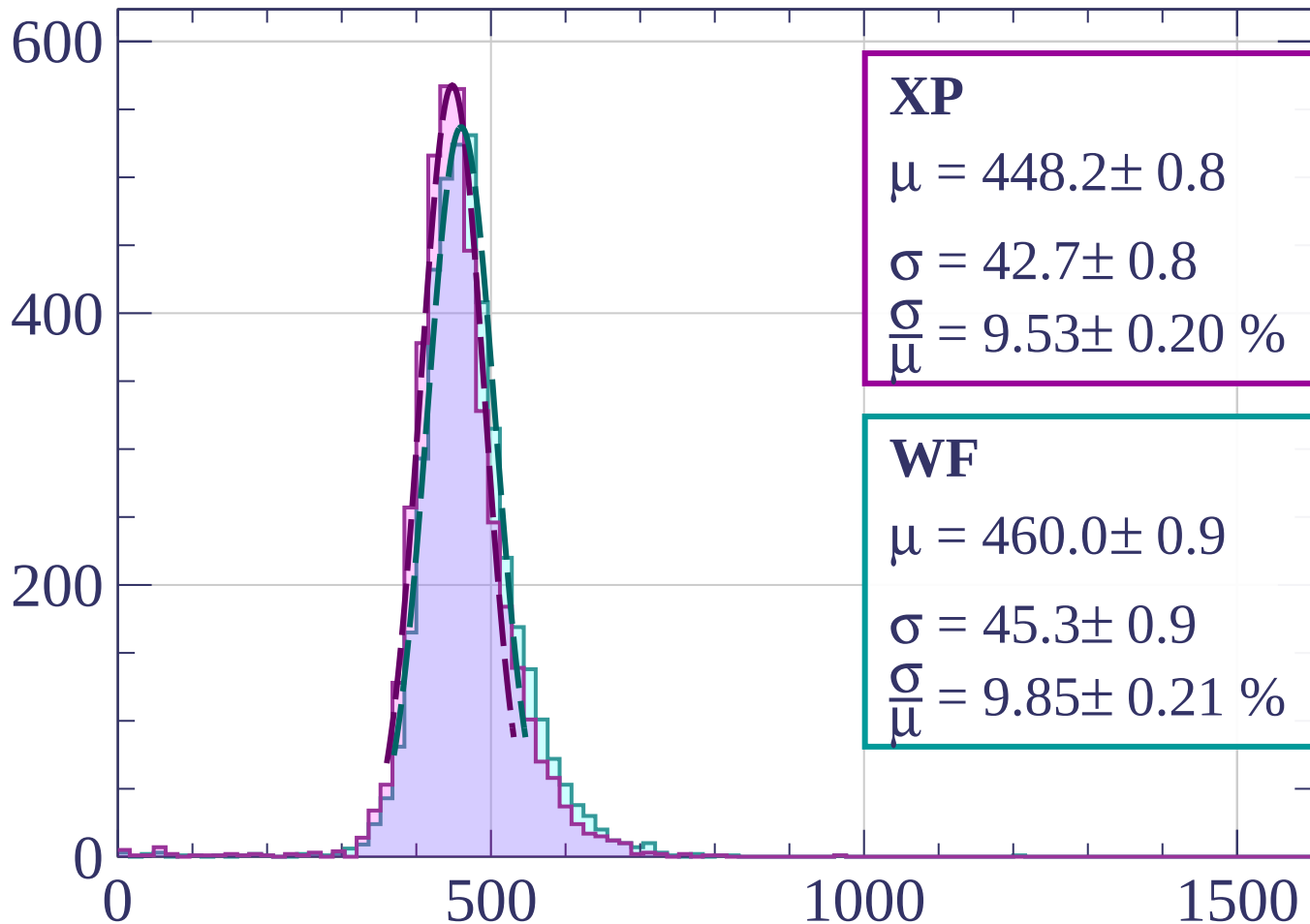
$$\sigma = 47.3 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.34 \pm 0.24 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1280 < p < 1300$

Count



XP

$$\mu = 448.2 \pm 0.8$$

$$\sigma = 42.7 \pm 0.8$$

$$\frac{\sigma}{\mu} = 9.53 \pm 0.20 \%$$

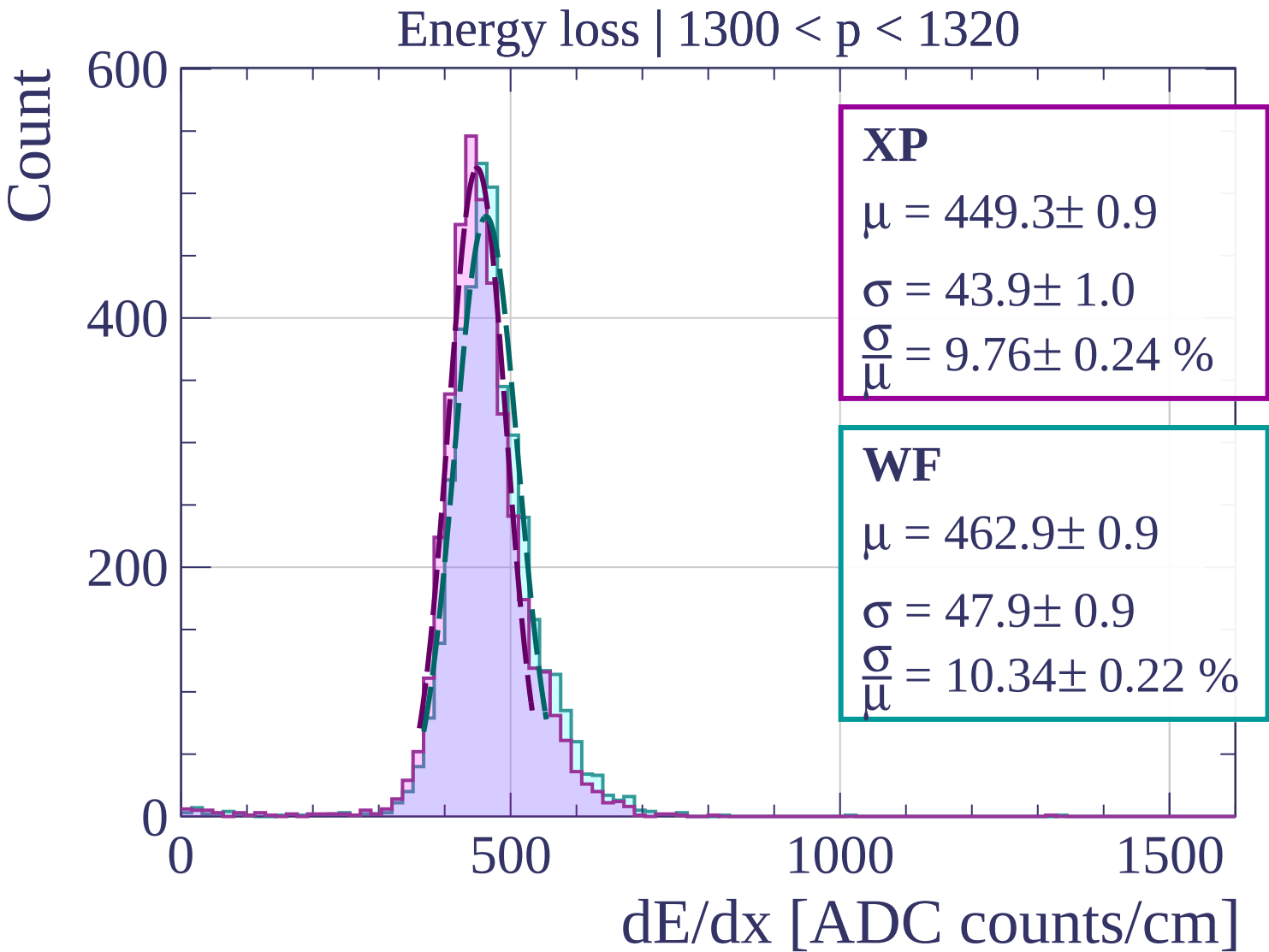
WF

$$\mu = 460.0 \pm 0.9$$

$$\sigma = 45.3 \pm 0.9$$

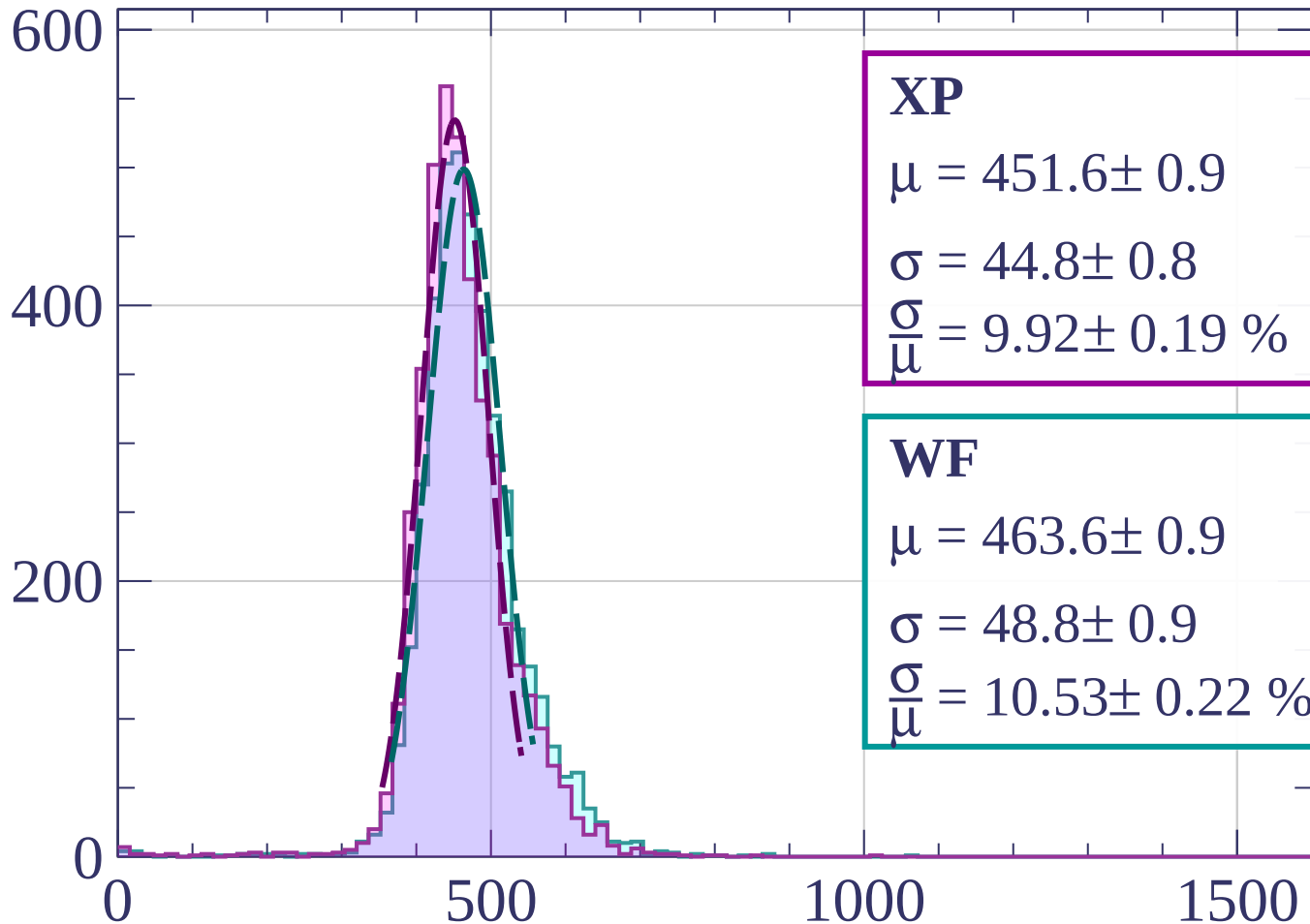
$$\frac{\sigma}{\mu} = 9.85 \pm 0.21 \%$$

dE/dx [ADC counts/cm]



Energy loss | $1320 < p < 1340$

Count



XP

$$\mu = 451.6 \pm 0.9$$

$$\sigma = 44.8 \pm 0.8$$

$$\frac{\sigma}{\mu} = 9.92 \pm 0.19 \%$$

WF

$$\mu = 463.6 \pm 0.9$$

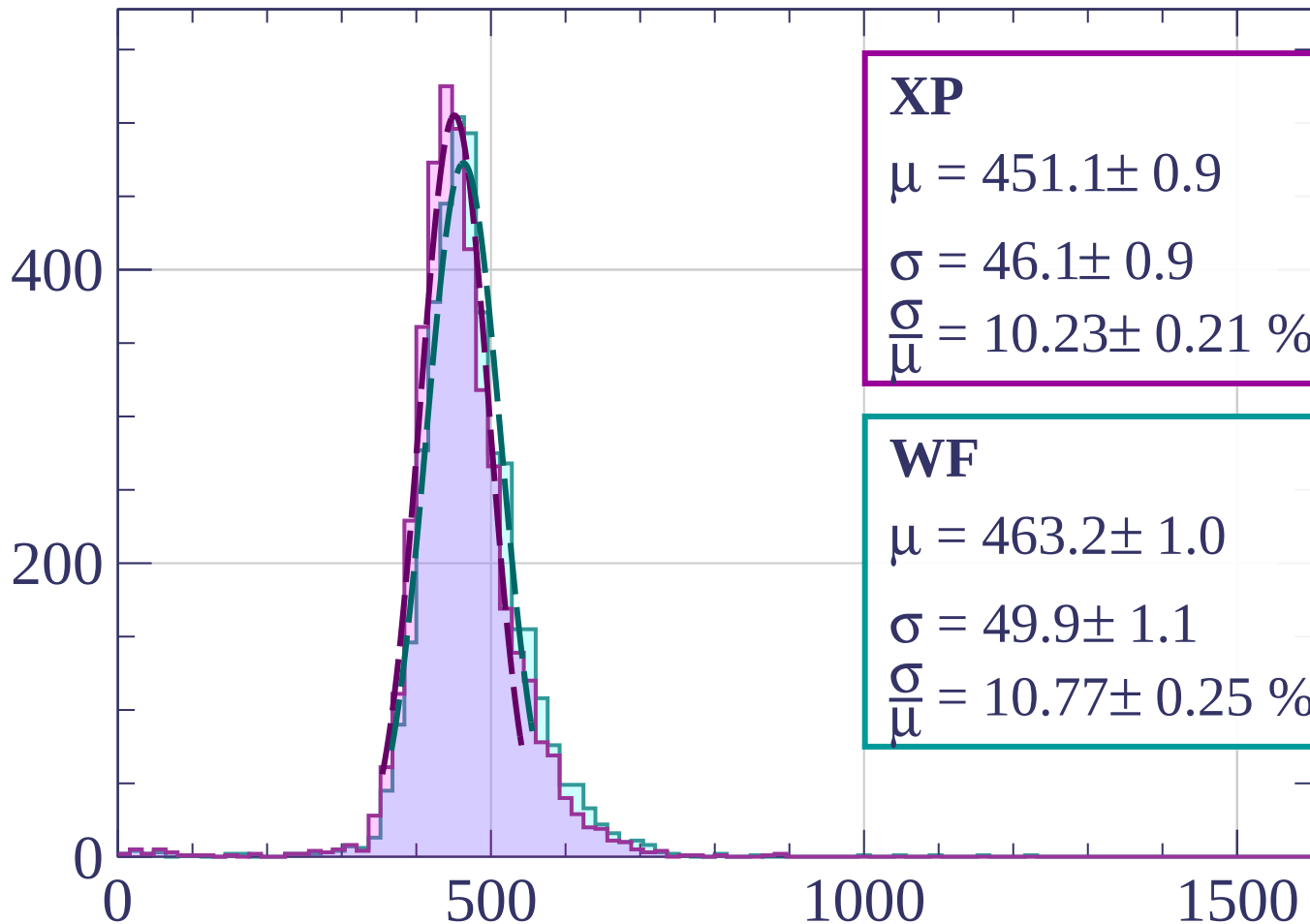
$$\sigma = 48.8 \pm 0.9$$

$$\frac{\sigma}{\mu} = 10.53 \pm 0.22 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1340 < p < 1360$

Count



XP

$$\mu = 451.1 \pm 0.9$$

$$\sigma = 46.1 \pm 0.9$$

$$\frac{\sigma}{\mu} = 10.23 \pm 0.21 \%$$

WF

$$\mu = 463.2 \pm 1.0$$

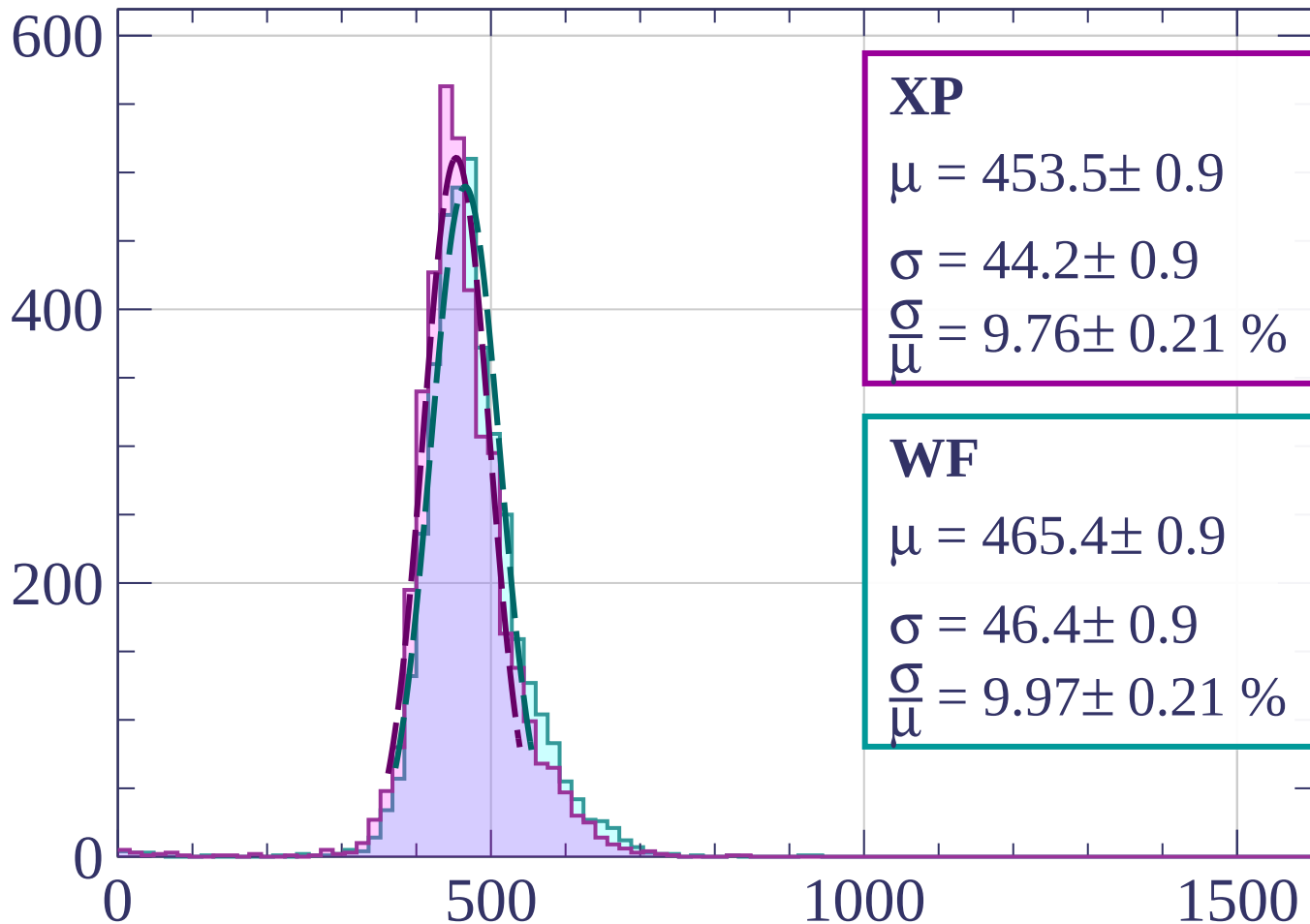
$$\sigma = 49.9 \pm 1.1$$

$$\frac{\sigma}{\mu} = 10.77 \pm 0.25 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1360 < p < 1380$

Count



XP

$$\mu = 453.5 \pm 0.9$$

$$\sigma = 44.2 \pm 0.9$$

$$\frac{\sigma}{\mu} = 9.76 \pm 0.21 \%$$

WF

$$\mu = 465.4 \pm 0.9$$

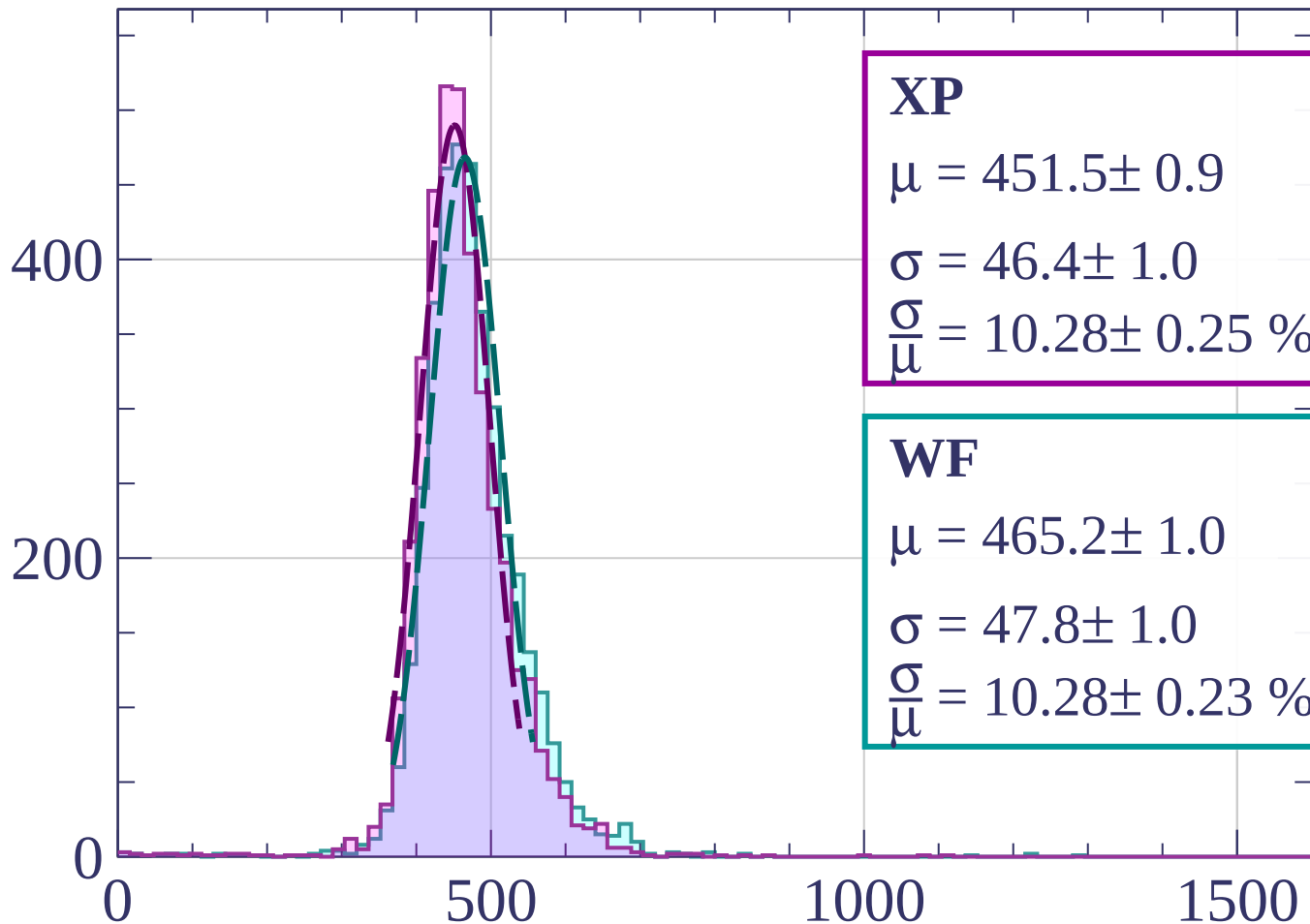
$$\sigma = 46.4 \pm 0.9$$

$$\frac{\sigma}{\mu} = 9.97 \pm 0.21 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1380 < p < 1400$

Count



XP

$$\mu = 451.5 \pm 0.9$$

$$\sigma = 46.4 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.28 \pm 0.25 \%$$

WF

$$\mu = 465.2 \pm 1.0$$

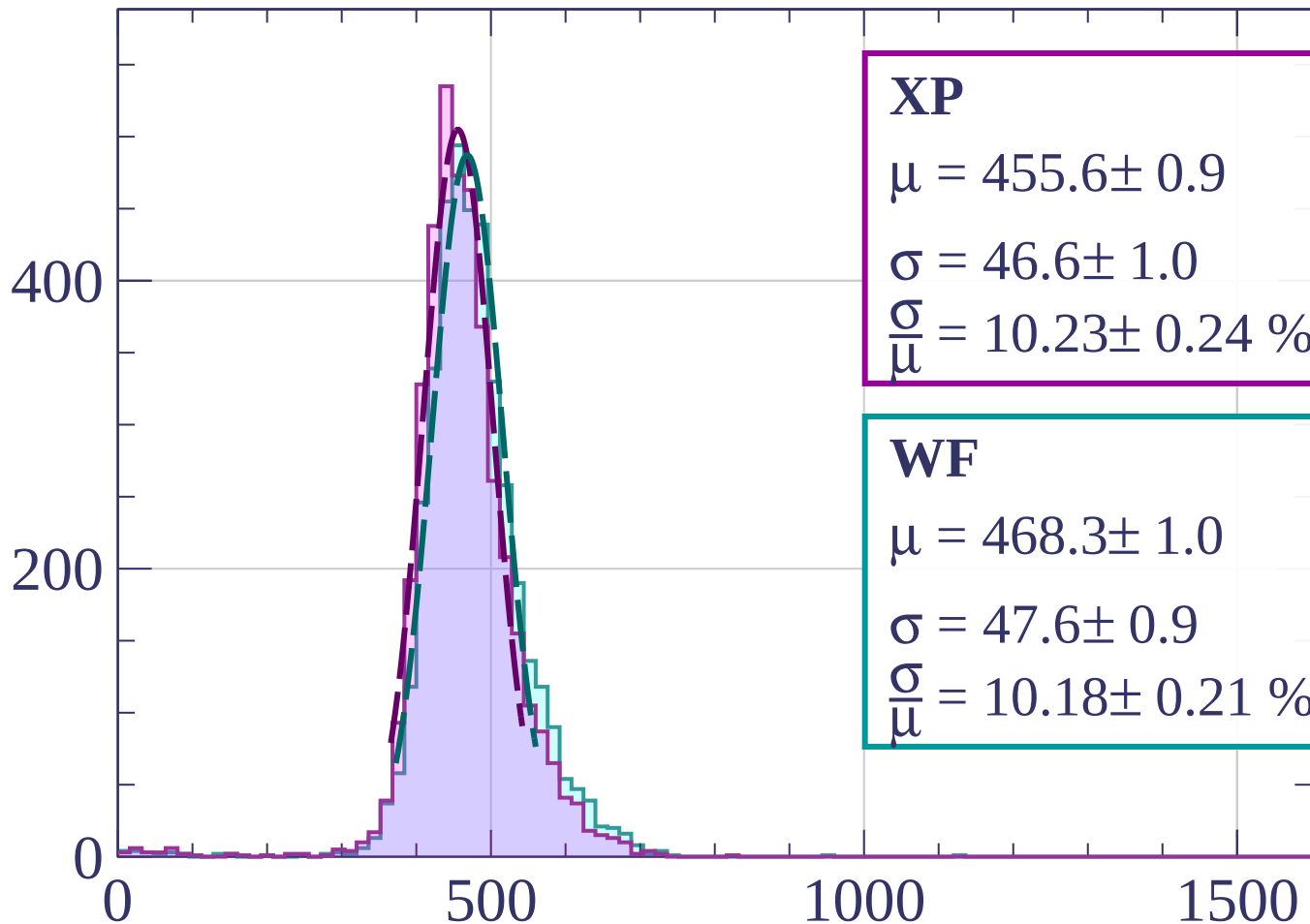
$$\sigma = 47.8 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.28 \pm 0.23 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1400 < p < 1420$

Count



XP

$$\mu = 455.6 \pm 0.9$$

$$\sigma = 46.6 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.23 \pm 0.24 \%$$

WF

$$\mu = 468.3 \pm 1.0$$

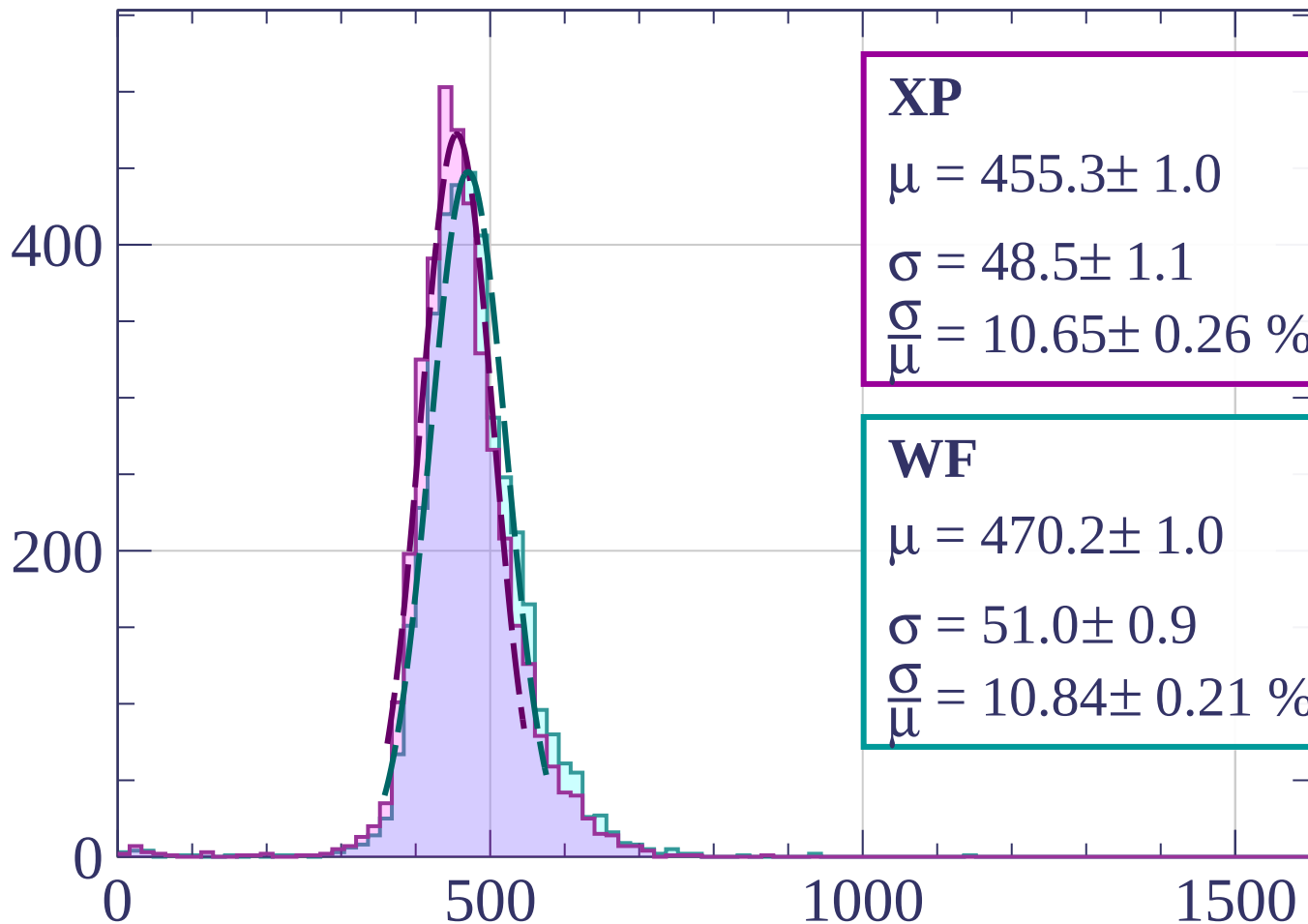
$$\sigma = 47.6 \pm 0.9$$

$$\frac{\sigma}{\mu} = 10.18 \pm 0.21 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1420 < p < 1440$

Count



XP

$$\mu = 455.3 \pm 1.0$$

$$\sigma = 48.5 \pm 1.1$$

$$\frac{\sigma}{\mu} = 10.65 \pm 0.26 \%$$

WF

$$\mu = 470.2 \pm 1.0$$

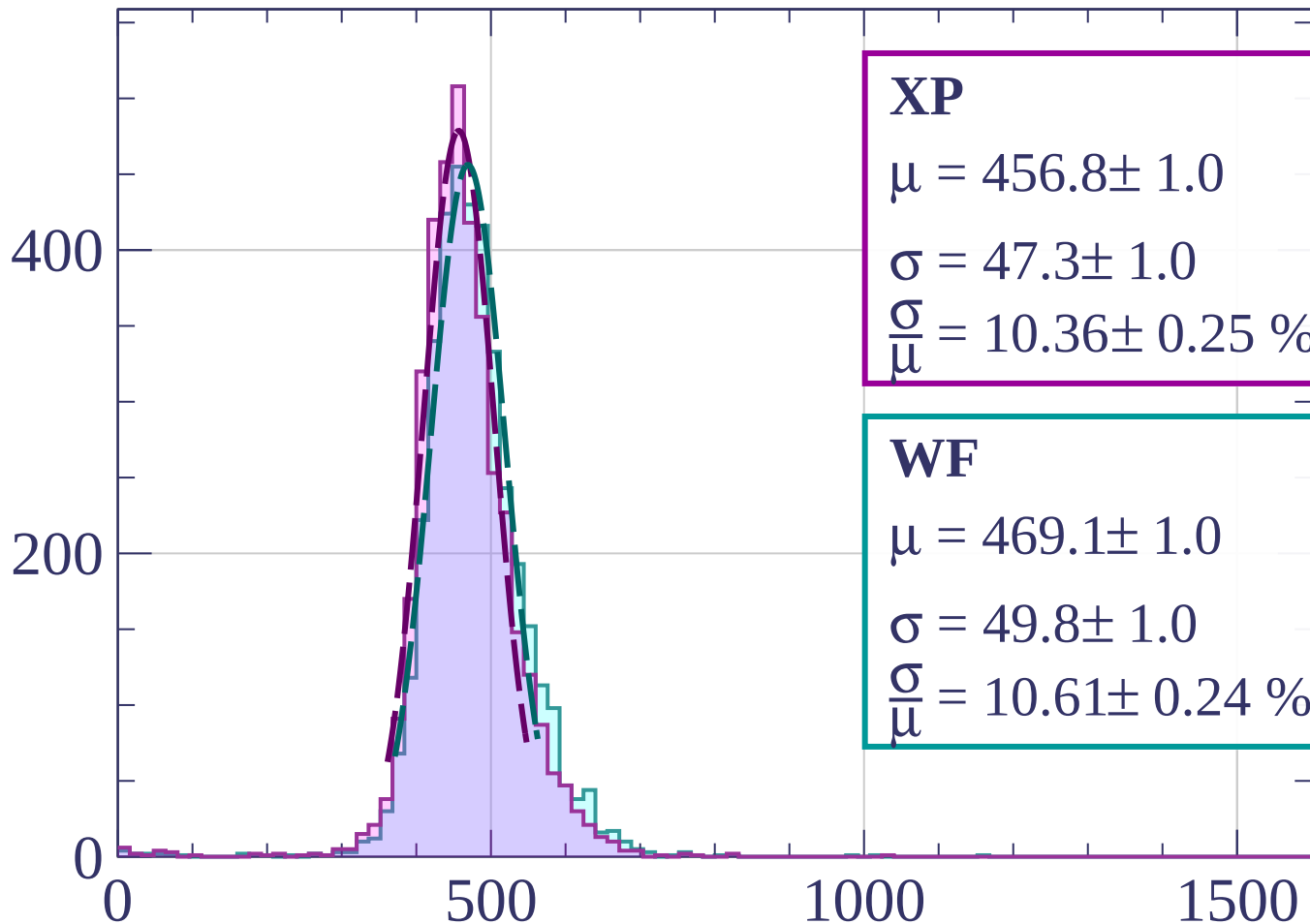
$$\sigma = 51.0 \pm 0.9$$

$$\frac{\sigma}{\mu} = 10.84 \pm 0.21 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1440 < p < 1460$

Count



XP

$$\mu = 456.8 \pm 1.0$$

$$\sigma = 47.3 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.36 \pm 0.25 \%$$

WF

$$\mu = 469.1 \pm 1.0$$

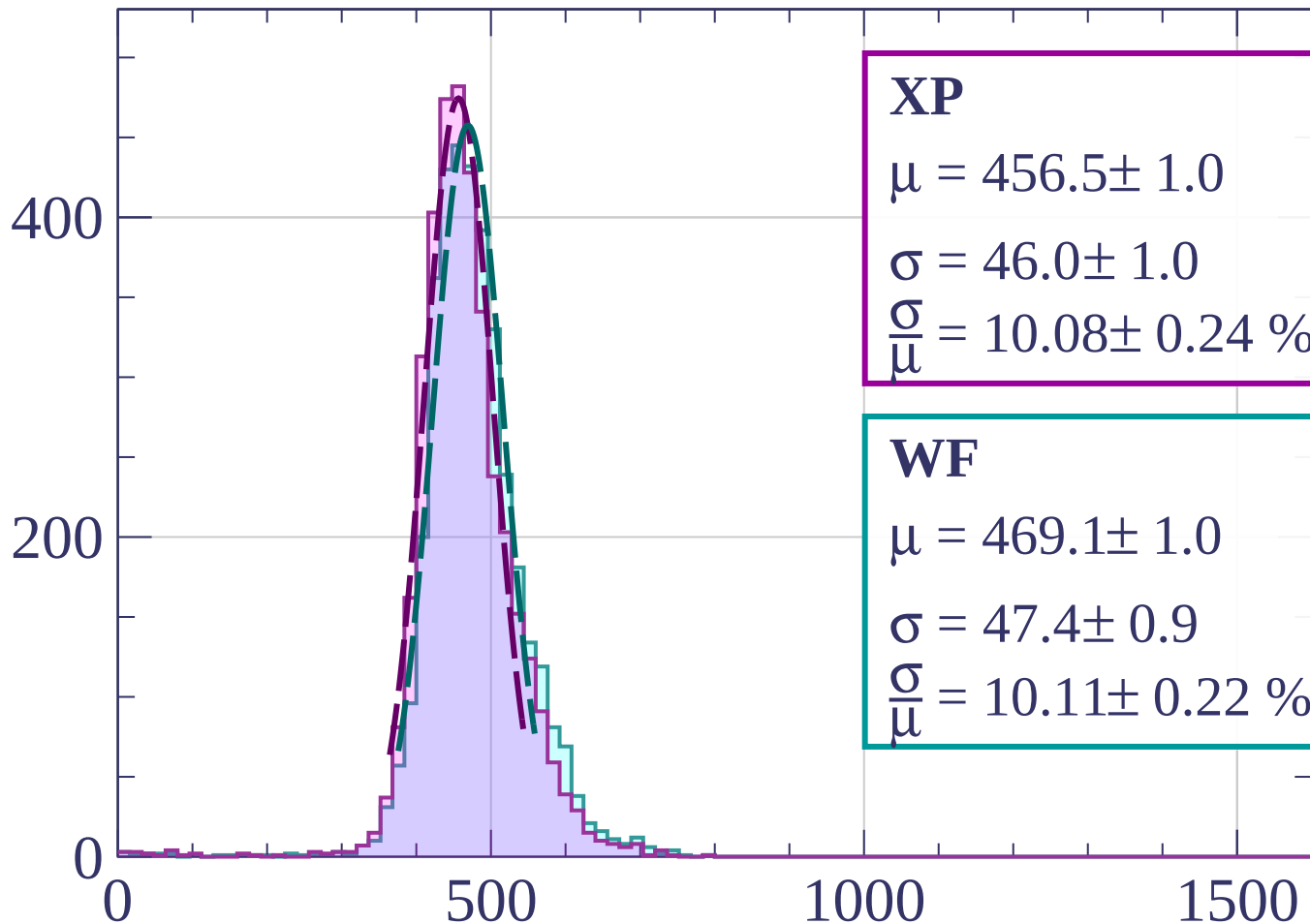
$$\sigma = 49.8 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.61 \pm 0.24 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1460 < p < 1480$

Count



XP

$$\mu = 456.5 \pm 1.0$$

$$\sigma = 46.0 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.08 \pm 0.24 \%$$

WF

$$\mu = 469.1 \pm 1.0$$

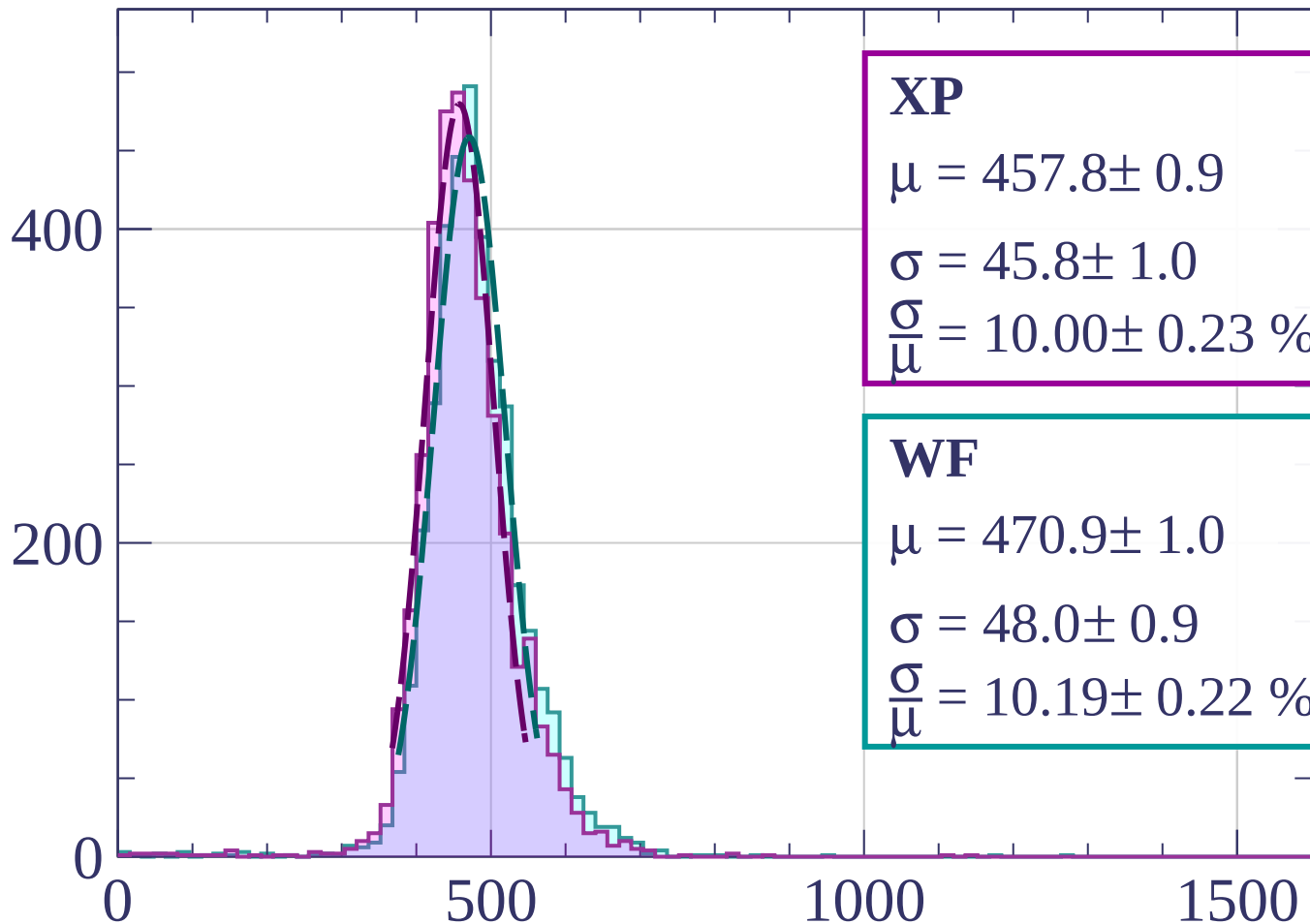
$$\sigma = 47.4 \pm 0.9$$

$$\frac{\sigma}{\mu} = 10.11 \pm 0.22 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1480 < p < 1500$

Count



XP

$$\mu = 457.8 \pm 0.9$$

$$\sigma = 45.8 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.00 \pm 0.23 \%$$

WF

$$\mu = 470.9 \pm 1.0$$

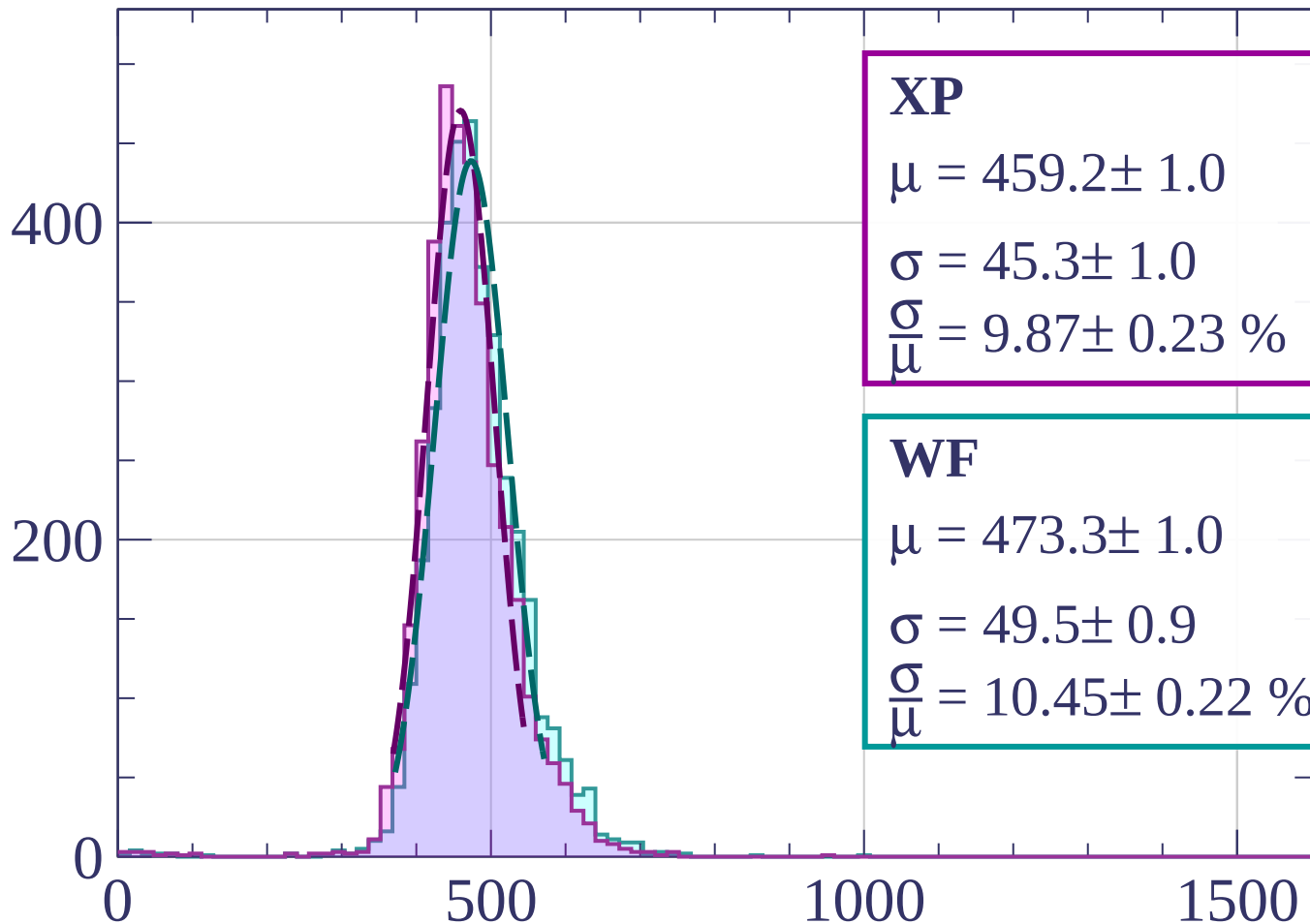
$$\sigma = 48.0 \pm 0.9$$

$$\frac{\sigma}{\mu} = 10.19 \pm 0.22 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1500 < p < 1520$

Count



XP

$$\mu = 459.2 \pm 1.0$$

$$\sigma = 45.3 \pm 1.0$$

$$\frac{\sigma}{\mu} = 9.87 \pm 0.23 \%$$

WF

$$\mu = 473.3 \pm 1.0$$

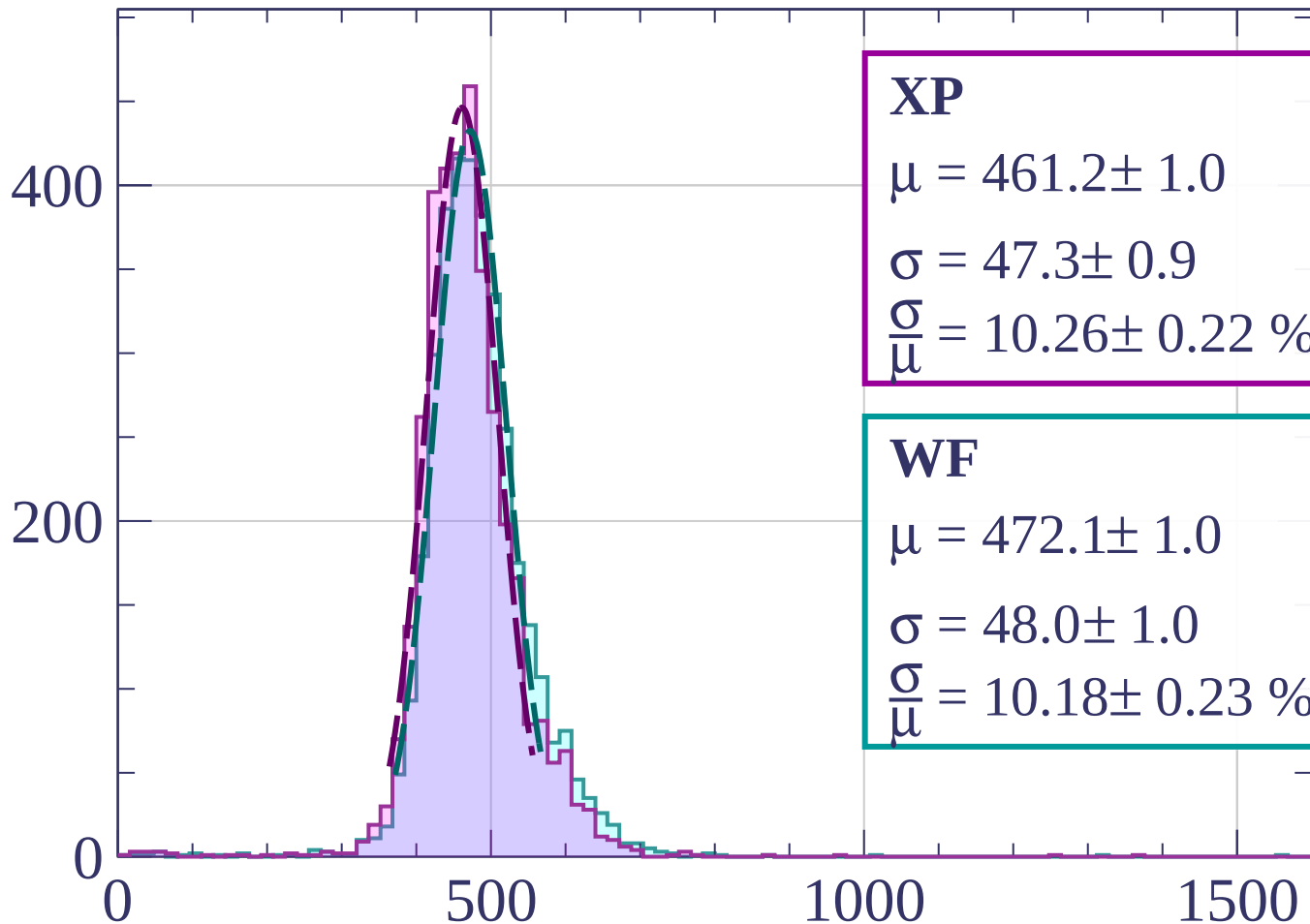
$$\sigma = 49.5 \pm 0.9$$

$$\frac{\sigma}{\mu} = 10.45 \pm 0.22 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1520 < p < 1540$

Count



XP

$$\mu = 461.2 \pm 1.0$$

$$\sigma = 47.3 \pm 0.9$$

$$\frac{\sigma}{\mu} = 10.26 \pm 0.22 \%$$

WF

$$\mu = 472.1 \pm 1.0$$

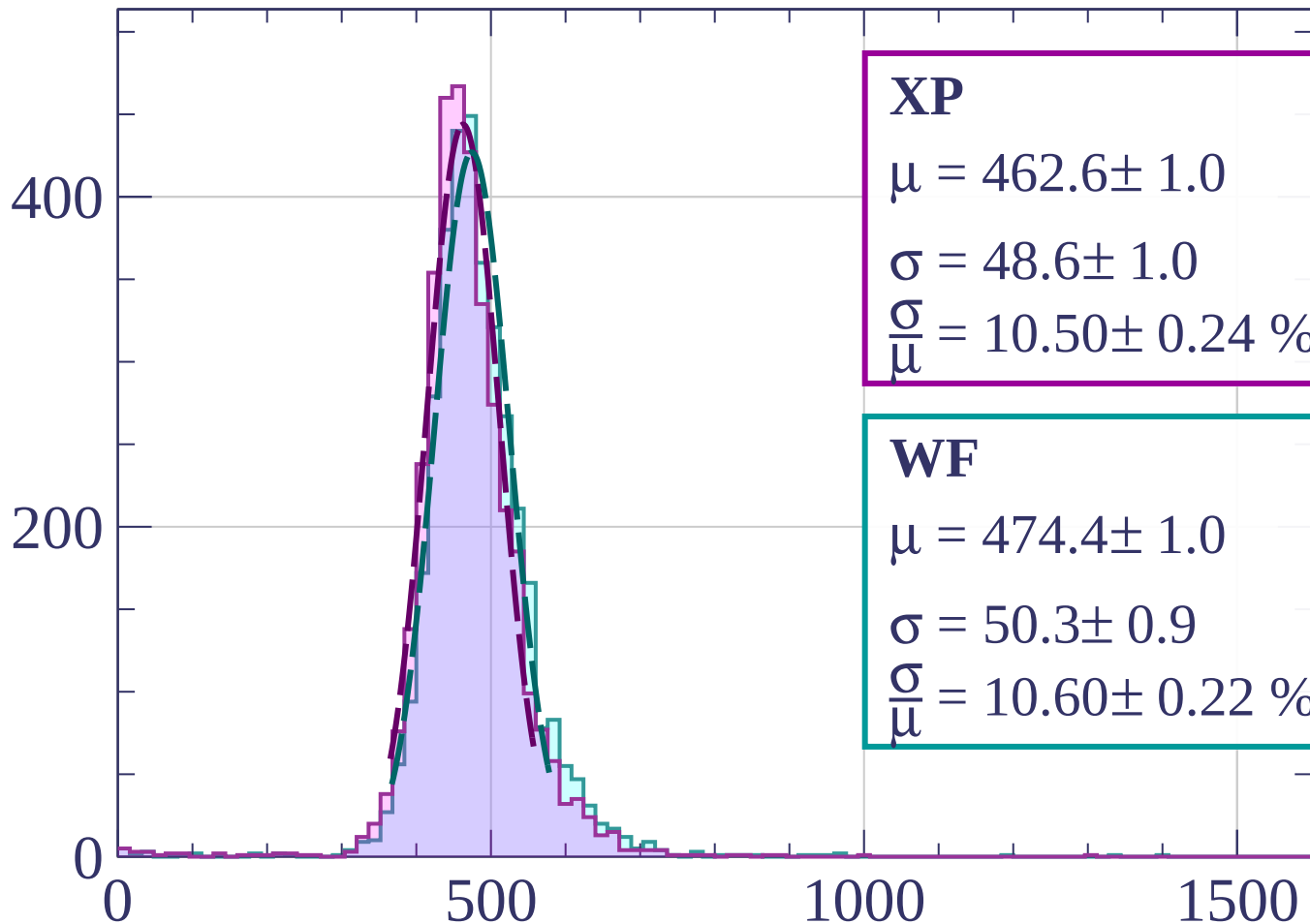
$$\sigma = 48.0 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.18 \pm 0.23 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1540 < p < 1560$

Count



XP

$$\mu = 462.6 \pm 1.0$$

$$\sigma = 48.6 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.50 \pm 0.24 \%$$

WF

$$\mu = 474.4 \pm 1.0$$

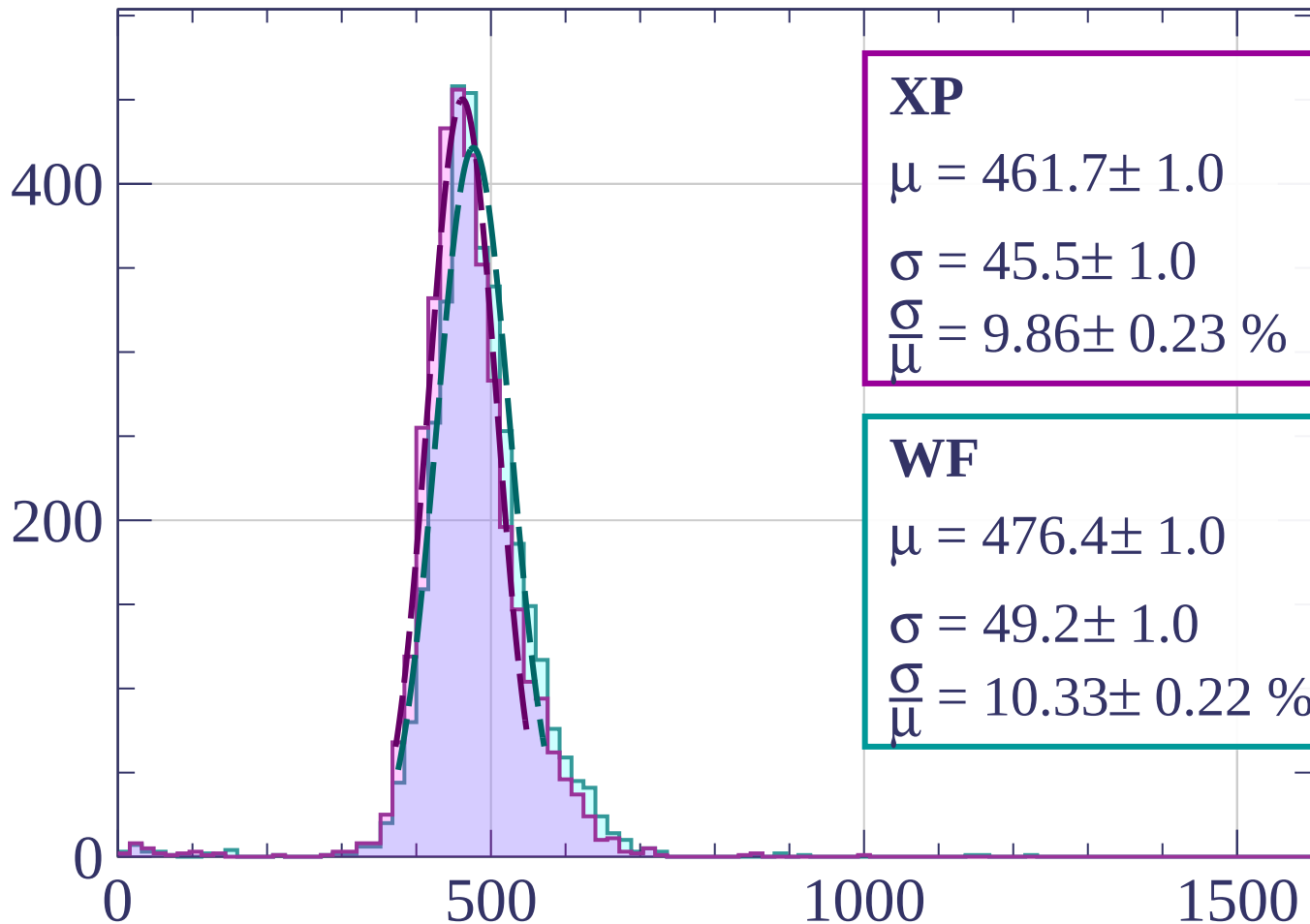
$$\sigma = 50.3 \pm 0.9$$

$$\frac{\sigma}{\mu} = 10.60 \pm 0.22 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1560 < p < 1580$

Count



XP

$$\mu = 461.7 \pm 1.0$$

$$\sigma = 45.5 \pm 1.0$$

$$\frac{\sigma}{\mu} = 9.86 \pm 0.23 \%$$

WF

$$\mu = 476.4 \pm 1.0$$

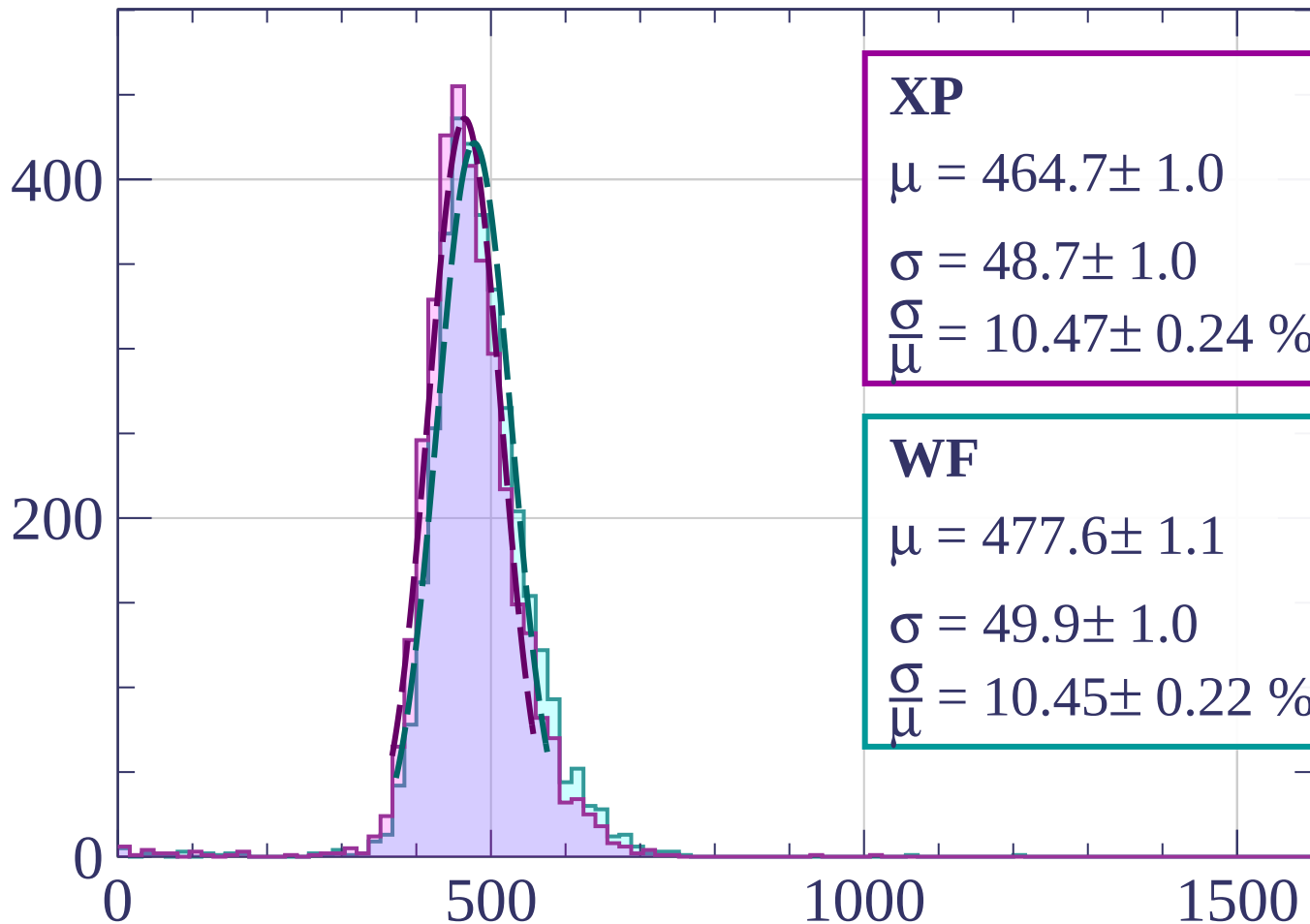
$$\sigma = 49.2 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.33 \pm 0.22 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1580 < p < 1600$

Count



XP

$$\mu = 464.7 \pm 1.0$$

$$\sigma = 48.7 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.47 \pm 0.24 \%$$

WF

$$\mu = 477.6 \pm 1.1$$

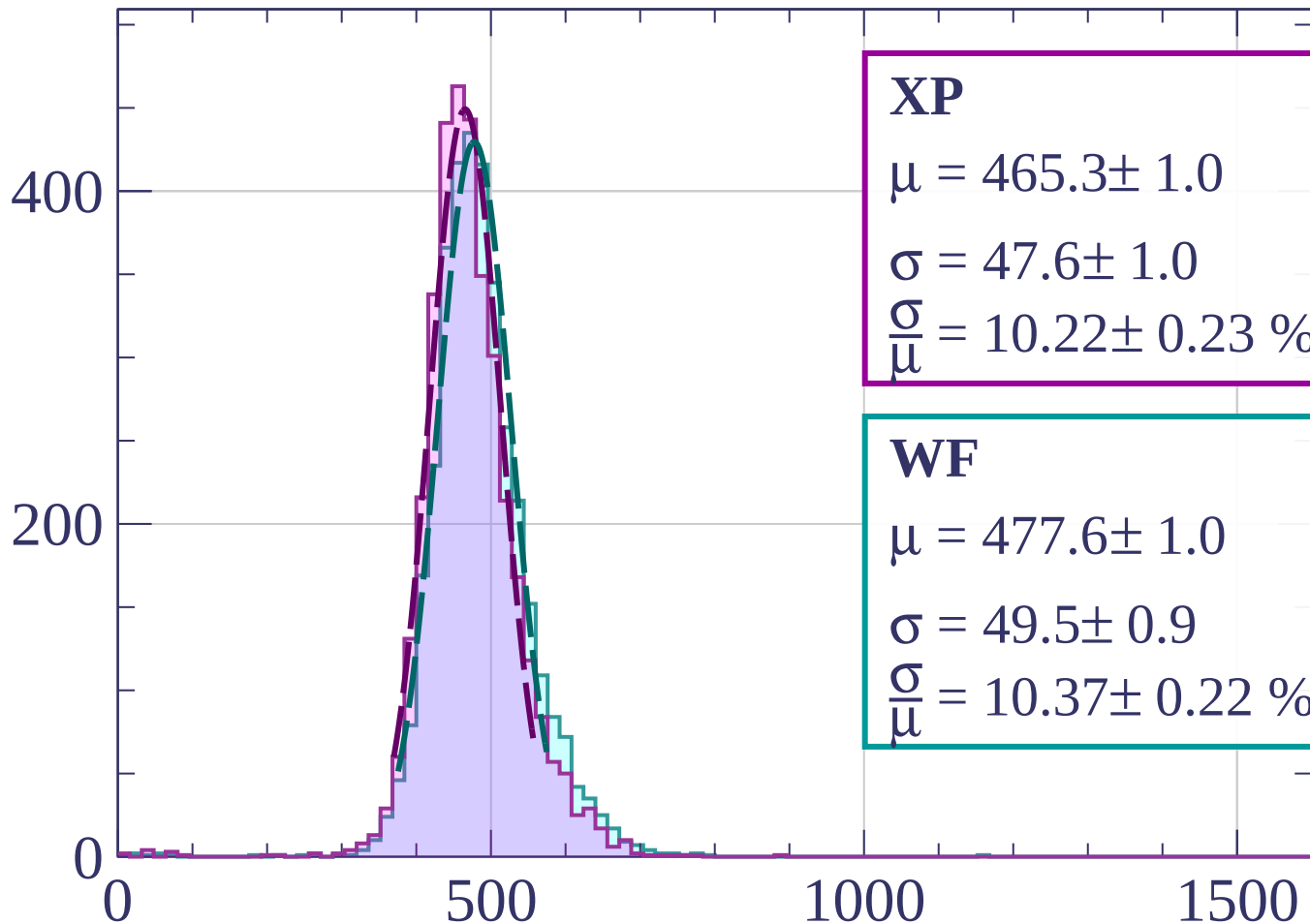
$$\sigma = 49.9 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.45 \pm 0.22 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1600 < p < 1620$

Count



XP

$$\mu = 465.3 \pm 1.0$$

$$\sigma = 47.6 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.22 \pm 0.23 \%$$

WF

$$\mu = 477.6 \pm 1.0$$

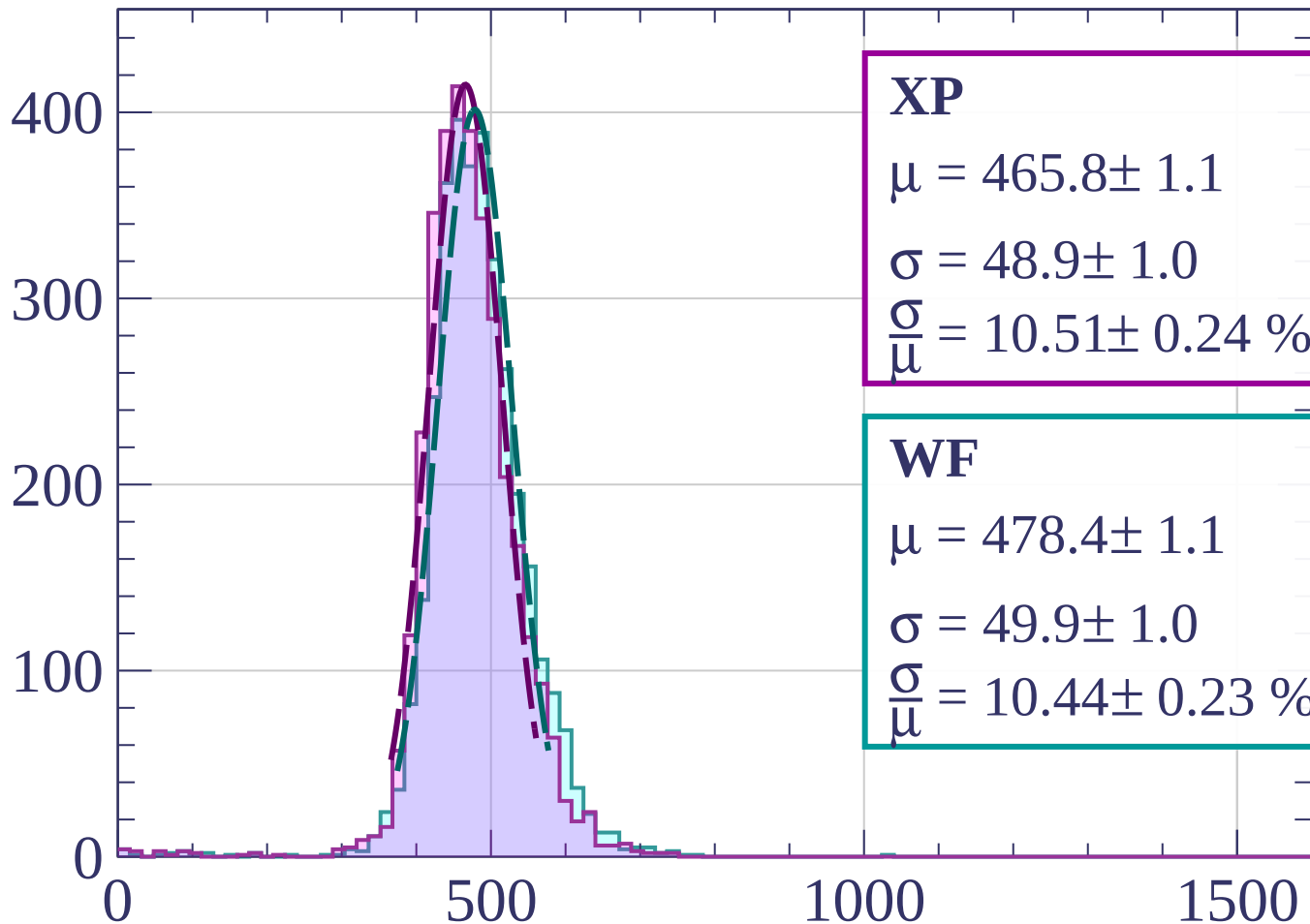
$$\sigma = 49.5 \pm 0.9$$

$$\frac{\sigma}{\mu} = 10.37 \pm 0.22 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1620 < p < 1640$

Count



XP

$$\mu = 465.8 \pm 1.1$$

$$\sigma = 48.9 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.51 \pm 0.24 \%$$

WF

$$\mu = 478.4 \pm 1.1$$

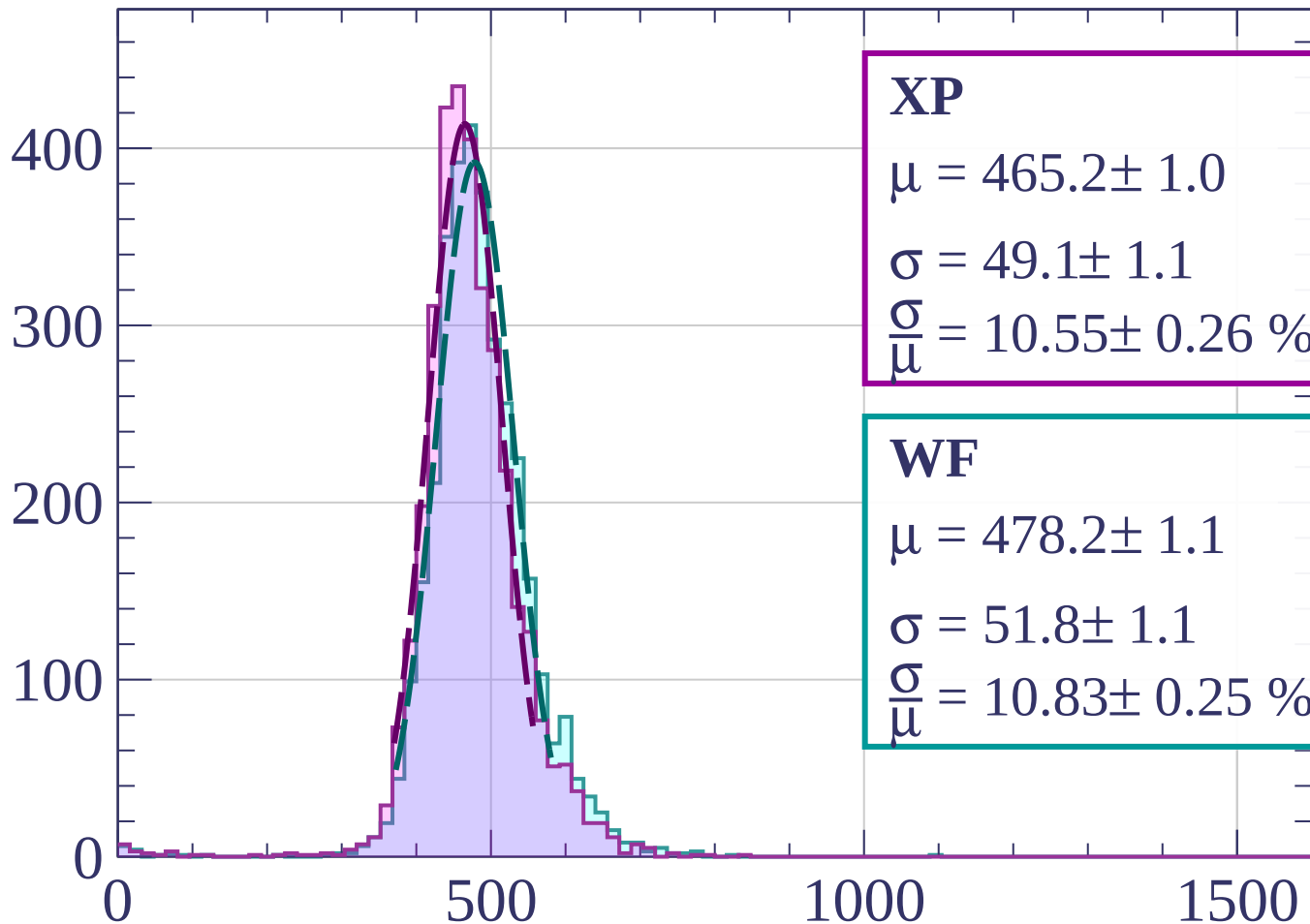
$$\sigma = 49.9 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.44 \pm 0.23 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1640 < p < 1660$

Count



XP

$$\mu = 465.2 \pm 1.0$$

$$\sigma = 49.1 \pm 1.1$$

$$\frac{\sigma}{\mu} = 10.55 \pm 0.26 \%$$

WF

$$\mu = 478.2 \pm 1.1$$

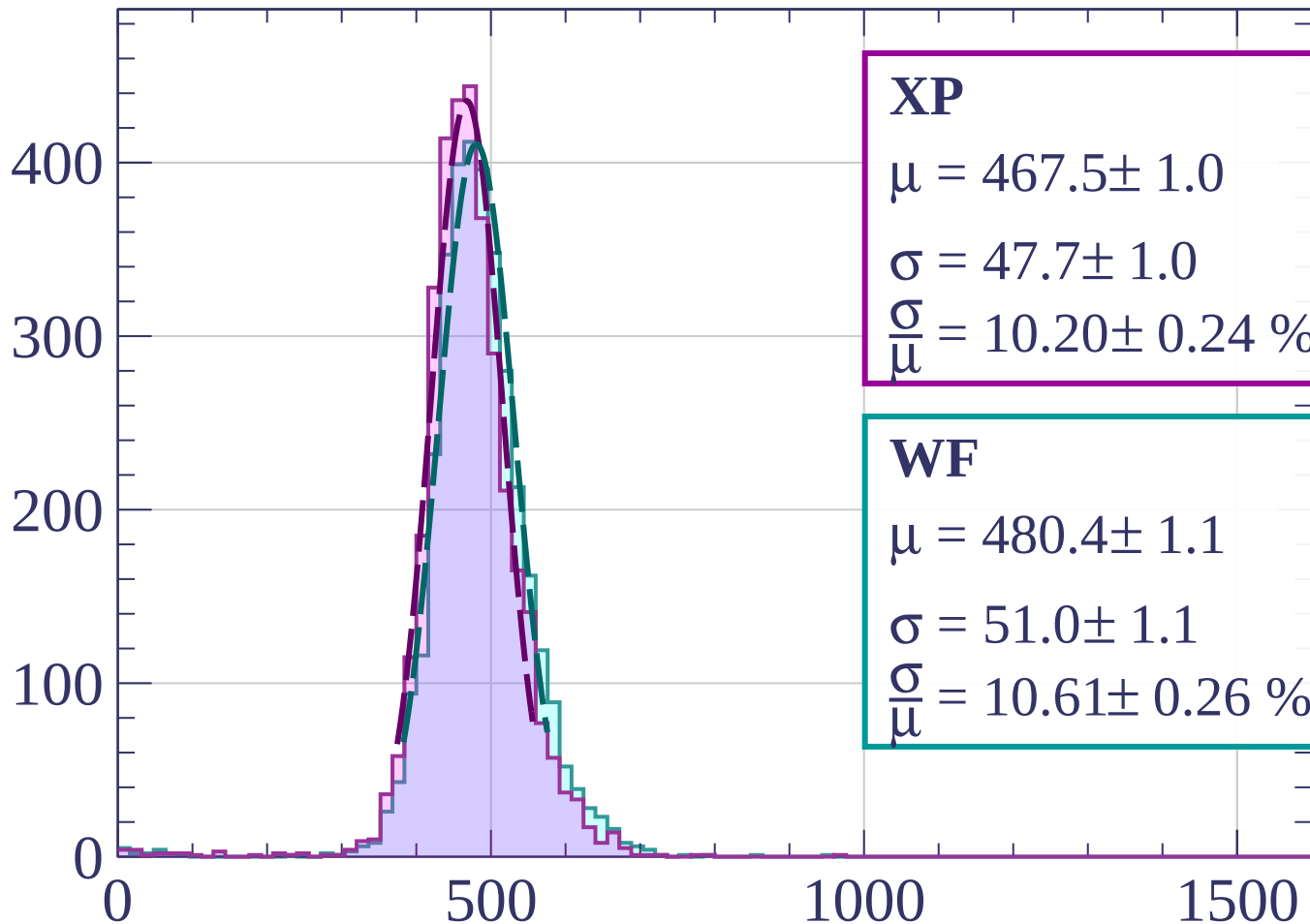
$$\sigma = 51.8 \pm 1.1$$

$$\frac{\sigma}{\mu} = 10.83 \pm 0.25 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1660 < p < 1680$

Count



XP

$$\mu = 467.5 \pm 1.0$$

$$\sigma = 47.7 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.20 \pm 0.24 \%$$

WF

$$\mu = 480.4 \pm 1.1$$

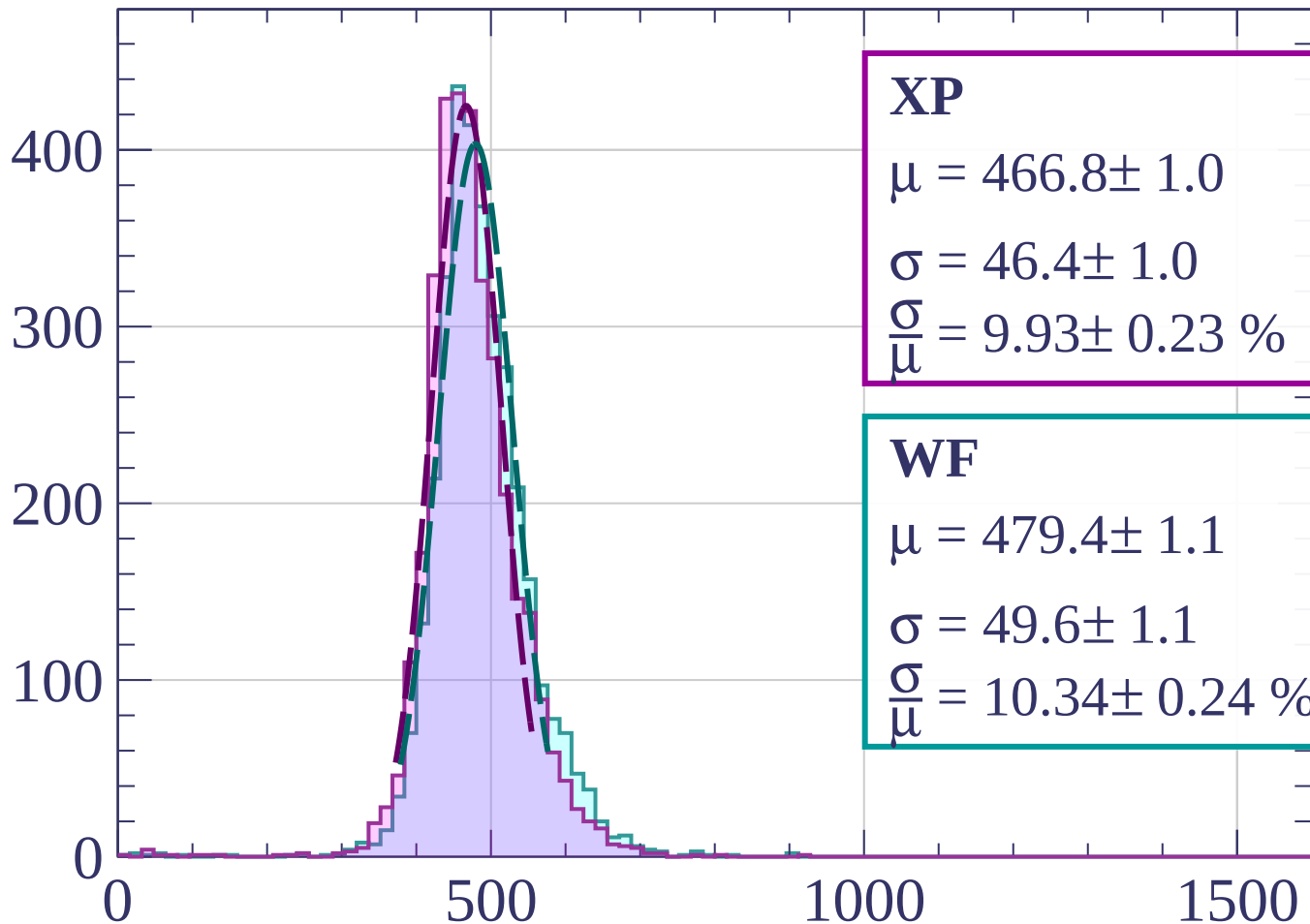
$$\sigma = 51.0 \pm 1.1$$

$$\frac{\sigma}{\mu} = 10.61 \pm 0.26 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1680 < p < 1700$

Count



XP

$$\mu = 466.8 \pm 1.0$$

$$\sigma = 46.4 \pm 1.0$$

$$\frac{\sigma}{\mu} = 9.93 \pm 0.23 \%$$

WF

$$\mu = 479.4 \pm 1.1$$

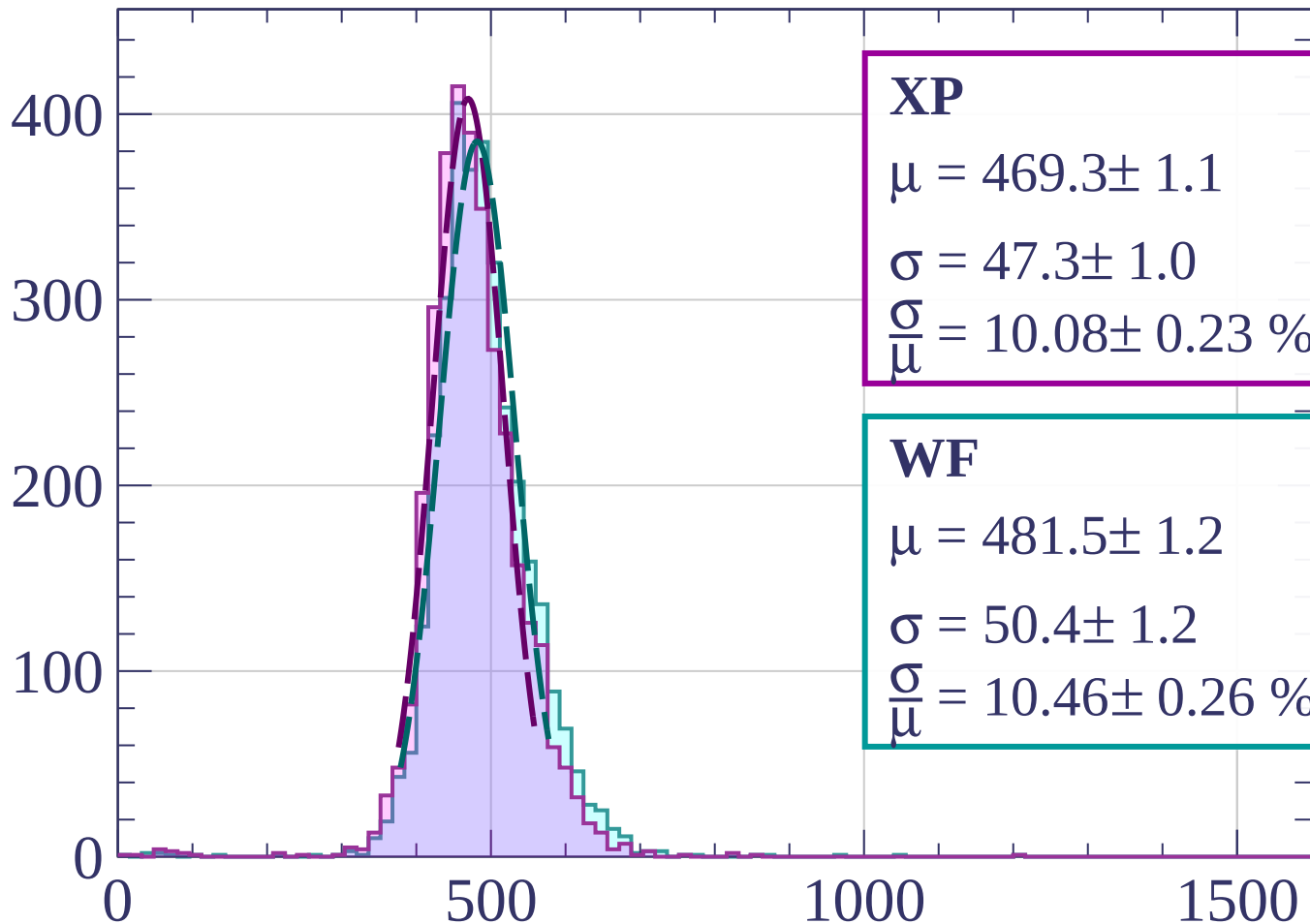
$$\sigma = 49.6 \pm 1.1$$

$$\frac{\sigma}{\mu} = 10.34 \pm 0.24 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1700 < p < 1720$

Count



XP

$$\mu = 469.3 \pm 1.1$$

$$\sigma = 47.3 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.08 \pm 0.23 \%$$

WF

$$\mu = 481.5 \pm 1.2$$

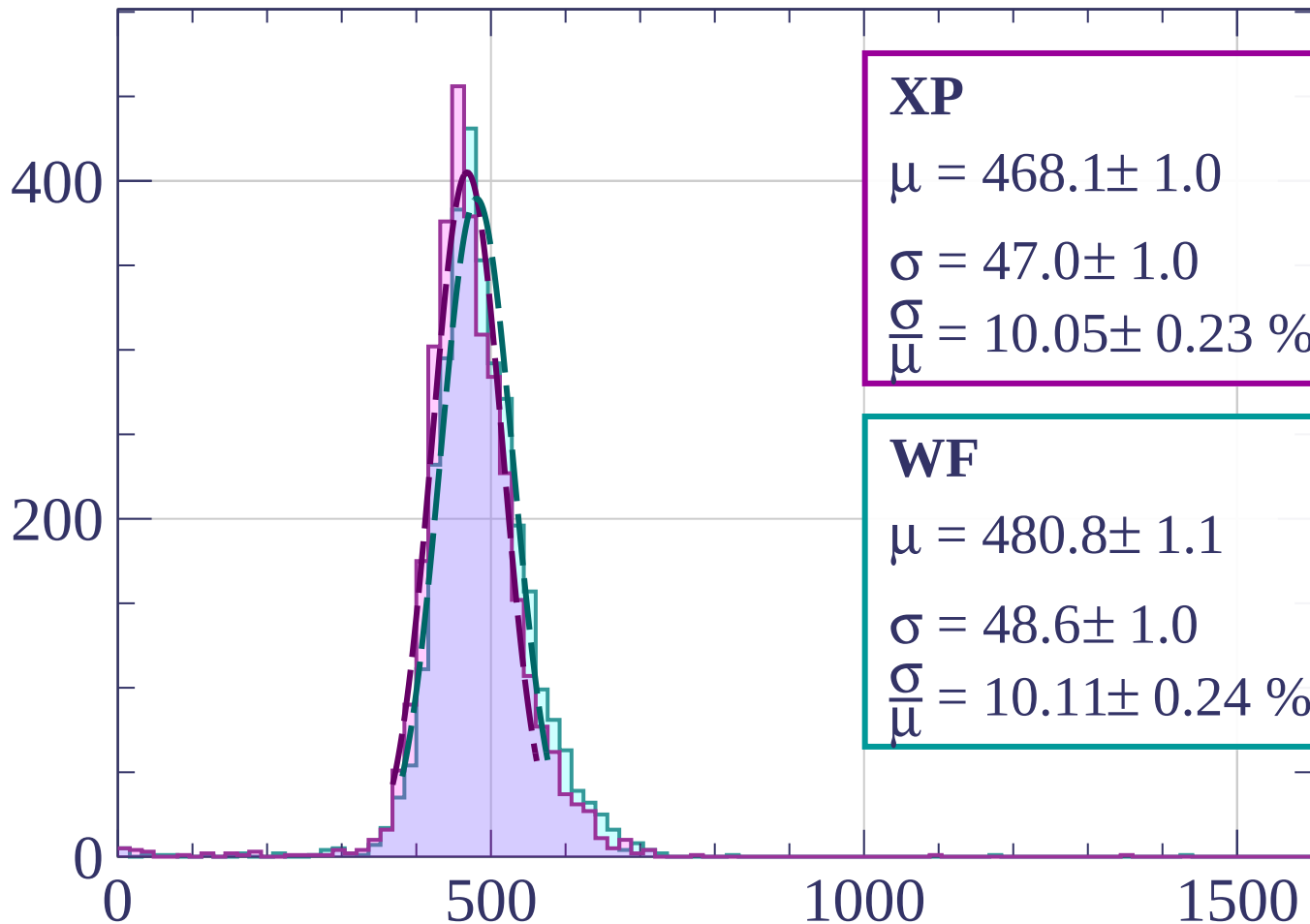
$$\sigma = 50.4 \pm 1.2$$

$$\frac{\sigma}{\mu} = 10.46 \pm 0.26 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1720 < p < 1740$

Count



XP

$$\mu = 468.1 \pm 1.0$$

$$\sigma = 47.0 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.05 \pm 0.23 \%$$

WF

$$\mu = 480.8 \pm 1.1$$

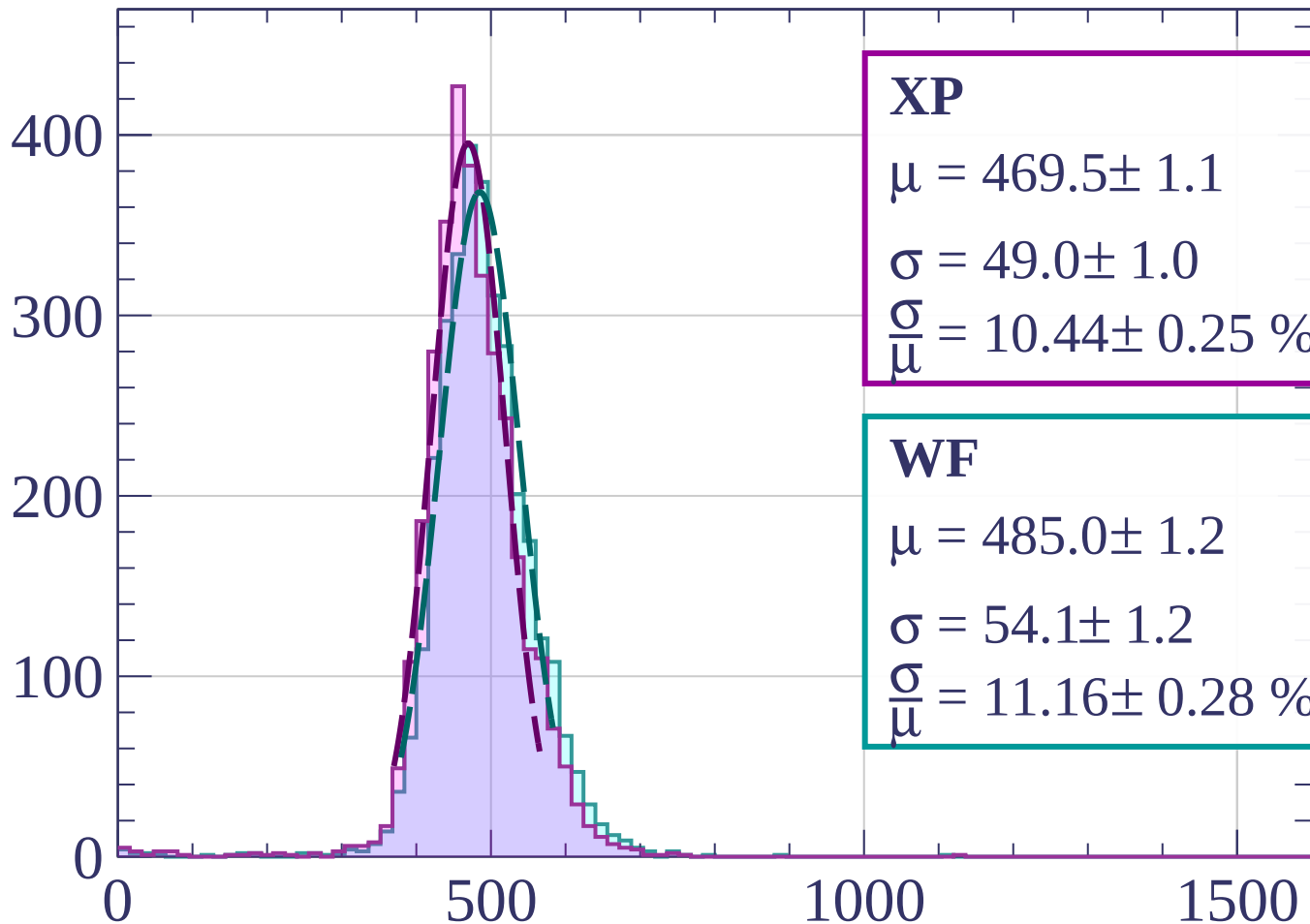
$$\sigma = 48.6 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.11 \pm 0.24 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1740 < p < 1760$

Count



XP

$$\mu = 469.5 \pm 1.1$$

$$\sigma = 49.0 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.44 \pm 0.25 \%$$

WF

$$\mu = 485.0 \pm 1.2$$

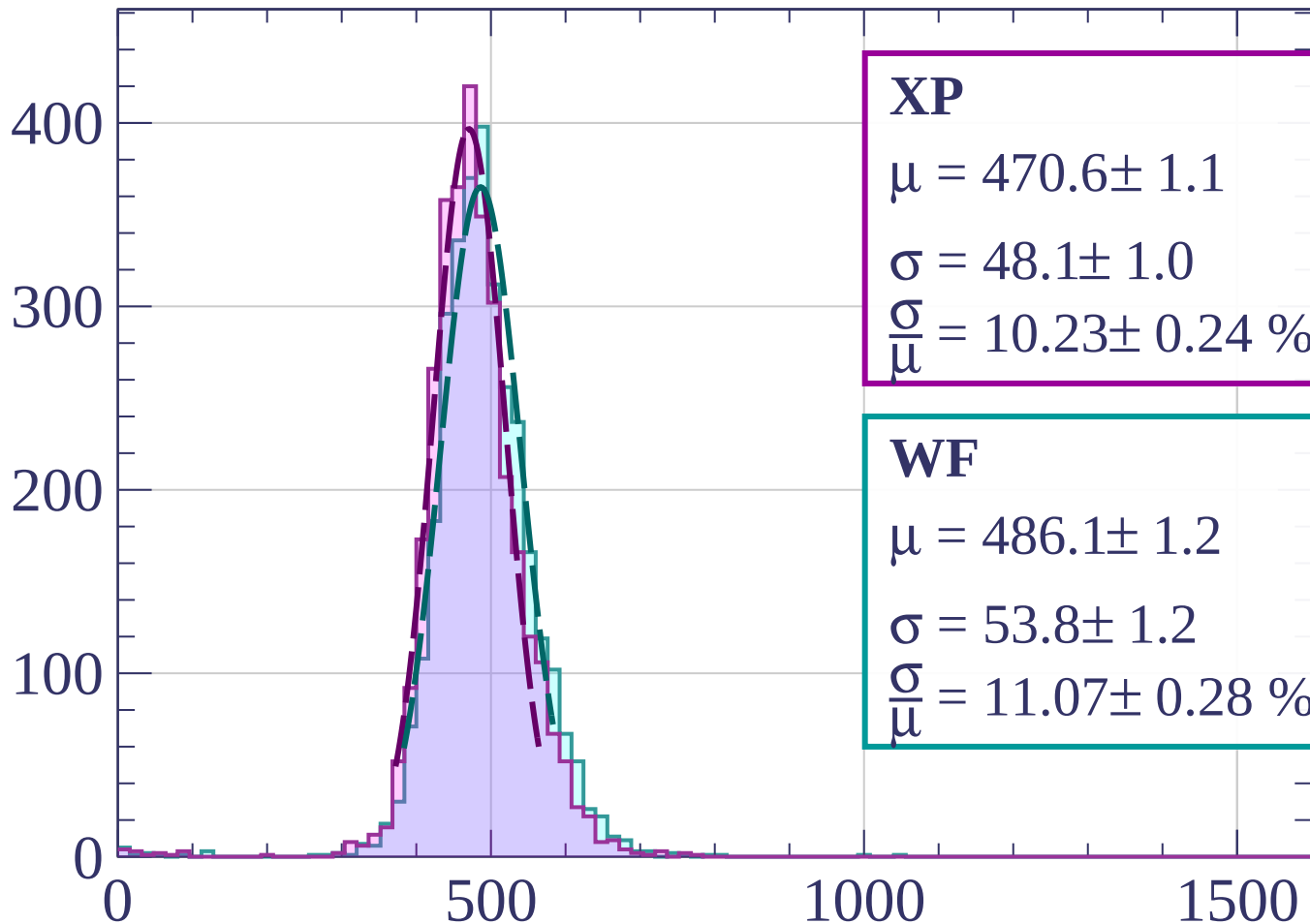
$$\sigma = 54.1 \pm 1.2$$

$$\frac{\sigma}{\mu} = 11.16 \pm 0.28 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1760 < p < 1780$

Count



XP

$$\mu = 470.6 \pm 1.1$$

$$\sigma = 48.1 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.23 \pm 0.24 \%$$

WF

$$\mu = 486.1 \pm 1.2$$

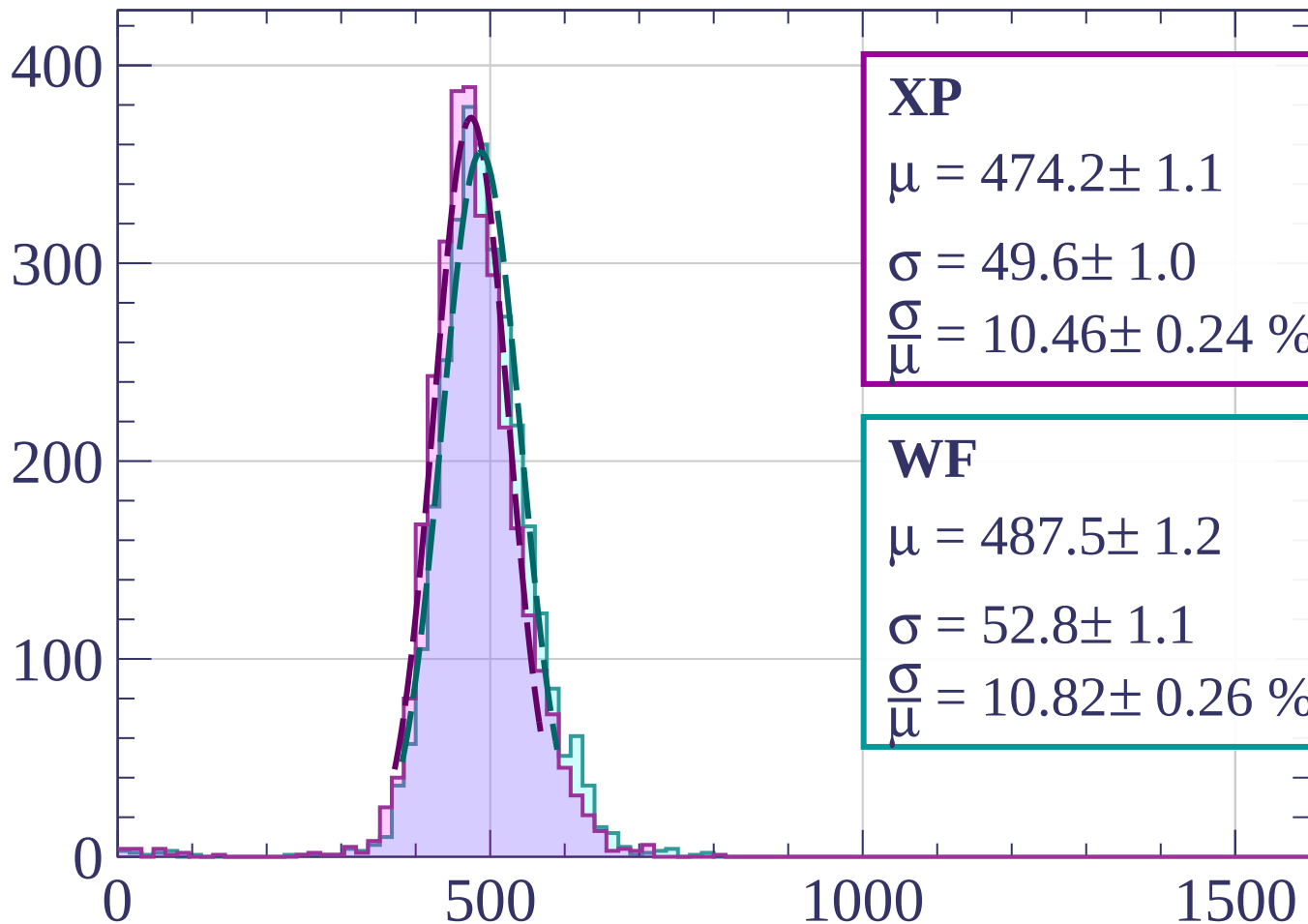
$$\sigma = 53.8 \pm 1.2$$

$$\frac{\sigma}{\mu} = 11.07 \pm 0.28 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1780 < p < 1800$

Count



XP

$$\mu = 474.2 \pm 1.1$$

$$\sigma = 49.6 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.46 \pm 0.24 \%$$

WF

$$\mu = 487.5 \pm 1.2$$

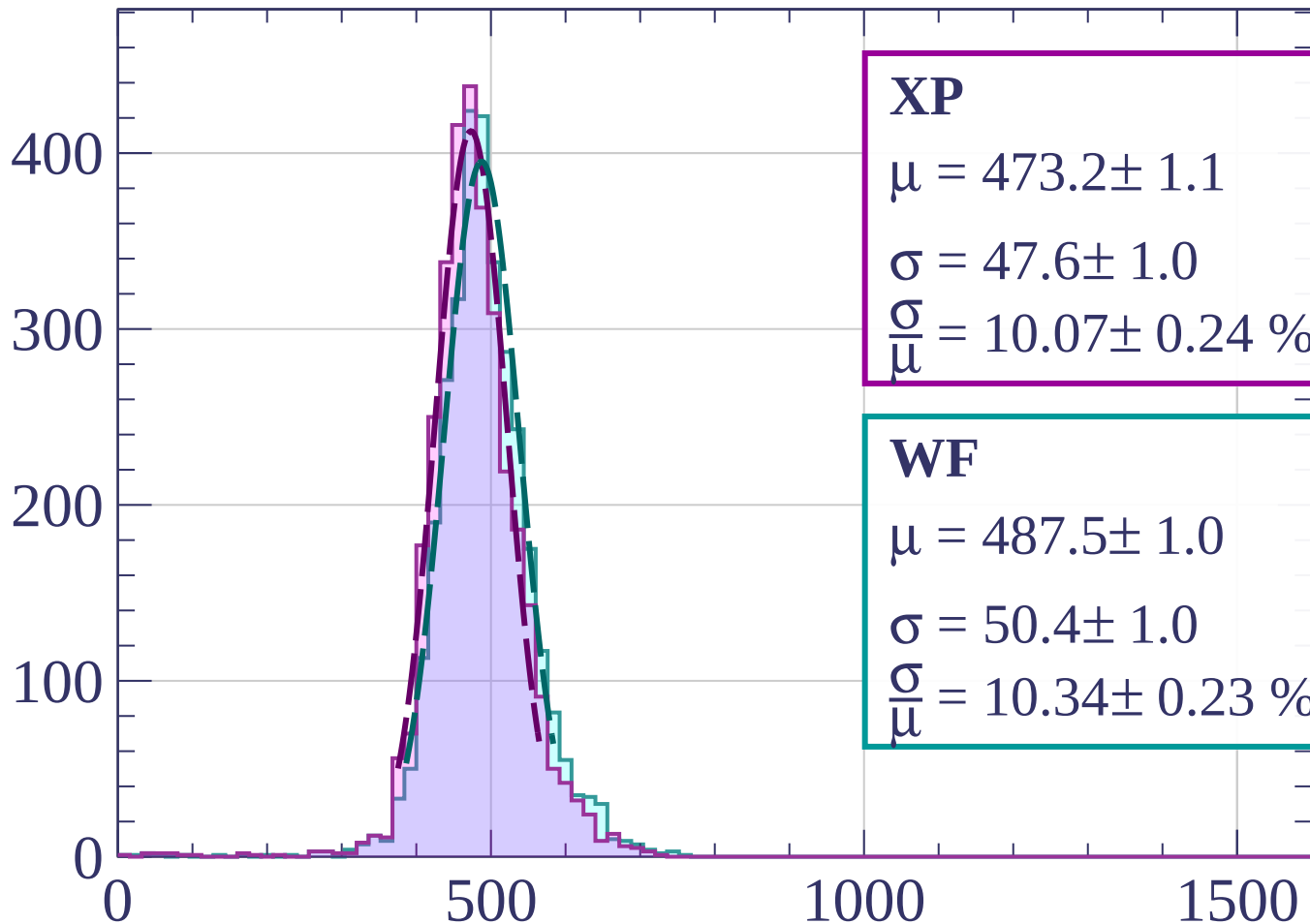
$$\sigma = 52.8 \pm 1.1$$

$$\frac{\sigma}{\mu} = 10.82 \pm 0.26 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1800 < p < 1820$

Count



XP

$$\mu = 473.2 \pm 1.1$$

$$\sigma = 47.6 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.07 \pm 0.24 \%$$

WF

$$\mu = 487.5 \pm 1.0$$

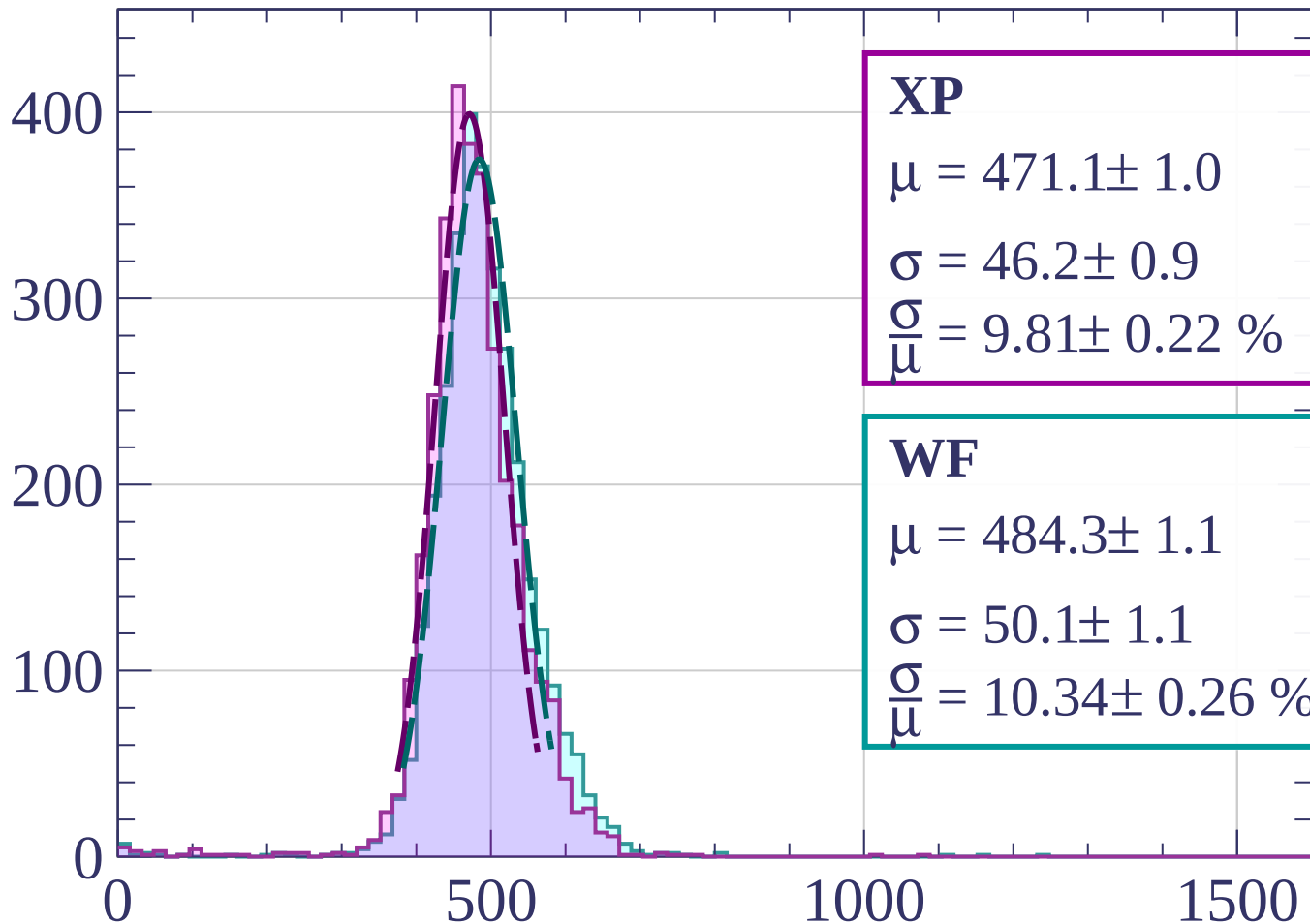
$$\sigma = 50.4 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.34 \pm 0.23 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1820 < p < 1840$

Count



XP

$$\mu = 471.1 \pm 1.0$$

$$\sigma = 46.2 \pm 0.9$$

$$\frac{\sigma}{\mu} = 9.81 \pm 0.22 \%$$

WF

$$\mu = 484.3 \pm 1.1$$

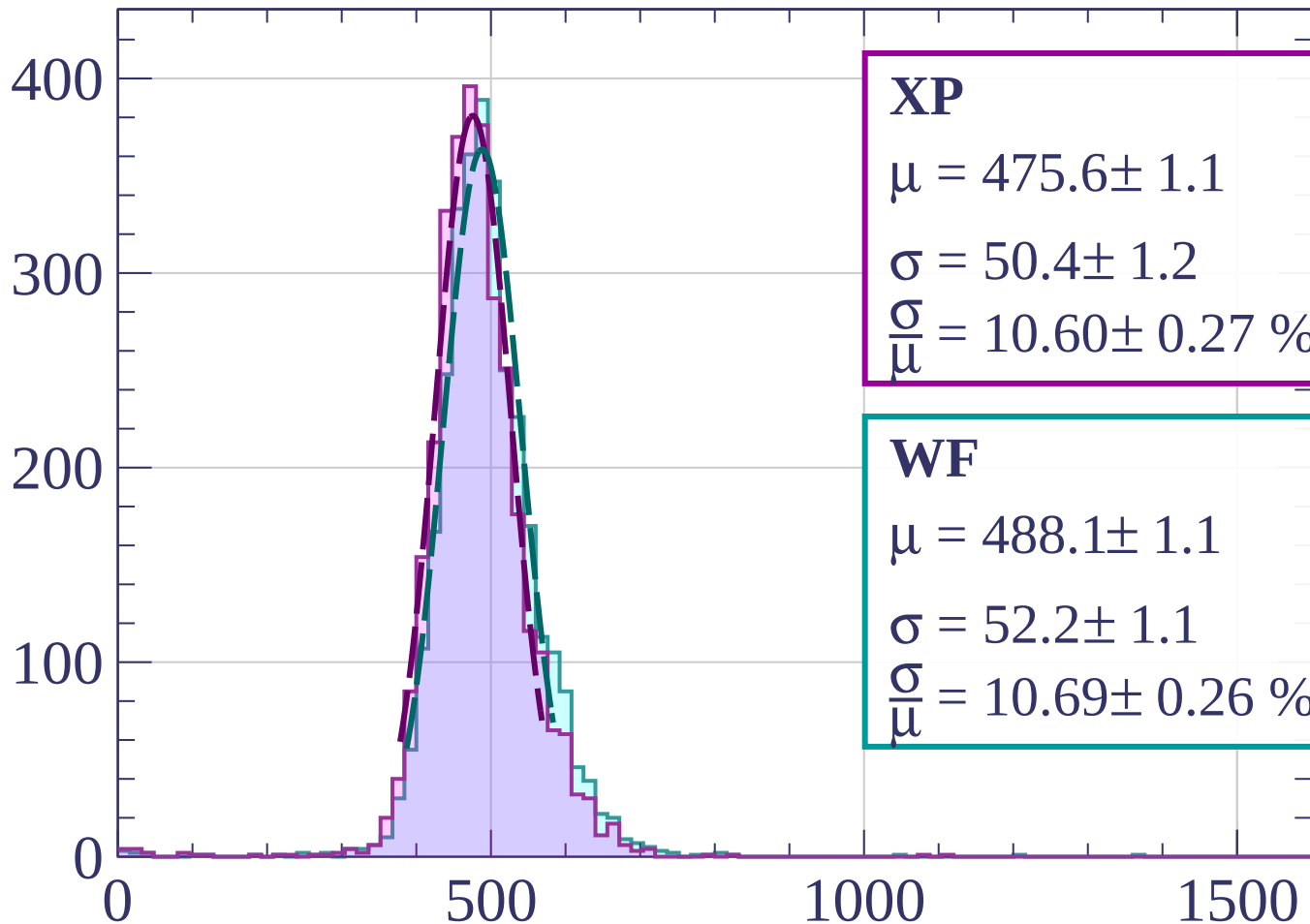
$$\sigma = 50.1 \pm 1.1$$

$$\frac{\sigma}{\mu} = 10.34 \pm 0.26 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1840 < p < 1860$

Count



XP

$$\mu = 475.6 \pm 1.1$$

$$\sigma = 50.4 \pm 1.2$$

$$\frac{\sigma}{\mu} = 10.60 \pm 0.27 \%$$

WF

$$\mu = 488.1 \pm 1.1$$

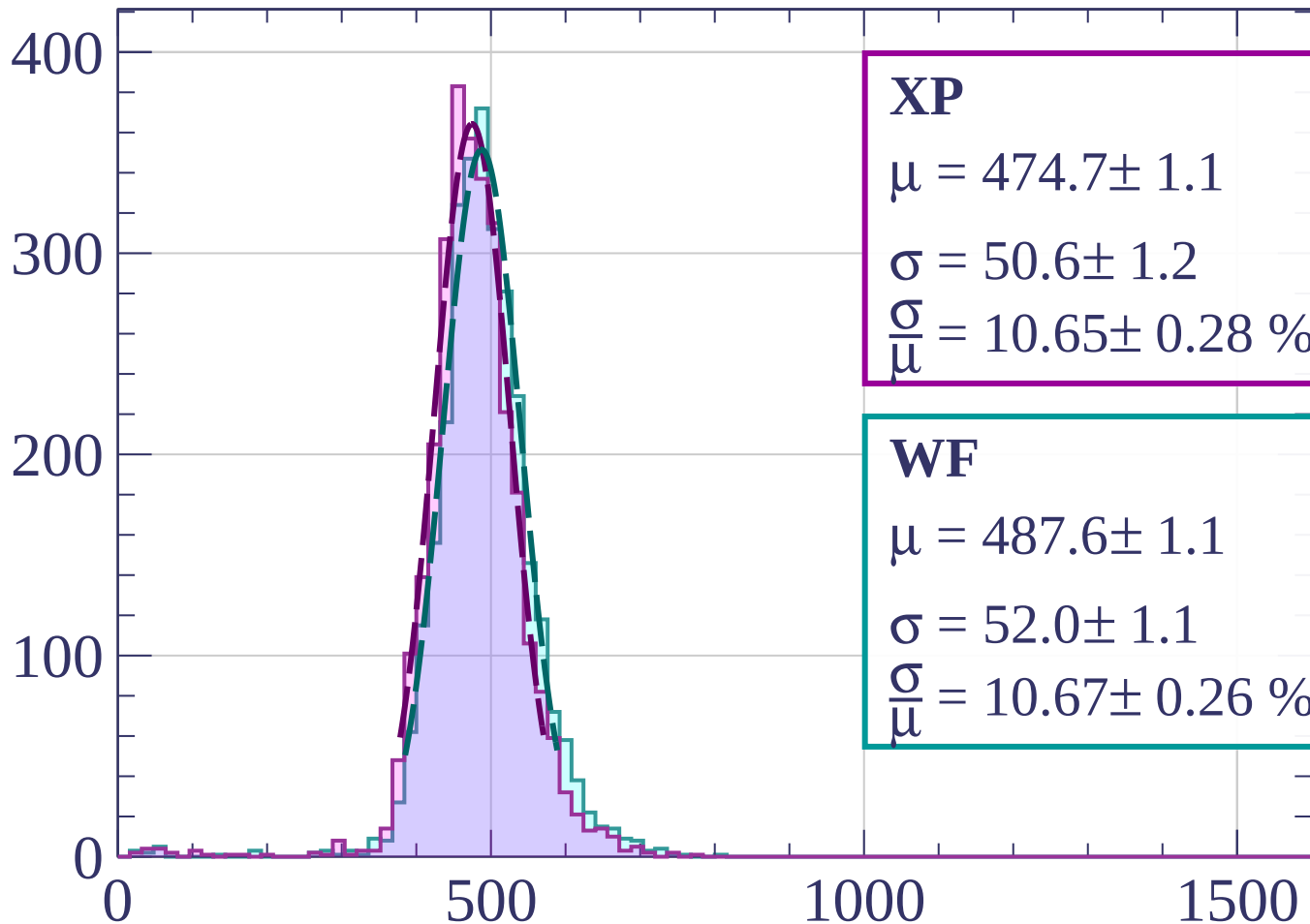
$$\sigma = 52.2 \pm 1.1$$

$$\frac{\sigma}{\mu} = 10.69 \pm 0.26 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1860 < p < 1880$

Count



XP

$$\mu = 474.7 \pm 1.1$$

$$\sigma = 50.6 \pm 1.2$$

$$\frac{\sigma}{\mu} = 10.65 \pm 0.28 \%$$

WF

$$\mu = 487.6 \pm 1.1$$

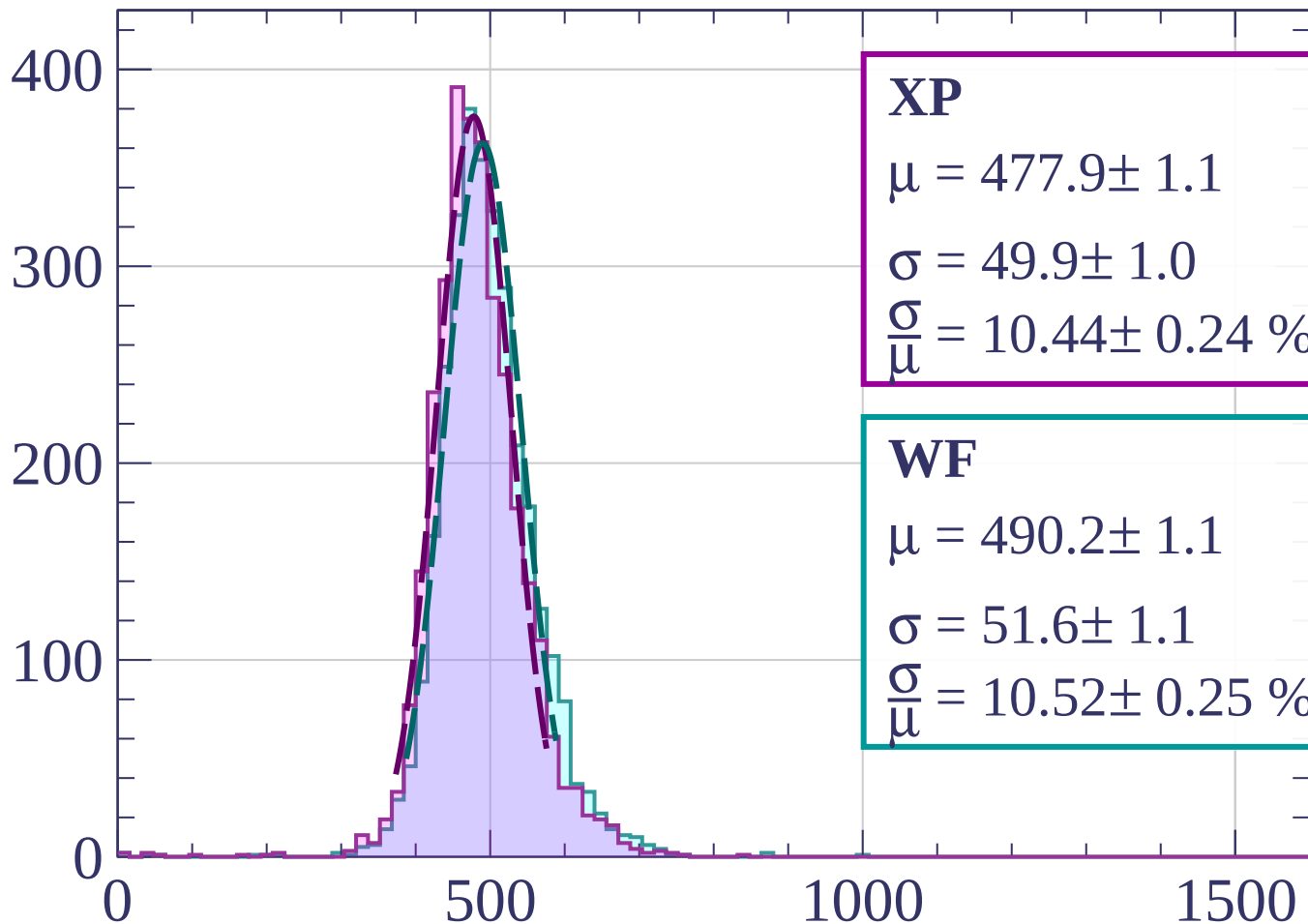
$$\sigma = 52.0 \pm 1.1$$

$$\frac{\sigma}{\mu} = 10.67 \pm 0.26 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1880 < p < 1900$

Count



XP

$$\mu = 477.9 \pm 1.1$$

$$\sigma = 49.9 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.44 \pm 0.24 \%$$

WF

$$\mu = 490.2 \pm 1.1$$

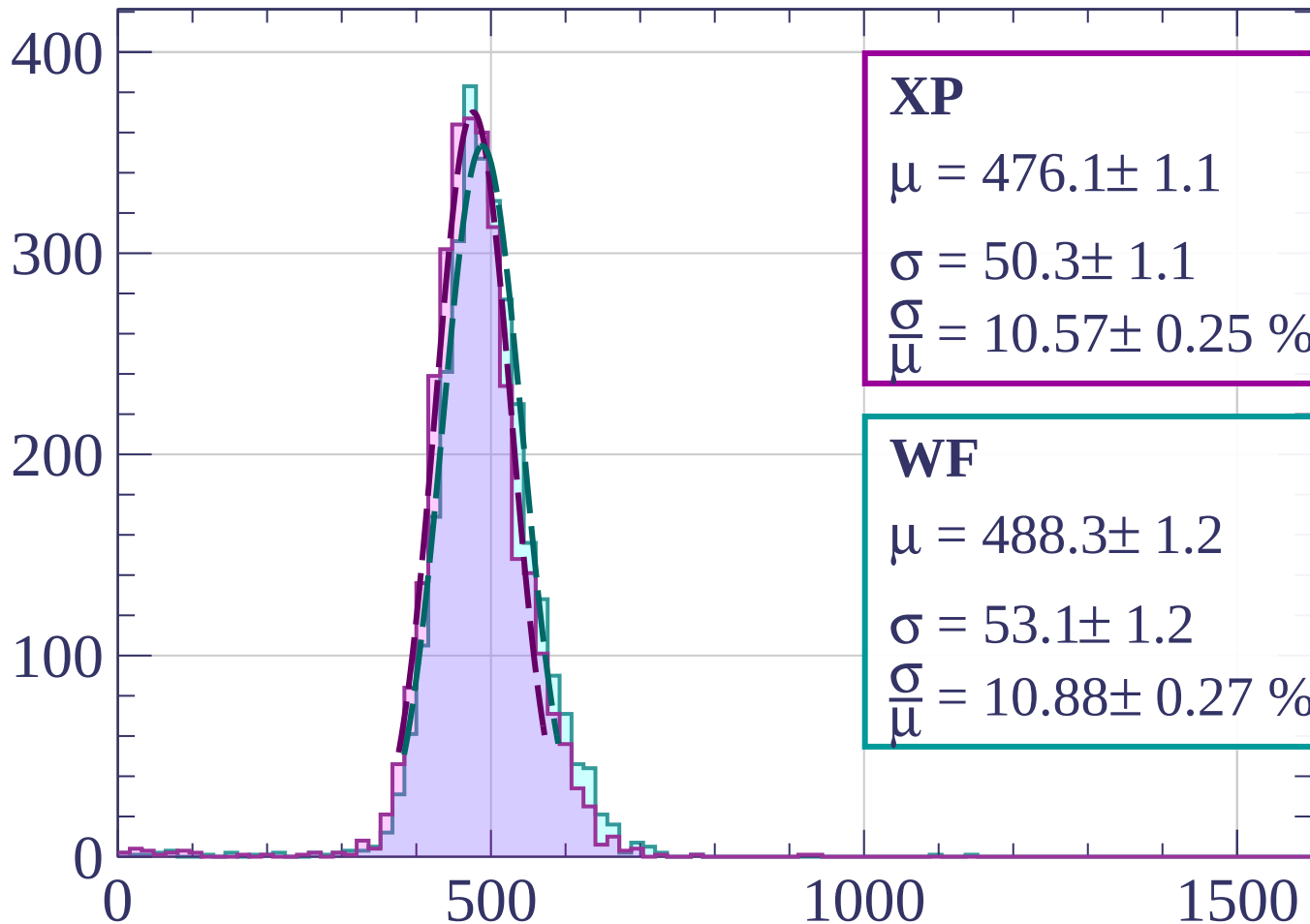
$$\sigma = 51.6 \pm 1.1$$

$$\frac{\sigma}{\mu} = 10.52 \pm 0.25 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1900 < p < 1920$

Count



XP

$$\mu = 476.1 \pm 1.1$$

$$\sigma = 50.3 \pm 1.1$$

$$\frac{\sigma}{\mu} = 10.57 \pm 0.25 \%$$

WF

$$\mu = 488.3 \pm 1.2$$

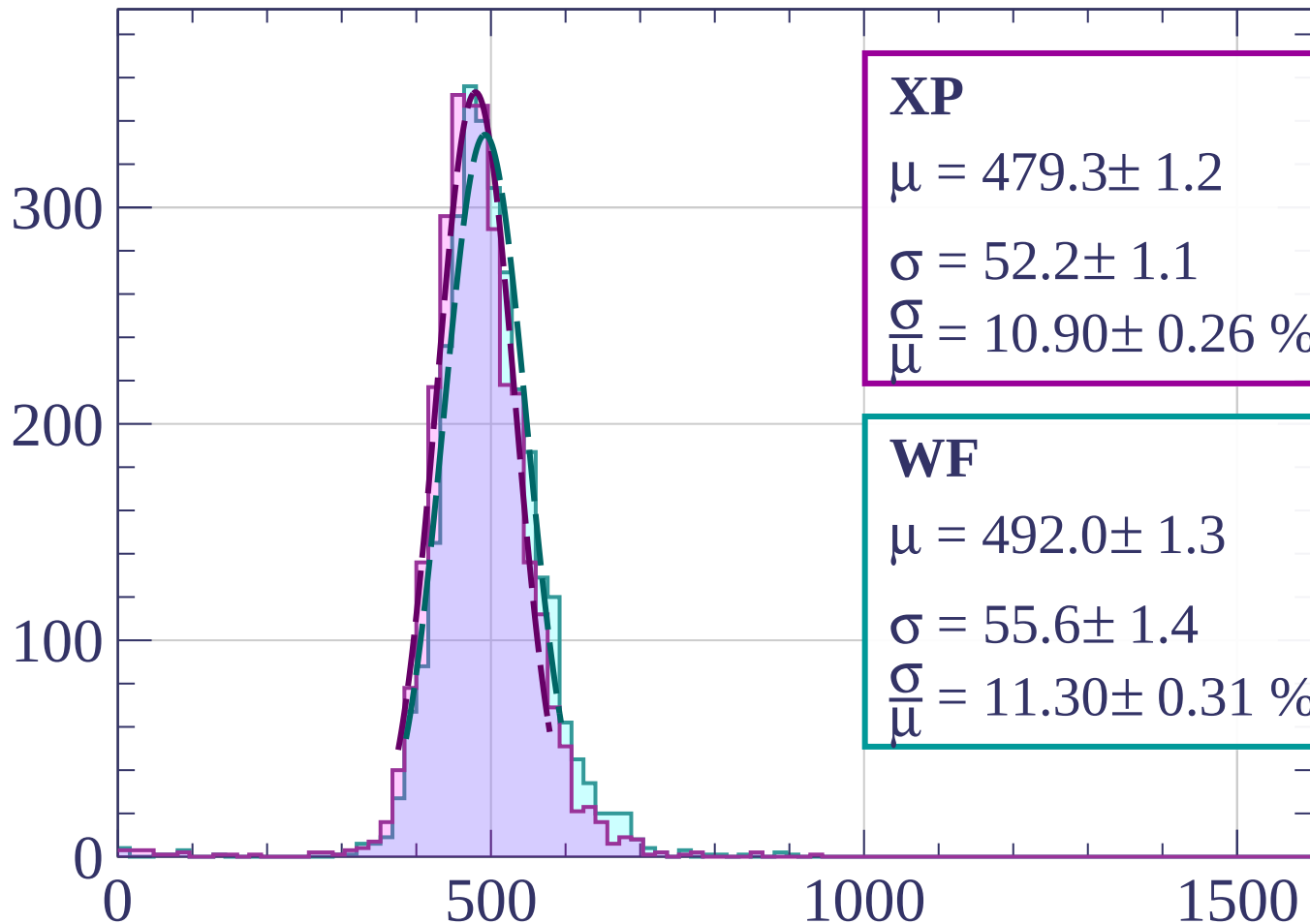
$$\sigma = 53.1 \pm 1.2$$

$$\frac{\sigma}{\mu} = 10.88 \pm 0.27 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1920 < p < 1940$

Count



XP

$$\mu = 479.3 \pm 1.2$$

$$\sigma = 52.2 \pm 1.1$$

$$\frac{\sigma}{\mu} = 10.90 \pm 0.26 \%$$

WF

$$\mu = 492.0 \pm 1.3$$

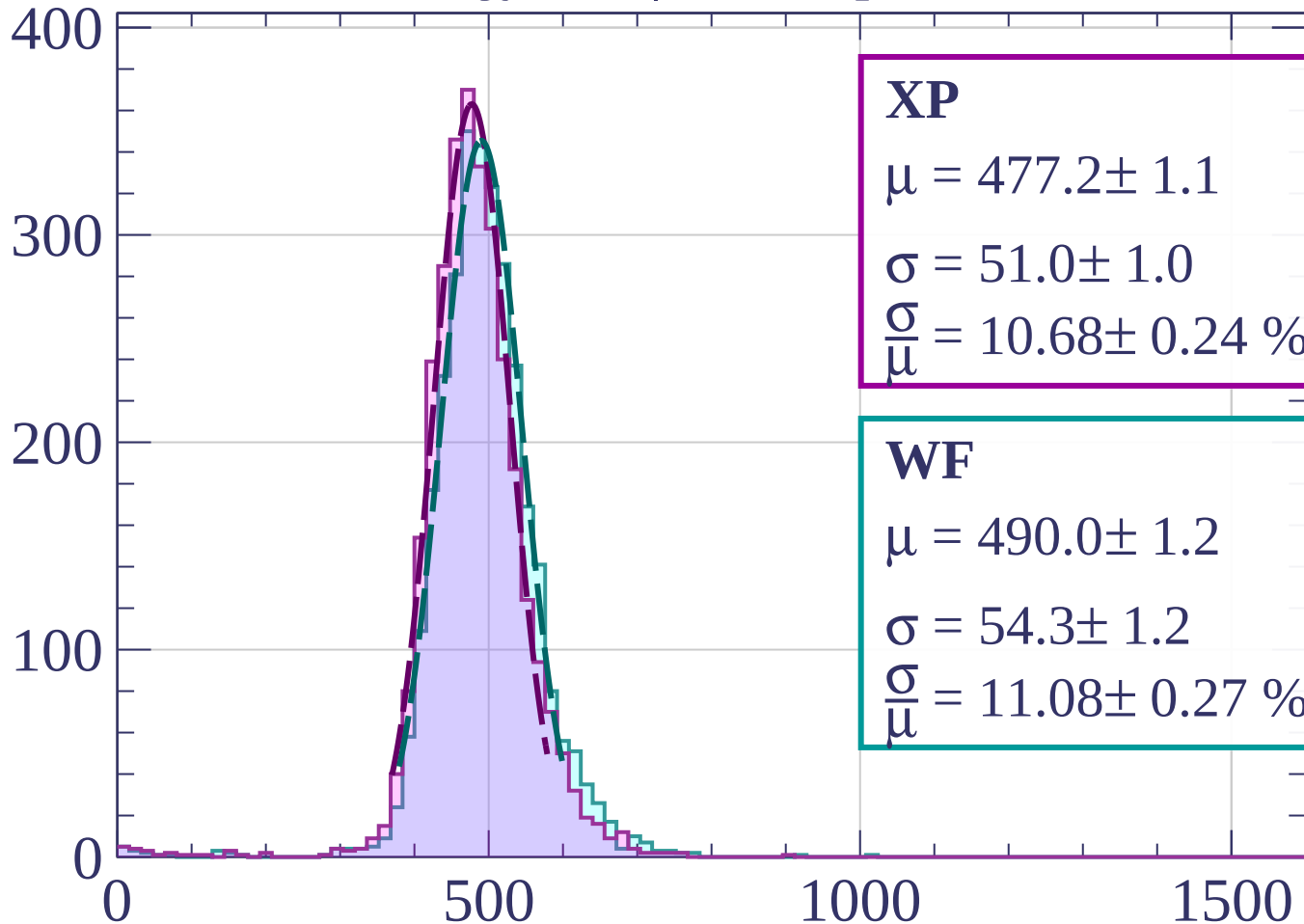
$$\sigma = 55.6 \pm 1.4$$

$$\frac{\sigma}{\mu} = 11.30 \pm 0.31 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1940 < p < 1960$

Count



XP

$$\mu = 477.2 \pm 1.1$$

$$\sigma = 51.0 \pm 1.0$$

$$\frac{\sigma}{\mu} = 10.68 \pm 0.24 \%$$

WF

$$\mu = 490.0 \pm 1.2$$

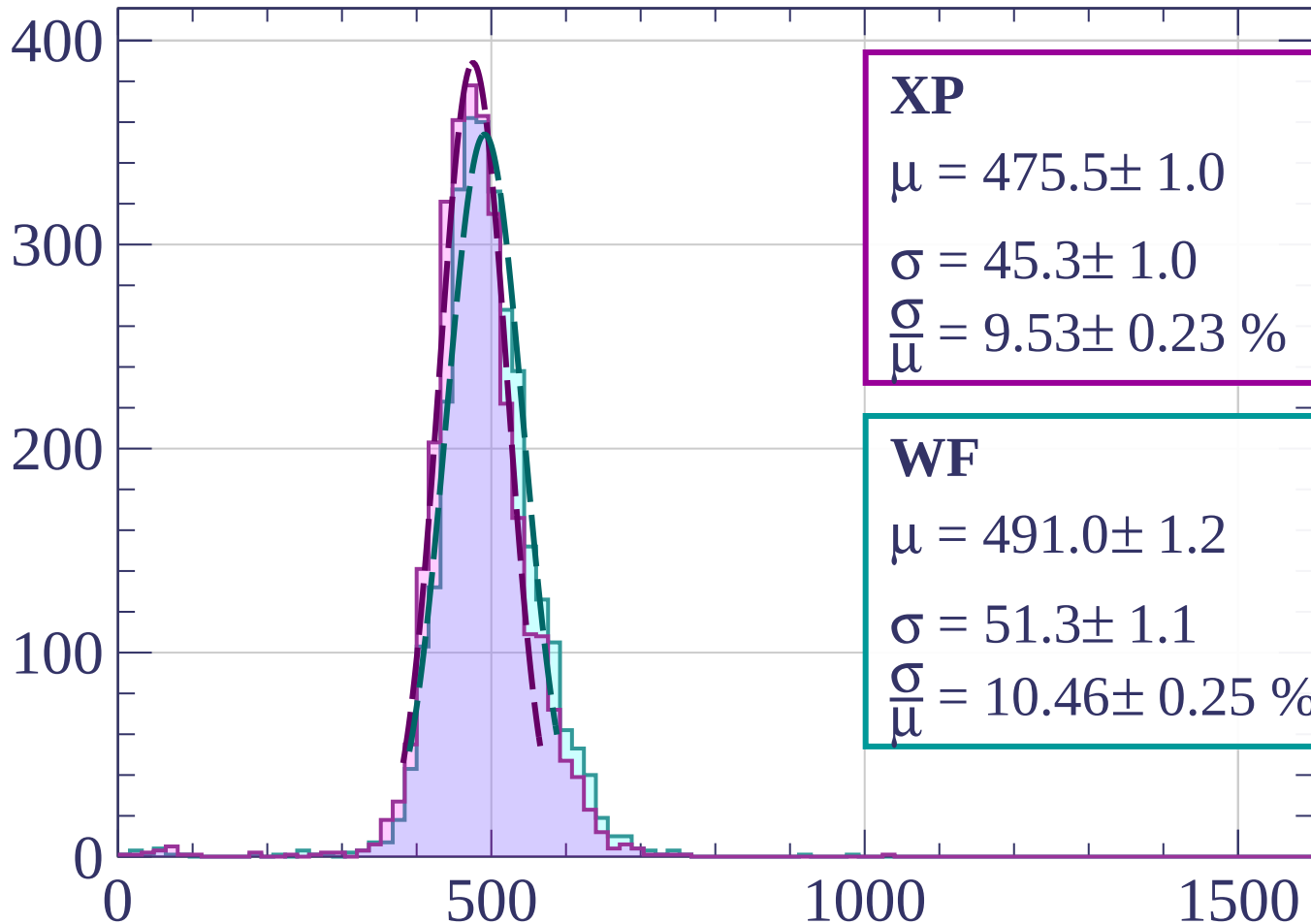
$$\sigma = 54.3 \pm 1.2$$

$$\frac{\sigma}{\mu} = 11.08 \pm 0.27 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1960 < p < 1980$

Count



XP

$$\mu = 475.5 \pm 1.0$$

$$\sigma = 45.3 \pm 1.0$$

$$\frac{\sigma}{\mu} = 9.53 \pm 0.23 \%$$

WF

$$\mu = 491.0 \pm 1.2$$

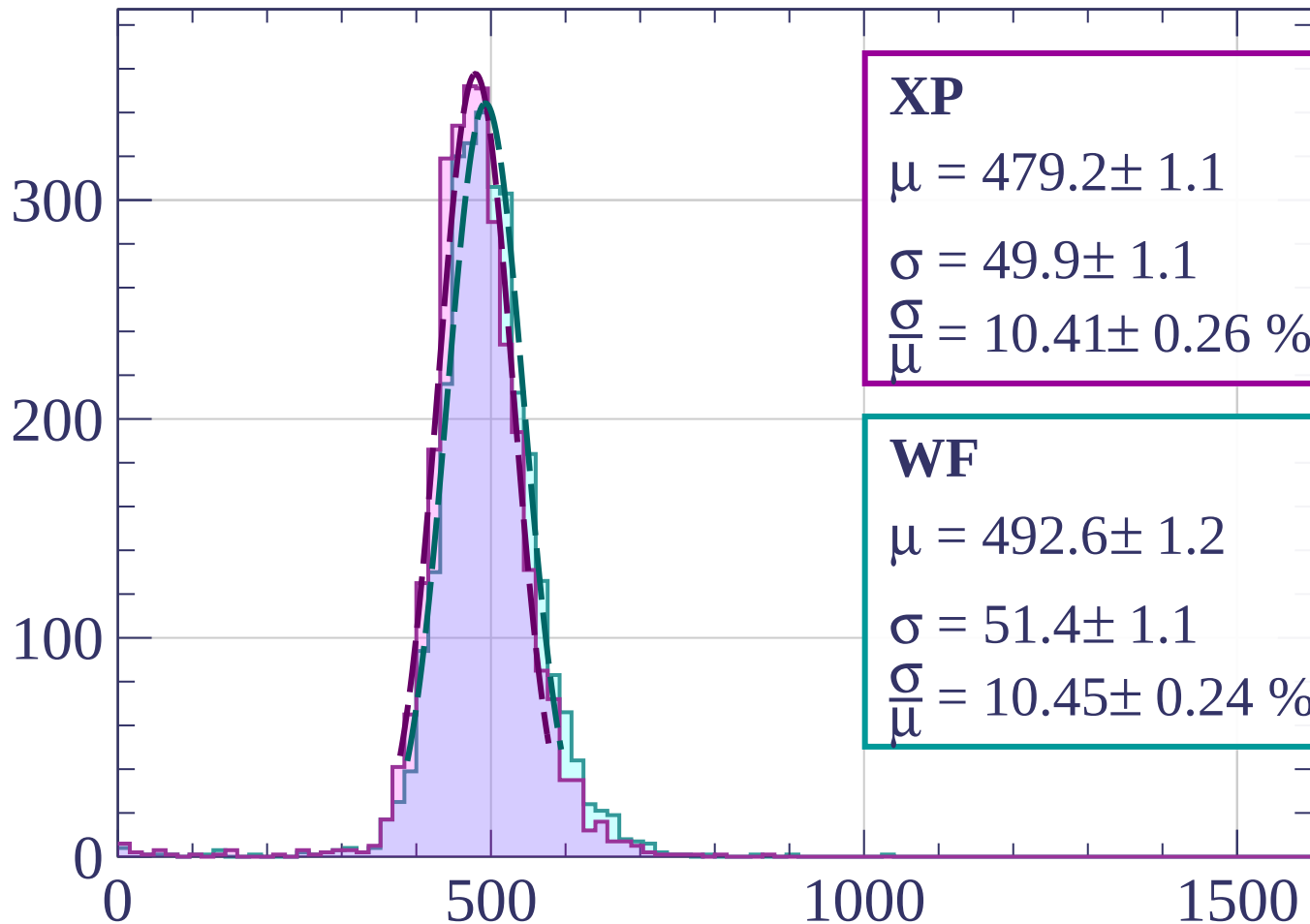
$$\sigma = 51.3 \pm 1.1$$

$$\frac{\sigma}{\mu} = 10.46 \pm 0.25 \%$$

dE/dx [ADC counts/cm]

Energy loss | $1980 < p < 2000$

Count



XP

$$\mu = 479.2 \pm 1.1$$

$$\sigma = 49.9 \pm 1.1$$

$$\frac{\sigma}{\mu} = 10.41 \pm 0.26 \%$$

WF

$$\mu = 492.6 \pm 1.2$$

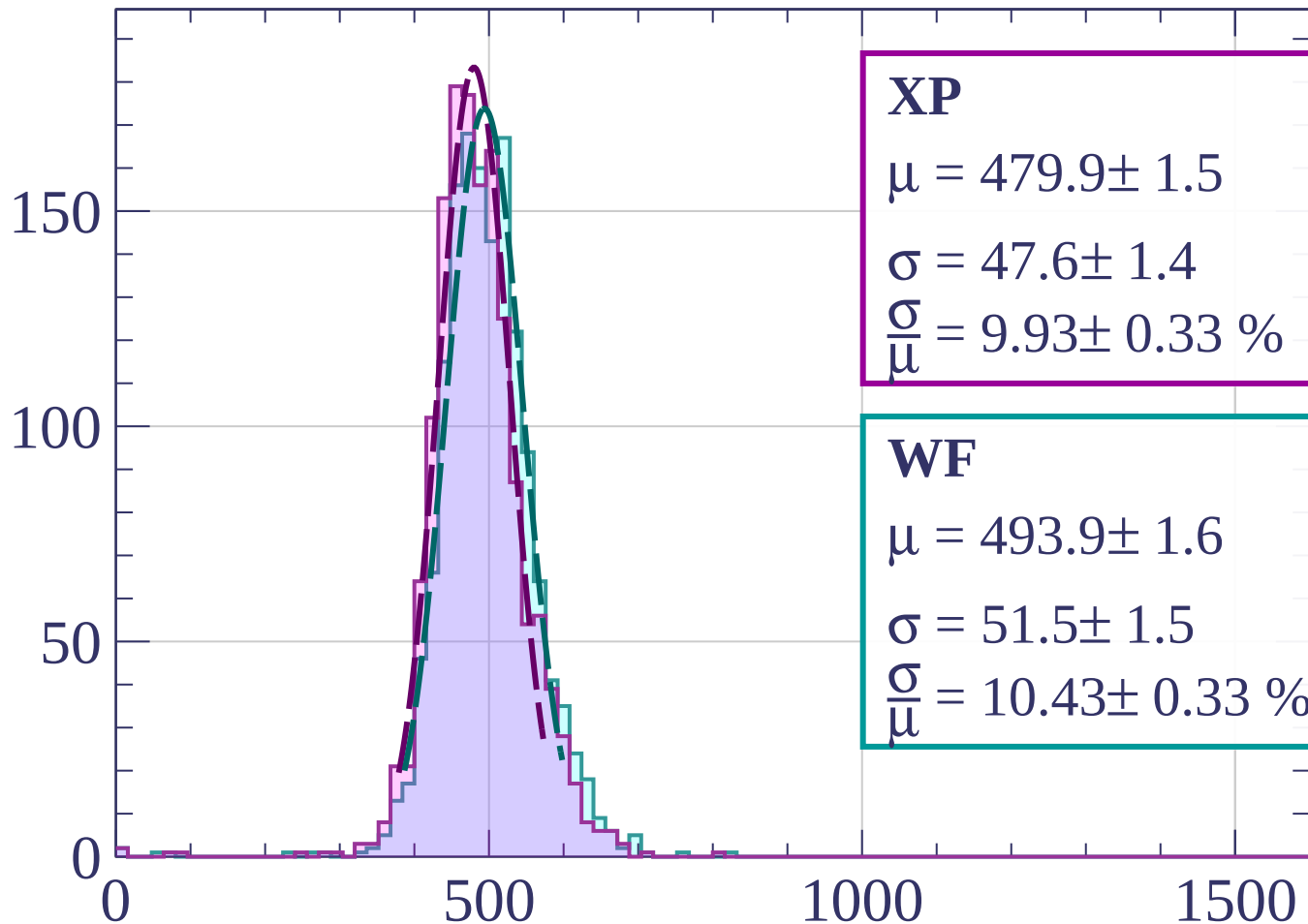
$$\sigma = 51.4 \pm 1.1$$

$$\frac{\sigma}{\mu} = 10.45 \pm 0.24 \%$$

dE/dx [ADC counts/cm]

Energy loss | $2000 < p < 2020$

Count



XP

$$\mu = 479.9 \pm 1.5$$

$$\sigma = 47.6 \pm 1.4$$

$$\frac{\sigma}{\mu} = 9.93 \pm 0.33 \%$$

WF

$$\mu = 493.9 \pm 1.6$$

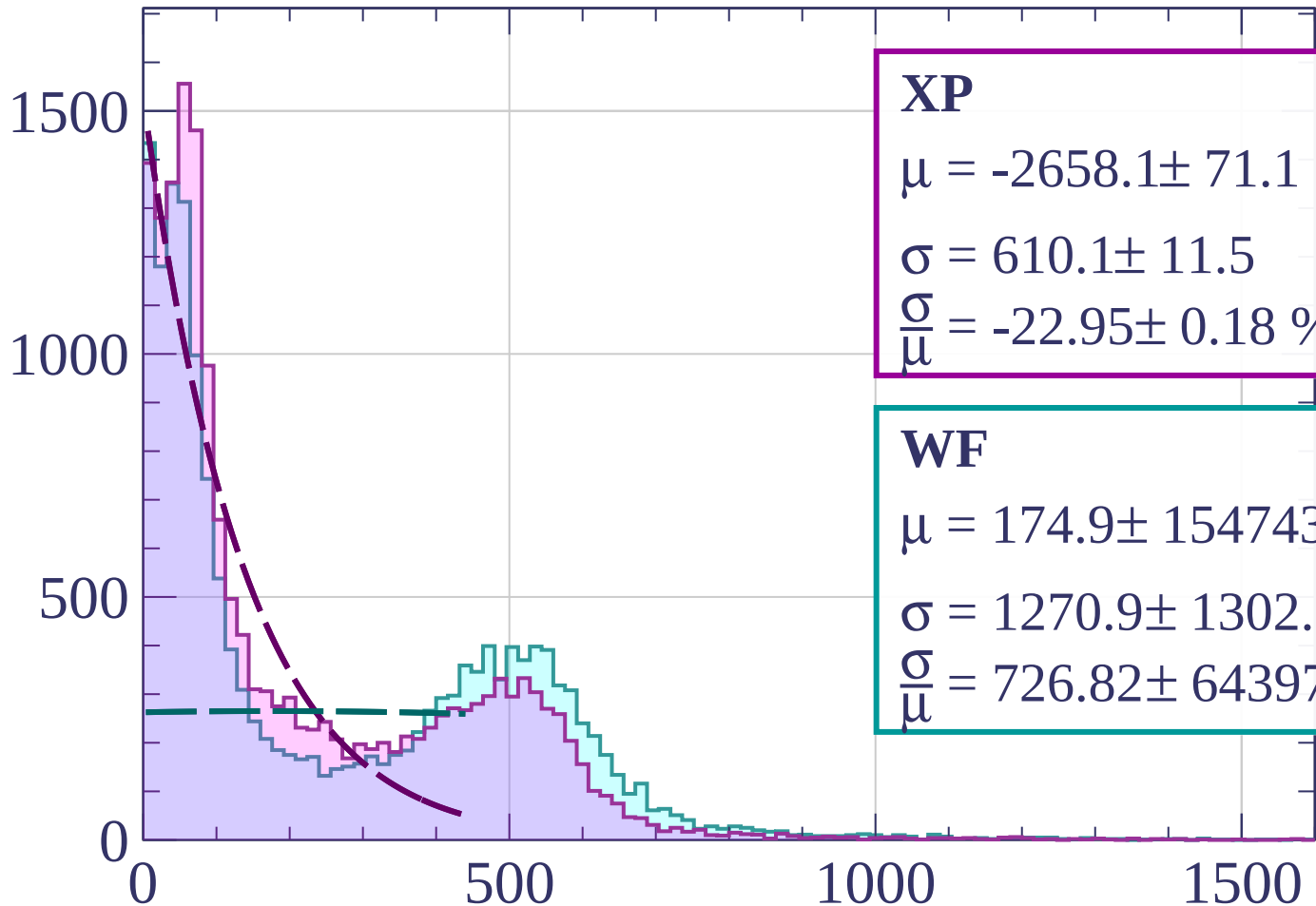
$$\sigma = 51.5 \pm 1.5$$

$$\frac{\sigma}{\mu} = 10.43 \pm 0.33 \%$$

dE/dx [ADC counts/cm]

Energy loss | $0 < \phi < 3$

Count



XP

$$\mu = -2658.1 \pm 71.1$$

$$\sigma = 610.1 \pm 11.5$$

$$\frac{\sigma}{\mu} = -22.95 \pm 0.18 \%$$

WF

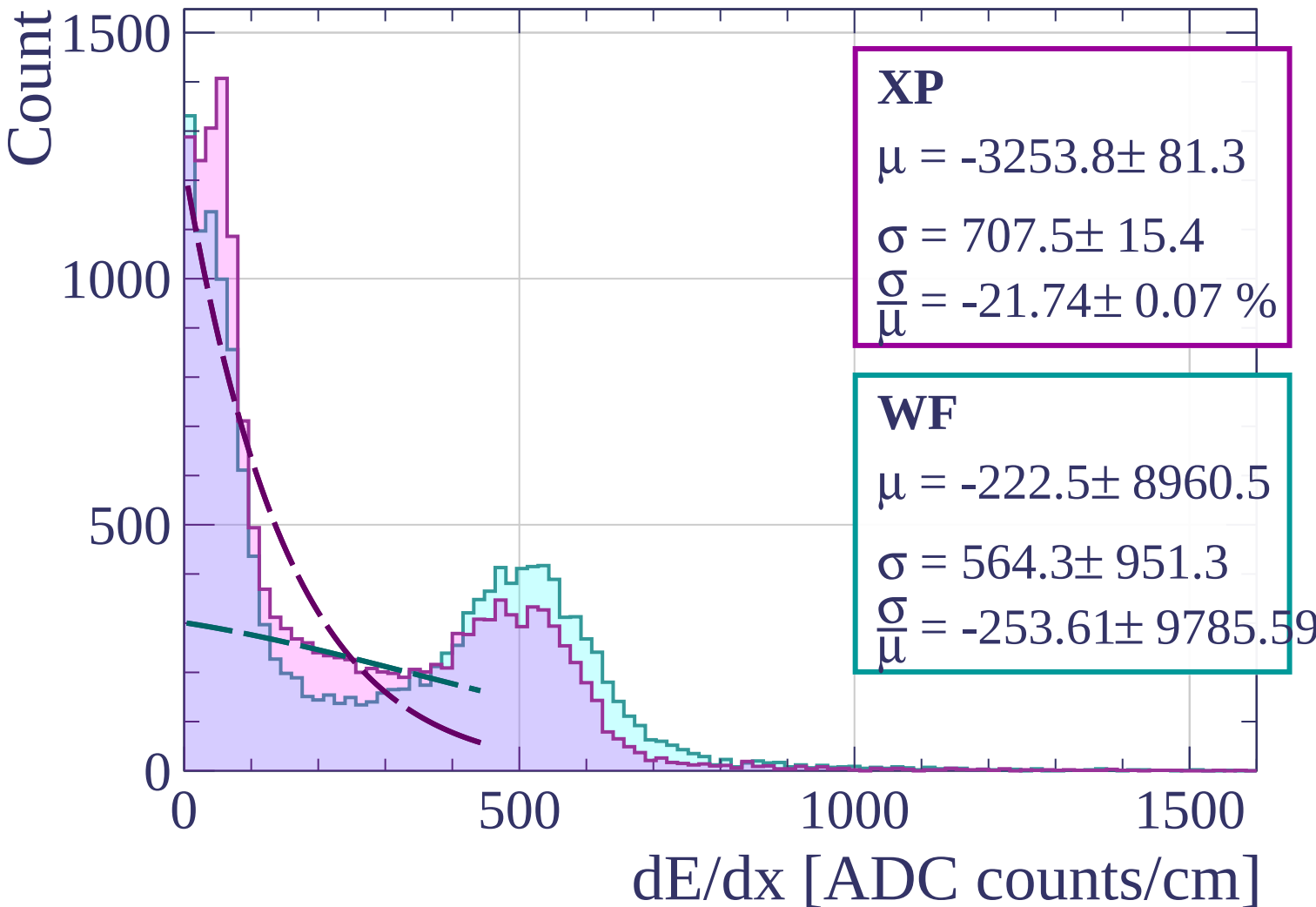
$$\mu = 174.9 \pm 154743.5$$

$$\sigma = 1270.9 \pm 1302.5$$

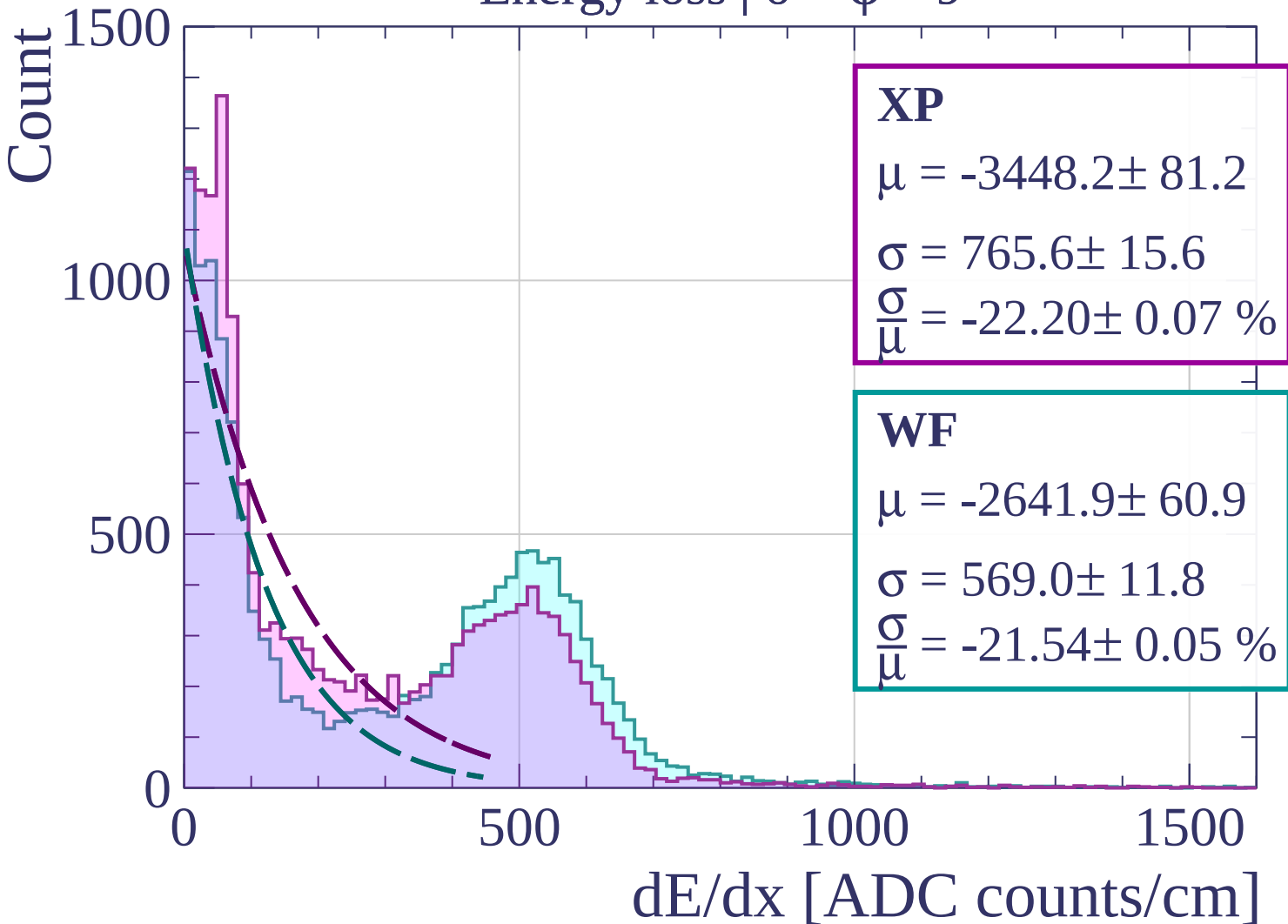
$$\frac{\sigma}{\mu} = 726.82 \pm 643971.3$$

dE/dx [ADC counts/cm]

Energy loss | $3 < \phi < 6$

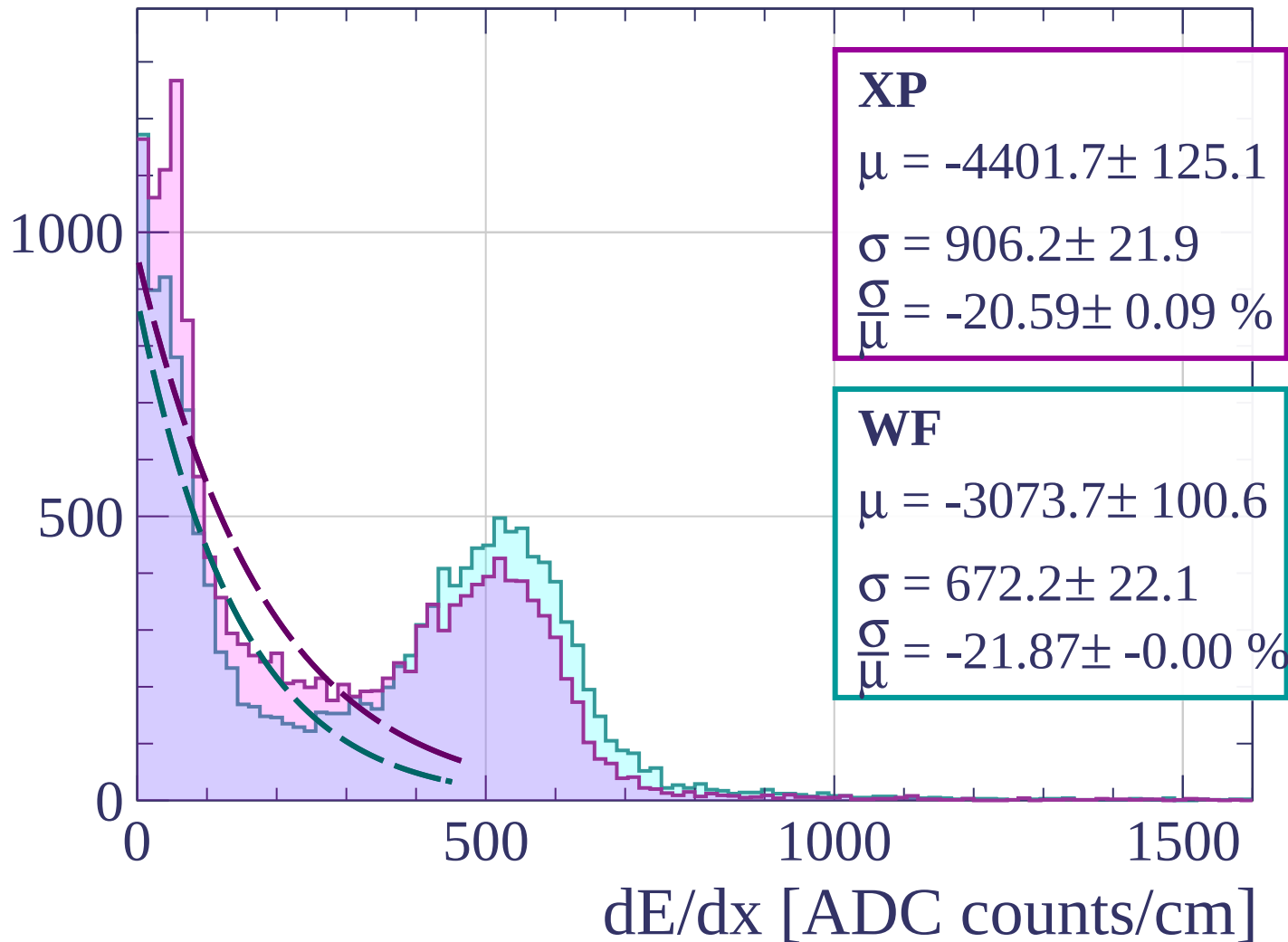


Energy loss | $6 < \phi < 9$



Energy loss | $9 < \phi < 12$

Count



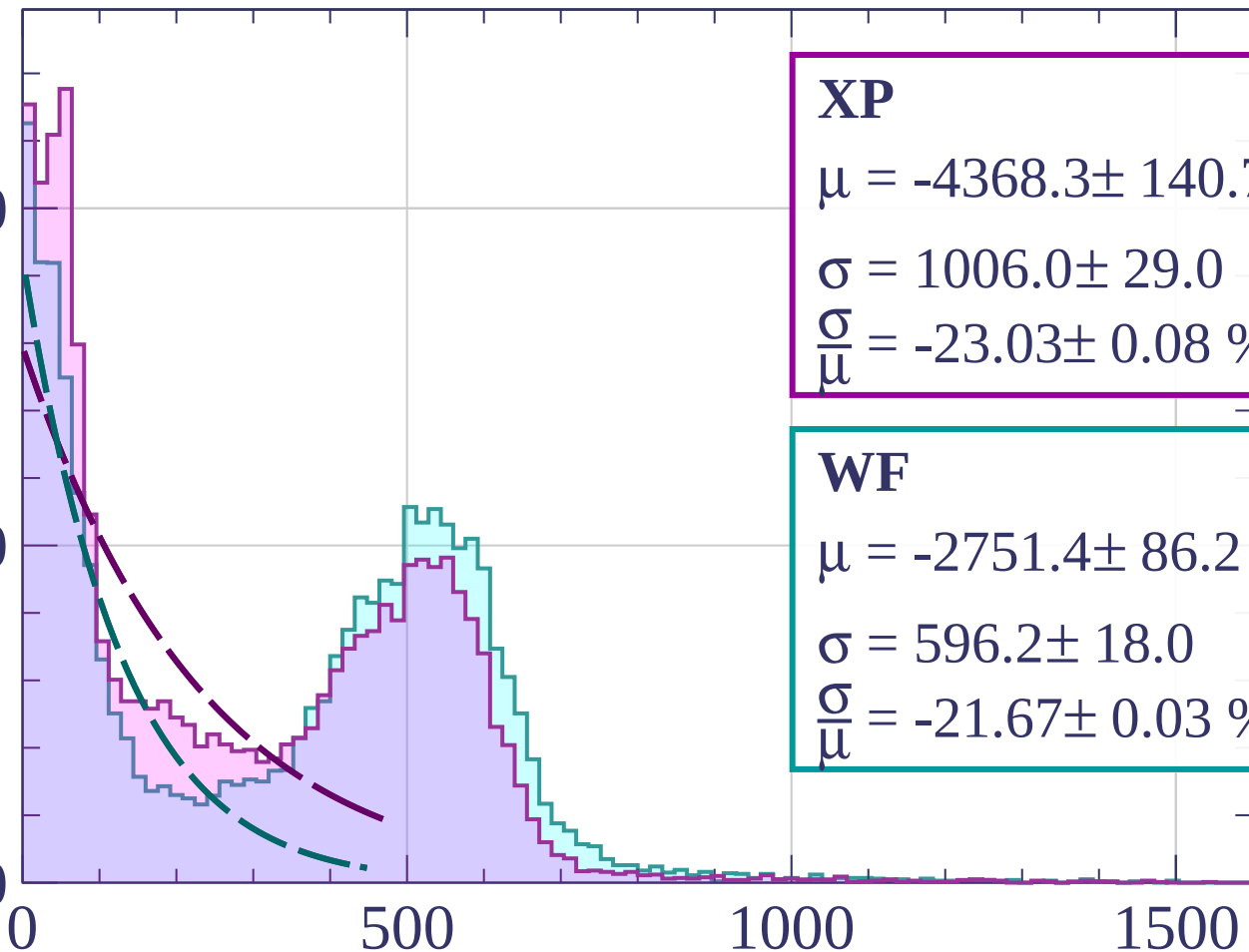
Energy loss | $12 < \varphi < 15$

Count

1000

500

0



dE/dx [ADC counts/cm]

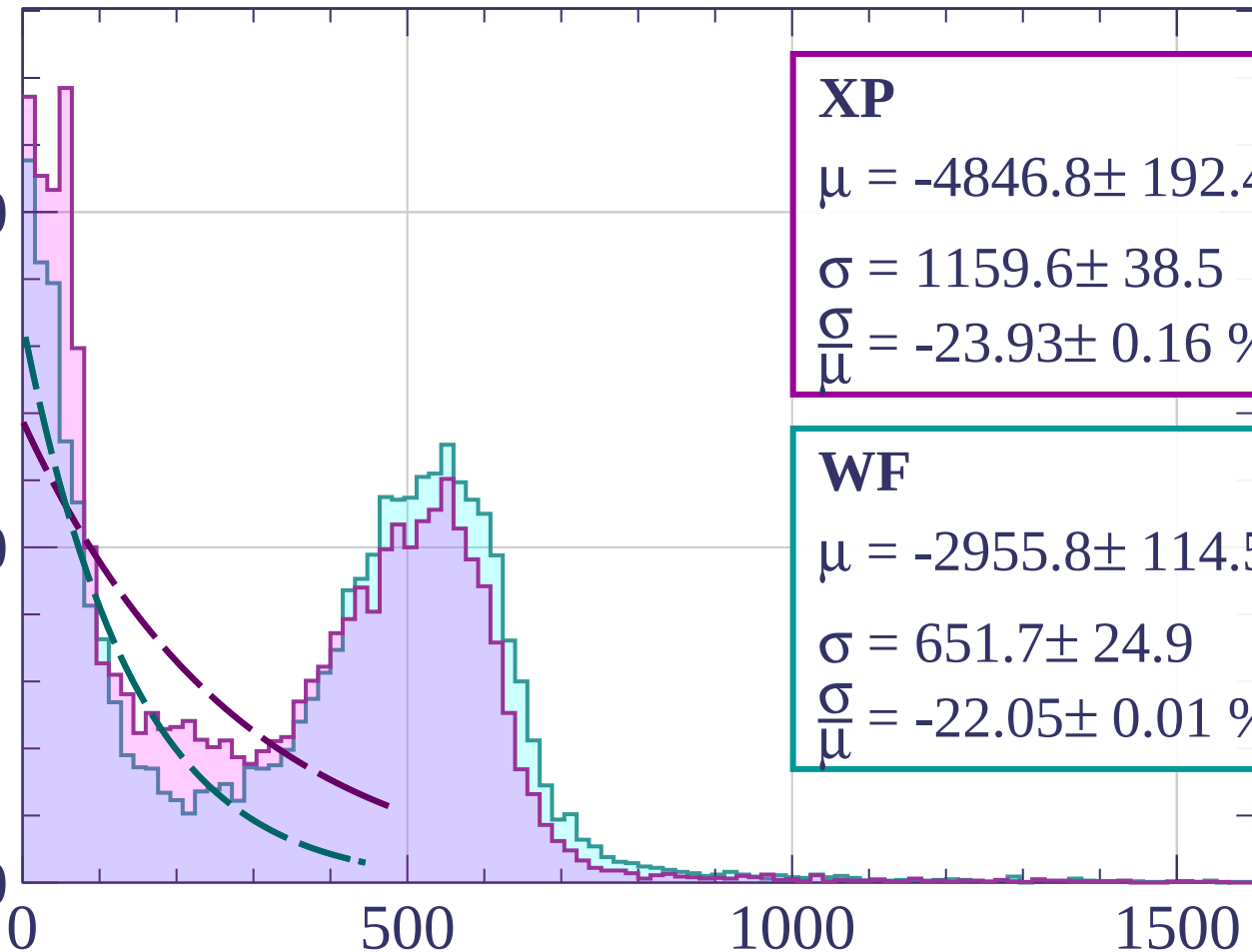
Energy loss | $15 < \varphi < 18$

Count

1000

500

0



XP

$$\mu = -4846.8 \pm 192.4$$

$$\sigma = 1159.6 \pm 38.5$$

$$\frac{\sigma}{\mu} = -23.93 \pm 0.16 \%$$

WF

$$\mu = -2955.8 \pm 114.5$$

$$\sigma = 651.7 \pm 24.9$$

$$\frac{\sigma}{\mu} = -22.05 \pm 0.01 \%$$

dE/dx [ADC counts/cm]

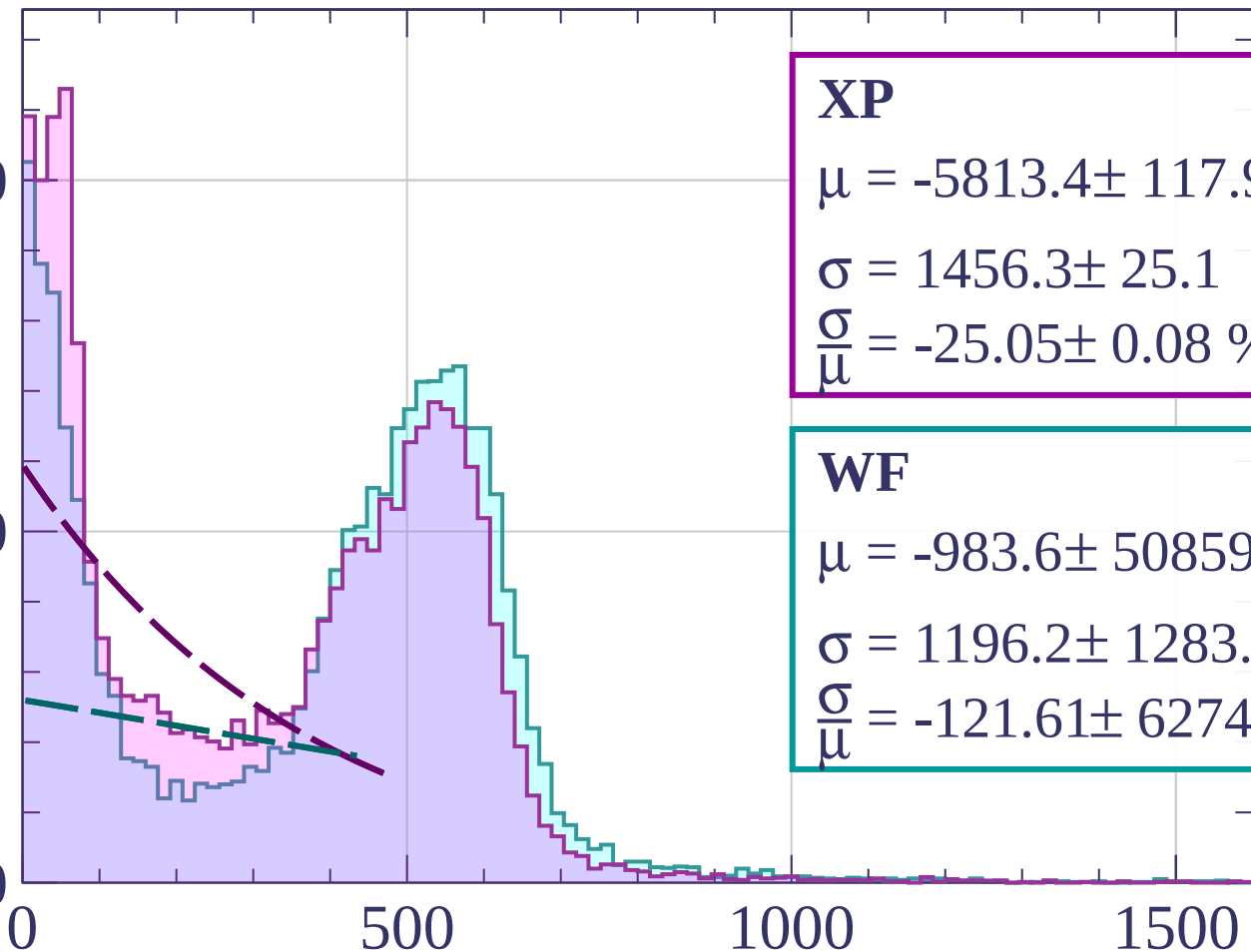
Energy loss | $18 < \phi < 21$

Count

1000

500

0



XP

$$\mu = -5813.4 \pm 117.9$$

$$\sigma = 1456.3 \pm 25.1$$

$$\frac{\sigma}{\mu} = -25.05 \pm 0.08 \%$$

WF

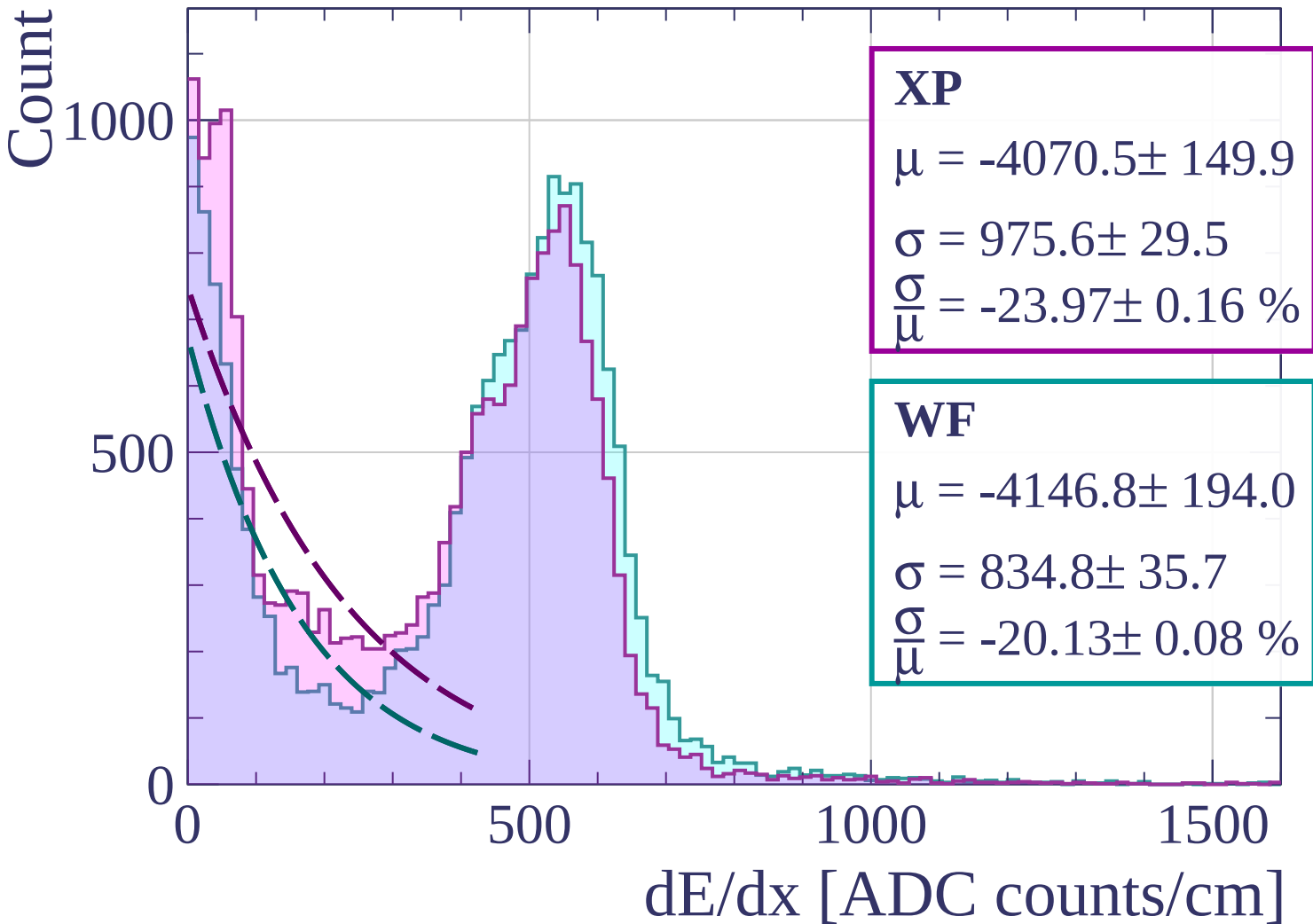
$$\mu = -983.6 \pm 508598.8$$

$$\sigma = 1196.2 \pm 1283.3$$

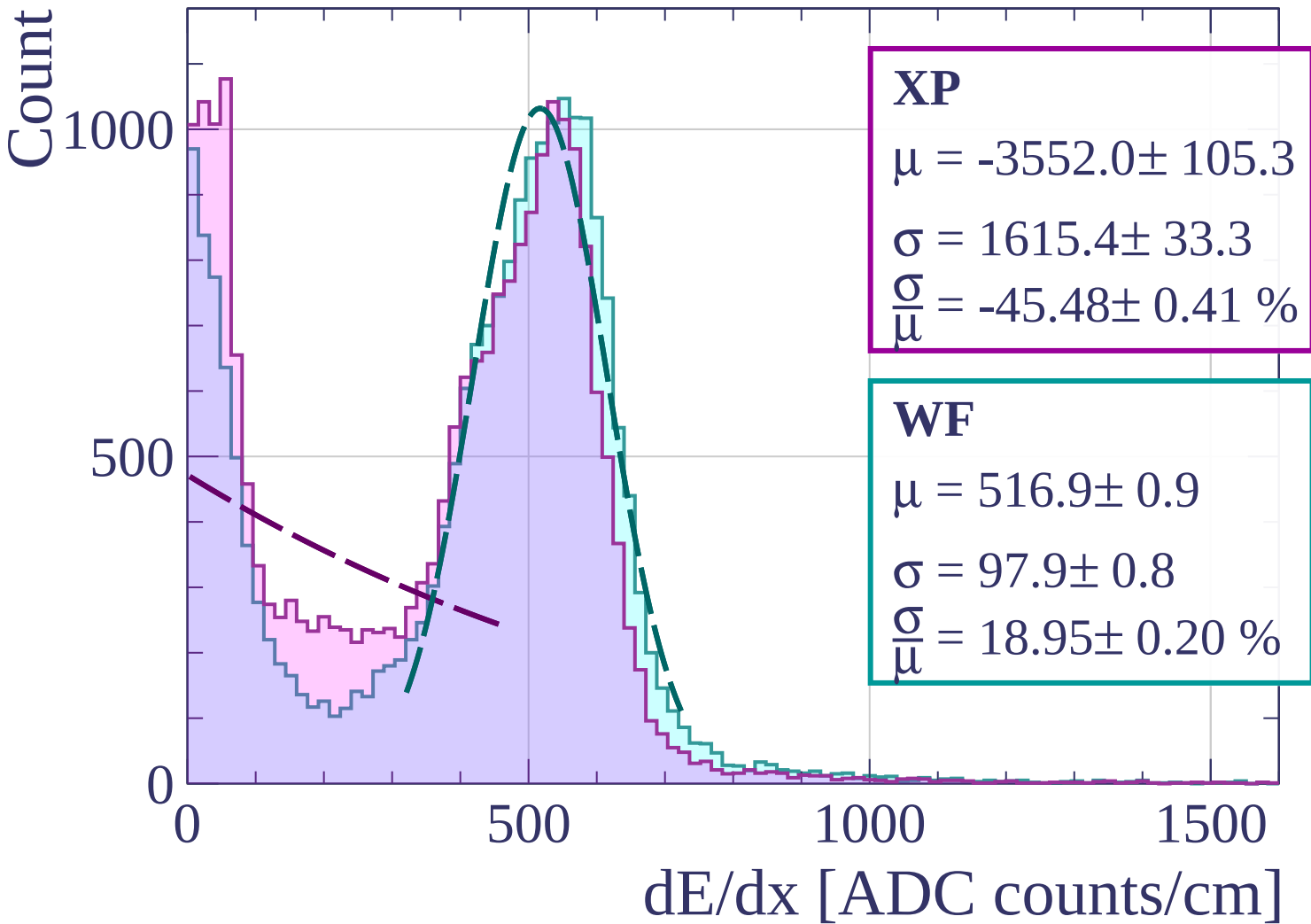
$$\frac{\sigma}{\mu} = -121.61 \pm 62749.5$$

dE/dx [ADC counts/cm]

Energy loss | $21 < \phi < 24$

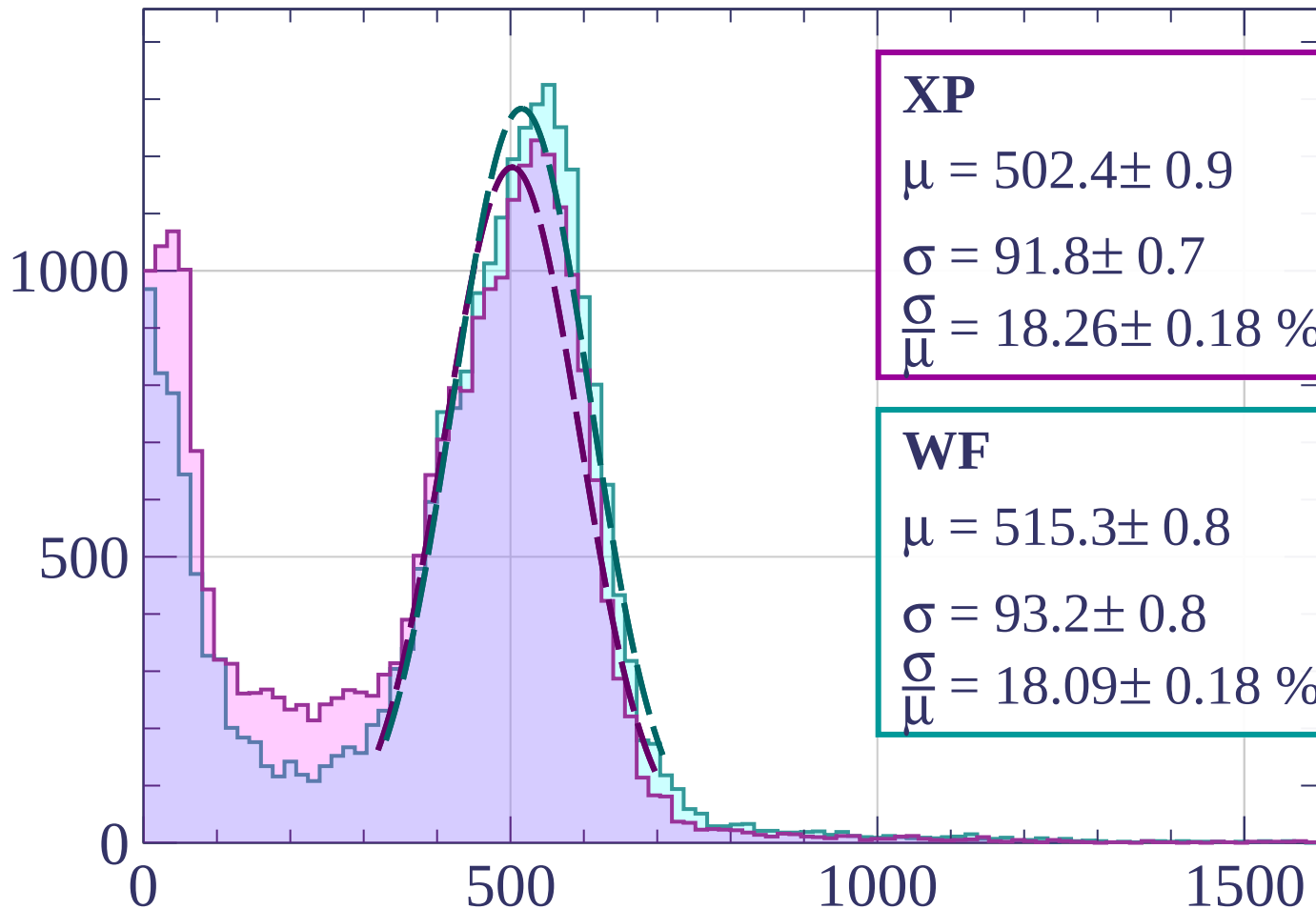


Energy loss | $24 < \phi < 27$



Energy loss | $27 < \phi < 30$

Count



XP

$$\mu = 502.4 \pm 0.9$$

$$\sigma = 91.8 \pm 0.7$$

$$\frac{\sigma}{\mu} = 18.26 \pm 0.18 \%$$

WF

$$\mu = 515.3 \pm 0.8$$

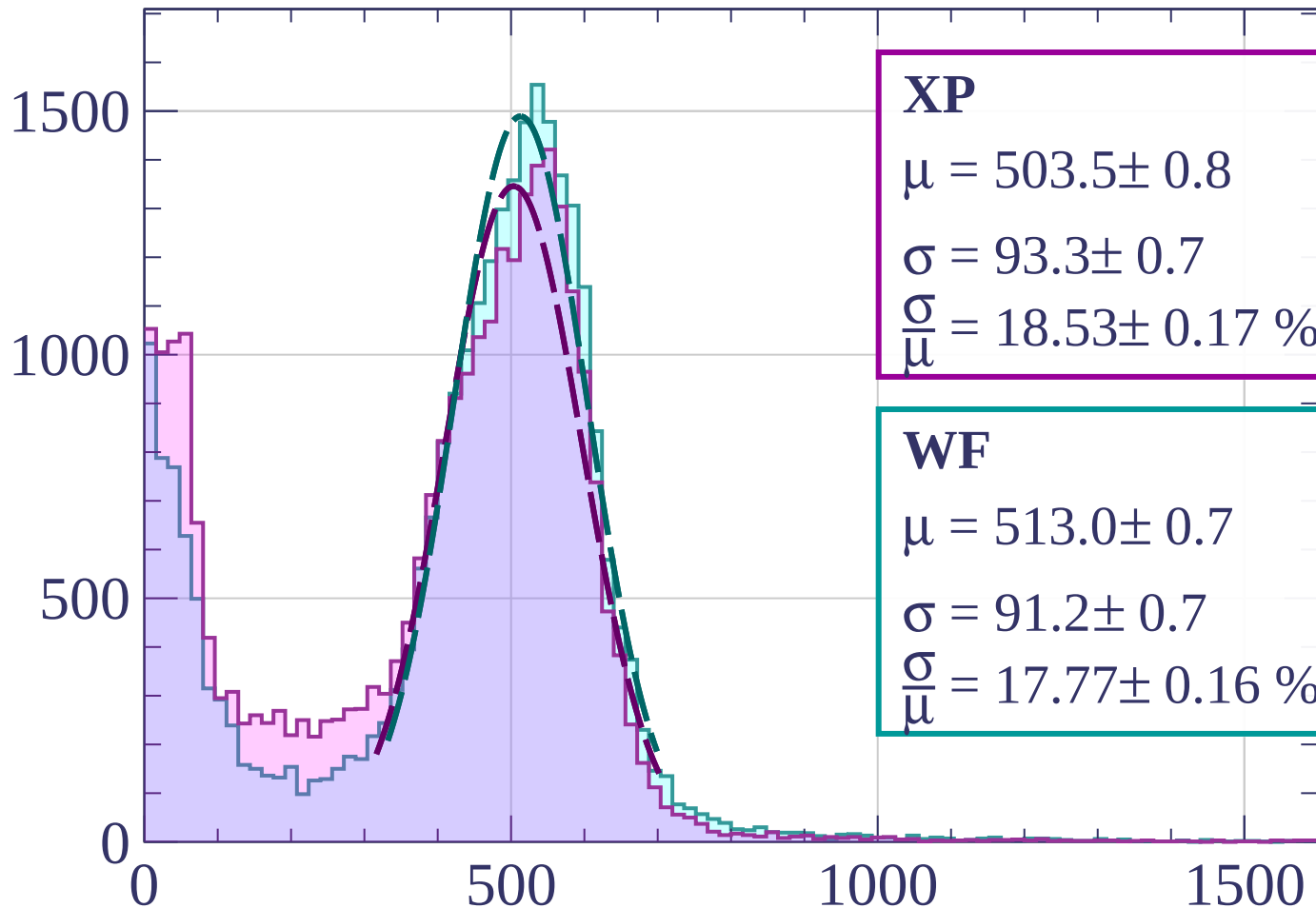
$$\sigma = 93.2 \pm 0.8$$

$$\frac{\sigma}{\mu} = 18.09 \pm 0.18 \%$$

dE/dx [ADC counts/cm]

Energy loss | $30 < \phi < 33$

Count



XP

$$\mu = 503.5 \pm 0.8$$

$$\sigma = 93.3 \pm 0.7$$

$$\frac{\sigma}{\mu} = 18.53 \pm 0.17 \%$$

WF

$$\mu = 513.0 \pm 0.7$$

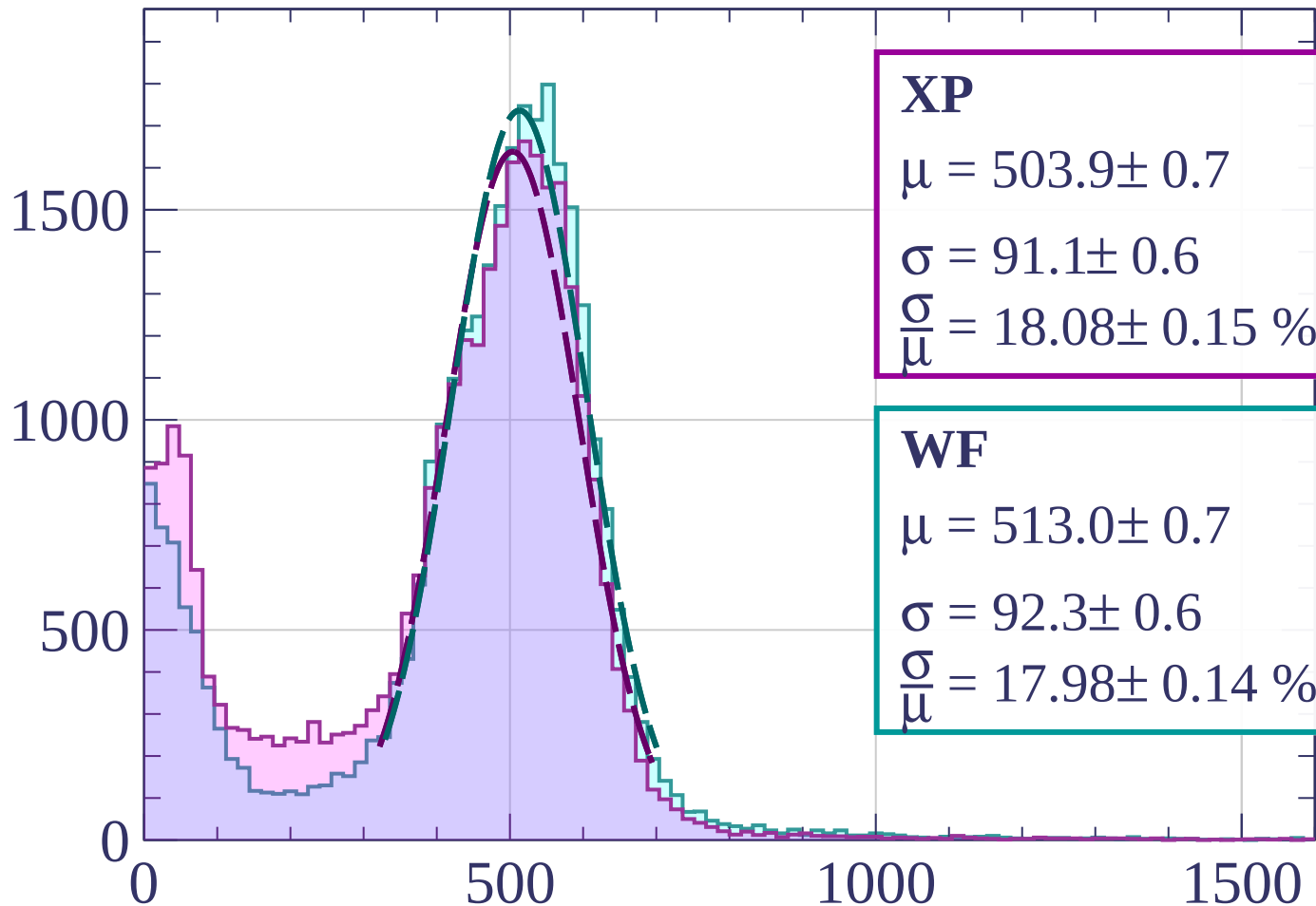
$$\sigma = 91.2 \pm 0.7$$

$$\frac{\sigma}{\mu} = 17.77 \pm 0.16 \%$$

dE/dx [ADC counts/cm]

Energy loss | $33 < \phi < 36$

Count



XP

$$\mu = 503.9 \pm 0.7$$

$$\sigma = 91.1 \pm 0.6$$

$$\frac{\sigma}{\mu} = 18.08 \pm 0.15 \%$$

WF

$$\mu = 513.0 \pm 0.7$$

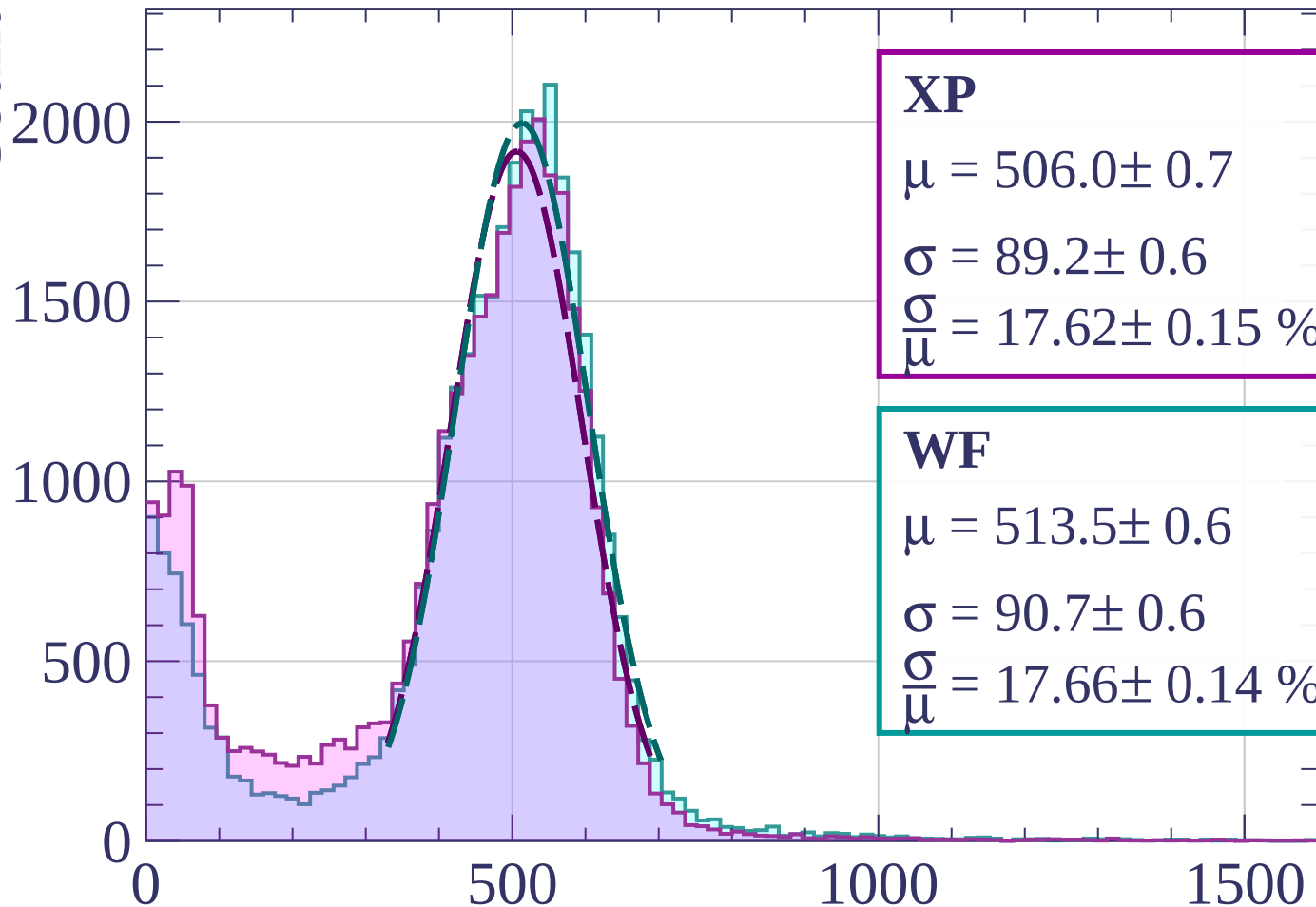
$$\sigma = 92.3 \pm 0.6$$

$$\frac{\sigma}{\mu} = 17.98 \pm 0.14 \%$$

dE/dx [ADC counts/cm]

Energy loss | $36 < \phi < 39$

Count



XP

$$\mu = 506.0 \pm 0.7$$

$$\sigma = 89.2 \pm 0.6$$

$$\frac{\sigma}{\mu} = 17.62 \pm 0.15 \%$$

WF

$$\mu = 513.5 \pm 0.6$$

$$\sigma = 90.7 \pm 0.6$$

$$\frac{\sigma}{\mu} = 17.66 \pm 0.14 \%$$

dE/dx [ADC counts/cm]

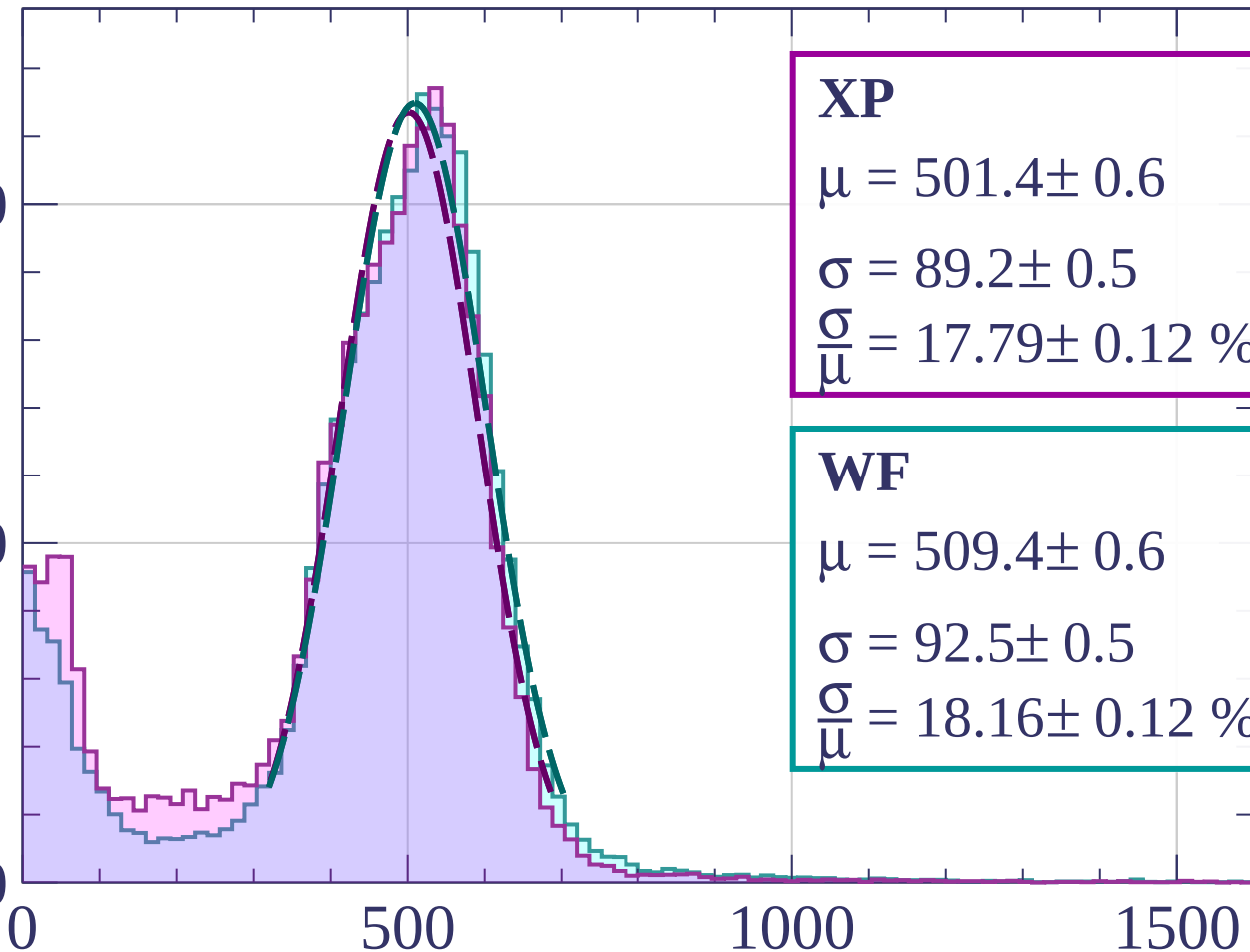
Energy loss | $39 < \phi < 42$

Count

2000

1000

0



XP

$$\mu = 501.4 \pm 0.6$$

$$\sigma = 89.2 \pm 0.5$$

$$\frac{\sigma}{\mu} = 17.79 \pm 0.12 \%$$

WF

$$\mu = 509.4 \pm 0.6$$

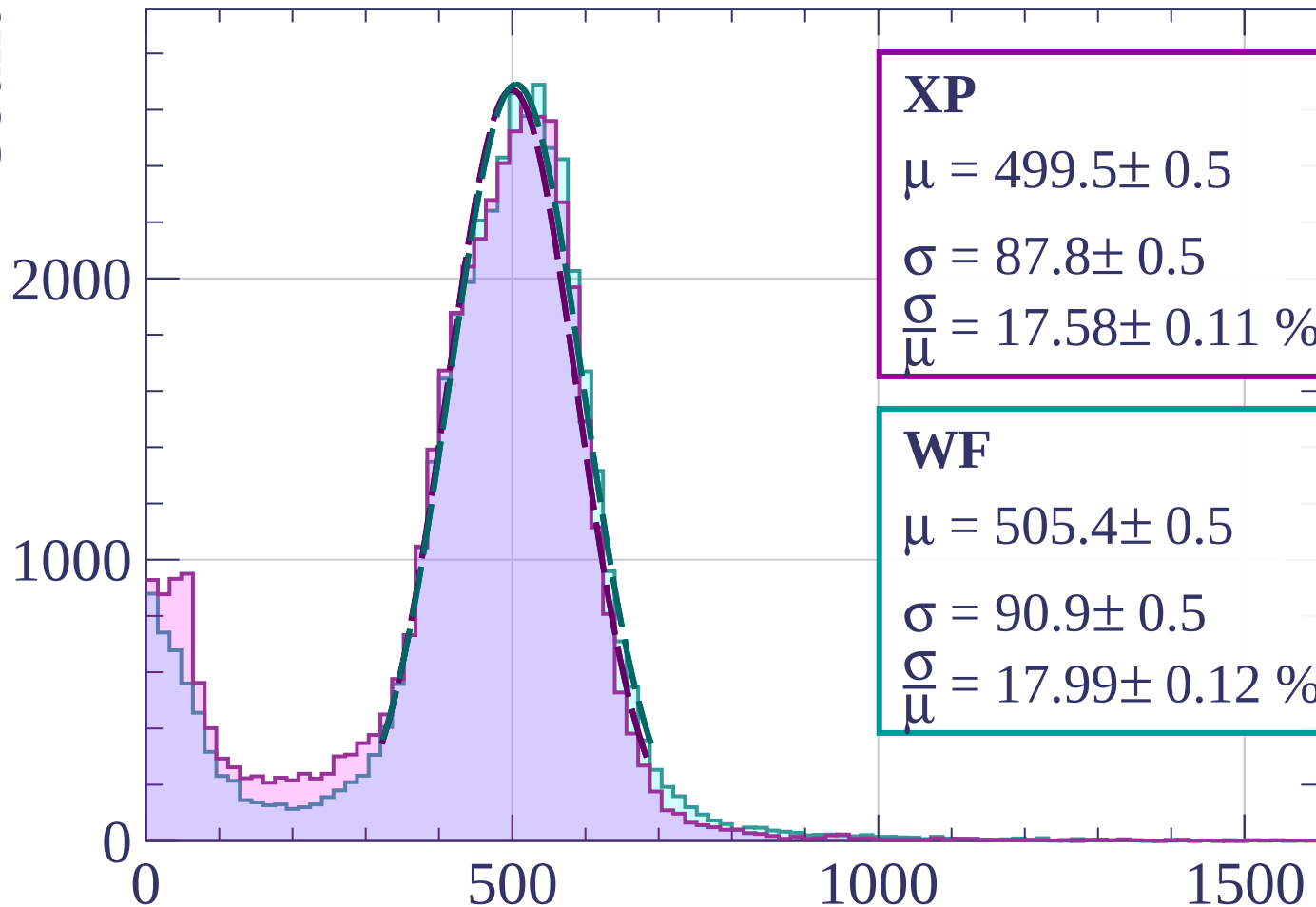
$$\sigma = 92.5 \pm 0.5$$

$$\frac{\sigma}{\mu} = 18.16 \pm 0.12 \%$$

dE/dx [ADC counts/cm]

Energy loss | $42 < \varphi < 45$

Count



XP

$$\mu = 499.5 \pm 0.5$$

$$\sigma = 87.8 \pm 0.5$$

$$\frac{\sigma}{\mu} = 17.58 \pm 0.11 \%$$

WF

$$\mu = 505.4 \pm 0.5$$

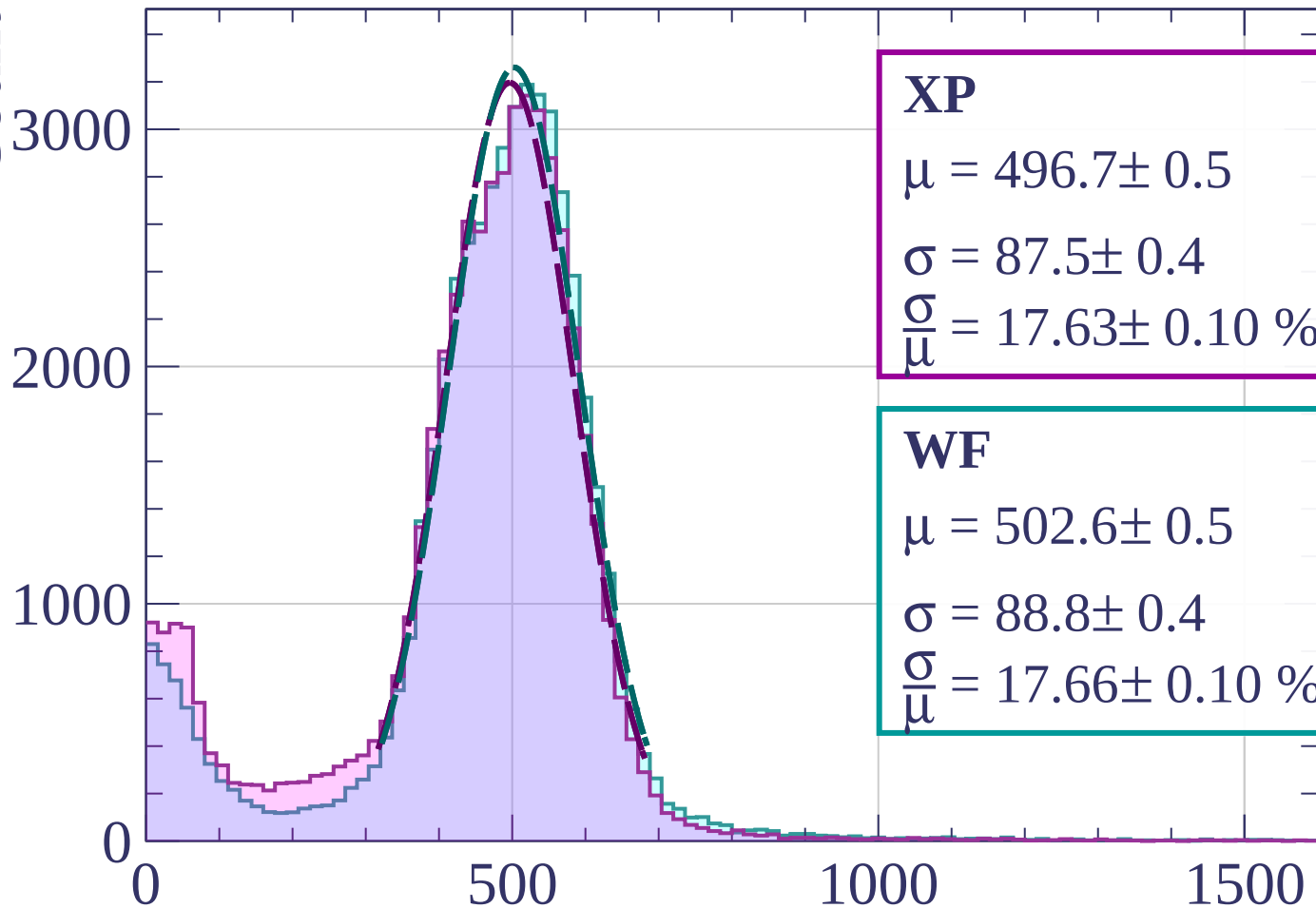
$$\sigma = 90.9 \pm 0.5$$

$$\frac{\sigma}{\mu} = 17.99 \pm 0.12 \%$$

dE/dx [ADC counts/cm]

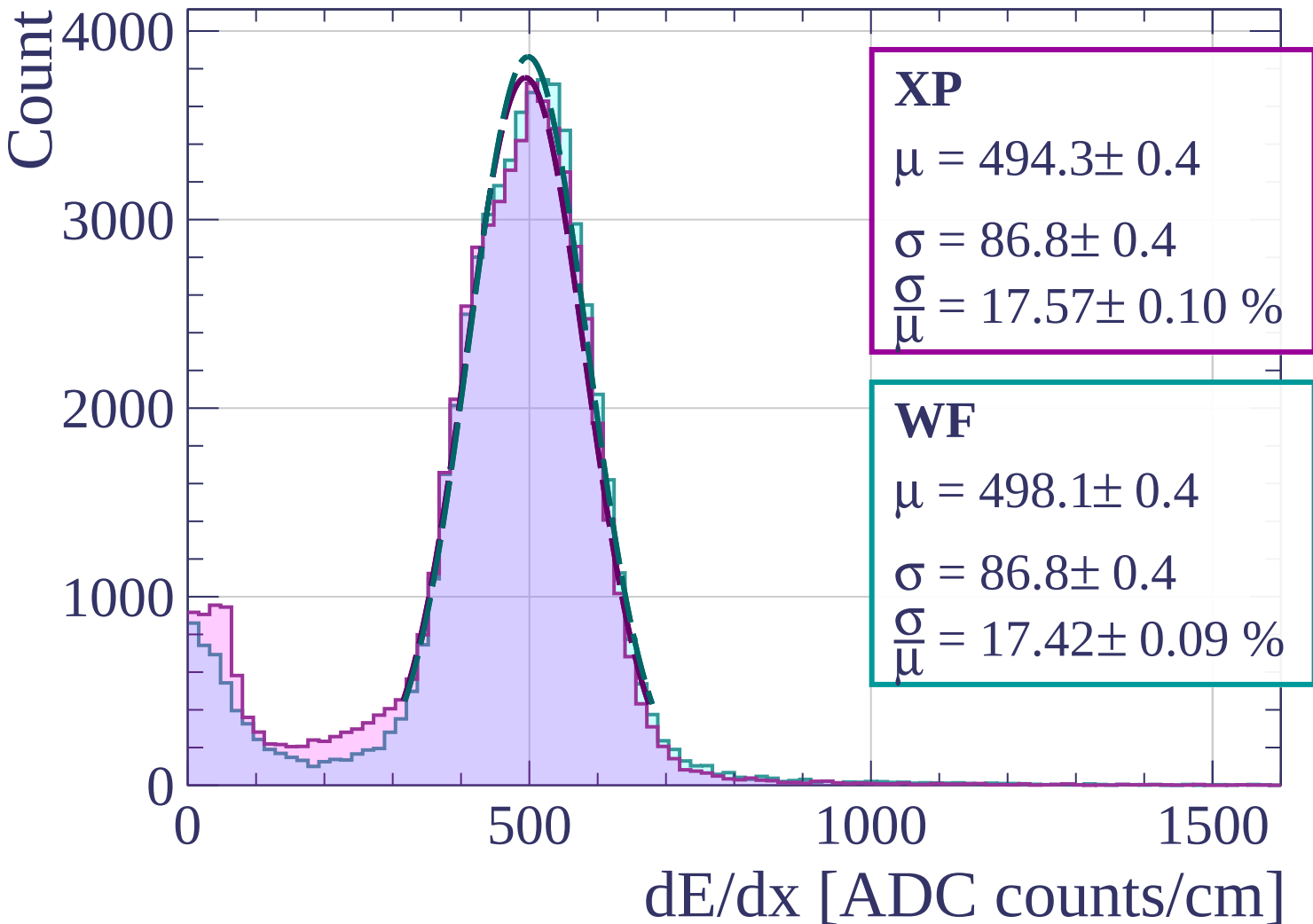
Energy loss | $45 < \varphi < 48$

Count

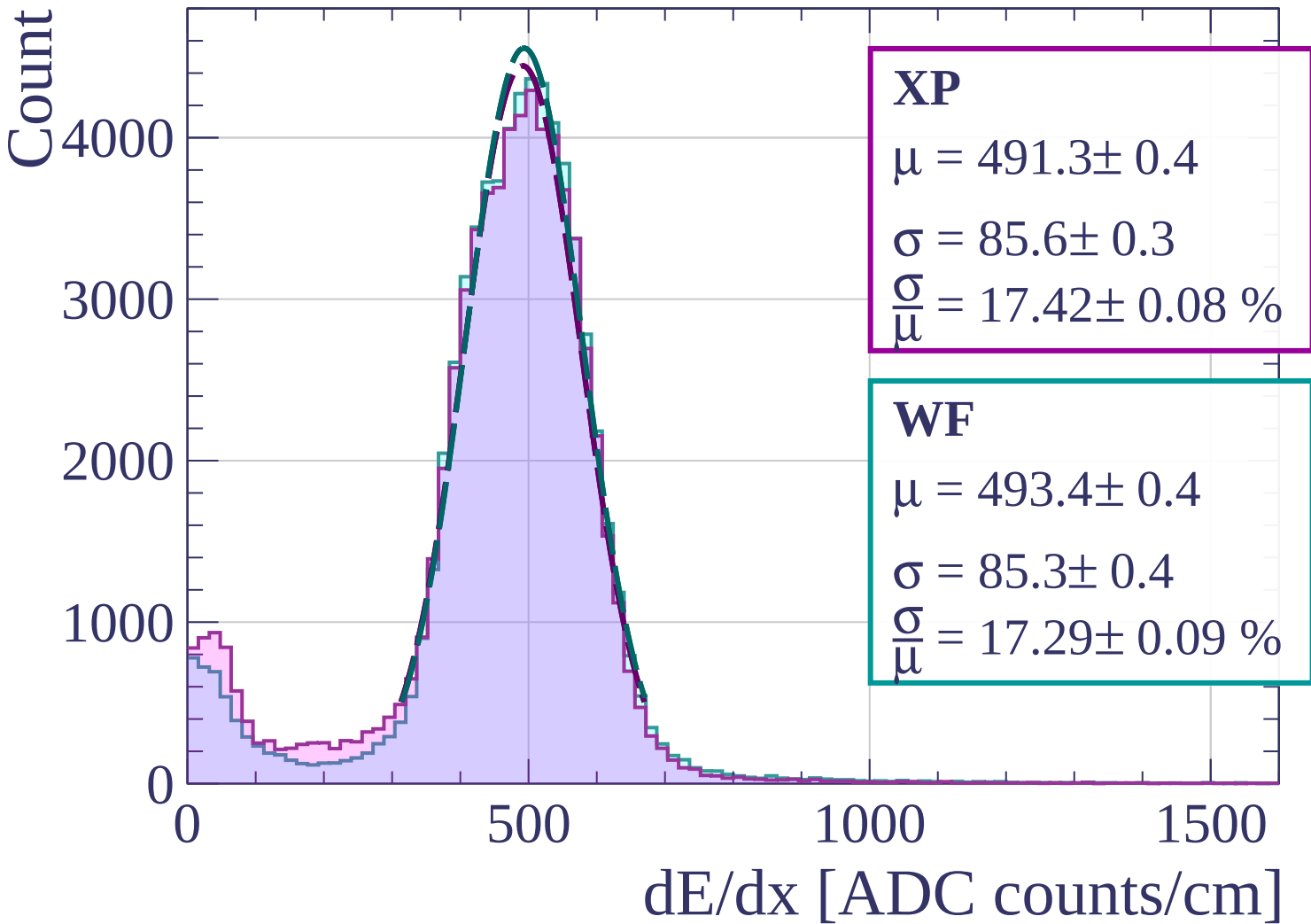


dE/dx [ADC counts/cm]

Energy loss | $48 < \varphi < 51$

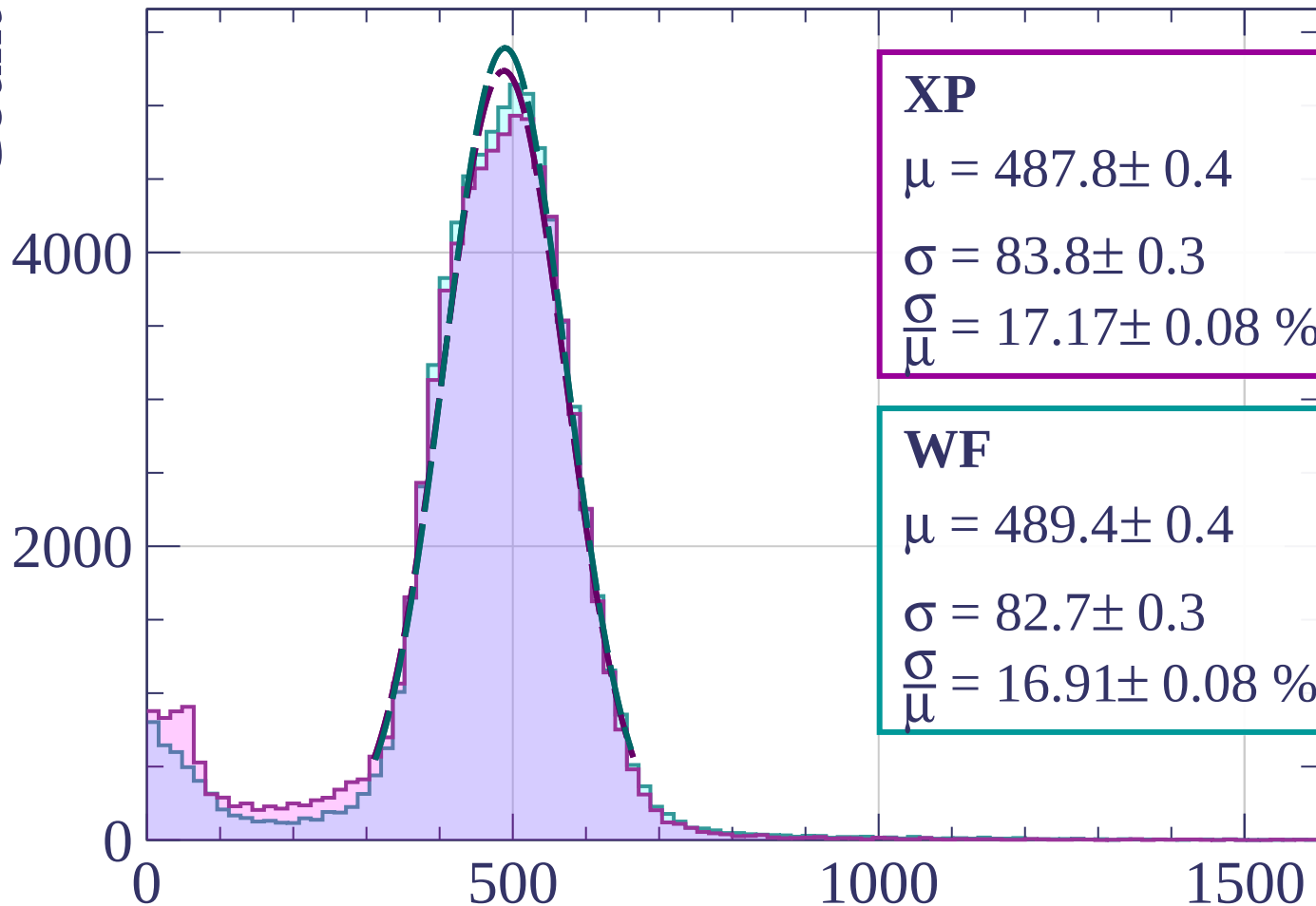


Energy loss | $51 < \phi < 54$



Energy loss | $54 < \varphi < 57$

Count



XP

$$\mu = 487.8 \pm 0.4$$

$$\sigma = 83.8 \pm 0.3$$

$$\frac{\sigma}{\mu} = 17.17 \pm 0.08 \%$$

WF

$$\mu = 489.4 \pm 0.4$$

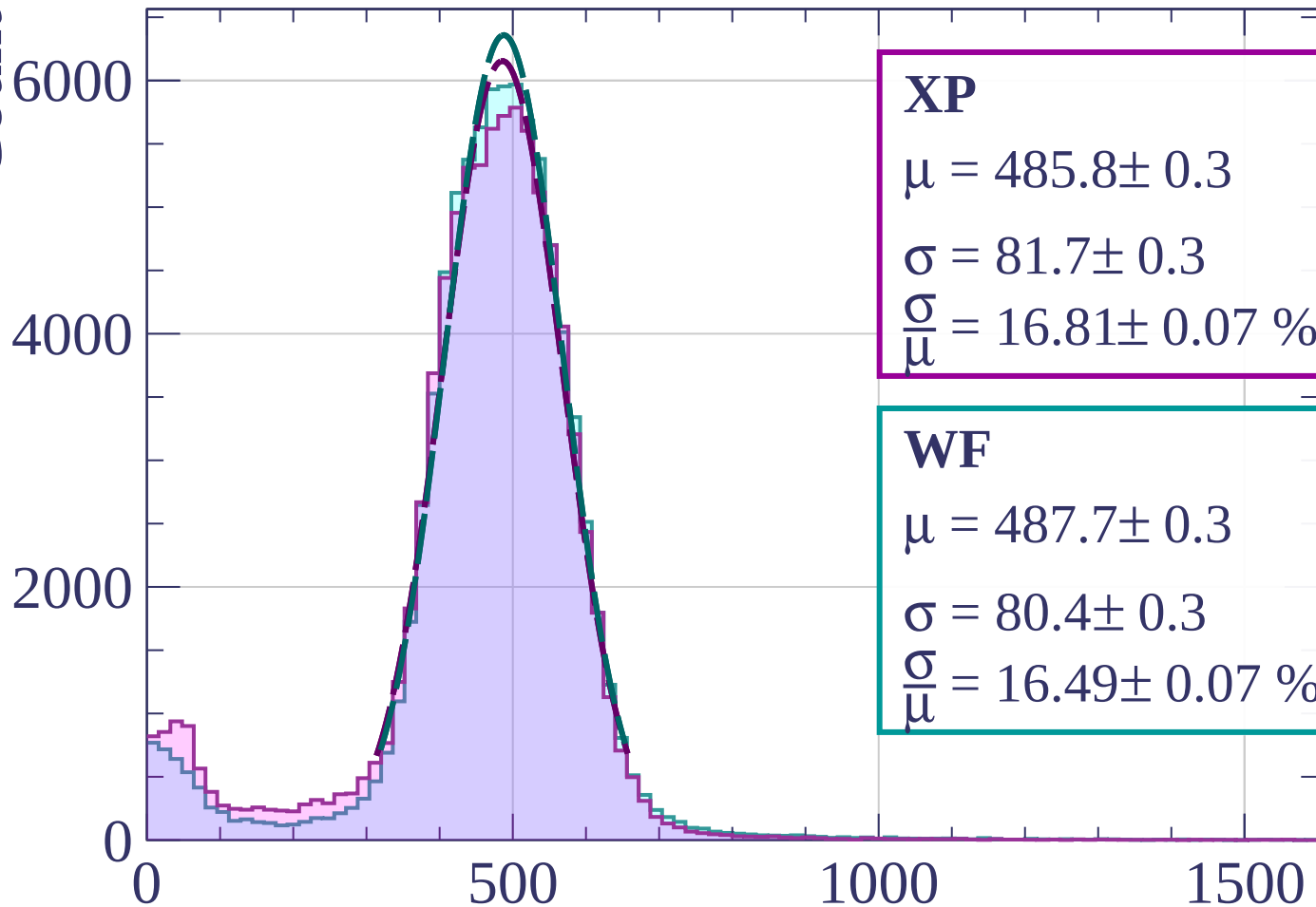
$$\sigma = 82.7 \pm 0.3$$

$$\frac{\sigma}{\mu} = 16.91 \pm 0.08 \%$$

dE/dx [ADC counts/cm]

Energy loss | $57 < \phi < 60$

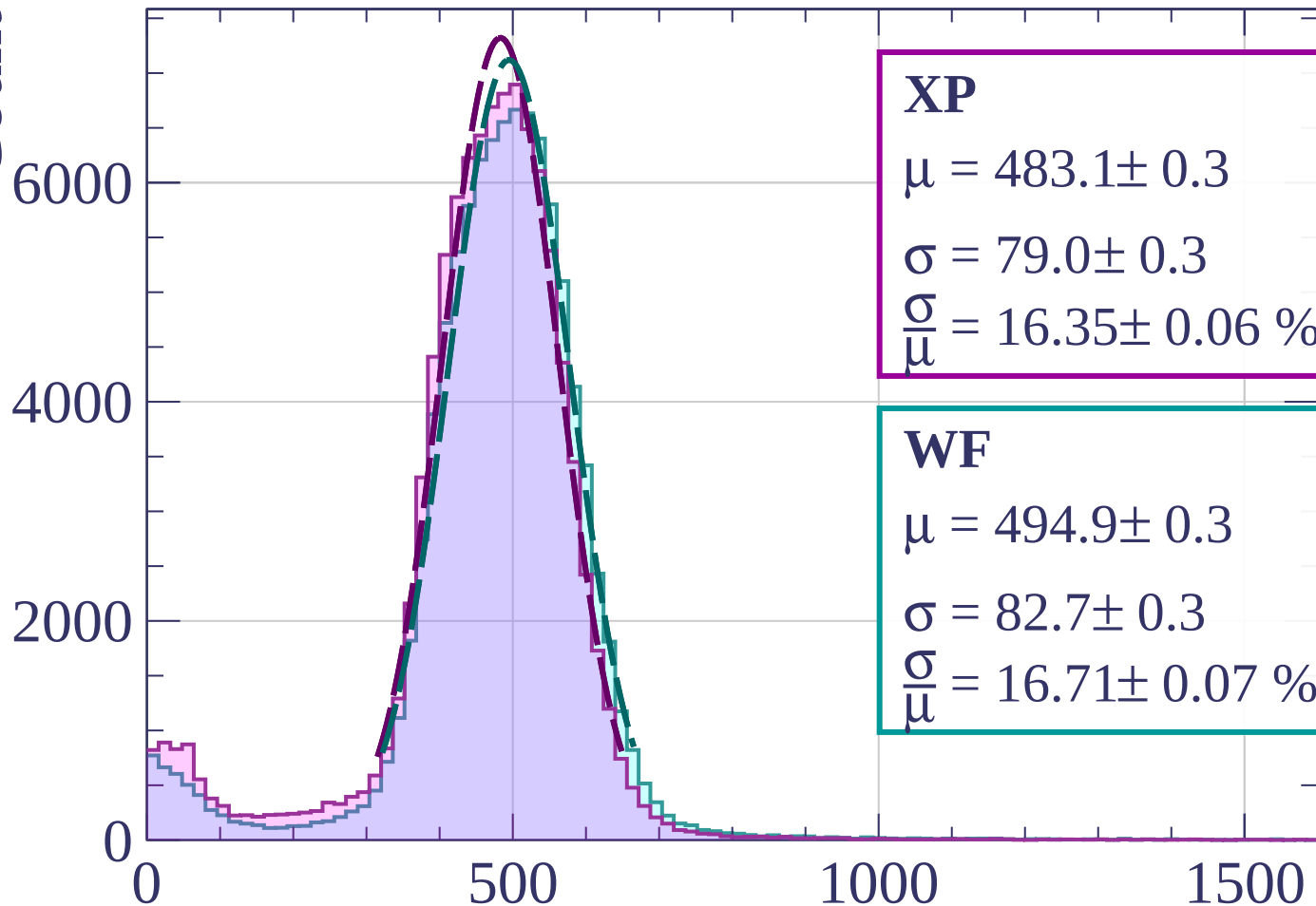
Count



dE/dx [ADC counts/cm]

Energy loss | $60 < \phi < 63$

Count



XP

$$\mu = 483.1 \pm 0.3$$

$$\sigma = 79.0 \pm 0.3$$

$$\frac{\sigma}{\mu} = 16.35 \pm 0.06 \%$$

WF

$$\mu = 494.9 \pm 0.3$$

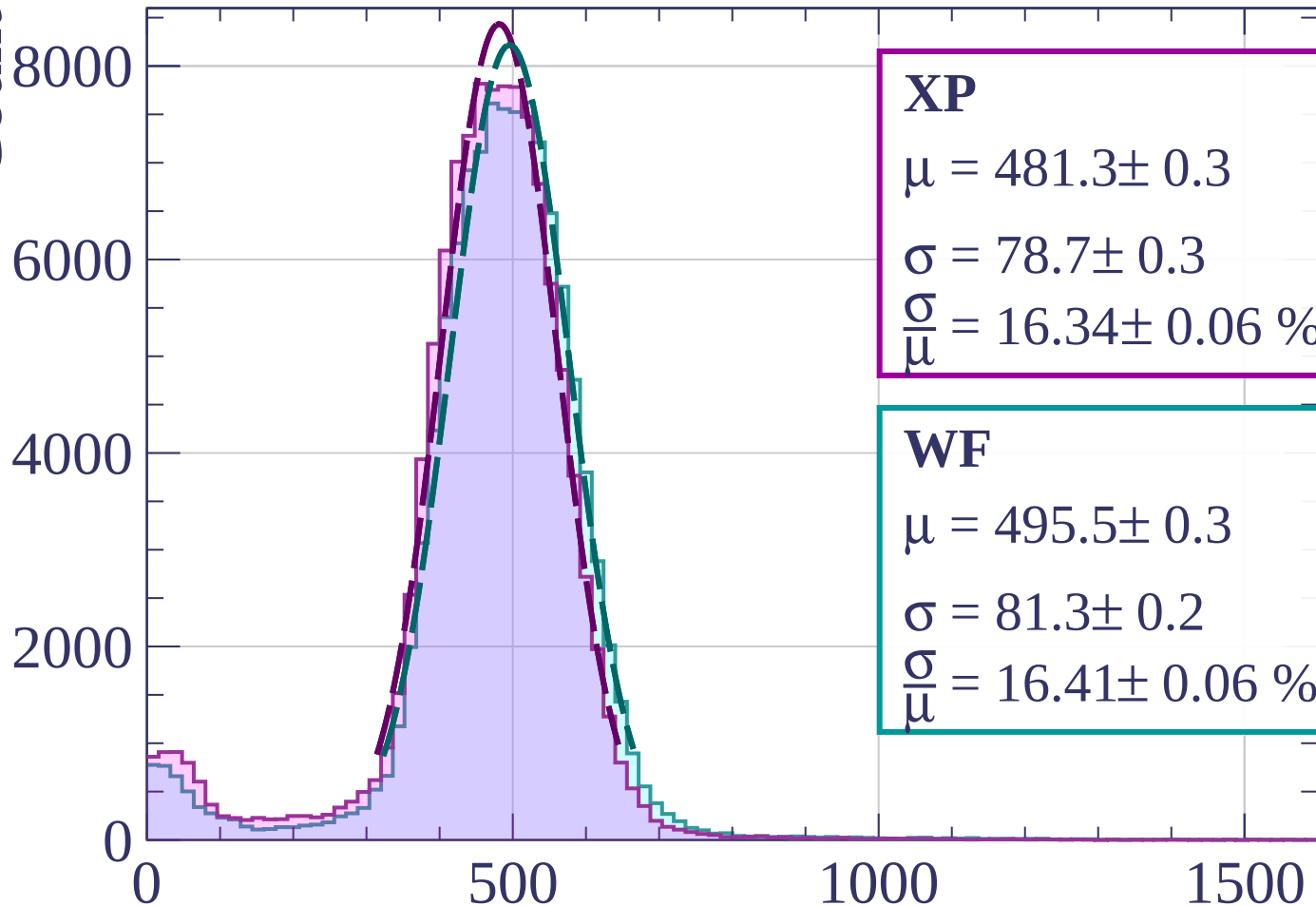
$$\sigma = 82.7 \pm 0.3$$

$$\frac{\sigma}{\mu} = 16.71 \pm 0.07 \%$$

dE/dx [ADC counts/cm]

Energy loss | $63 < \phi < 66$

Count



XP

$$\mu = 481.3 \pm 0.3$$

$$\sigma = 78.7 \pm 0.3$$

$$\frac{\sigma}{\mu} = 16.34 \pm 0.06 \%$$

WF

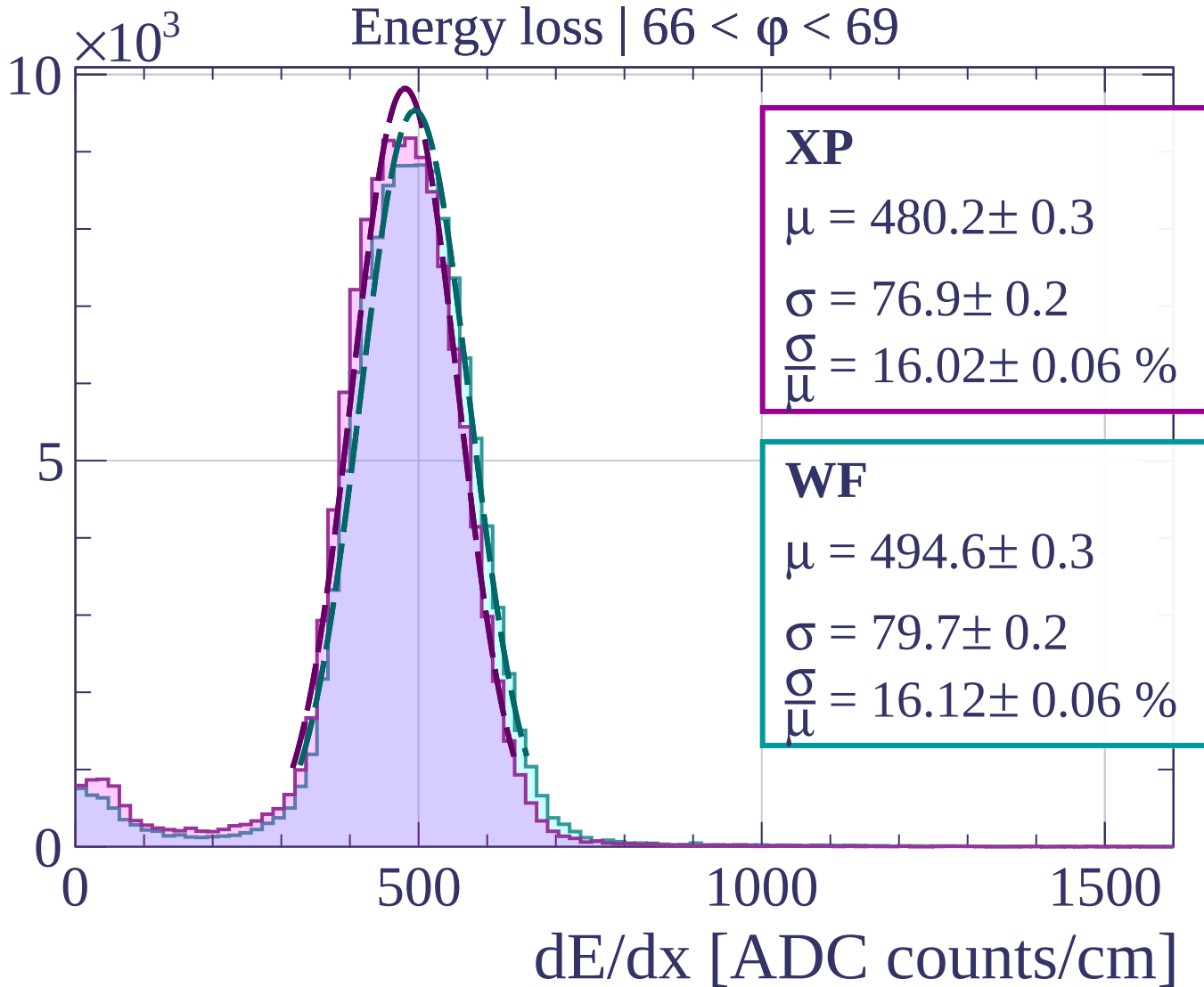
$$\mu = 495.5 \pm 0.3$$

$$\sigma = 81.3 \pm 0.2$$

$$\frac{\sigma}{\mu} = 16.41 \pm 0.06 \%$$

dE/dx [ADC counts/cm]

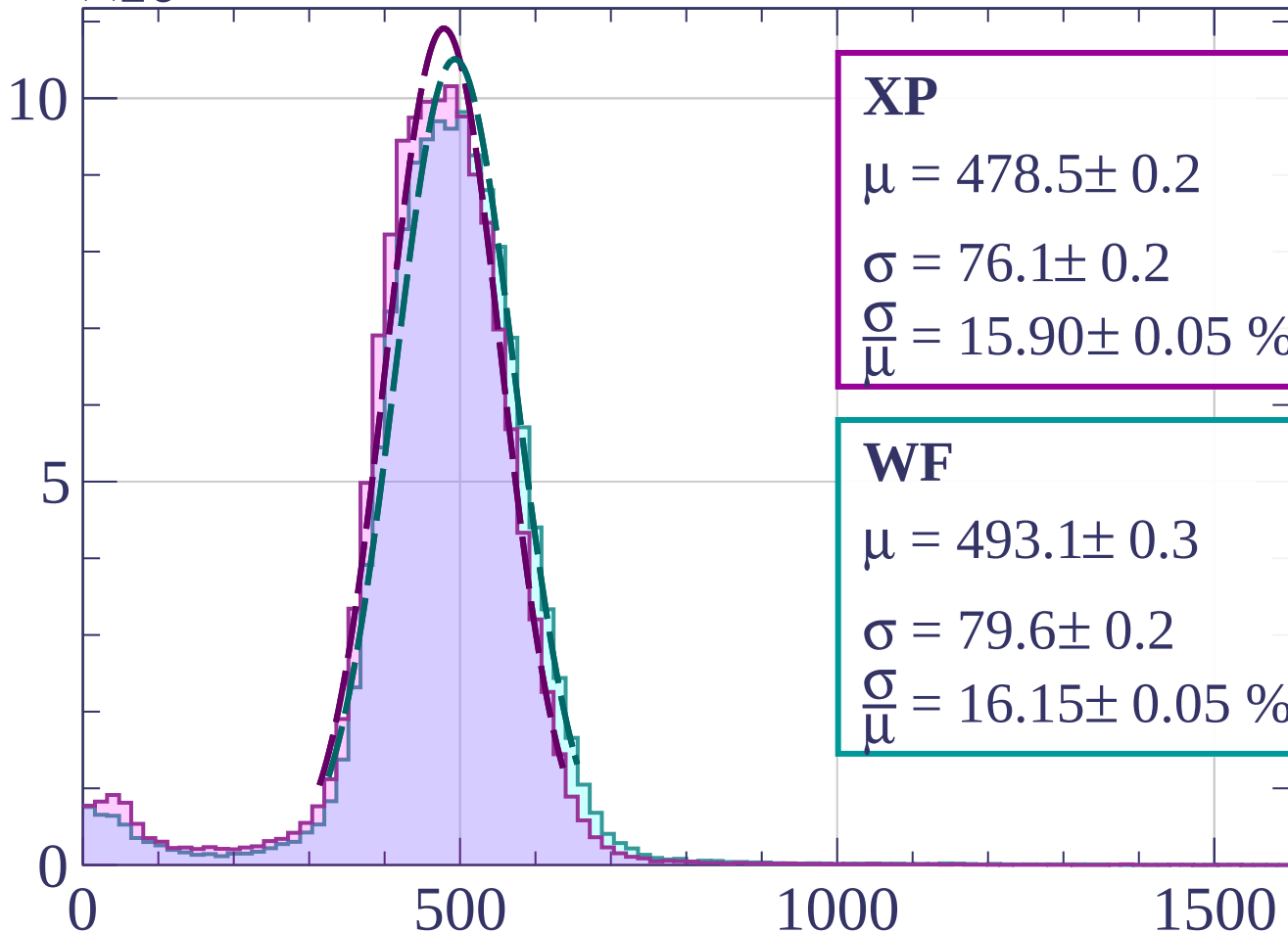
Count



Count

$\times 10^3$

Energy loss | $69 < \phi < 72$



dE/dx [ADC counts/cm]

Count

$\times 10^3$

Energy loss | $72 < \varphi < 75$

XP

$$\mu = 476.1 \pm 0.2$$

$$\sigma = 76.2 \pm 0.2$$

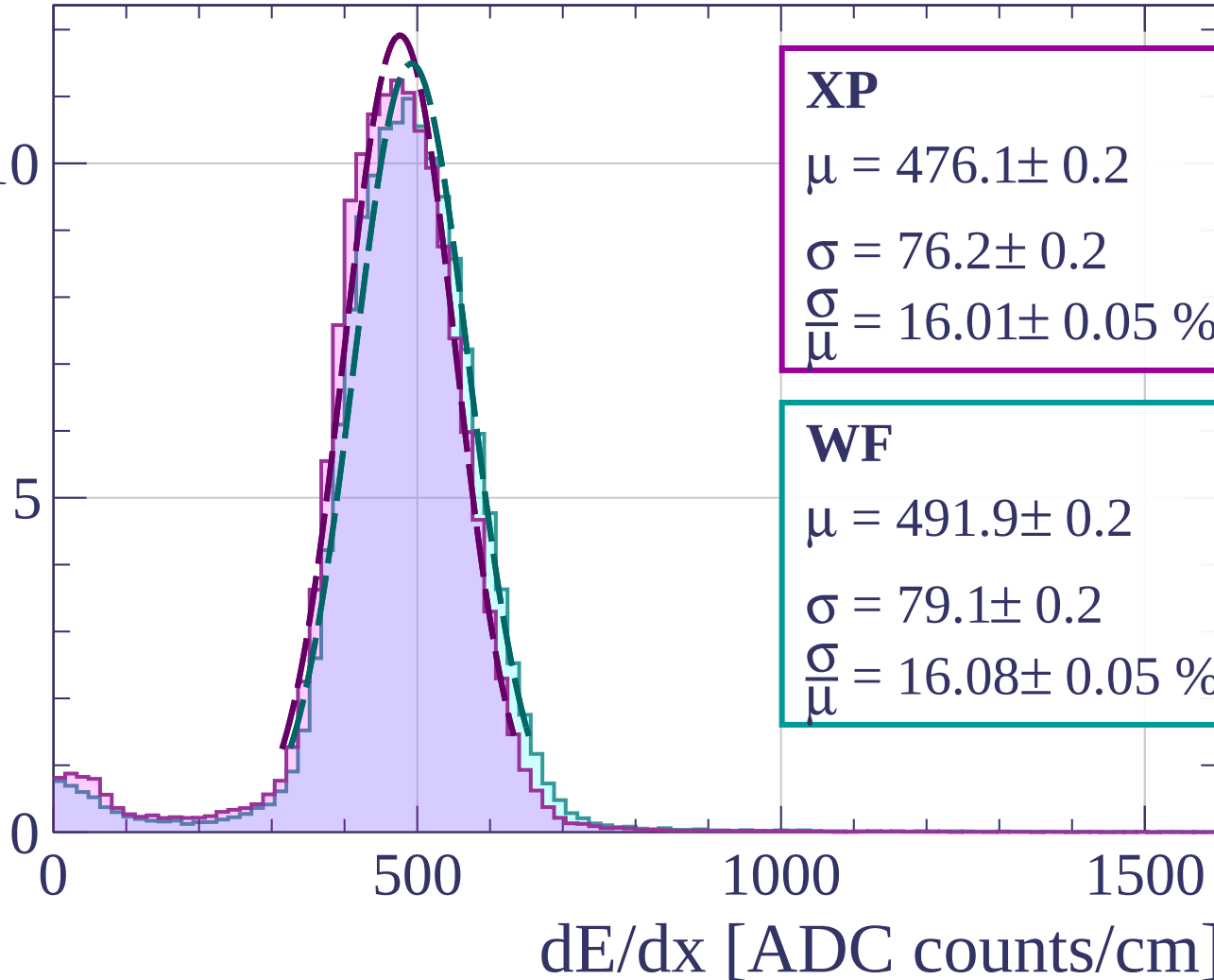
$$\frac{\sigma}{\mu} = 16.01 \pm 0.05 \%$$

WF

$$\mu = 491.9 \pm 0.2$$

$$\sigma = 79.1 \pm 0.2$$

$$\frac{\sigma}{\mu} = 16.08 \pm 0.05 \%$$



Count

$\times 10^3$

Energy loss | $75 < \varphi < 78$

XP

$$\mu = 475.0 \pm 0.2$$

$$\sigma = 76.1 \pm 0.2$$

$$\frac{\sigma}{\mu} = 16.02 \pm 0.05 \%$$

WF

$$\mu = 491.2 \pm 0.2$$

$$\sigma = 79.2 \pm 0.2$$

$$\frac{\sigma}{\mu} = 16.12 \pm 0.05 \%$$

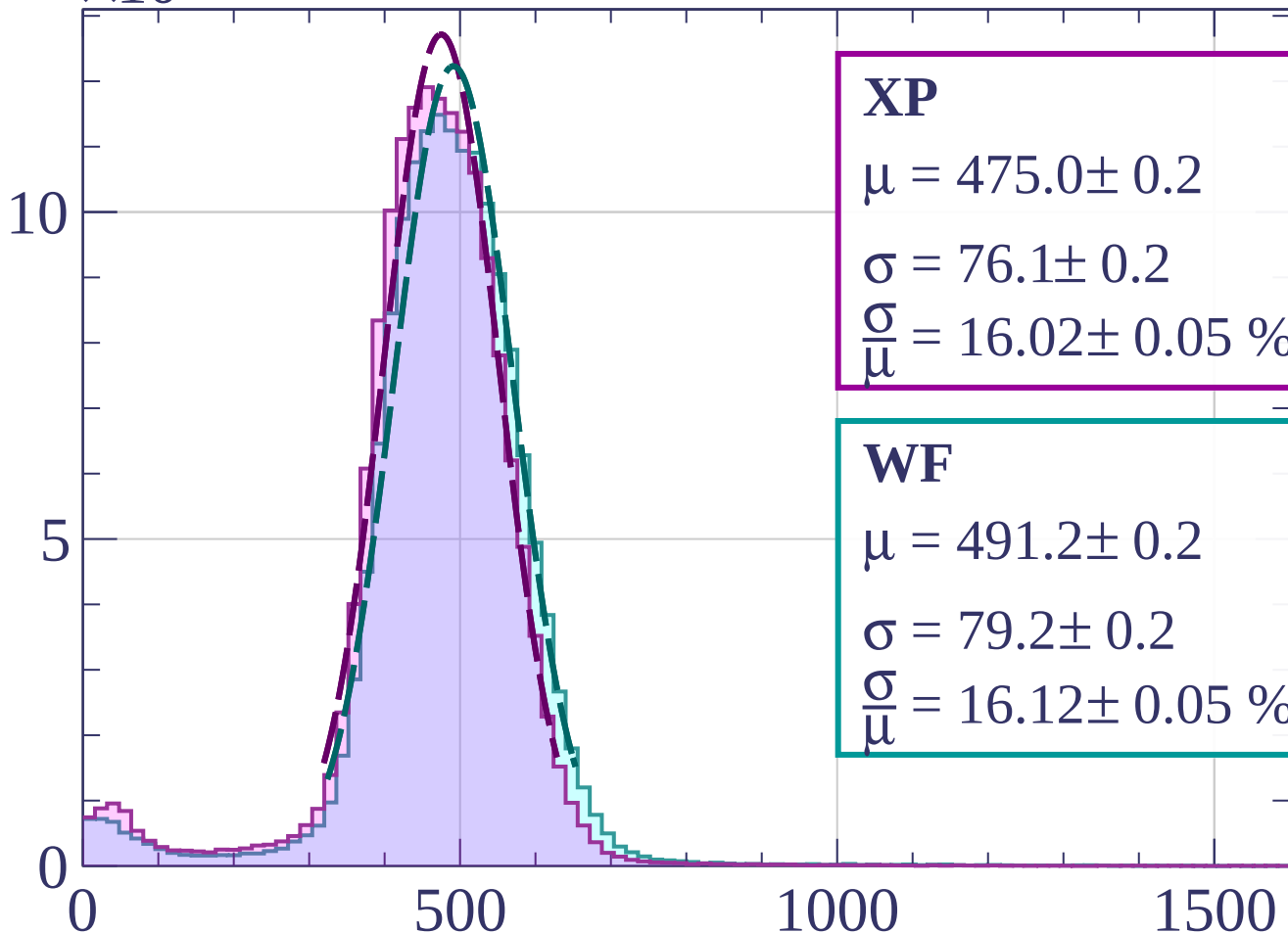
0

500

1000

1500

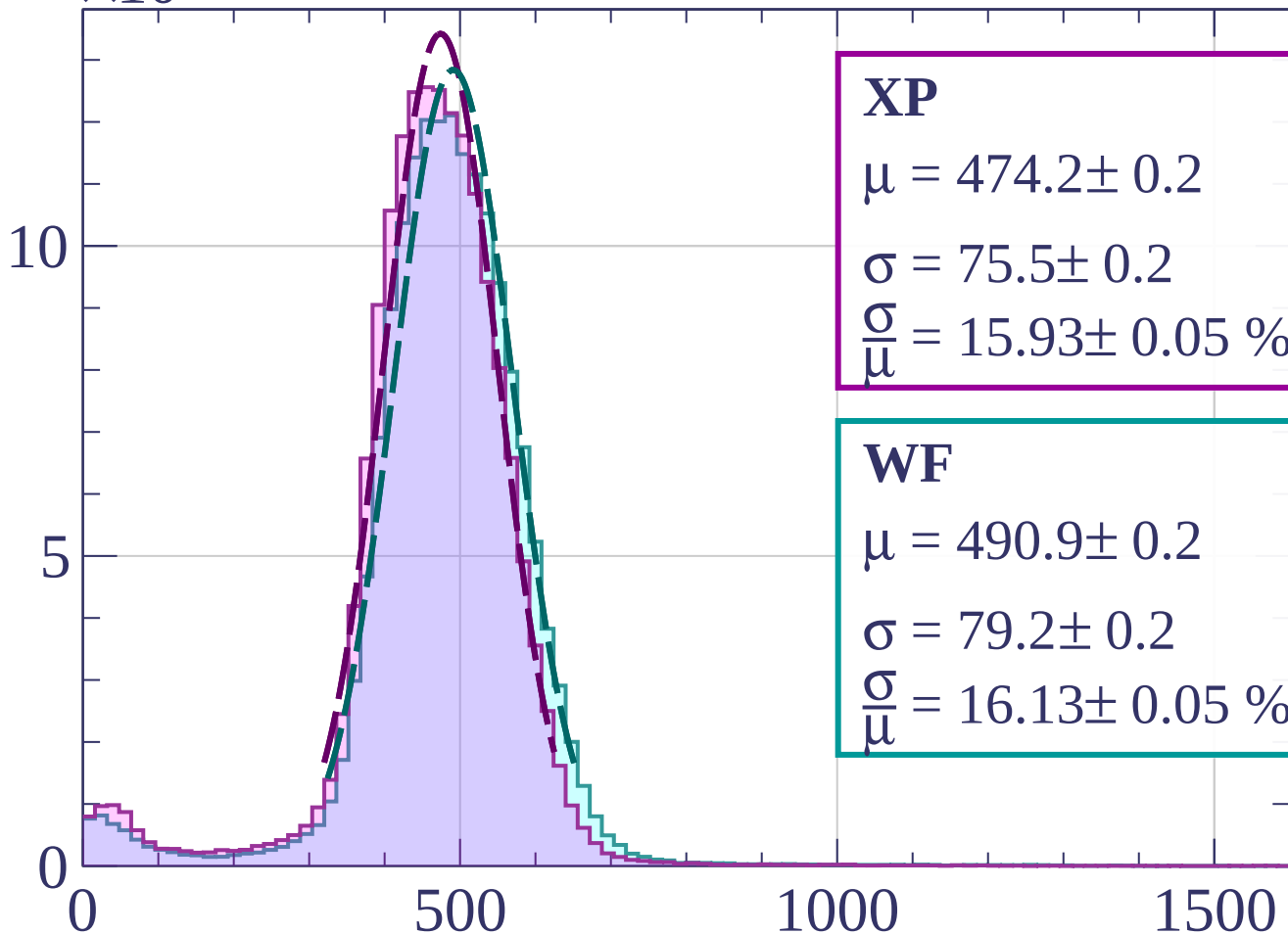
dE/dx [ADC counts/cm]



Count

$\times 10^3$

Energy loss | $78 < \varphi < 81$



XP

$$\mu = 474.2 \pm 0.2$$

$$\sigma = 75.5 \pm 0.2$$

$$\frac{\sigma}{\mu} = 15.93 \pm 0.05 \%$$

WF

$$\mu = 490.9 \pm 0.2$$

$$\sigma = 79.2 \pm 0.2$$

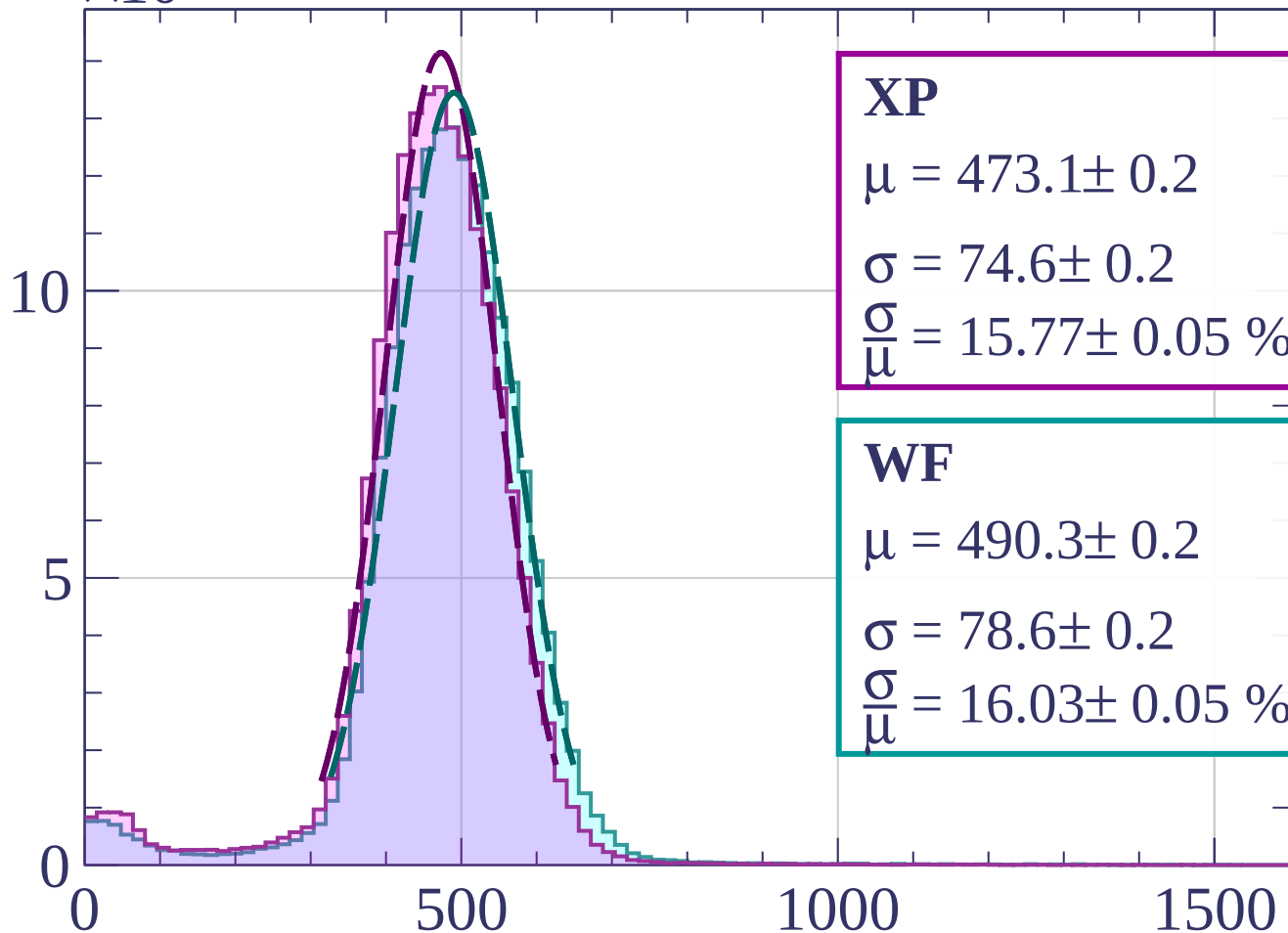
$$\frac{\sigma}{\mu} = 16.13 \pm 0.05 \%$$

dE/dx [ADC counts/cm]

Count

$\times 10^3$

Energy loss | $81 < \phi < 84$



XP

$$\mu = 473.1 \pm 0.2$$

$$\sigma = 74.6 \pm 0.2$$

$$\frac{\sigma}{\mu} = 15.77 \pm 0.05 \%$$

WF

$$\mu = 490.3 \pm 0.2$$

$$\sigma = 78.6 \pm 0.2$$

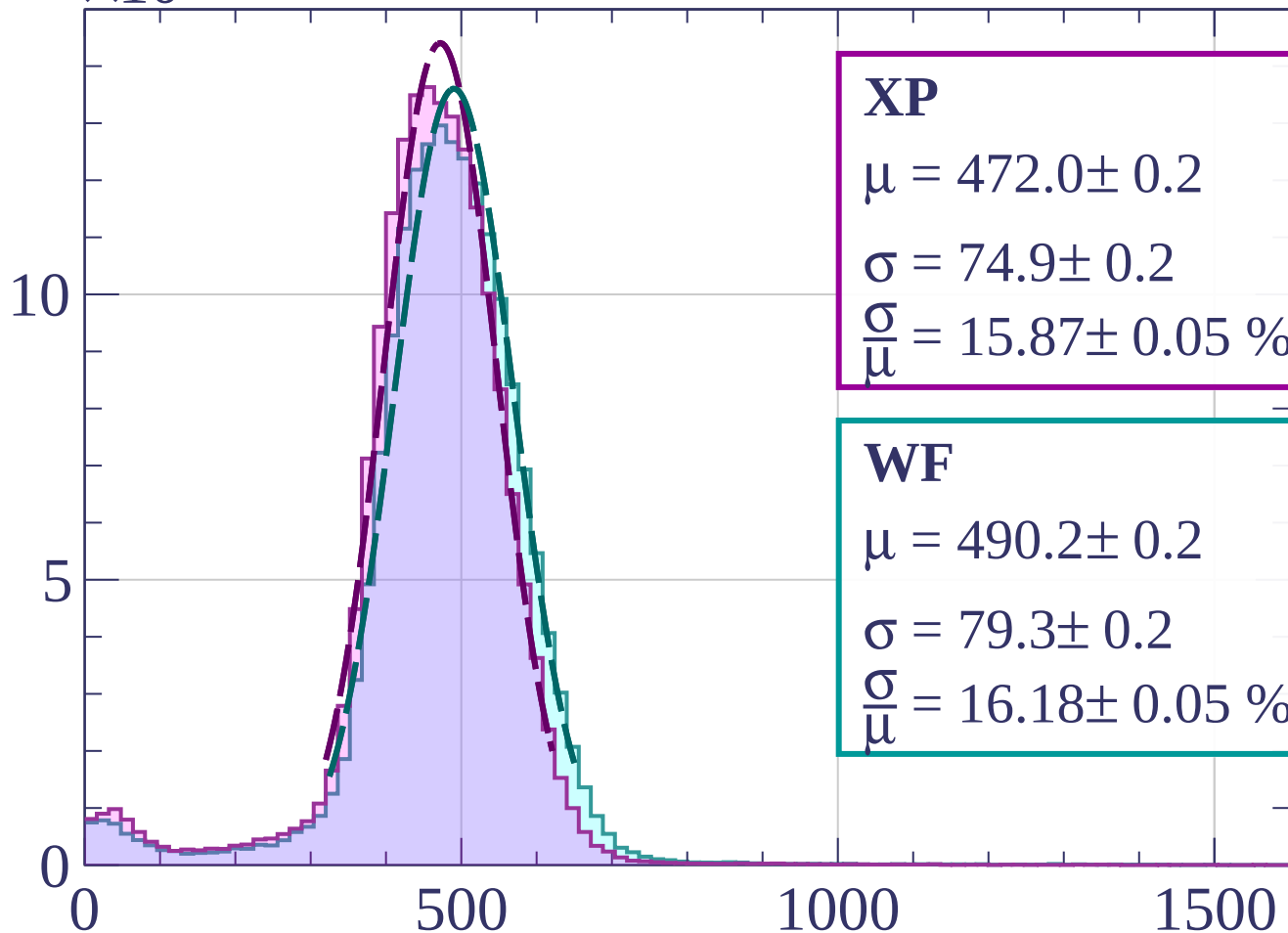
$$\frac{\sigma}{\mu} = 16.03 \pm 0.05 \%$$

dE/dx [ADC counts/cm]

Count

$\times 10^3$

Energy loss | $84 < \phi < 87$



XP

$$\mu = 472.0 \pm 0.2$$

$$\sigma = 74.9 \pm 0.2$$

$$\frac{\sigma}{\mu} = 15.87 \pm 0.05 \%$$

WF

$$\mu = 490.2 \pm 0.2$$

$$\sigma = 79.3 \pm 0.2$$

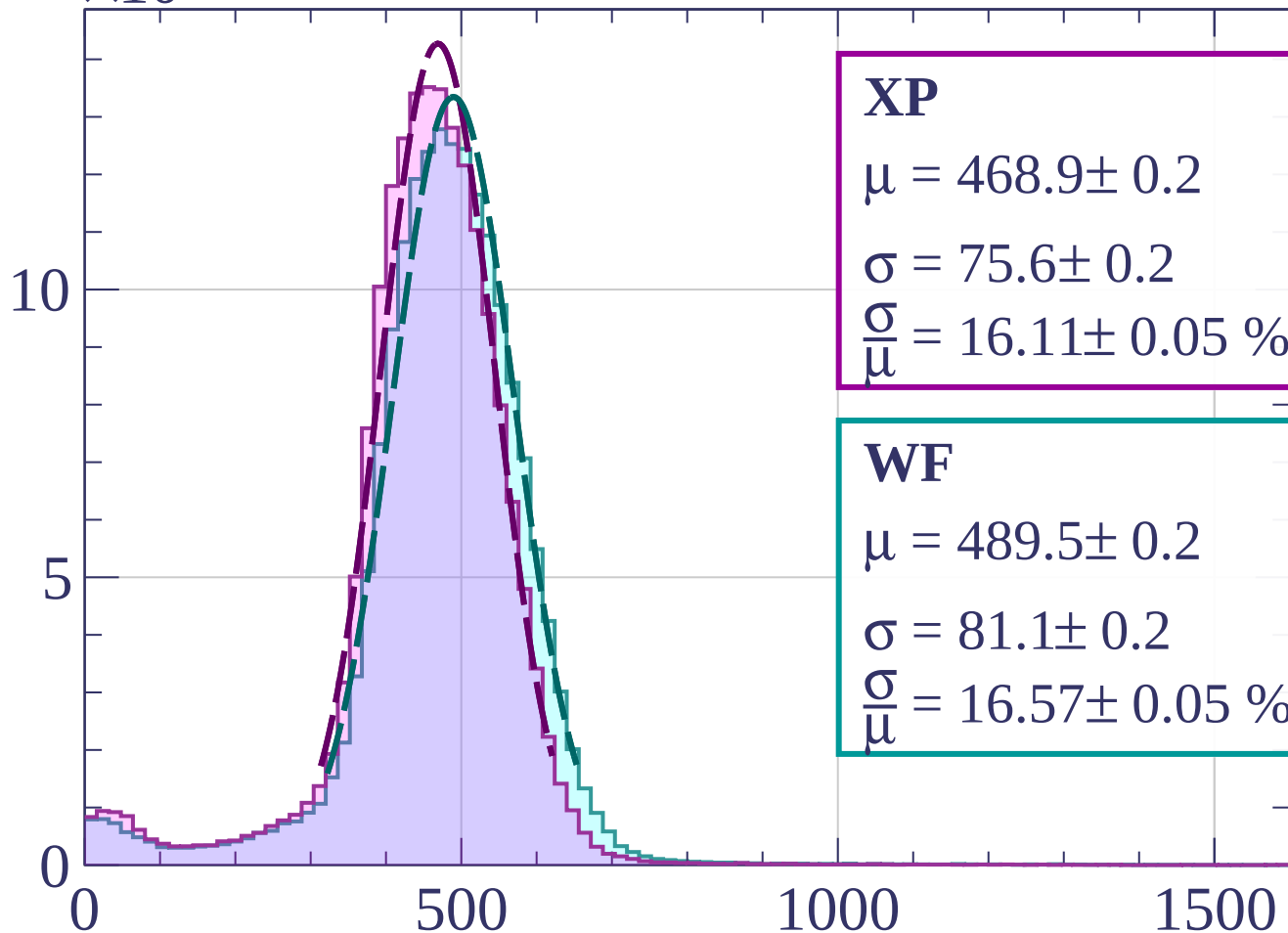
$$\frac{\sigma}{\mu} = 16.18 \pm 0.05 \%$$

dE/dx [ADC counts/cm]

Count

$\times 10^3$

Energy loss | $87 < \phi < 90$



XP

$$\mu = 468.9 \pm 0.2$$

$$\sigma = 75.6 \pm 0.2$$

$$\frac{\sigma}{\mu} = 16.11 \pm 0.05 \%$$

WF

$$\mu = 489.5 \pm 0.2$$

$$\sigma = 81.1 \pm 0.2$$

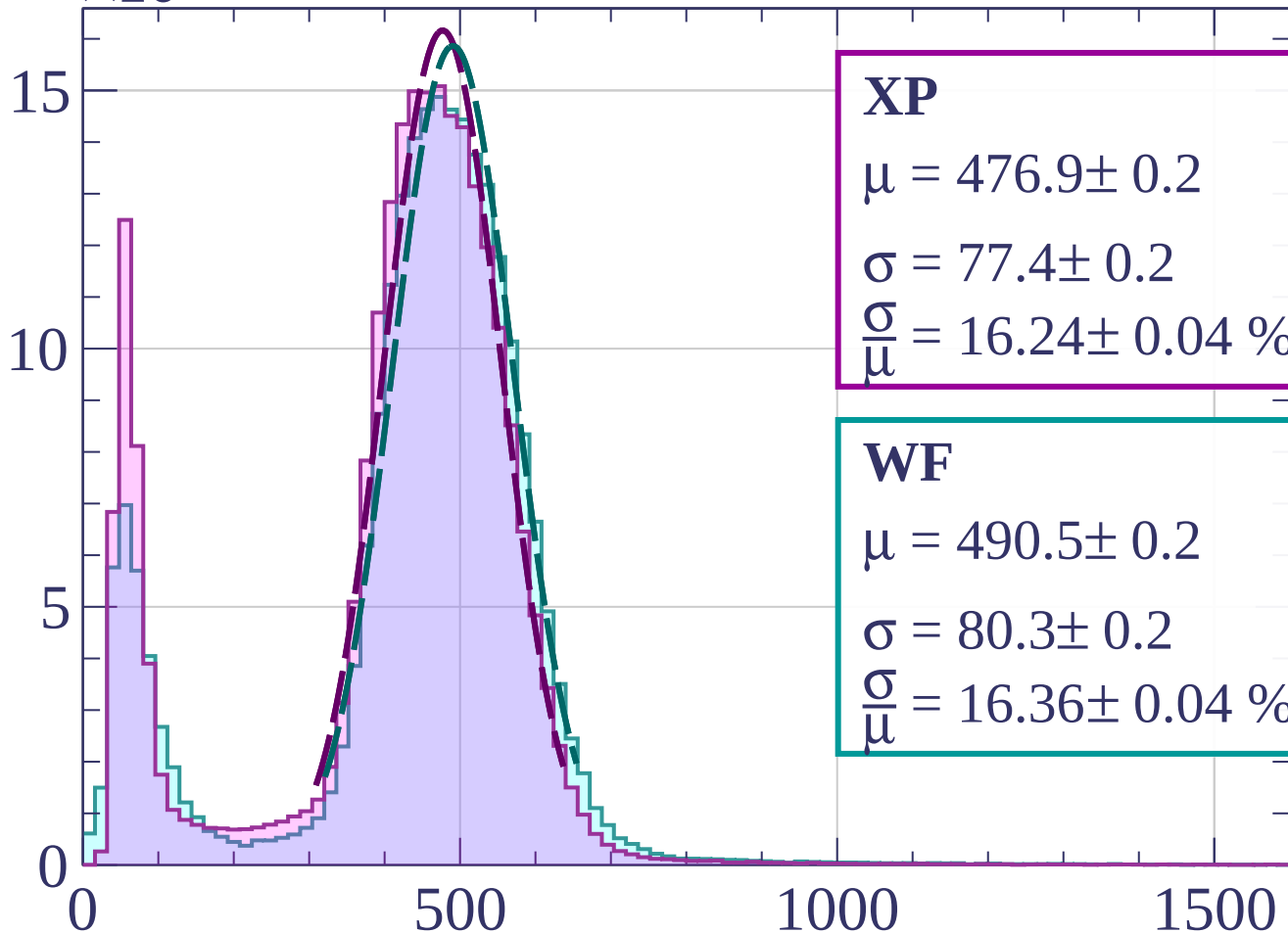
$$\frac{\sigma}{\mu} = 16.57 \pm 0.05 \%$$

dE/dx [ADC counts/cm]

Count

$\times 10^3$

Energy loss | $0 < \theta < 3$



XP

$$\mu = 476.9 \pm 0.2$$

$$\sigma = 77.4 \pm 0.2$$

$$\frac{\sigma}{\mu} = 16.24 \pm 0.04 \%$$

WF

$$\mu = 490.5 \pm 0.2$$

$$\sigma = 80.3 \pm 0.2$$

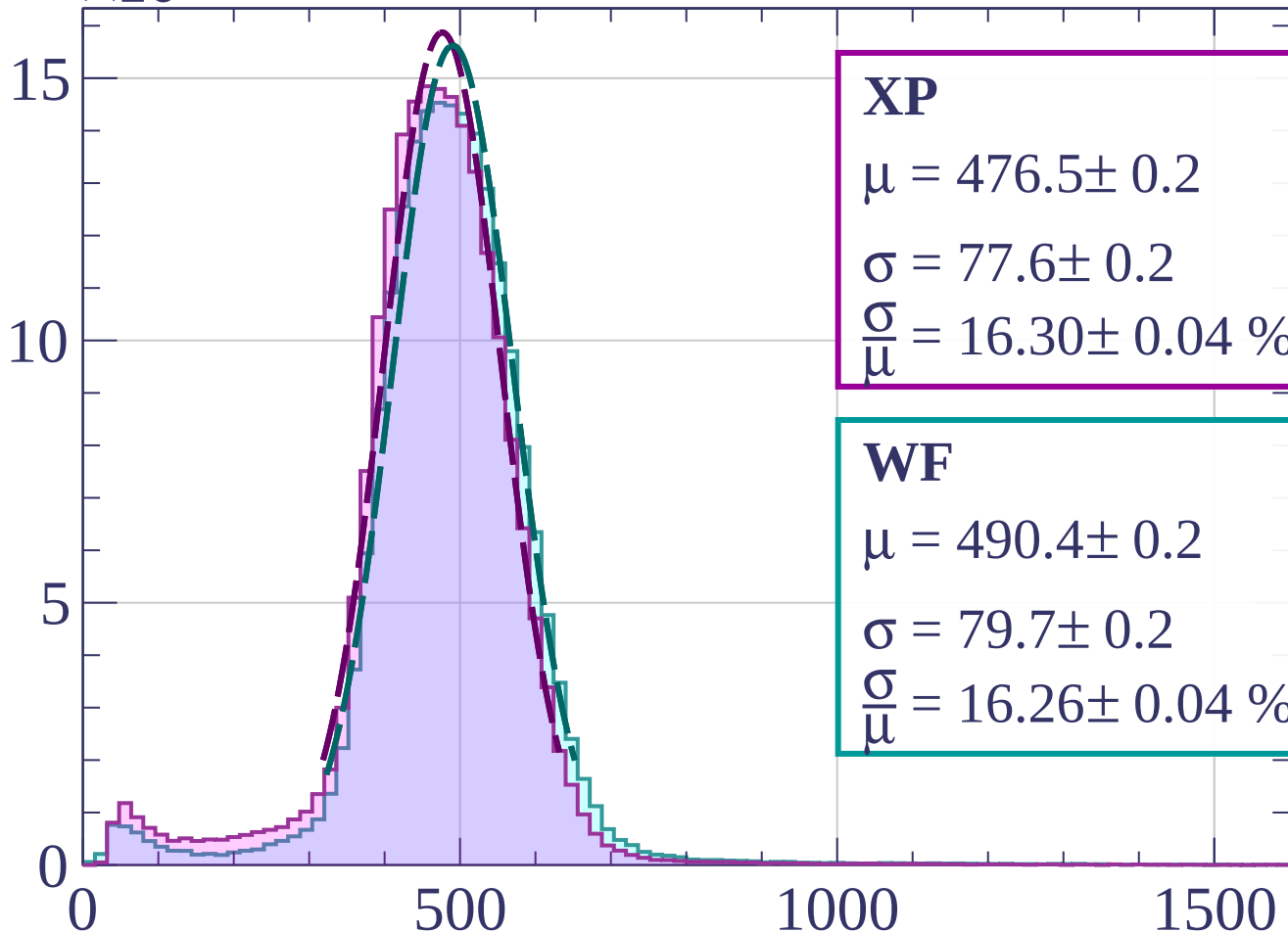
$$\frac{\sigma}{\mu} = 16.36 \pm 0.04 \%$$

dE/dx [ADC counts/cm]

Count

$\times 10^3$

Energy loss | $3 < \theta < 6$



XP

$$\mu = 476.5 \pm 0.2$$

$$\sigma = 77.6 \pm 0.2$$

$$\frac{\sigma}{\mu} = 16.30 \pm 0.04 \%$$

WF

$$\mu = 490.4 \pm 0.2$$

$$\sigma = 79.7 \pm 0.2$$

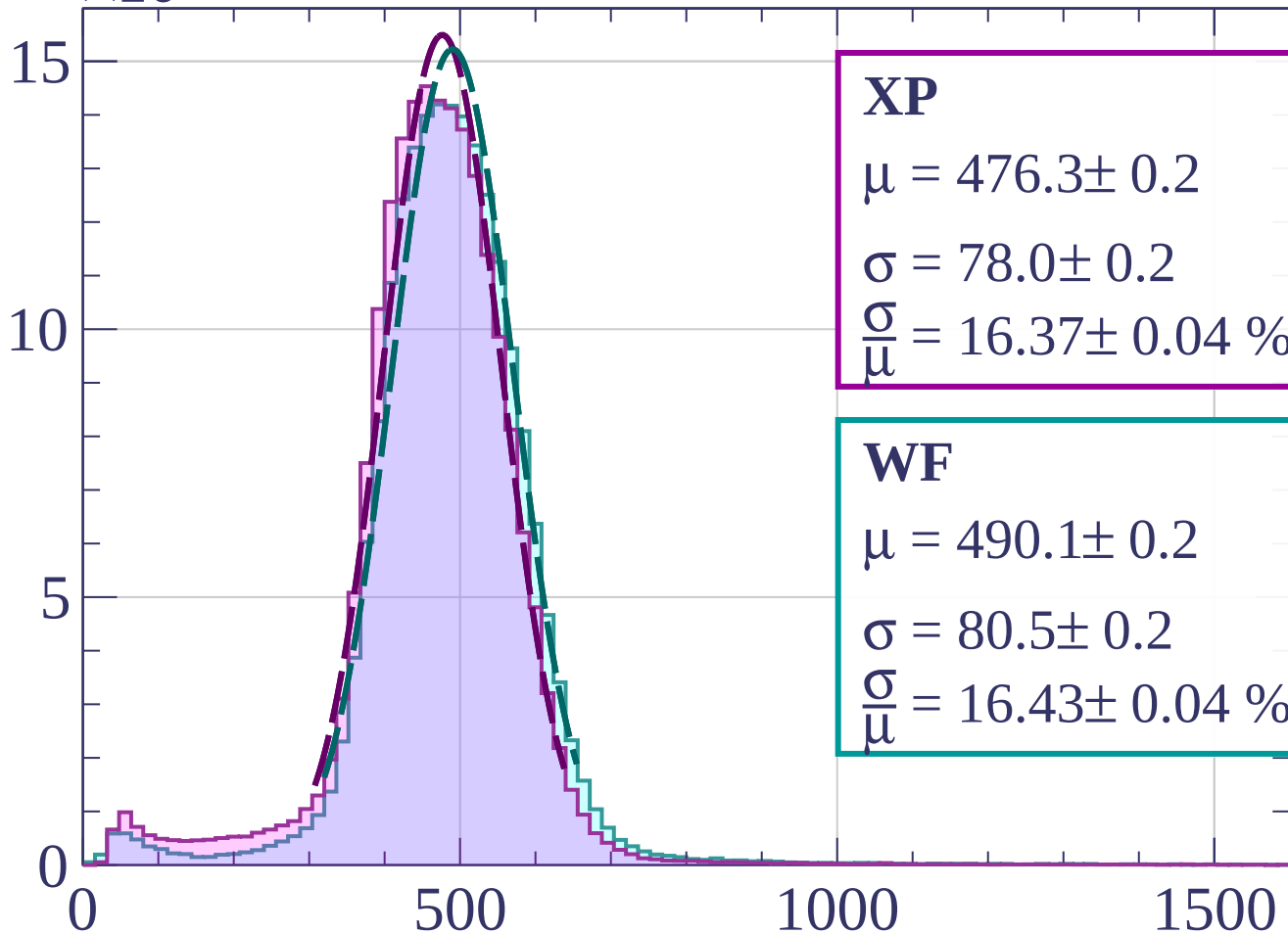
$$\frac{\sigma}{\mu} = 16.26 \pm 0.04 \%$$

dE/dx [ADC counts/cm]

Count

$\times 10^3$

Energy loss | $6 < \theta < 9$



XP

$$\mu = 476.3 \pm 0.2$$

$$\sigma = 78.0 \pm 0.2$$

$$\frac{\sigma}{\mu} = 16.37 \pm 0.04 \%$$

WF

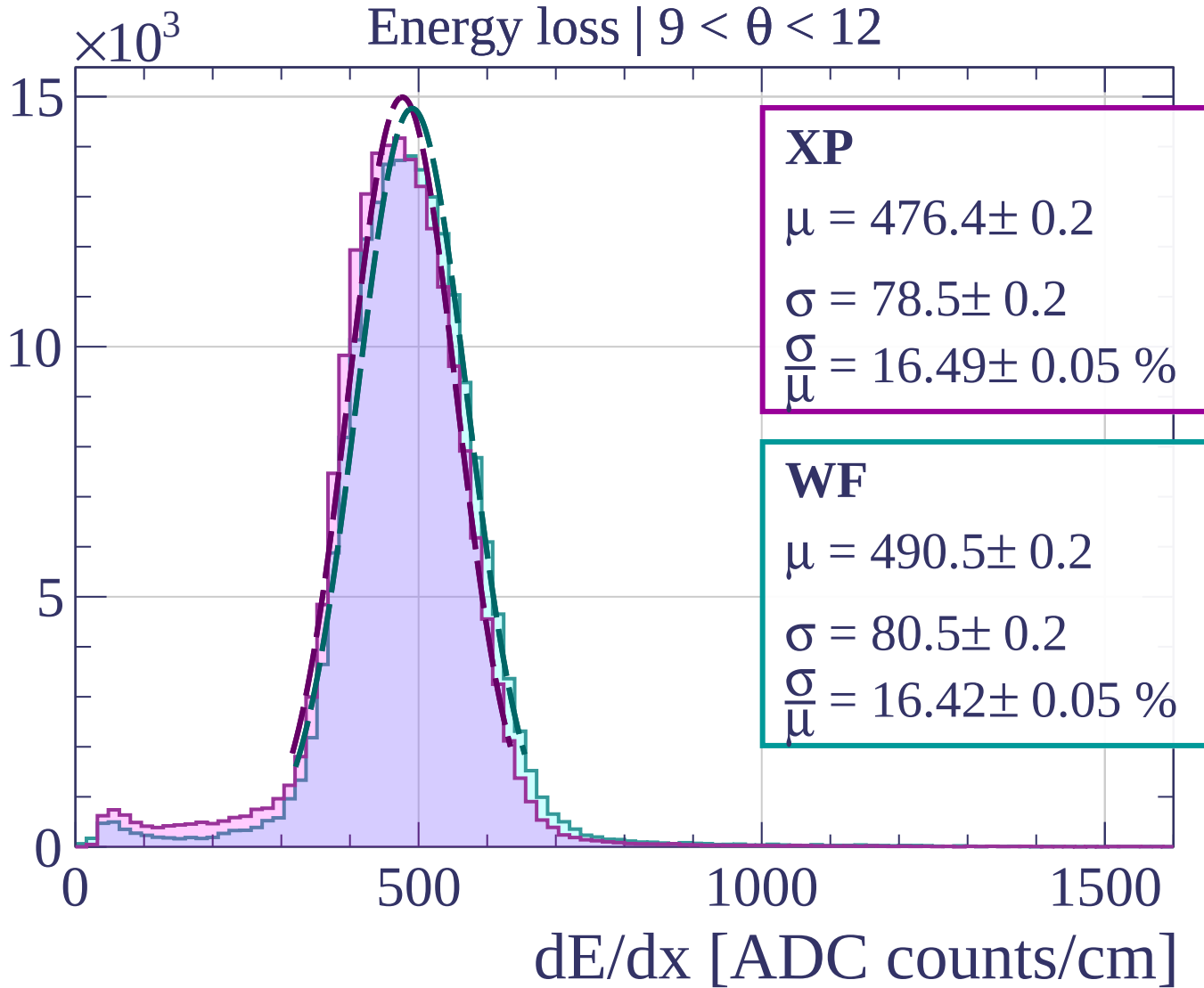
$$\mu = 490.1 \pm 0.2$$

$$\sigma = 80.5 \pm 0.2$$

$$\frac{\sigma}{\mu} = 16.43 \pm 0.04 \%$$

dE/dx [ADC counts/cm]

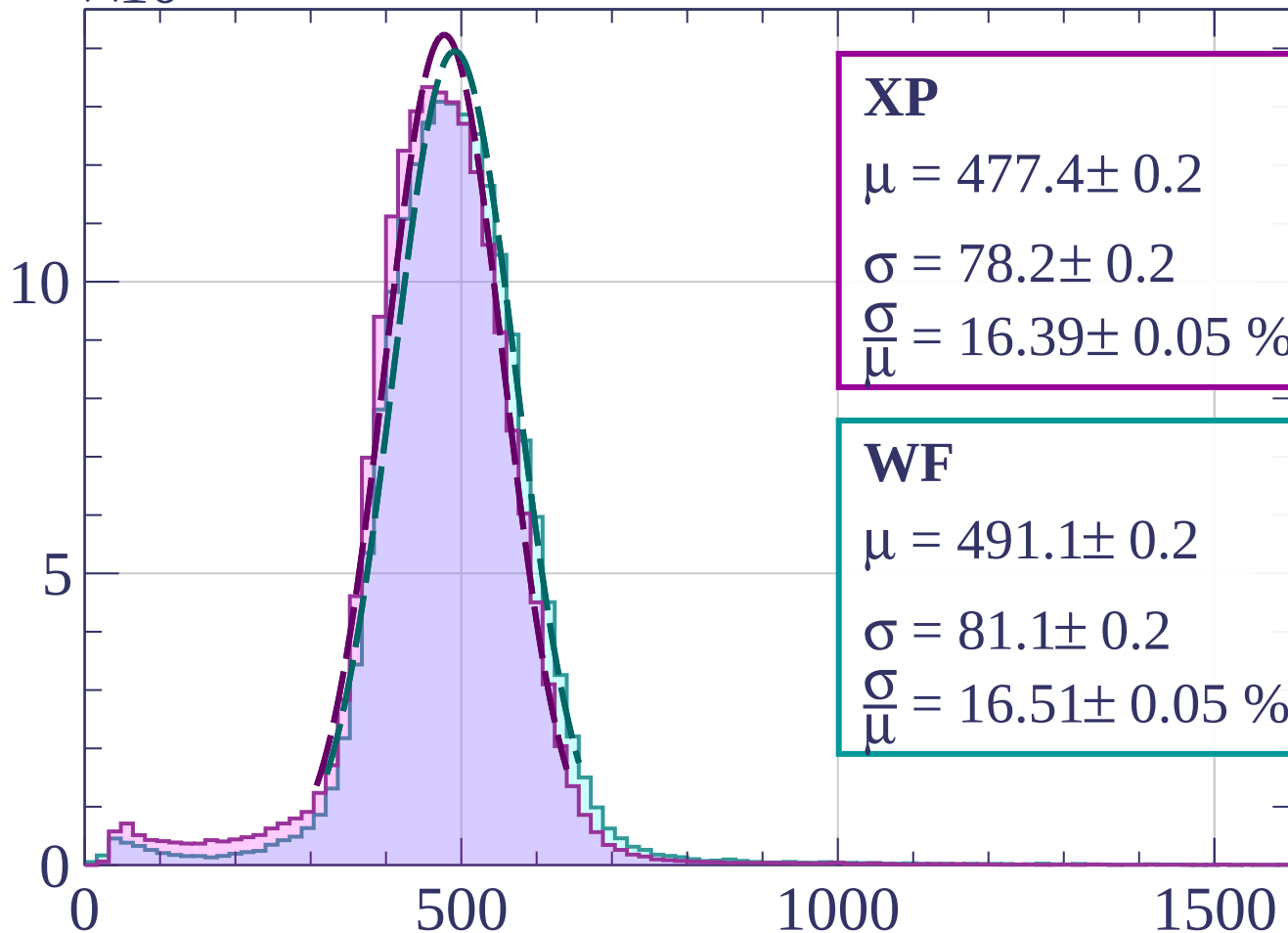
Count



Count

$\times 10^3$

Energy loss | $12 < \theta < 15$



XP

$$\mu = 477.4 \pm 0.2$$

$$\sigma = 78.2 \pm 0.2$$

$$\frac{\sigma}{\mu} = 16.39 \pm 0.05 \%$$

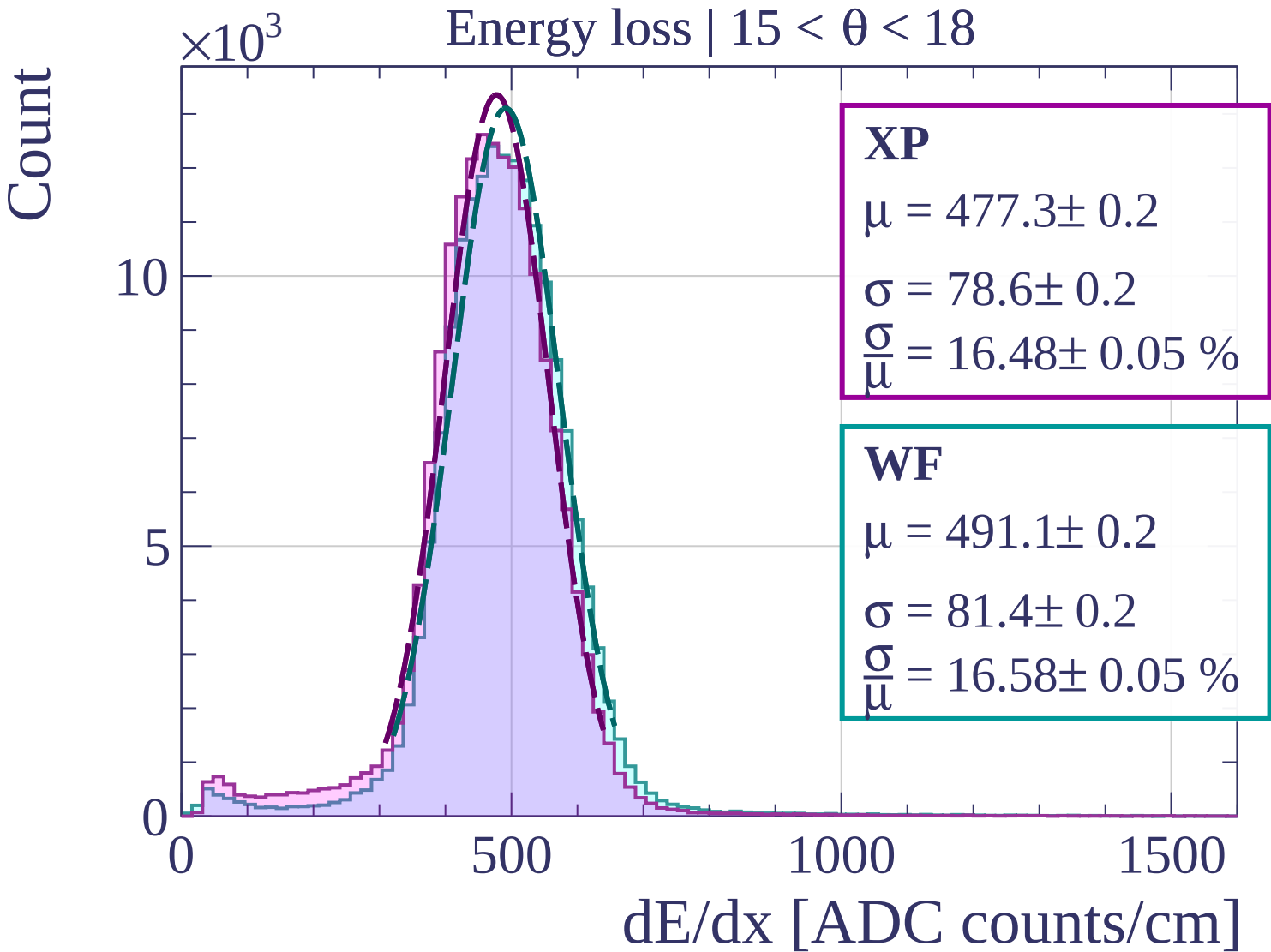
WF

$$\mu = 491.1 \pm 0.2$$

$$\sigma = 81.1 \pm 0.2$$

$$\frac{\sigma}{\mu} = 16.51 \pm 0.05 \%$$

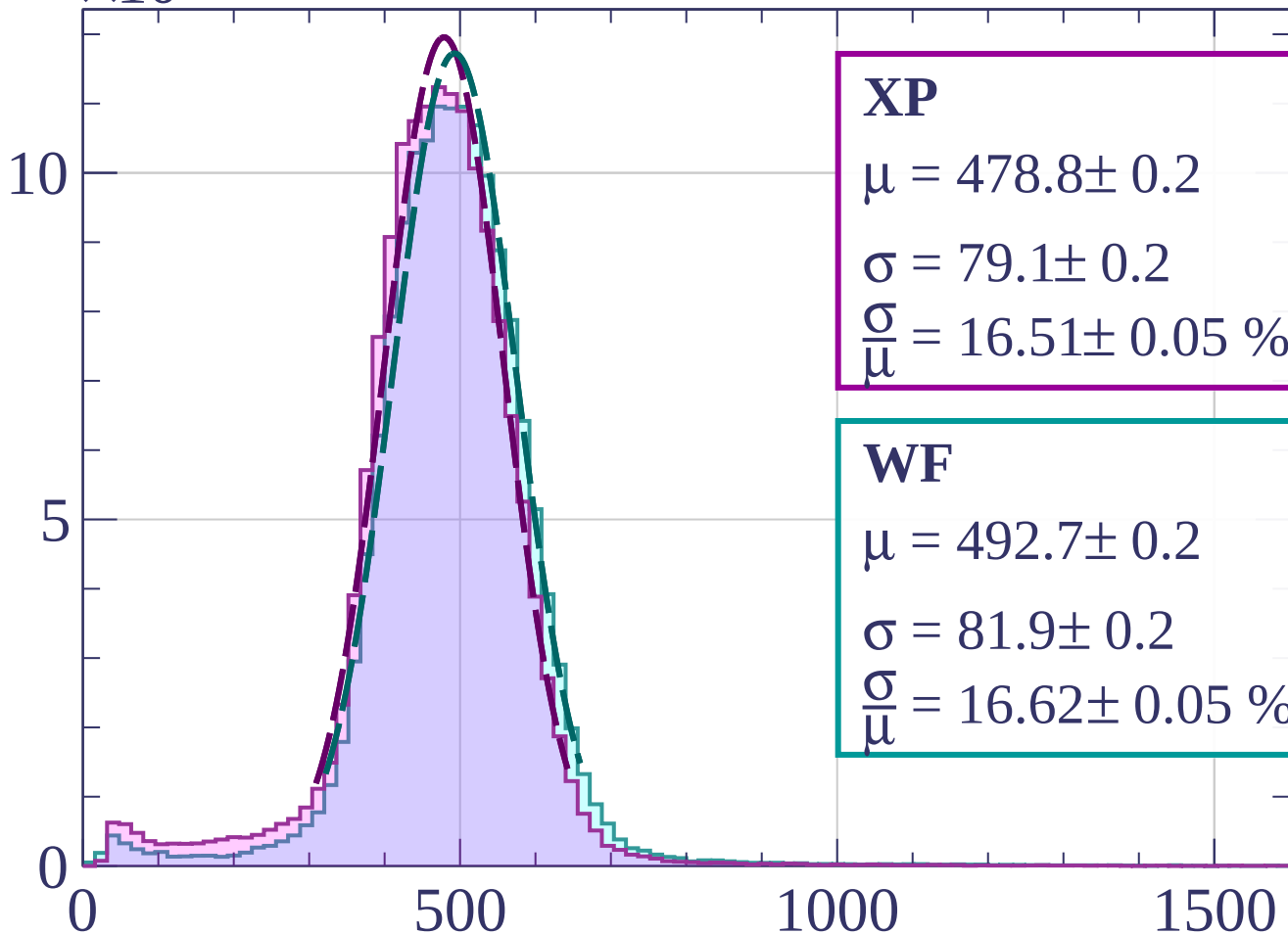
dE/dx [ADC counts/cm]



Count

$\times 10^3$

Energy loss | $18 < \theta < 21$



XP

$$\mu = 478.8 \pm 0.2$$

$$\sigma = 79.1 \pm 0.2$$

$$\frac{\sigma}{\mu} = 16.51 \pm 0.05 \%$$

WF

$$\mu = 492.7 \pm 0.2$$

$$\sigma = 81.9 \pm 0.2$$

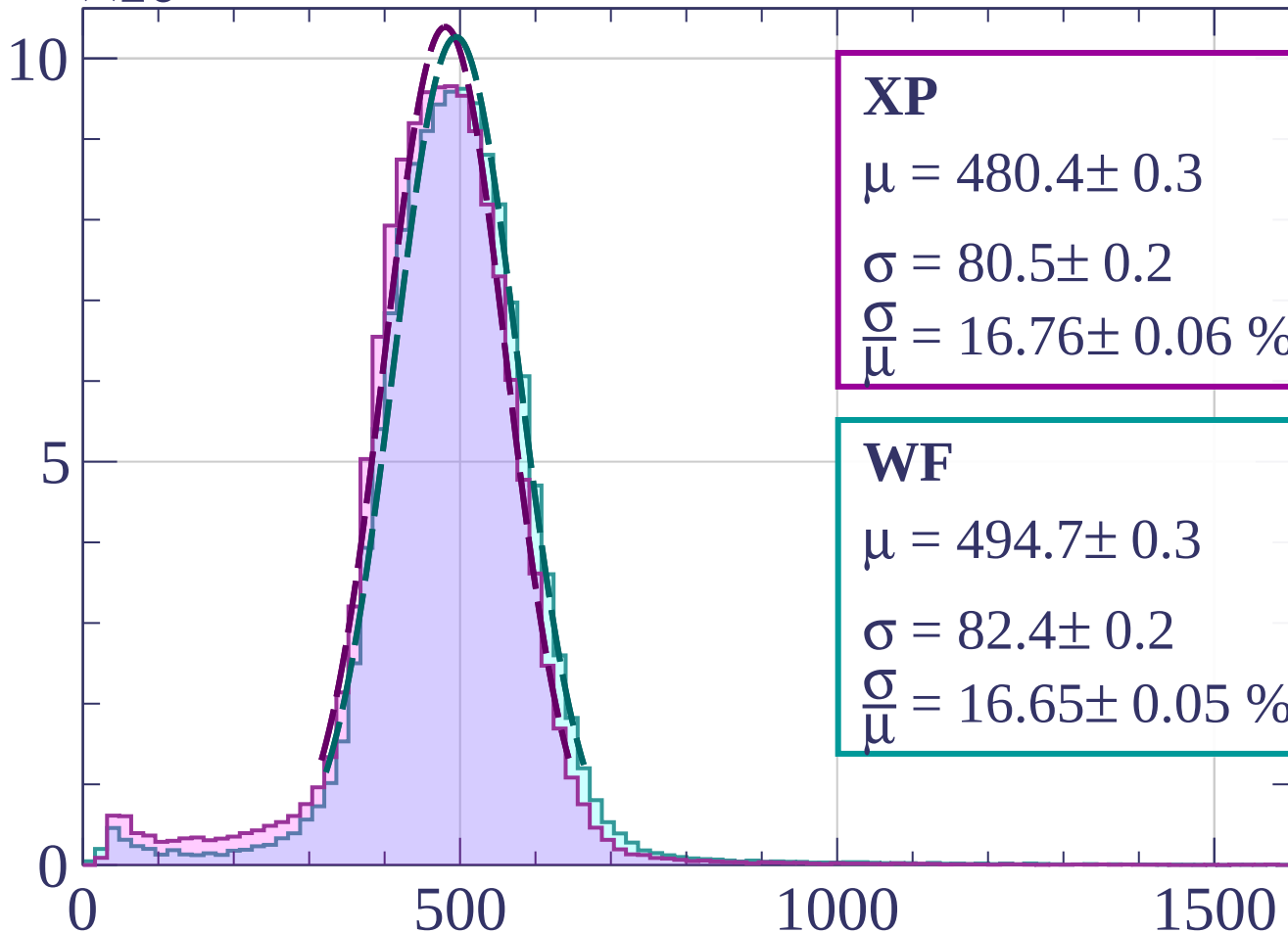
$$\frac{\sigma}{\mu} = 16.62 \pm 0.05 \%$$

dE/dx [ADC counts/cm]

Count

$\times 10^3$

Energy loss | $21 < \theta < 24$



XP

$$\mu = 480.4 \pm 0.3$$

$$\sigma = 80.5 \pm 0.2$$

$$\frac{\sigma}{\mu} = 16.76 \pm 0.06 \%$$

WF

$$\mu = 494.7 \pm 0.3$$

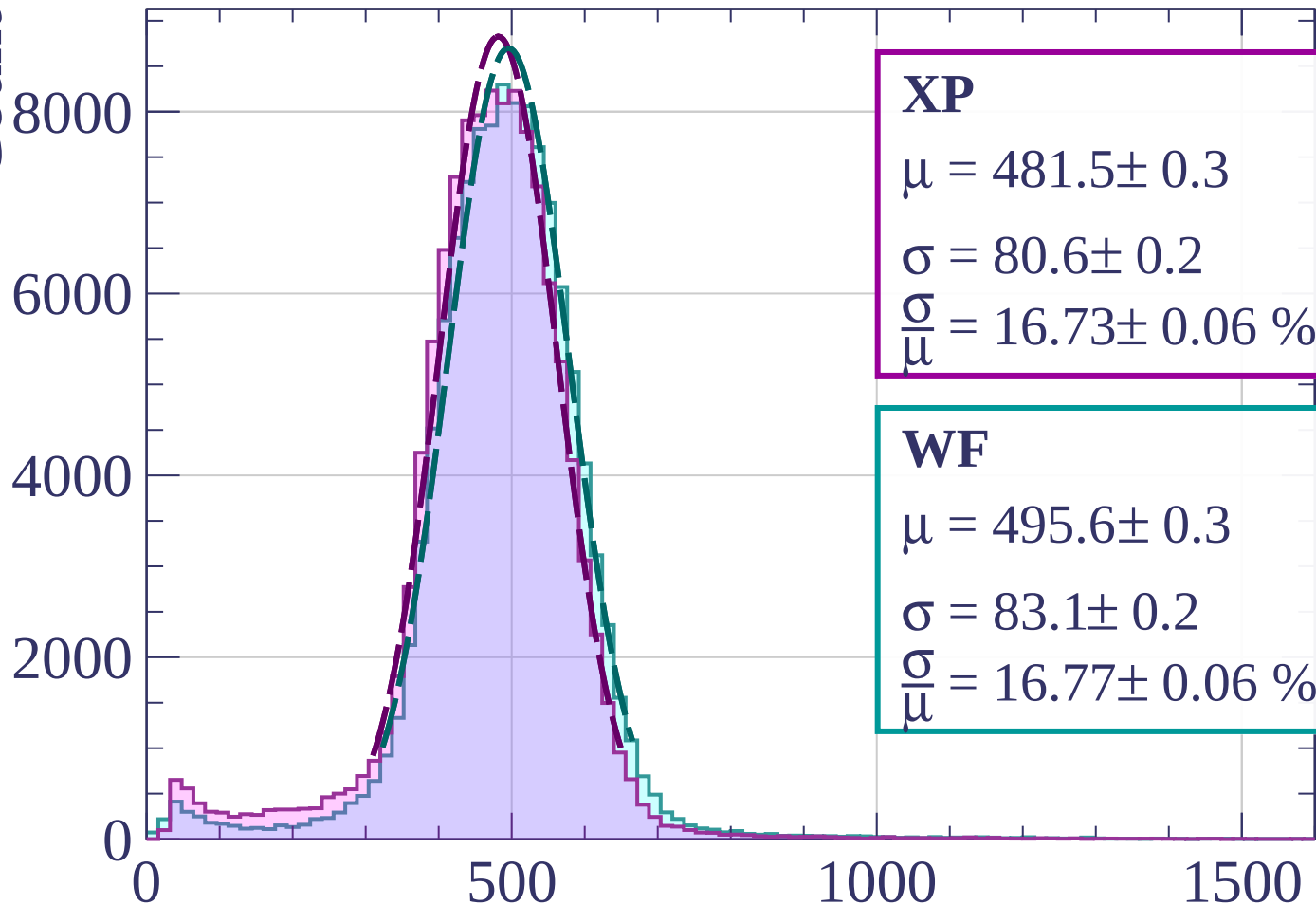
$$\sigma = 82.4 \pm 0.2$$

$$\frac{\sigma}{\mu} = 16.65 \pm 0.05 \%$$

dE/dx [ADC counts/cm]

Energy loss | $24 < \theta < 27$

Count



XP

$$\mu = 481.5 \pm 0.3$$

$$\sigma = 80.6 \pm 0.2$$

$$\frac{\sigma}{\mu} = 16.73 \pm 0.06 \%$$

WF

$$\mu = 495.6 \pm 0.3$$

$$\sigma = 83.1 \pm 0.2$$

$$\frac{\sigma}{\mu} = 16.77 \pm 0.06 \%$$

dE/dx [ADC counts/cm]

Energy loss | $27 < \theta < 30$

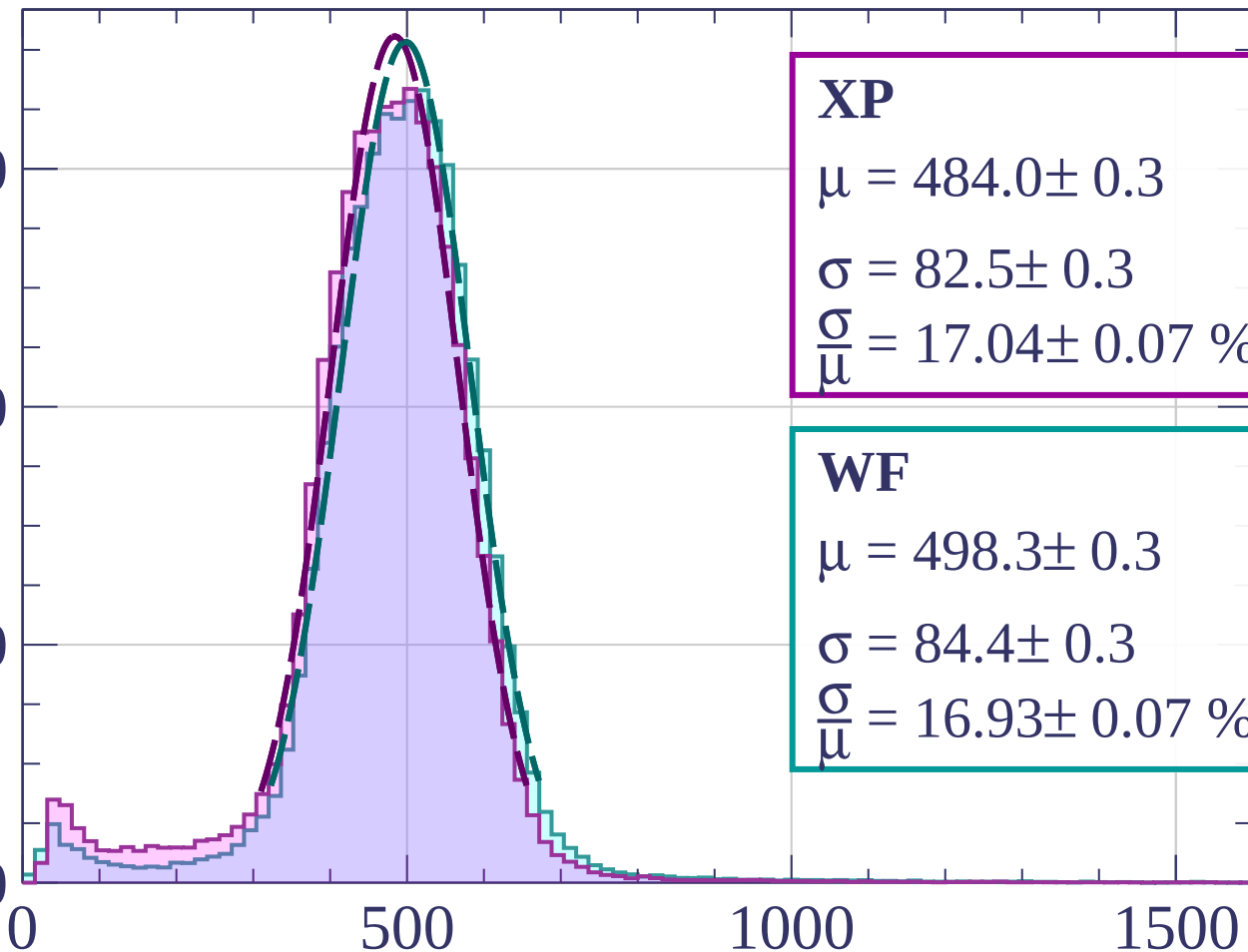
Count

6000

4000

2000

0



XP

$$\mu = 484.0 \pm 0.3$$

$$\sigma = 82.5 \pm 0.3$$

$$\frac{\sigma}{\mu} = 17.04 \pm 0.07 \%$$

WF

$$\mu = 498.3 \pm 0.3$$

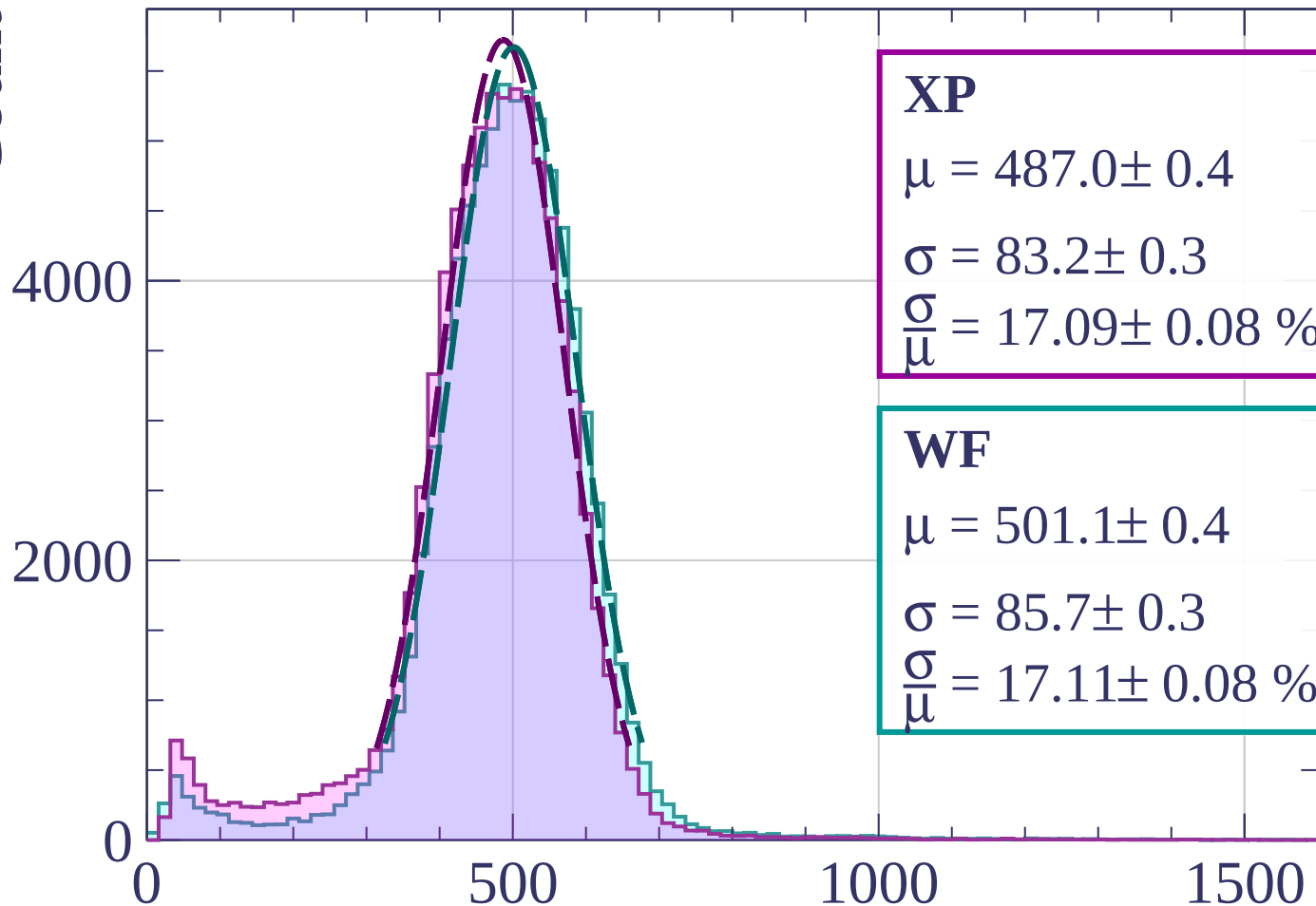
$$\sigma = 84.4 \pm 0.3$$

$$\frac{\sigma}{\mu} = 16.93 \pm 0.07 \%$$

dE/dx [ADC counts/cm]

Energy loss | $30 < \theta < 33$

Count



XP

$$\mu = 487.0 \pm 0.4$$

$$\sigma = 83.2 \pm 0.3$$

$$\frac{\sigma}{\mu} = 17.09 \pm 0.08 \%$$

WF

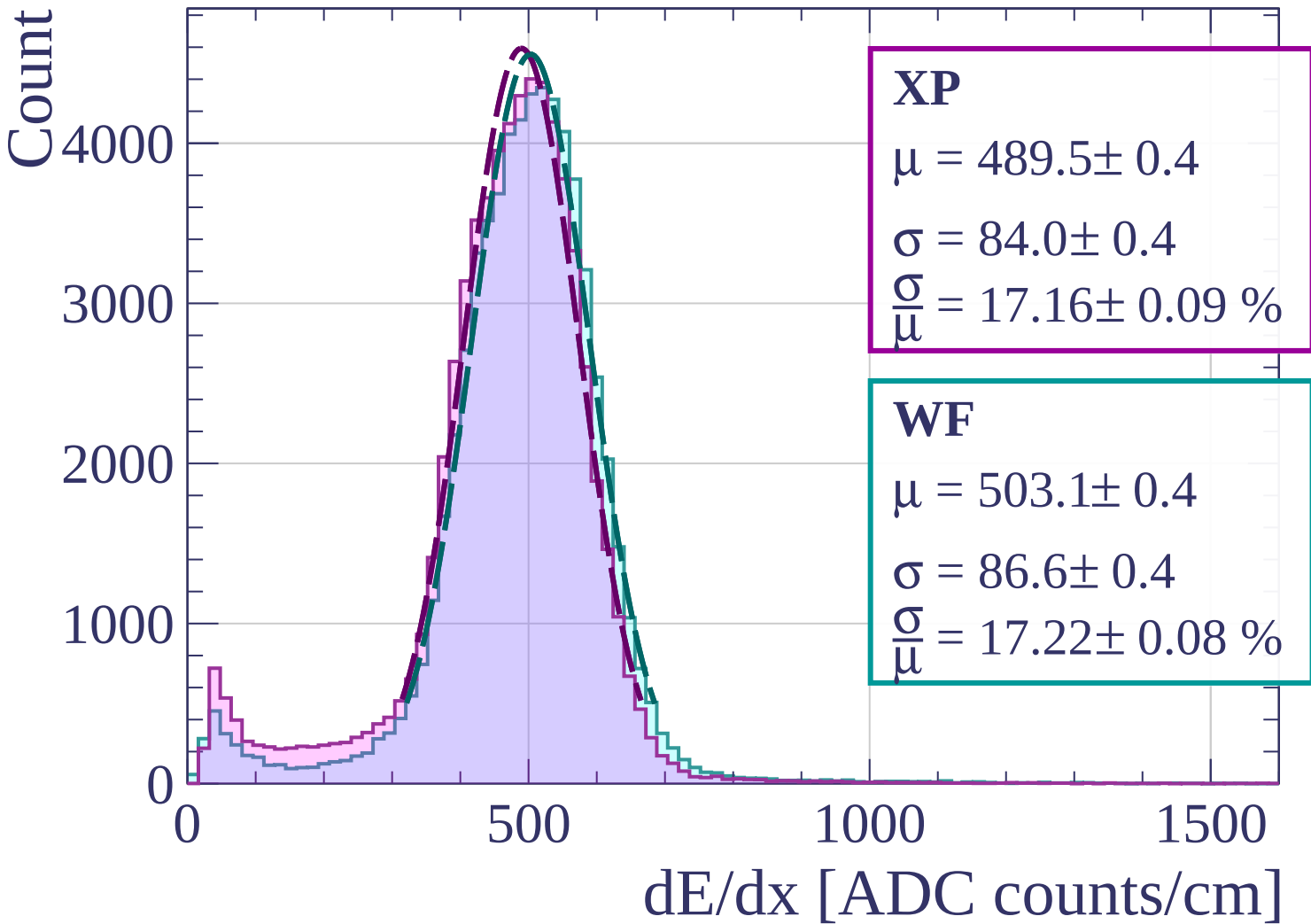
$$\mu = 501.1 \pm 0.4$$

$$\sigma = 85.7 \pm 0.3$$

$$\frac{\sigma}{\mu} = 17.11 \pm 0.08 \%$$

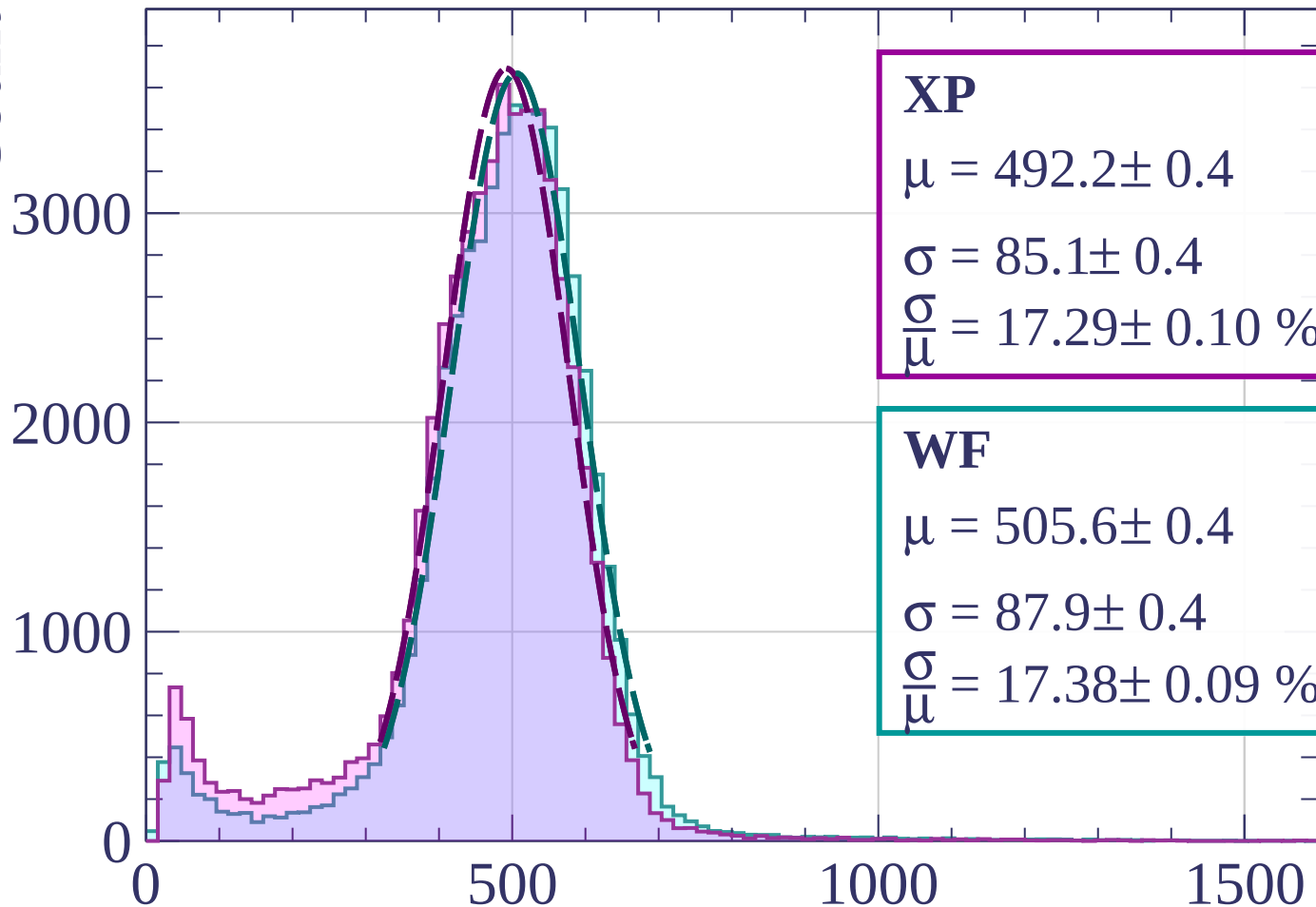
dE/dx [ADC counts/cm]

Energy loss | $33 < \theta < 36$



Energy loss | $36 < \theta < 39$

Count



XP

$$\mu = 492.2 \pm 0.4$$

$$\sigma = 85.1 \pm 0.4$$

$$\frac{\sigma}{\mu} = 17.29 \pm 0.10 \%$$

WF

$$\mu = 505.6 \pm 0.4$$

$$\sigma = 87.9 \pm 0.4$$

$$\frac{\sigma}{\mu} = 17.38 \pm 0.09 \%$$

dE/dx [ADC counts/cm]

Energy loss | $39 < \theta < 42$

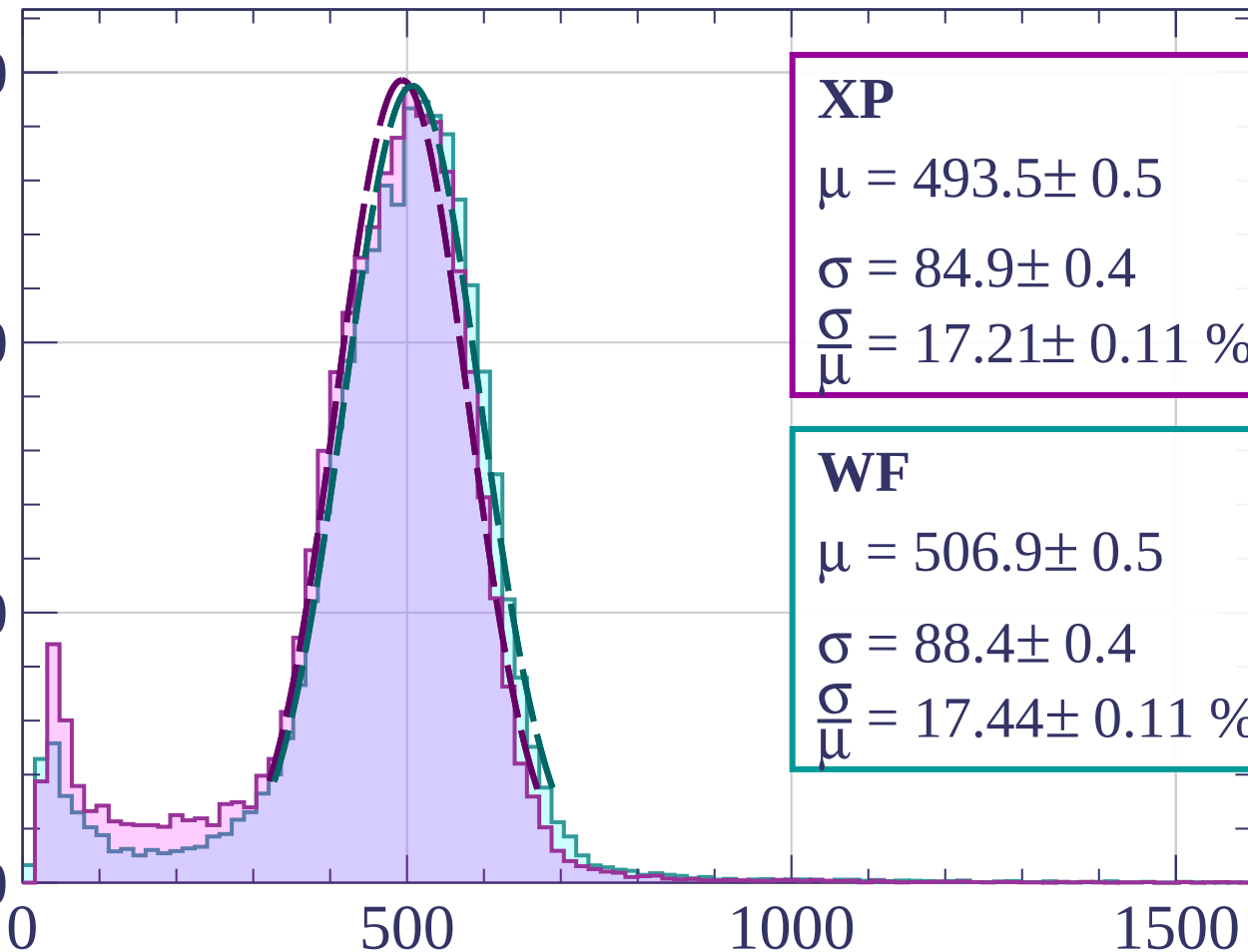
Count

3000

2000

1000

0



XP

$$\mu = 493.5 \pm 0.5$$

$$\sigma = 84.9 \pm 0.4$$

$$\frac{\sigma}{\mu} = 17.21 \pm 0.11 \%$$

WF

$$\mu = 506.9 \pm 0.5$$

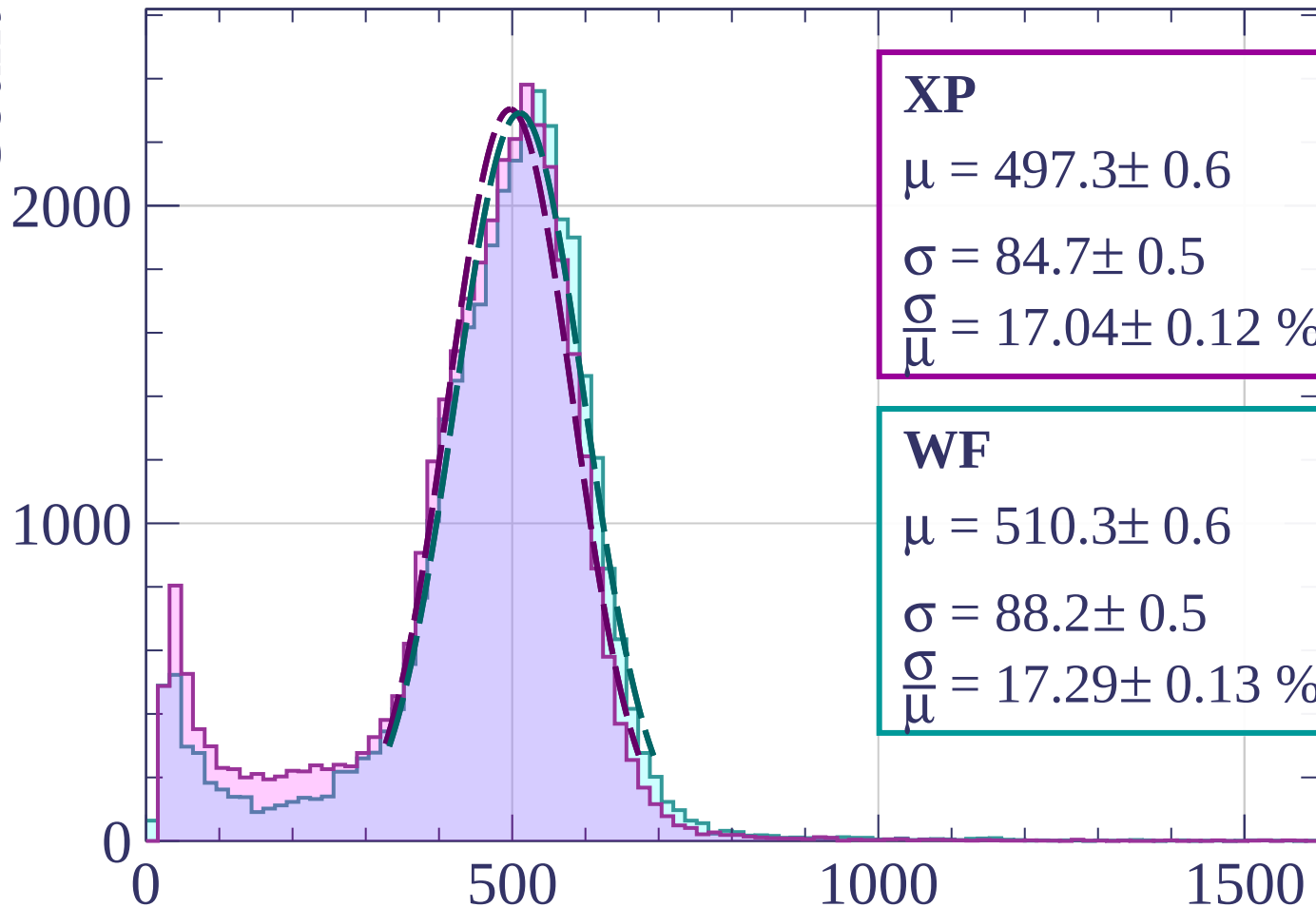
$$\sigma = 88.4 \pm 0.4$$

$$\frac{\sigma}{\mu} = 17.44 \pm 0.11 \%$$

dE/dx [ADC counts/cm]

Energy loss | $42 < \theta < 45$

Count



XP

$$\mu = 497.3 \pm 0.6$$

$$\sigma = 84.7 \pm 0.5$$

$$\frac{\sigma}{\mu} = 17.04 \pm 0.12 \%$$

WF

$$\mu = 510.3 \pm 0.6$$

$$\sigma = 88.2 \pm 0.5$$

$$\frac{\sigma}{\mu} = 17.29 \pm 0.13 \%$$

dE/dx [ADC counts/cm]

Energy loss | $45 < \theta < 48$

Count

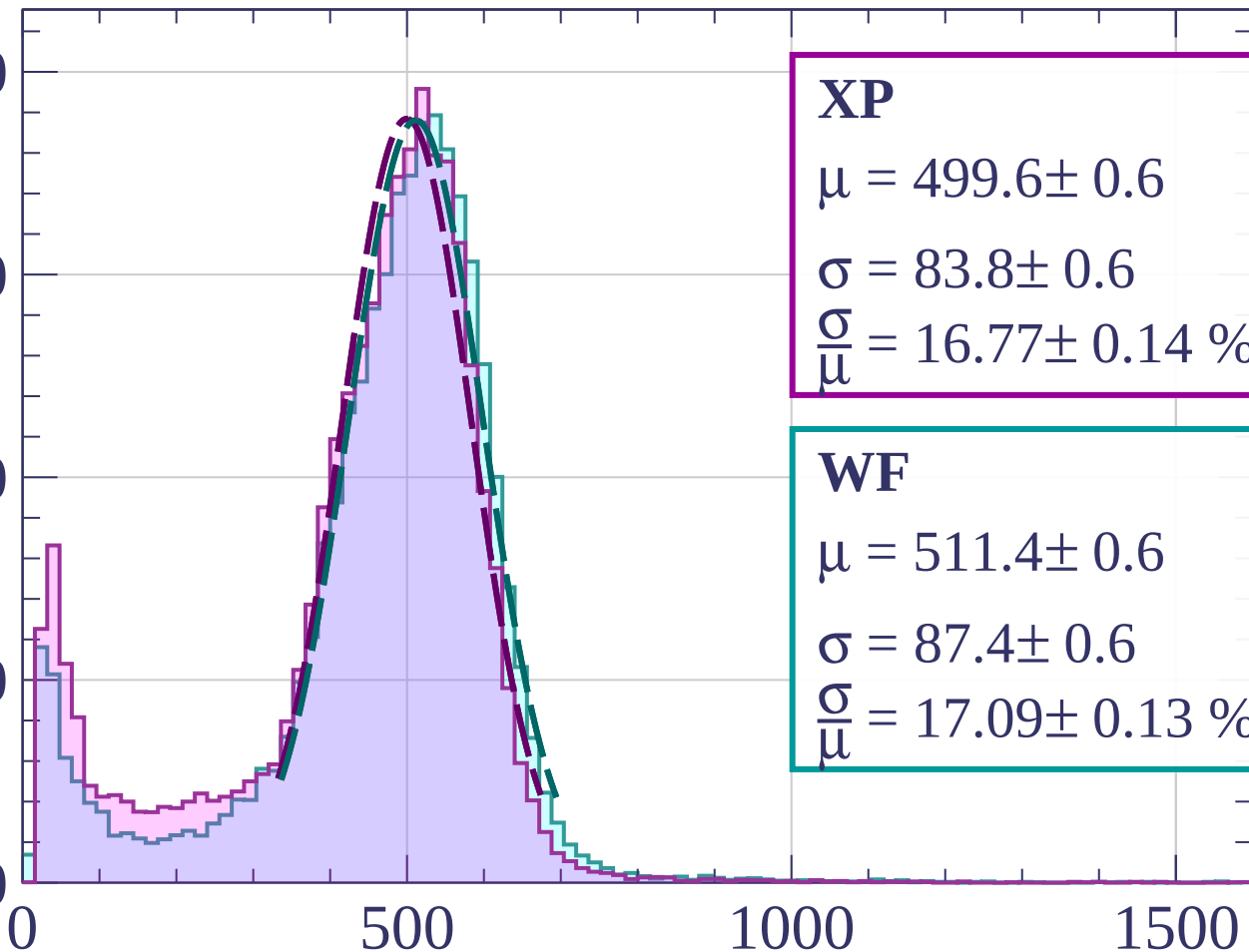
2000

1500

1000

500

0



XP

$$\mu = 499.6 \pm 0.6$$

$$\sigma = 83.8 \pm 0.6$$

$$\frac{\sigma}{\mu} = 16.77 \pm 0.14 \%$$

WF

$$\mu = 511.4 \pm 0.6$$

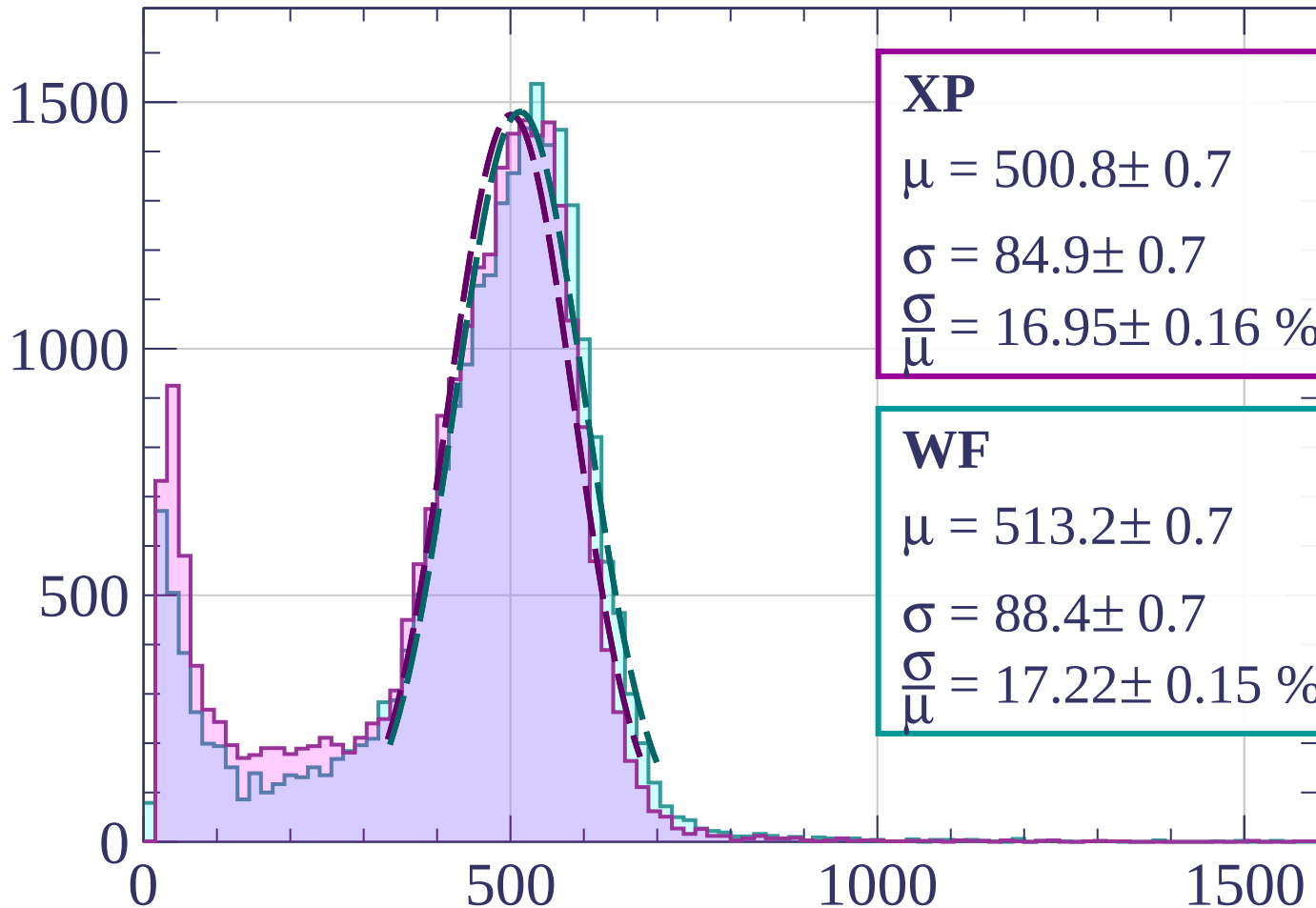
$$\sigma = 87.4 \pm 0.6$$

$$\frac{\sigma}{\mu} = 17.09 \pm 0.13 \%$$

dE/dx [ADC counts/cm]

Energy loss | $48 < \theta < 51$

Count



XP

$$\mu = 500.8 \pm 0.7$$

$$\sigma = 84.9 \pm 0.7$$

$$\frac{\sigma}{\mu} = 16.95 \pm 0.16 \%$$

WF

$$\mu = 513.2 \pm 0.7$$

$$\sigma = 88.4 \pm 0.7$$

$$\frac{\sigma}{\mu} = 17.22 \pm 0.15 \%$$

dE/dx [ADC counts/cm]

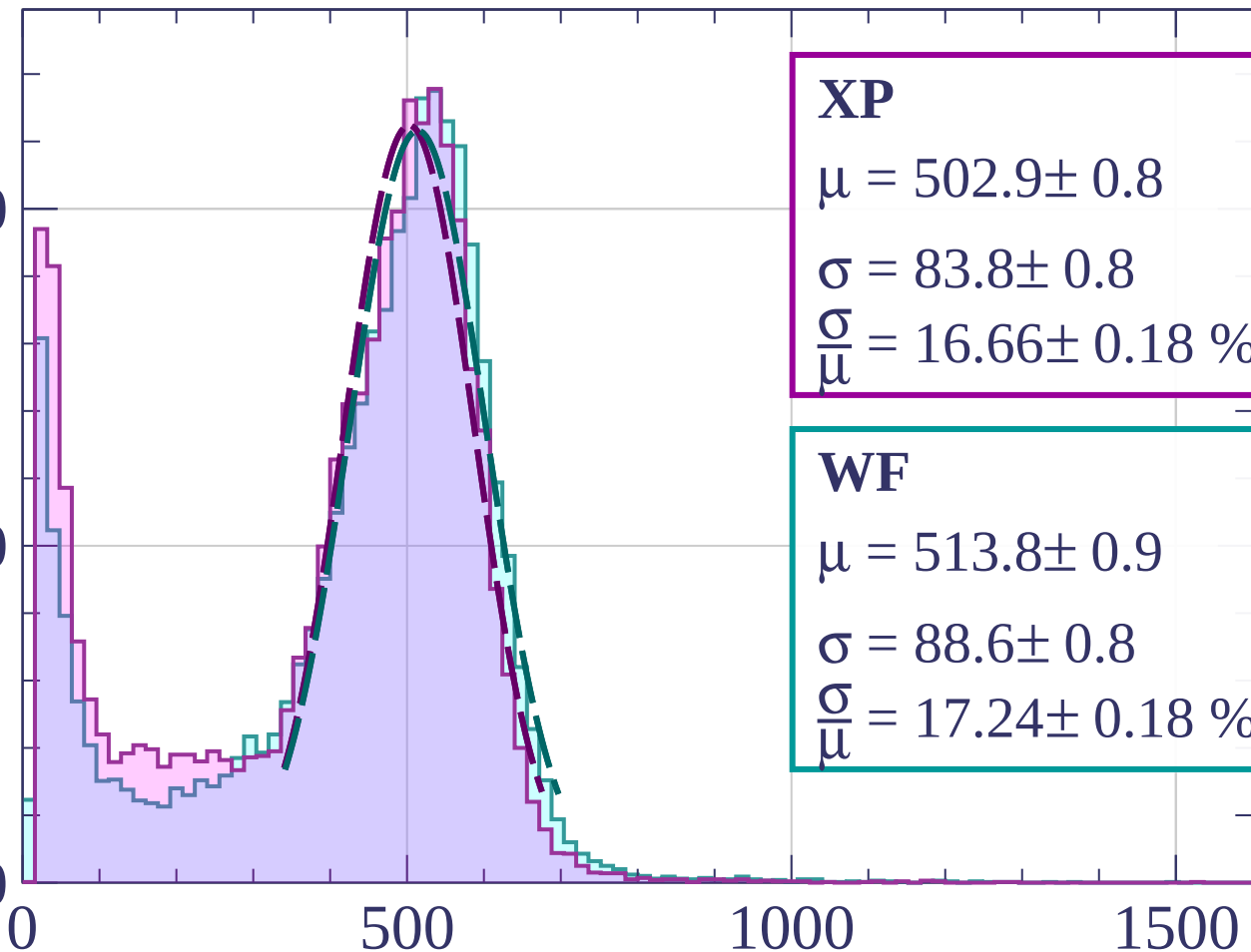
Energy loss | $51 < \theta < 54$

Count

1000

500

0



XP

$$\mu = 502.9 \pm 0.8$$

$$\sigma = 83.8 \pm 0.8$$

$$\frac{\sigma}{\mu} = 16.66 \pm 0.18 \%$$

WF

$$\mu = 513.8 \pm 0.9$$

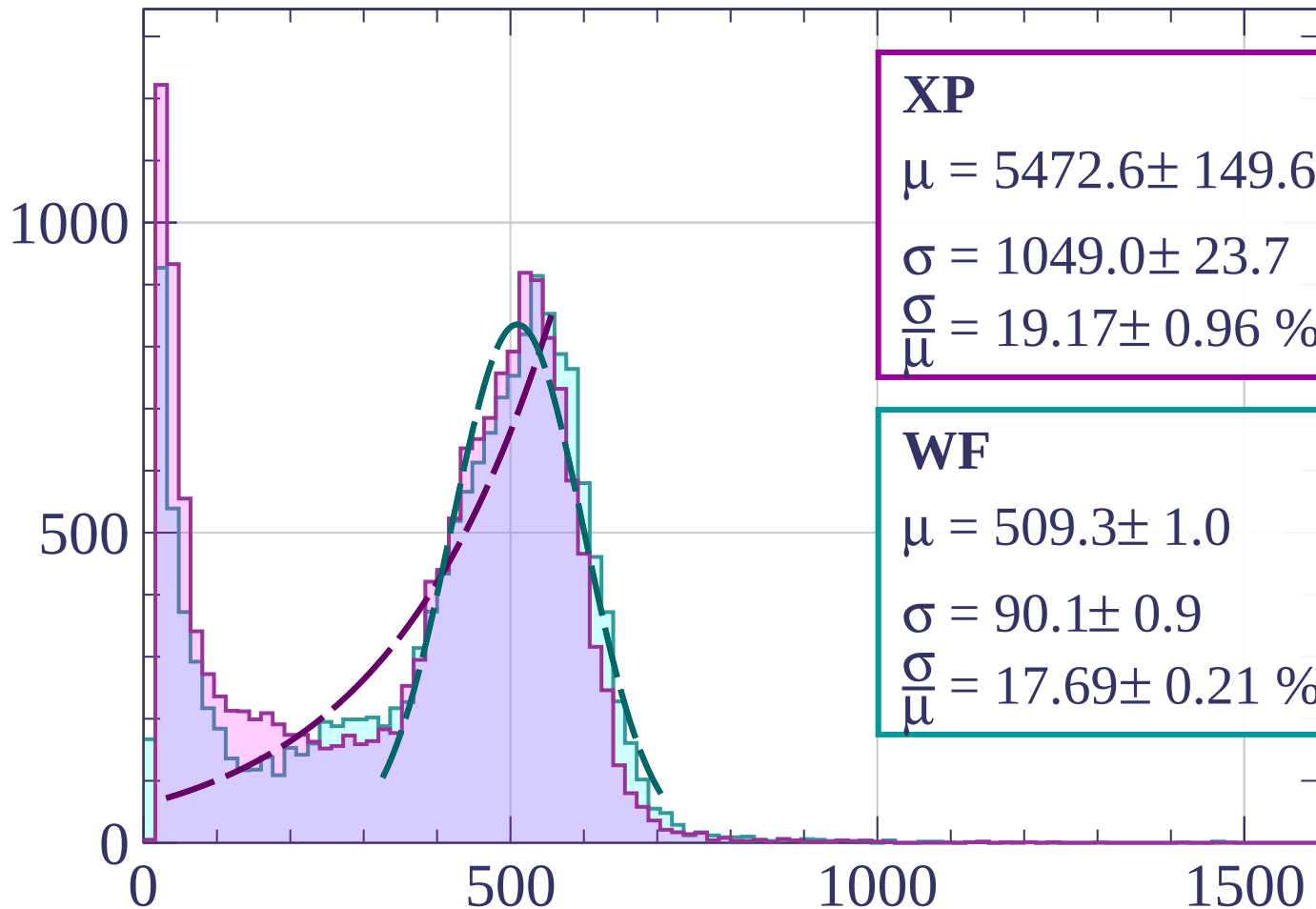
$$\sigma = 88.6 \pm 0.8$$

$$\frac{\sigma}{\mu} = 17.24 \pm 0.18 \%$$

dE/dx [ADC counts/cm]

Energy loss | $54 < \theta < 57$

Count



XP

$$\mu = 5472.6 \pm 149.6$$

$$\sigma = 1049.0 \pm 23.7$$

$$\frac{\sigma}{\mu} = 19.17 \pm 0.96 \%$$

WF

$$\mu = 509.3 \pm 1.0$$

$$\sigma = 90.1 \pm 0.9$$

$$\frac{\sigma}{\mu} = 17.69 \pm 0.21 \%$$

dE/dx [ADC counts/cm]

Energy loss | $57 < \theta < 60$

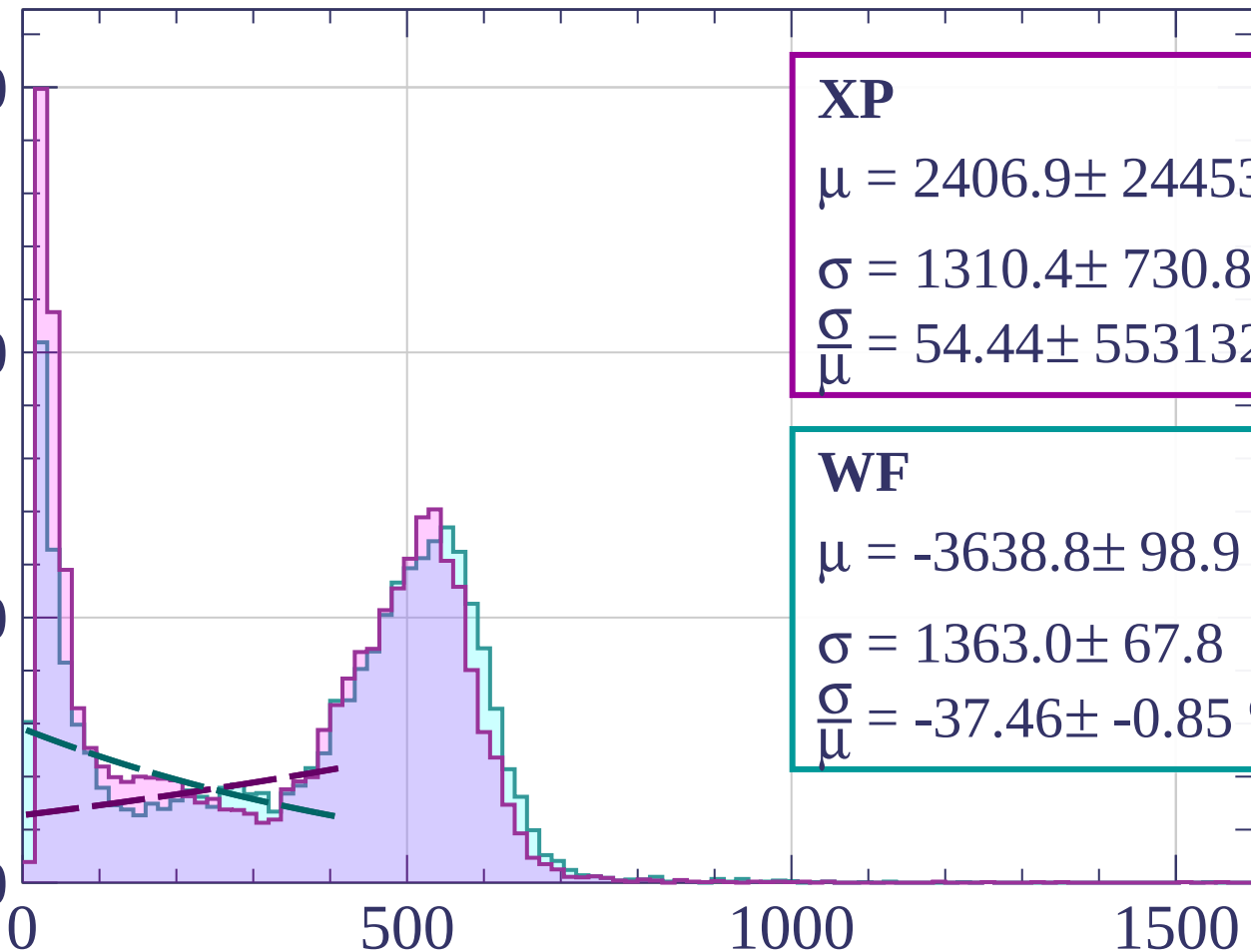
Count

1500

1000

500

0



XP

$$\mu = 2406.9 \pm 2445321$$

$$\sigma = 1310.4 \pm 730.8$$

$$\frac{\sigma}{\mu} = 54.44 \pm 553132.25$$

WF

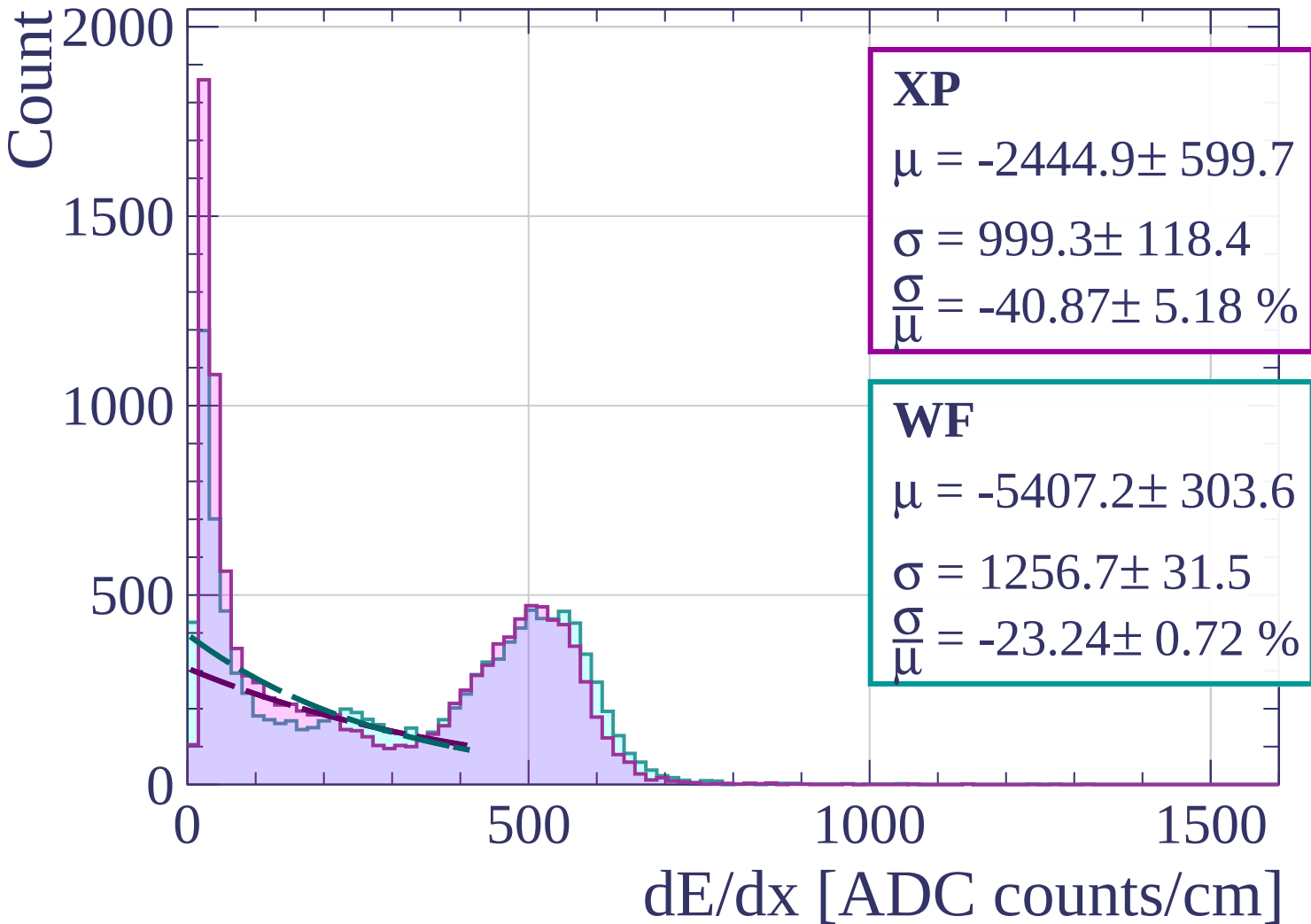
$$\mu = -3638.8 \pm 98.9$$

$$\sigma = 1363.0 \pm 67.8$$

$$\frac{\sigma}{\mu} = -37.46 \pm -0.85 \%$$

dE/dx [ADC counts/cm]

Energy loss | $60 < \theta < 63$



Energy loss | $63 < \theta < 66$

Count

2000

1000

0

500

1000

1500

dE/dx [ADC counts/cm]

XP

$$\mu = -2291.3 \pm 104.5$$

$$\sigma = 613.0 \pm 17.6$$

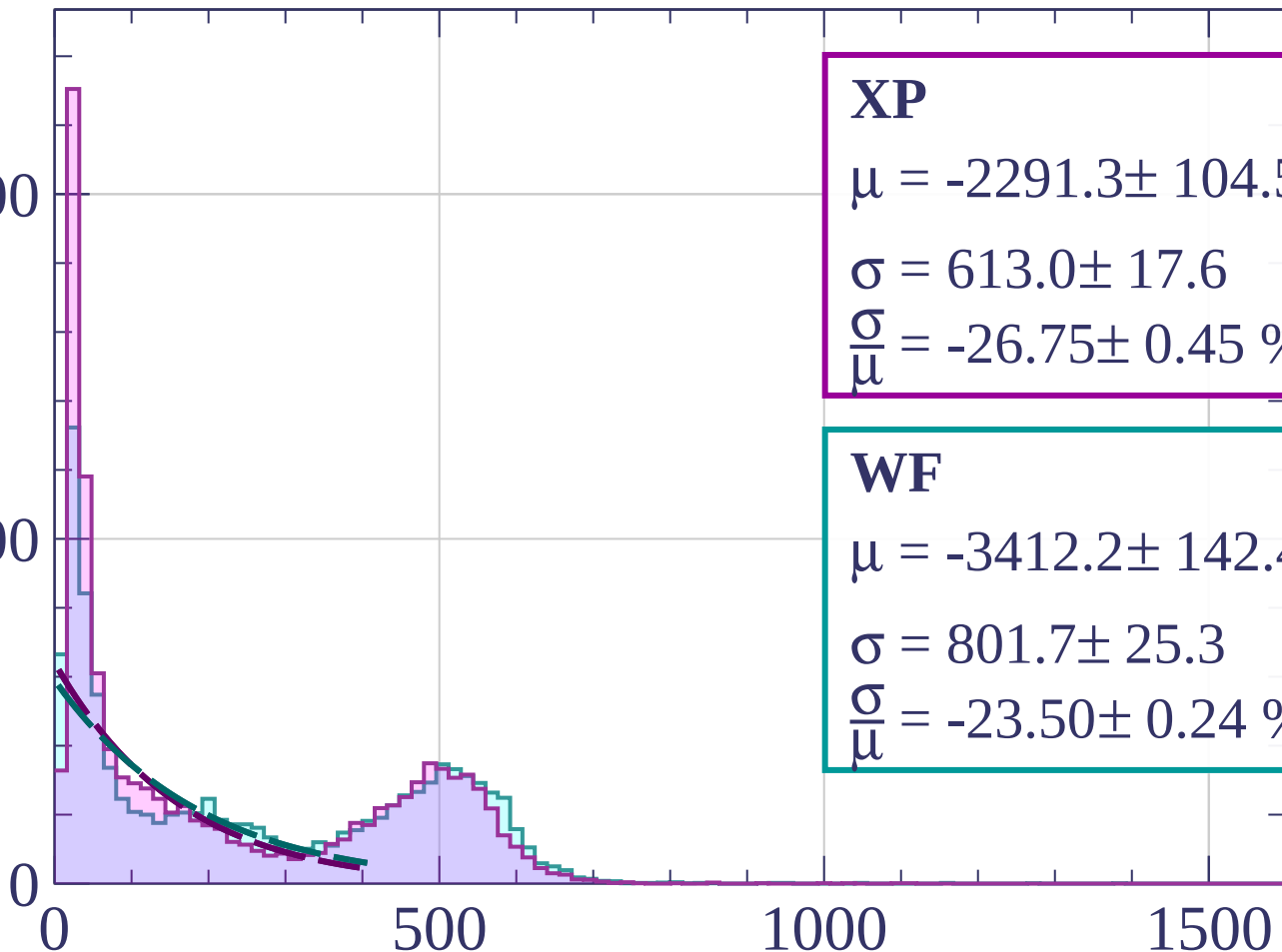
$$\frac{\sigma}{\mu} = -26.75 \pm 0.45 \%$$

WF

$$\mu = -3412.2 \pm 142.4$$

$$\sigma = 801.7 \pm 25.3$$

$$\frac{\sigma}{\mu} = -23.50 \pm 0.24 \%$$



Energy loss | $66 < \theta < 69$

Count

2000

1000

0

500

1000

1500

dE/dx [ADC counts/cm]

XP

$$\mu = -1885.3 \pm 57.9$$

$$\sigma = 440.2 \pm 9.4$$

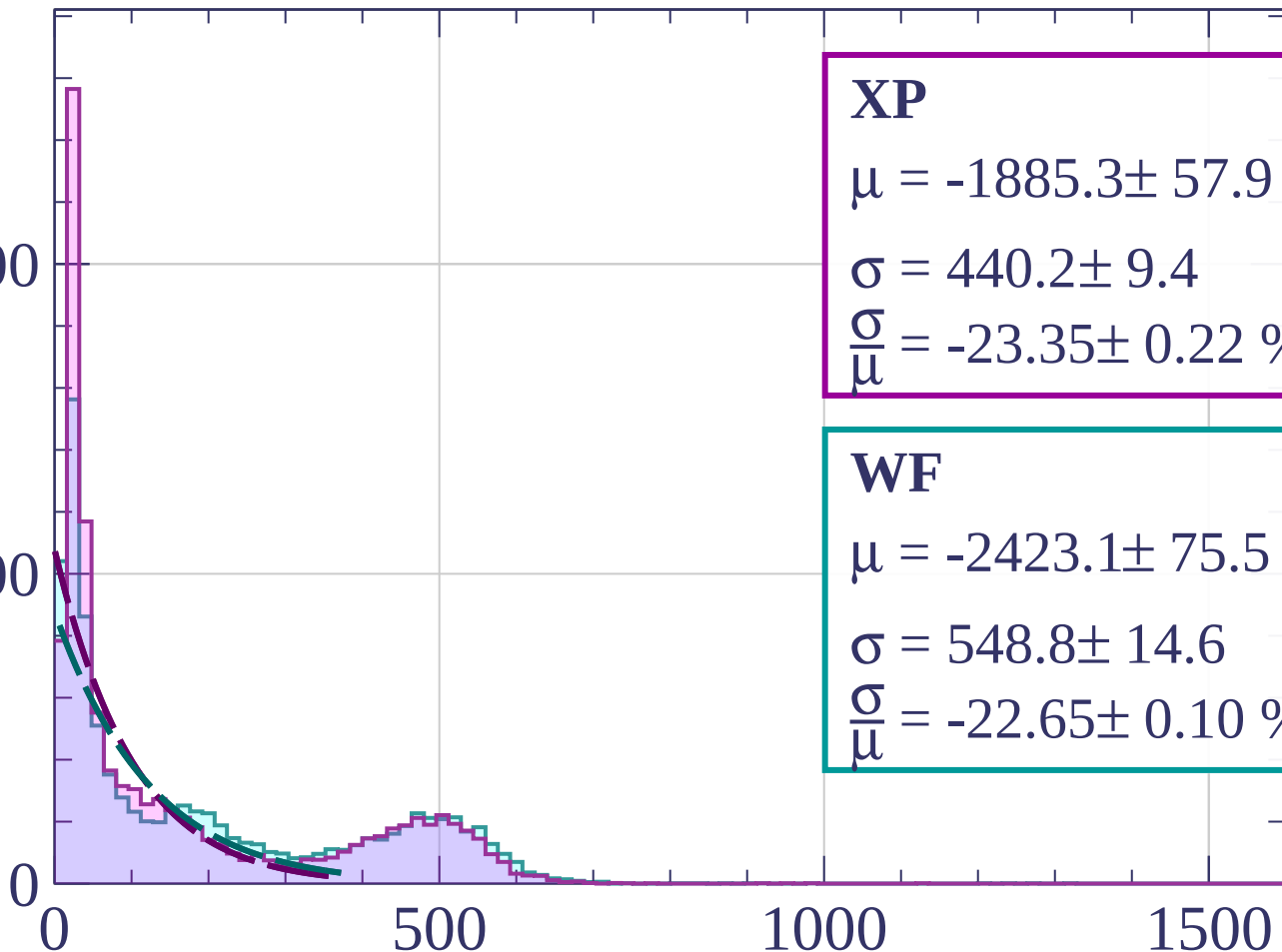
$$\frac{\sigma}{\mu} = -23.35 \pm 0.22 \%$$

WF

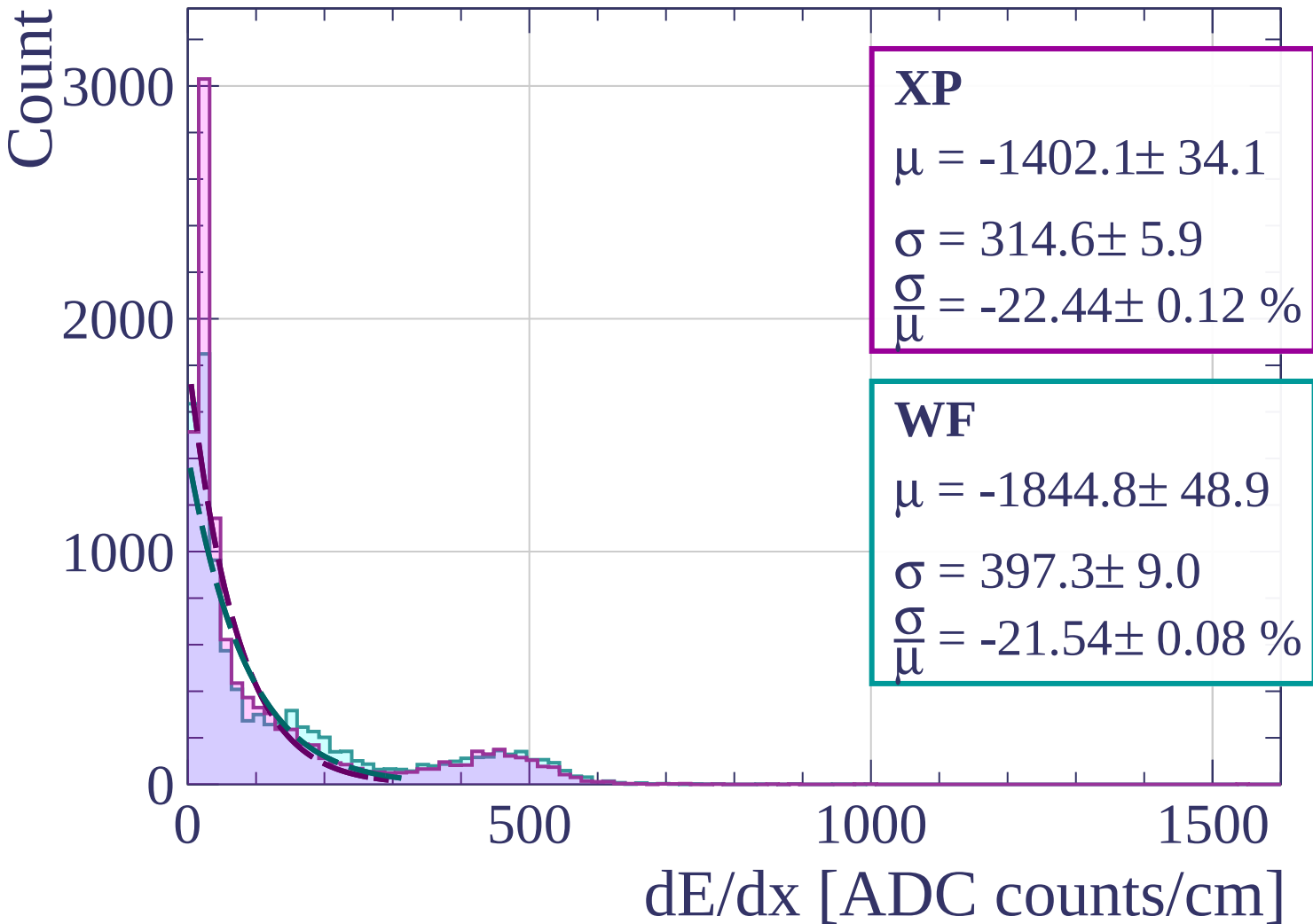
$$\mu = -2423.1 \pm 75.5$$

$$\sigma = 548.8 \pm 14.6$$

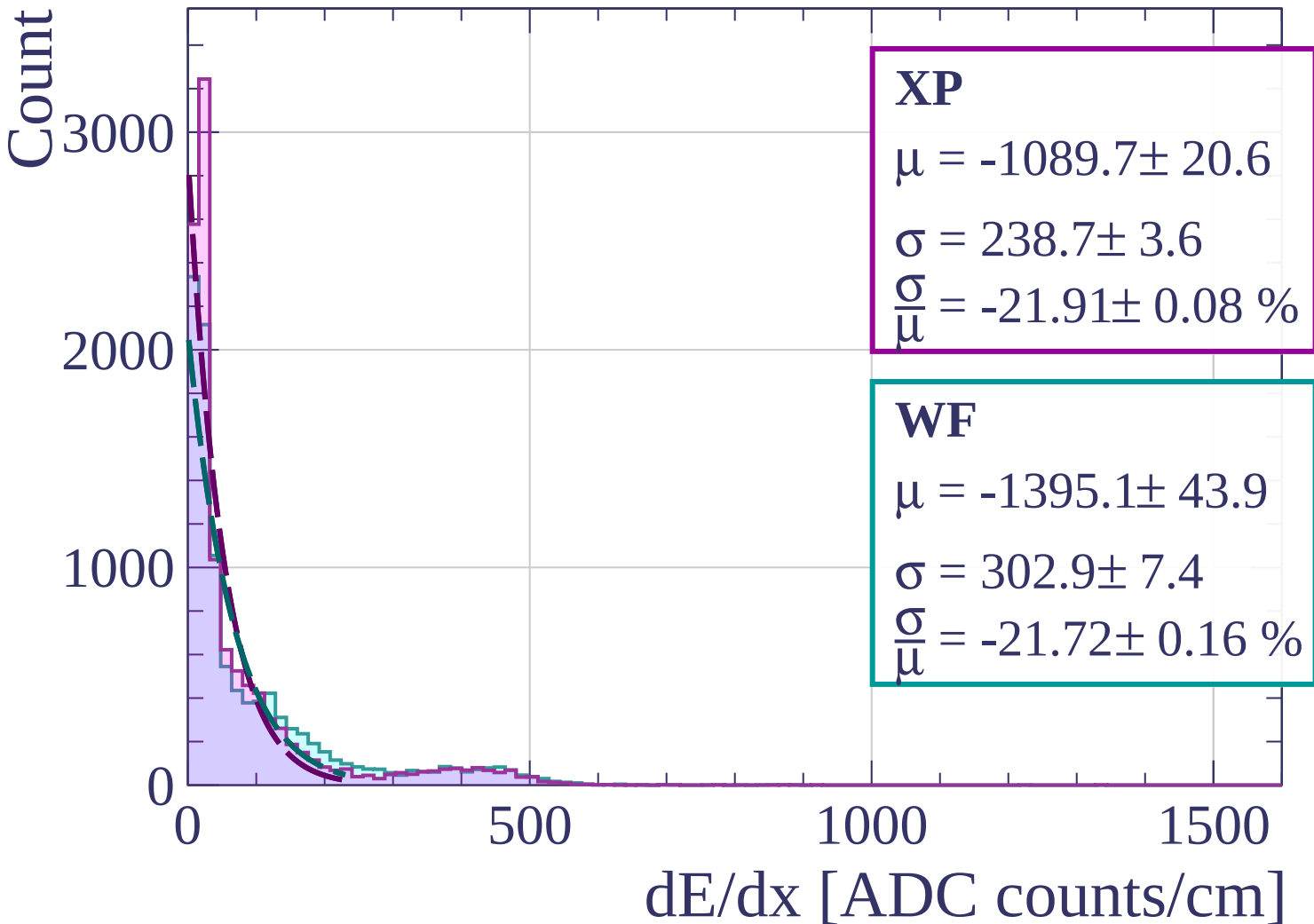
$$\frac{\sigma}{\mu} = -22.65 \pm 0.10 \%$$



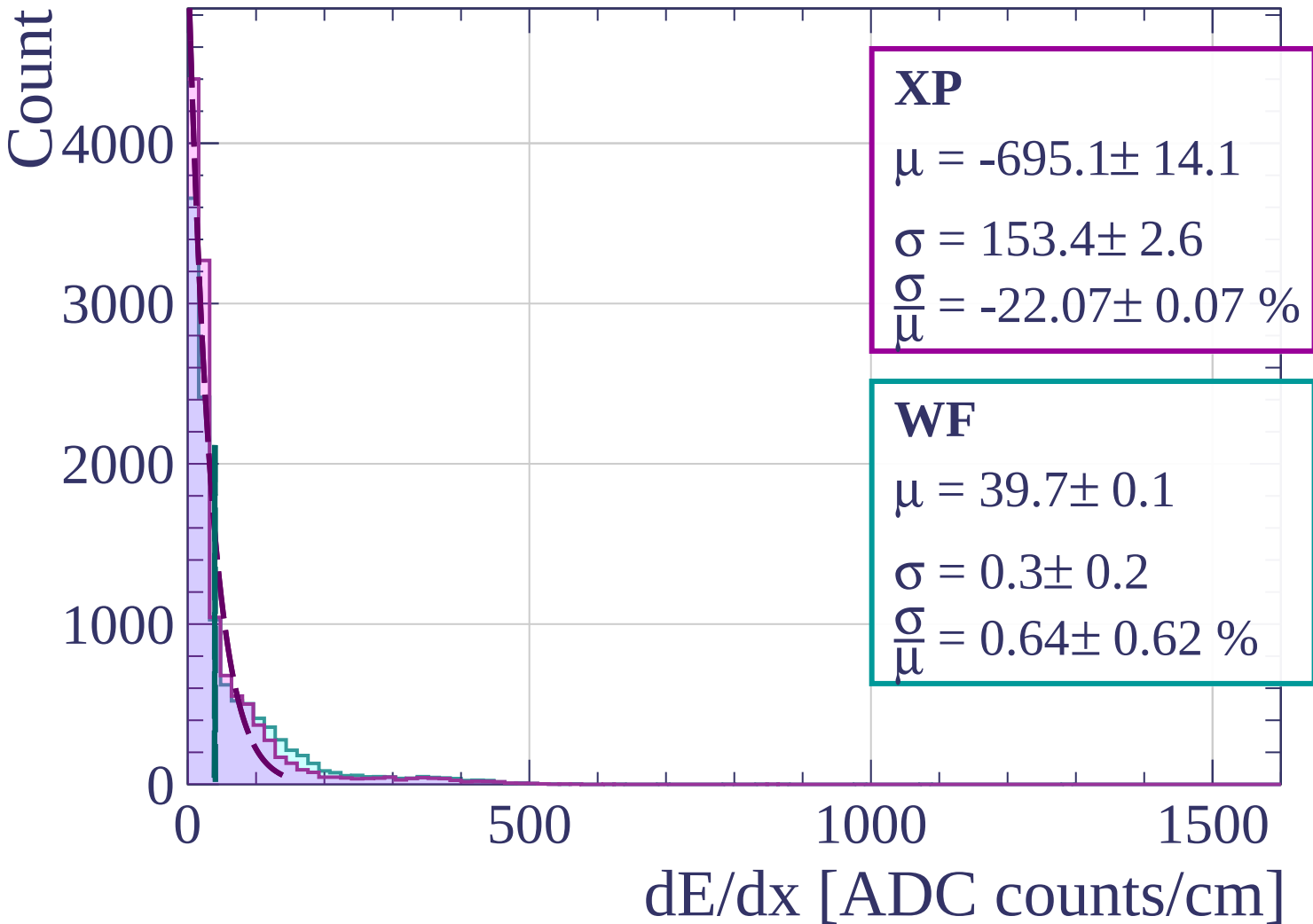
Energy loss | $69 < \theta < 72$



Energy loss | $72 < \theta < 75$

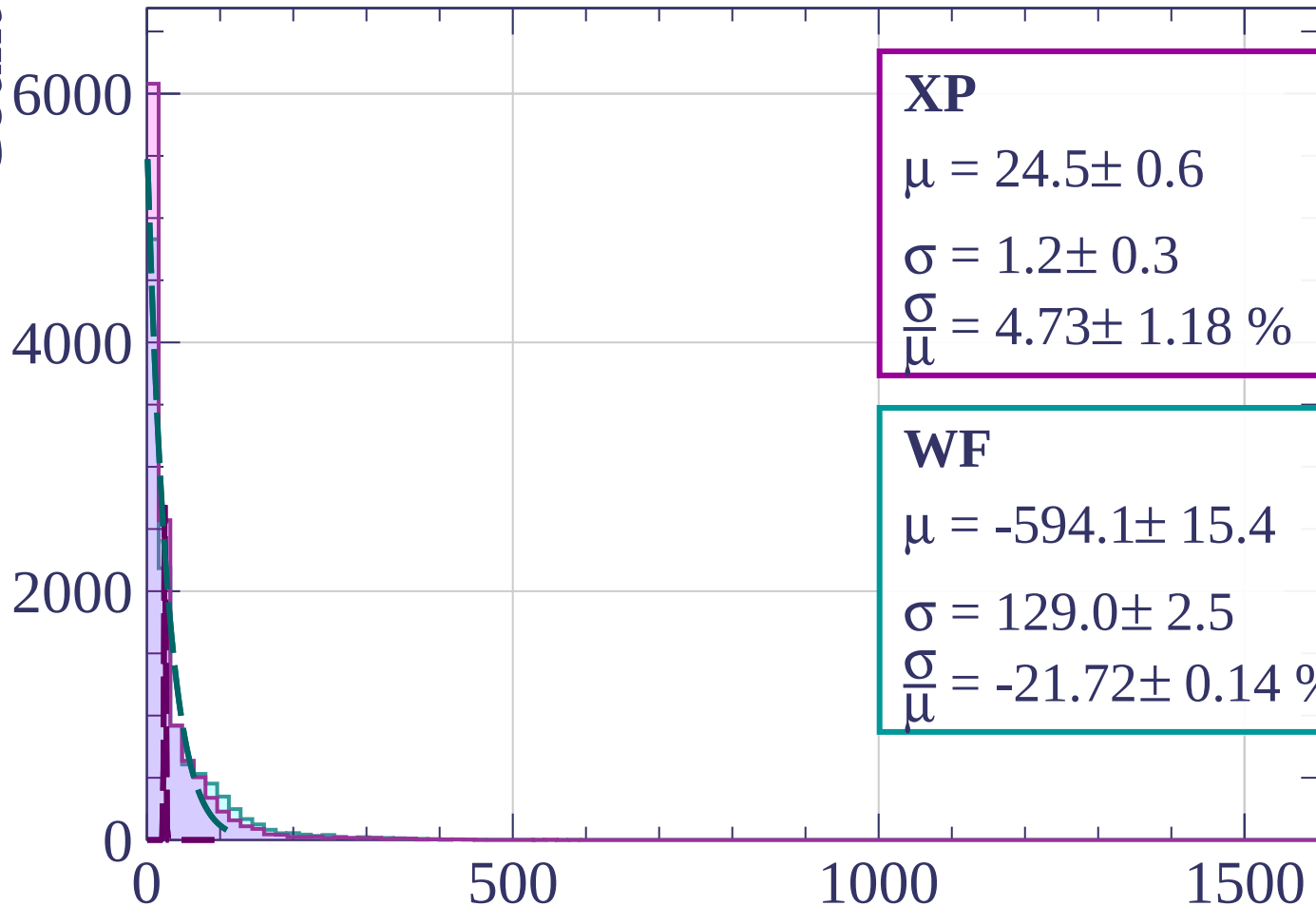


Energy loss | $75 < \theta < 78$



Energy loss | $78 < \theta < 81$

Count



XP

$$\mu = 24.5 \pm 0.6$$

$$\sigma = 1.2 \pm 0.3$$

$$\frac{\sigma}{\mu} = 4.73 \pm 1.18 \%$$

WF

$$\mu = -594.1 \pm 15.4$$

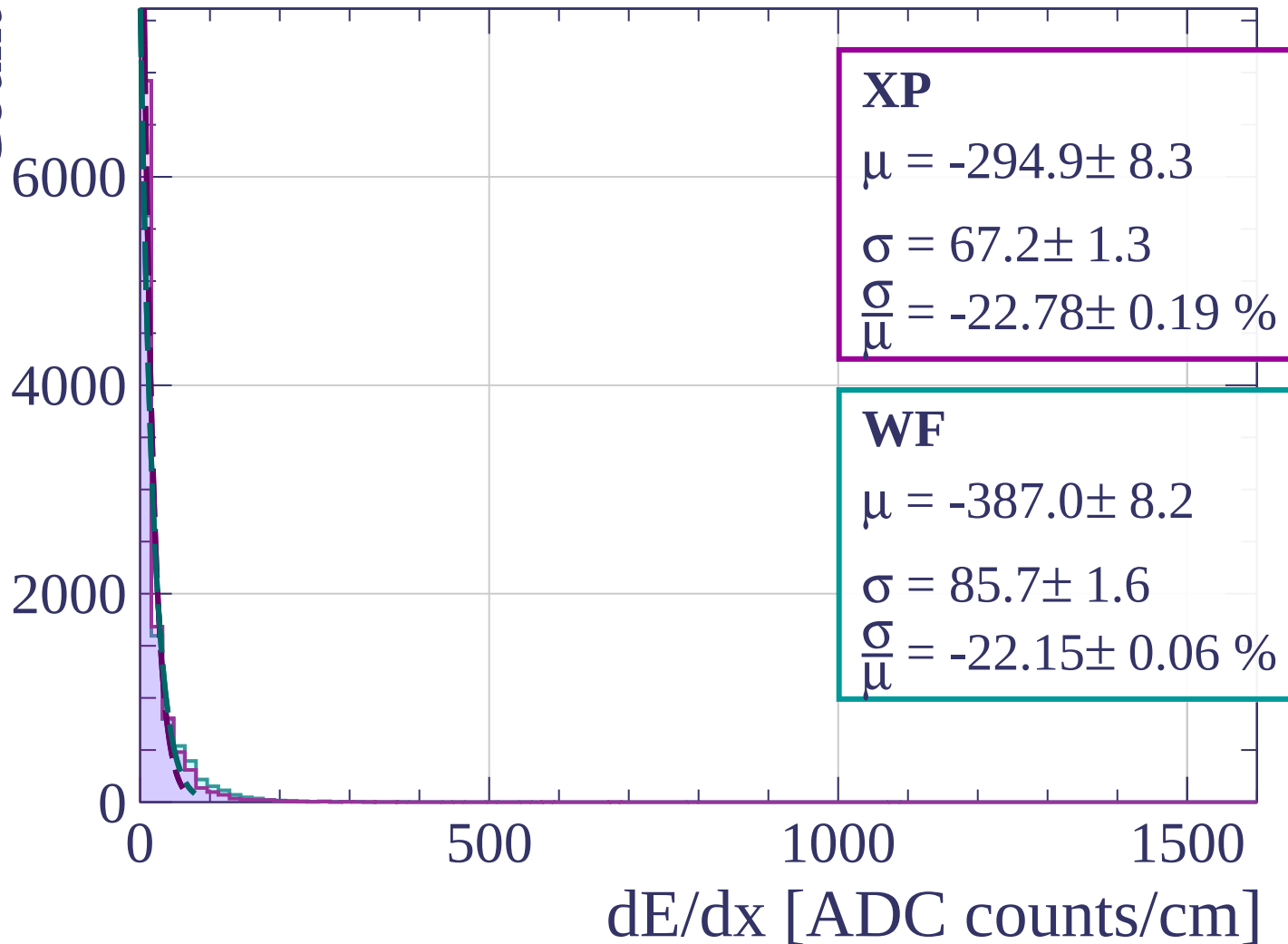
$$\sigma = 129.0 \pm 2.5$$

$$\frac{\sigma}{\mu} = -21.72 \pm 0.14 \%$$

dE/dx [ADC counts/cm]

Energy loss | $81 < \theta < 84$

Count



Energy loss | $84 < \theta < 87$

Count

4000

2000

0

0

500

1000

1500

dE/dx [ADC counts/cm]

XP

$$\mu = 7.2 \pm 0.9$$

$$\sigma = 1.9 \pm 0.5$$

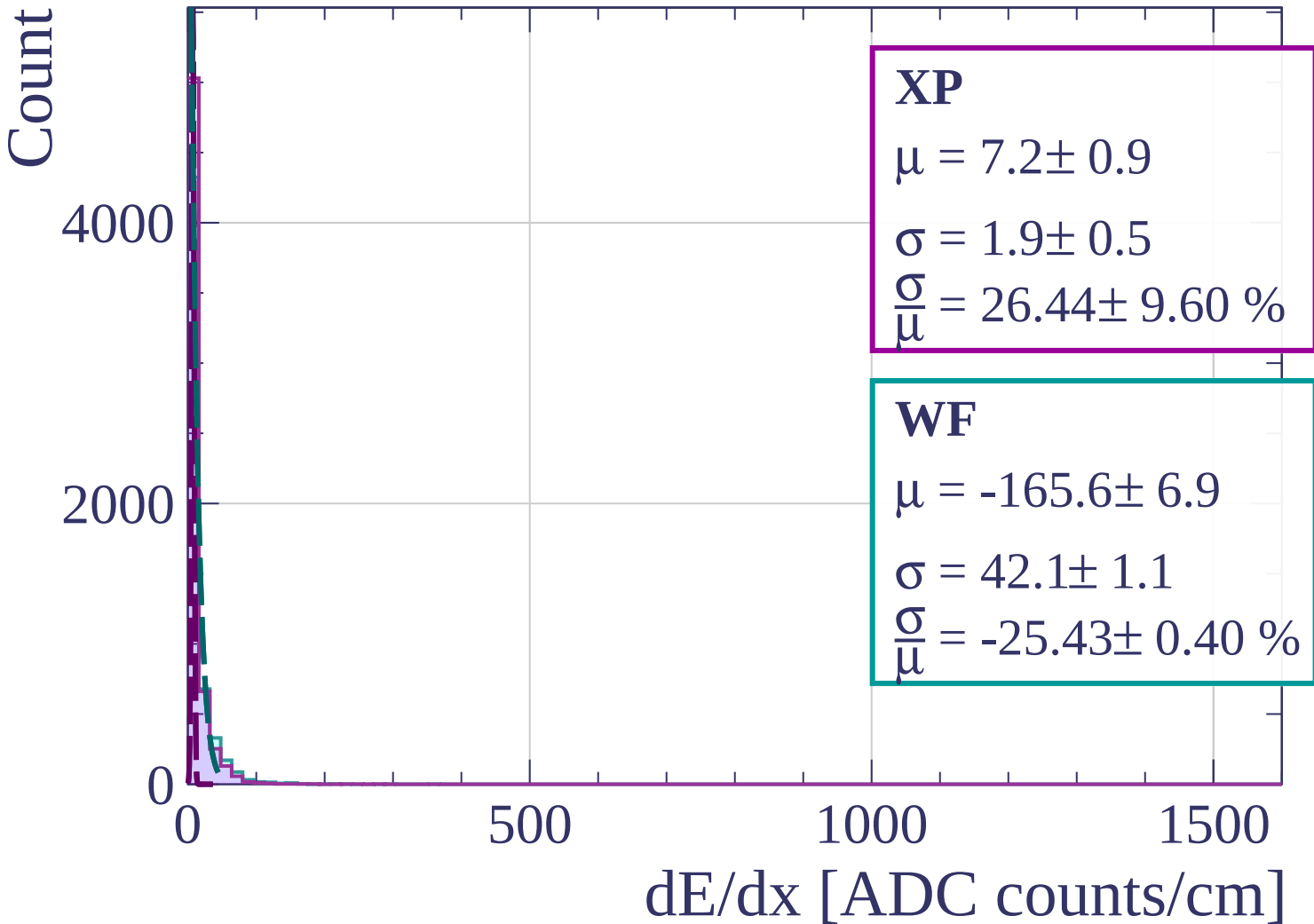
$$\frac{\sigma}{\mu} = 26.44 \pm 9.60 \%$$

WF

$$\mu = -165.6 \pm 6.9$$

$$\sigma = 42.1 \pm 1.1$$

$$\frac{\sigma}{\mu} = -25.43 \pm 0.40 \%$$



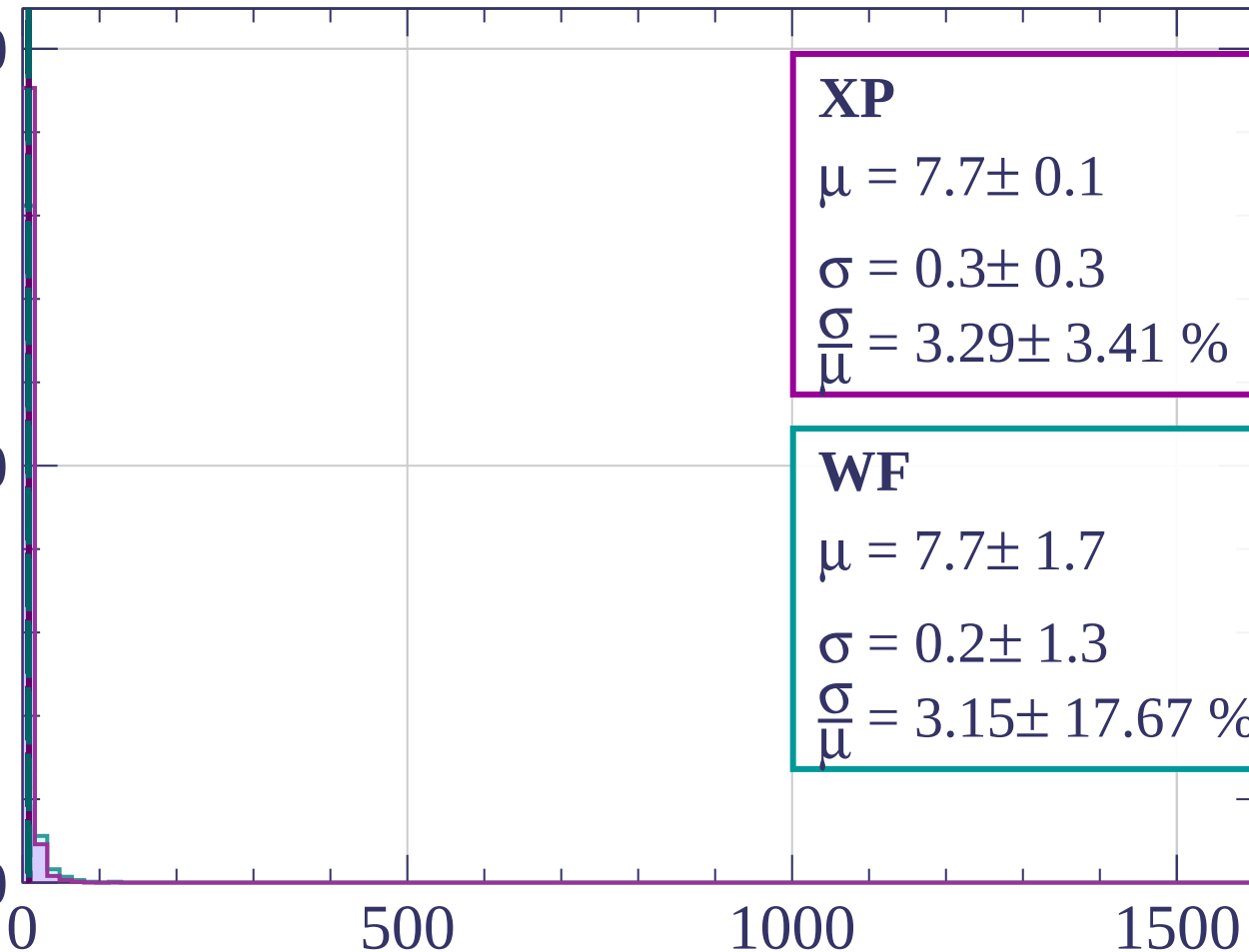
Energy loss | $87 < \theta < 90$

Count

1000

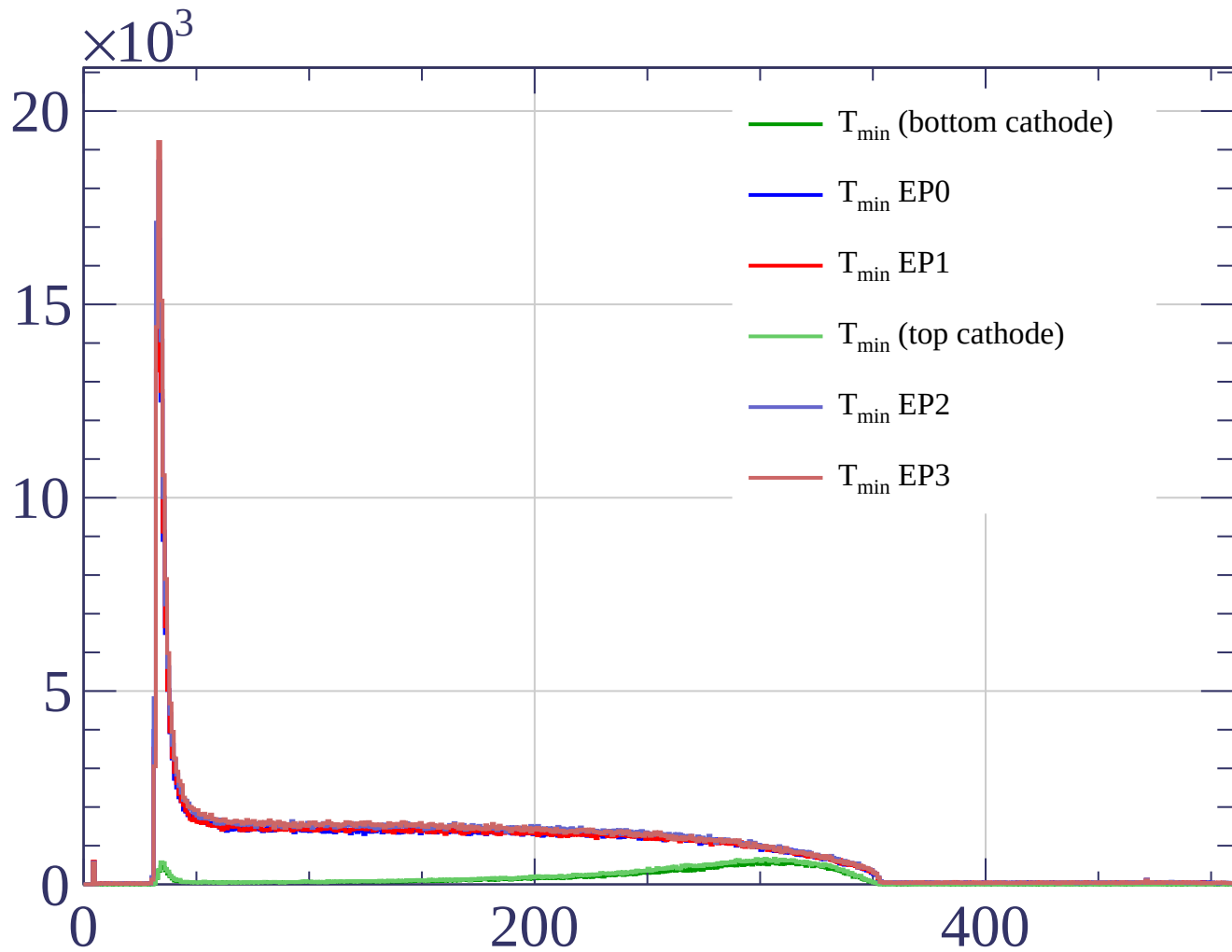
500

0



dE/dx [ADC counts/cm]

Count



Count

